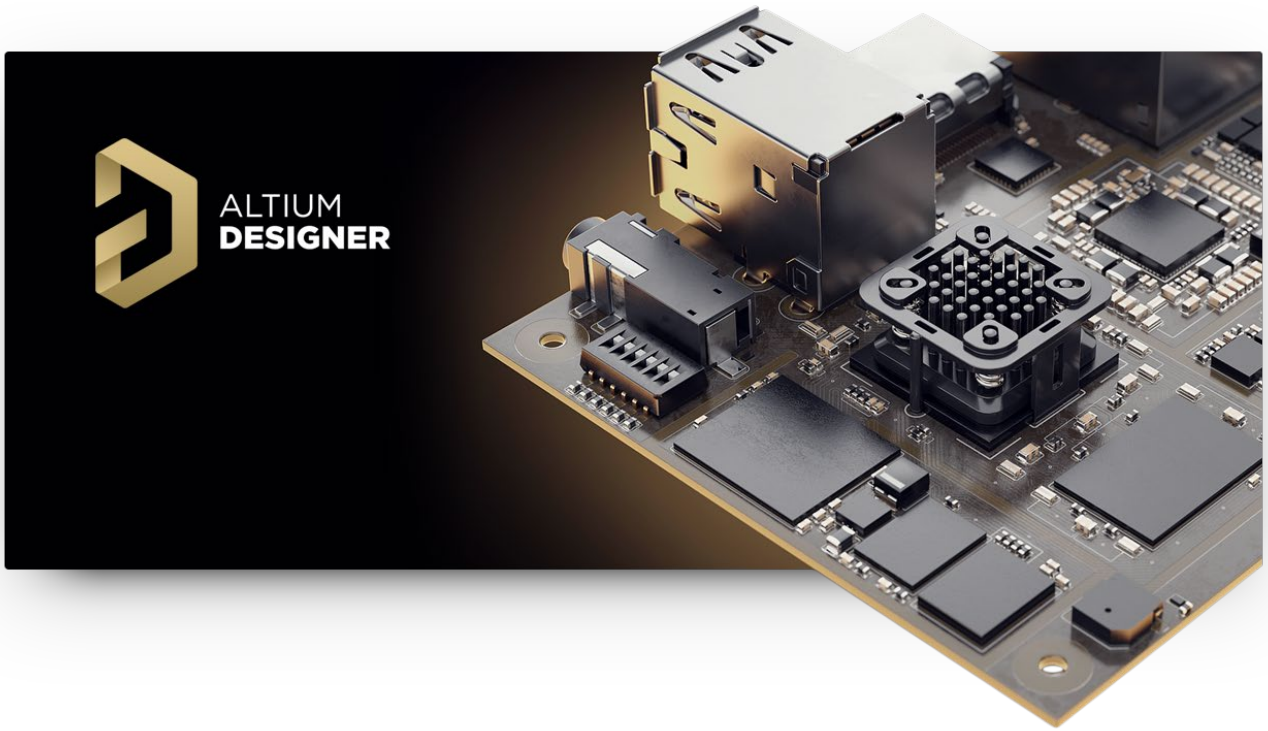

Altium Designer 20

Essentials Course

Software, documentation and related materials:

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Table of Contents

Table of Contents 3

Module 1: Design Environment 14

1.1 Purpose 14

1.2 Opening Files 14

1.3 Documents & Panels 17

1.3.1 Document Bar 17

1.3.2 Panel Access 18

1.3.3 Panel Display Modes 18

1.3.3.1 Docked Mode 18
1.3.3.2 Pop-up Mode 19
1.3.3.3 Floating Mode 20

1.3.4 Grouping Panels 21

1.3.4.1 Tabbed Panel Grouping 21
1.3.4.2 Fractal Panel Grouping 22

1.3.5 Closing Panels 22

1.3.6 Maximizing/Restore Panels 22

1.3.7 Desktop Layouts 23

1.4 Toolbars 23

1.4.1 Active Bar 23

1.4.2 Classic Toolbars 24

1.5 Menu Operations 25

1.5.1 Right-Click Context Menu 25

1.5.2 Shortcut Keys 25

1.6 Status Bar and Command Status 26

1.7 Undo and Redo 26

Module 2: It Starts with a Project 29

2.1 Purpose 29

2.2 Create a New Project 29

2.2.1 Project Wizard 29

2.2.2 Adding New Documents 30

2.2.3 Project Options 32

2.3 Package a Project for Easy Transfer 33

Module 3: Schematic Preferences 34
3.1 Purpose 34
3.2 Preparation 34
3.3 Setting Preferences 34
3.4 Graphical Editing 39
3.4.1 Always Drag 39
3.4.2 Convert Special Strings 39
3.5 Grids 42
3.6 Schematic Defaults 43
3.6.1 Default Primitives 43
Module 4: Navigating Schematics 45
4.1 Purpose 45
4.2 Preparation 45
4.3 Basic Schematic Preference Settings 45
4.3.1 Setting the Default Template & Sheet Size 45
4.3.2 Project Insight 47
4.3.3 Connectivity Insight 49
4.3.4 Schematic Navigator 53
4.4 Navigation within the Schematic Workspace 54
4.4.1 Basic Navigation using the Mouse 54
4.4.2 Basic Navigation using the Keyboard 54
4.4.3 View Menu Hotkey Sequences 55
4.4.4 Jump Menu Hotkey Sequences 55
4.4.5 Search with the Search Field 56
Module 5: Schematic Capture 57
5.1 Purpose 57
5.2 Preparation 57
5.3 Create A New Project 57
5.3.1 Adding New Documents 58
5.3.2 Adding an Integrated Library 59
5.4 Component Placement 62

5.4.1 Setting the grid 62
5.4.2 Processor Interface Schematic Component Placement 62
5.4.3 CAN_Interface Schematic Component Placement 65
5.4.4 Relay_IO Schematic Component Placement 69
5.5 Adding an Existing Schematic to Your Design 71
Module 6: Making the Connection 72
6.1 Purpose 72
6.2 Preparation 72
6.3 Wiring the Schematic 72
6.3.1 Shortcuts 72
6.3.2 Processor_Interface Schematic 72
6.3.2.1 Wire Connections 72
6.3.2.2 Smart Paste Wires 74
6.3.2.3 Power Ports 74
6.3.2.4 Net Labels 75
6.3.2.5 Ports 76
6.3.3 CAN_Interface Schematic 78
6.3.3.1 Net Labels 78
6.3.3.2 Smart Paste Ports 78
6.3.4 Digital_IO Schematic 79
6.3.4.1 Auto-Generated Wires 79
Module 7: Schematic Updating 81
7.1 Purpose 81
7.2 Align 81
7.2.1 Align Right 81
7.3 Formatting Net Labels 81
7.3.1 Selecting All Net Labels 81
7.3.2 Apply Formatting to Net Labels 82
7.4 Locking Components Designators 83
7.4.1 Selecting only 100nF 50V Capacitors 83
7.4.2 Lock the 100nF 50V Capacitors Designators 84
7.5 Net Color Highlighting 85
7.6 Schematic Annotation 87
Module 8: Schematic Graphics 89

8.1 Purpose 89
8.2 Preparation 89
8.3 CAN Interface Schematic 89
8.3.1 Text String 89
8.3.2 Round Rectangle 90
8.4 Number Schematic Sheets 94
8.5 Smart PDF 96
Module 9: Compiling in Altium Designer 98
9.1 Purpose 98
9.2 Preparation 98
9.3 Project Options 98
9.3.1 Error Reporting Tab 98
9.3.2 Connection Matrix Tab 100
9.3.3 Options Tab 101
9.4 Running the Compiler 102
9.4.1 Correcting Errors and Warnings 102
9.5 Dynamic Compiler 106
9.5.1 Enabling the Dynamic Compiler 106
Module 10: Adding Hierarchy 109
10.1 Purpose 109
10.2 Preparation 109
10.3 Create Top-Level Schematic 109
10.4 Sheet Symbols 109
10.4.1 Creating a Sheet Symbol from Scratch 109
10.4.2 Adding Sheet Entries 111
10.4.2.1 Use the Smart Paste to convert Ports to Sheet Entries 111
10.4.3 Creating a Sheet Symbol from Existing Sheet 113
10.5 Connecting the Sheet Symbols 114
10.5.1 Wiring the Sheet Symbols 114
10.5.2 Wiring the Relay Bus 114
10.5.3 Labeling the Bus and Bus Slices 114

10.6 Creating a Schematic Sheet from Sheet Symbol 115
10.6.1 Create Sheet from Sheet Symbol 115
10.6.2 Completing the Newly Created Schematic Sheet 115
Module 11: Transfer to PCB 118
11.1 Purpose 118
11.2 Preparation 118
11.3 Before the Transfer 118
11.3.1 Footprint Validation 118
11.3.1.1 Connect to the Content Vault 119
11.3.1.2 Install the WCT.IntLib library 120
11.4 Transferring the Design 121
11.4.1 The Initial Transfer 121
11.4.2 Cross Select Mode 121
11.4.3 Assigning Directives 122
Module 12: Updating the PCB 126
12.1 Purpose 126
12.2 Preparation 126
12.3 Component Links 126
Module 13: PCB Introduction 128
13.1 Purpose 128
13.2 Preparation 128
13.3 Moving PCB View 128
13.4 View Commands 129
13.5 General 129
13.6 DRC Violations Display 132
13.7 Interactive Routing 133
Module 14: PCB Viewing and Navigation 135
14.1 Purpose 135
14.2 Preparation 135
14.3 Jump Command 135

14.4 Cross Probe Mode 136
14.5 Cross Select Mode 137
14.6 3D Layout Mode 138
14.7 PCB Layers 139
14.8 View Configurations 140
14.9 Board Insight Display 142
14.10 Display 145
14.11 Board Insight Mode 146
Module 15: PCB Panels and Global Editing 147
15.1 Purpose 147
15.2 Preparation 147
15.3 Properties Panel 147
15.3.1 Modify PCB Traces 147
15.4 Filter Panel 149
15.4.1 Object Checkboxes 150
15.4.2 Query Selection 151
15.5 Find Similar Objects 154
15.5.1 PCB Find Similar Objects Dialog 154
15.5.1.1 Selecting Vias 154
15.5.1.2 Changing Via Size 155
15.6 List Panel 157
15.6.1 Selecting Components 157
15.6.2 Modifying Properties 158
Module 16: PCB Grids 160
16.1 Purpose 160
16.2 Preparation 160
16.3 Predefined Default Grids 160
16.4 Grid Manager 161
16.4.1 Global Board Snap Grid 161
16.4.2 Custom Cartesian Grid 163

16.4.3 Custom Polar Grid 166
Module 17: PCB Layers 168
17.1 Purpose 168
17.2 Preparation 168
17.3 Adding a new PCB 168
17.4 Layer Stack Manager 169
17.4.1 Modify Default Layer Stack 169
17.5 Layer Stack Table 172
Module 18: Shaping the PCB 174
18.1 Purpose 174
18.2 Preparation 174
18.3 Board Planning Mode 174
18.4 Define Board Shape from Selected Objects 178
18.4.1 Circle 178
18.4.2 Rectangle 180
18.5 Defining Board Shape from DXF/DWG 183
Module 19: PCB Design Rules 186
19.1 Purpose 186
19.2 Preparation 186
19.3 Basic Design Rule 186
19.3.1 PCB Rules and Constraints Editor 186
19.3.2 Rule Priorities 187
19.4 Design Rule Using Net Classes 189
19.5 Design Rule Using Custom Queries 192
19.5.1 Create Queries Using Find Similar Objects 192
19.5.2 Verifying Applied Design Rules in the PCB 194
19.6 Design Rule Wizard 196
Module 20: PCB Placement 198
20.1 Purpose 198

20.2 Preparation 198
20.3 Shortcuts 198
20.4 Placement of Components Within Rooms 199
20.4.1 Placing within an Area 200
20.4.2 Placing Individual Components 202
20.4.3 Cross Select Mode 203
20.4.4 Position Remaining Components 206
Module 21: PCB Routing 208
21.1 Purpose 208
21.2 Preparation 208
21.3 Shortcuts 208
21.4 PCB Routing 209
21.4.1 Setting Preferences 209
21.4.2 Quick Routing for Simple Nets 209
21.4.3 Interactive Routing Nets 209
21.4.4 Vias 211
21.4.4.1 Setting Via Sizes 211
21.4.4.2 Adding Vias 211
21.4.5 Multi-Trace Routing 212
21.4.6 Adjusting Tracks 214
21.4.7 Contour Follow Mode 214
21.4.8 Track Glossing 214
21.4.9 Active Route 215
Module 22: Polygon Pours 218
22.1 Purpose 218
22.2 Preparation 218
22.3 Placing a Polygon Pour 218
22.4 Polygon Properties 220
22.5 Polygon Pour Manager 222
22.6 Flooding a Layer 224
22.7 Editing Polygons 226
22.7.1 Editing an Edge 226

22.7.2 Adding a Vertex 227
22.7.3 Deleting a Vertex..... 227
22.8 Controlling the Connection Style - Polygon 228
22.9 Power Planes 230
22.9.1 Controlling the Connection Style – Power Plane..... 231
Module 23: DRCs on PCBs 232
23.1 Purpose 232
23.2 Preparation 232
23.3 Design Rule Check 232
23.4 PCB Rules and Violations Panel 234
23.5 Resolving Violations 236
Module 24: Draftsman 240
24.1 Purpose 240
24.2 Preparation 240
24.3 Creating a Draftsman Document 240
24.3.1 Creating Draftsman From a Template..... 240
24.3.2 Creating Draftsman Without a Template 241
24.4 Page Configuration 241
24.5 Assembly View 243
24.6 Fabrication View 245
24.7 Additional Views and Dimensions 246
24.8 Tables 248
24.8.1 Optional..... 248
Module 25: PCB Fabrication Files 249
25.1 Purpose 249
25.2 Preparation 249
25.3 Bill of Materials 249
25.4 Output Job File 251
25.4.1 Creating a New Output Job File 251

25.4.2 Defining Outputs **251**
25.4.2.1 Documentation Outputs 251
25.4.2.2 Assembly Outputs 252
25.4.2.3 Fabrication Outputs 253
25.4.2.4 Report Outputs 253
25.4.2.5 Validation Outputs 253
25.4.3 Defining Containers **254**
25.4.3.1 PDF Container 254
25.4.3.2 Folder Container 254
25.4.4 Generating the Outputs **255**
25.4.4.1 Generating the PDF 255
25.4.4.2 Generating the Folder Structure 255

Module 26: Schematic Library **257**

26.1 Purpose **257**

26.2 Create a Crystal Symbol **257**
26.2.1 Creating a Schematic Library **257**
26.2.2 Symbol Properties **257**
26.2.3 Defining the Pins **259**
26.2.4 Adding Symbol Body **261**

26.3 Using the Symbol **263**
26.3.1 Installing the Library **263**
26.3.2 Adding Symbol to Schematic **264**

26.4 Revising the Symbol **265**
26.4.1 Making a Change to the Symbol **265**
26.4.2 Updating the Schematic **265**

Module 27: PCB Library **266**

27.1 Purpose **266**

27.2 Creating a PCB Library **266**

27.3 Footprint Creation **267**
27.3.1 Setting the Grid **268**
27.3.2 Adding Pads **269**
27.3.3 Setting Component Origin **270**
27.3.4 Adding Top Overlay Outline **271**
27.3.5 Add Pin 1 Marker **272**
27.3.6 3D Crystal Body **273**

27.4 Use the Footprint **275**

27.4.1 Installing the Footprint Library 275
27.4.2 Adding Crystal Footprint to PCB Document 276
Module 28: Component Creation 277
28.1 Purpose 277
28.2 Linking Footprint to the Symbol 277
28.3 Using Linked Component 279
Module 29: Integrated Library 282
29.1 Purpose 282
29.2 Preparation 282
29.3 Create Integrated Library Project 282
29.4 Using the New Library 286

Module 1: Design Environment

1.1 Purpose

In this exercise, you will learn to use some general GUI and navigation functions. You will have mastered the basic operations of the Altium Designer Environment after the completion of this exercise. This will be the foundation for future studies.

Altium Designer is a unified design environment. When you run Altium Designer for the first time, it will become readily apparent through your first steps of working in the software that all EDA tools are included in a single application. This includes schematic design, simulation, signal integrity analysis, PCB design tools, documentation/fabrication/assembly outputs, and library creation.

1.2 Opening Files

1. Click **File » Open Project.** Select **Local Projects** and browse to the example projects folder, which is installed by default in `C:\Users\Public\Public Documents\Altium\<AD Version>\Examples\ Bluetooth Sentinel.`
2. Open the `Bluetooth_Sentinel.PrjPcb` from its Bluetooth Sentinel sub-folder as shown in Figure 1 below.

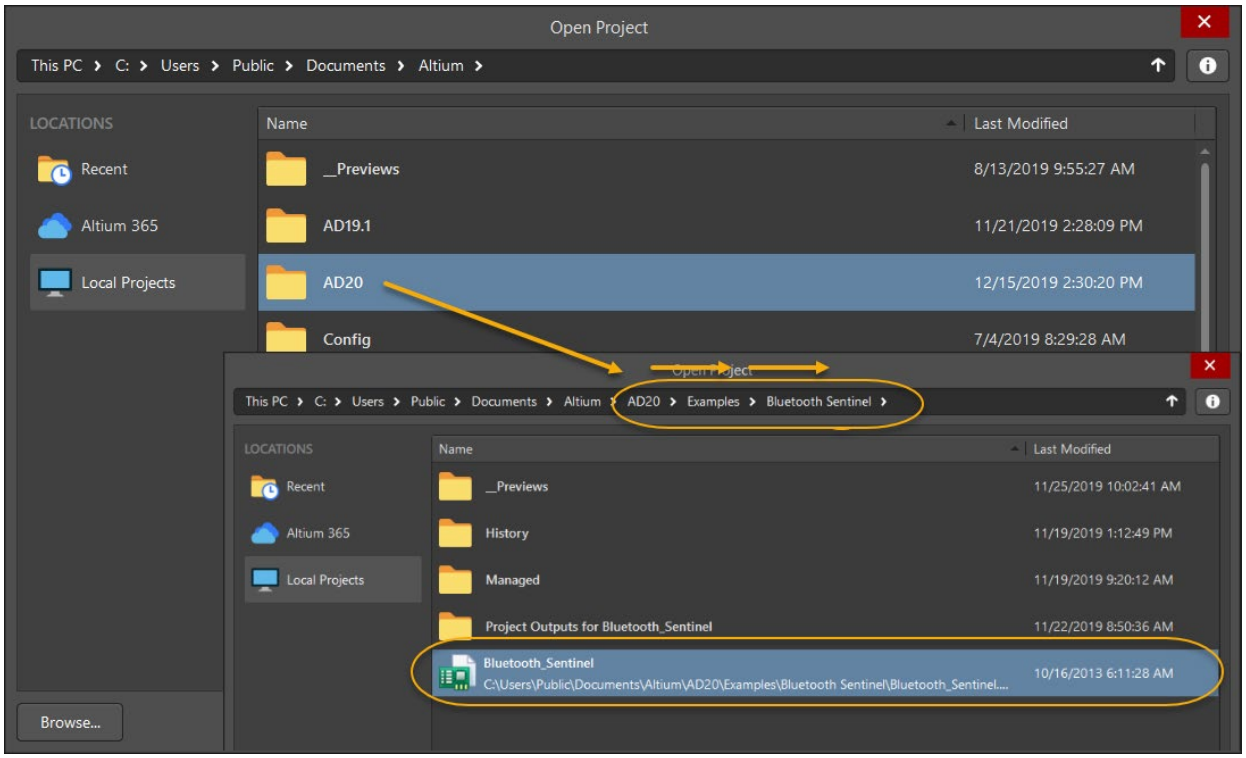


Figure 1. Open Project Dialog

NOTE: If this is your first time opening Altium Designer, some panels may appear in the middle of your screen. They can be closed by clicking the **"x"** in the top-right corner of the panel, or by right-clicking the title bar of the panel, and **Close.**

3. Locate the *Projects* panel on the left of the window as shown in Figure 2 below and double-click on the schematic file `Microcontroller_STM32F101.SchDoc`. This will open the schematic document within the active workspace and provide the working view of the schematic.
4. In the Projects panel, double-click on the PCB file `Bluetooth_Sentinel.PcbDoc`. This will open the PCB document within the active workspace and provide the working view of the PCB.

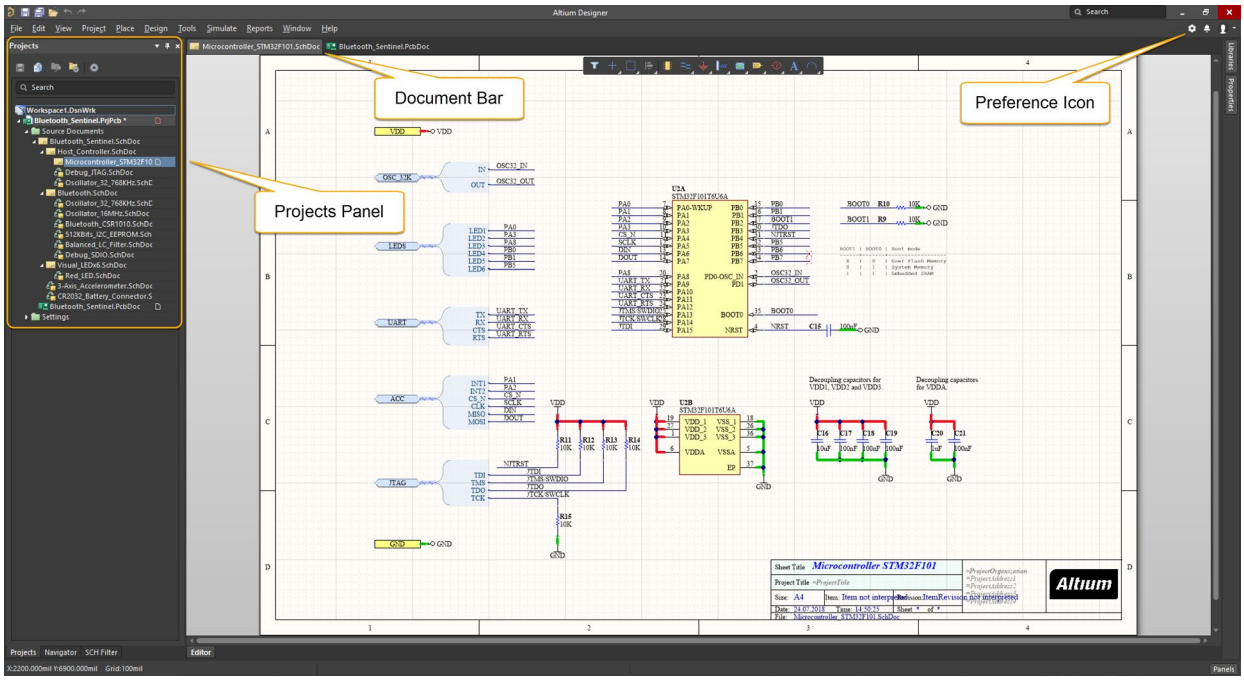


Figure 2. Altium Designer Environment

5. Notice that there are now two tabs on the Document Bar at the top of the editor. Click on each of these tabs and observe the workspace change to match the document that you are focused on.
6. Hit the shortcut keys **Ctrl+Tab** or **Ctrl+Shift+Tab** to cycle between the documents.
7. If we open multiple documents to the point where the tabs fill up the entire bar, we can configure its grouping method. Click the *Preferences* icon on the upper right corner of Altium Designer. Expand the *System* branch and open the *View* page. Within the *Documents Bar* settings, you can enable the **Group documents if need** checkbox and set it to **By document kind** as shown in Figure 3.

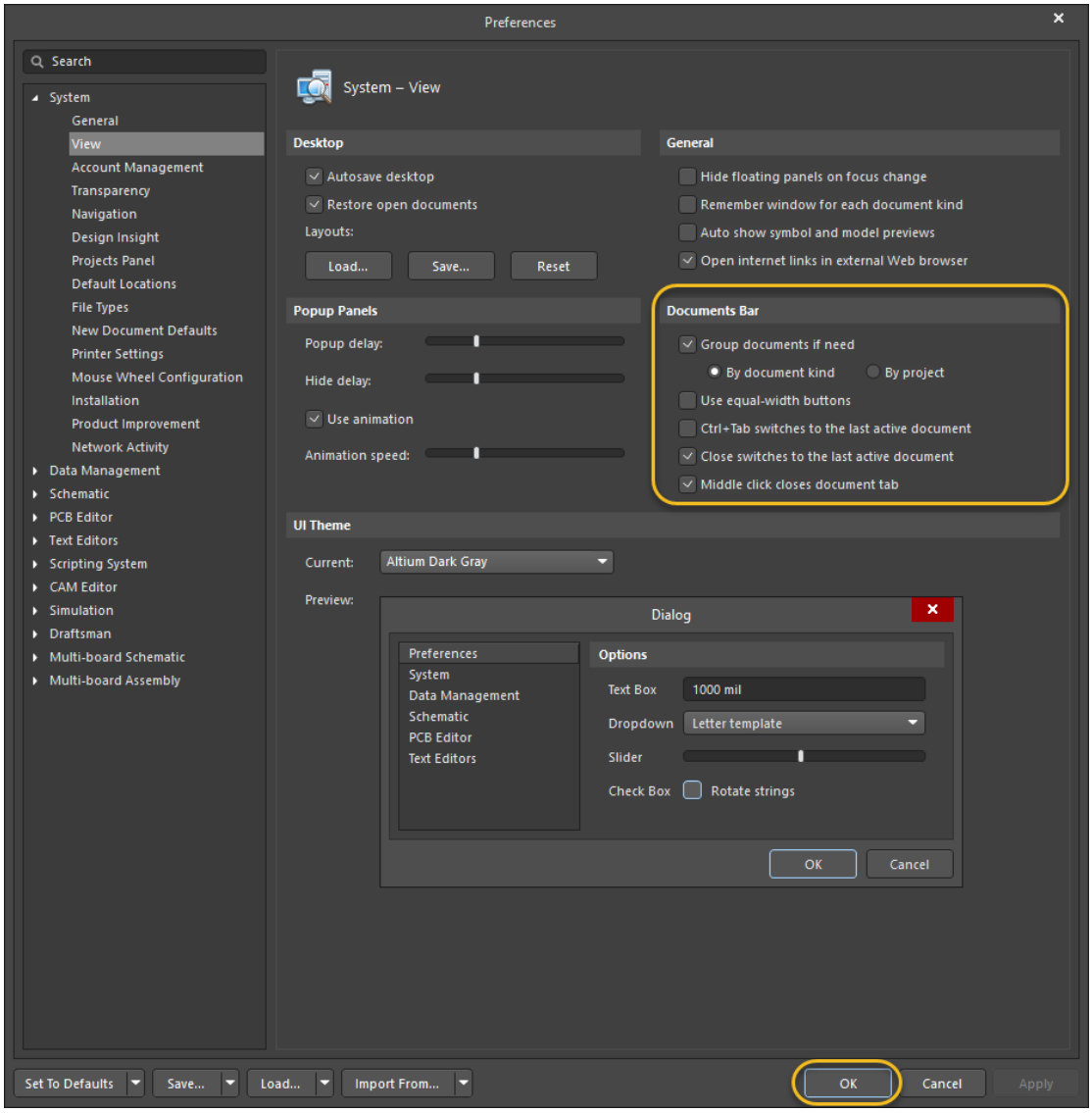


Figure 3. Preferences for document grouping

8. Click **OK** to close the *Preferences* window.

If enabled, the documents will be grouped as needed. If there is adequate space within your documents bar, the documents will not be grouped.

1.3 Documents & Panels

1.3.1 Document Bar

9. Right-click on a Document tab in the Document Bar (as shown in Figure 4 below) and select **Tile All** in the pop-up menu. All the currently open documents will be split into individual windows and will be displayed within the main design window.

To resize the window, simply move the cursor to the divider of the two split windows. When a bi-directional arrow appears, left-click and drag to resize the split windows.

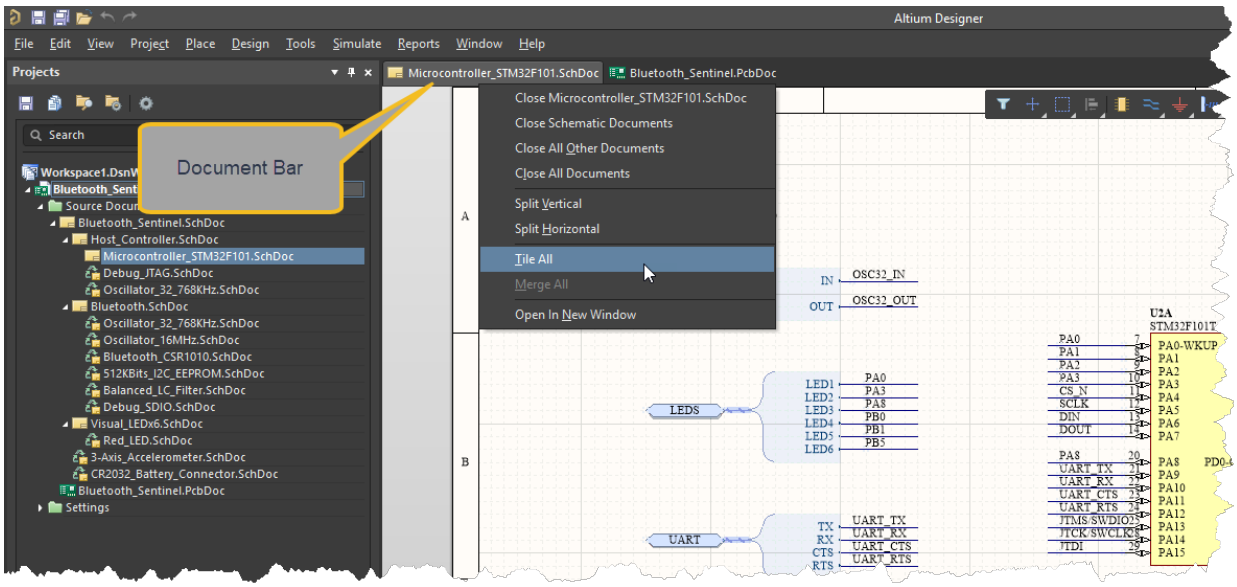


Figure 4. Document Bar

10. While documents are tiled or split within the editing workspace, right-click on any document tab in the Document Bar and select **Merge All** from the pop-up menu. The workspace is restored to a single document display.
11. Feel free to experiment with the other options in the right-click menu like **Split Vertically** and **Split Horizontally**. For multi-monitor display, click on **Open in New Window** in the right-click menu to open a new instance or simply drag a document tab to the secondary display.

1.3.2 Panel Access

12. Locate the **Panels** button in the lower right corner as shown in Figure 5 below.
13. Select **Components** from the popup to open the *Components* panel. The panels are also listed in the menu under **View » Panels.**

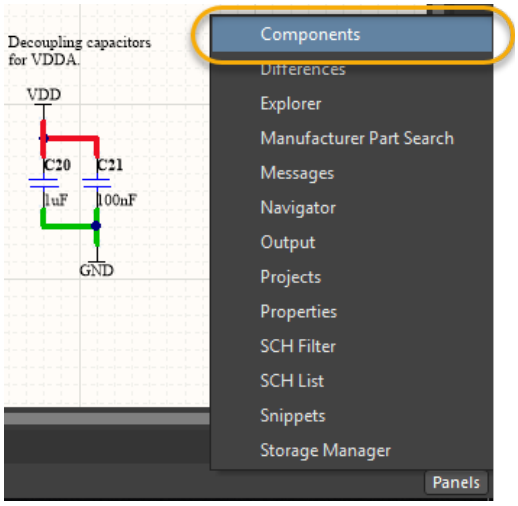


Figure 5. Panel access

1.3.3 Panel Display Modes

Three different display modes are supported for panels - Docked, Pop-up and Floating.

1.3.3.1 Docked Mode

14. Right-click on the title bar of the *Projects* panel. Click on **Allow Dock** in the pop-up menu as shown in Figure 6 below and ensure that it is enabled for both **Horizontally** and **Vertically**.

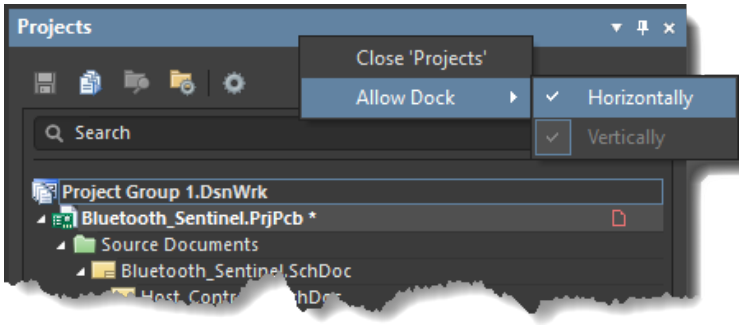


Figure 6. Allowed docking mode settings

15. Left-click on the title bar of the *Projects* panel to move it. Drag it to the center of the window and notice that while the panel is attached to the cursor, we have both horizontal and vertical docking option as shown in Figure 7 below. Hover the cursor above the left docking option and release the left-click hold.

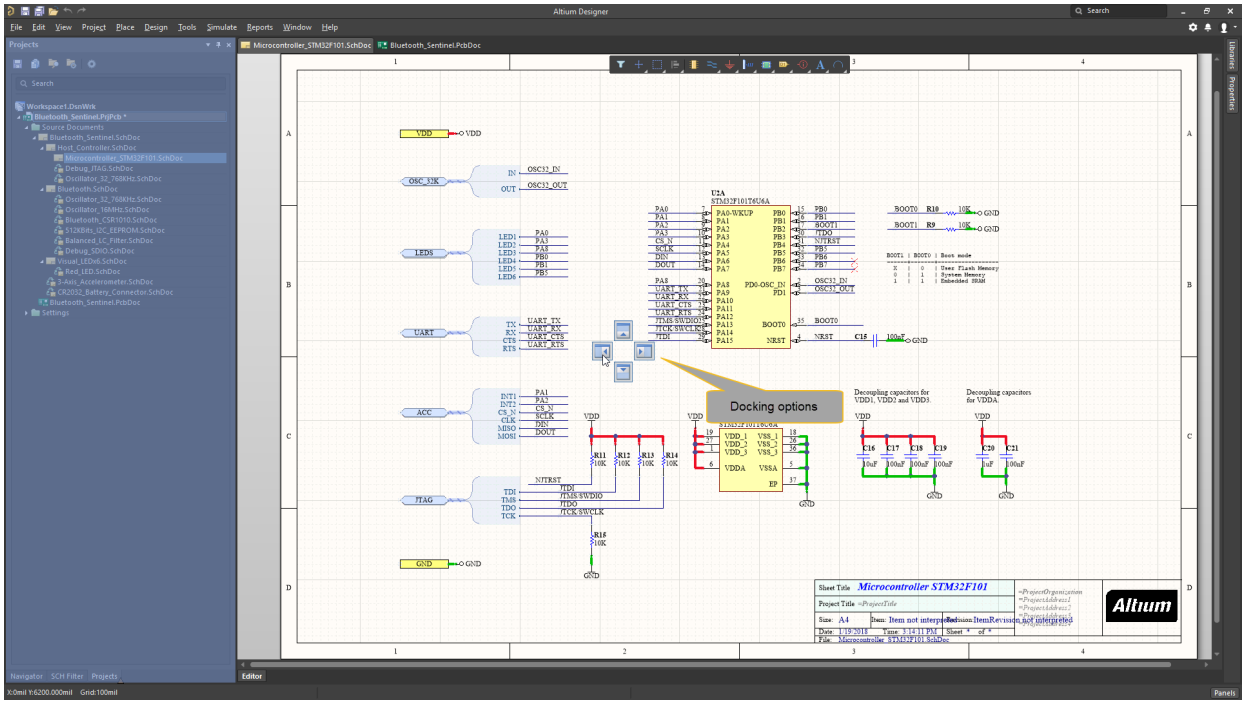


Figure 7. Docking Options

1.3.3.2 Pop-up Mode

16. Locate the *Components* panel tab on the right of the window. Hover above its tab to open the panel. This mode is what we refer to as Pop-up mode, and is often used to maximize screen's real estate especially when working on a small monitor. If the *Components* panel is not in pop-up mode, head to the **Panels** button to the lower right side of Altium Designer to open it.
17. To change the *Components* panel to a docked mode, click on the pin symbol while the panel is open. The pin symbol will then change to indicating that it is in docked mode.
18. To revert it back to a pop-up mode, click on the pin symbol again. While in this mode, moving the cursor away would hide then panel after a brief amount of time.
19. Note that the pop-up and hide delay is controlled in Preferences under the *System* branch, in the *View* page as shown in Figure 8.

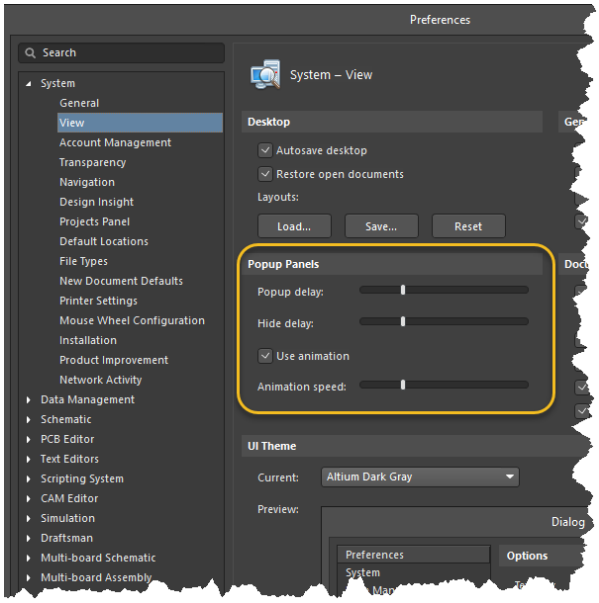


Figure 8. Pop-up panel delay settings

1.3.3.3 Floating Mode

20. In the bottom right corner of Altium Designer, click on **Panels** and select **Storage Manager** to open the *Storage Manager* panel shown in Figure 9 below. By default, this is a floating panel and handy to explain Floating Mode.

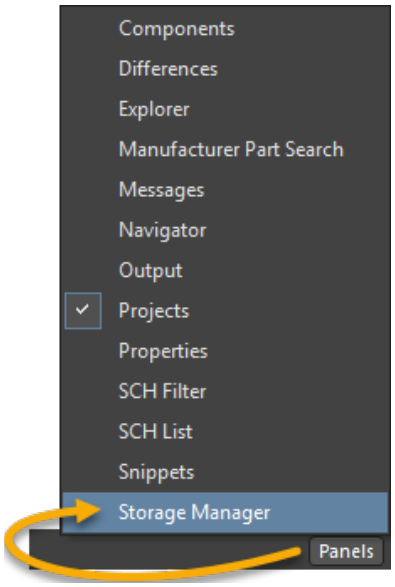


Figure 9. Opening the Storage Manager panel

21. A panel in this mode can be placed anywhere within or outside the Altium Designer environment. For example, the *Components* panel is commonly made to be floating and placed on another monitor. Move the *Storage Manager* panel to the middle of the window.
22. To observe the transparency effect of this floating panel, invoke a command by going to **Place » Wire**. Move the command crosshair towards the panel and notice the change of its transparency. Note that the transparency settings can be controlled in Preferences under the *System* branch, in the *Transparency* page as shown in Figure 10 below.
23. End the **Place » Wire command** using the **ESC key** and close the *Storage Manager* panel by clicking on the close cross button.

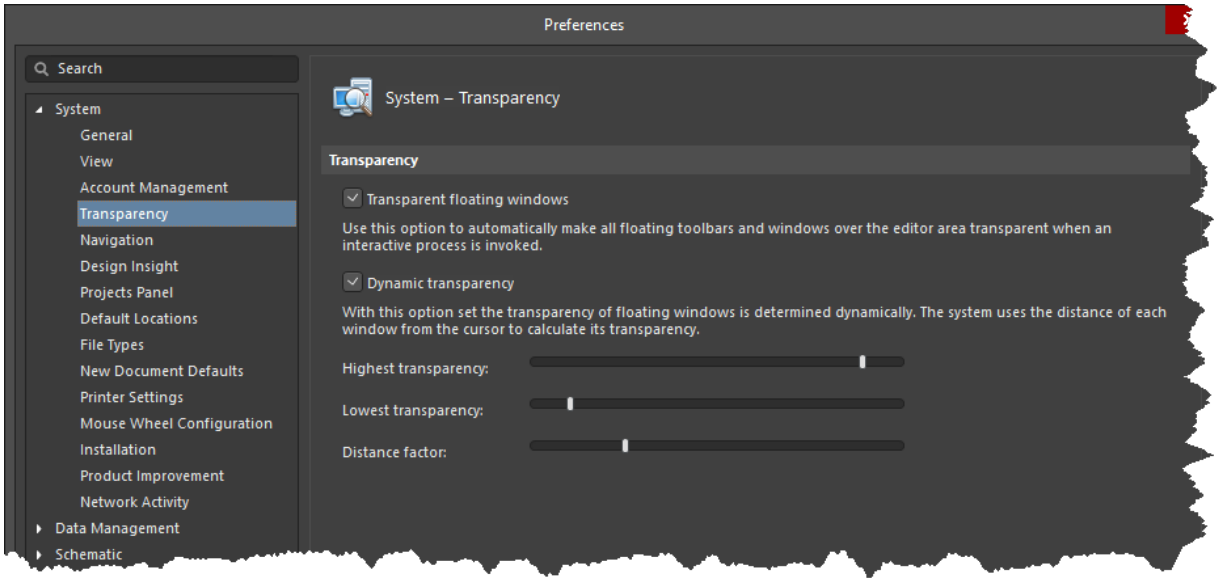


Figure 10. Transparency setting for Floating panels

1.3.4 Grouping Panels

Two modes of grouping panels are supported: Tabbed Grouping and Fractal Grouping.

1.3.4.1 Tabbed Panel Grouping

24. By default, the *Projects*, *Navigator* and *SCH Filter* panels are grouped together in one panel. To open any one of them, click on the tabs at the bottom of the panel shown in Figure 11. This is what we refer to as Tabbed Panel Grouping.

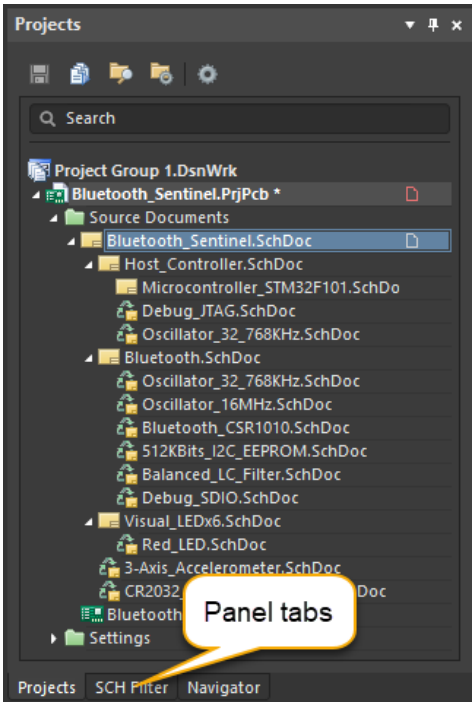


Figure 11. Tabbed Panel Grouping

1.3.4.2 Fractal Panel Grouping

25. Left-click on Projects tab of the panel group and drag it out to enter Floating mode. Drag it back into the original panel area and notice that as we hover above any of the 4 directional icons, it will display a blue highlight as shown in Figure 12.
26. Release the click on the top directional icon and notice that the panel will be split into fractal segments.

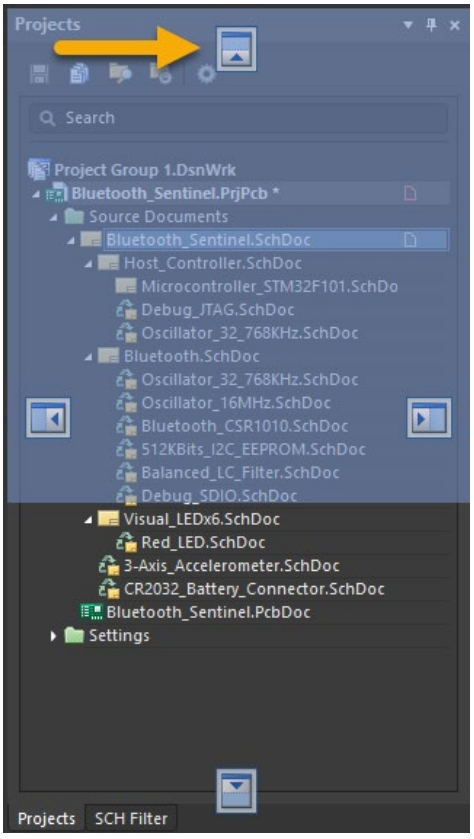


Figure 12. Fractal Panel Grouping

To prevent a moving panel from automatically docking, grouping or snapping, hold the **CTRL** key

1.3.5 Closing Panels

27. Close any panel by right-clicking in its title bar or on its tab, then choose **Close** from the pop-up menu. Alternatively, click the **close cross** ✕ at the far right of a panel's caption bar, if it is not in tabbed mode.

If the panel is part of a group, clicking the ✕ entry closes all panels in the group. To close a particular panel, right-click on its name in the title bar and close.

1.3.6 Maximizing/Restore Panels

28. When in floating mode, a panel can be maximized by right-clicking in its title bar (or on its tab, where available) and choosing the **Maximize** entry from the pop-up menu.
29. To restore a maximized panel back to its original size, simply right-click in its title bar or tab and choose the **Restore** command from the pop-up menu. Alternatively, you can double-click the title bar to toggle between maximized and restored states.

1.3.7 Desktop Layouts

30. The panel arrangements are what we refer to as Desktop Layouts. It can be saved, loaded and reset to default in Preferences. Open the *Preferences*, accessible from the upper right corner of Altium Designer.
31. Open the *System* branch and open the *View* page as shown in Figure 13 below. Within the Desktop pane, desktop layouts can be saved, loaded or reset to default.
32. Click on **Reset**. Close the *Storage Manager* and *SCH List* panel if they're open. Then, click **OK** to exit the Preferences.

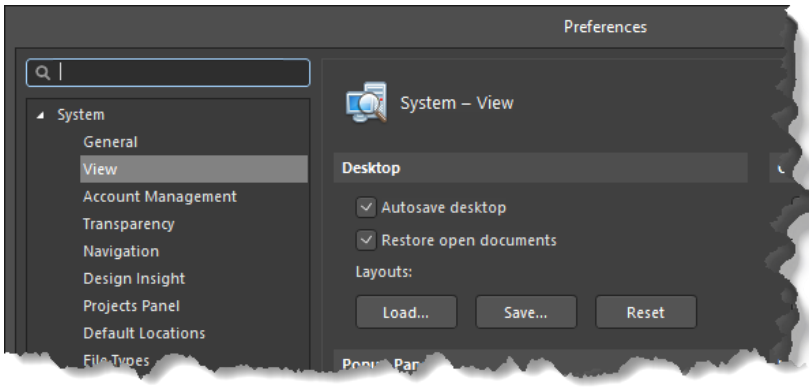


Figure 13. Desktop Layouts

1.4 Toolbars

Like menus and panels, both the *Active Bar* and Classic Toolbars are context sensitive, so only commands that are relevant to the document of focus are displayed.

1.4.1 Active Bar

33. Make the schematic the current document of focus. At the top of the document, the *Active Bar* appears, as shown in Figure 14 below.
34. Move the cursor to any of the icons in the *Active Bar*. Wait a second for a hint to appear describing the function of the tool.

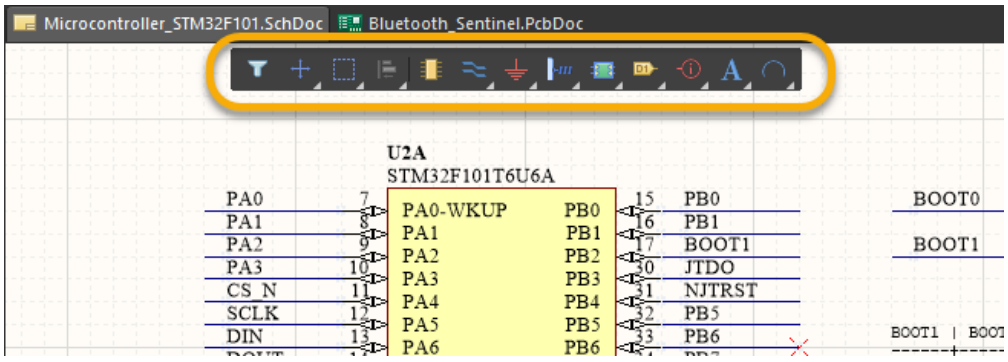


Figure 14. Active Bar in the Schematic Editor

35. Open the PCB document. Observe the icons as shown in the *Active Bar* and note the changes shown in Figure 15 below.

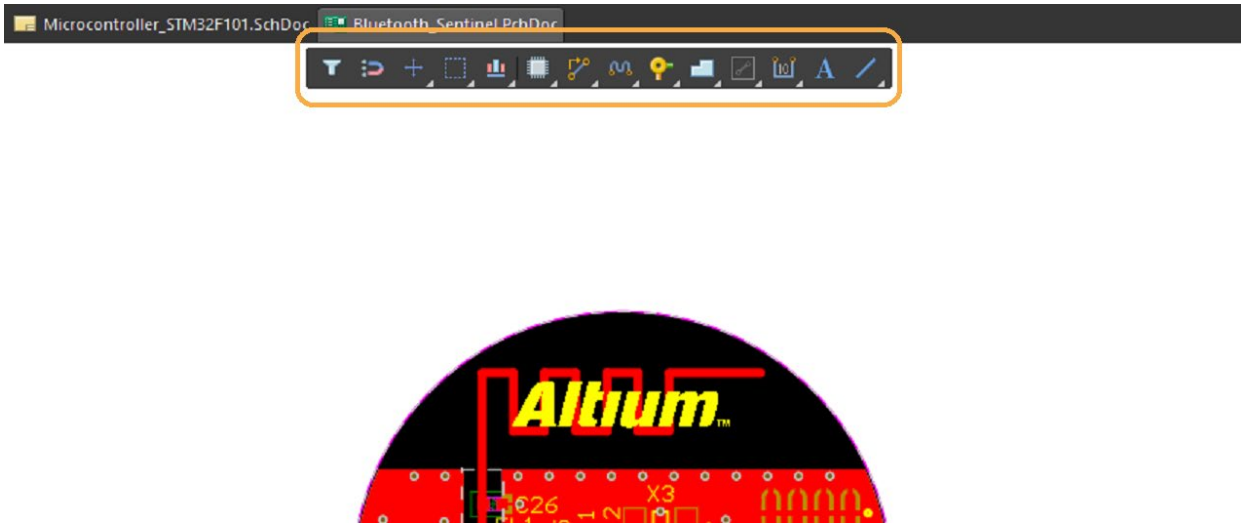


Figure 15. Active Bar in the PCB Editor

36. Return to the schematic document.

1.4.2 Classic Toolbars

In addition to the Active Bar, classic toolbars are also available.

37. To open a classic toolbar, go to **View » Toolbars** and select **Schematic Standard** to enable it.
38. Notice that the toolbar will then be added underneath the menu list on the top of the window as shown in Figure 16 below**.** To disable it, simply return to **View » Toolbars** and uncheck on **Schematic Standard.**

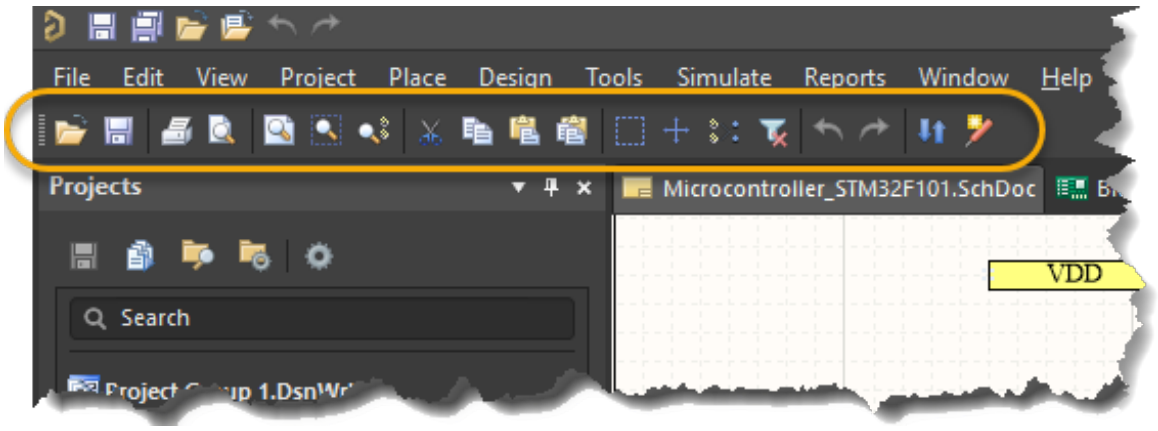


Figure 16. Classic Toolbar

1.5 Menu Operations

1.5.1 Right-Click Context Menu

39. Right-click any free space in the schematic editor and observe the changes to the menu as shown on the left side of Figure 17 below.
40. In the `Microcontroller_STM32F101.SchDoc`, left-click to select any component, then right-click to observe the content in the context menu as shown on the right side of Figure 17 below.

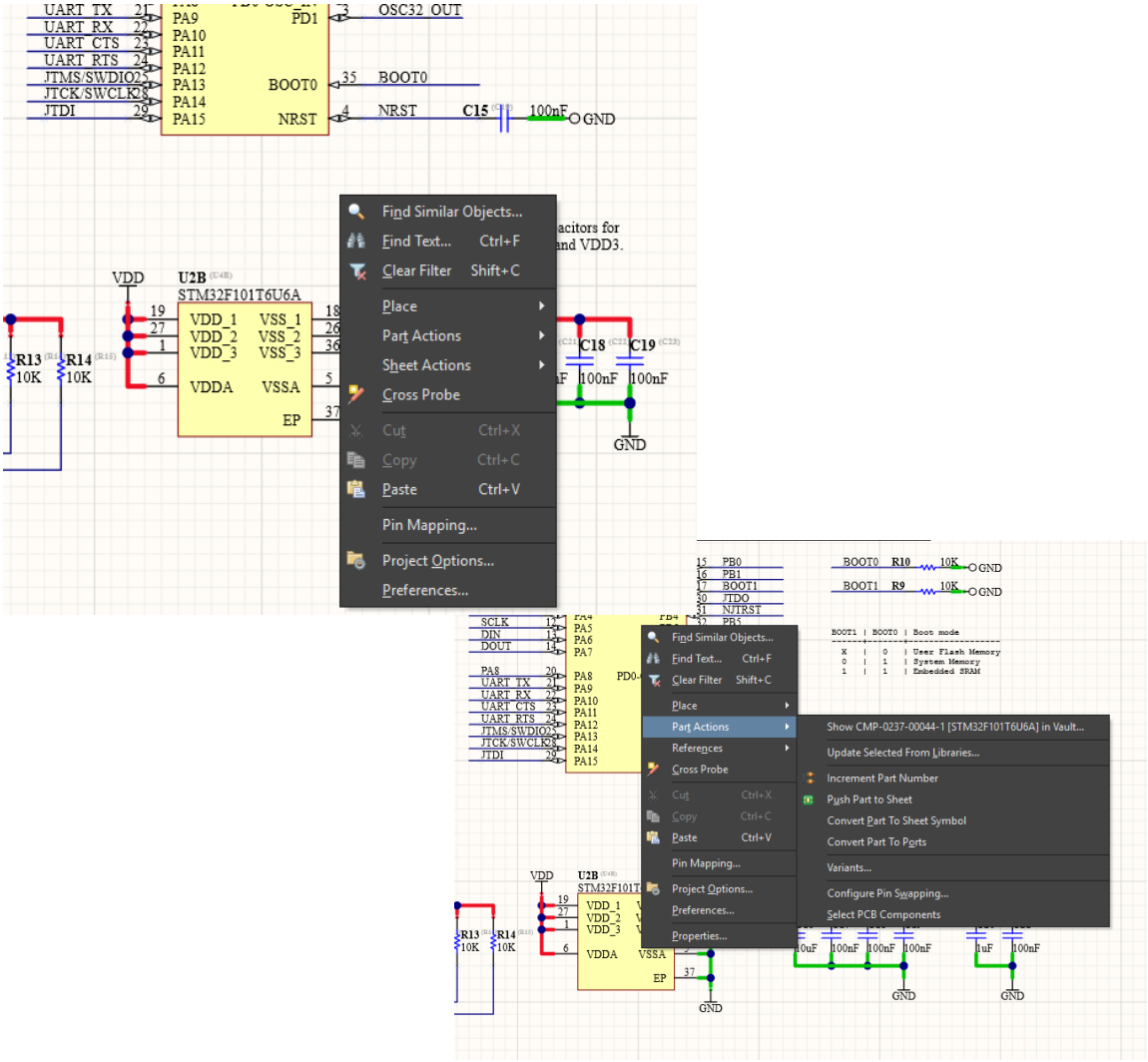


Figure 17. Right-click context menu

41. Feel free to explore the menus and options available while in the right-click menu.

1.5.2 Shortcut Keys

42. Some menu and menu entries have underlined letters.
    a) With the schematic in focus, access the **File** Menu by pressing the **F** key directly. The File menu opens at the location of your cursor.
    b) From the popup, press the **S** key to access the **Save** command.

1.6 Status Bar and Command Status

43. Click the **View** menu and locate the **Status Bar** and **Command Status** entries as shown in Figure 18 below.

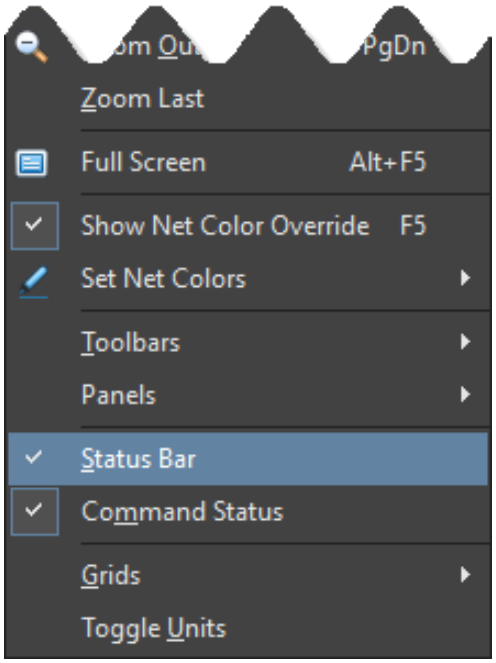


Figure 18. Activation status for the Status Bar and Command Status

44. Ensure that both entries are enabled with a *check* next to their name. To enable the options, click the item to toggle it.
45. Move the cursor in the schematic editor area and you'll notice the X and Y values changing in the Status Bar, as shown in Figure 19 below.
46. Go to **Place » Wire.** The active command is displayed in the command status area shown in Figure 19 below. Hit **Escape** or right-click to exit from the command.

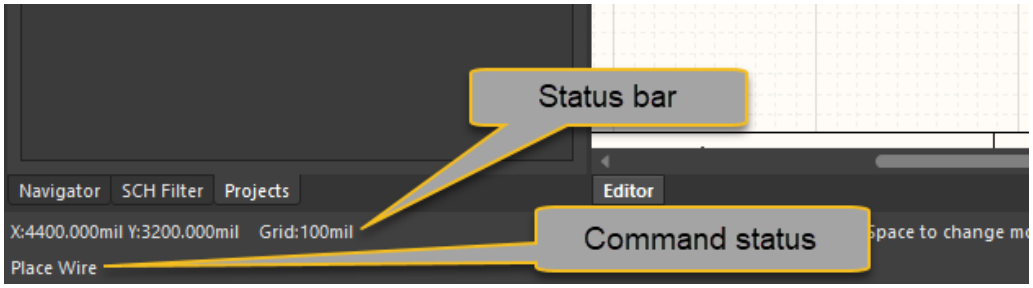


*Figure 19. Status Bar and Command Bar information with in the **Place » Wire** command*

1.7 Undo and Redo

47. Select a component and hit **Delete** to delete a component in the schematic.

48. Click the Undo and Redo icons in the top left corner as shown in Figure 20 below.

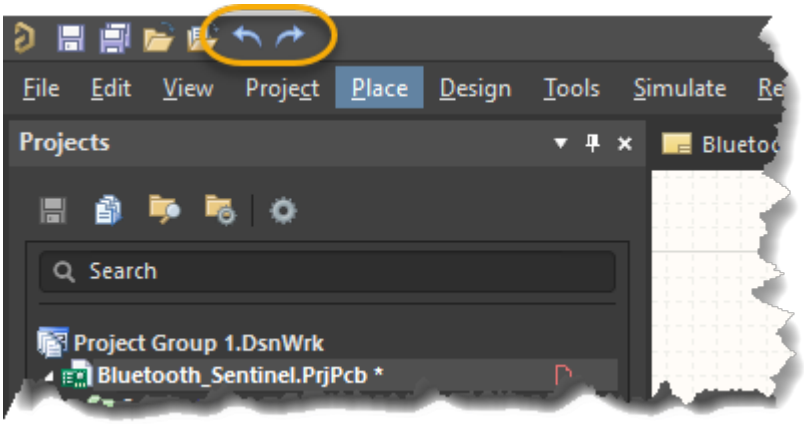


Figure 20. Undo and Redo icons

49. As an alternative, use the shortcut key **Ctrl+Z** or **Alt+Backspace** for Undo. For redo, the shortcut key is **Ctrl+Y** or **Ctrl+Backspace**.

Feel free to experiment with your panels, opening them, closing them, docking, undocking, creating tabbed and fractal groups. You can always revert to defaults so give them a try and become comfortable with their controls.

50. Open the *Projects* panel. Right click on the project and click on **Close Project** on shown in Figure 21. Click **Save None** and click **OK**.

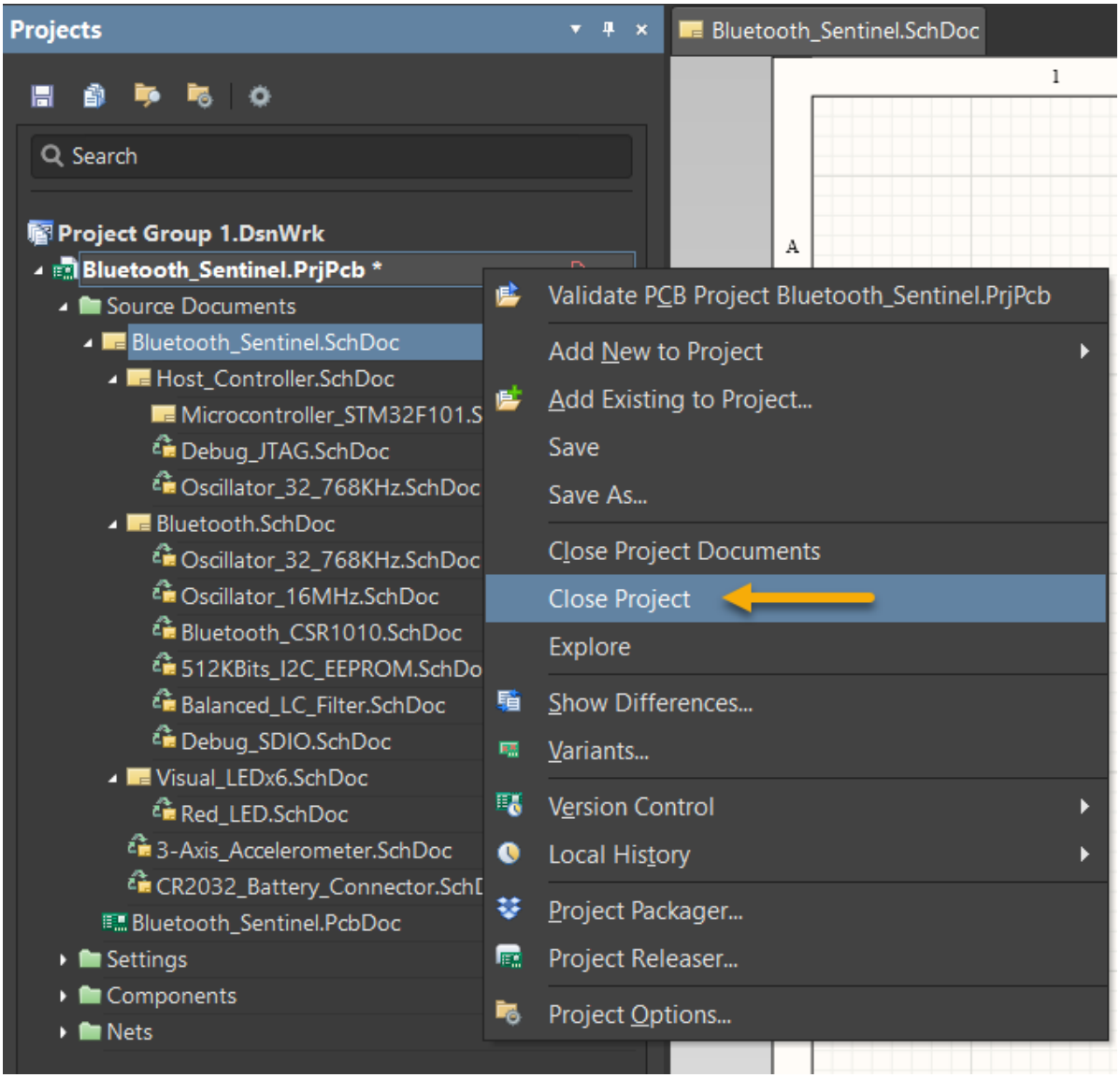


Figure 21. Close Project

Module 2: It Starts with a Project

2.1 Purpose

In this exercise, you will learn to create a new project, add new and existing files to the project and save the project. You will then package the project for archiving or to transfer to external parties.

A project file contains references to its design documents such as schematic and PCB documents. It also stores project-specific options and parameters defined. In this exercise, we will be creating a new PCB project.

2.2 Create a New Project

2.2.1 Project Wizard

1. Close all open projects in the *Project* panel.
2. Go to **File » New » Project...** to launch the *Create Project* dialog as shown in Figure 22 below.

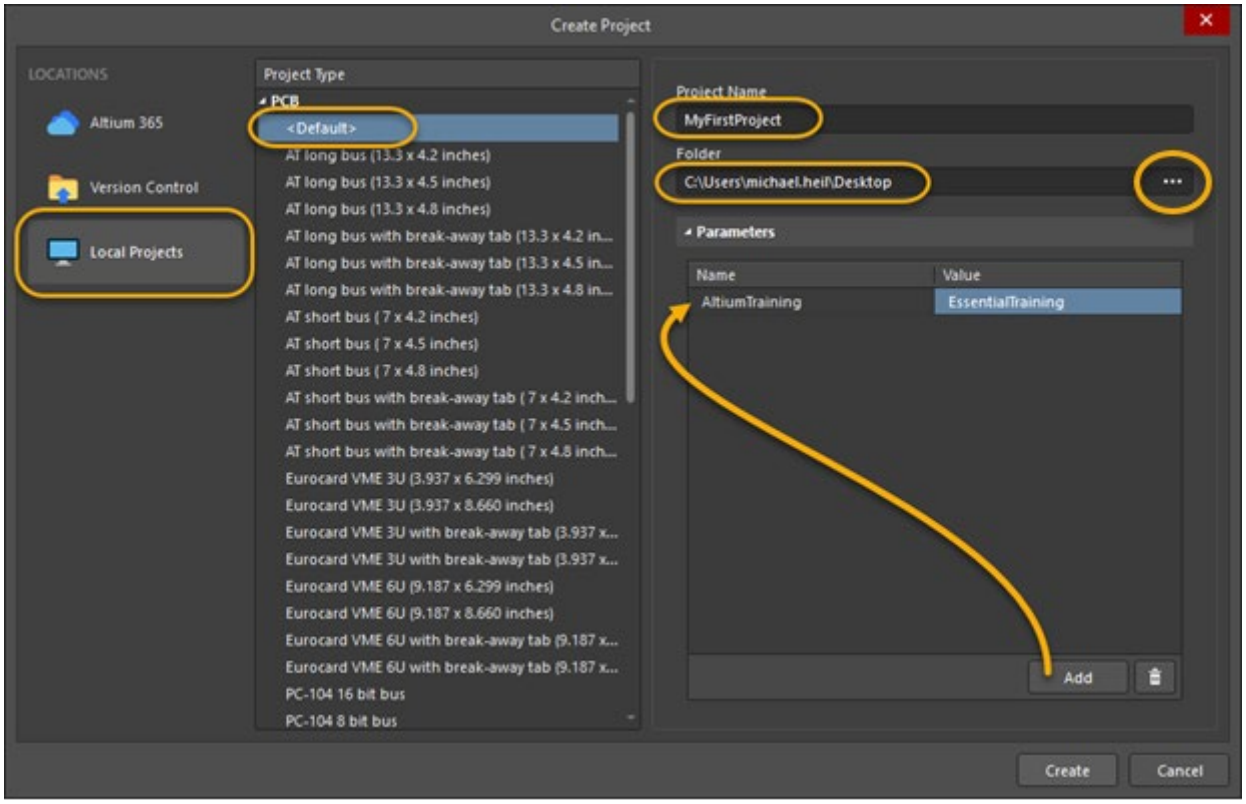


Figure 22. Project Wizard

3. Select **Local Projects** under the *Locations* area as shown in Figure 22 above.
4. In the *Project Type* area, under *PCB*, select **<Default>** to create a blank project.
5. Enter a project name of `MyFirstProject`.
6. Click on the **Triple Dot** button in the Folder area (as shown in Figure 22) and select your Desktop so this project will be easy to find later on.

7. Expand the **Parameters** area and add a project parameter. This parameter will be available throughout all of the schematic documents of our project later on. Use the same name and value as shown in Figure 22 on the previous page.

Projects can be directly saved to a Version Control repository if VCS is chosen under Locations.

8. Click the **Create** button to create the new project. Go to your Desktop to confirm that a `MyFirstProject` folder has been created on your desktop.
9. In Altium Designer, notice there is now a project called `MyFirstProject.PrjPcb` in the *Projects* panel as shown in Figure 23 below.

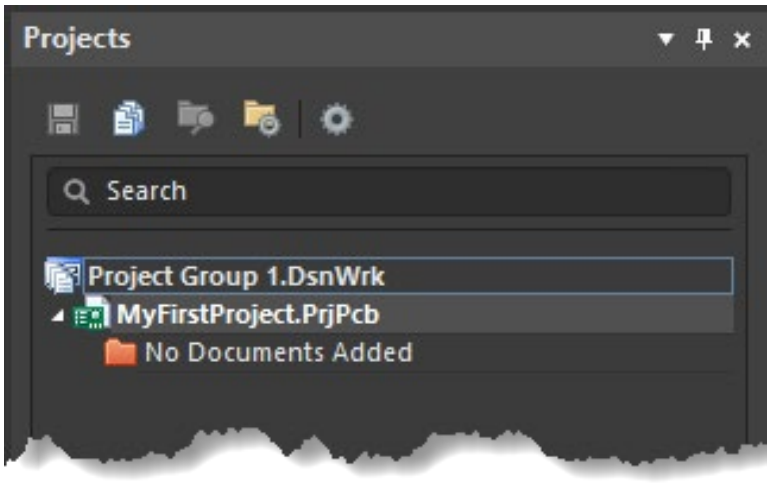


Figure 23. Projects panel with the new project you just created

2.2.2 Adding New Documents

10. **Right-click** on `MyFirstProject.PrjPcb` in the *Projects* panel.
    a) Select **Add New to Project** and choose **Schematic** to add a new schematic to the Project as shown in Figure 24. The default name for the schematic is `Sheet1.SchDoc`
    b) Repeat this operation to add a second schematic. The default name for the second schematic is `Sheet2.SchDoc`.

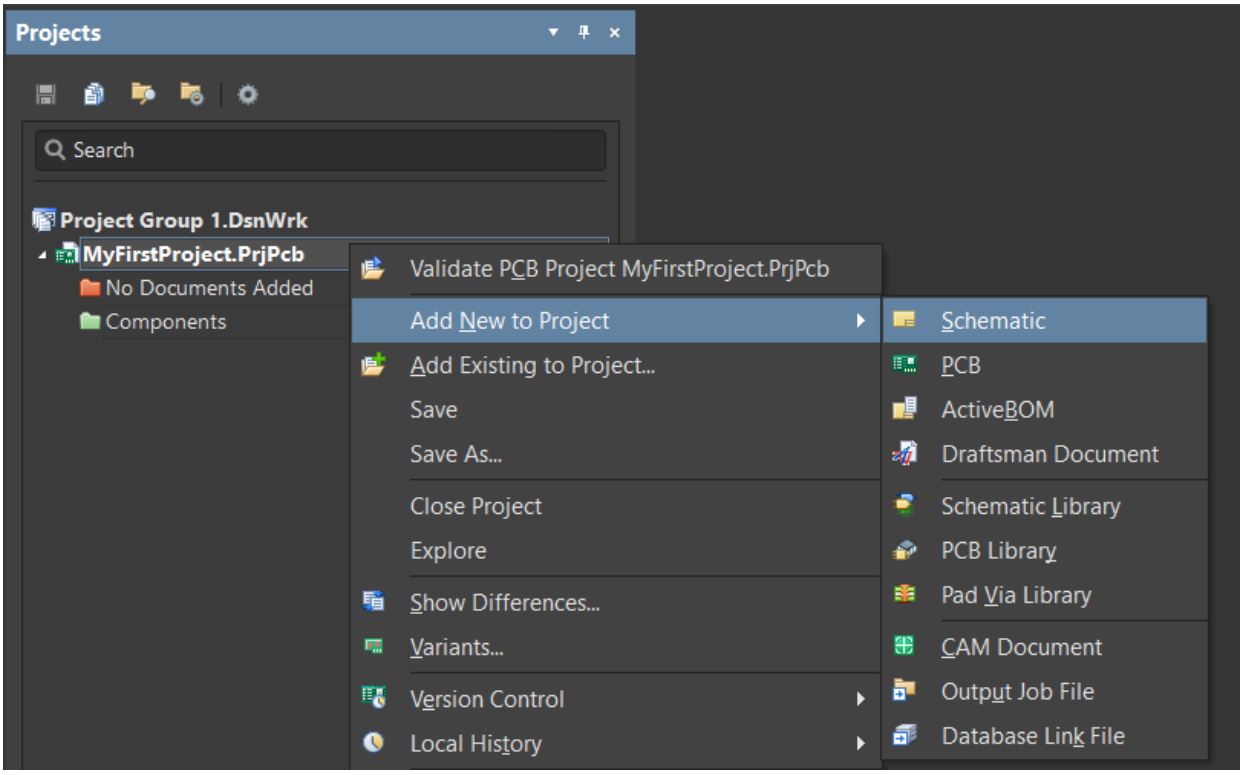


Figure 24. Adding new Schematics to the Project

11. Go to **File » Save All**. A dialog box will open, prompting you to save the new files you created. Save the files in the default desktop folder `MyFirstProject` that you created earlier:

- 1st schematic as `Hello`
- 2nd schematic as `World`

12. The *Projects* panel should now look like Figure 25 below.

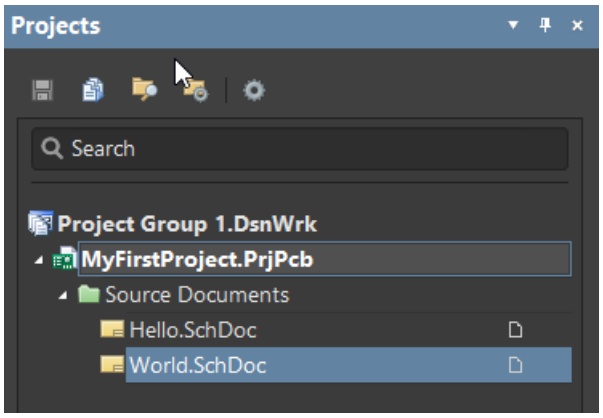


Figure 25. Project panel with new files names

13. We will navigate to the Desktop project folder directly by right-clicking on `MyFirstProject.PrjPcb` in the *Projects* panel and clicking on **Explore** as shown in Figure 26 below.

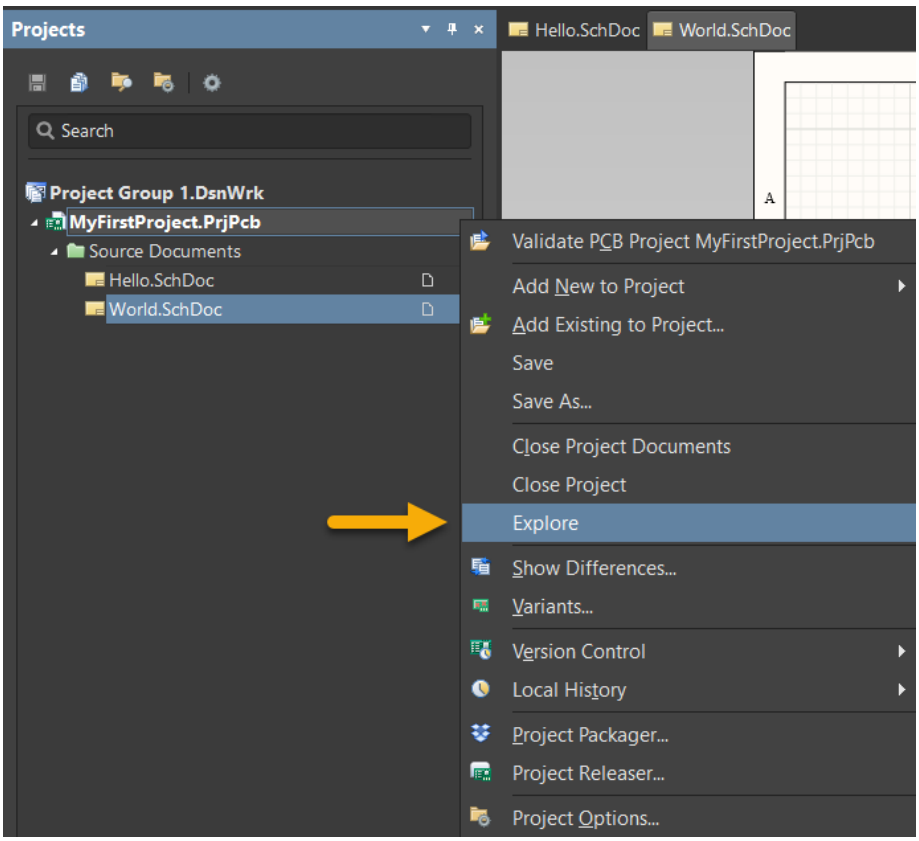


Figure 26. Navigate to the File Explorer from the Projects Panel

Add existing files to a project by right-clicking on the project name in the *Projects* panel and clicking on **Add Existing to Project**. Files located outside the project's directory structure are displayed in the *Projects* panel with a linked icon.

2.2.3 Project Options

14. Go to the menu **Project** and select **Project Options.**
    a) Select the *Options* tab.
    b) Select the *Parameters* tab, and you'll see the parameter you added during the project creation.
    c) Click **OK.** Go to **File » Save All**.

2.3 Package a Project for Easy Transfer

15. We can use the Project Packager to generate an archive of the project and its documents:
    a) Right-click the project from the *Projects* panel, and select the **Project Packager.**
    b) Use the default settings and click next to progress through the *Project Packager* as shown in Figure 27 below.
    c) Click **Finish** when all the settings are defined.

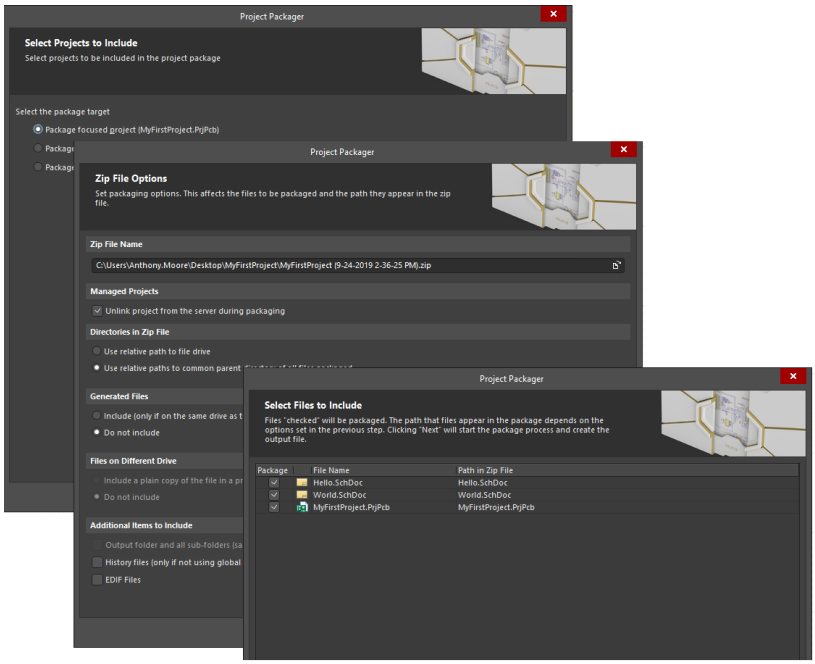


Figure 27. Project Packager

16. In the *Projects* panel, right-click on `MyFirstProject.PrjPcb` and click **Explore** to see the `MyFirstProject.zip` file added to the project folder (Figure 28 below). The Name and Location of the ZIP file were defined in the *Zip File Options* page of the *Project Packager.* The default ZIP name includes the project name with the date and time stamps.

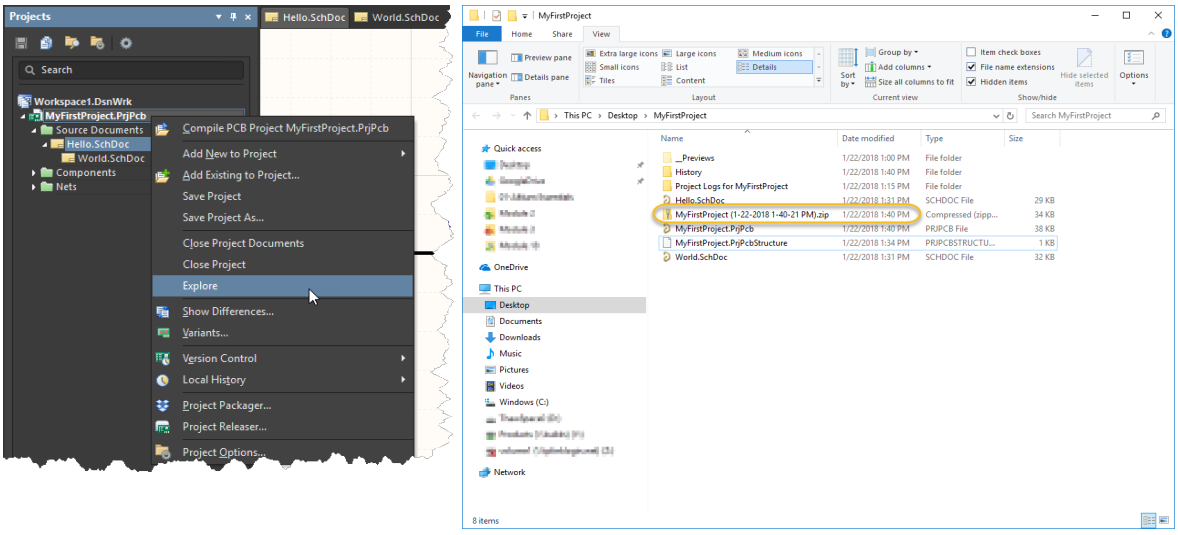


Figure 28. Navigate to the Packaged Project

Module 3: Schematic Preferences

3.1 Purpose

In this exercise, you will learn about schematic preferences and how to customize them. These preferences will allow you to customize your schematic environment and ensure document clarity and standardization.

3.2 Preparation

1. Close all open projects in the *Projects* panel.
2. Go to **File » Open Project** and browse to the installed Example projects folder. Open `<AD Version>\Examples\Bluetooth Sentinel\Bluetooth_Sentinel.PrjPcb`. This project may also be available in the **Recent Projects** menu.

The default path to your installed examples is `C:\Users\Public\Public Documents\Altium\<AD Version>\Examples\`

3.3 Setting Preferences

3. Go to Preferences in the upper right corner of Altium Designer.
4. Expand the *Schematic* branch and open the *General* page. Verify that **Components Cut Wires** is enabled as shown in Figure 29.
5. Open the *Graphical Editing* page. Verify that **Display Strings as Rotated** is unchecked (it's enabled by default). It can be found in Figure 29.

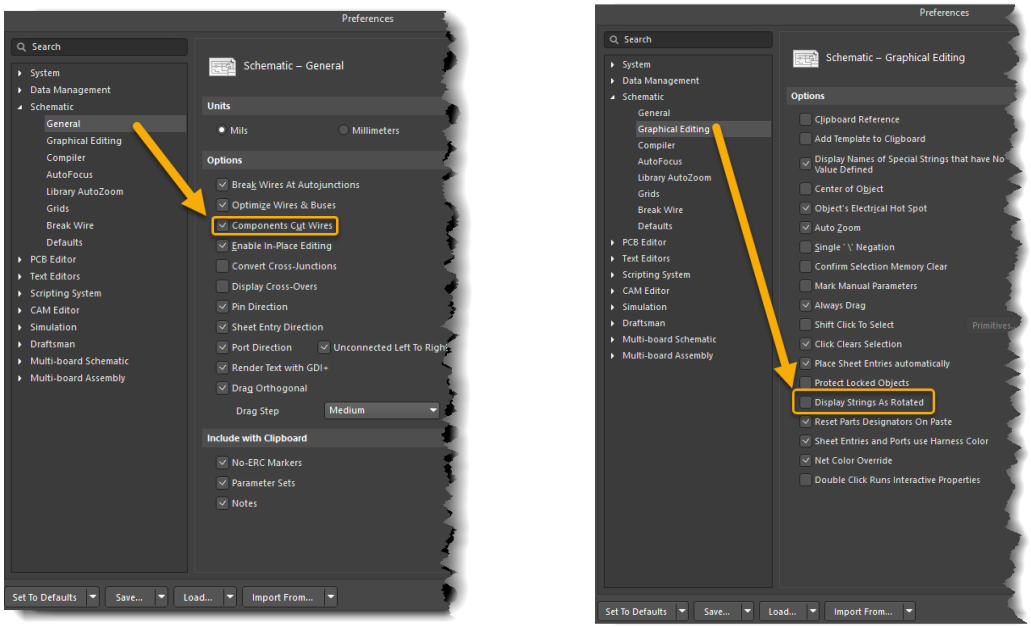


Figure 29. General and Graphical Editing Preference Settings

6. Click **OK** to save and close Preferences.

7. In the *Projects* Panel, double-click on `Microcontroller_STM32F101.SchDoc` to open it (Figure 30 below).

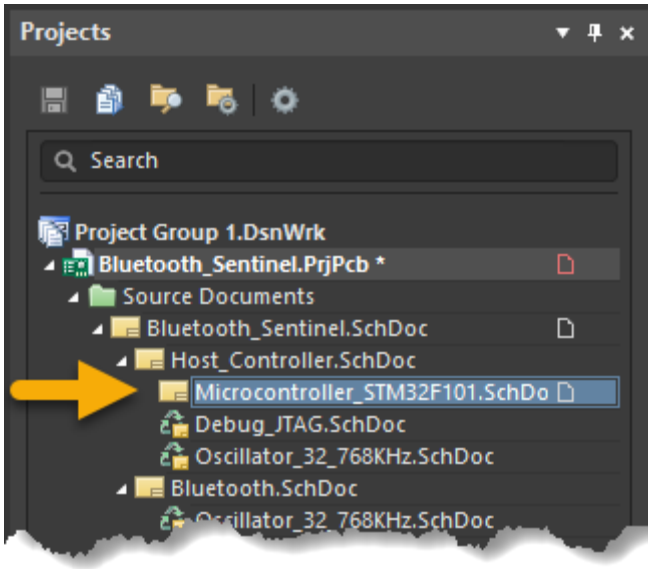


Figure 30. Projects Panel

8. Open the *Components* Panel from the **Panels** button. In this panel, set the drop-down to select the `Miscellaneous Devices.IntLib` as shown in Figure 31 below.

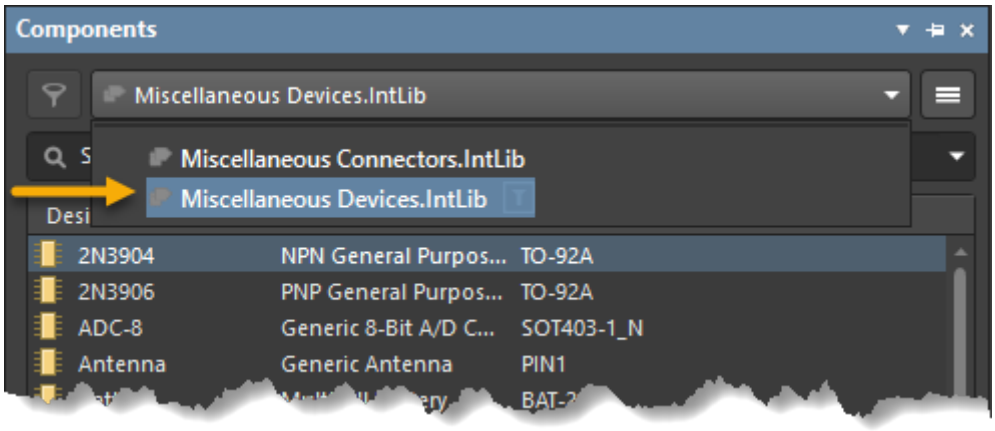


Figure 31. Library Selection from the Components panel

9. Type `cap` in the keyword search to reduce the list to capacitors and select `Cap Semi` from the library as shown in Figure 32 below.

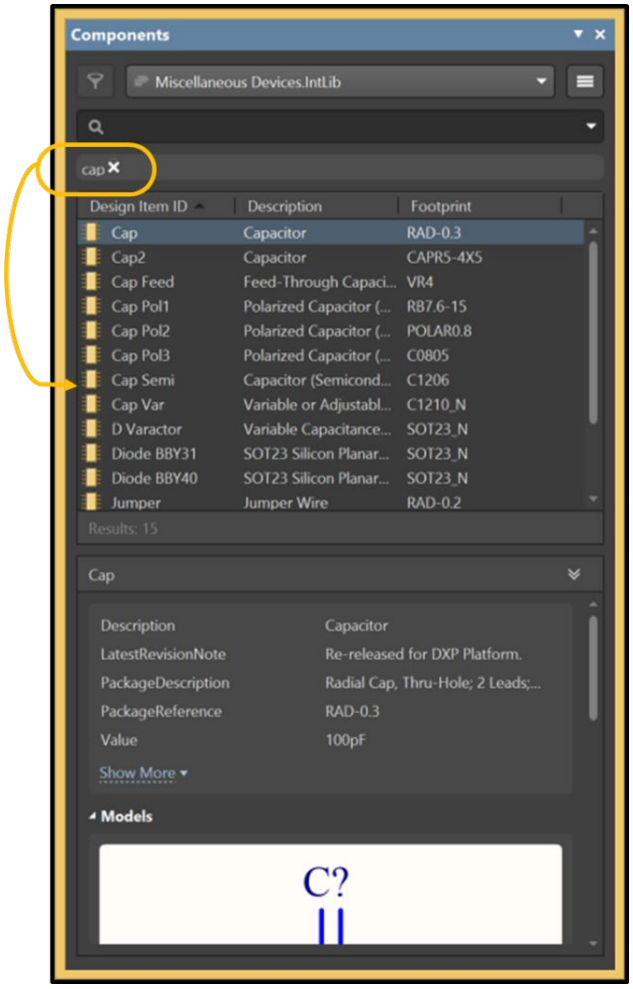


*Figure 32. Select Cap Semi from the Miscellaneous Devices.IntLib^*

10. Right click and select **Place Cap Semi** and then rotate the capacitor using the **Spacebar**.
11. **Left-click** to place the capacitor on existing wires as shown in Figure 33 below.
12. After you place the capacitor, **right-click** to exit the **Place Part** command.
    Notice that the capacitor will be dropped on the wire which will automatically be cut into two segments, which are terminated at the components' pins. This is because we enabled the *Component Cut Wire* preference

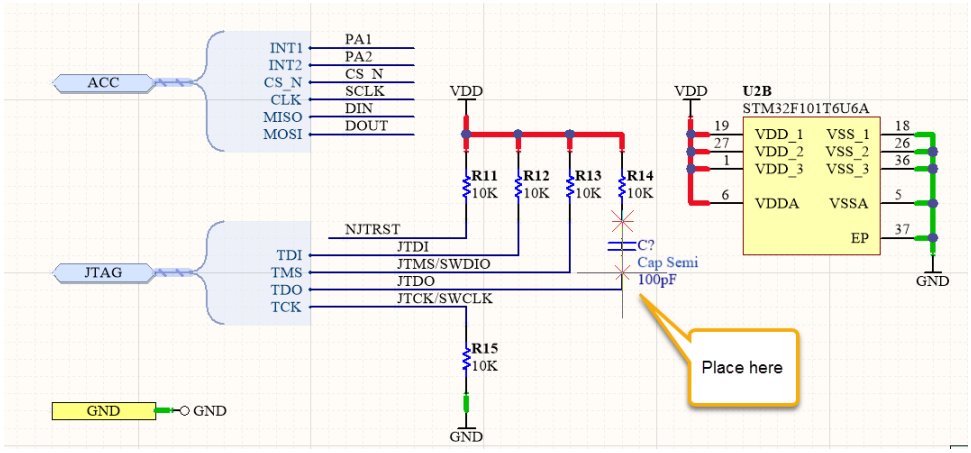


Figure 33. Place Cap Semi onto the existing wire

13. Before continuing to the next schematic, open the *Components* panel and reset the *'cap'* search as shown Figure 34 below.

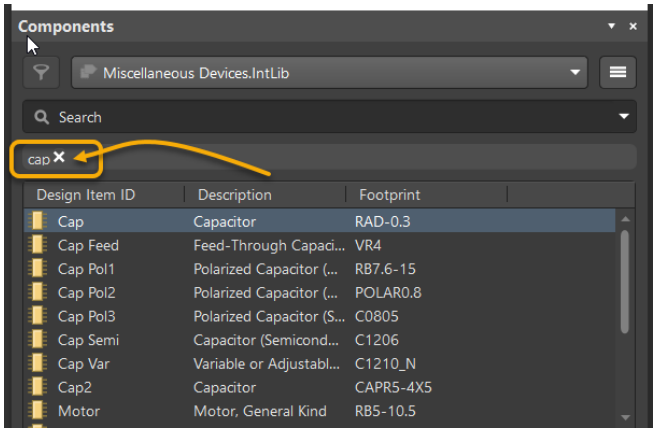


Figure 34. Reset "cap" search in the Components panel

14. Open the schematic `Oscillator_16MHz.SchDoc`.
    Observe how the wires are currently displayed on the left of component `X1`, displayed in Figure 35 below. For document clarity, it is usually recommended for *Cross-Overs* to be displayed. This can be enabled in Preferences.

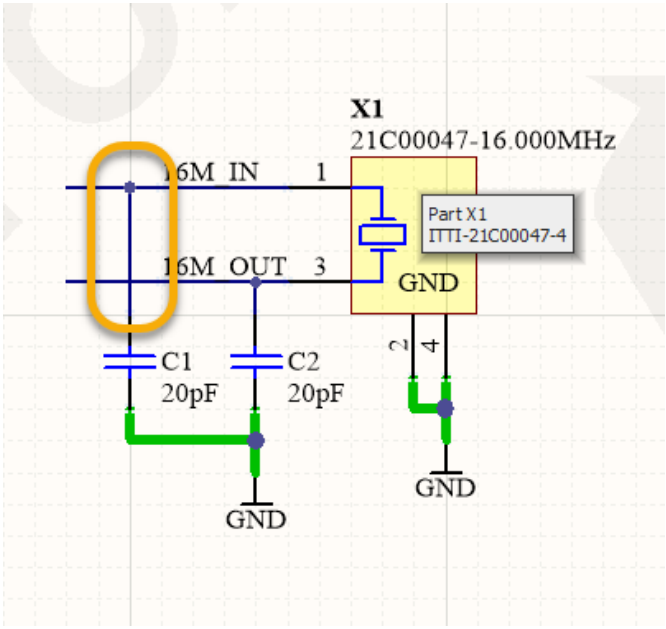


Figure 35. Schematic Connections for X1

15. Click Preferences, in the upper right corner of Altium Designer. Expand the *Schematic* branch and open the *General* page. Enable **Display Cross-Overs** as shown in Figure 36 below.
16. Click **OK.**

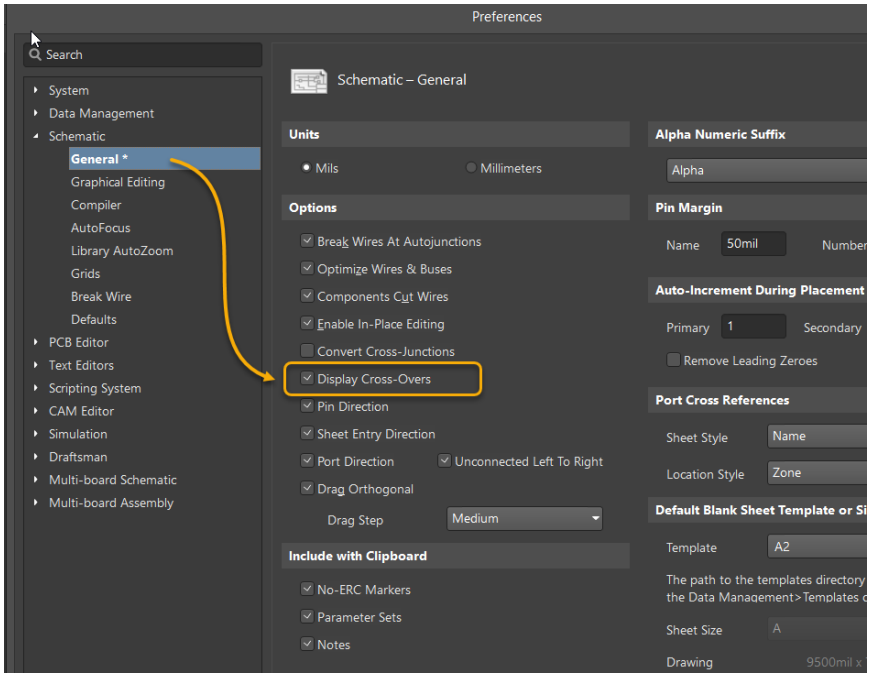


Figure 36. Display Cross-Over Preference

17. Notice in Figure 37 below that the cross-overs will then be displayed in the schematics.

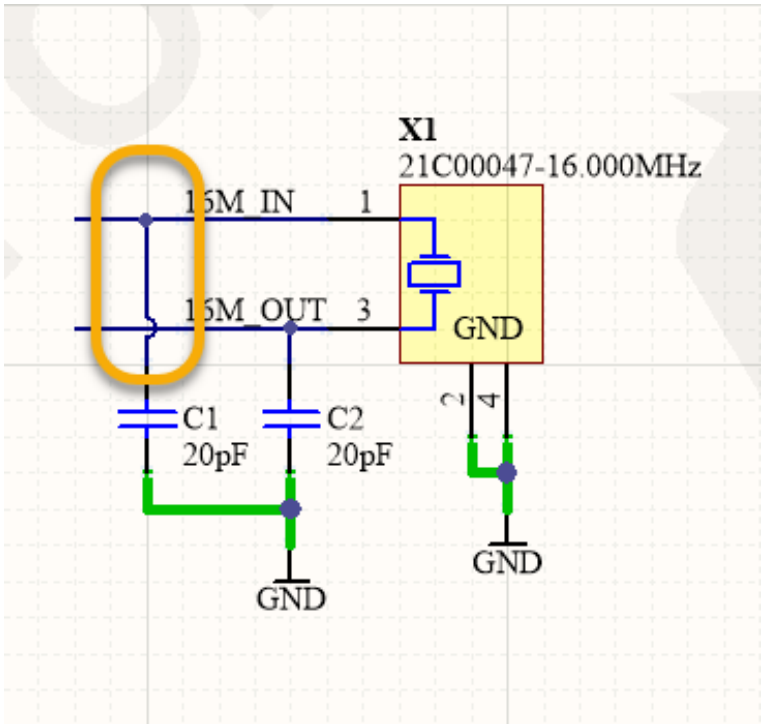


Figure 37. Cross-Overs being displayed after enabling it in Preferences

3.4 Graphical Editing

3.4.1 Always Drag

18. Make the schematic file `Microcontroller_STM32F101.SchDoc` the active document.
19. Click Preferences, in the upper right corner of Altium Designer. Expand the *Schematic* branch and open the *Graphical Editing* page. Check if **Always Drag** is active. Click **OK** to close the Preferences dialog.
20. Left-click on *U2B*. While holding the left-click, move the cursor to drag the component around. Notice the behavior of the connections where the wires will remain connected to the components.
21. Hit **Escape** or **Right-Click** to abort the drag.
22. To move the component without maintaining connections, hold the **Ctrl** key, left-click and hold it on *U2B*. The component should now move independently of the previously connected tracks.
23. Go to **Edit » Undo** or hit **Ctrl+Z** to undo any changes if necessary.

While dragging a component, ensure to check the connections that it is about to create. Before releasing the left mouse button, an indicator will show if there is a netlist modification or not.

3.4.2 Convert Special Strings

24. Ensure `Microcontroller_STM32F101.SchDoc` schematic is the active document. Open Preferences, in the *Schematic* branch, go to *Graphical Editing*. Ensure that the option **Display Names of Special Strings that have No Value Defined** is enabled. Click **OK** to exit the Preferences.
25. Navigate to the title block at the bottom right of the schematic sheet and notice the placed text strings. These strings are pointing to parameters which currently do not have values yet. Hence, the parameter names are shown, instead of their values.
26. We will make some changes to the Parameters as shown in Figure 38 below.
    a) Open the *Properties* panel from the **Panels** button and open the **Parameters** tab.
    b) Try changing the `Title` parameter value to `Test` and hit enter. Notice the change on the title block.
    c) Change the `Title` parameter value back to `Microcontroller STM32F101`.

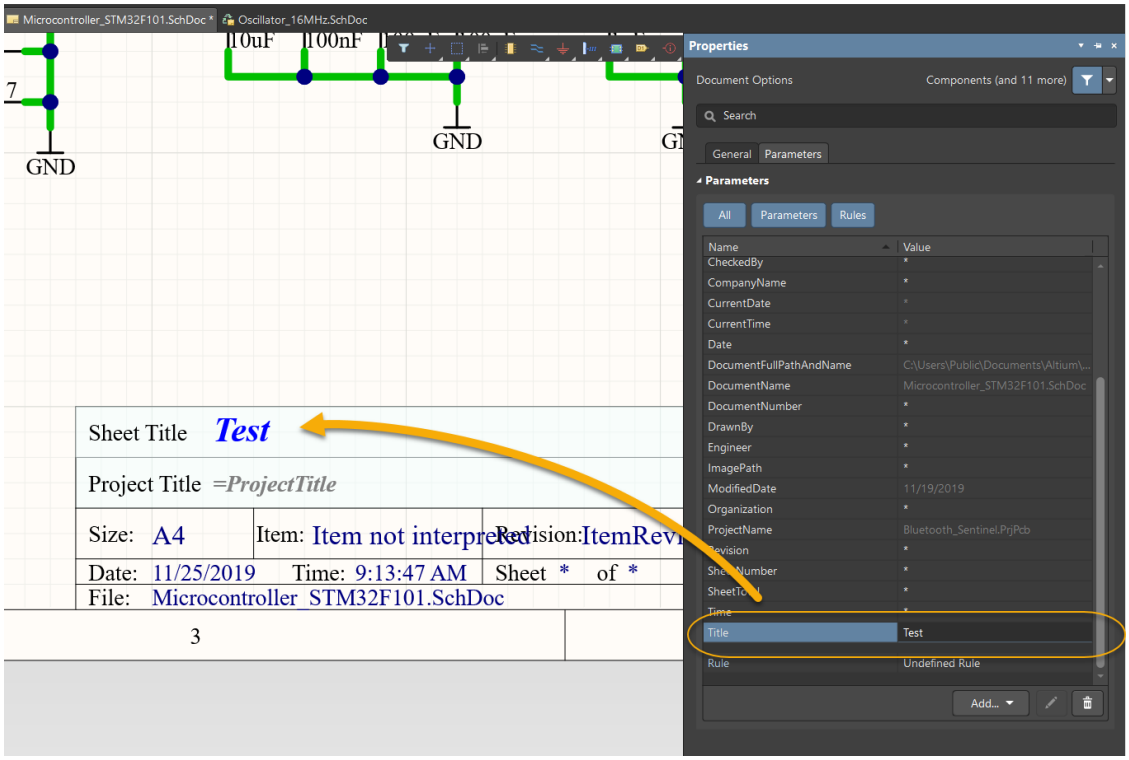


Figure 38. Document Parameters

27. Other parameters like `=ProjectTitle` are not in the document's parameters list yet.
28. However, instead of adding a parameter as a *Document Parameter*, we will add a *Project Parameter*. Note that the *Document Parameters* are local and unique per document, while *Project Parameters* are global and common for all project documents.
29. To modify the project parameter list, we will refer to Figure 39 below.
    a) Go to **Project » Project Options**. Then, open the **Parameters** tab.
    b) Click **Add…**. Type `ProjectTitle` as the Parameter Name and `Training Project` as its Value. Click **OK**.
    c) The newly created parameter will then be listed. Click **OK**.

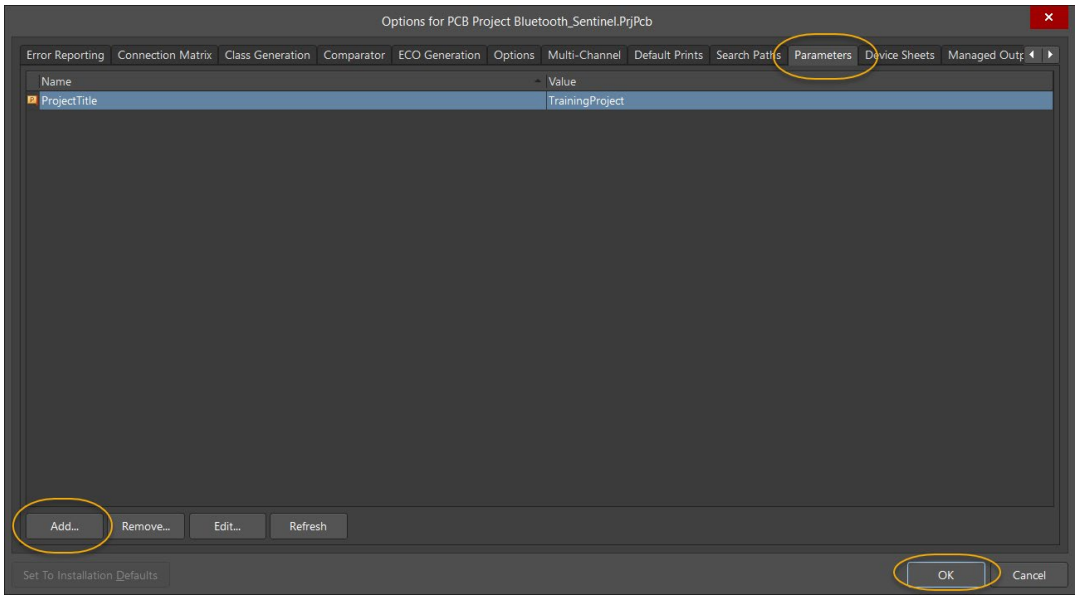


Figure 39. Project Options

30. Notice that in Figure 40 below, the schematic should now show the newly created parameter value in the Title Block. Open `Bluetooth.SchDoc` and notice that it will also share the same value for the project parameter.

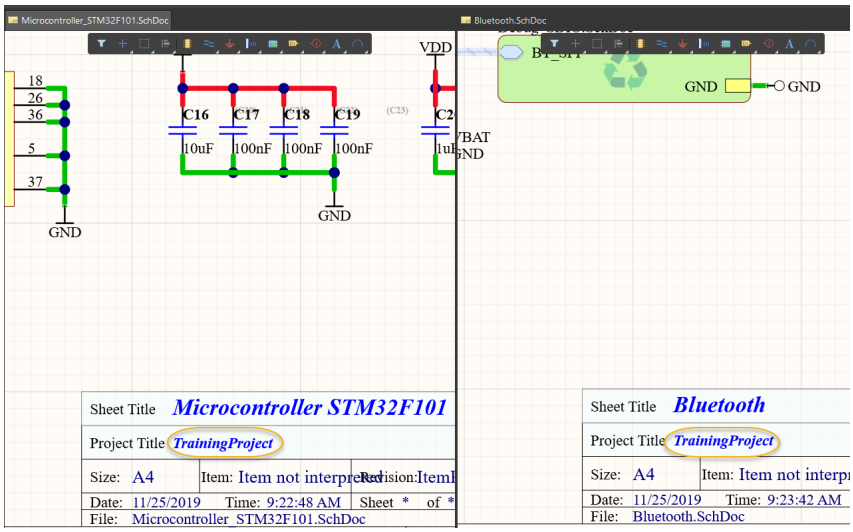


Figure 40. Project Parameter added to each Schematic

Special strings of the title block are defined & modified within the Schematic Template file *.SchDot.

3.5 Grids

31. Click Preferences and expand *Schematic* branch and open the *Grids* page.
32. Under *Imperial Grid Presets,* click **Altium Presets** and select **Course (3 settings)** as shown in Figure 41 below.
33. Click on the color box to the right of **Grid Color** if you wish to change the color of the grid.
34. Click **OK** to save and close the dialog.

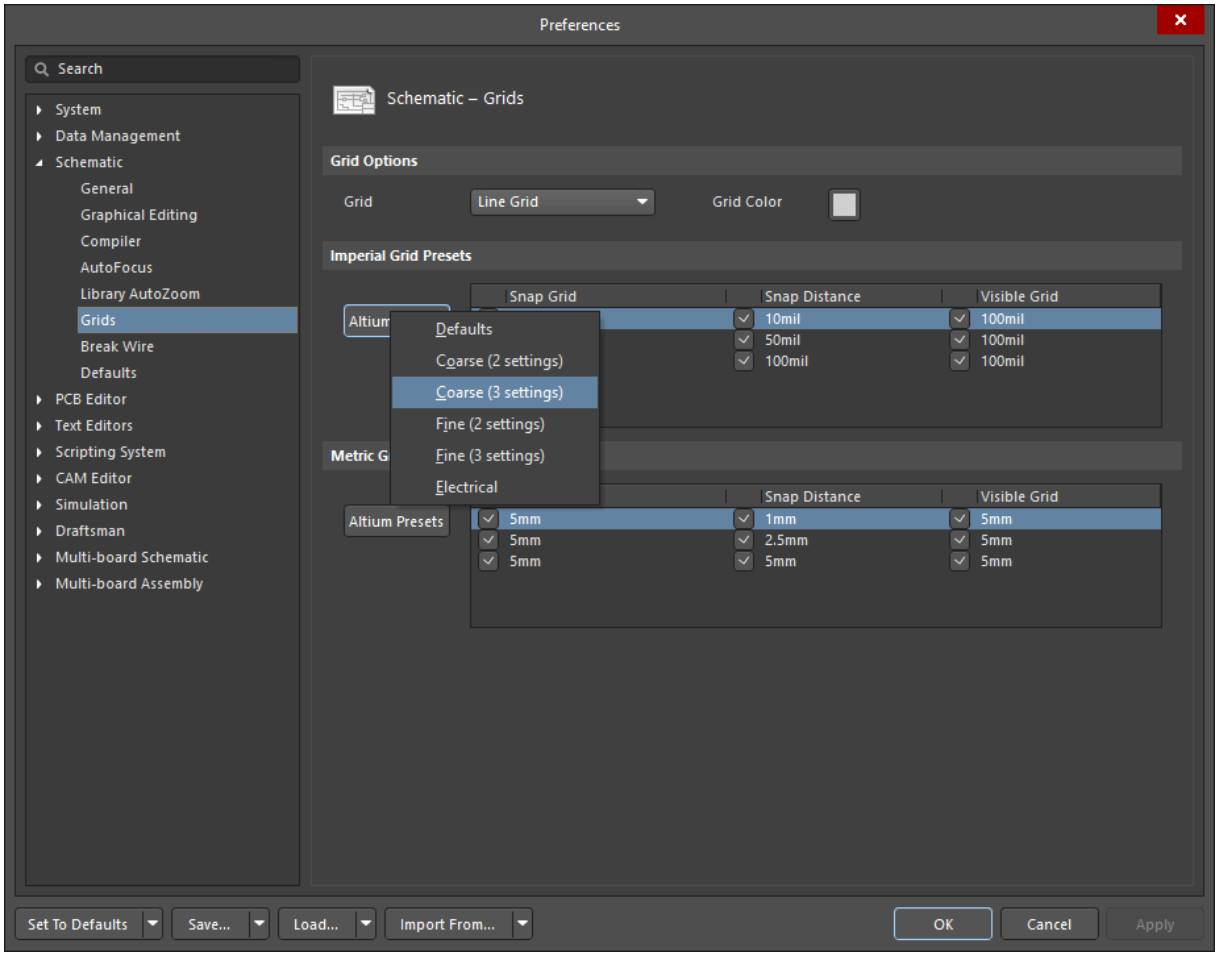


Figure 41. Grid Settings in Preferences

35. In the schematics, toggle the snap grid through your new presets by hitting the **G** key. You can see the grid change between 50mil, 100mil, and 200mil on the *Status Bar* as shown in Figure 42 below.

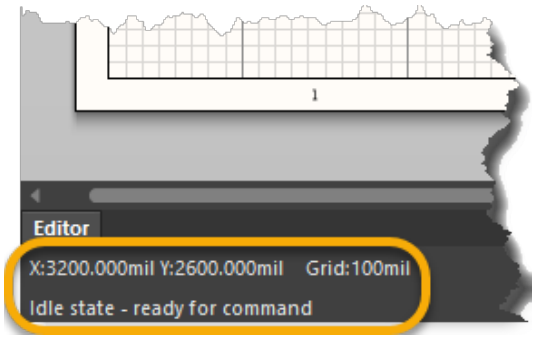


Figure 42. Grid Information from the Status Bar

3.6 Schematic Defaults

3.6.1 Default Primitives

Currently, when placing a wire, it will be a blue, small-width wire. In this section, we will change the default settings for wires so that it defaults to green, large-width wire when placed.

36. Click Preferences and expand the *Schematic* branch and open the *Defaults* page.
37. In the *Primitives List*, select **Wire** as shown in Figure 43 below. Change the default Wire width to **Large** and the color to **Green**.

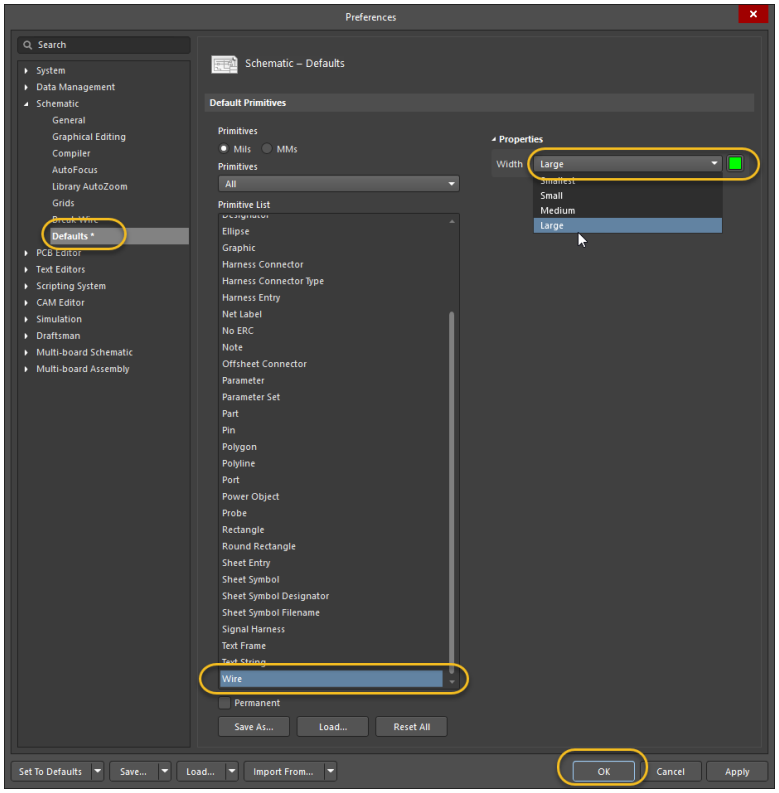


Figure 43. Default Primitives for Wire

38. Click **OK** to save and close the preferences dialog.
39. Open a schematic document. Go to **Place » Wire**. Try placing some wires on the schematic. Notice that the wires placed will now be default to green, large wires as shown in Figure 44 below.

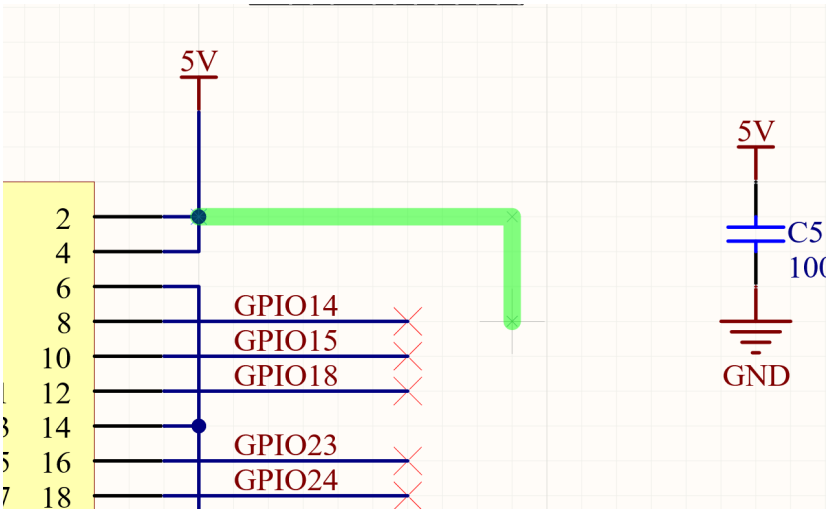


Figure 44. Large Green default Wires

40. Hit **Esc** to end command.
41. Click Preferences, expand the *Schematic* branch and open the *Defaults* page. To reset all primitive defaults, click **Reset All**.

The reset is applied by clicking the "OK" button. Also, we can hit F1 while in a Preferences page or any general menu to open the corresponding online documentation regarding the active window or menu.

42. In the *Projects* panel, close all projects and files without saving.

Module 4: Navigating Schematics

4.1 Purpose

This module will teach you some basic operations in the schematic editor such as preference settings, mouse actions, frequently used shortcut keys and search actions in the schematic environment.

4.2 Preparation

1. If not already open, click **File » Open Project** and browse to your installed Example projects folder. Select `<AD version>\Examples\Bluetooth Sentinel\Bluetooth_Sentinel.PrjPcb`. This project may also be available in the **Recent Projects** menu.

The default path to your installed examples is C:\Users\Public\Public Documents\Altium\<AD Version>\Examples\

4.3 Basic Schematic Preference Settings

Schematic Preferences are system specific settings which help you control the behavior and display of the Schematic Editor to suit your working style and needs. Let's take a look at some basic preferences you may want to change.

4.3.1 Setting the Default Template & Sheet Size

2. Select Preferences, in the upper-right corner of the interface. Open the *Schematic* branch and highlight the *General* page as shown in Figure 45. From within a focused schematic document, you can also gain access to the command **Tools » Preferences…** .

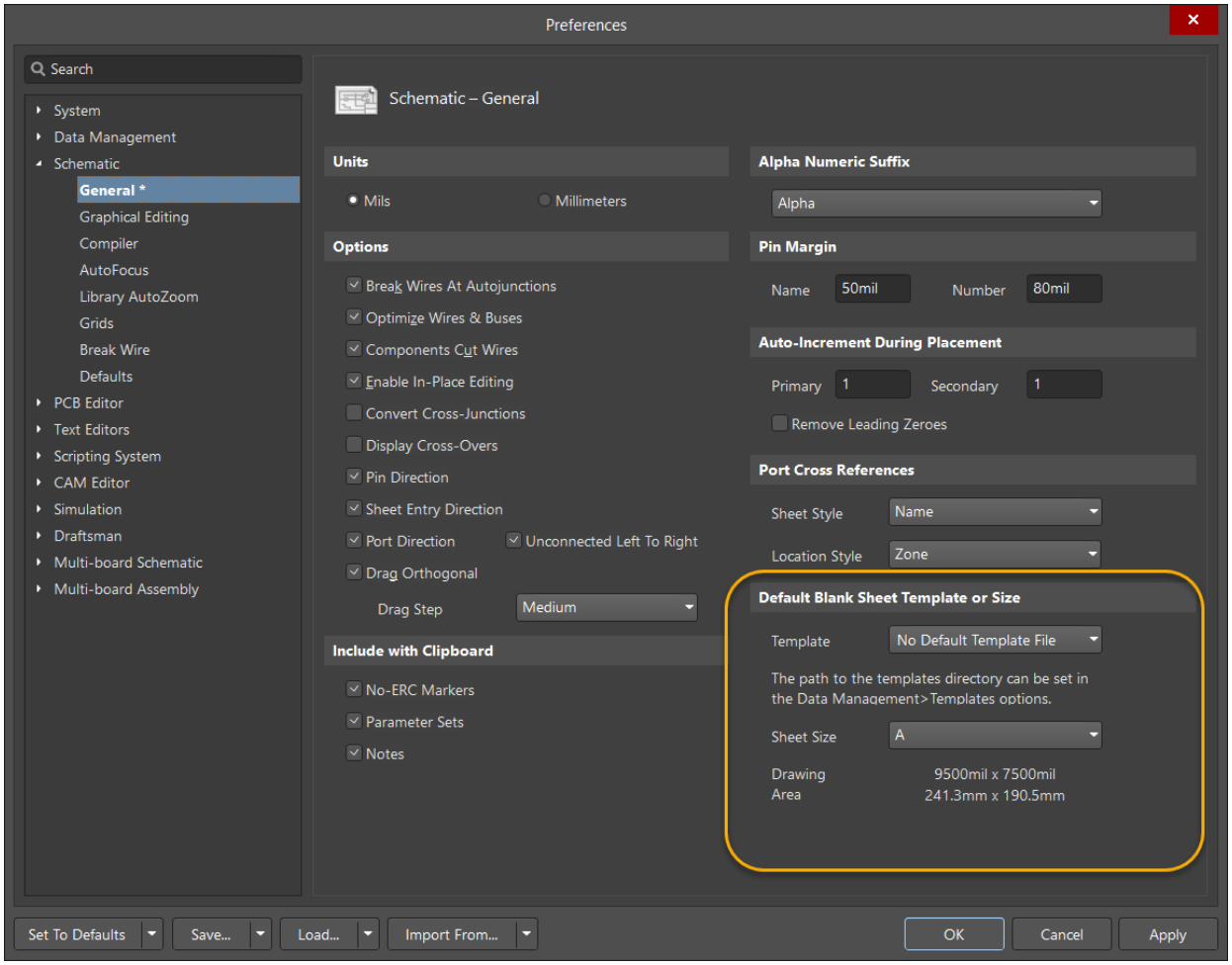


Figure 45. General Page from the Schematic Preferences

3. Navigate to **Sheet Size** in the bottom right, and a drop-down menu will appear as highlighted in Figure 45 above. You can set the default schematic size in this menu.
4. The default setting is A4 or A – depending on the region you are working in. **Cancel** the dialog.
5. Use the **File » New » Schematic** command to add a schematic sheet to the project. The **Sheet Size** setting according to your preference is used on the new schematic sheet (e.g. A4).
6. Reopen Preferences, click **Template** and a drop-down menu will appear. You can set the default template file for new schematics in this menu. By default, **No Default Template File** is defined.
7. Change the setting to a template of your choice and **OK** the dialog.
8. Add a second schematic sheet using the **File » New » Schematic** command and compare the two newly added sheets. Now the chosen template is used when adding a schematic sheet.
9. Close both schematic sheets by selecting them and using the **Right Mouse Click Button » Close** command.

4.3.2 Project Insight

10. To view the connectivity insight of our design, the project needs to be compiled. Use **Project » Validate PCB Project Bluetooth_Sentinel.PrjPcb** to compile the project. Close the *Messages* panel if it appears.

If you have **Dynamic Compiling** enabled, there is no need to compile the project to show the Design Insight, as it would do it automatically. We will discuss Dynamic Compiling in a later module.

11. In the Preferences, open the *System* branch and look at the *Design Insight* page. Check **Enable Project Insight** and click **OK** as shown in Figure 46 below.

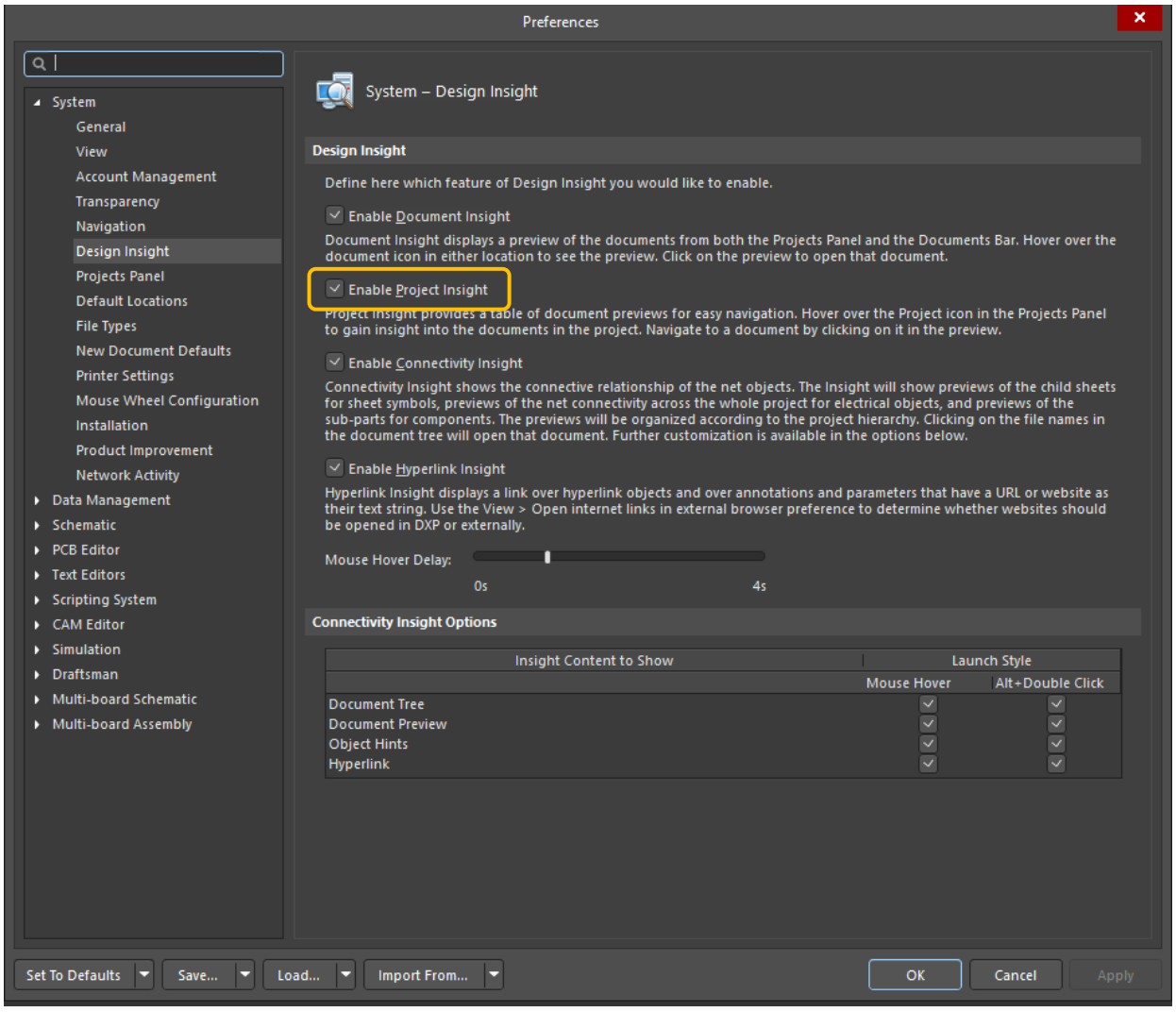


Figure 46. Preferences for Design Insight

12. In the *Projects* panel, hover the mouse cursor over the `Bluetooth_Sentinel.PrjPcb` project icon as shown in Figure 47 below, and a thumbnail view of the documents within the project will appear at the cursor position.

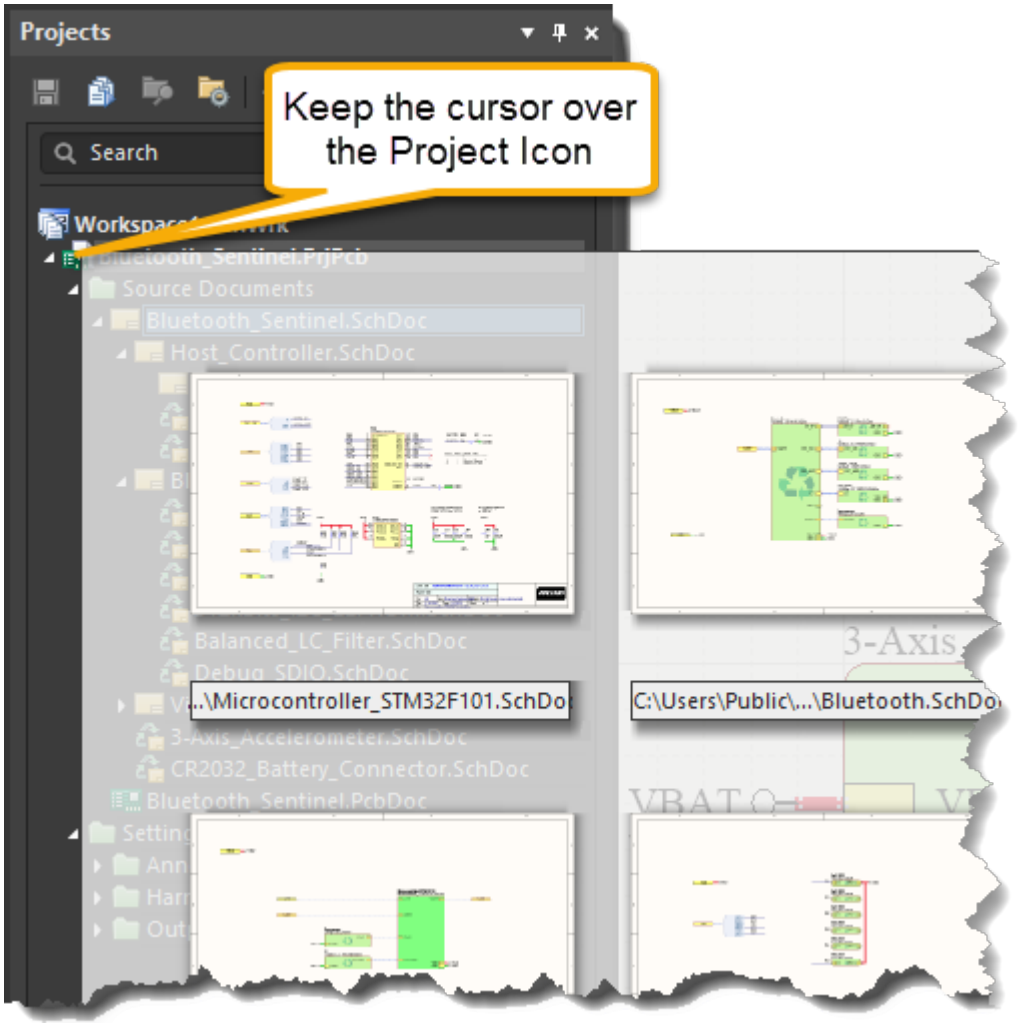


Figure 47. Project Insight thumbnail when hovering over the Project icon

4.3.3 Connectivity Insight

13. In the Preferences within the *Design Insight* page, enable the checkbox for **Enable Connectivity Insight** and in the *Document Tree* for *Mouse Hover* as shown in Figure 48 below.
14. Click **OK**.

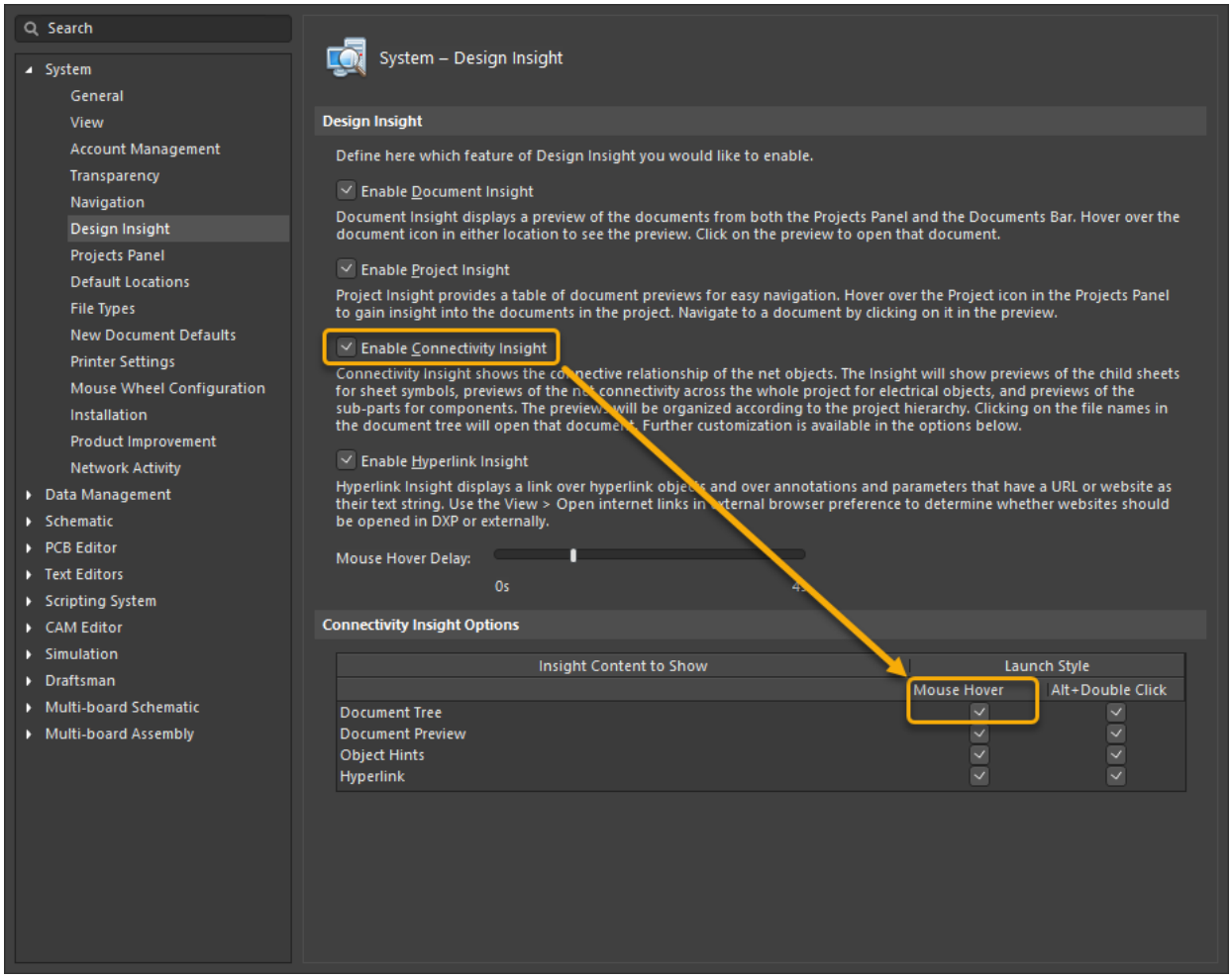


Figure 48. Connectivity Insight Preferences

15. Open `Bluetooth_Sentinel.SchDoc` and `Host_Controller.SchDoc`.
16. Make `Bluetooth_Sentinel.SchDoc` the focused document.
17. Hover the mouse over the `Processor` Sheet Symbol block as shown in Figure 49. A large thumbnail of the `Host_Controller.SchDoc` will appear next to the cursor.

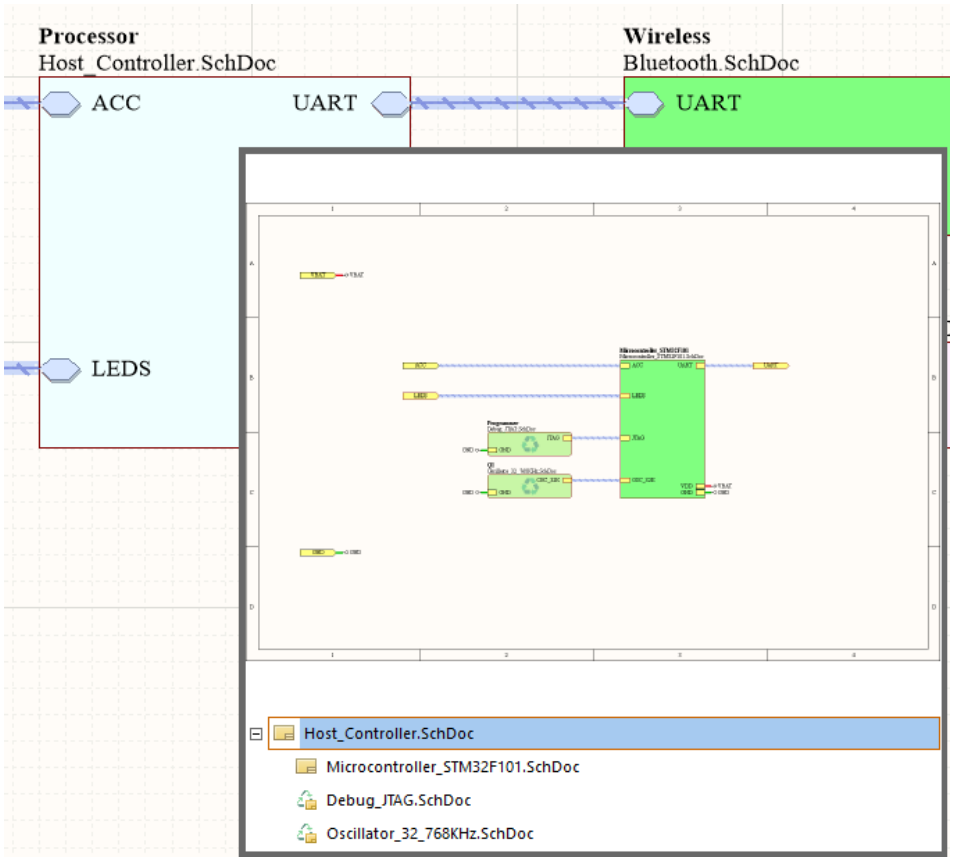


Figure 49. Hover over the Sheet Symbol to view the Schematic thumbnail

18. Next, move the cursor to hover over the *Harness* as shown in Figure 50 below.

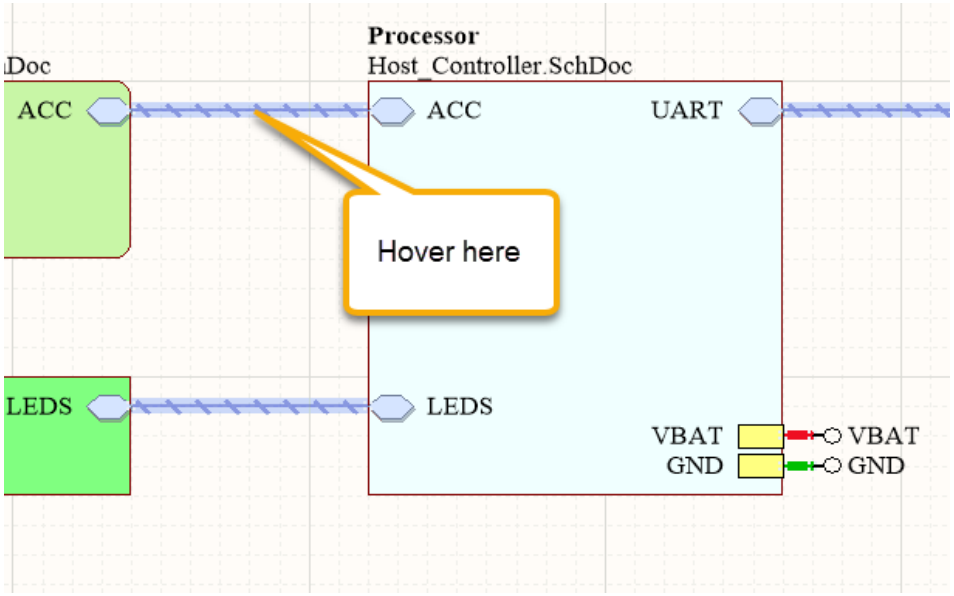


Figure 50. Hover over the Bus to view connectivity

19. A large thumbnail will appear next to the cursor, as shown in Figure 51 below, which shows the connections in the design.

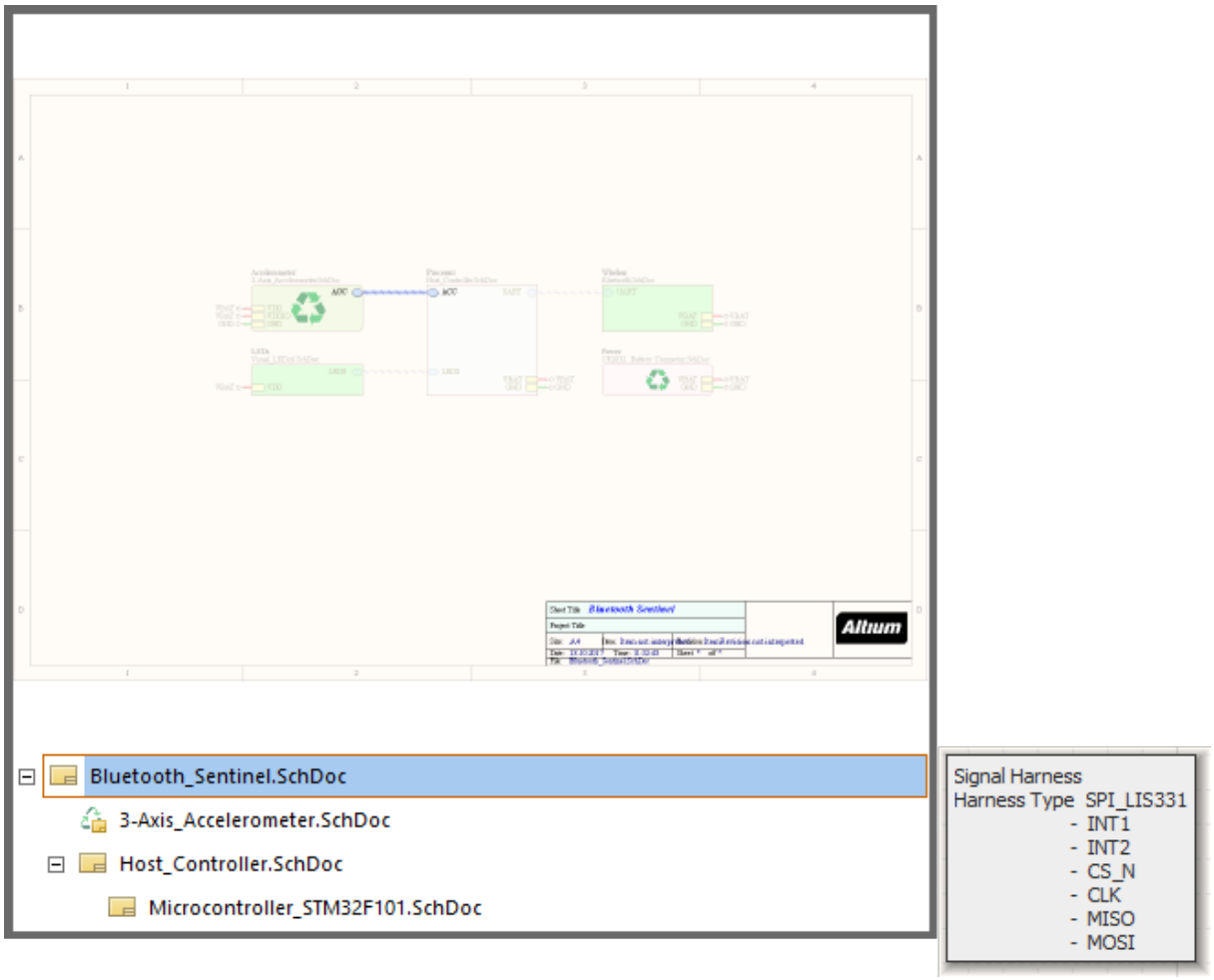


Figure 51. Connectivity Insight after hovering over the Harness

20. Slowly move your cursor down to `Host_Controller.SchDoc` at the bottom of the thumbnail as shown in Figure 52 below, to change the thumbnail view to the lower level schematic.
21. Left-click `Host_Controller.SchDoc` from the pop-up insight to open the schematic.

The schematic preview will be blank and only show a name and icon if the corresponding schematic is not currently opened.

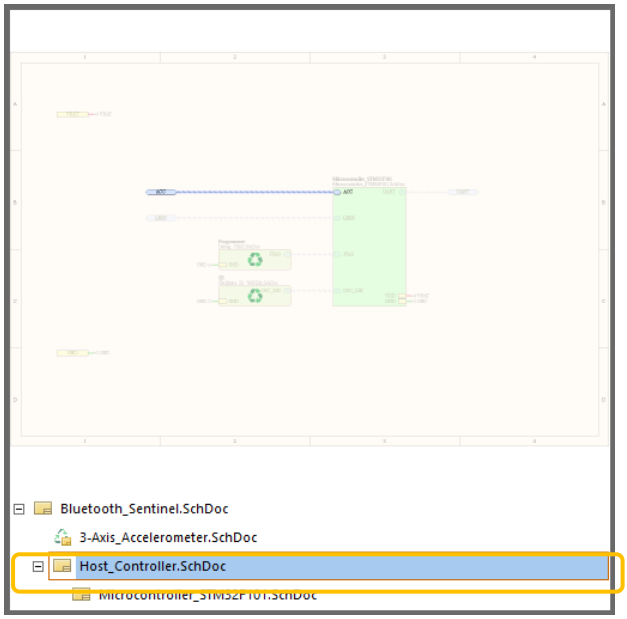


Figure 52. Connectivity Insight with Host_Controller.SchDoc

22. Open Preferences and click **Set to Defaults » Page** in the lower left corner of the *Design Insight* page as shown in Figure 53 below. Click **OK** to exit the dialog.

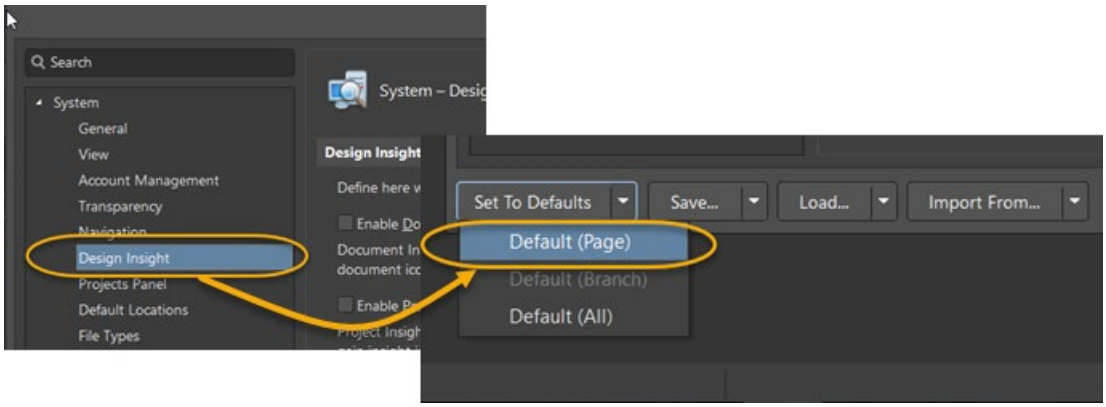


Figure 53. Set Design Insight Preferences back to Default

In the preferences within the **Navigation** page, in the section *Highlight Methods*, you'll find a slider that allows you to control the mask level for the objects. The mask level is used for navigation, cross-probing and exploring differences. To clear any masking, hit **Shift+C** to clear any selections or masking.

4.3.4 Schematic Navigator

23. Use the *Navigator* panel to browse nets, components, or pins across the entire design.
24. Open the *Navigator* panel by clicking **Panels** button and select **Navigator** as shown in Figure 54 below.

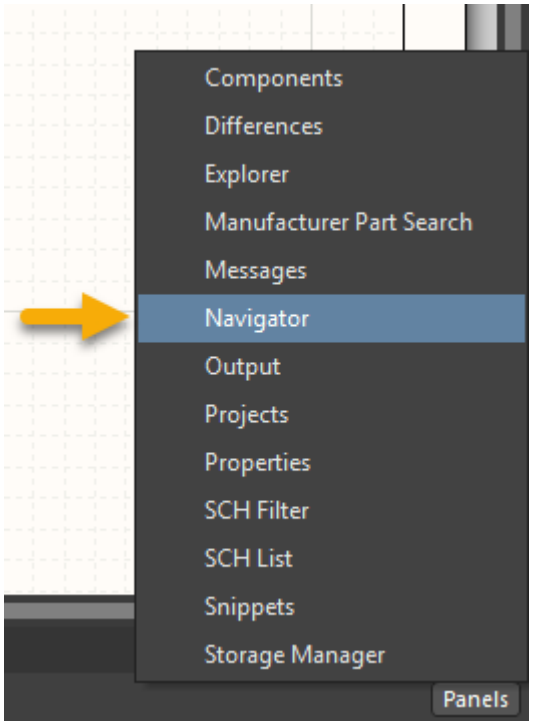


Figure 54. Opening the Navigator panel

25. The *Navigator* panel will open and docked to the left by default. Use Figure 55 as a reference for the following steps.
26. Click on a document name in the *Navigator* panel to see nets and components just for that page.

If no components or nets are displayed inside the *Navigator* panel, compile the project by selecting **Project » Validate PCB Project Bluetooth_Sentinel.PcbDoc.** If **Dynamic Compiling** is enabled, the components and nets will be populated automatically.

27. Click on the **Flattened Hierarchy** to see the nets and components for all schematics in the project.
28. With the **Flattened Hierarchy** tab selected, search for component R9 to jump to that component as shown in Figure 55.

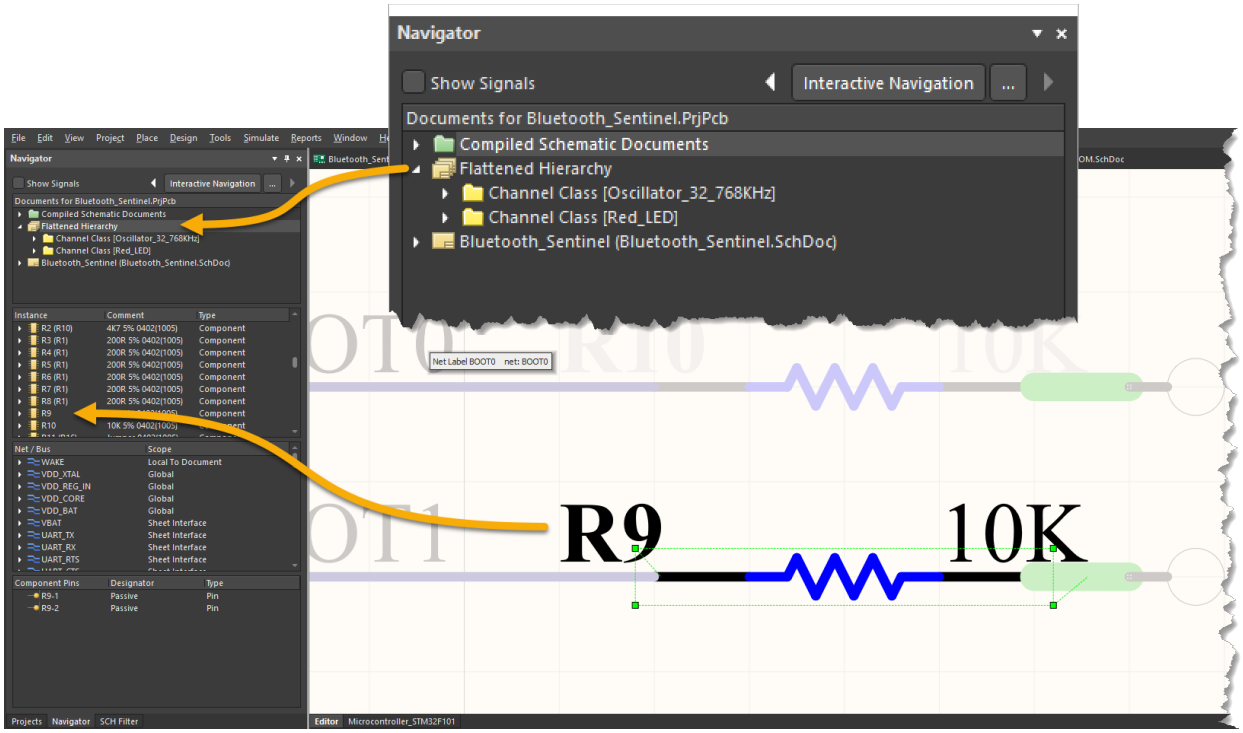


Figure 55. The Navigator Panel docked to the left

4.4 Navigation within the Schematic Workspace

4.4.1 Basic Navigation using the Mouse

29. Make the schematic file `Microcontroller_STM32F101.SchDoc` the focused document by left-clicking on its document tab or from the *Projects* panel.
    a) From the **View** menu, select **Fit All Objects**
    b) In the schematic, right-click and hold until the panning hand cursor appears.
    c) Drag the schematic around while holding the right mouse button down.
    d) Release the right mouse button to end.
30. To change the zoom, click & hold mouse wheel, then slide mouse forward/back.
31. To Zoom in or out click and hold **Ctrl** + roll mouse wheel
32. Up/Down: roll mouse wheel (scrolling)
33. Left/Right: click and hold **Shift** + roll mouse wheel

4.4.2 Basic Navigation using the Keyboard

34. Maintain the focus on the Schematic Workspace and observe the behavior of the following Shortcut Keys:
35. To Zoom out press PAGE DOWN.
36. To Zoom in press **PAGE UP.**
37. To Fit All Objects, press **CTRL+PAGE DOWN.**
38. To pan and re-center your workspace based on your cursor position press **HOME.**
39. To move in a direction by one Grid Step press the **ARROW KEYS.**
40. To move in a direction by 10 Grid Steps press **SHIFT+ARROW KEYS**.

4.4.3 View Menu Hotkey Sequences

41. Maintain focus on the Schematic Workspace and try the following Hotkey Sequences:

Follow the underlined letters within menus and sub-menus to learn hotkey sequences.

- To **View » Fit Document** press the **V » D** keys in succession
- To **View » Fit All Objects** press the **V » F** keys in succession
- To **View » Area** press the **V » A** keys in succession, define the viewing area by drawing the diagonal corners of a rectangular area
- To **View » Around Point** press the **V » P** keys in succession
- To **View » Selected Objects** press the **V » E** keys in succession

4.4.4 Jump Menu Hotkey Sequences

42. Maintain the focus on the Schematic Workspace and observe the behavior of the following jump commands. The **Edit » Jump** menu is commonly used, so the status of this sub-menu has been elevated to a popup menu with its hotkey **J.**
43. To **Jump » Origin**, press the **J » O** keys in succession
44. To **Jump » Location**, press the **J » L** keys in succession. Then enter the X, Y coordinates in the *Jump to Location* dialog as shown in Figure 56 below and click **OK** to move the cursor to the defined coordinate location. Zooming and panning will occur when necessary if the target location is outside the visible area.

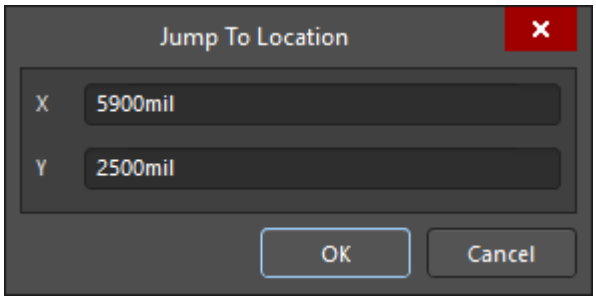


Figure 56. Jump to Location (Shortcut Keys J » L)

45. Maintain the focus on `Microcontroller_STM32F101.SchDoc`
    a) To **Jump » Component**, press the **J » C** keys in succession.
    b) Enter `U2` into the resulting *Component Designator* dialog.
    c) Press **OK** and the cursor will jump to the component `U2`.
    In this example `U2` is a component with two parts. You have the option to jump to the next part of `U2` when the *Find Text – Jump* dialog opens.
    d) Zoom and pan will occur only if necessary to reach the location of the component in the event it is currently outside the visible area.
    e) Click **Close** for the *Find Text – Jump* dialog, then close the *Messages* panel.

4.4.5 Search with the Search Field

46. Make the schematic file `Bluetooth_Sentinel.SchDoc` the focused document by left-clicking on its document tab or open it from the *Projects* panel.
47. With the *Projects* panel open, enter `R10` in the search field at the top of the *Projects* panel as seen in Figure 57 below.
48. You will notice that the search result will highlight all fields that matches the input of the search field. Locate the component R10 by opening the Components folder and its subdirectory.

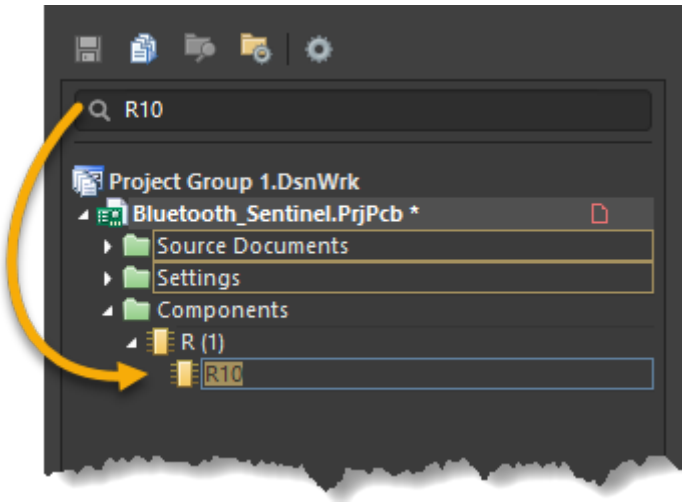


Figure 57. Component Search from the Projects panel

49. Close all documents from the **Window** menu and **Close All.** Do **not** save any changes if prompted.
50. Then, delete the "`R10`" text in the search field at the top of the *Projects* panel, which can be seen in Figure 58 below.

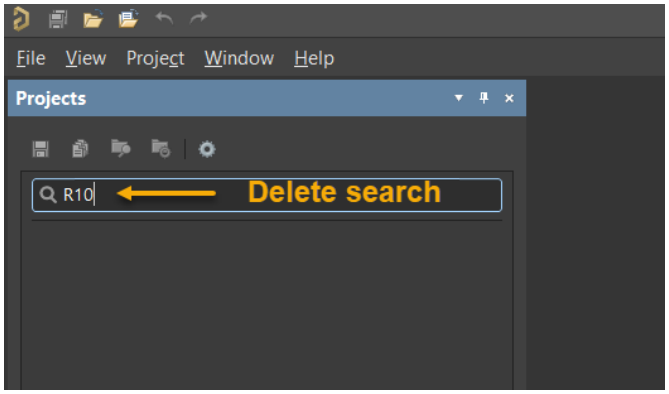


Figure 58. Delete search field from the Projects panel

Module 5: Schematic Capture

5.1 Purpose

In this module, you'll learn how to add components in the schematic editor, along with learn the best practices of schematic component placement.

In this exercise, you will start a project called `WCTopping.PrjPcb` and place components on three schematics in this design. This is a daughter board providing expansion of a typical controller board via a 40-pin connector.

5.2 Preparation

1. Close all existing projects you may have open.
2. Ensure that you have extracted the `AD20 Essentials Exercises.ZIP` file to your `Desktop`. This will make the project easy to find for later modules.
3. Find the folder `AD20 Essentials Exercises\Module 5 - Schematic Capture`. This folder contains files you will use and can reference while you complete this module.

5.3 Create A New Project

4. Select **File » New » Project…** to launch the *Create Project* dialog. The dialog shown in Figure 59 will appear.

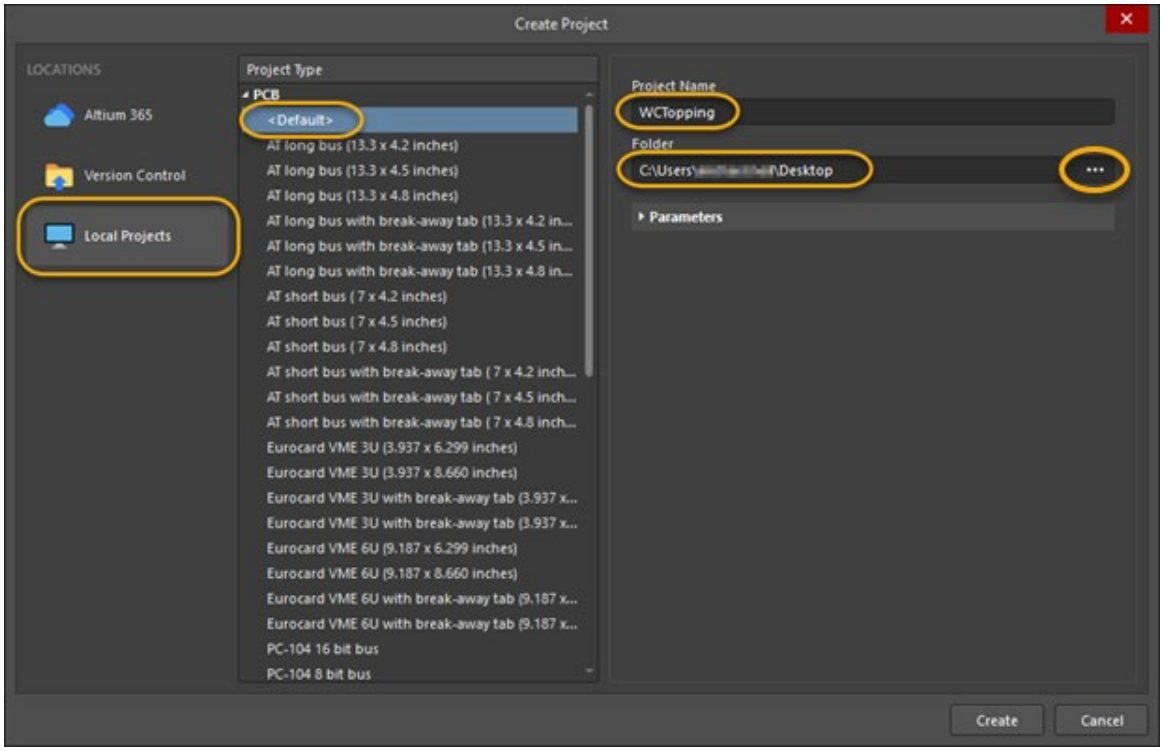


Figure 59. Create a New Project

5. Select **Local Projects** in the left pane. Use Figure 59 above as a reference for the following steps.
6. Select **<Default>** in the *Project Type* pane, so that we create a new blank project.
7. Enter a project name of `WCTopping`.
8. Under the **Folder** area, click on the **Triple Dot Icon** and navigate to your `Desktop`. This is where your project will be saved.
9. Click the **Create** button to create the new project. Feel free to go to your desktop to confirm that a `WCTopping` folder has been created.
10. Return to Altium Designer and notice that in the *Projects* panel there is a project called `WCTopping.PrjPCB` as shown in Figure 60 below.

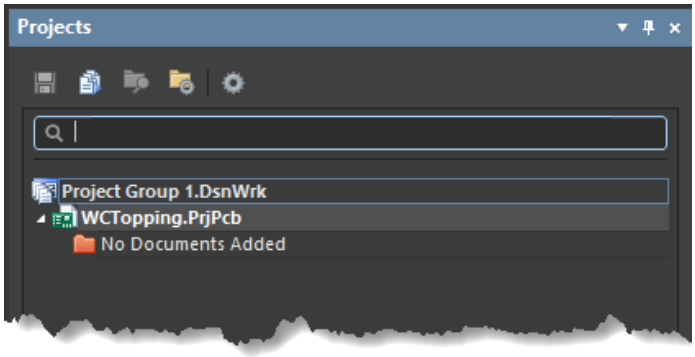


Figure 60. Projects panel with the WCTopping project recently created

5.3.1 Adding New Documents

11. Right-click the project and select **Add New to Project » Schematic** from the pop-up menu as shown in Figure 61 below. This will add a new schematic to project. The default name is `Sheet1.SchDoc`
12. Repeat this operation and add **three additional** schematic files.

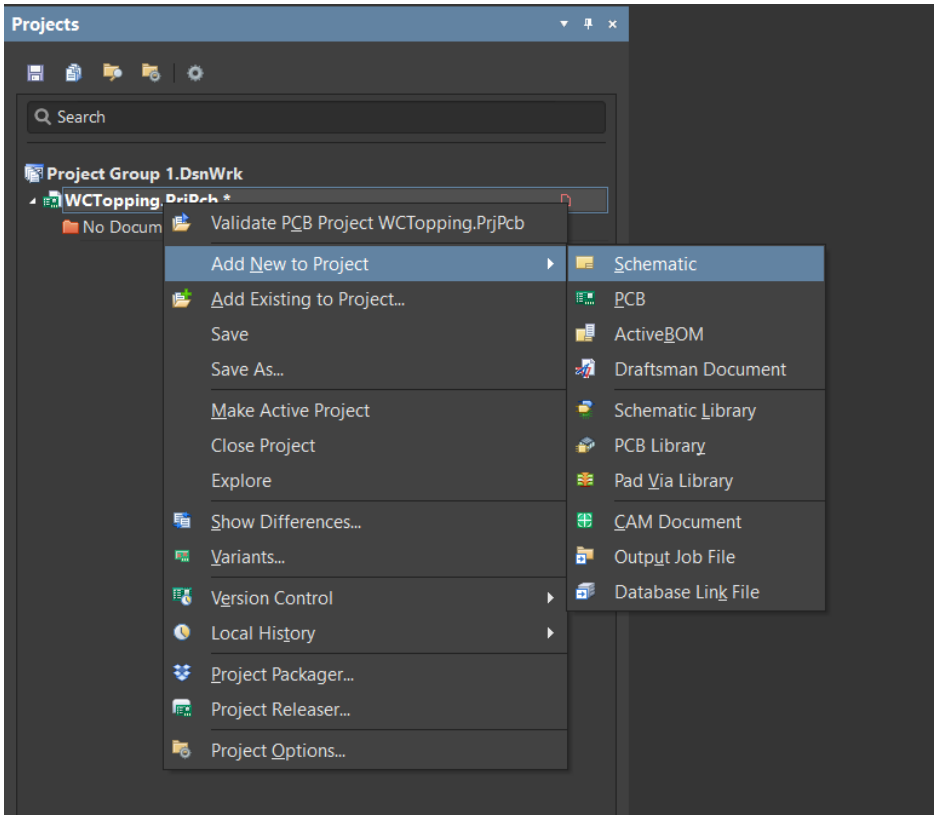


Figure 61. Added new Schematics to the Project

13. Select menu **File » Save All**. A dialog box will pop-up prompting you to save the new schematics you added.
14. Name the schematics in your project using the same names below:
    - 1st schematic as `Processor_Interface.SchDoc`
    - 2nd schematic as `Digital_IO.SchDoc`
    - 3rd schematic as `Relay_IO.SchDoc`
    - 4th schematic as `CAN_Interface.SchDoc`
15. The *Projects* panel should now look like the one in Figure 62 below. Double check this by navigating to the Desktop project folder. To do this, **right-click** on the project name and select the **Explore** menu.

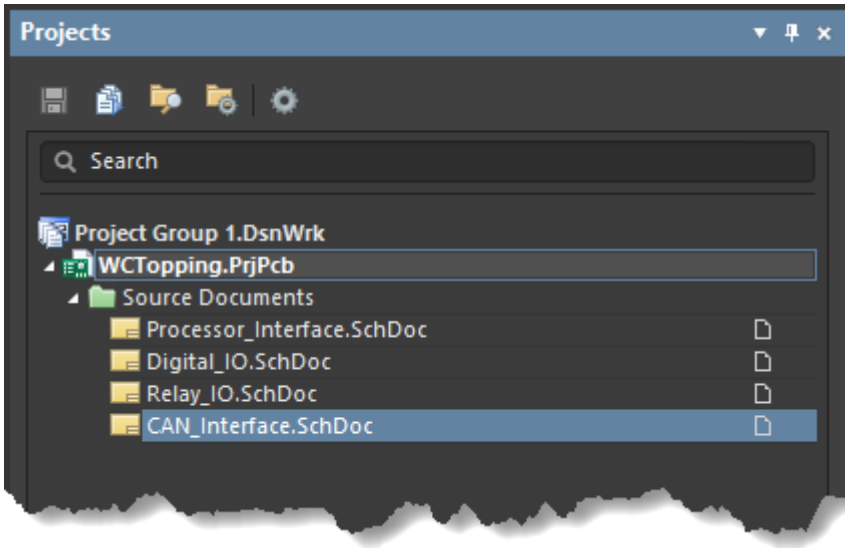


Figure 62. Renamed Project Files

5.3.2 Adding an Integrated Library

The schematic library can be a local library or a library of components contained in an Integrated Library stored on a server. For this exercise, we will use the Integrated Library located in the folder `AD20 Essentials Exercises\Module 5 - Schematic Capture`.

16. Open the *Components* panel by clicking on the **Panels** button in the bottom right corner.
    a) The *Components* panel should open with the `Miscellaneous Devices.IntLib` selected as shown in Figure 63. If not, select the drop-down menu from the top of the *Components* panel and then select the required library.
    b) Note that the *Components* panel is divided into several panes. The topmost pane displays a list of library components for the selected library. The subsequent panes have information related to the selected component, e.g., symbol and footprint previews and models linked to the component.
17. Click the button in the top right corner of the *Components* panel and select **File-based Libraries Preferences…**. The *Available File-based libraries* dialog will open as shown in Figure 63 and Figure 64.

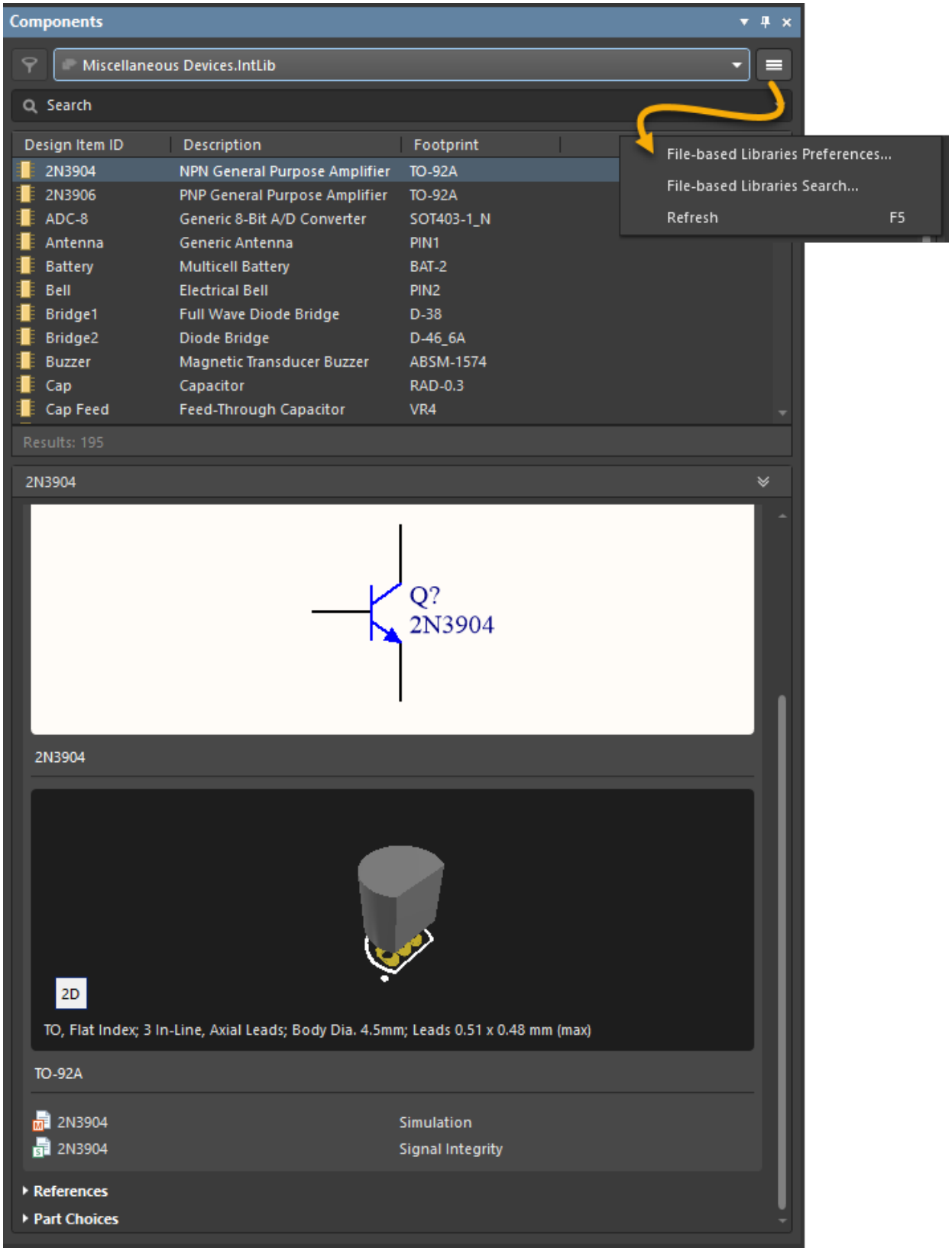


Figure 63. Components Panel with the Miscellaneous Devices.IntLib selected

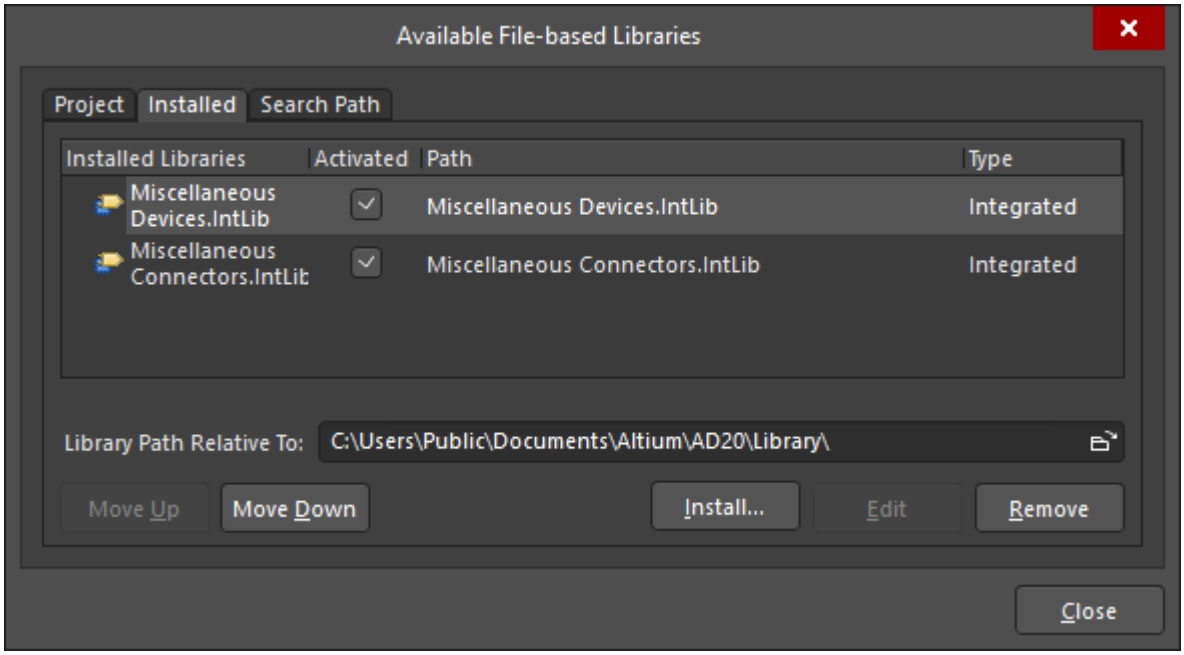


Figure 64. Installed Libraries currently available

18. Select the *Installed* tab.
19. Click on the **Install…** button. This is where we will install a component Library for us to access the components from that Library.
20. Navigate to the training folder `AD20 Essentials Exercises\Module 5 - Schematic Capture` and select `WCT.IntLib`, click the **Open** button.
21. The `WCT.IntLib` library should now appear in the *Available File-based libraries* dialog along with the previously installed libraries as shown in Figure 65.

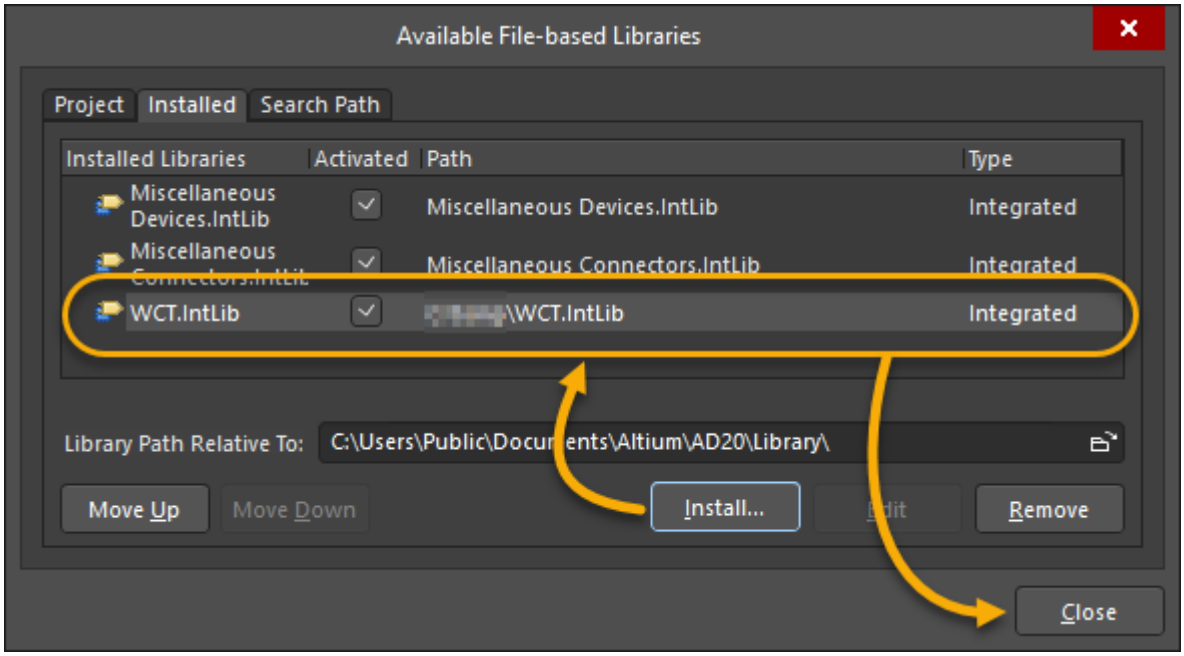


Figure 65. WCT.IntLib added to Installation

22. Click **Close** to close the *Available File-based libraries* dialog.

5.4 Component Placement

5.4.1 Setting the grid

23. Start by double clicking on the `Processor_Interface.SchDoc` schematic to make it the focused schematic.
    a) Open the *Properties* panel from the **Panels** button.
    b) In the *General* tab, set the Snap Grid value to `200mil`, and press **Enter** to apply the new value as shown in Figure 66.

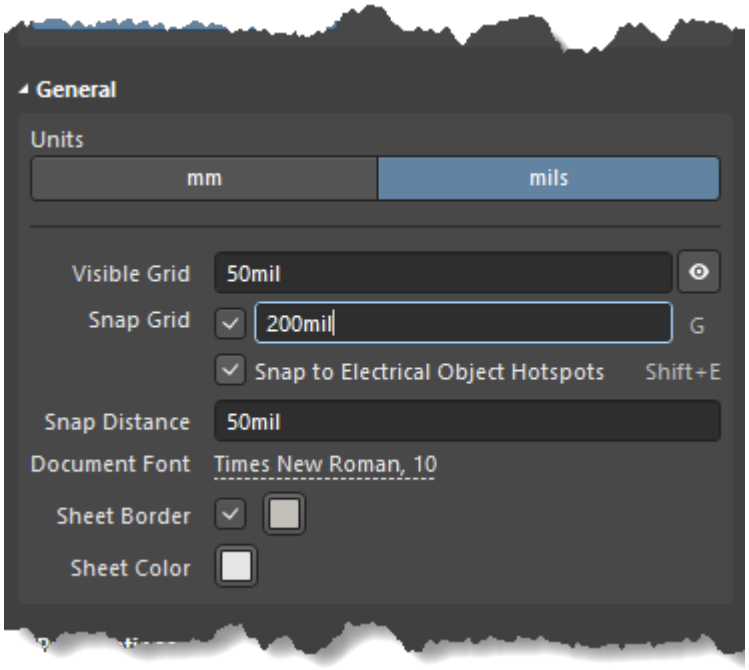


Figure 66. Set the Snap Grid to 200mil

5.4.2 Processor Interface Schematic Component Placement

24. Ensure that the `Processor_Interface.SchDoc` is the active schematic document. Figure 67 below will be used a reference for the following section.

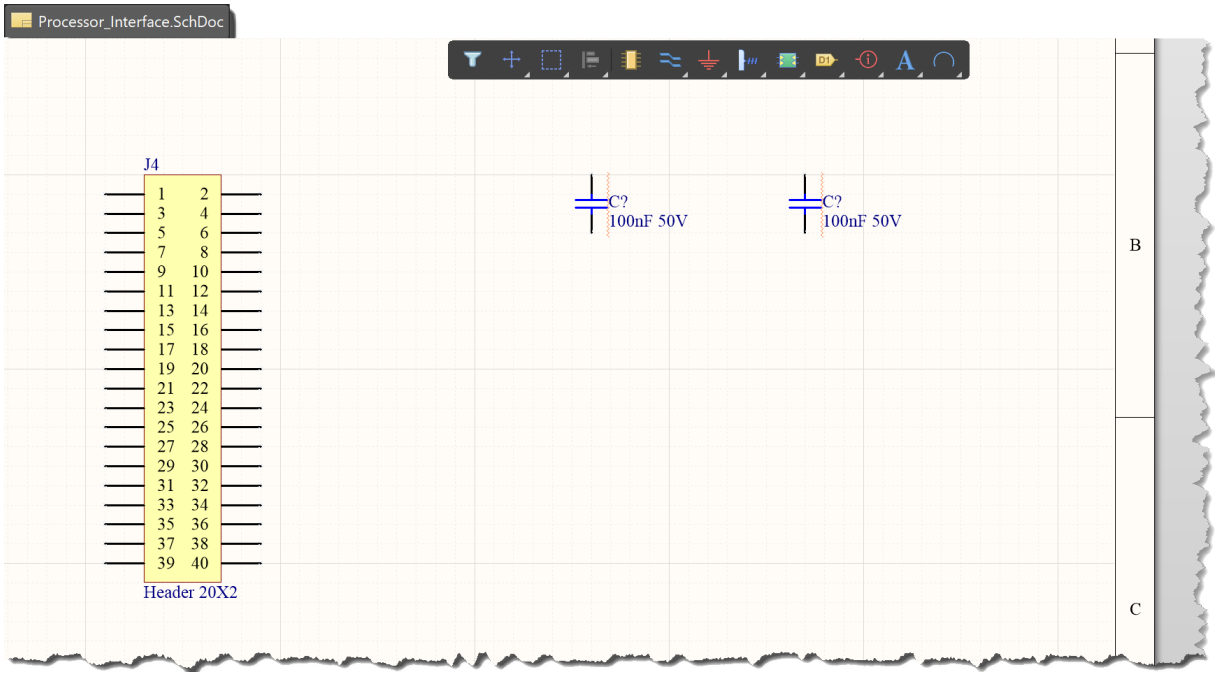


Figure 67. Processor_Interface Schematic

25. Select the `Miscellaneous Connectors.IntLib` from the drop-down list of libraries in the *Components* as shown in Figure 68 below.

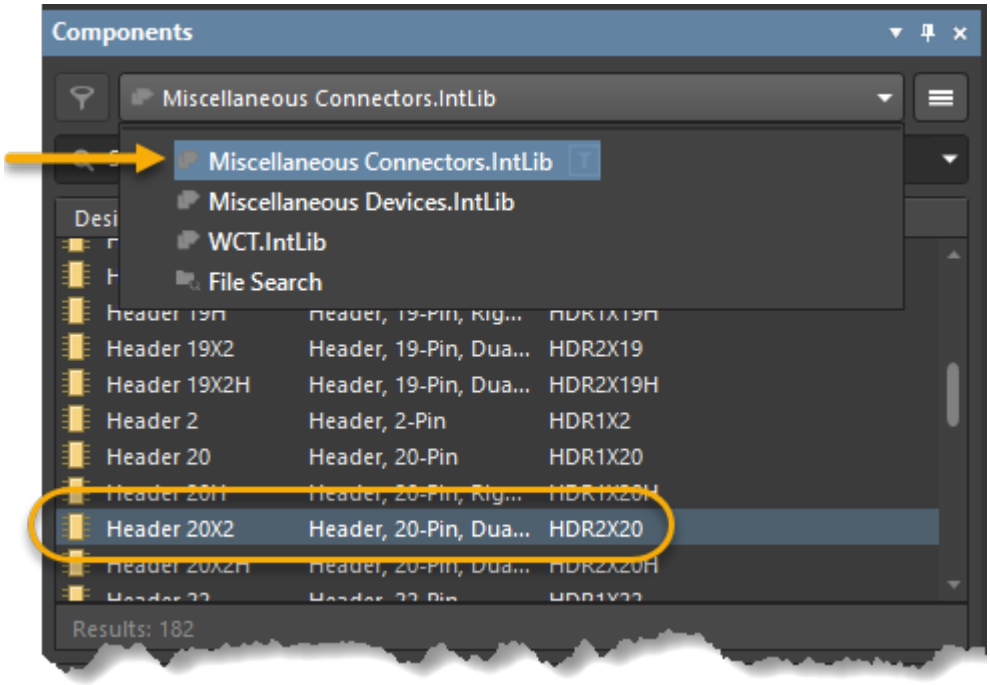


Figure 68. Select the Miscellaneous Connectors library

26. Scroll down the list of components to find the 40-pin I/O Connector called `Header 20x2` as shown in Figure 68 above.
27. Right-click to **Place Header 20x2** and move the cursor with the attached symbol inside the schematic page.

28. Before placing the connector on the schematic, press the **TAB** key:
    a) In the *Properties* panel that opens, any changes to the default properties will be applied to subsequently placed objects.
    b) Change the reference designator to `J1` as shown in Figure 69.
    c) To apply the changes press, **Enter** and continue with the placement.

The **TAB** key is used to pause the command that you are currently in. This allows you to make changes to an item's properties. To resume the command, hit the **Enter** key, or the ***"Pause"*** icon in the middle of the editor to continue with the command.

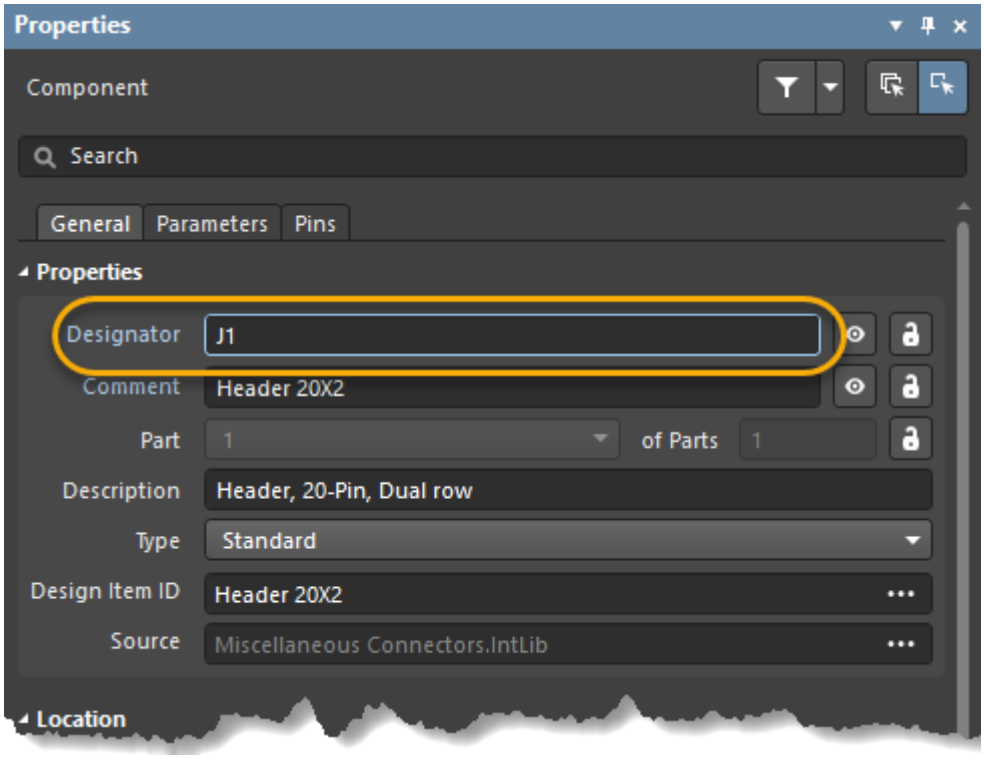


Figure 69. Component Properties dialog

29. While in placement mode with a connector attached to the mouse pointer, experiment with commands to orient the component.
    a) **Spacebar** to rotate counterclockwise 90°
    b) **Shift+Spacebar** to rotate clockwise 90°
    c) **X** to flip along the X-axis
    d) **Y** to flip along the Y-axis

If a placed component should be rotated, select the component and press **Spacebar** to rotate counterclockwise 90°) or **Shift+Spacebar** to rotate clockwise 90°**.**

30. Left-click anywhere in the schematic to place the component.
31. Continue placing multiple components, up to at least `J4`. The reference designators will increment automatically. You will remain in placement mode and be able to place multiple components until you **right-click** or press **ESC.**

32. Delete all the connectors on the schematic except for `J4`. This will give you the opportunity to experiment with the **Smart Drag Selection** feature:
    a) Drag a selection window from **left-to-right** using the **left-mouse-click**, around the 3 connectors (J1 to J3). This will select all objects that fall completely within the bounds of the selection area. This behavior is the same as using the **Edit » Select » Inside Area** command.
    b) Now drag a selection window from **right-to-left**. This will select all objects that fall completely inside the selection area and are touched by its boundary. This behavior is the same as using the **Edit » Select » Touching Rectangle** command.
    c) The **Green** or **Blue** coloring is used to distinguish which mode of selection is being used visually. Dragging **left-to-right** uses a **blue rectangle** while dragging **right-to-left** uses a **green rectangle** as shown in Figure 70 below.

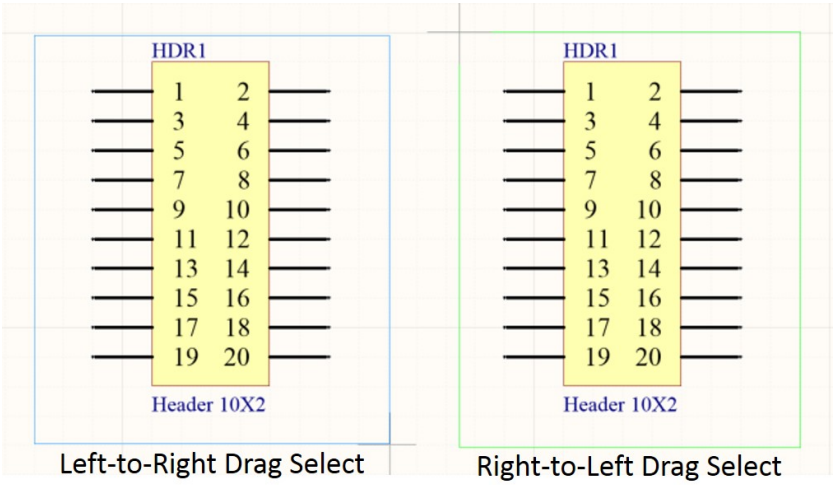


Figure 70. Colored boxes indicating the drag-select method used

33. Make sure you hit the **Delete** key to delete the appropriate connectors from the schematic.
34. Open the *Components* panel, and make sure the ***WCT.IntLib*** is the active library. It can be found from the topmost drop-down menu of the *Components* panel.
35. Use the library search, or simply navigate down the list of components, and place two **Cap100nF_50V_0603** capacitors from the `WCT.IntLib` library, into to the `Processor_Interface.SchDoc`. Making the *Components* panel larger may help you see the full name of the component you wish to select and place.
36. Save the schematic by right-clicking on the schematic in the *Projects* panel, and click **Save.**

5.4.3 CAN_Interface Schematic Component Placement

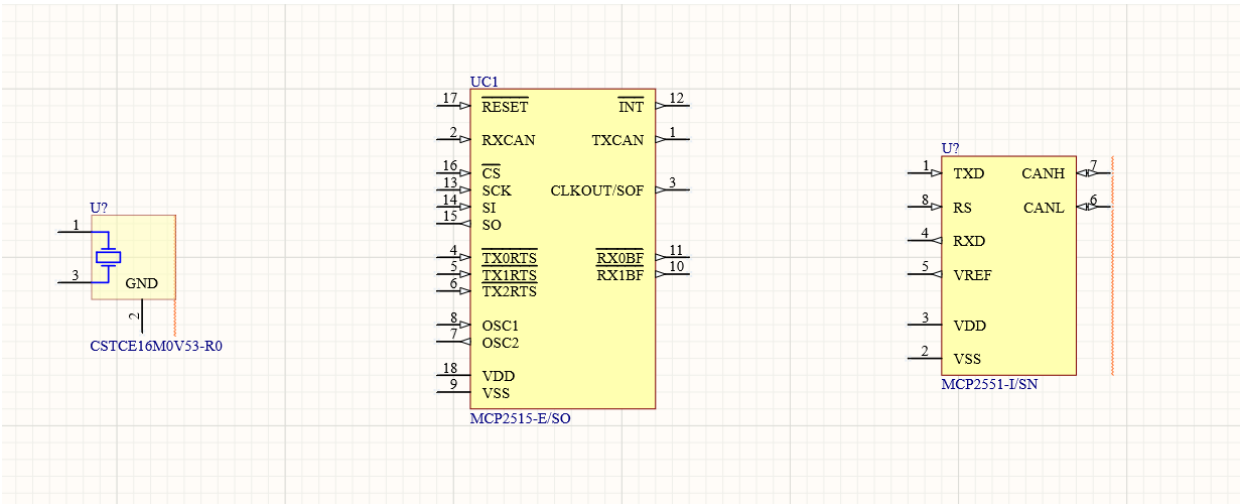


Figure 71. CAN_Interface Schematic

37. Open the `CAN_Interface.SchDoc`.
38. Since we would now like to place components from the *Manufacturer Part Search* panel, we will need to connect to the *Altium Content Vault* to yield the maximum component search results.
    a) Ensure that you are signed into your *AltiumLive* account, by going to the icon in the upper right corner as shown in Figure 72 below, or click **Licenses** to open the *Licenses Management* page, and enter your *AltiumLive* credentials.

If you already see your name in the upper-right corner instead of the icon shown in Figure 72, this means you're already signed into your AltiumLive account. If you don't remember your login information, please contact your in-house Altium administrator, or contact Altium Technical Support.

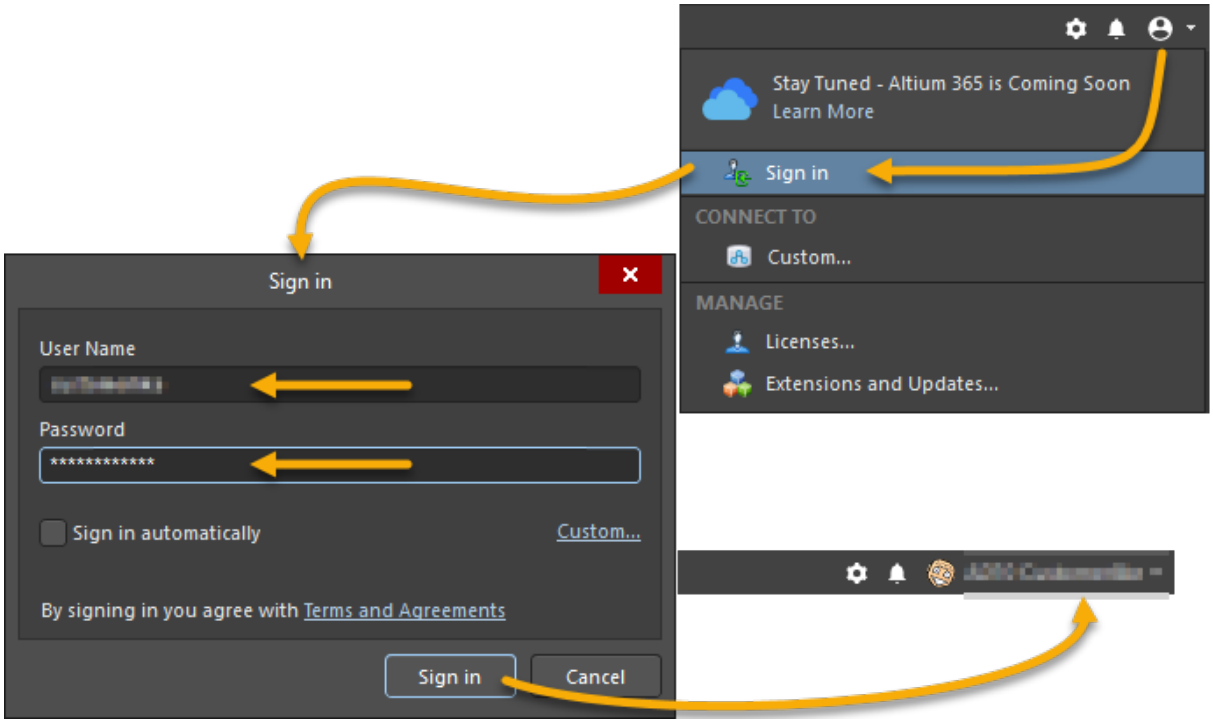


Figure 72. Signing into your AltiumLive account to access the Altium Content Vault

b) Select Preferences, in the upper right corner of Altium Designer.
    i) Open the page *Data Management - Servers*
    ii) Expand the "*Known Servers*" option as shown in Figure 73.
    iii) Ensure that **Automatically connect Altium Content** Vault is enabled, and that the **"On"** option is also enabled. You will know that you are connected when the **"Connected**" word is visible.
    iv) Close the preferences by clicking **OK.**

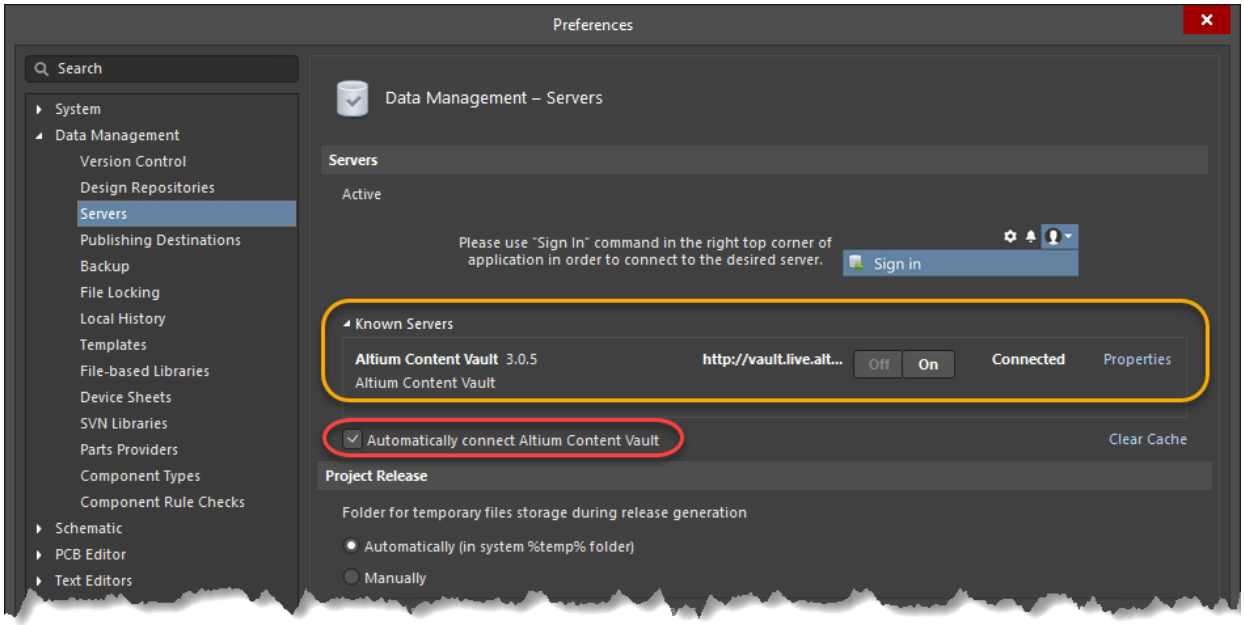


Figure 73. Connecting to the Altium Content Vault in Preferences

c) Make the `CAN_Interface` schematic the active document.

39. Placing components from the *Manufacturer Part Search* panel is possible, allowing you to view Supplier Information prior to placement. Open the *Manufacturer Part Search* panel by clicking on the *Panels* button and choosing *Manufacturer Part Search.*

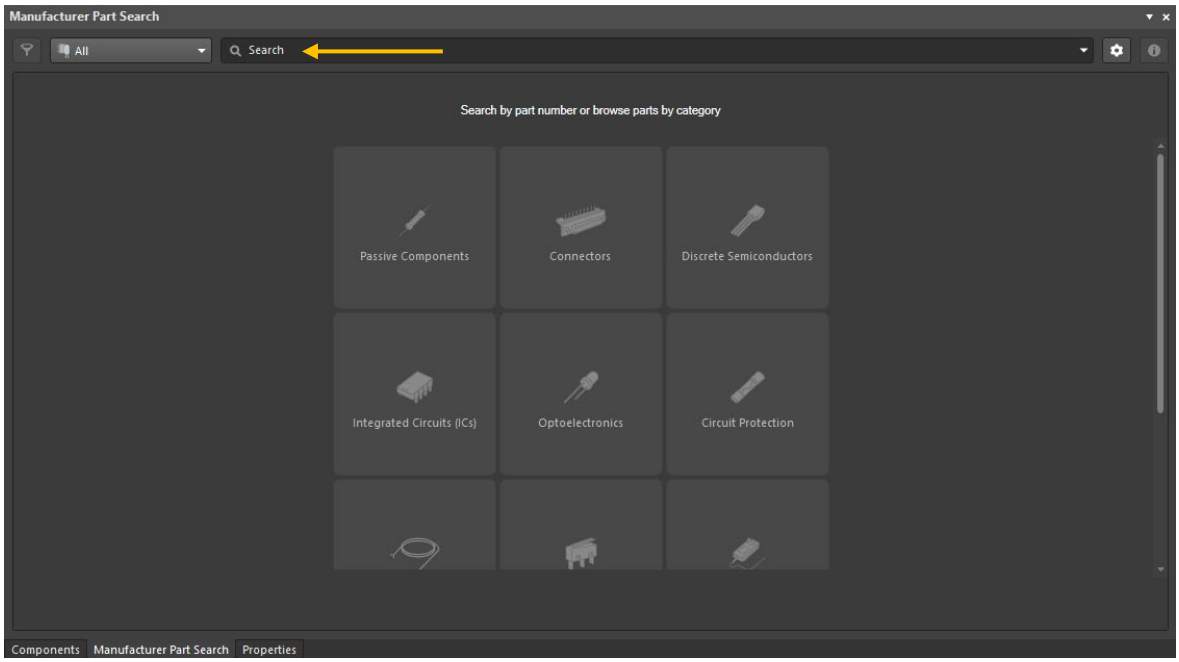


Figure 74. Manufacturer Part Search panel

40. With the `CAN_Interface` schematic open, search for `MCP2515` by typing in the upper main field of the *Manufacturer Part Search* panel as shown in Figure 74 above and pressing **Enter**.
41. In the search results, **right-click** `MCP2515-E/SO` and select **Place** as shown in Figure 75 below. You may need to expand the panel for easier viewing.
42. With the component still on your cursor, before clicking to place on the schematic, press **Tab** and change the reference designator to `UC1` in the *Properties* panel.

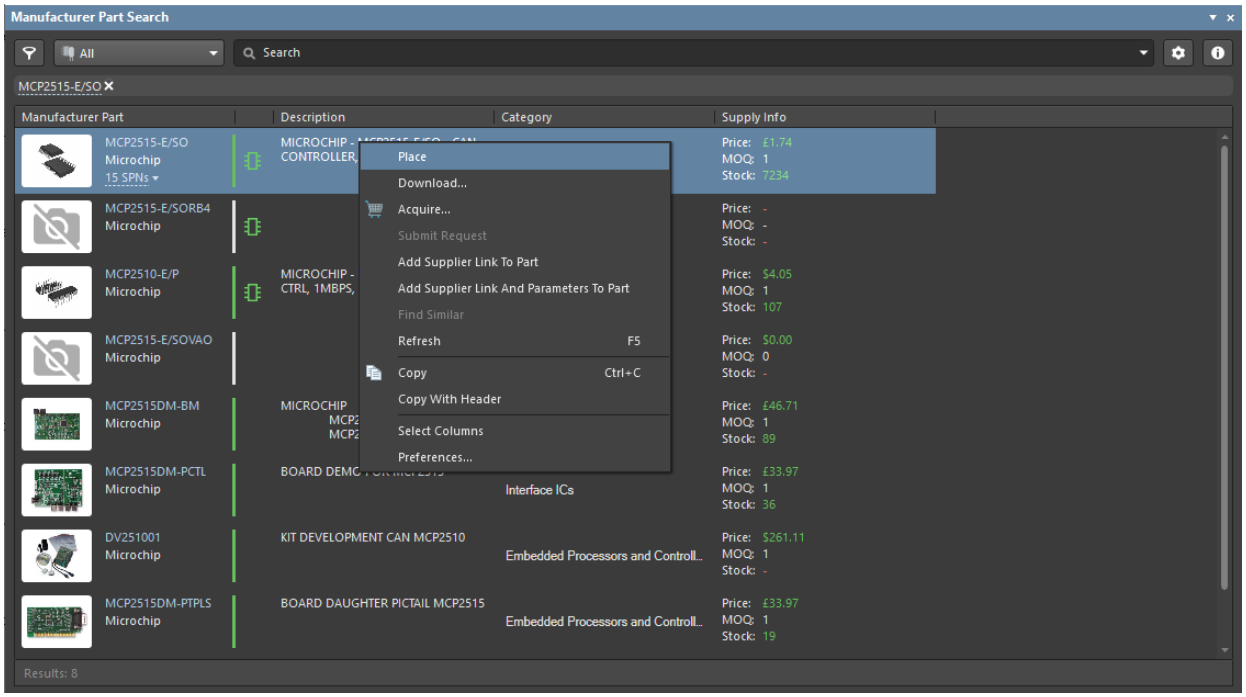


Figure 75. Select and Place component from the Manufacturer Part Search panel

43. Right-click once the component has been placed to return focus to *the Manufacturer Part Search* panel. If you closed the *Manufacturer Part Search* panel, it can be re-opened from the **Panels** button.
44. Using the search bar in the *Manufacturer Part Search* Panel, search for `MCP2551`. **Right-click** and **place** `MCP2551-I/SN` to the right of `UC1`. During placement, press **TAB** to set the reference designator to `UC2`. Once again, you may need to move or close the Manufacturer Part Search panel before changing the designator of the component.
45. Return to the *Manufacturer Part Search* panel and search for `CSTCE16M0V53-R0` (a 16MHz Crystal).
46. **Right-click** and **place** `CSTCE16M0V53-R0` to the left of the `UC1` `MCP2515` component, as shown in Figure 71 on a previous page. Press **TAB** before placing it and set the reference designator to `X1`. You can always open the *Properties* panel to access the components properties.
47. Hit **ESC** to end the command, and **Save** the `CAN_Interface` schematic from the *Projects* panel.

5.4.4 Relay_IO Schematic Component Placement

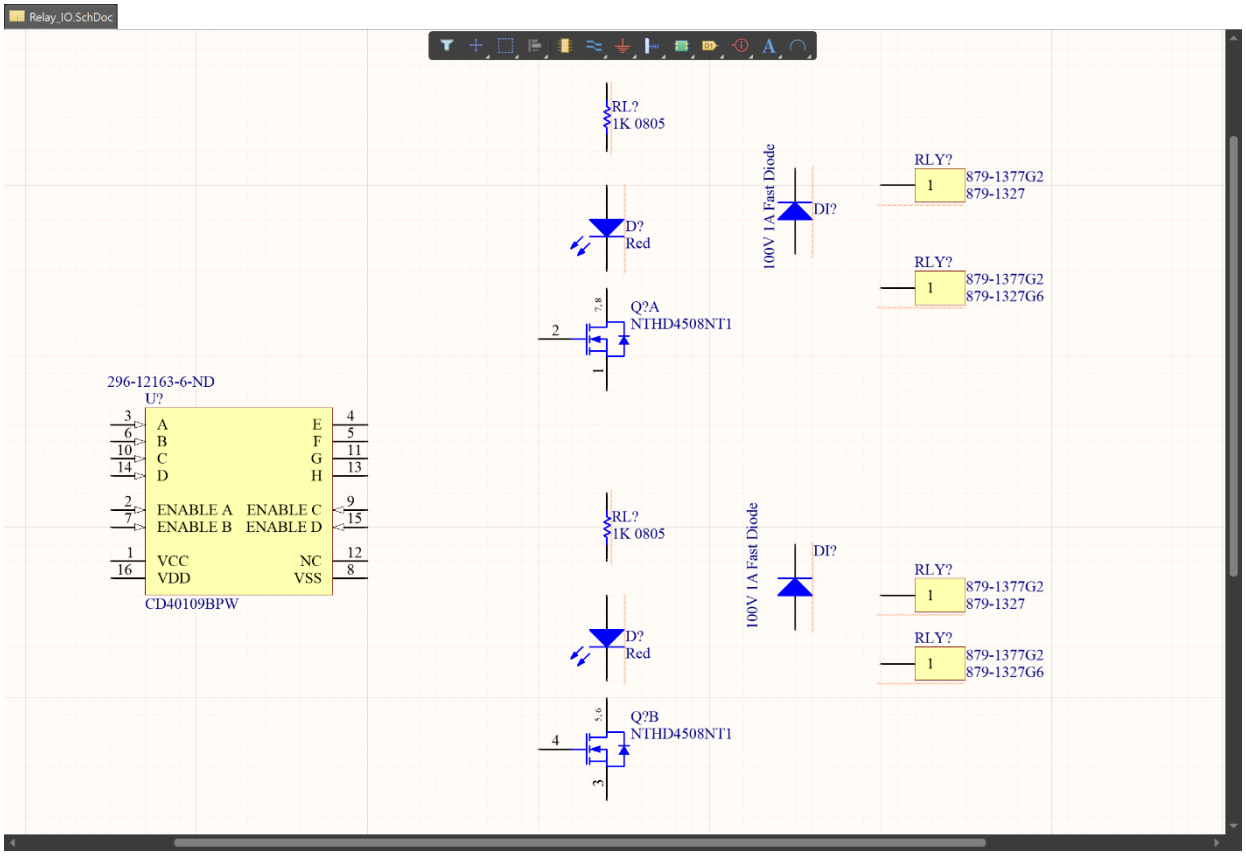


Figure 76. Relay I/O Schematic

48. Open the `Relay_IO.SchDoc`
49. From the *Components* panel, place the `NTHD4508NT1` from the `WCT.IntLib` library. Use the search bar, or navigate to the component to find it as shown in Figure 77.
    a) This is a multi-part component as indicated by the part information below the Symbol preview as shown in Figure 77. A multi-part component is represented by multiple schematic symbols. Expand the *Models* pane if necessary.

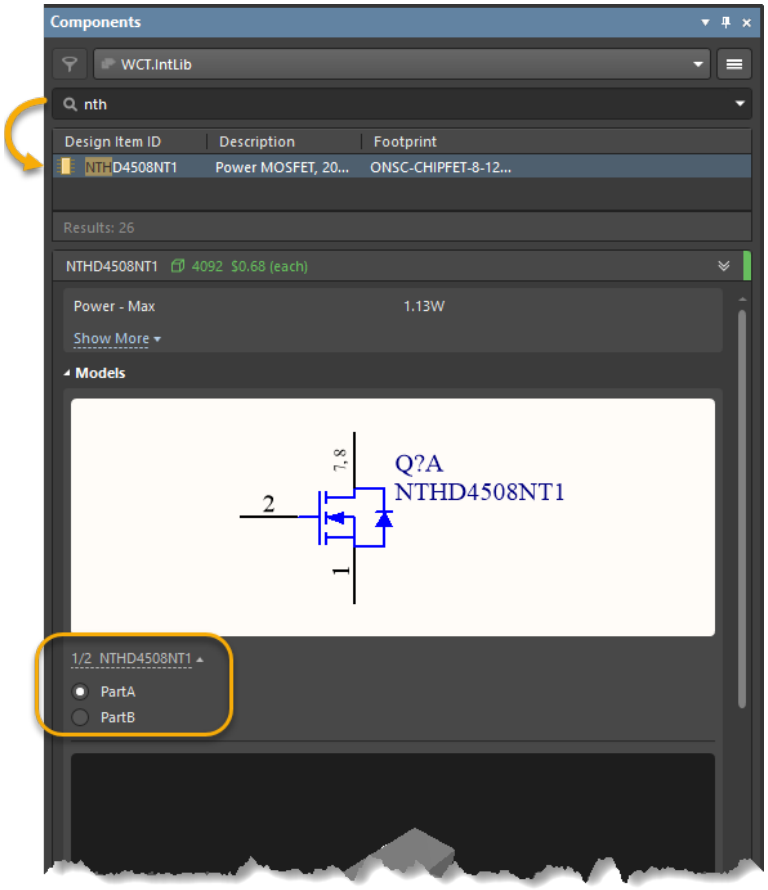


Figure 77. Placing NTH Multi-Part into the Relay I/O schematic

b) Select `Part A`.

c) Right-click and place the component. During placement, press **TAB** to change the designator to `Q1`, hit **Enter** and place the component.

d) It will place `Part A`, and now `Part B` will be on your cursor. Place that part as well. This is considered a Multi-Part component that is represented by more than 1 schematic symbol.

e) Right-click to end the command after placing `Q1A` and `Q1B`.

50. We will now place another component from the *Manufacturer Part Search* panel.

a) Open the *Manufacturer Part Search* panel from the **Panels** button.

b) Search for `CD40109BPW`.

c) Right-click and **Place** `CD40109BPW`, which should be the first option that appears.

d) Right-click to end the command once the component is placed, and close the *Manufacturer Part Search* panel.

51. Search and place the remaining components for the `Relay_IO.SchDoc` from the `WCT.IntLib` found in the *Components* panel. Press **TAB** to set the designators accordingly. Refer to Figure 76 on a previous page to reference the component placement.

a) `Res1K_0805` (2)

b) `LED_Red0603` (2)

c) `Diode 1N4148` (2)

d) `Anderson-1Pin BLK_PCB_RHT` (2)

e) `Anderson-1Pin Red_PCB_RHT` (2)

52. Save the `Relay_IO.SchDoc` schematic.

5.5 Adding an Existing Schematic to Your Design

53. In Altium Designer, right-click the `WCTopping.PrjPcb` project in the *Project* panel and select **Add Existing to Project…** as shown in Figure 78 below.

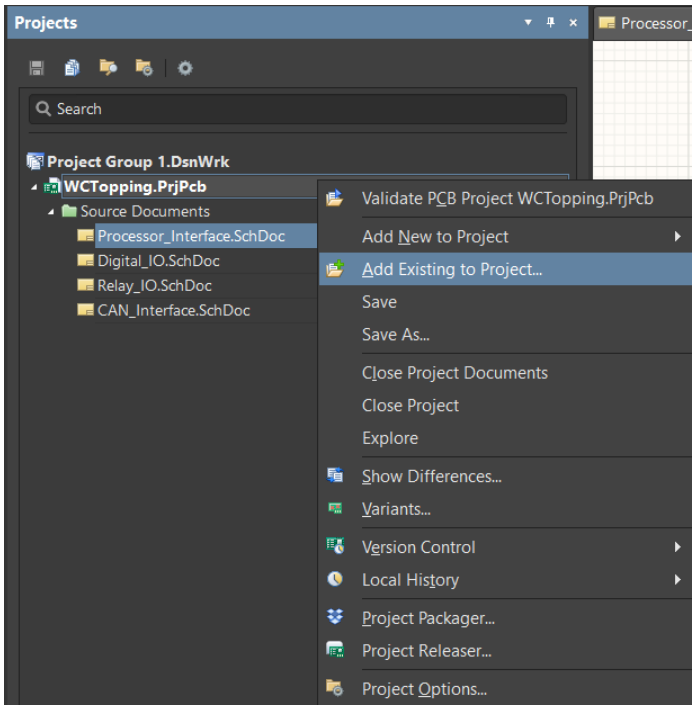


Figure 78. Right-click the Project to Add Existing to Project

54. In the *Choose Documents to Add to Project* dialog, locate the schematic `Power Supply.SchDoc` found in the `AD20 Essentials Exercises\Module 5 - Schematic Capture` folder on your desktop.
55. Click **Open** to add the file to your project.
56. Open the `Power Supply` schematic from the *Projects* panel and ensure its focused document.
57. Right-click the `Power Supply` schematic in the *Projects* panel, and click **Save As**. Browse to your `WCTopping` folder (stored on your desktop) to save the schematic into our project folder.
58. Save all documents using **File » Save All.**

Module 6: Making the Connection

6.1 Purpose

In this exercise, you will complete the Processor_Interface, CAN_Interface and Digital_IO schematics by adding wires, net labels, power ports and ports to the design for connecting the components.

6.2 Preparation

1. Close your current `WCTopping.PrjPcB` project.
2. Open the provided `WCTopping.PrjPcb` project from the `AD20 Essentials Exercises\ Module 6 - Making the Connection` folder.

6.3 Wiring the Schematic

6.3.1 Shortcuts

Place » Wire (shortcut **P » W**)
Place » Power Port (shortcut **P » O**)
Place » Port (shortcut **P » R**)
Place » Net Label (shortcut **P » N**)

6.3.2 Processor_Interface Schematic

6.3.2.1 Wire Connections

3. In your `WCTopping.PrjPcb` project, open the `Processor_Interface.SchDoc`. It should only contain 3 components without any connections. By the end of this section, this schematic will be populated with wires, port, power port and net label connections as shown in Figure 79.

The negate overscore shown on ports *CS* and *IOexpand_ CS* can be created by entering a backslash "\" character after the character to negate.

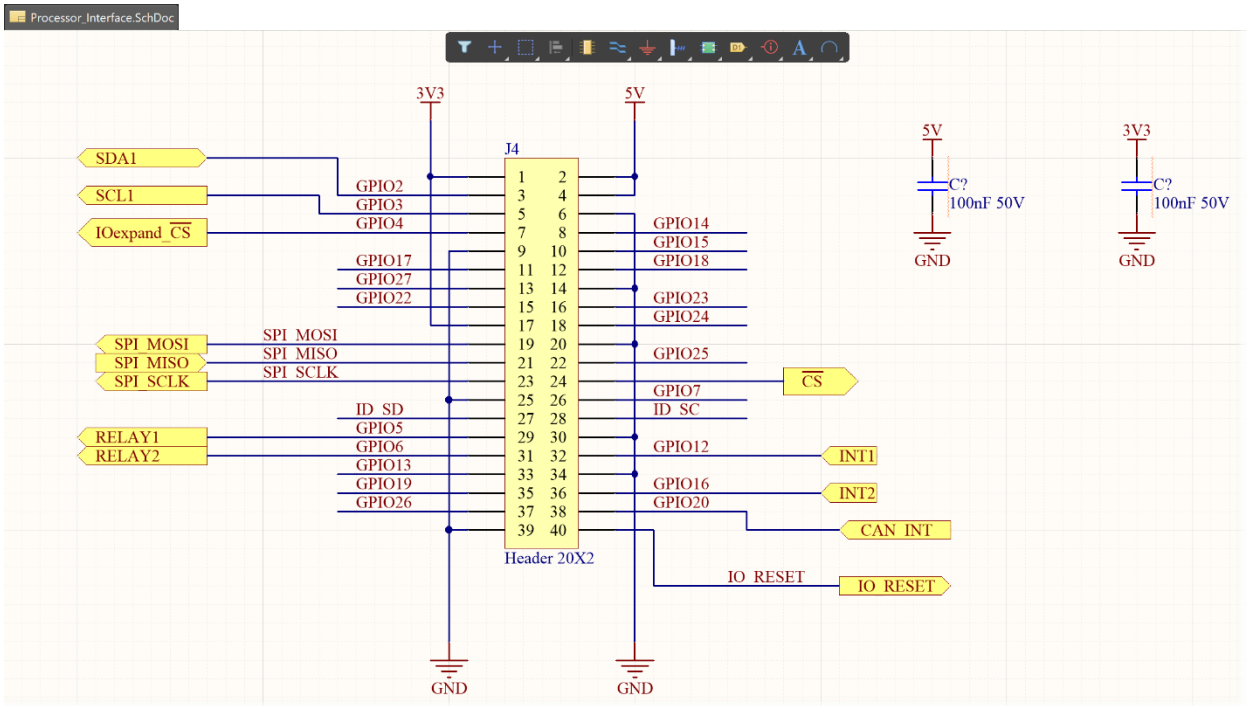


Figure 79. Processor_Interface Schematic

4. We will connect Pin 1 and Pin 17 of the header (J4). Use Figure 79 above as a reference.
   a) Use **Place » Wire** (shortcut **P » W**).
   b) Move the cursor near the tip of the Pin 1 of J4. You will see a cross appear, indicating you are on the pin's electrical hotspot as seen in Figure 80.
   c) **Left-click** when the hotspot is active to begin placing a wire segment.
   d) Press **Spacebar** to flip the bending side of wire if needed.
   e) A new segment will start with each left-click until another electrical hotspot is clicked on (such as another pin) to complete the placement of this first wire.
   f) Complete the connection from Pin 1 to Pin 17
5. During placement of a segment, **right-click** once to stop the current segment but remain in placement mode.
6. Connect Pin 9, Pin 25 and Pin 39 using the method described in Step 4.
7. **Right-click** twice (while there is no current wire being placed) to exit the command.

Cross-Overs may be displayed depending on the settings in your Schematic Preferences; Section Schematic – General and the option **Display Cross Overs**

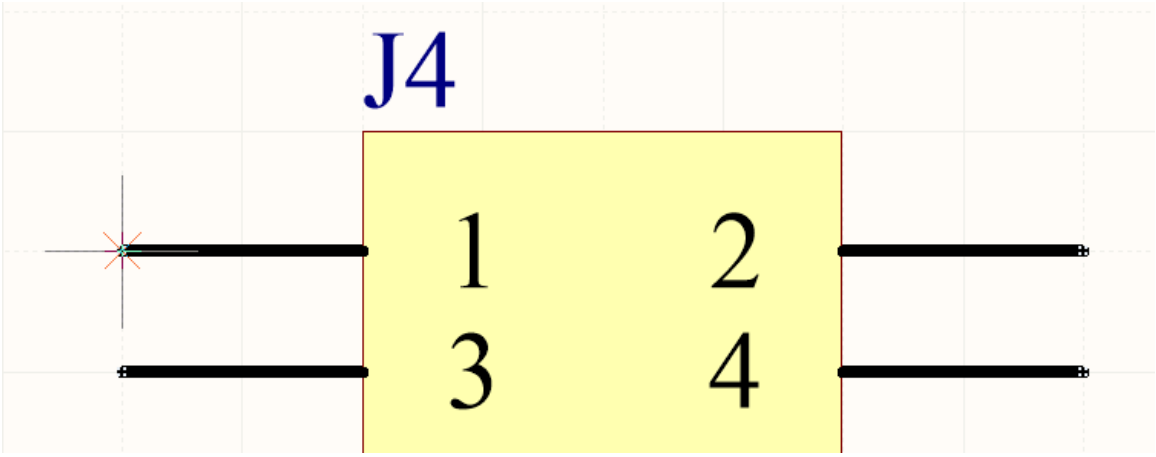


Figure 80. Electrical Hotspot of a pin for schematic connections

Use Spacebar to change the angle of a wire corner during placement.
Use Shift+Spacebar to change the corner mode of a wire during placement.
Use Backspace to unwind an inadvertently placed segment during placement**.**
Use Right-click or Escape to end placement mode.

6.3.2.2 Smart Paste Wires

8. Place a short wire on Pin 19 of the header.
9. Select the wire stub and copy it with **Edit » Copy** or **CTRL+C**.
10. Use **Edit » Paste** or **CTRL+V** to paste the wire stub on the remaining unconnected pins on the left side of the connector.
11. Select **Edit » Smart Paste** to open the *Smart Paste* dialog. This can also be done using the **CTRL+Shift+V** shortcut keys.
12. Smart Paste knows that a wire object has been copied. Copied objects are shown in the left-most list box.
13. In this case, you will be pasting the Wires as Themselves, so make sure **Themselves** is selected in the *Choose Paste Action* list box.
14. Check the **Enable Paste Array** option on the right side. Create `20` rows of wires by setting **Column** to Count `1`, the **Rows** to Count `20`, with a row spacing of `100mil` and pressing **OK**.
15. Place the `20` wire stubs attached to the pins of the header on the right side of the connector.

6.3.2.3 Power Ports

16. Let's place a Power Port from **Place » Power Port,** or from the **Active Bar** to place a power port.
17. Press **Tab** to access the properties as shown in Figure 81.

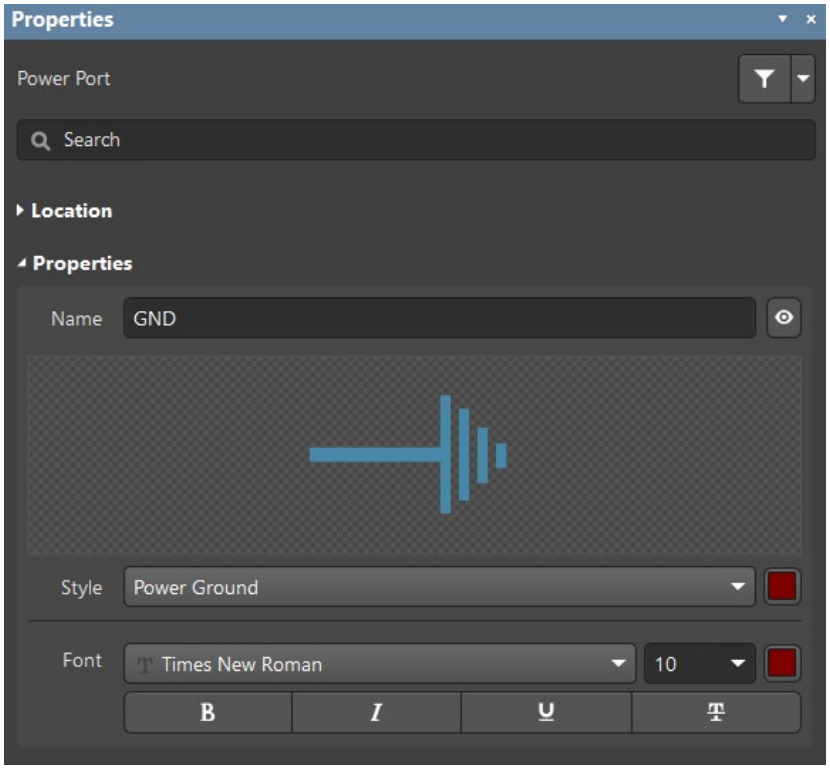


Figure 81. Changing the Power Port style in its Properties

18. Change the **Net** to `GND` and set the **Style** to `Power Ground`. Press **Enter** or click on the Pause symbol to dismiss the dialog and continue placement of the power port.
19. Place two **GND Power Ports** below the header (J4), using Figure 79 on a previous page as a guide.
20. Using Figure 79 for reference, add wires from the **GND Power Ports** to your existing circuit.
21. Using Figure 79 for reference, repeat the above process to add, modify and connect the **3V3** and **5V Power Ports** above the header (J4). The power ports can be rotated by pressing **Spacebar** while moving them. They can be changed to **"Bar"** type, by changing the *"Style"* in the *Properties* panel.
22. Power ports may be placed directly on the pins of symbols. Place **GND**, **5V and 3V3 Power Ports** directly on the capacitors in the upper right corner of the `Processor_Interface.SchDoc` schematic using Figure 79 as reference.
23. Using Figure 79 for reference, draw the remaining wires for the entire schematic. In the next steps, we will describe how to place Net Labels and Ports.

6.3.2.4 Net Labels

24. Right-click and select **Place » Net Label** or use the shortcuts **P » N**
25. With the **Place Net Label** command active, press **Tab** to open the *Properties* panel, so that a name can be entered.
26. Enter the name `GPIO2`, press **Enter** and place the net label on the wire leading to pin 3 of the header using Figure 79 for reference.
27. Continue to place all of the same Net Labels using Figure 79 for reference. Notice the numeric values automatically incrementing. The properties of a Net Label can always be changed by hitting **TAB** during placement, or double-clicking the Net Label and change its name manually.

Some Net Labels contain an **Underscore** that could be mistaken as a space, and not clearly seen if placed on a Wire. These Net Labels include **SPI_xx**, **ID_xx** and **IO_RESET** group of Net Labels. This naming syntax is necessary for proper net connectivity.

28. Double-check and make sure all Wires and Net Labels are place as shown in Figure 79. We will be placing the Ports in the next step.

Moving Net Labels after they are placed on wires can be difficult because moving the net label drags the attached wire. Selecting the Net Labels, then going to **Edit** » **Move » Move Selection** can be helpful when moving net labels. Also, if 'Always Drag' is enabled in Preferences, you may use **Ctrl+left mouse button** to drag a net label off of a wire.

6.3.2.5 Ports

29. Use the **Place » Port** command to place a port that will connect wires to another sheet in the design. This can also be done by using the **P»R** shortcut keys.
30. Press the **Tab** key with the port active and change the name of the port to `SDA1`. Set the **I/O Type** to *Bidirectional* as shown in Figure 82. Press **Enter** or click on the Pause symbol to dismiss the dialog and continue placement of the port.
31. Place the **SDA1 Port** near **Pin 3** of the header as shown in Figure 79 on a previous page. The port will require two left-clicks. The first left-click defines the left side of the port, and the second left-click defines the right-side of the port and its length.

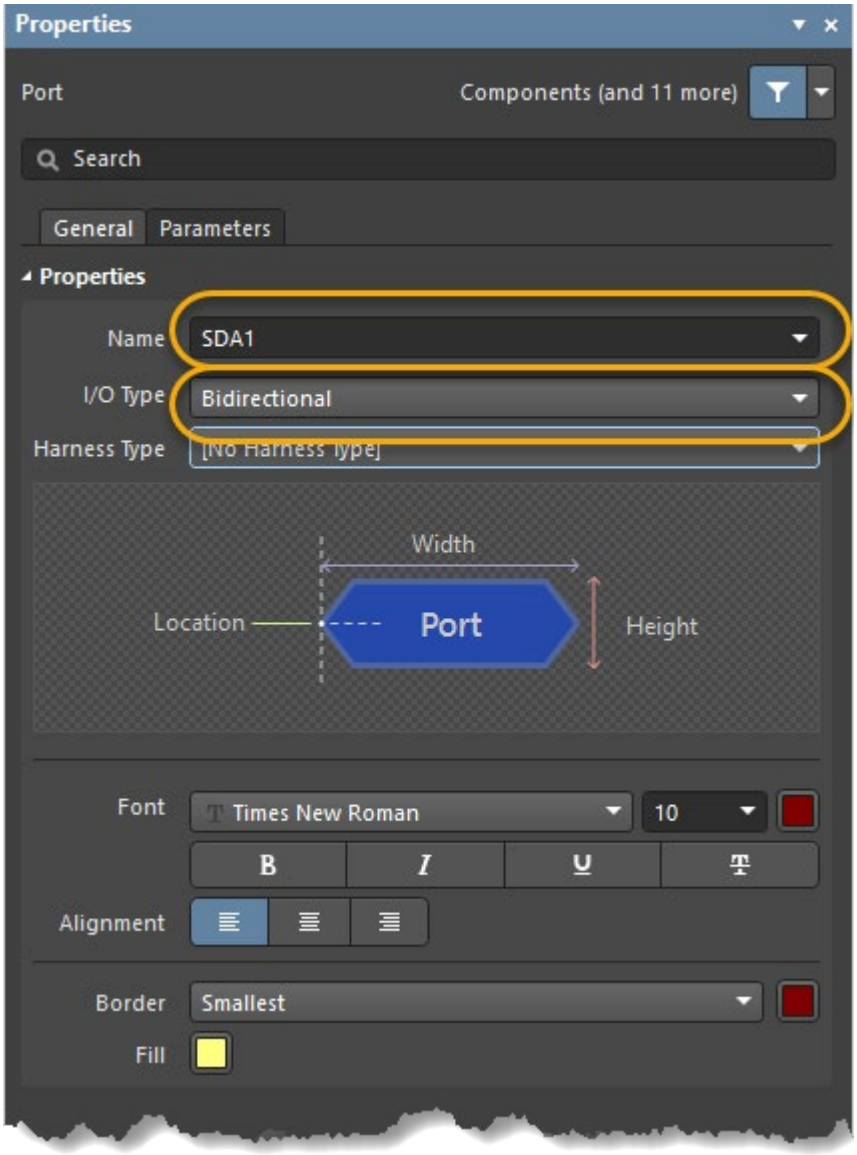


Figure 82. Port Properties dialog

32. Continue to place the ports as shown in Figure 79 on the previous page. Ports named `SPI_MISO`, `INT1`, `INT2`, and `CAN_INT` should be set as **Input** for *I/O Type*, and the rest should be set as **Output**.

Ports will automatically show their I/O type direction (such as Input, Output) once its snapped to a Wire. The best way to place them, is to set their I/O Type in the Properties, then snap it to a Wire.

33. **Save** the schematic.

6.3.3 CAN_Interface Schematic

6.3.3.1 Net Labels

34. Open `CAN_Interface.SchDoc` from your *Projects* panel.
35. **Place » Net Label** and press **Tab** to set the name to `CAN_RXD`. Place the net label between the two resistors attached to `UC2-4` (`Pin 4` of `UC2`) as shown in Figure 83 below.

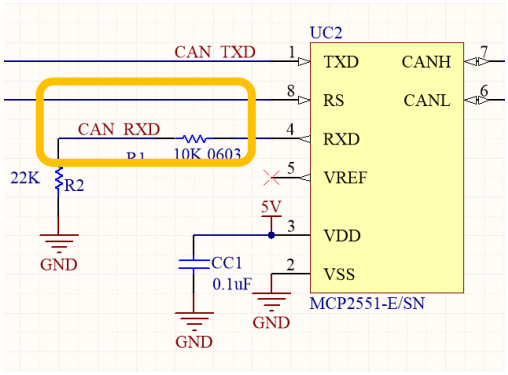


Figure 83. CAN_RXD Net Label placement for UC2

36. Place another instance of the `CAN_RXD` on the wire attached to `UC1-2` to create a connection without needing to place a wire as shown in Figure 84 below.

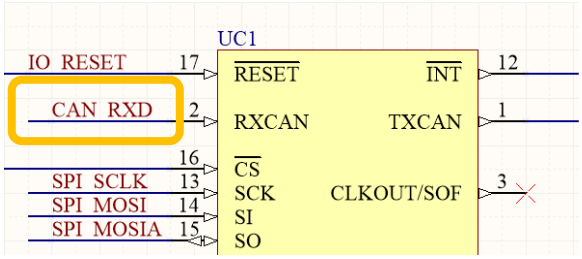


Figure 84. CAN_RXD Net Label placement for UC1

6.3.3.2 Smart Paste Ports

37. Select the `SPI_SCLK, SPI_MOSI, and SPI_MOSIA` net labels, choosing the first one then holding **Shift** down to select the remaining.
38. Copy the net labels with **CTRL+C**.
39. Select the menu **Edit » Smart Paste** or use the shortcut keys **CTRL+SHIFT+V**.
40. Select **Ports** for the *Paste As* option.
41. Making sure **Enable Paste Array** is not checked, that *Sort Order* is set to **By Location**, *Signal Names* is set to **Keep** and *Port Width* is set to **Set Width To Widest.**
42. Press **OK**.
43. Place the ports on the `SPI_SCLK, SPI_MOSI` and `SPI_MOSIA` wires as shown in Figure 85 below.
44. Ensure that the `SPI_MOSIA Port` is set to **Output**.
45. Add a `CAN_RXD` port and set it to an Input Port, and wire it to Pin 2 of `UC1` as shown in Figure 85 below.
46. Save the schematic.

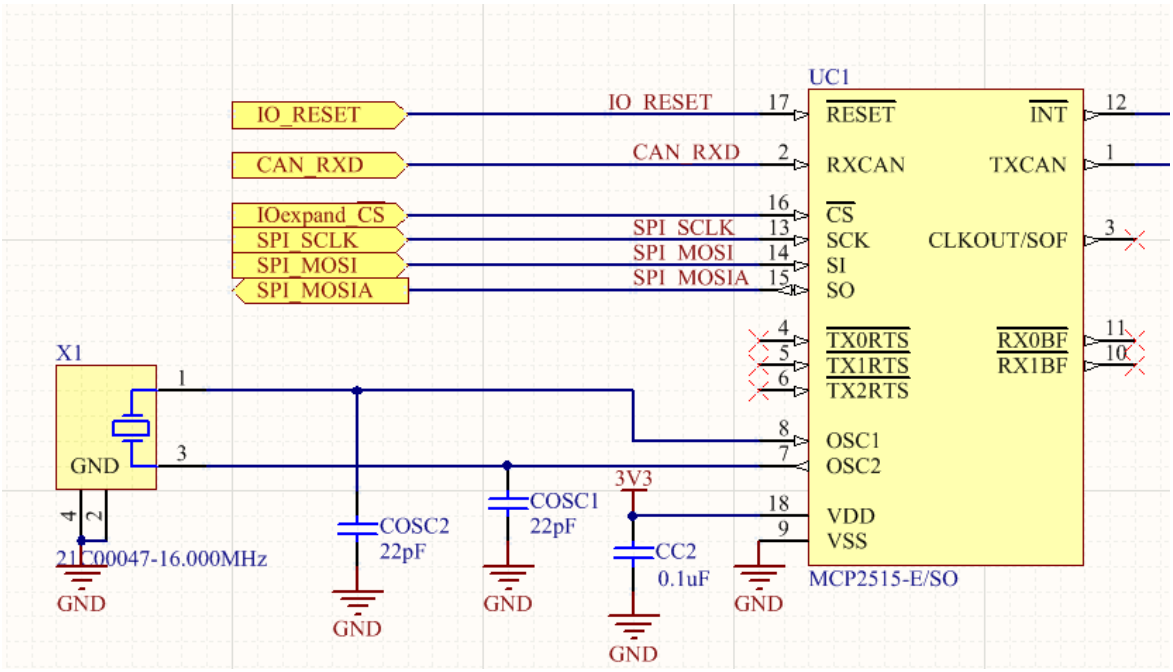


Figure 85. Finalized section of the CAN_Interface schematic

6.3.4 Digital_IO Schematic

6.3.4.1 Auto-Generated Wires

47. Open `Digital_IO.SchDoc` from the *Projects* panel.
48. Select the `IOS1` header.
49. Drag the `IOS1` header over to `UI1` so that pins of the `IOS1` header bump up against `Pins 21-28` of `UI1` as shown in Figure 86 below. Do not overlap them by too much.

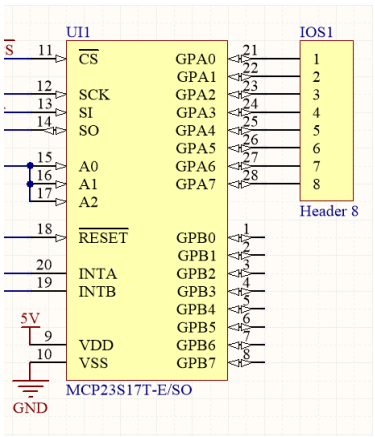


Figure 86. Overlap pins for UI1 and IOS1

50. Drag the `IOS1` away from `UI1` to create wire connections between the pins automatically as shown in Figure 87 below.

If **Always Drag** is not enabled in Schematic Preferences, holding the **CTRL** key will be required for automatically creating connections.

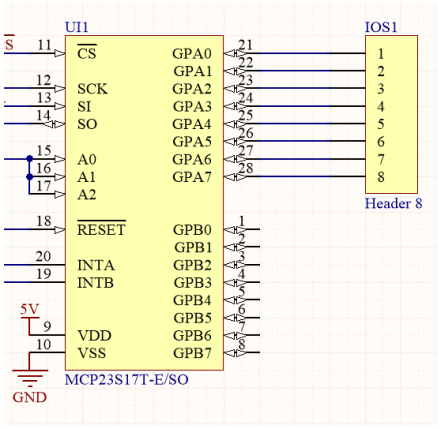


Figure 87. Auto-Generated wires when pulling IOS1 away from UI1

51. Repeat the above steps to generate wires for the `IOS2` header so that it connects to Pins `GPB0-GPB7` of `UI1`.
52. Save the `Processor_Interface`, `CAN_Interface`, and `Digital_IO` schematics to complete your work.

Module 7: Schematic Updating

7.1 Purpose

In this exercise, we'll perform modifications to multiple schematic objects as once, such as alignment, formatting and net color highlighting. These operations are what we refer to as Global Editing. We'll also look at the schematic annotation of components.

1. Close the current `WCTopping.PrjPcb` project.
2. Open the `WCTopping.PrjPcb` project from the `AD20 Essentials Exercises\Module 7 - Schematic Updating` folder.

7.2 Align

7.2.1 Align Right

Alignment tools are used to ensure that objects are aligned to one another, or to the grid to achieve a clean document.

3. Open the `Processor_Interface.schdoc` from the Project Panel.
4. Left-click and drag a rectangular selection from left to right around the ports on the left of the schematic.
5. Go to **Edit » Align » Align Right** or use the **Active Bar** to align the ports.

Use the **Edit » Align » Align to Grid** to align objects within inherited/imported designs that were not captured on grid. This is to maintain proper connectivity between electrical objects.

It is also possible by right mouse click on the Source Documents to **Open All** documents or **Open All Schematic Documents**.

7.3 Formatting Net Labels

7.3.1 Selecting All Net Labels

In this section, we'll modify all the Net Labels in the project to ensure they have the same font, font size, and color.

6. Open the `Processor_Interface.SchDoc`.
7. Right-click on one of the net labels such as GPIO2. In the right-click menu, click on **Find Similar Objects…** (also referred to as "FSO" for short). This will open the *Find Similar Objects* window.
8. Set the selection options as defined in Figure 88.

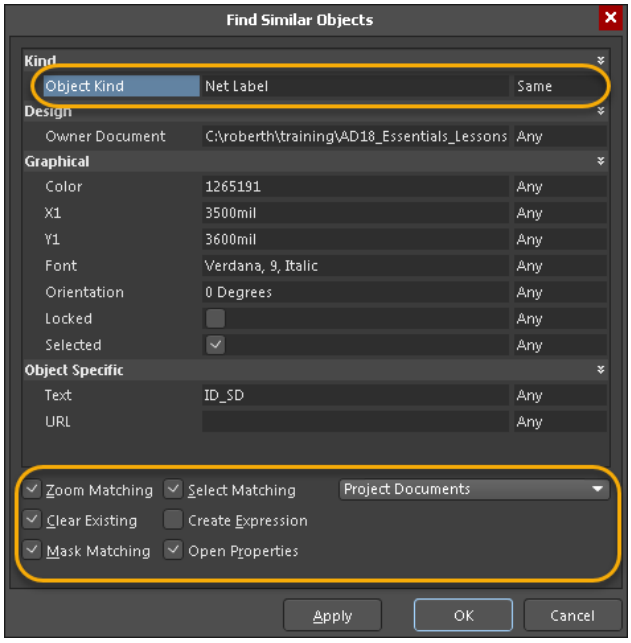


Figure 88. Find Similar Objects configuration for Net Label selection

9. Click **OK**. Notice that all the net labels in the schematic are selected.
10. Without clicking anywhere on the schematics, open `Digital_IO.SchDoc`. Notice that the Net Labels in this schematic are selected as well. Return to `Processor_Interface.SchDoc`.

7.3.2 Apply Formatting to Net Labels

11. While the net labels remain selected, open the *Properties* panel if not already opened.
12. Change the text font to `Verdana` change the text size to `9`.
13. Click on the color box within this panel and click on a dark green color. Then, click on *I* to italicize the net labels shown in Figure 89 below.

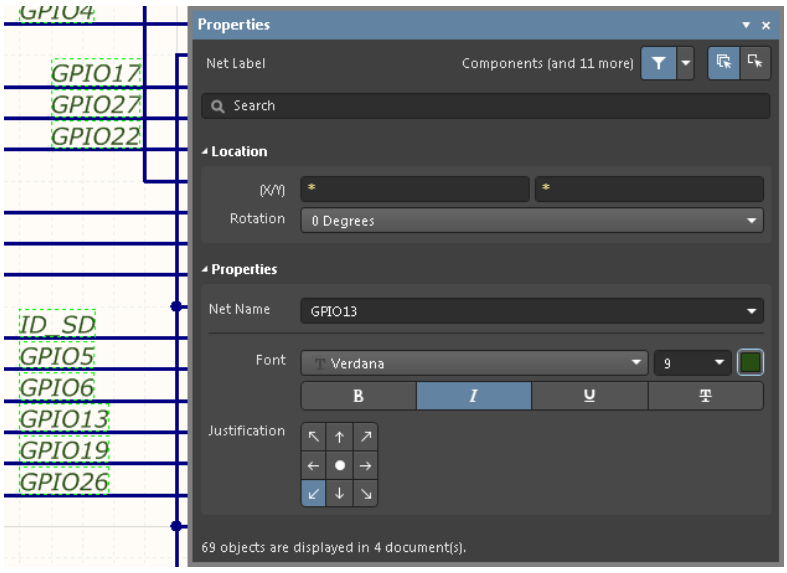


Figure 89. Net Label formatting defined in the Properties panel

14. The formatting defined in this panel will be applied on all the Net Labels selected in the project. Click once in the main workspace and hit **Shift+C** to clear the highlight and mask.

7.4 Locking Components Designators

7.4.1 Selecting only 100nF 50V Capacitors

In the previous section, we selected all Net Labels by using a single selection filter (*Object Kind is Net Label*). In this section, we will use three filters to select components of a specific name.

15. Hit **J** then **C** to jump to a component. Type C5 and click **OK**. Close the *Messages* panel if it appears.
16. Right click on C5 and click on **Find Similar Objects**.
17. In the *Find Similar Objects* window, set the three filters as shown in Figure 90 below.

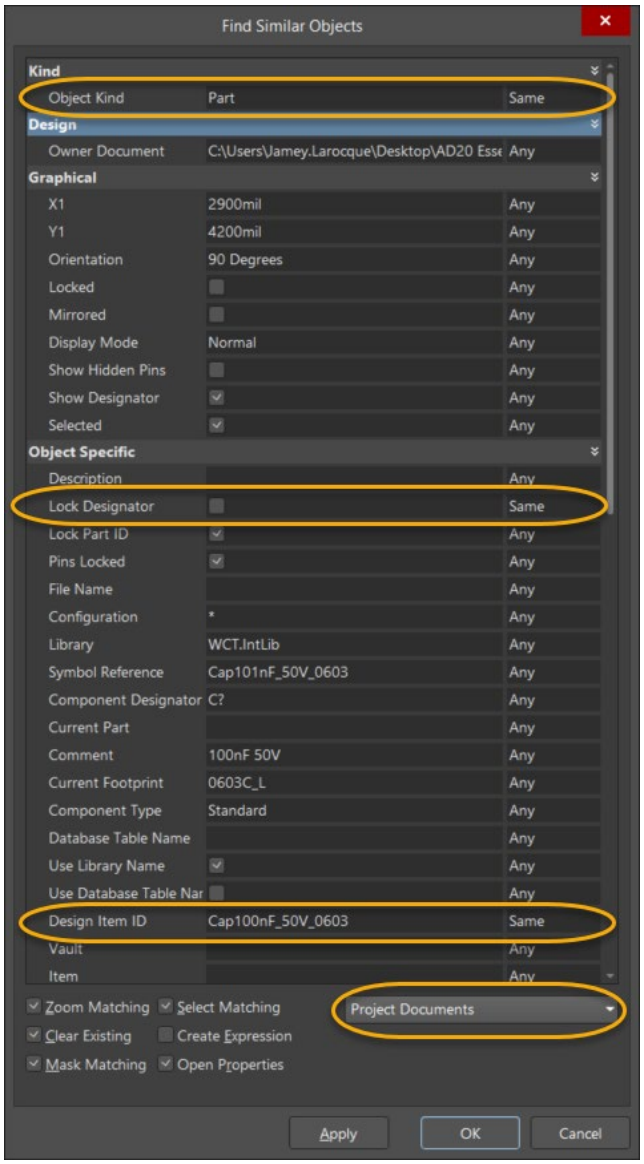


Figure 90. Three filters to ensure we select only the 100nF 50V Capacitors with unlocked designators

18. Click **OK**. Notice that only 100nF 50V capacitors with unlocked designators are selected.
19. Without clicking anywhere on this schematic, open the `Digital_IO.SchDoc`. Similarly, only the 100nF 50V capacitors are selected and not the other components.

The total number of selected objects and the document count are displayed at the bottom of the Properties panel. In this case, five components are selected, and their common attributes are shown and can be modified through this *Properties* panel.

7.4.2 Lock the 100nF 50V Capacitors Designators

20. While the capacitors remain selected, open the *Properties* panel.
21. To lock the component designators, simply click on the padlock icon beside Designator in the panel, as shown in Figure 91 below.

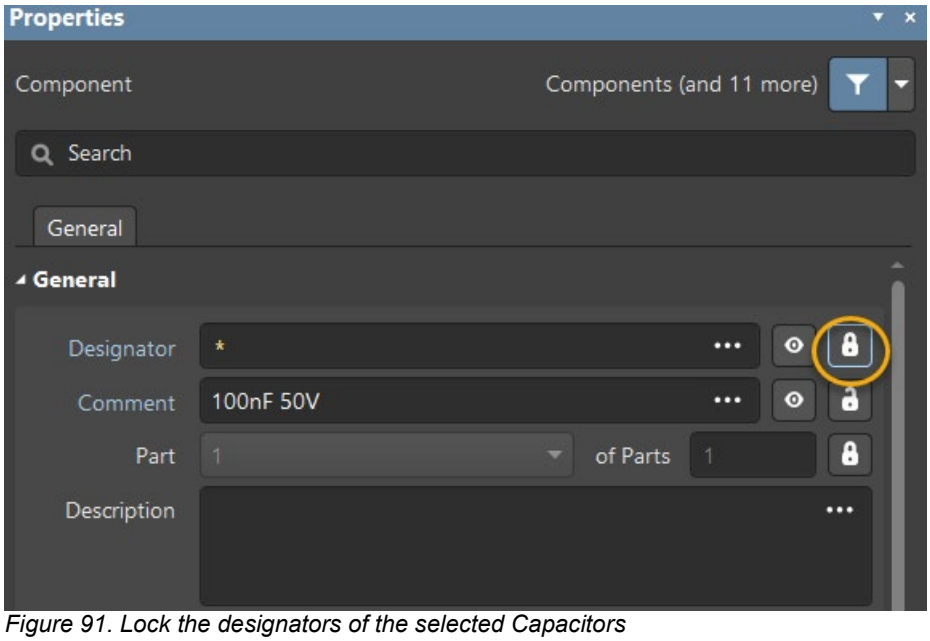


Figure 91. Lock the designators of the selected Capacitors

22. Note that once the component designators are locked, they will not be modified through either the annotate schematics window or the *Properties* panel. They must be unlocked first by clicking on the padlock icon again.
23. Click once in the main workspace and hit **Shift+C** to clear the highlight and mask.

7.5 Net Color Highlighting

24. Open the `Processor_Interface.SchDoc` from the *Projects Panel*.
25. Select the color **Red** from **View » Set Net Color » Red** as shown in Figure 92 below.

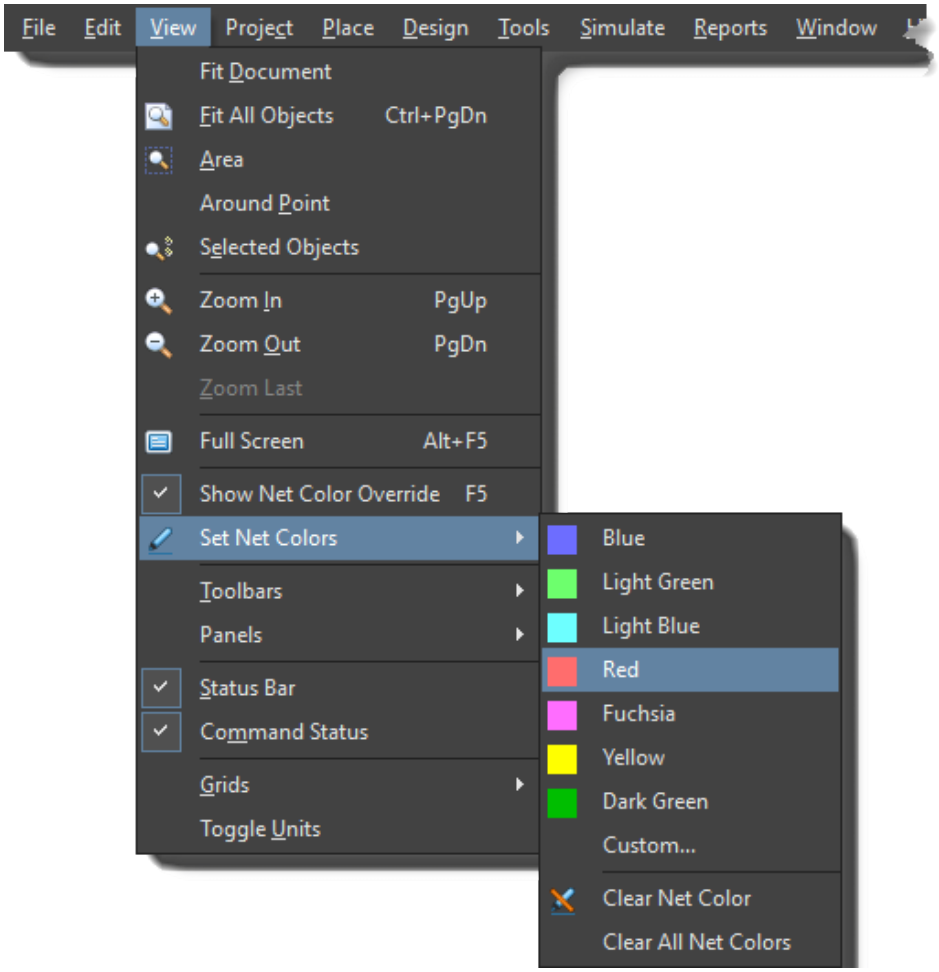


Figure 92. Set Net Colors menu

26. Click on a wire connected to a `5V Power Port` to highlight the entire net in **Red** as shown in Figure 93.
27. Then, **right-click** to exit the command.

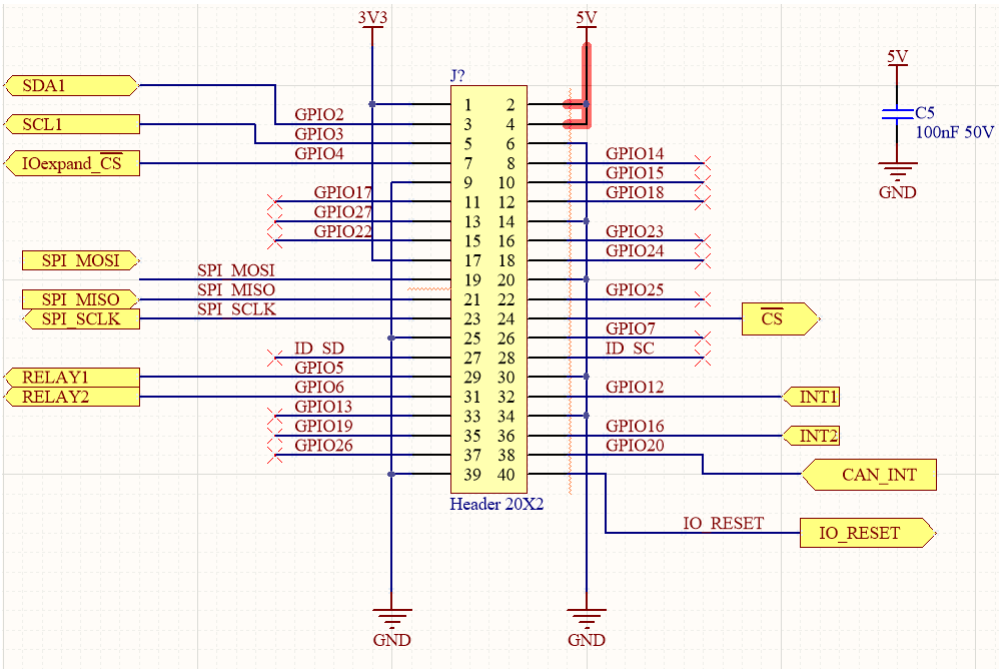


Figure 93. Processor_Interface.SchDoc with 5V net highlighted in red

28. Notice that power ports that are directly connected to a component will not have a **Red** highlight because there are no wires to highlight on. For example, the power port connected to C5.
29. Drag the 5V power port away from C5 to create a wire between them. We will then see that wire being highlighted in **Red**, as shown in Figure 94 below.

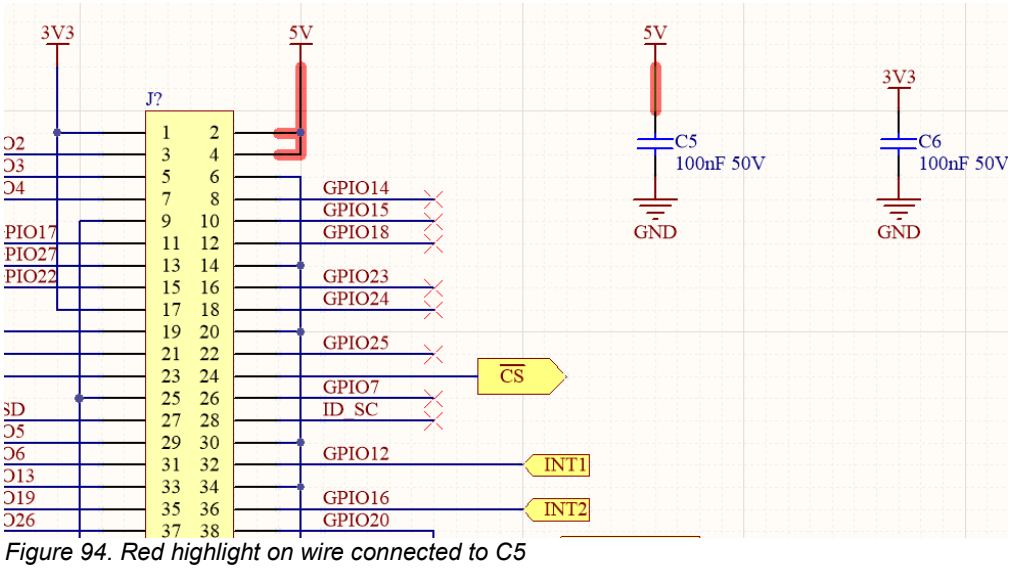


Figure 94. Red highlight on wire connected to C5

30. All the wires connected to 5V in the other schematics such as in `Digital_IO.SchDoc` and `Power Supply.SchDoc` are highlighted in **Red** as well. They may be hiding by the yellow rectangles drawn over the circuitry.

7.6 Schematic Annotation

31. Head to **Tools » Annotation » Annotate Schematics…** to open the *Annotate* dialog. The window shown in Figure 95 below, will appear.

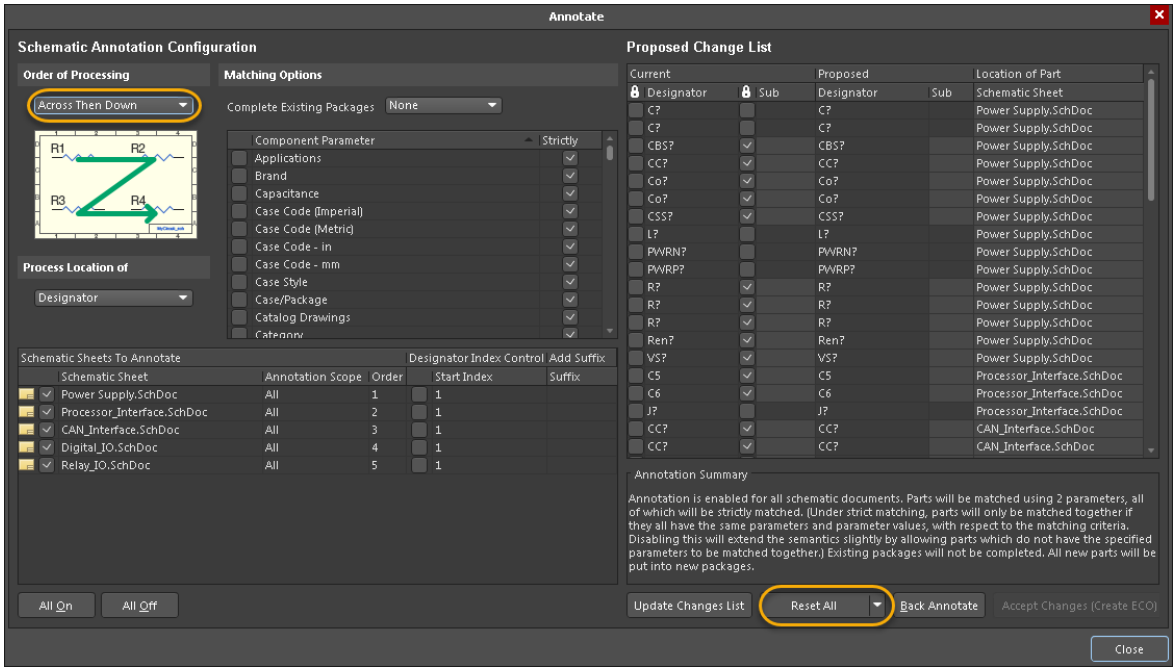


Figure 95. Annotate Dialog

a) Set the *Order of Processing* using the pull-down menu. Set it to **Across Then Down**.

b) Click **Reset All**.

c) Click **OK** on the pop-up dialog that appears.

32. Notice that the capacitors which designators were locked earlier will not be reset, as shown in the Proposed Change List. Next, while still in the Annotate window, perform the following:

a) Click the **Update Changes List** button

b) Click **OK** on the pop-up dialog that appears.

c) A preview of the proposed designator values will be displayed similar to what is shown in Figure 96.

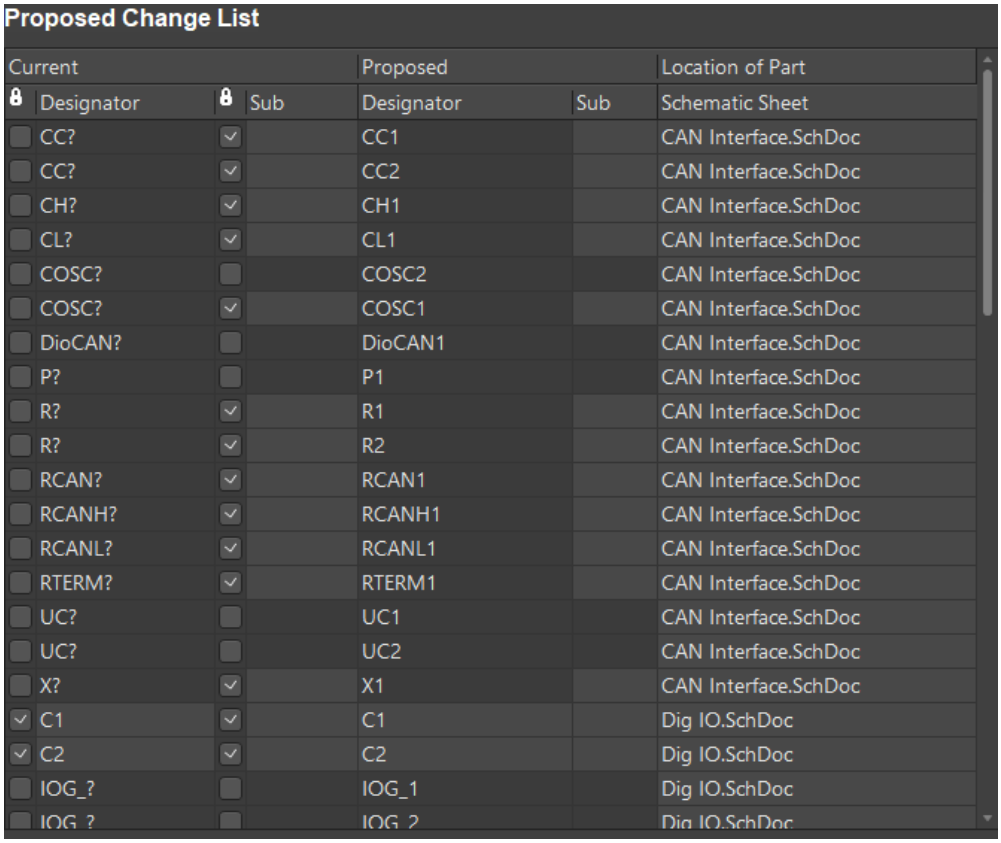


Figure 96. Proposed Designator changes based on the Annotation settings

d) These changes are only proposed so far. To commit this change to the design, press the **Accept Changes (Create ECO)** button, and then **Execute Changes** within the resulting *Engineering Change Order*. All undesignated parts should now have a unique designator.

e) Press the **Close** button to close the completed *ECO*, and then again press the **Close** button to close the *Annotate* dialog.

33. View the schematics to confirm that they now have assigned designators. Right-click on the project in Projects panel and click on close project. Click on **Save All** and click on **OK**.

If there are any “old” designators, they are referenced as a superscript in light gray next to the newly named designators. **Compiling** will remove the superscript of the “old” designators.

Module 8: Schematic Graphics

8.1 Purpose

In this exercise you will add text and graphics to the schematics to provide documentation and organization. You will also number schematic sheets and print them to PDF using the SmartPDF Wizard.

8.2 Preparation

1. Close the current `WCTopping.PrjPcb` project.
2. Open the `WCTopping.PrjPcb` project from the `AD20 Essentials Exercises\Module 8 - Schematic Graphics` folder.

8.3 CAN Interface Schematic

8.3.1 Text String

3. Open the `CAN_Interface.SchDoc` from the *Projects* panel.
4. Place a text string by selecting **Place » Text String**.
5. While in the middle of string placement, press the **Tab** key to enter the string's properties.
6. Enter the text `3.3V Logic Levels`, change the size of the font size to `16` by typing in 16, as shown in Figure 97 below and press **Enter** to continue the text placement.

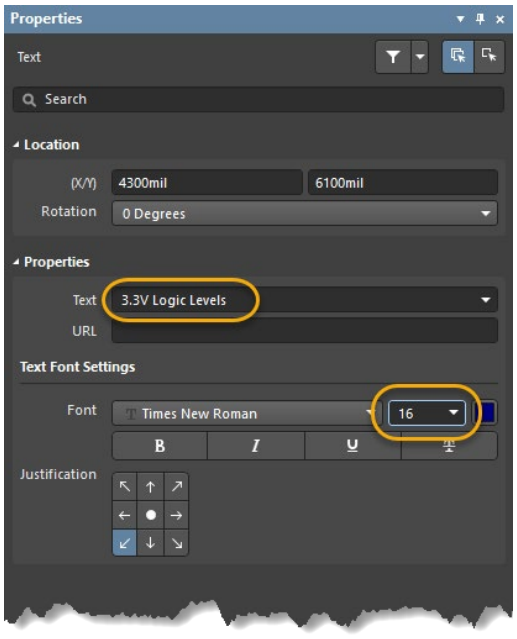


Figure 97. Font dialog in the Properties panel

7. Place the string on top of `UC1` as shown in Figure 98.
8. With the place string command still active, we will place another text string.
9. Press **Tab** to enter the String's properties.
10. Enter the text `5V Logic Levels`.
11. Change the size of the font to `22`.

12. Press the **Enter** to continue the text placement
13. Place the `5V Logic Levels` string on top of UC2 as shown in Figure 98 below.

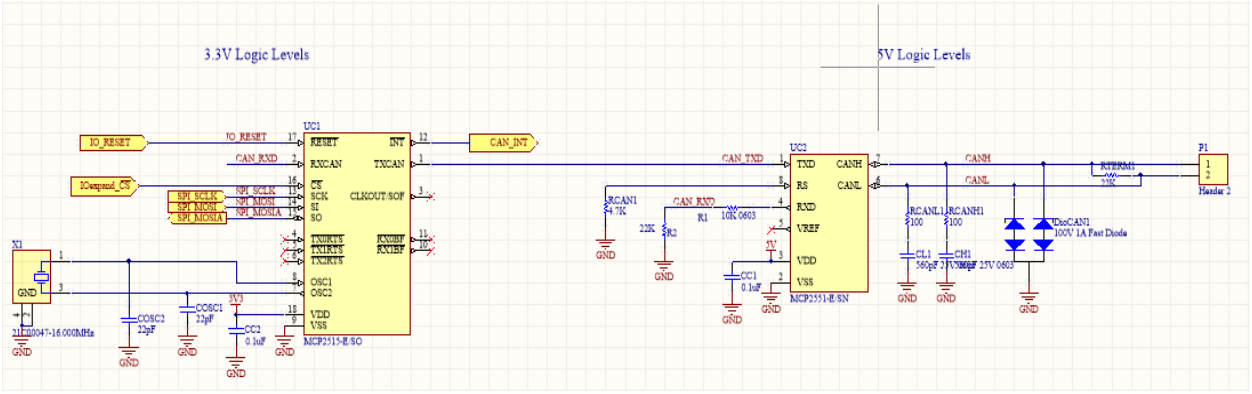


Figure 98. Placing a Text string

14. Double-click on the `5V Logic Levels` string to open the *Properties* panel.
15. In the *Properties* pane, decrease the size of the font size to `16` to match the other string.

8.3.2 Round Rectangle

16. Place a Round Rectangle by selecting **Place » Drawing Tools » Round Rectangle** or use the **Active Bar**.
17. Left-click to place the top left corner of the Round Rectangle similar to what is shown in Figure 99 below.
18. Move the mouse down and to the right.
19. Left-click again to place the lower right corner of the Round Rectangle. Right click to escape the command.

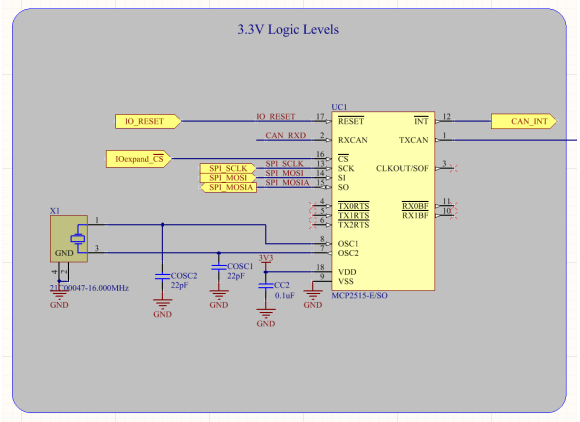


Figure 99. Placing a Round Rectangle

20. The Round Rectangle will appear in front of the schematic objects. Select **Edit » Move » Send to Back** as shown in Figure 100 below.

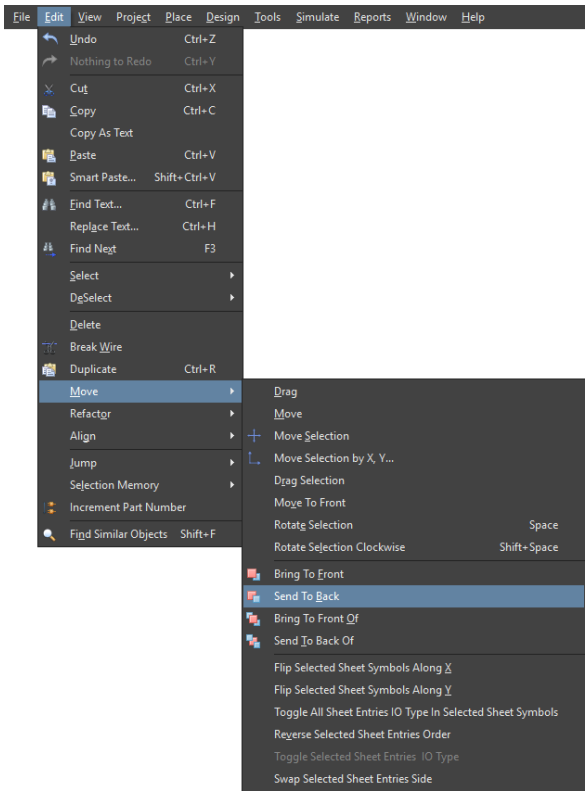


Figure 100. Move » Send to Back menu

21. With the **Send to Back** command active, left-click on the Round Rectangle with the crosshair cursor to send it behind the other objects. Right click to escape the command.
22. The result should look similar to Figure 101 below.

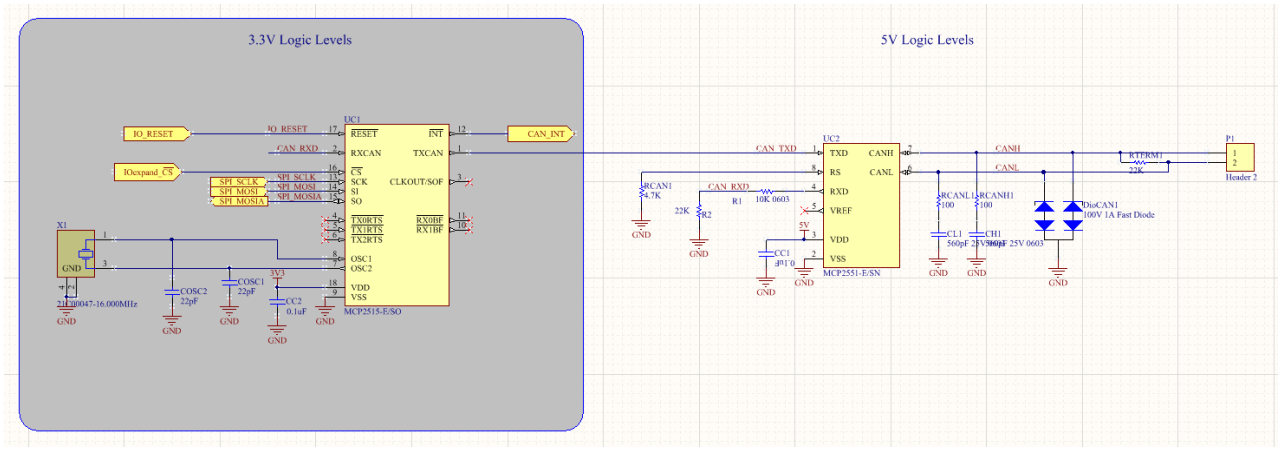


Figure 101. Round Rectangle moved to the Front

23. Using the same steps, place another Round Rectangle for the right part of the schematic similar to Figure 102 below.

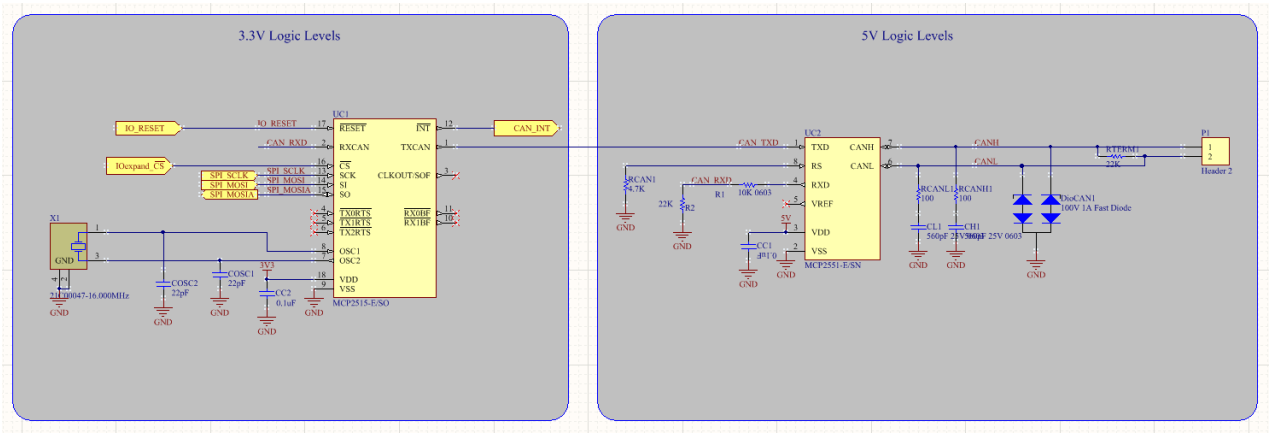


Figure 102. Placing a second Round Rectangle

24. With both Round Rectangles selected, open the *Properties* panel.
25. Select **Smallest** for the *Border* width.
26. Click on *Fill Color* field and set the fill color to **light blue** as shown in Figure 103 below.

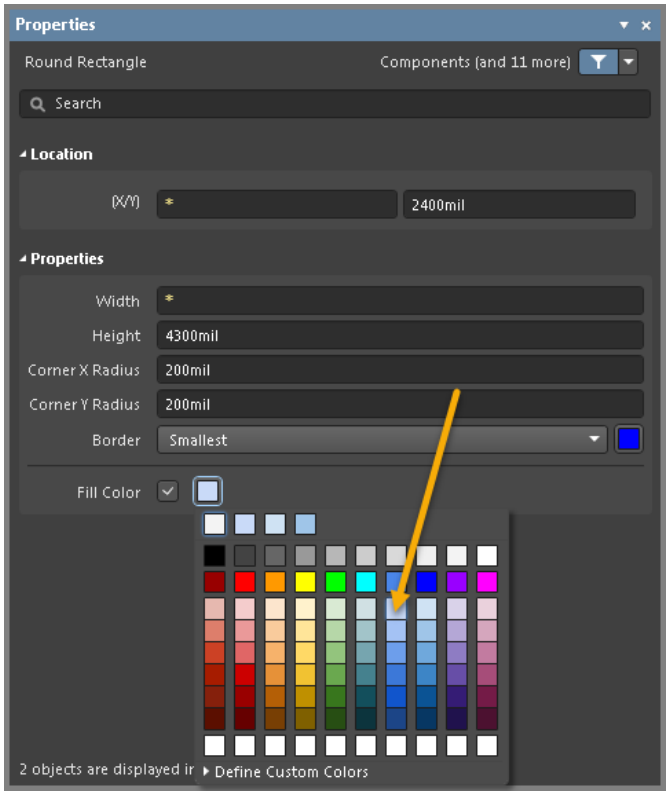


Figure 103. Select Light Blue for the Fill Color

27. Click on the *Border Color* field and set the border color to **dark blue** as shown in Figure 104 below.

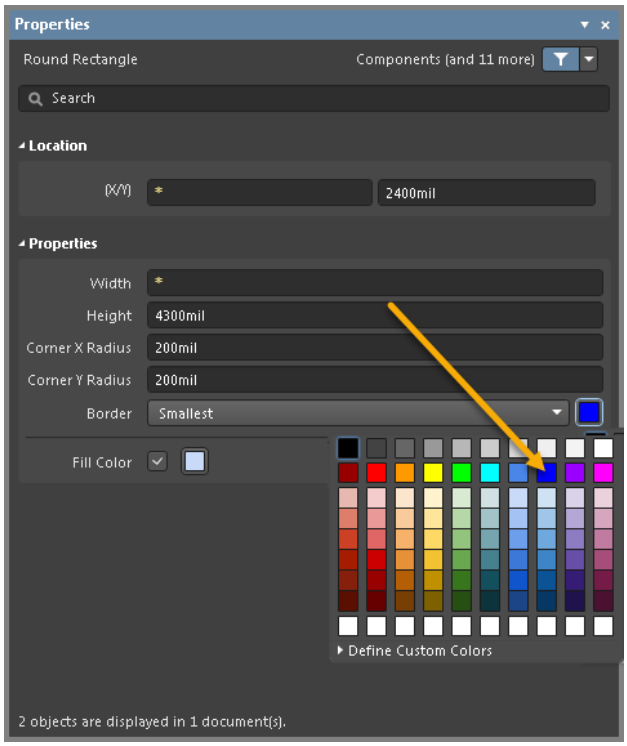


Figure 104. Select Dark Blue for the Border Color

28. Set the *Corner Y radius* to `200mil`.
29. Set the *Corner X radius* to `200mil`.
30. Once you've changed the properties, the schematic should look similar to Figure 105 below.

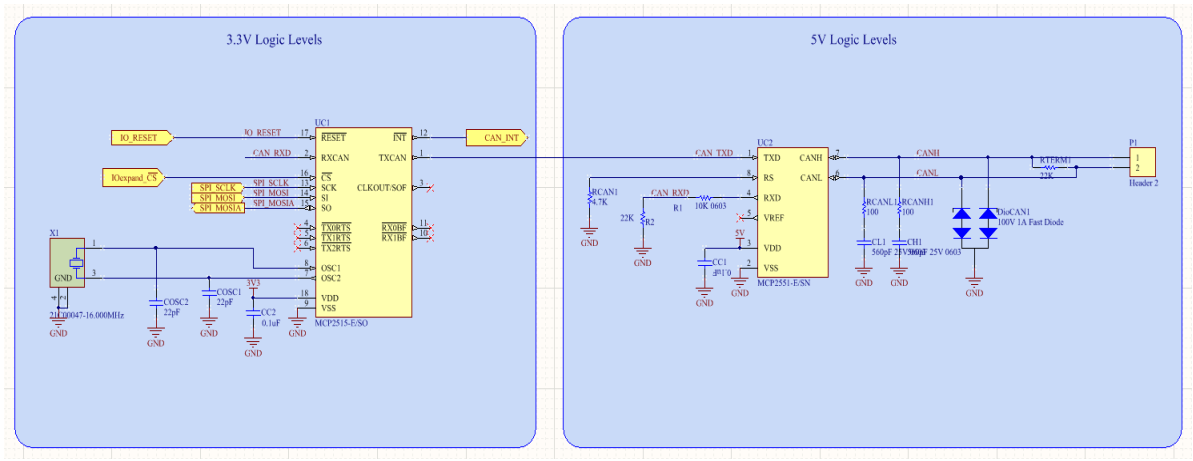


Figure 105. CAN_Interface schematic with text and rectangle placement

8.4 Number Schematic Sheets

31. Select **Tools » Annotation » Number Schematic Sheets** as shown in Figure 106 below.

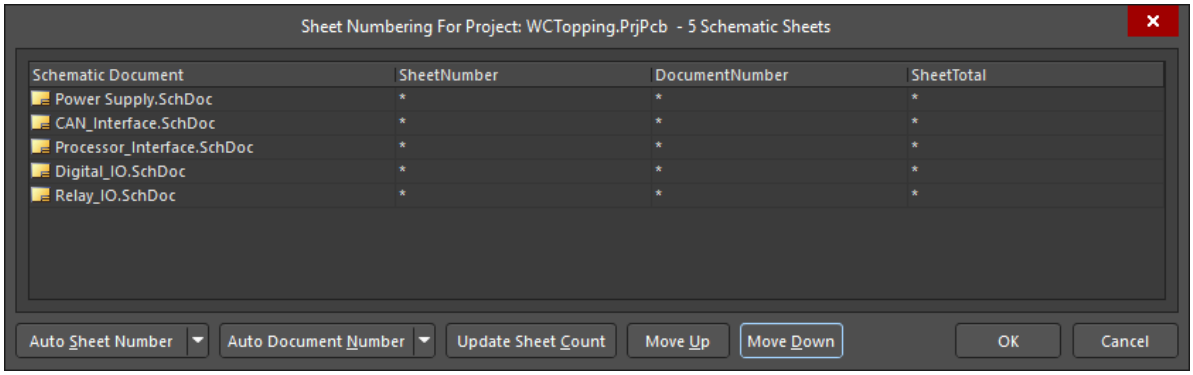


Figure 106. Sheet Annotation dialog

32. Click **Auto Sheet Number** to assign numbers to the sheets in the order of the list you see in the table as shown in Figure 107 below.

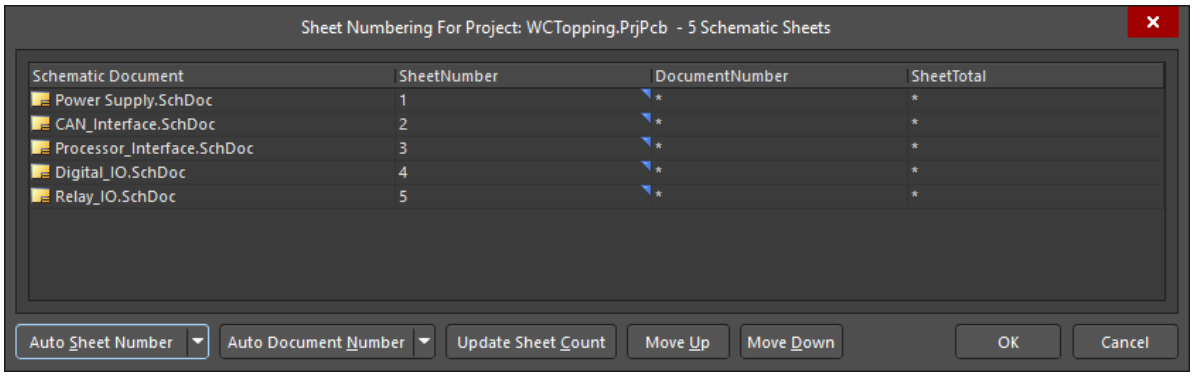


Figure 107. Sheet Annotation dialog with Sheet Numbering

33. Select `Processor_Interface.Schdoc`.
34. Move `Processor_Interface.Schdoc` up to the top by pressing the **Move Up** button. Only the order has changed. Left-click **Auto Sheet Number** to renumber everything as shown in Figure 108 below.

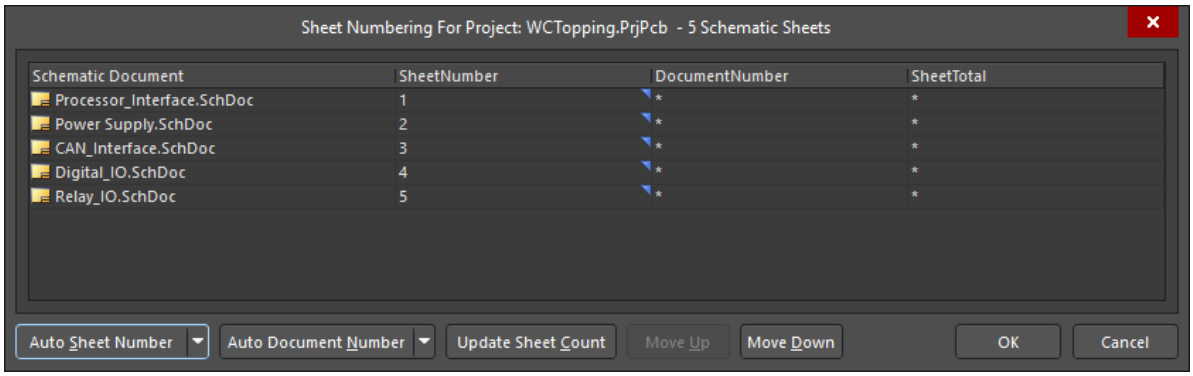


Figure 108. Sheet Annotation dialog after Sheet Numbering

35. Click **Update Sheet Count** to fill in the *SheetTotal* column as shown in *Figure 109* below.

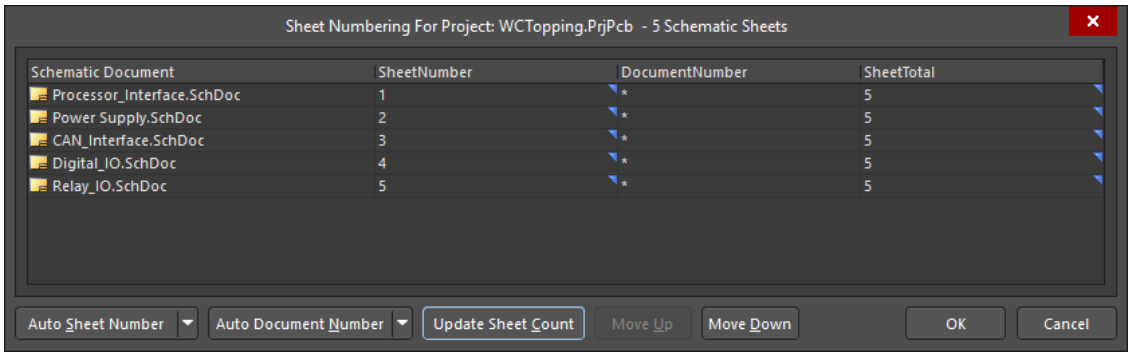


Figure 109. Annotation dialog after updating the sheet count

36. Click the **Auto Document Number** drop-down as shown in Figure 110 below.
37. Add prefix `DOC_`.

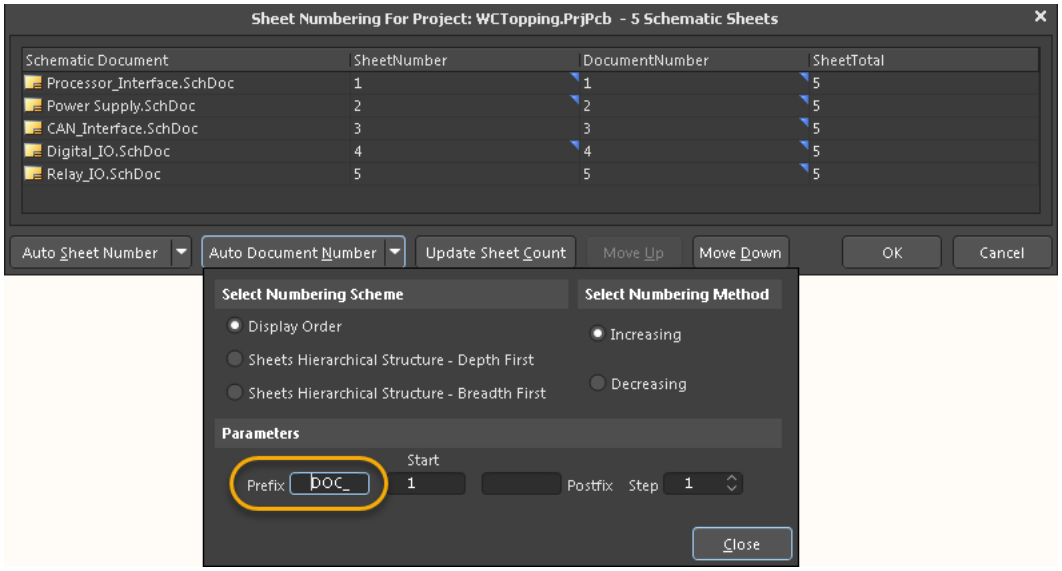


Figure 110. Annotation dialog for Auto Document Number options

38. Click **Auto Document Number** again as shown in Figure 111 below.

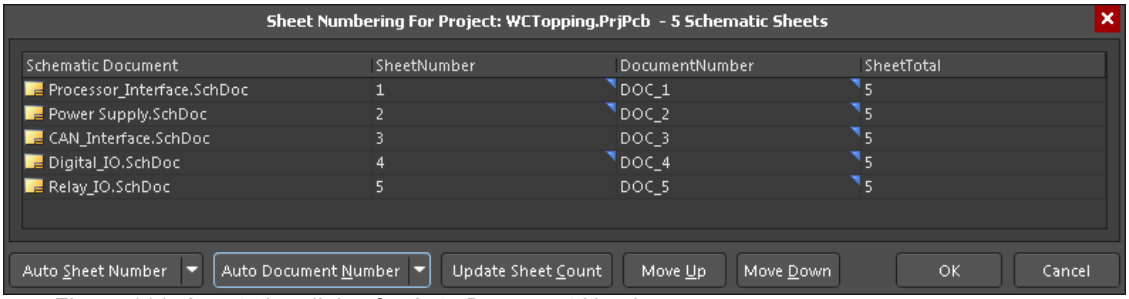


Figure 111. Annotation dialog for Auto Document Number

39. Click **OK** to close the *Sheet Numbering* window.

40. The title block in each schematic will now show the proper sheet number according to how it was numbered as shown in Figure 112 below.

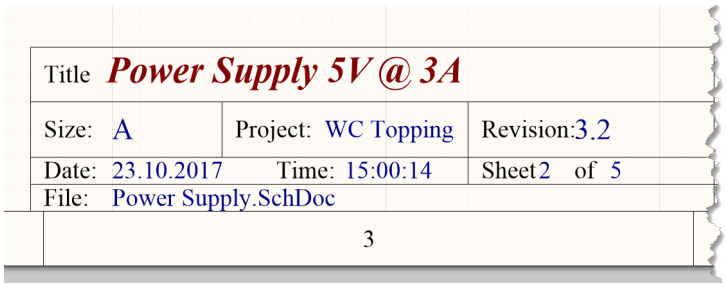


Figure 112. Title Block with Sheet number

8.5 Smart PDF

41. Select **File » Smart PDF** to open the *Smart PDF* wizard.
42. Click **Next** to begin the wizard.
43. Select **Current Project,** enter the name of the output file and the location on where you want to save the PDF as shown in Figure 113 below. Click **Next.**

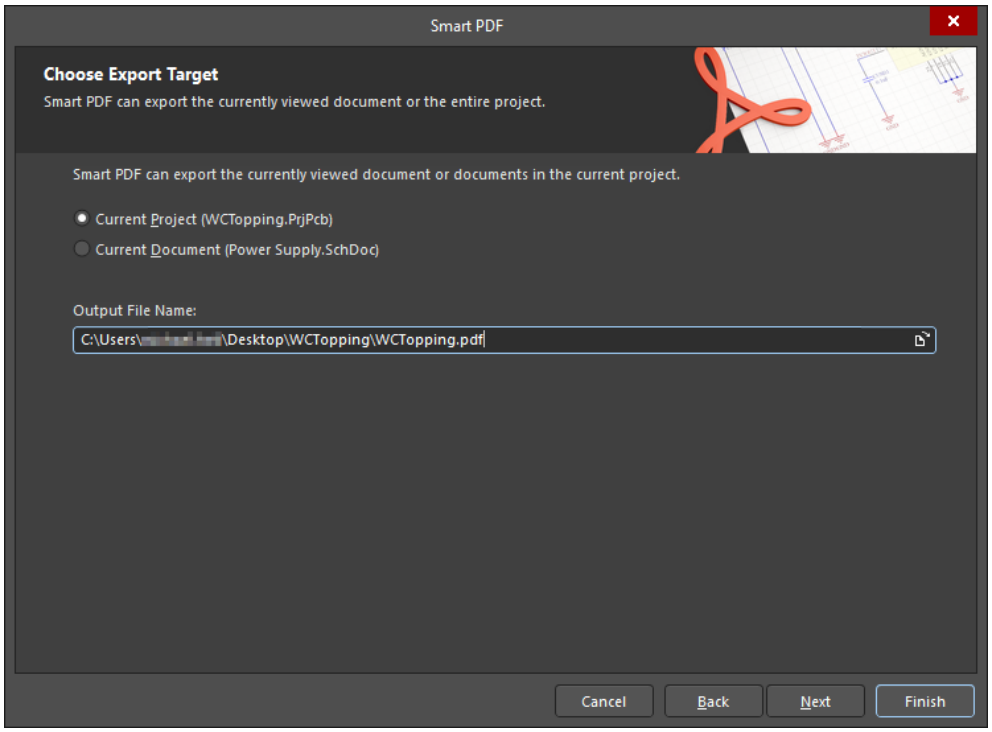


Figure 113 Choose SmartPDF Export target

44. All project files will be selected by default on the next page. Press the **Next** button with all files selected.
45. Ensure that ☐ Export a Bill of Materials is unchecked. The *Bill of Materials (BOM)* generation is explored fully in a later exercise. Click **Next.**
46. Leave the *PCB Printout Setting*s options as is, and click **Next.**
47. Leave the *Additional PDF Settings* options as is, as shown in Figure 114 below.
48. Click **Next** to continue the wizard.

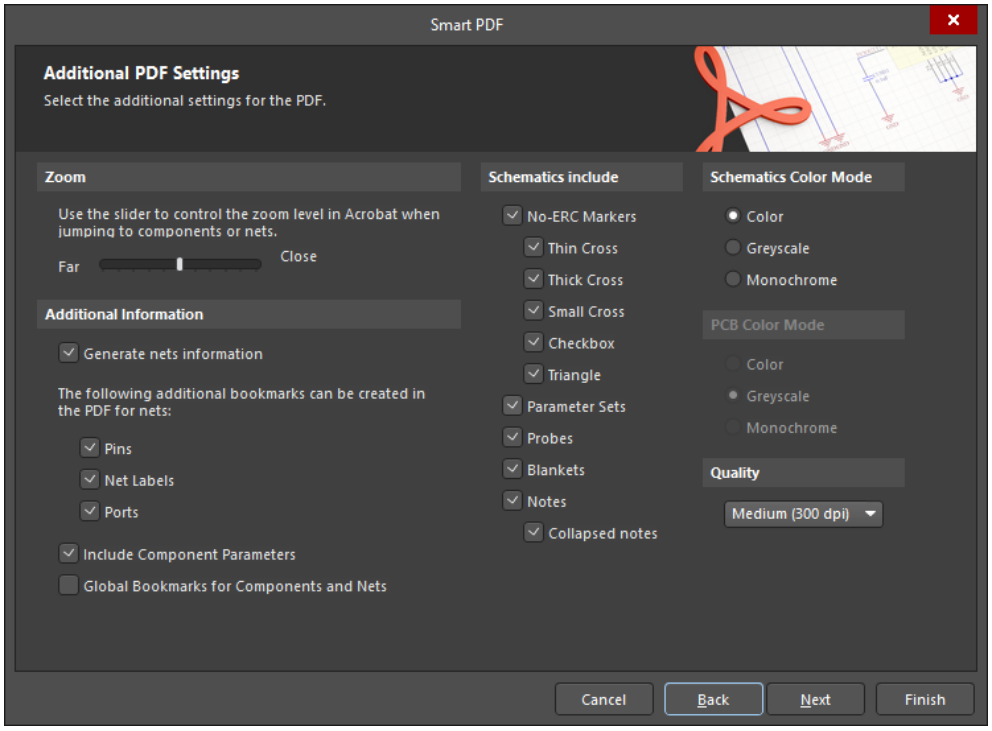


Figure 114. SmartPDF settings

49. Because we have a flat design, structure settings are not relevant. Leave ☐ Use Physical Structure unchecked and click next.
50. Ensure that ☑ Open PDF file after export is enabled and click **Finish**.
51. Review the pages of the PDF by expanding the different sections under the bookmarks tab as shown in Figure 115 below. You can now select components, nets, and ports from *Bookmarks*.

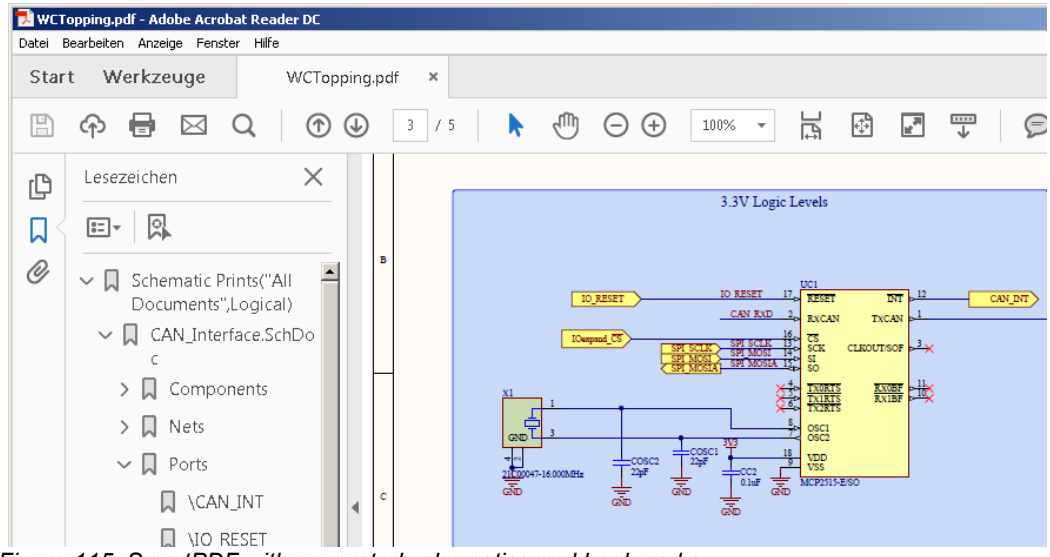


Figure 115. SmartPDF with generated schematics and bookmarks

Module 9: Compiling in Altium Designer

9.1 Purpose

In this exercise, you will learn how to compile a PCB Project and learn how to review and resolve any reported messages after compiling, such as warnings and errors.

Compiling is an important step that should not be overlooked. Compiling builds a netlist for each schematic sheet, creates sheet-to-sheet connectivity and builds the Unified Data Model. Running the compiler analyzes the complete design to check for drafting and electrical errors.

In addition to traditional compiling, we have the option to enable Dynamic Compiling. The design connectivity model is incrementally updated after each user operation. You do not need to manually compile the project to see the contents of the Navigator panel, run the BOM, perform an ECO, etc.

9.2 Preparation

1. Close your current `WCTopping.PrjPcb` project.
2. Open the `WCTopping PrjPcb` project from the `AD20 Essentials Exercises\Module 9 - Compiling in Altium Designer` folder.

9.3 Project Options

Before compiling the project, you should first check the *Error Reporting* settings to be sure they are defined properly for what you are designing.

9.3.1 Error Reporting Tab

3. Right-click the `WCTopping` project from the *Projects* panel, and select **Project Options**. Click on the *Error Reporting* tab as shown in Figure 116.

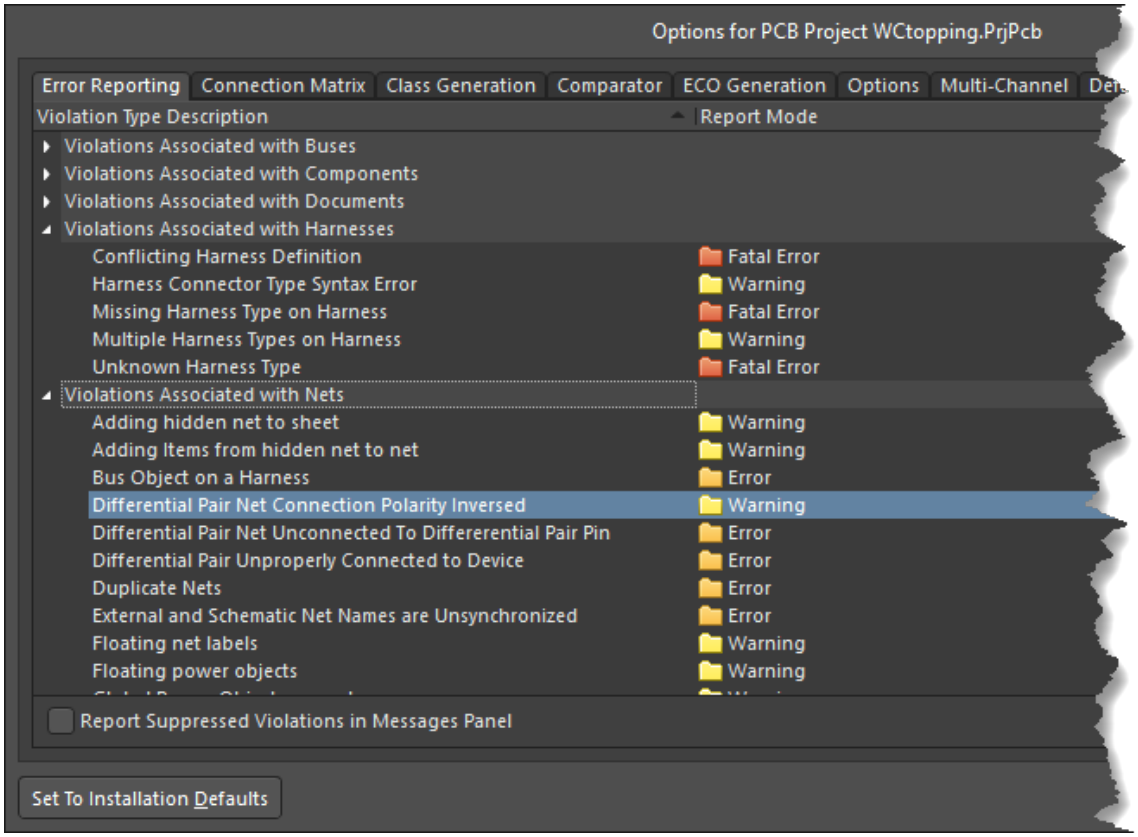


Figure 116. Project Options Error Reporting Tab

4. Scroll down to view the different categories of violation checks that the compiler performs.
5. Left-click one of the *Report Mode* icons (such as Warning or Error) and change it to a different value. Error reporting for all of these violations can be changed at any time, and will only apply to the specific project.
6. Click on **Set to Installation Defaults** button in the bottom left corner to reset, confirm the reset with **YES.**

9.3.2 Connection Matrix Tab

7. Click the **Connection Matrix** tab. You can set the reporting mode for the connections between the different pin, port, and sheet entry types. Left-click on some of the boxes to change the level of reporting as shown in Figure 117 as shown below.

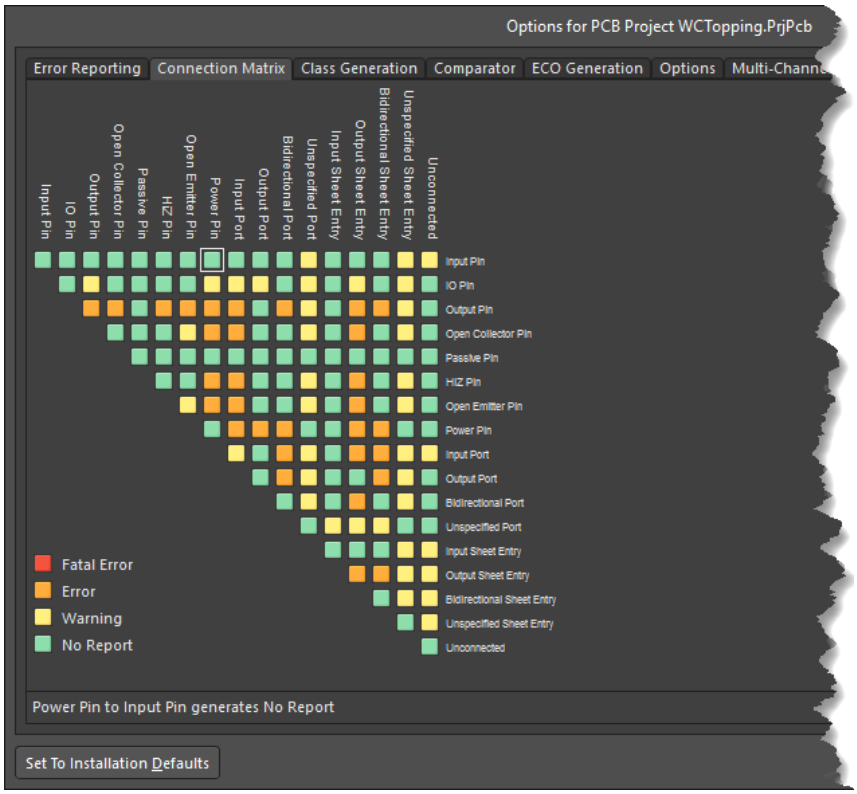


Figure 117. Connection Matrix Tab

8. Click the **Set to Installation Defaults** button in the bottom left corner to reset the *Connection Matrix* settings, confirm the reset with **YES**.

A common setting that is user for the default ERC matrix setting, is to set the **Unconnected** information to Warning instead of No Report, to report all open Pins, Ports, and Entries.

9.3.3 Options Tab

9. Open the *Options* tab as shown in Figure 118 below.

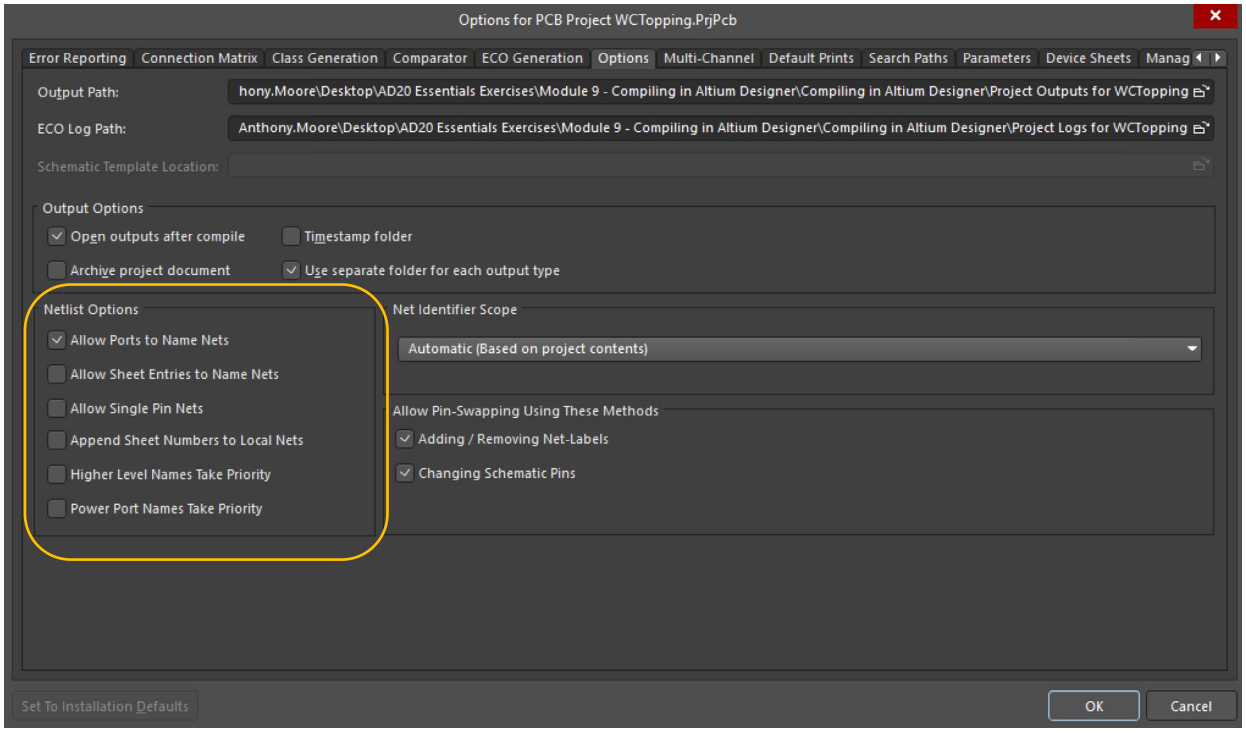


Figure 118. Options Tab showing Netlist Options

10. In the bottom left corner of this dialog, ensure that the **Netlist Options** checkboxes are all disabled, except for **Allow Ports To Name Nets**. These choices modify the net names based on your selection of checkboxes.

If you do not disable the other features; in a hierarchical project the **Sheet Entries** would generate different nets from connecting schematics, resulting in warning violations stating that there are ***Multiple Net Names,*** even if the net labels are the same.

11. Click **OK** to accept the changes.
12. Go to **Preferences** » **System** » **Navigation**. In the Highlight Methods section, set the **Far-Close** slider to mid-range. Click **OK** to save and close Preferences.

9.4 Running the Compiler

13. Compile the `WCTopping` design using the menu **Project » Validate PCB Project WCTopping.PrjPcb** or by right-clicking on the project in the *Project* panel and selecting **Validate PCB Project WCTopping.PrjPcb**.
14. After you compile/validate, the *Messages* panel will appear. If no window appears, the project has either passed the Electrical Rules Check (ERC), or there are only **Warnings** and no **Errors** or **Fatal Errors**.

If you cannot find the *Messages* Panel, click the **Panels** button in the bottom right corner of Altium Designer and select **Messages** from the popup menu.

9.4.1 Correcting Errors and Warnings

15. Upon compilation, you should see many **Errors** and **Warnings** in the *Messages Panel*. Double-clicking a warning or error message in the *Messages Panel* will navigate to the location where the warning or error occurs. It will display the affected elements in the *Details* section of the *Messages* panel.
16. Sort the listings in the *Message* panel to group the *Warnings* and *Errors* by clicking the **Class** column header until the Errors are brought to the top of the list.
17. You should see 3 `"Net GPIOXX has only one pin"` Errors in the *Messages Panel* as shown in Figure 119 below.
18. Double-clicking on each of these 3 errors (`"Net <GPIO##> has only one pin)` will highlight the affected pins associated to that error. In this case, it's Pin 11, Pin 15 and Pin 13 of the J4 component.

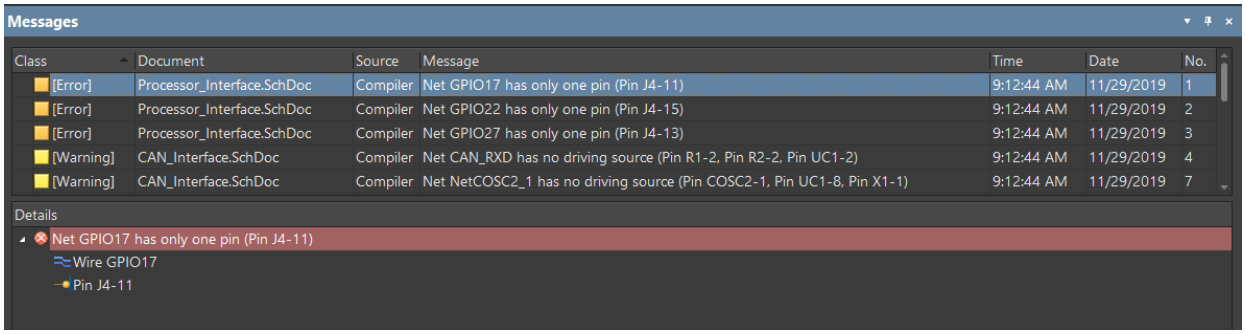


Figure 119. Net <name> only has one Pin error in the Messages panel

19. For suppressing warning and errors, we will assign a **No ERC directive** as follows:
    a) Right-click on the error in the *Messages* panel called *Net GPIO17 has only one pin (Pin J4-11)*
    b) Select **Place Specific No ERC for this Violation…** from the menu as shown in Figure 120.

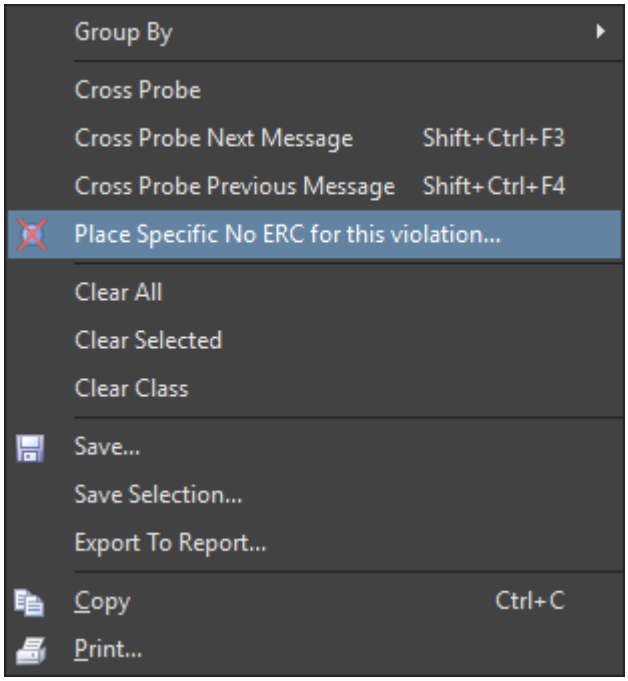


Figure 120. Place No ERC Directive directly from the Messages panel

c) The appropriate schematic will open and the **No ERC** directive will be on your cursor. **Place** the directive at the end of the wire for that pin (*GPIO17*).

d) Continue to add **Specific No ERC directives** to the 2nd & 3rd errors as shown in Figure 121 below.

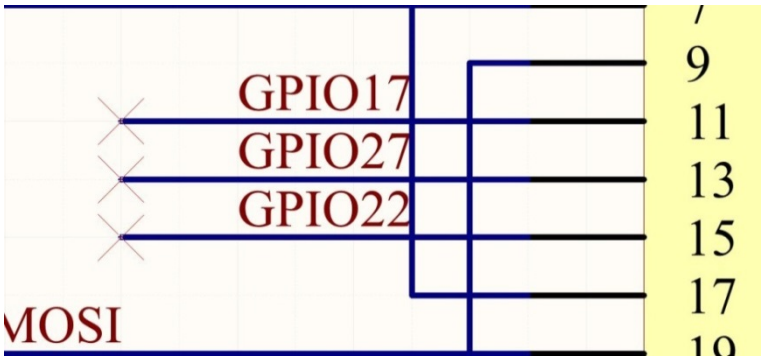


Figure 121. Placed No ERC directives

20. Compile the project once more using **Project » Validate PCB Project WCTopping.PrjPcb**.
21. Verify the list of messages no longer reports the "`Net <GPIO##> has only one pin (Pin J4-##)`" errors.
22. Open the *Messages* panel notice that there are still other warnings that need to be addressed.

NOTE: Placing a "No Specific ERC" will mask the Error or Warning associated with that object. This directive should only be used if you want to purposely disregard that Error or Warning (e.g. pin not connected or being used)

It is best practice to resolve the Warning or Error correctly, instead of placing a No Specific ERC directive. Be cautious when using them.

23. The next group of warnings `Net <Net ##> has no driving source` occurs when an input of an IC is connected to a passive component as shown in Figure 122 below.
24. Open the Project Options using the **Project » Project Options** menu.

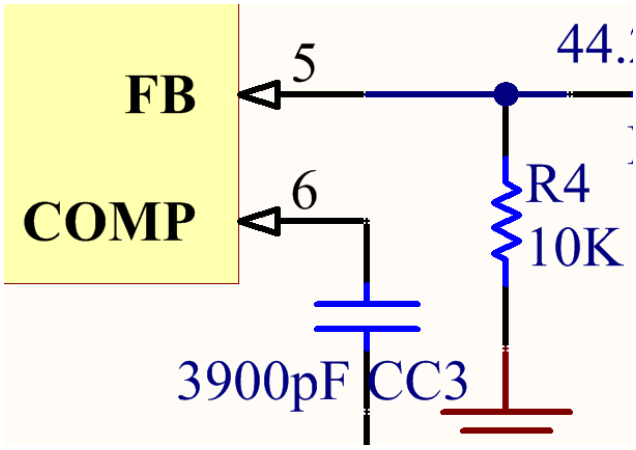


Figure 122. Input of an IC with no driving source

In the *Error Reporting* tab find the *Violations Associated with Nets* section

a) Locate the *Nets with no driving source* error listing.

b) Set this to **No Report** in the *Report Mode* column.

c) Press the **OK** button to exit the dialog.

25. Re-Run the compiler to verify there are no more *Nets with no driving source* Warnings.
26. To resolve the next set of warnings for "`SPI_xxxx contains IO Pin and Output Port`", open the Project Options again using the **Project » Project Options** menu.

    a) In the *Connection Matrix* tab; locate the *IO Pin* to *Output Port* square and click the square until it is green. This will be set to *No Report*.

    b) Click the **OK** button to exit the dialog. Re-run the compiler to verify there is only 2 Warnings remaining.
27. Finally, to resolve the final 2 warnings for "`Nets Element[x]:RELAY has multiple names`", place a no **ERC directive** on the bus with the RELAY[1..2] port on it located in the *Relay_IO.SchDoc* as shown in Figure 123 below.

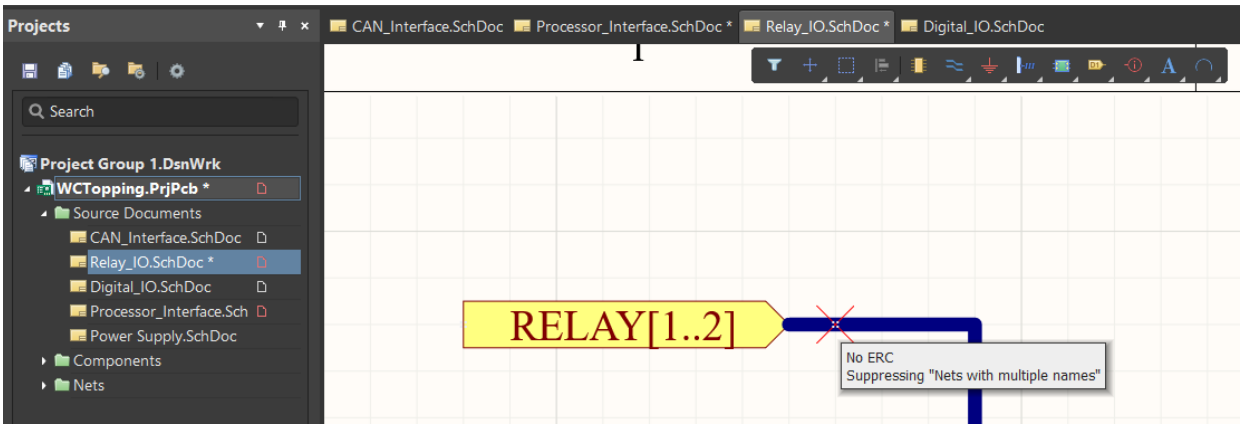


Figure 123. Placing a no ERC on the RELAY [1..2] Bus

This violation occurs when a net in the design has been detected to have multiple names associated with it. This could be by intention, e.g. connecting two different pages in a hierarchical design or it could be an indication for a short circuit.

28. Re-run the compiler by going to **Project** » **Validate PCB Project WCTopping.PrjPcb**. There should not be any Errors or Warnings, indicating a clean compile as shown in Figure 124 below.

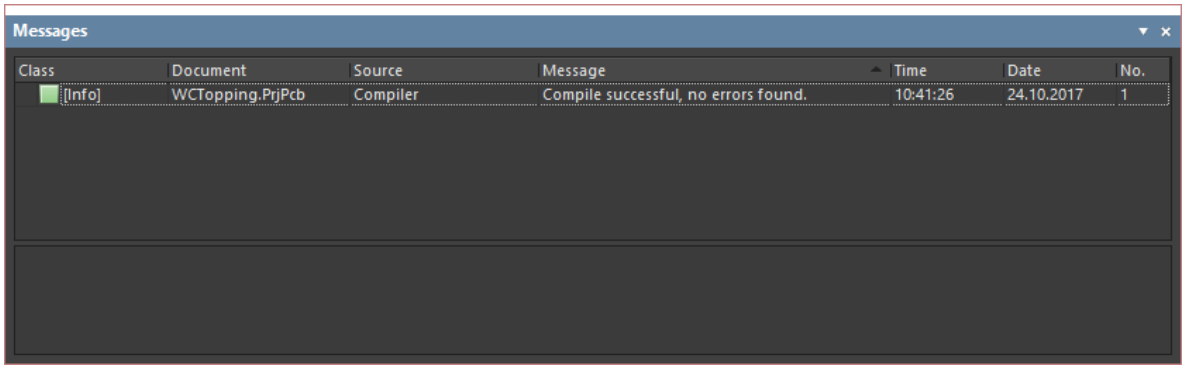


Figure 124. Messages panel with no errors or warnings

29. Save your work with **File » Save All**.

9.5 Dynamic Compiler

As mentioned at the start of this module, we have the option to enable Dynamic Compiling. The design connectivity model is incrementally updated after each user operation. You do not need to manually compile the project to see the contents of the Navigator panel, run the BOM, perform an ECO, etc.

9.5.1 Enabling the Dynamic Compiler

30. To enable the Dynamic Compiler, go into **Preferences** » **System** » **General** and click the **Advanced…** button as shown in Figure 125 below.

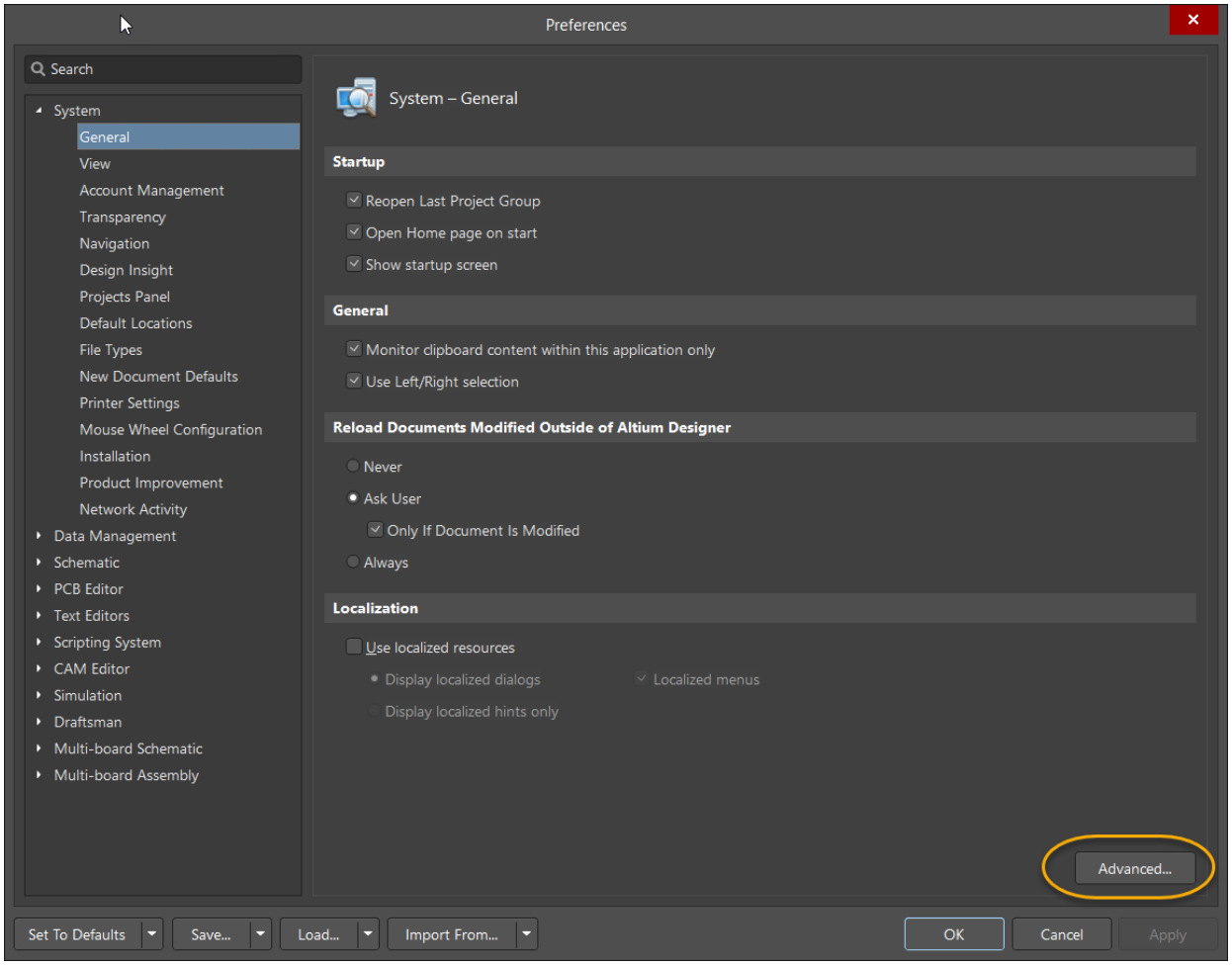


Figure 125. Dynamic Compiling settings

31. With the *Advanced Settings* dialog open, use the search bar at the top of the window, and search for "*Dynamic*" as shown in Figure 126.

32. Enable the **Schematic.Dynamic.Compiler.Navigator.Autorefresh** option to enable the Dynamic Compiler as shown in Figure 126 below.

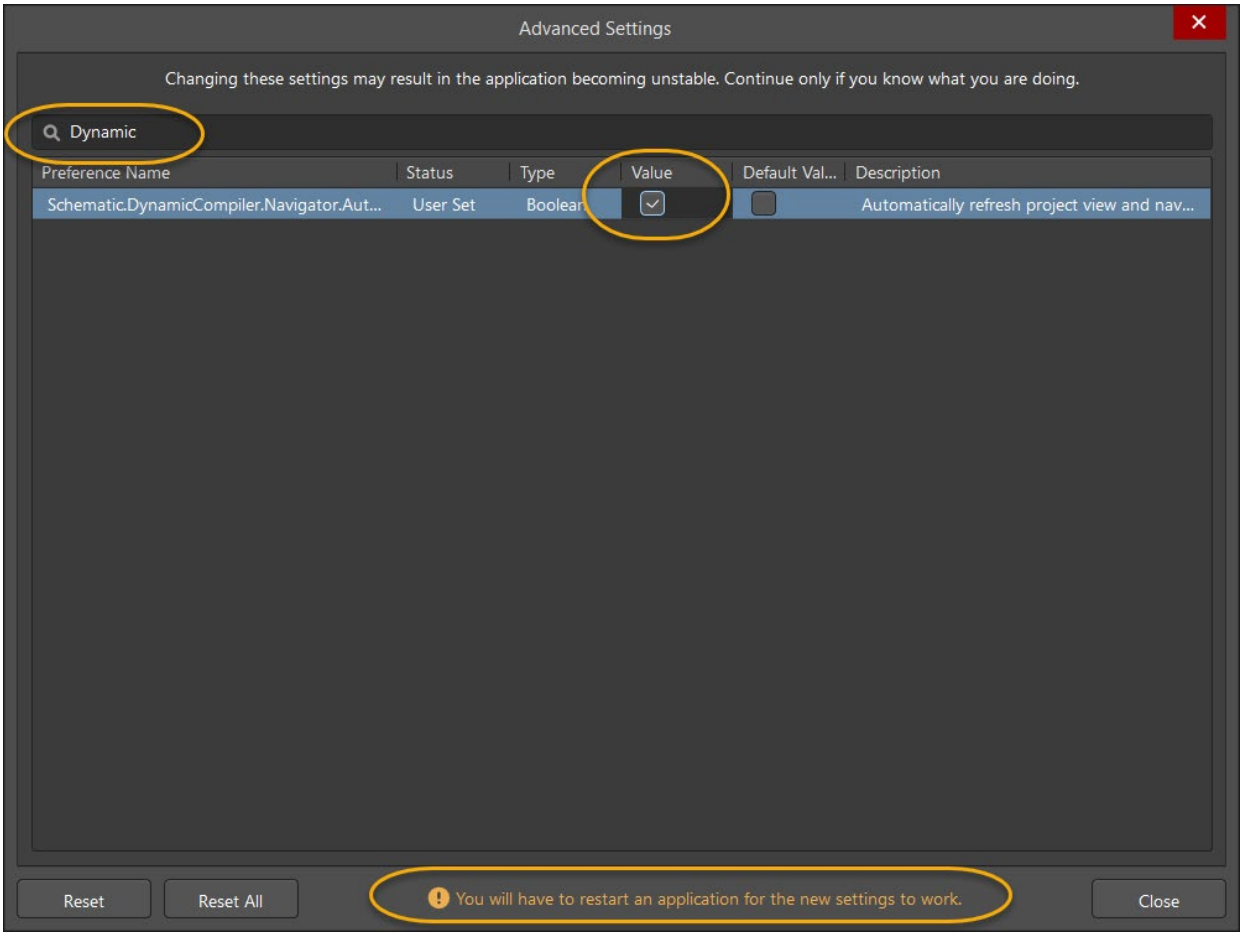


Figure 126. Enabling the Dynamic Compiler in Advanced Settings

For the Dynamic Compiler to take effect, Altium Designer must be closed and re-launched. Ensure that you save any important documents before closing the application.

33. Close the *Advanced Settings* dialog box, and click **OK** to close the Preferences.
34. Close Altium Designer completely, and re-launch the Application. This is necessary for the Dynamic Compiler to take effect.
35. When Altium Designer has re-opened, the `WCTopping.PrjPcb` should still be present in the projects panel. If you closed the project, it can be re-opened by going to **File » Recent Projects** and the `WCTopping.PrjPcb` project for this exercise should be the 1st option in the pop-up menu.
36. Open the `Processor_Interface.SchDoc`.
37. Expand the *Components* sub-folder, as well as the *Capacitors* sub-folder from the *Projects* panel as shown in Figure 127. You will see there is 17 components with the “C” designator.

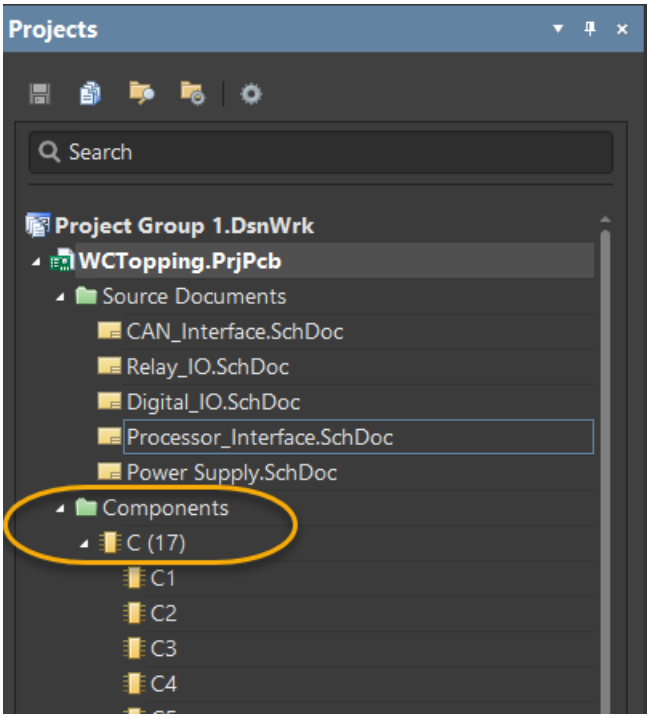


Figure 127. 17 Capacitor components in the design

38. Select and Copy **(CTRL+C)** C6 from the `Processor_Interface.SchDoc` and paste **(CTRL+V)** a new capacitor component beside it as shown in Figure 128 below.
39. You will then notice that the Components list in the *Projects* panel will automatically be updated, and will now show 18 components with the "C" designator. This automatic updating also updates all nets and sheet connectivity for the entire project when Dynamic Compiling is enabled.

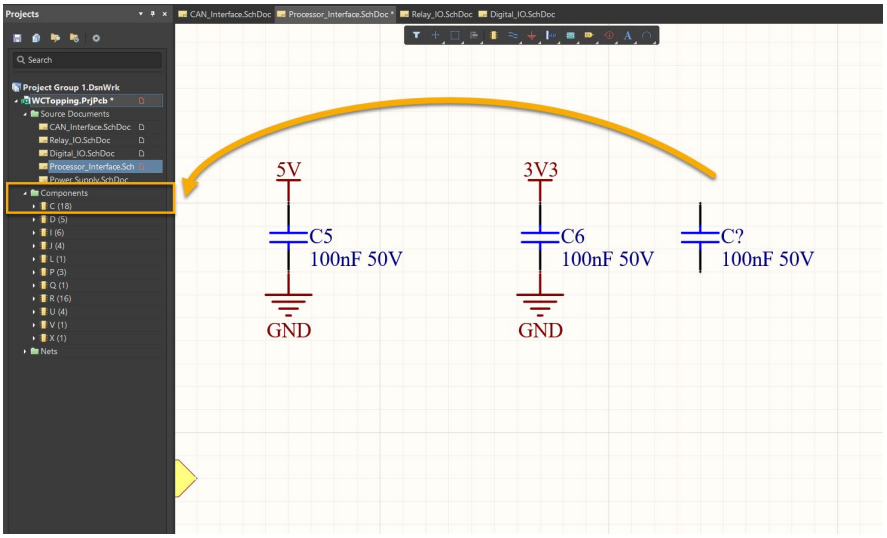


Figure 128. Dynamic Compiler automatically updating the Components folder after adding a Capacitor

40. Close the Project and click **NO** to save any changes when prompted.

Module 10: Adding Hierarchy

10.1 Purpose

In this exercise, the basics of a hierarchical design will be explored. Creating a Hierarchical design allows you to control connectivity from sheet to sheet in a multi-sheet design.

The top-level schematic will be completed by populating it with Sheet Symbols corresponding to each of the lower level schematics and creating lower level schematic sheets from newly created Sheet Symbols.

10.2 Preparation

1. Close the current `WCTopping.PrjPcb` project.
2. Open the `WCTopping.PrjPcb` project located in the `AD20 Essentials Exercises\Module 10 - Adding Hierarchy` folder.

10.3 Create Top-Level Schematic

3. Right-Click on `WCTopping.PrjPcb` in *Project* panel and select **Add New to Project » Schematic.**
4. Save the new schematic using **File » Save.** Name the file `Top_Level.SchDoc` when prompted.

10.4 Sheet Symbols

Adding sheet symbols on the Top Level Schematic establishes the hierarchical order. The next steps cover two methods of creating sheet symbols.

10.4.1 Creating a Sheet Symbol from Scratch

5. From the **Place** menu, select **Place » Sheet Symbol**.
6. With the sheet symbol on your cursor, press **Tab** to open the *Properties* panel.
   a) Set the *Designator* property to MID1 as shown in Figure 129.
   b) Type the value for the Field File name to *Processor_Power* as shown in Figure 129.

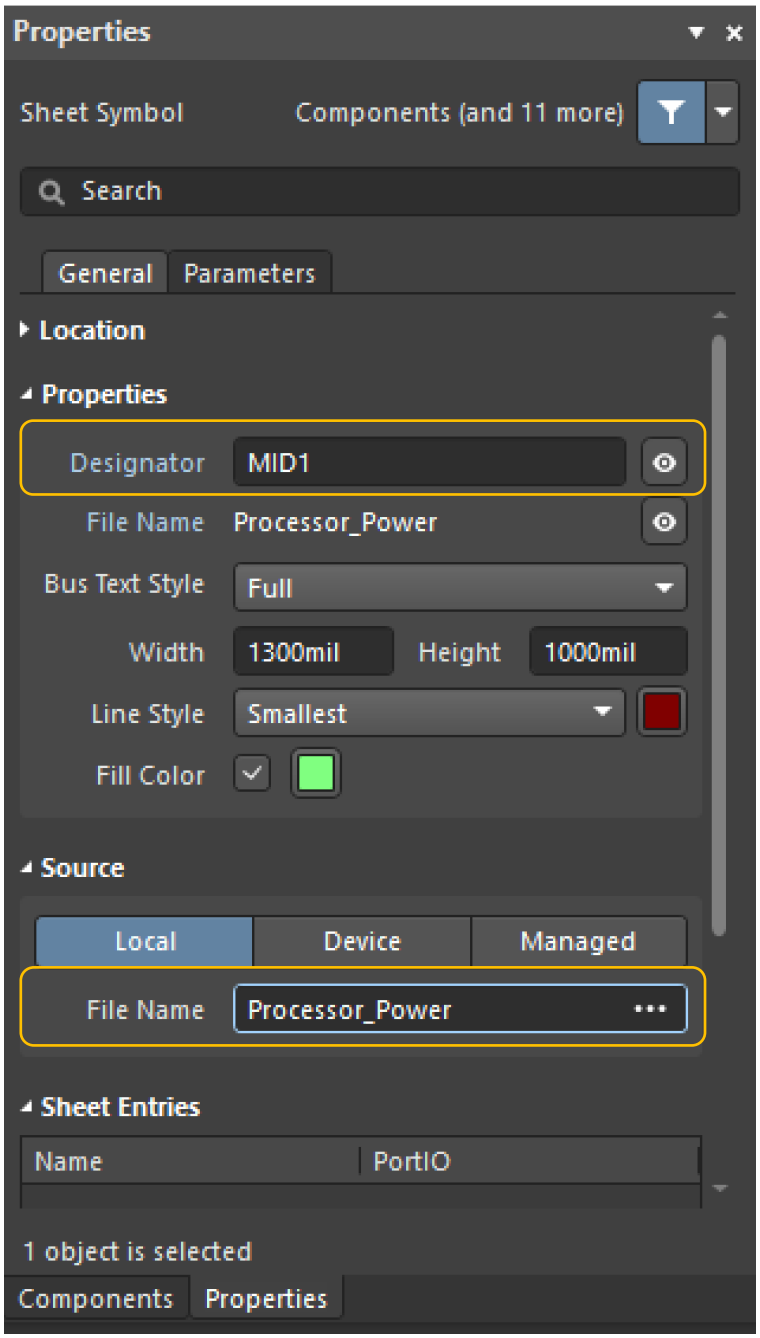


Figure 129. Sheet Symbol properties dialog

Processor_Power will be a new schematic that we will use in step 10.6 Creating a Schematic Sheet from Sheet Symbol. **Do not** link this value to the `Top_Level.SchDoc`. Simply type in **Processor_Power** in the *File Name* field.

c) Press **Enter** or hit the icon to continue the placement of the sheet symbol.

7. Position the Sheet Symbol with two left-clicks:
   a) Click once to define the upper left corner of the sheet symbol.
   b) Click a second time to define the lower right corner.
   c) Right-click to exit the command.

10.4.2 Adding Sheet Entries

10.4.2.1 Use the Smart Paste to convert Ports to Sheet Entries

8. Open the `Processor_Interface.SchDoc` from the Projects Panel.
9. Group select the ports in the `Processor_Interface.SchDoc`
   a) Open the *Properties Panel*
   b) At the pane *Selection Filter*, deactivate the selection filter for **All-On** and activate only the **Ports** as shown in Figure 130 below.
   c) Select the Ports with the command **Edit » Select » Inside Area**
   d) Right-click the selection and select **Copy**.

In this scenario, we're configuring the Selection Filter to only select Ports. Don't forget to activate the **All Objects** button after you're finished your current task to be able to select any objects moving forward.

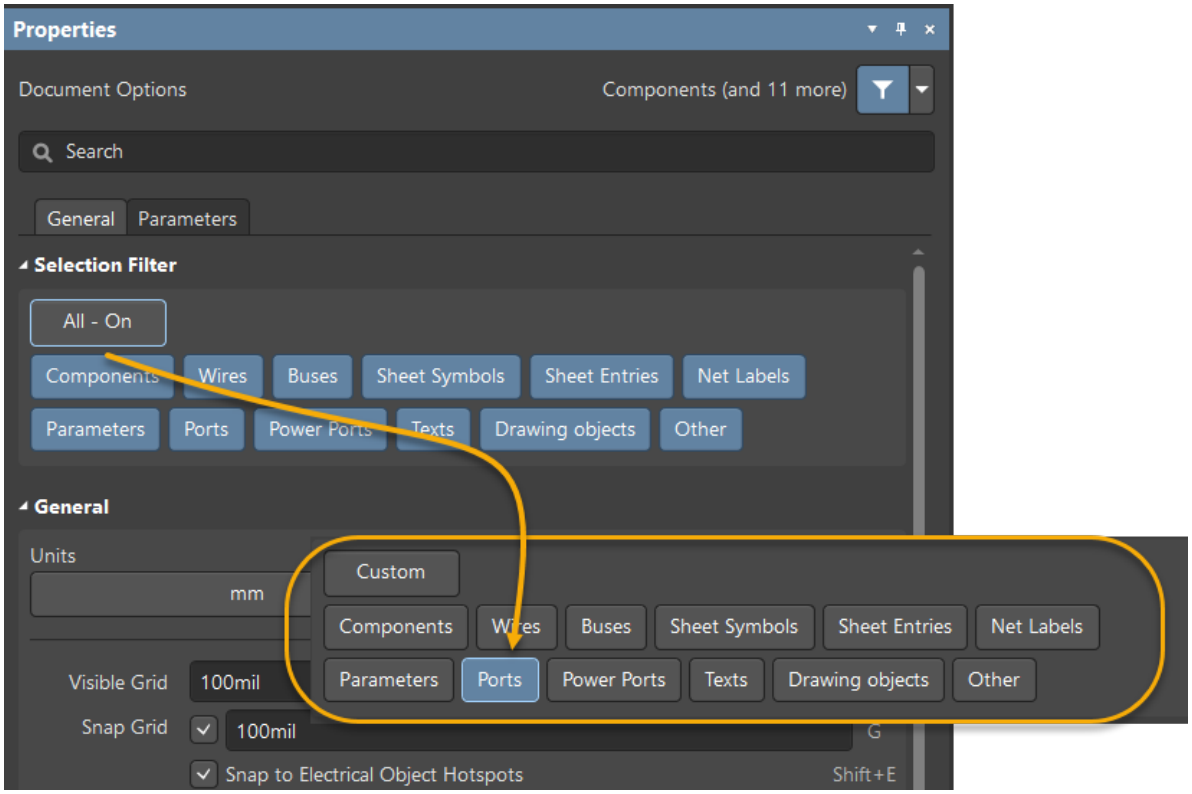


Figure 130. Selection Filter for Ports

10. Change focus back to the `Top_Level.SchDoc` by clicking on its document tab.
11. Run **Edit » Smart Paste**
    a) Select **Sheet Entries** in the pane *Choose Paste Action*, *Paste As* list box as shown in Figure 131.
    b) Press the **OK** button

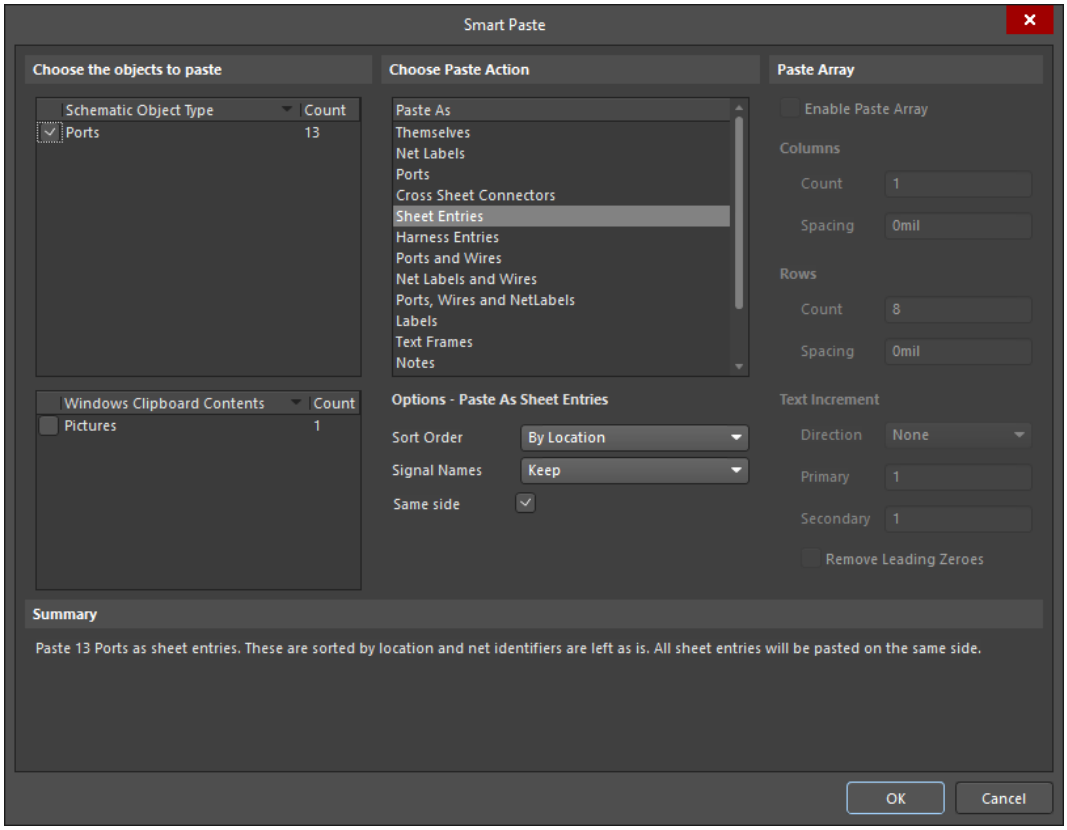


Figure 131. Smart Paste dialog

12. Click on the MID1 sheet symbol and place the sheet entries on the right side of the sheet symbol. You'll notice the sheet entries will snap to the side of the sheet symbol shown in Figure 132 below.
13. Position the sheet entries and adjust the size of the sheet symbol similar to Figure 132 below.
    a) Setting the grid to 100mil will help with placement, press **G** to change the grid
    b) Change the value of *Snap Distance* if needed. (*Properties* panel)

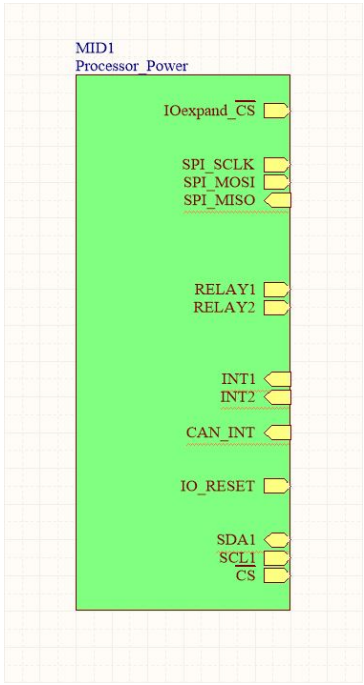


Figure 132. Sheet Symbol and Sheet Entry placement

10.4.3 Creating a Sheet Symbol from Existing Sheet

14. Right-click anywhere on the schematic and select **Sheet Actions » Create Sheet Symbol from Sheet**.
15. Select the `IO Logic Block.SchDoc` from the *Choose Document to Place* dialog and select **OK** as shown in Figure 133 below.

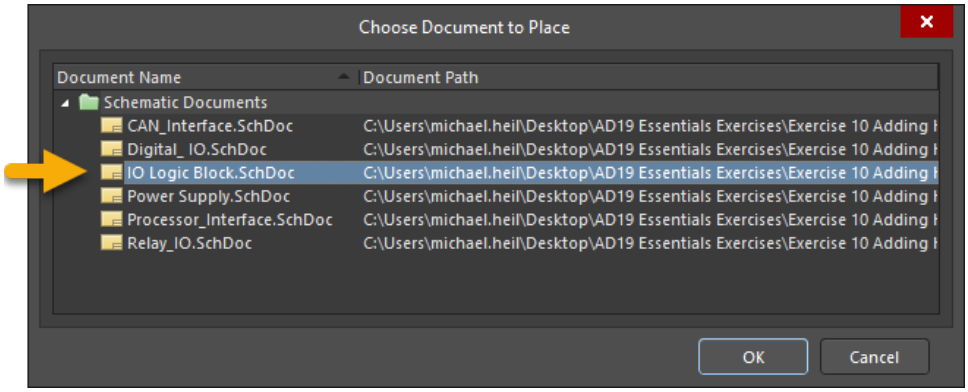


Figure 133. Choose Document to Place as a Sheet Symbol

16. With the Sheet Symbol still attached to your cursor, press **Tab** to bring up its properties
17. Enter `MID2` in the *Designator* properties and press **Enter**.
18. Place the Sheet Symbol to the right of the `MID1` Sheet Symbol and align the Sheet Entries adjacent to the sheet entries in Sheet Symbol MID1 as shown in Figure 134 below.

Activate the large 90° cursor (Preferences, section Schematic – Graphical Editing). With an active Large Curser 90, it is easier to position the entries.

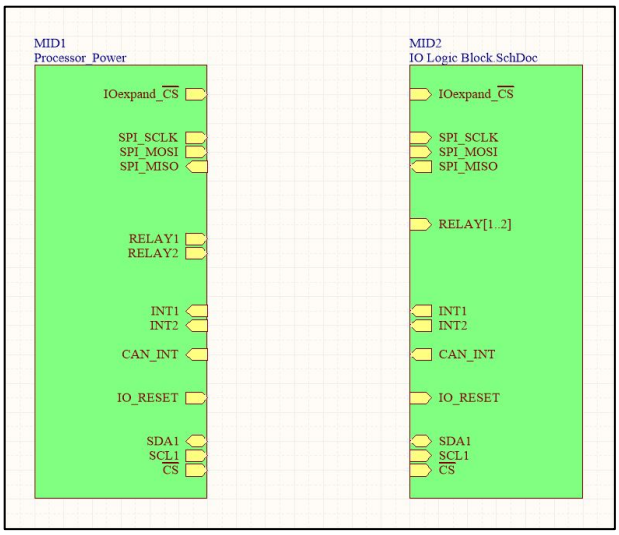


Figure 134. Sheet Symbol Alignment

10.5 Connecting the Sheet Symbols

Now that the sheet symbols have been placed, you will wire the connections between the two sheet symbols

10.5.1 Wiring the Sheet Symbols

19. Move the `Processor_Power` sheet symbol directly in contact with the `IO Logic Block` sheet symbol so that the sheet entries line up beside each other. You will notice a small green checkmark that will appear.
20. Drag the `Processor_Power` sheet symbol back to its original position. Wires should appear connecting the two blocks (this is dependent on your preferences; you may need to try again while holding down the **Ctrl** key).

10.5.2 Wiring the Relay Bus

21. From the **Place** menu, select **Place » Bus**.
    a) Start the Bus at the `RELAY[1..2]` sheet entry in the `MID2` sheet symbol and extend half way to the `MID1` sheet symbol.
    b) Proceed downwards so that there is a sufficient amount of bus wire available for wires to extend from the `RELAY1` and `RELAY2` sheet entries of the `MID1` sheet symbol and left-click.
    c) Right-click twice to exit the command.
22. Wire the `RELAY1` and `RELAY2` sheet entries of the `MID1` sheet symbol to the bus using **Place » Wire**, as shown in Figure 135 below.

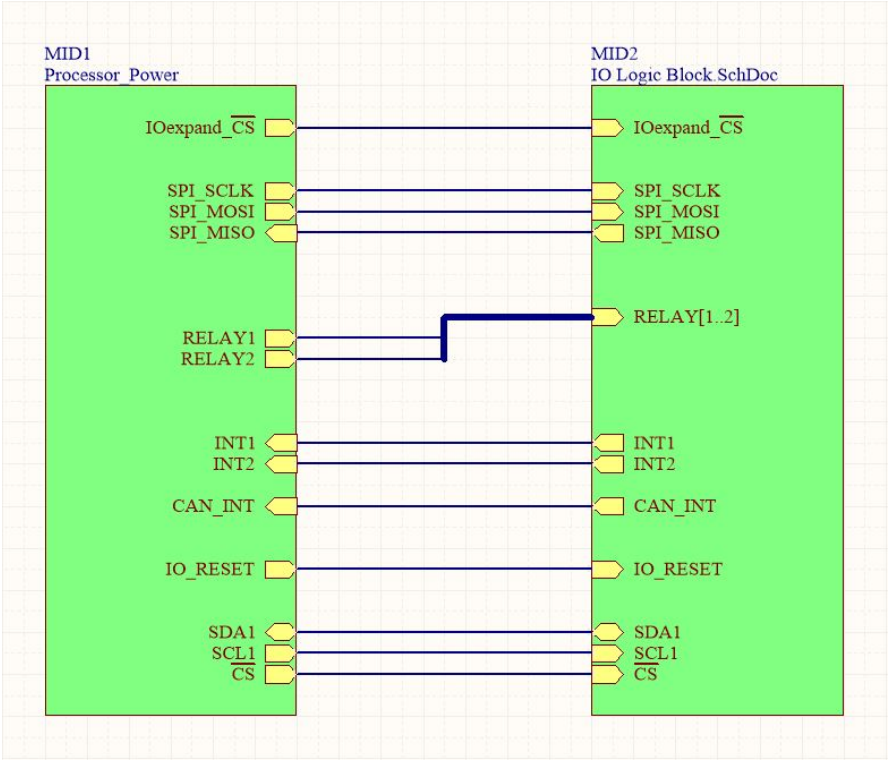


Figure 135. Wired Sheet Symbols

10.5.3 Labeling the Bus and Bus Slices

23. Label the Bus and Bus Slices using the **Place » Net Label** command.
    a) With the net label attached to your cursor, hover over the `RELAY[1..2]` sheet entry and press **Insert** key. The new net label inherits the `RELAY[1..2]` name. Alternatively, press **Tab** and manually change the value.

b) Left-click to place the `RELAY[1..2]` net label on the Bus.

c) With a net label attached to your cursor, hover your cursor over the `RELAY1` Sheet Entry in the `MID1` Sheet Symbol.

d) Press **Insert** so that the net label picks up the `RELAY1` Sheet Entry name.

e) Left-click to place the net label on the extended wire from the `RELAY1` sheet entry. (After placing the net label, the net label on the cursor automatically increments to `RELAY2`).

f) Left-click to place the `RELAY2` net label on the wire extending from the `RELAY2` Sheet Entry. Use Figure 136 as a reference.

g) Right-click to exit the command.

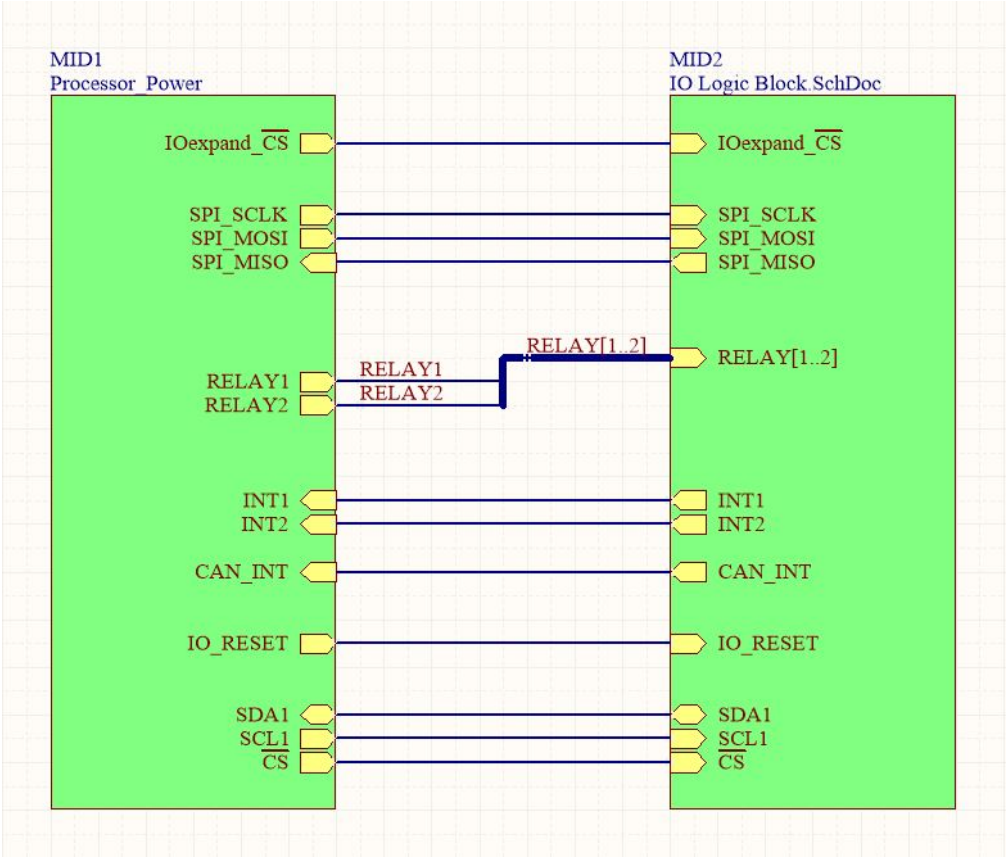


Figure 136. Net Labeled Bus and Bus Slices

10.6 Creating a Schematic Sheet from Sheet Symbol

The MID1 Processor_Power Sheet Symbol is not referencing an existing schematic sheet. In this exercise, you will create the `Processor_Power` Schematic Sheet from the existing Sheet Symbol.

10.6.1 Create Sheet from Sheet Symbol

24. Right-click on the `MID1 Processor_Power` sheet symbol in the `Top_Level.SchDoc` and execute the **Sheet Symbol Actions » Create Sheet From Sheet Symbol** command.

10.6.2 Completing the Newly Created Schematic Sheet

25. In the newly created `Processor_Power.SchDoc` you will see ports that reflect the sheet entries of the created `MID1` sheet symbol in the `Top_Level` schematic.

    a) Hit **Shift+C** to clear all selections from multiple documents in the project. Group-select the ports by dragging a selection box around them.

b) Once selected, move them off to the left of the schematic sheet. The ports will still be selected after the move.

26. With the ports selected, open the *Properties Panel*.
    a) Change the width to `700mil`
    b) Press **Enter** to accept the new value
    c) Click on the schematic to clear the selection or use **Shift + C**.
27. Still in the `Processor_Power.schdoc` create a new sheet symbol from a schematic by selecting the menu command, **Design » Create Sheet Symbol From Sheet**.
    a) Select the `Processor_Interface.SchDoc` from the list in the *Choose Document to Place* dialog.
    b) Click **OK** and place the sheet symbol into the center of the schematic sheet.
28. Create a second sheet symbol.
    a) Right-click an empty space in the schematic and select **Sheet Actions » Create Sheet Symbol From Sheet.**
    b) Select the `Power Supply.SchDoc` from the list in the *Choose Document to Place* dialog.
    c) Click **OK** and place the sheet symbol into the lower right of the schematic sheet.
29. Arrange and position the sheet entries of the `Processor_Interface.SchDoc` sheet symbol as shown in Figure 137 below. (You may need to toggle your grid settings using the **G** key for easy positioning)

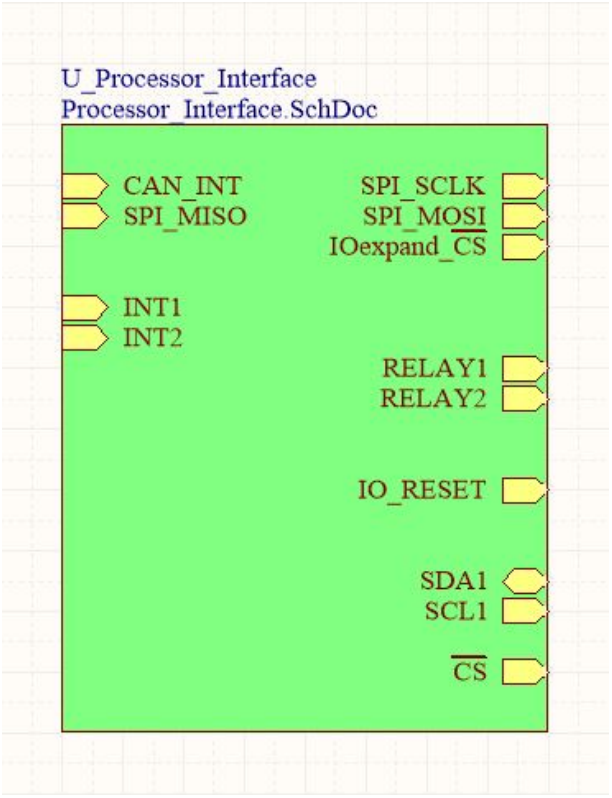


Figure 137. Sheet Entry position

30. Place wire connections from the sheet entries of the `Processor_Interface.SchDoc` sheet symbol to the corresponding ports as shown in Figure 138.
    a) Move the port so that it makes direct contact with the corresponding sheet entry.
    b) Drag the port away from the sheet to the desired location. Dragging to create wires can be done simultaneously with multiple ports. First, move all of the ports to be in direct contact with the sheet entries. Then group select the ports and drag.

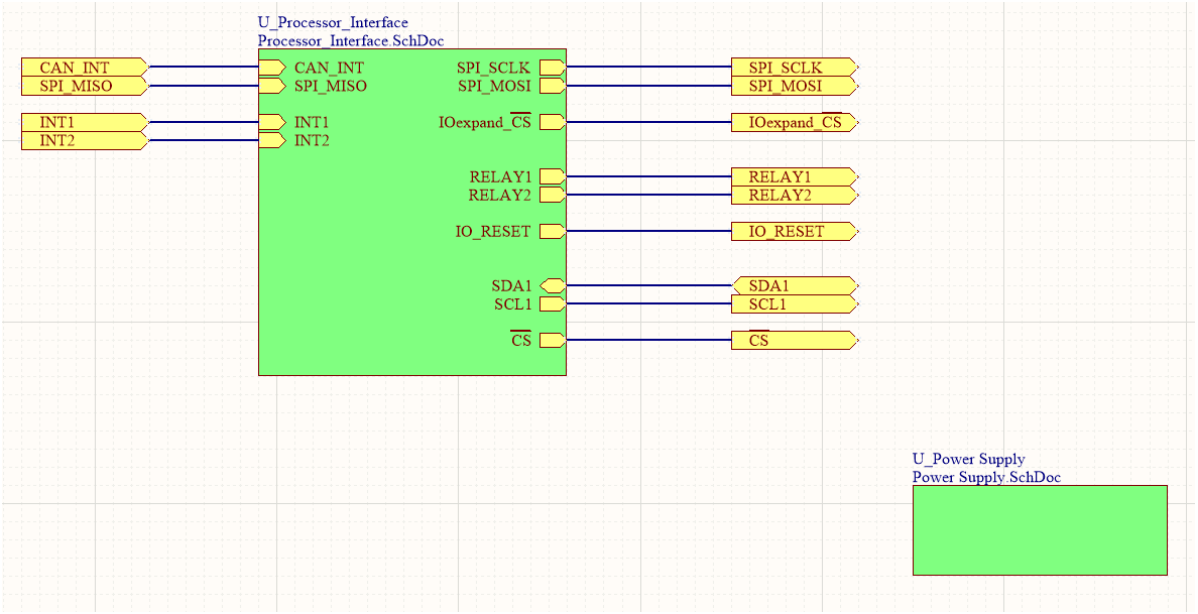


Figure 138. Sheet Symbol wired to Ports

31. Right-click on the project in the *Projects* panel and select `Validate PCB Project WCTopping.PrjPcb`. If you have Dynamic Compiling enabled, manually compiling is not necessary. Dynamic Compiling is covered in Module 9: Compiling in Altium Designer.
32. After compiling the project, the resulting hierarchical structure is visible in the *Projects* panel as shown in Figure 139 below.

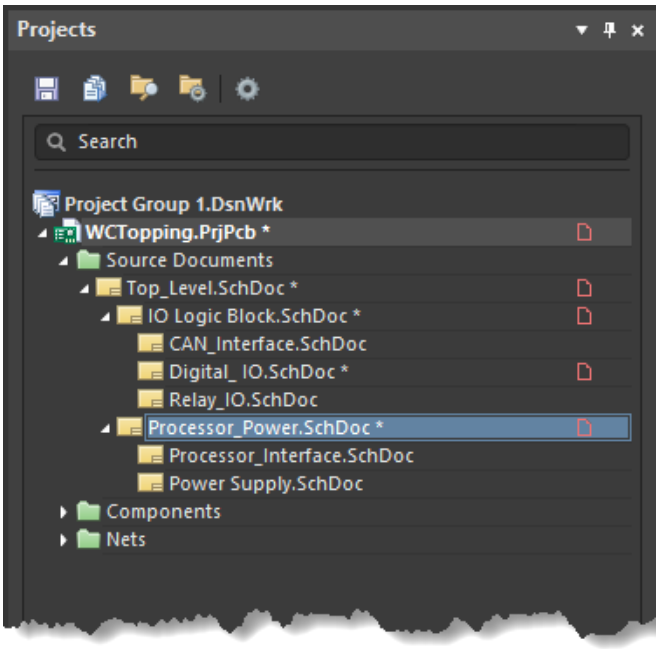


Figure 139. Hierarchical Design Project Structure

33. Disregard any Warnings or Errors and select **File » Save All**.

Module 11: Transfer to PCB

11.1 Purpose

In this exercise, you will be introduced to transferring the design from schematic to PCB. An additional part of this exercise is the placement, configuration, and transfer of directives from the schematic to the PCB.

Using a comparison engine, Altium Designer creates an *Engineering Change Order (ECO)*. This is used to synchronize the schematic to PCB, as well as update the relevant changes from the PCB back to the schematic. This bi-directional synchronization guarantees a high level of design integrity and helps to ensure the PCB and schematics remain synchronized. If any items fail during the *ECO Execution*, you should locate and correct the errors before completing the transfer.

11.2 Preparation

1. Close the current `WCTopping.PrjPcb` project.
2. Open the `WCTopping.PrjPcb` project located in the `AD20 Essentials Exercises\Module 11 - Transfer to PCB` folder.

11.3 Before the Transfer

11.3.1 Footprint Validation

The Footprint Manager is used to ensure that all of the components in your schematic documents, have an associated footprint. This will prevent errors should as "*Footprint Not Found*" when pushing your design to the PCB.

3. From the *Projects* panel, open any schematic document.
4. Select the **Tools » Footprint Manager…** file menu.
5. Select all the components from the *Component List* with a right-click and **Select All**, or use **CTRL+A** as shown in Figure 140. All of selected components are now listed in the *View and Edit Footprints* list to the right side of the dialog.
6. Select all the footprints in the *View and Edit Footprints* list.
7. Click the **Validate** button below the *View and Edit Footprints* list.
8. After the Validation process is finished, review the *Found In* column for a listing of `Footprints Not Found`. Review the *PCB Library* column to see the library source for that footprint.

If you had not installed the `WCT.IntLib` library, you will see `Footprints Not Found` for components that originated from that library. In that case, it will need to be installed in the *Components* panel. A connection to the Altium Content Vault is also needed to find footprints that originated from the Altium Content Vault.

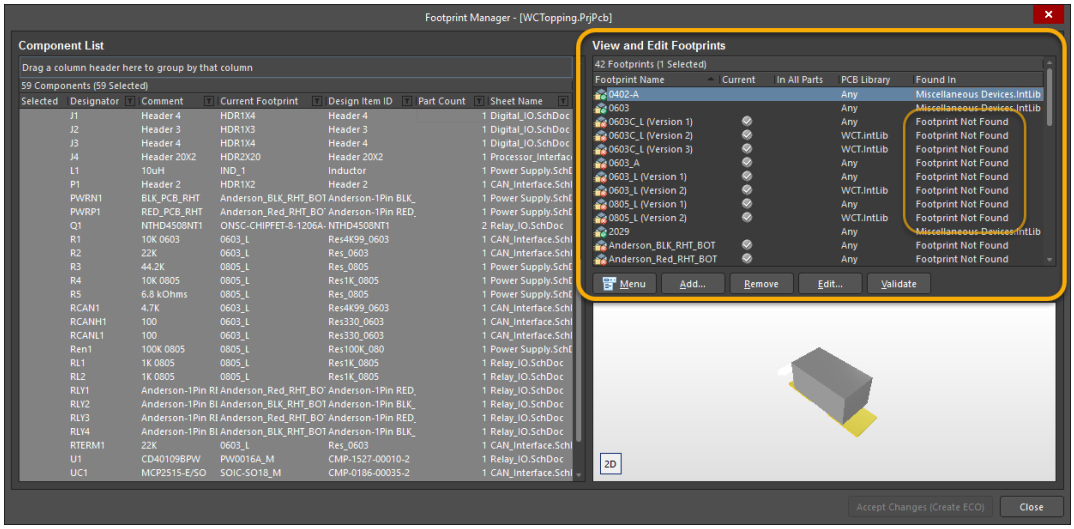


Figure 140. Footprint Manager after footprint validation

9. Close the *Footprint Manager*.
10. If all footprints are found, continue with ***11.4 Transferring the Design***. If not, please follow the steps below.

11.3.1.1 Connect to the Content Vault

11. This design uses components from the Altium Content Vault. Connection to the Altium Content Vault is required to load the footprint data into the design. If you're not signed in, we will need to connect to the Altium Content Vault.
    a) Open the *Sign in* dialog in the upper right corner as shown in Figure 141 below. If you are already signed in, your name with your login credentials will already be there. If this is the case, continue with step 11.3.1.1 c)

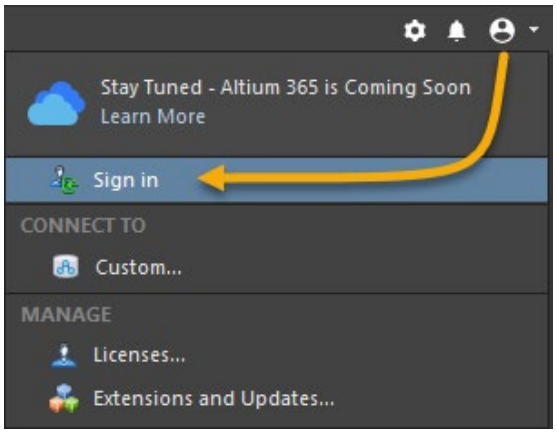


Figure 141. AltiumLive sign in dialog

    b) Sign in to Altium Live with your Altium Live account.
    c) Open the Preferences using the icon, in the upper right corner of Altium Designer.
    d) Under *Data Management - Servers*. Expand the *Known Server* section if necessary.
    e) Check the **Automatically connect Altium Content Vault** checkbox. Ensure you're connected to the *Altium Content Vault* as shown in Figure 142 and click **OK** to exit the Preferences.

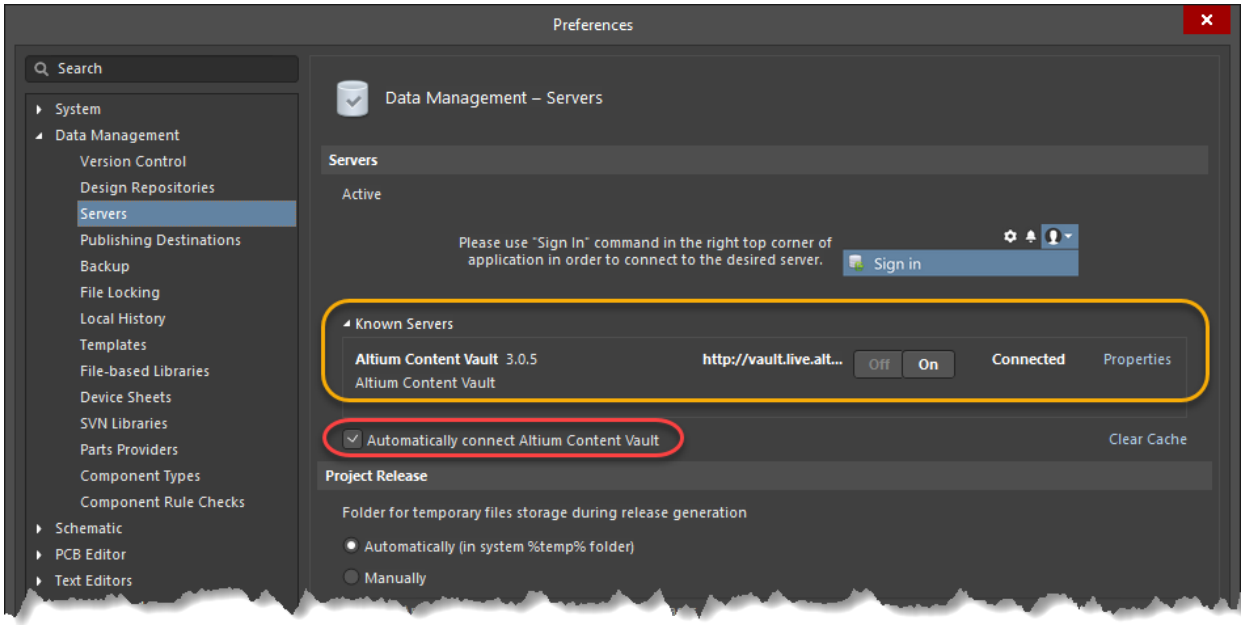


Figure 142. Altium Content Vault connection

11.3.1.2 Install the WCT.IntLib library

12. If you have not installed the WCT.IntLib library, please return to *Exercise* 5.3.2 Adding an Integrated Library for instructions to install this library. Access to this library is necessary to properly complete this exercise.
13. Reopen the *Footprint Manager* under the **Tools** menu and validate that all footprints can be found with a connection to the Altium Content Vault and the `WCT.IntLib` installed from the *Components Panel.* Close the *Footprint Manager.*

11.4 Transferring the Design

11.4.1 The Initial Transfer

14. Right-click the project name in the *Projects* panel and select **Add New to Project » PCB**.
15. Save the PCB and name it `WCTopping.PcbDoc`
16. From within a schematic, click on **Design » Update PCB Document WCTopping.PcbDoc.**
17. Click the **Execute Changes** button in the lower left of the dialog, as shown in Figure 143 below.

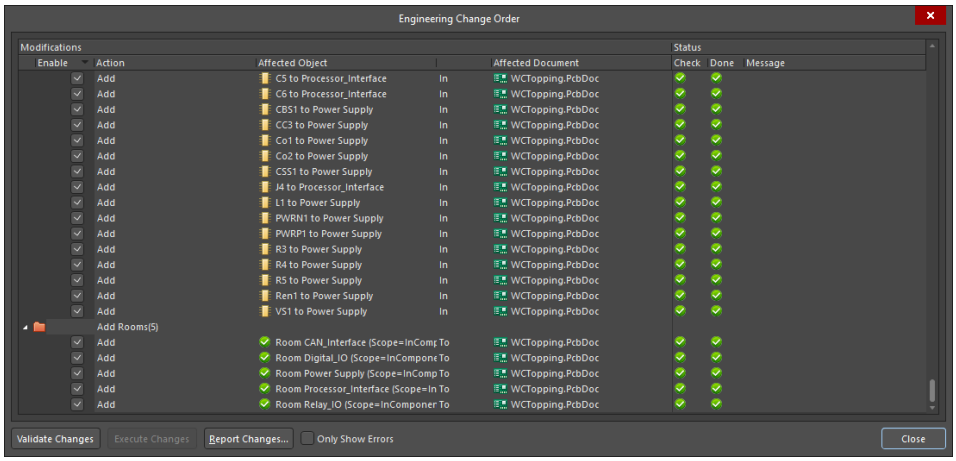


Figure 143. Engineering Change Order dialog

18. Enable the **Only Show Errors** checkbox to check for any errors.
19. Close the *Engineering Change Order* dialog.

11.4.2 Cross Select Mode

20. Right-click on the `WCTopping.PcbDoc` tab in the *Documents Bar* and select **Split Vertical**.
21. Select the `Relay_IO.SchDoc` tab in the *Documents Bar* so that you see that particular schematic, along with the PCB on the other side as shown in Figure 144 below.

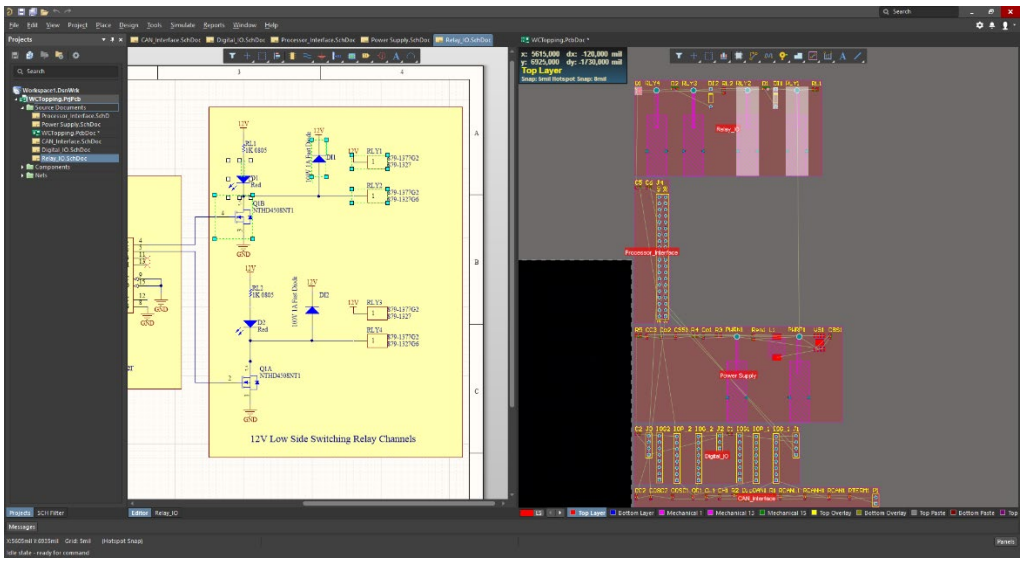


Figure 144. Split Vertical view with selected components in Cross Select mode

22. Check to see if cross selection mode is enabled in the schematic by going to **Tools » Cross Select Mode.** Ensure this option is enabled and highlighted as shown in Figure 145 below.

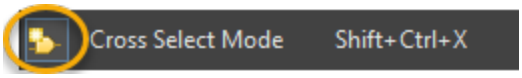


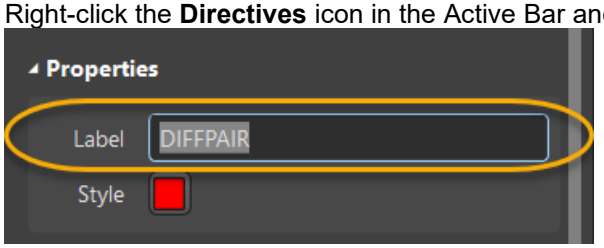
Figure 145. Cross-Select mode enabled

23. Group-select the following components within the `Relay_IO.SchDoc` in the following order by holding down the **Shift** key and selecting components: RL1, D1, Q1B, DI1, RLY1, RLY2.
    a) Right-click on the `WCTopping.PcbDoc` tab in the *Documents Bar* and select **Merge All** to condense the tiling.
    b) Under the **Tools** menu, select **Component Placement » Reposition Selected Components**.
    c) Click in the PCB workspace to place the component on your cursor. With each subsequent click, the next component will snap onto your cursor based on the order selected in the schematic.

11.4.3 Assigning Directives

24. If not already open, open the `Processor_Interface.SchDoc` and make it your active document.
25. Replace the net label `GPIO2` and the `GPIO3` net labels with `CLK_P` and `CLK_N`

Right-click the **Directives** icon in the Active Bar and select **Differential Pair**, as shown in



26. Figure 146 below. This can also be done from the **Place » Directives** menu.
27. With the Differential Pair directive on your cursor, hit the **Tab** key to access its Properties.

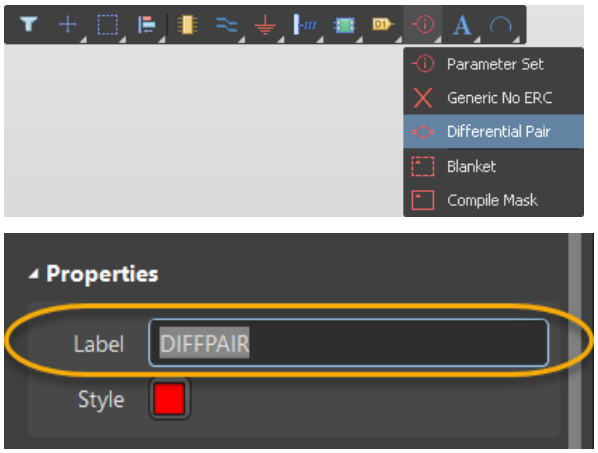
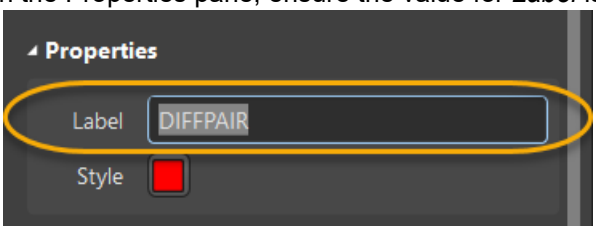


Figure 146. Placing a Differential Pair directive in the Schematic

a) In the Properties pane, ensure the value for *Label* is `DIFFPAIR` as shown in



b) Figure 146 above.
c) Click on the pause symbol in the center of the Schematic to continue the placement.
d) Place this directive on the `CLK_P` and `CLK_N` wires as shown in Figure 147.

e) Right-click to exit the command.

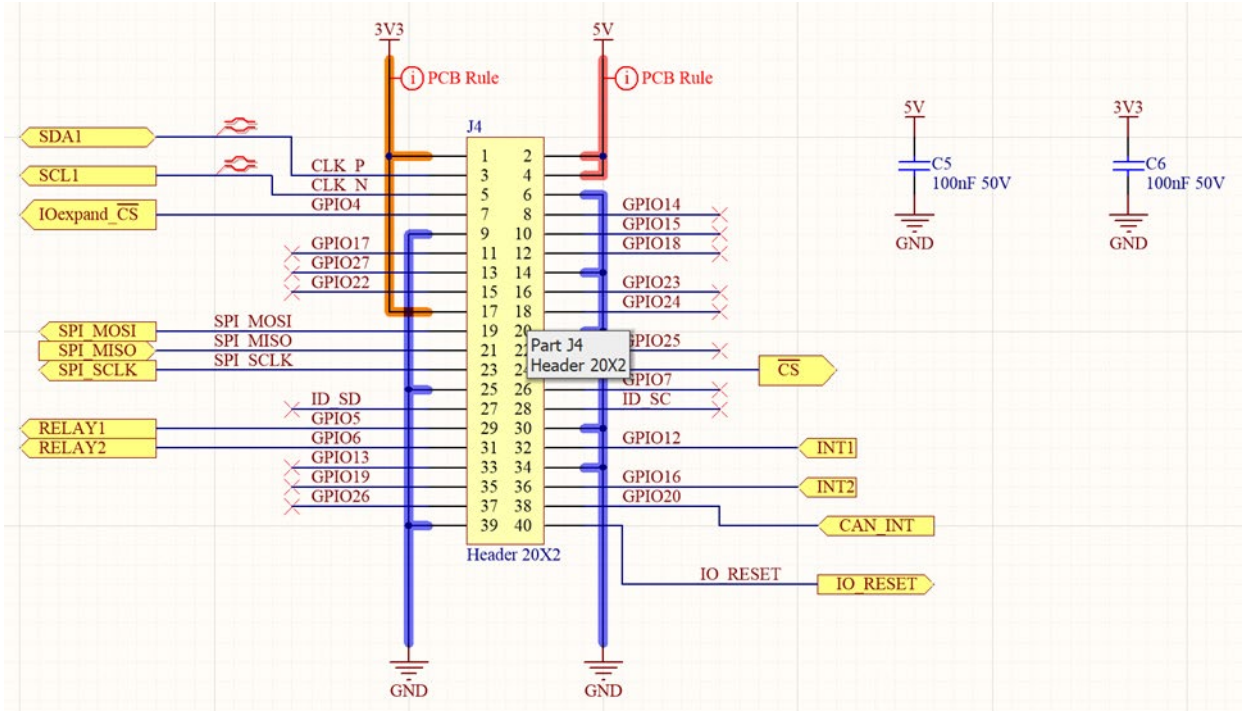


*Figure 147. Directives placed in the schematic*Use the command **Place » Directives » Parameter Set**.

a) With the directive on your cursor, press the **Tab** key.

b) In the *Properties* pane, ensure the value for *Label* is `PCB Rule`

c) Under the *Parameters* pane, click **Add... » Rule**, as shown in Figure 148.

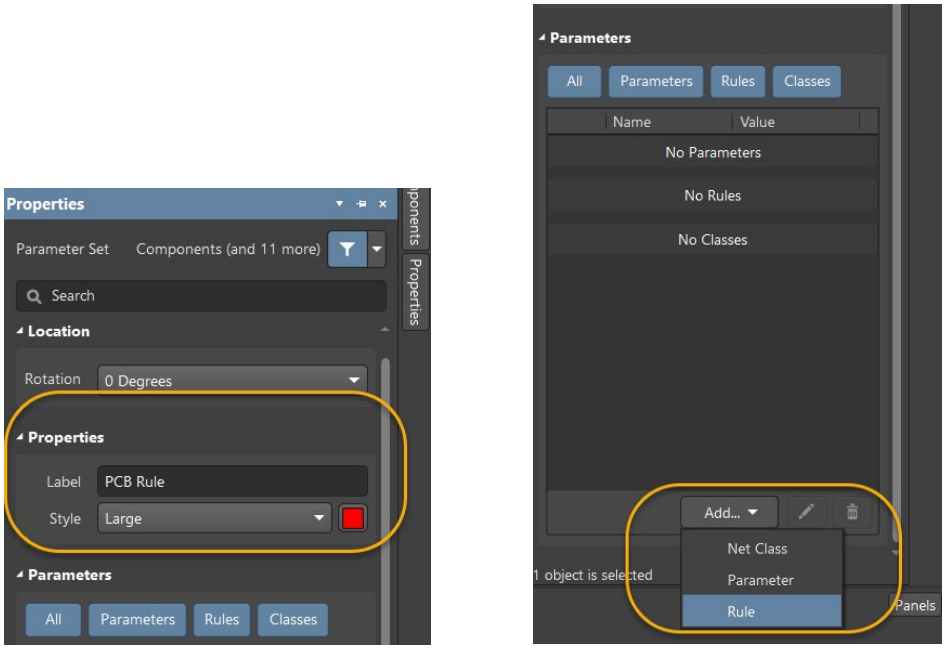


Figure 148. Adding PCB rule directive from the schematic

d) Select the *Width Constraint* rule from within the *Routing* section.

e) Press the **OK** button

f) In the upper section set:

i) *Min Width* to `30mil`

ii) *Preferred Width* to `50mil`

iii) *Max Width* to `100mil`.

iv) Press the **OK** button and the Pause symbol to continue the placement of the PCB Layout Directive.

g) Place the PCB Layout Directive on both the `3V3` and `5V` wires extending from the top of the `J4` component as shown in Figure 147.

h) Right-click to exit the command.

29. Select **Design » Update PCB Document WCTopping.PcbDoc** from the file menu.

a) The Design Rules are listed in the *Engineering Change Order*.

b) Click **Execute Changes.**

c) **Close** the *Engineering Change Order*.

30. With the PCB document active, open the *PCB* panel from the **Panels** button.

31. Click the drop-down menu to select *Differential Pairs Editor* as shown in Figure 149 below.

32. Then, click on *<All Differential Pairs>* to see all of the Differential Pairs in the design. You should see the CLK Differential Pair that you added from the schematic directive.

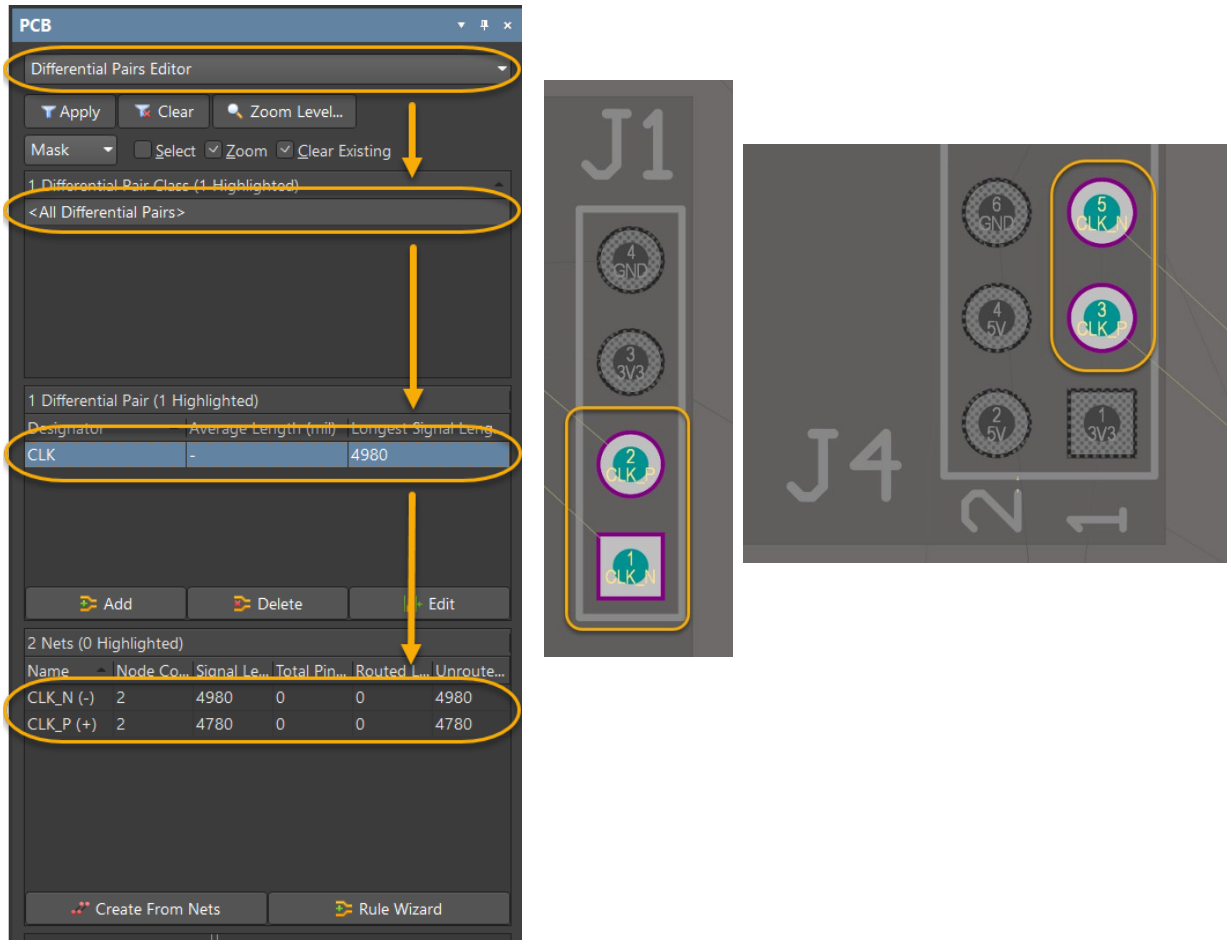


Figure 149. Differential Pair Editor in the PCB panel

33. Open the *PCB Design Rules and Constraints Editor* with **Design » Rules…** and check to see that the PCB Rule directive is transferred to the PCB as shown in Figure 150 below.

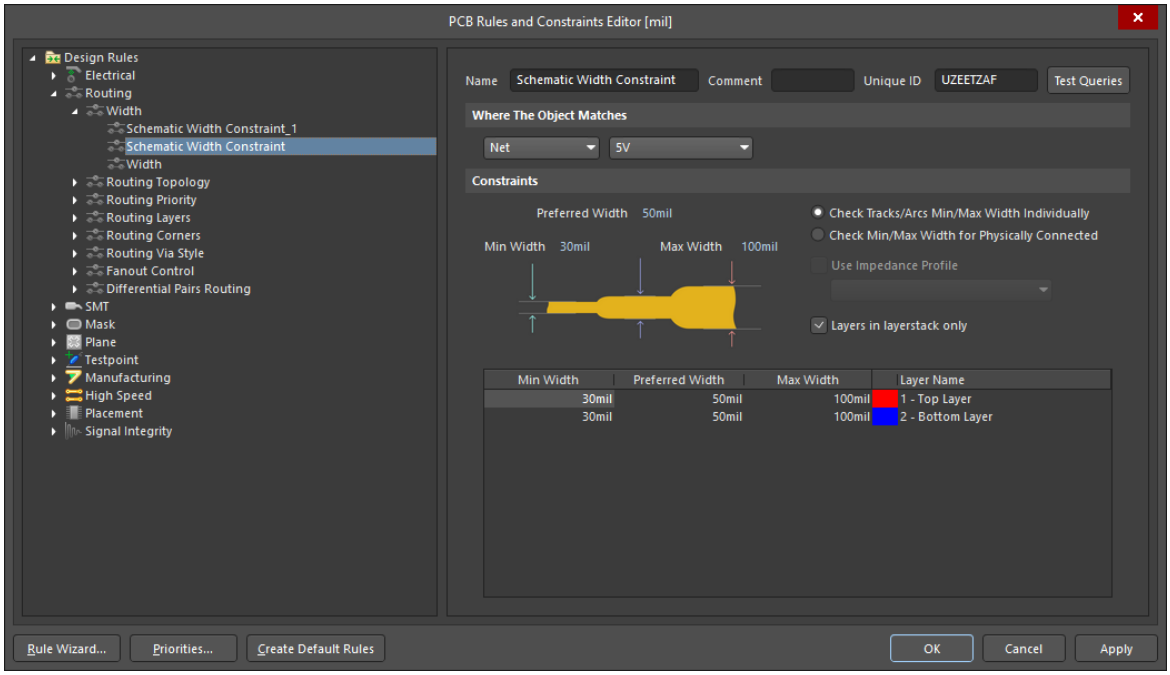


Figure 150. Design Rules showing the new Width rule that was added from the Schematic

34. Save all documents and close the project.

Module 12: Updating the PCB

12.1 Purpose

In this exercise, you will learn about component links and how it is used to synchronize the schematic components to the PCB document.

In order to synchronize schematic components to a PCB document within a project, each component contains a unique identifier which is used to link them together. If no link is defined or if the link is broken, the synchronization will fall back to the Component Designator. Matching the Component Links is crucial part to keeping your design synchronized at all times.

12.2 Preparation

1. Close all open projects.
2. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `AD20 Essentials Exercises\Module 12 - Updating the PCB` folder.
3. Open the `WCTopping.PrjPcb` project in Altium Designer.
4. Open and focus on the `WCTopping.PcbDoc`. Notice all components placed outside of the board towards the right.

12.3 Component Links

5. Go to **Design » Import Changes from WCTopping.PrjPcb**. Notice that there are unmatched components as shown in Figure 151 below. This can be easily resolved by clicking **Automatically Create Component Links.** However, this exercise will detail how it can be done manually.
6. Click **Cancel** and then close the *Engineering Change Order* window that pops up.

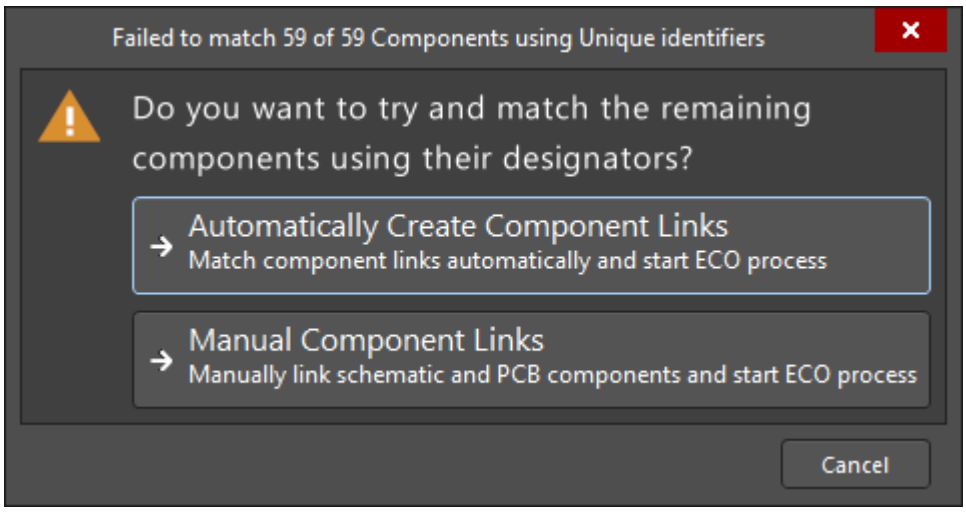


Figure 151. Component Links error dialog

7. Go to **Project » Component Links…** . Choosing “Manual Component Links” Step 5 would have loaded the same dialog.
8. Notice the unmatched components in the first two columns as shown in Figure 152 below.

9. Ensure that the *Designator* checkbox is ticked, then click **Add Pairs Matched By ».** This will match the schematic symbol with the PCB footprint for each component, based on their designator.

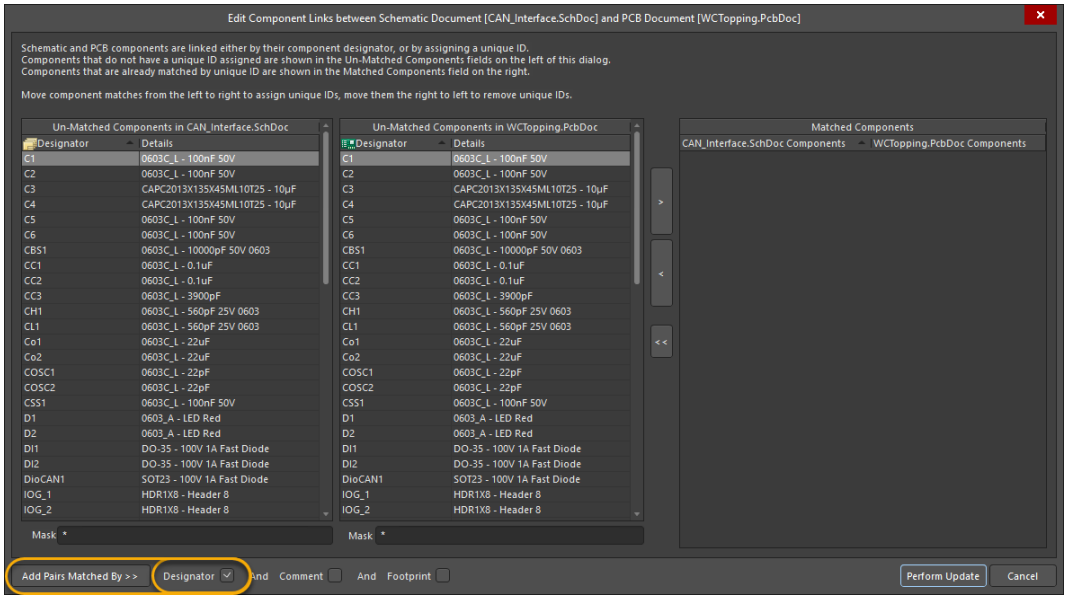


Figure 152. Component Link Setup

10. Click **Perform Update**. Notice the Information window as shown in Figure 153 below, indicating that 59 links have been modified. Click **OK**.

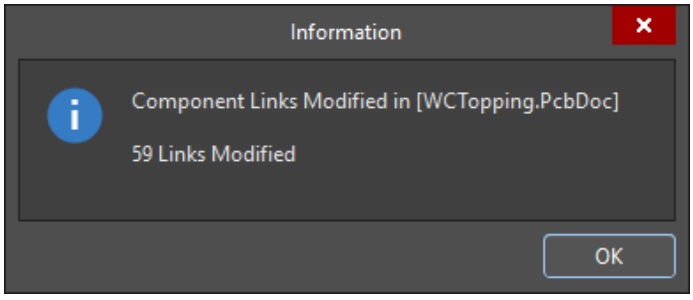


Figure 153. Component Links information dialog

11. To ensure all components are synchronized, run the **Design » Import Changes from WCTopping.PrjPcb**, if the design is synchronized and up to date you will see the dialog below (Figure 154). If any difference is detected it will be shown in an ECO dialog.

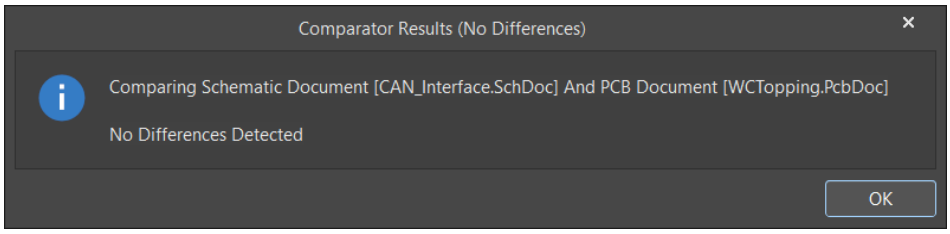


Figure 154. No differences detected, meaning that your Component Links are synchronized

Module 13: PCB Introduction

13.1 Purpose

In this exercise, you'll explore some of the PCB specific preference settings and learn to customize them. You will also briefly use basic commands, such as navigating to specific objects and more. We'll take a more in-depth look at some of these commands in later exercises.

13.2 Preparation

1. Close all open projects.
2. Open the PCB project using **File » Open Project** and browse to your default installed example projects folder. Select:
   `<AD Version>\Examples\Bluetooth Sentinel\Bluetooth_Sentinel.PrjPcb`.
   Since this project had been opened previously, it can also access using **File » Recent Projects**.

The default path to your installed examples is `C:\Users\Public\Public Documents\Altium\<AD Version>\Examples\`

3. Navigate to the *Projects* panel and double-click the icon for the PCB file `Bluetooth_Sentinel.PcbDoc` with the left mouse button so that the PCB document is open.

13.3 Moving PCB View

4. When the PCB is first opened, you may be in the 3D environment. To get into the 2D workspace, where we do the majority of our work, hit the **2** key.
5. Hold the Right-Mouse-Button until the dragging hand cursor appears. Move your mouse while holding the right mouse button to drag the PCB around. This is how we Pan. Release the right button to end the drag operation.
6. Scroll the mouse wheel to scroll the PCB vertically.
7. Hold **Shift** while scrolling the mouse wheel to scroll horizontally.
8. Hold **Ctrl** while scrolling the mouse wheel to zoom in and out.
9. Press and hold both right and left mouse buttons simultaneously to zoom in and out.
10. Press and hold the mouse wheel down to zoom in and out.

13.4 View Commands

11. Maintain the focus on the PCB Workspace and observe the behavior of the following shortcut keys:
    a) To **View » Fit Board** or press the **V » F** keys in succession.
    b) To **View » Fit Document** or press the **V » D** keys in succession.
    c) To **View » Zoom In** or press the **V » I** keys in succession or press **Page Up.**
    d) To **View » Zoom Out** or press the **V » O** keys in succession or press **Page Down.**
    e) To **View » Area** or press the **V » A** and draw a rectangle with the Mouse.
    f) To **View » Zoom Point** or press the **V » P** keys in succession, then draw a rectangle with the mouse somewhere on the PCB. The contents of the rectangle will be focused and shown with the zoom level adjusted.

13.5 General

12. Click in the upper right corner, to open the Preferences. Open the *PCB Editor* branch and look in the *General* page as shown in Figure 155 below.

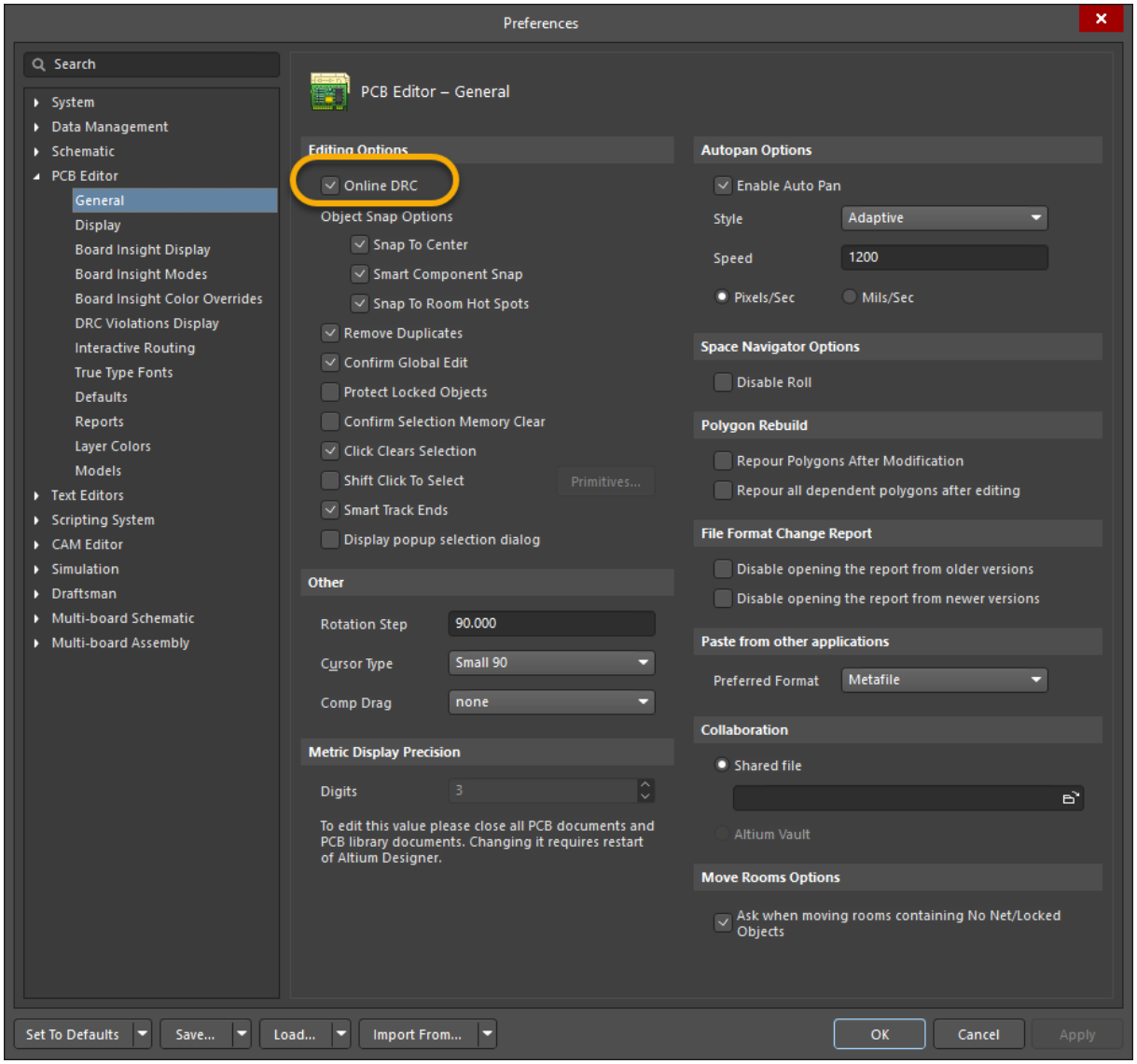


Figure 155. General Preferences for the PCB Editor

13. Verify that **Online DRC** is enabled. Click **OK** to close the preferences dialog.
14. Zoom to the left side of the PCB and select on component `R13` as shown in Figure 156 below.

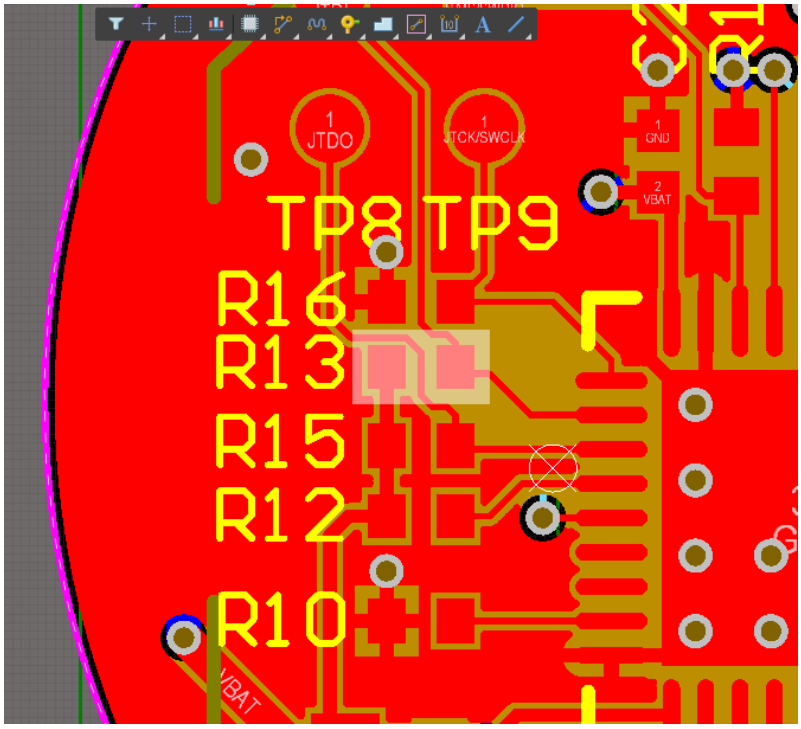


Figure 156. R13 from the Bluetooth_Sentinel.PcbDoc

Quickly find the component using the command:
Edit » Jump » Component (J»C).

15. Left-click and hold component `R13`.
    a) Drag and drop `R13` so it overlaps with component `R16`.
    b) Notice that the overlapping components highlight in green with error markers.
16. Undo the changes by selecting **Edit » Undo**.
17. Go back to the *Preferences*: Click in the upper right corner.
    a) In **PCB Editor** » **General,** enable the option **Display popup selection dialog** under the *Editing Options* category
    b) Press **OK** to exit *Preferences*
18. Locate component `X2`. You can use the **J** » **C** shortcut keys. Left-click component `X2` and a selection window may appear as shown in Figure 157. Select `X2` from the list.

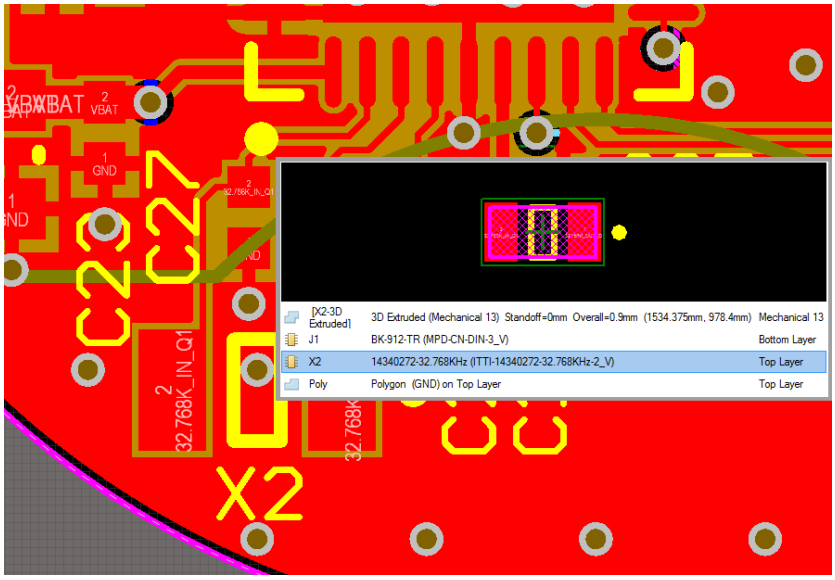


Figure 157. Selection Window

19. Press **Shift+C** to clear the selection.
20. Open the *Properties* panel using
    a) **View » Panels » Properties**
    b) or from the **Panels** menu accessed from the Panels button in the lower right of the workspace.
21. Expand the section *Selection Filter* pane as shown below.
22. Activate the selection filter for **Components** and deactivate it for all other elements as shown in Figure 158 below. This will allow us to only select components.

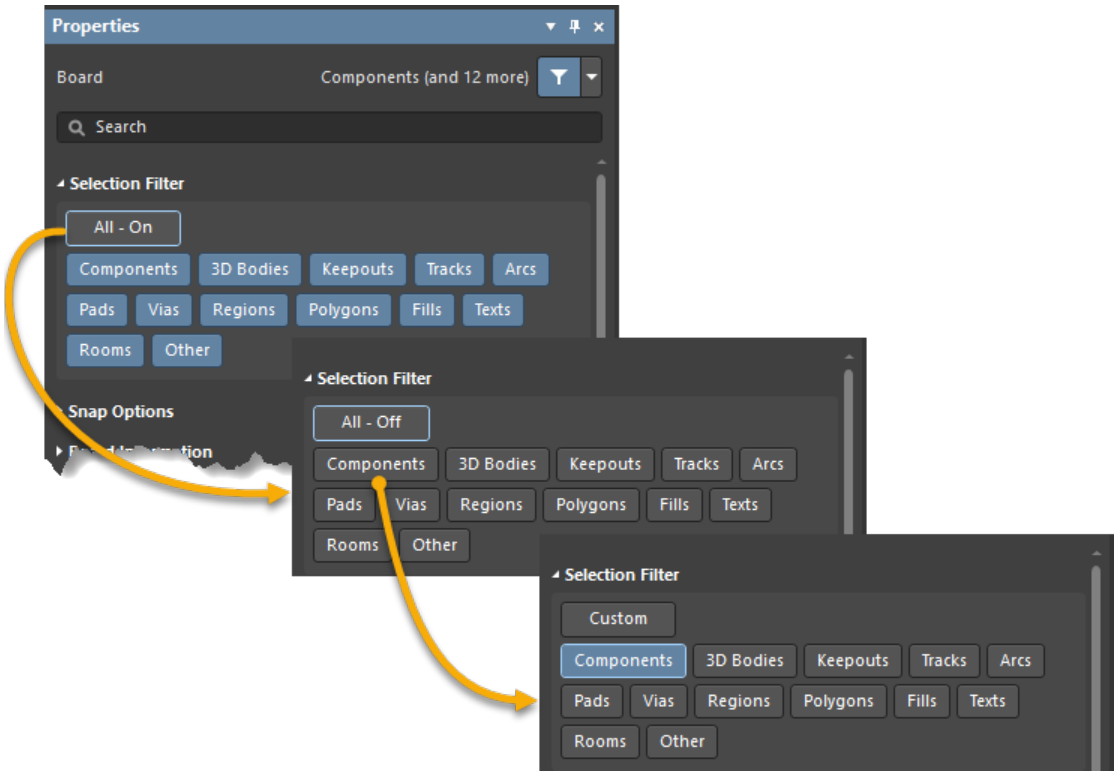


Figure 158. Selection Filter for Components

22. Left-click component `R13` and select it in the P*op-up Selection* dialog.
23. Move the resistor to the left, being sure to select the center of the component body, not the designator or comment. Notice a violation with the surrounding polygon.

24. Repour the Polygon with the command **Tools » Polygon Pours » Repour Violating Polygons** or by a **right-clicking** over the polygon and execute **Polygon Actions » Repour violating polygons…**.
25. Reopen the *Preferences*.
    a) Under *General*, enable **Repour Polygons After Modification** and **Repour all dependent polygons after editing**.
    b) Click **OK** to close the preferences.
26. Left-click and move component `R15` to the left near `R13`. Release the component to see the polygon automatically repour when modified.
27. Undo changes with **Edit » Undo**.
28. Use **Shift+C** to clear the selection of `R15`
29. Open the *Properties* panel and the *Selection filter* pane.
30. Click on **Custom** to allow the selection of all kinds of objects again. It should now say "**All-On**".

13.6 DRC Violations Display

31. Jump to component `R13` using **J»C** and drag the component to overlap component `R16`. It should now have violation overlays on the two components. Additional to the , Altium shows error markers, e.g. short circuits .
32. Open the *Preferences*:
    a) Select the *DRC Violations Display* section.
    b) Change the *Violation Overlay Style* from **Style B** to **Style A** and press **OK**.
    c) Notice the violations have now changed to violation markers shown in Figure 159 below.
    d) Repeat steps and try other *Violation Overlay Styles*.

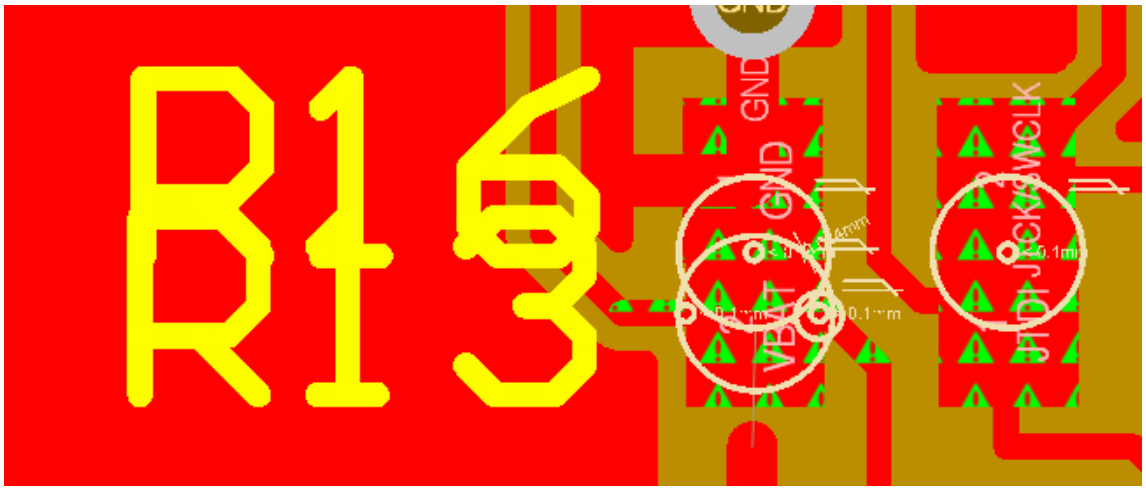


Figure 159. Violation Display Markers

33. For individual DRC violations, the section "*Choose DRC Violations Display Style"* in the *Preferences* allows you to configure the *Violation Detail* and *Violation Overlay* information.
    a) Access the *DRC Violations Display* in the preferences and deactivate the details information for the *Short Circuit* rule shown in Figure 160 below.
    b) Press **OK**
    c) Notice the violations detail for *Short Circuit* are now removed.

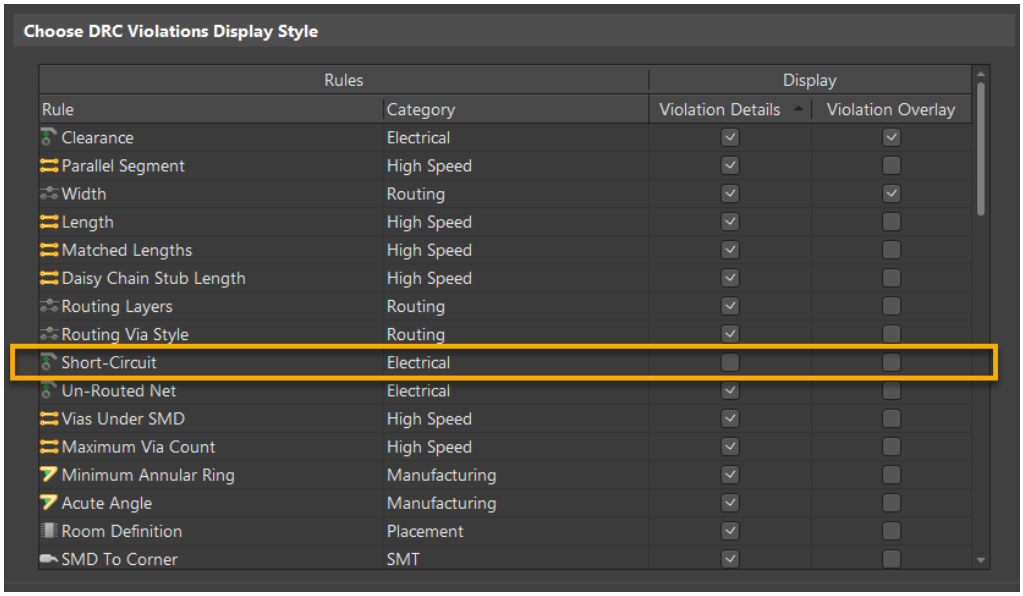


Figure 160. Configuration for the DRC details and violations

34. Undo the violation by selecting **Edit » Undo**.
35. Clear the selection using **Shift+C.**

13.7 Interactive Routing

Shelving the Polygons with **Tools » Polygon Pours » Shelve…** could make it easier to see the routing on a PCB. But, before sending the PCB out for fabrication, make sure you **Restore Shelved Polygons.** This is covered in a later module.

36. Try different Interactive Routing modes:
    a) Open the *Preferences* and select the *Interactive Routing* page.
        i) In the *Routing Conflict Resolution* options, set *Current Mode* to **Ignore Obstacles**.
        ii) Press **OK**.
    b) Zoom to `Pad U4-28` and activate the route by selecting **Route » Interactive Routing.**
    c) Place crosshair over `Pad U4-28` and route from the pad to the net JTDI on the pad below. Notice that the track can go past the JTDI, creating a violation due to the Ignore Obstacles option.
    d) **Right-click** twice to exit the routing mode. **Undo** any tracks that you may have made.
    e) Repeat to experiment with other Routing Conflict Resolution modes.

Quickly toggle through the routing modes while actively routing using **Shift + R**. The *Heads up display* and *Status bar* will show the active routing conflict resolution mode.

37. Place a Route from `Pad U4-28`:
    a) Press **Shift+Space** to toggle through routing angles. Notice that there are options to route at Any Angle and Arcs (This is indicated at the bottom of the workspace in the Status Bar).
    b) **Right-click** twice to exit routing mode.
38. Clear the selection using **Shift+C**.
39. Open *Preferences* and go to *Interactive Routing*.
    a) Enable **Display Clearance Boundaries** and select **OK** to confirm.

b) Route from `Pad U4-28` to see an outline around objects on the board as shown in Figure 161 below.

The clearance boundaries will not show if you have Ignore Obstacles as the active routing conflict mode. Press **Tab** while routing to access the routing preferences where you can enable **Display Clearance Boundaries.**

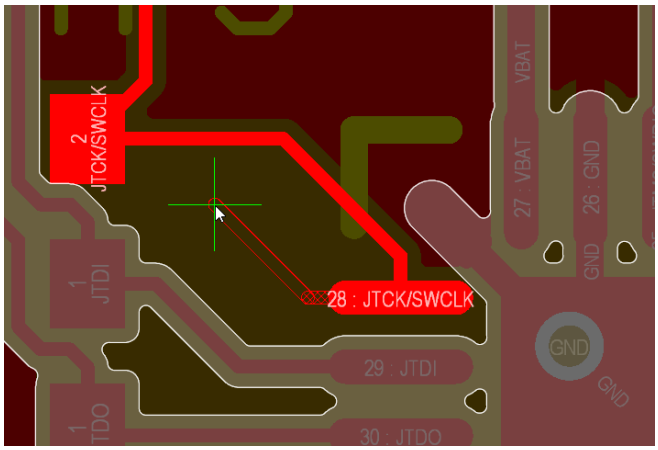


Figure 161. Clearance Boundaries

Reduce this outline by enabling **Reduce Clearance Display Area** in *Preferences*

40. Explore Routing Gloss Effort options by routing on the board.
    a) Try dragging some of the vias outside `U4`, e.g. Via close to `U4-32` and note the attached track behavior, avoiding the GND net vias.
    b) Open the *Preferences*, Page *Interactive Routing* pane *Routing Gloss Effort* and change the routing *Gloss Effort* from **Weak** to **Strong**.

If you finished the exercise and are looking for another position to do some test routing you can activate `Mid-Layer 2` and reroute the signal `Wake`.

41. Close the project without saving.

Module 14: PCB Viewing and Navigation

14.1 Purpose

In this exercise, users will gain a better understanding of the PCB Editor interface and learn about basic navigation and viewing tools.

This exercise is for users to become familiar with the PCB workspace. **Do not save** the project or design documents.

In this example, we use special objects called Device Sheets. The logical designators from the schematic do not match the designator information we see on the PCB. This is based on the specific Device Sheets function we use in this example. To see the physical designator on the schematic, you can change the view by selecting the second (compiled) tab at the lower left side of the schematic page. For further information about Device Sheets and logical/physical designators have a look at the documentation.

14.2 Preparation

1. If not already open use **File » Open Project** and browse to your installed Example projects folder. Select: `Examples\Bluetooth Sentinel\Bluetooth_Sentinel.PrjPcb`. Since this project had been opened previously, it can also access using **File » Recent Projects**.

The default path to your installed examples is `C:\Users\Public\Public Documents\Altium\<AD Version>\Examples\`

2. Navigate to the *Projects* panel and double-click the icon for the PCB file `Bluetooth_Sentinel.PcbDoc` with the left mouse button so that the PCB document is open.

14.3 Jump Command

The **Edit » Jump** command is commonly used so the status of this sub-menu has been elevated to a popup menu with its hotkey **J**.

3. Hit the **2** key to ensure you're in 2D mode.
4. To jump to the current origin: press the **J - O** keys in succession.
5. To jump to a specific location: press the **J - L** keys in succession, then enter the X, Y coordinates within the *Jump to Location* dialog as shown in Figure 162 below.

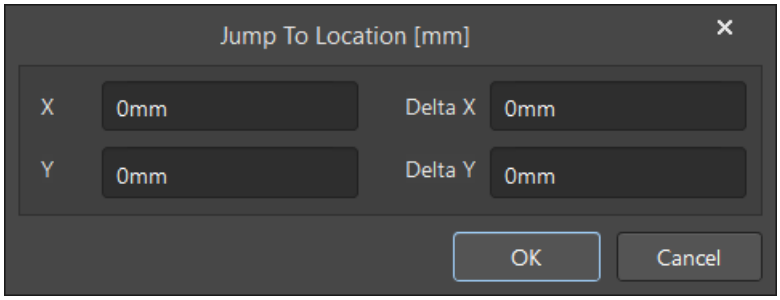


Figure 162. Jump to Location dialog

6. To **Jump » Component...** press the **J » C** keys in succession. Enter `U1` into the resulting *Component Designator* dialog as shown in Figure 163 below.

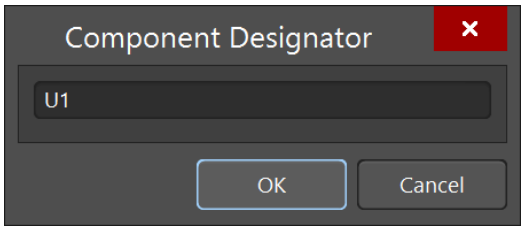


Figure 163. Jump to Component dialog

7. Press **OK** and the cursor will jump to the component `U1`, a zoom and pan will occur only if necessary to reach the location of the component, in the event it is currently outside the visible area.

14.4 Cross Probe Mode

8. Keep the PCB Document `Bluetooth_Sentinel.PrjPcb` open.
9. Navigate back to the *Projects* panel and open the `3-Axis_Accelerometer.SchDoc`.
10. Select from the menu **Tools » Cross Probe;** the mouse cursor should now be a crosshair.
11. Select component `U1`. The PCB should briefly pop up before returning to the schematic.
12. Press **ESC** to stop the cross probe mode.
13. Switch to the PCB Workspace to see footprint `LGA16-3X3X1_L` now masked.
14. If you are wondering why U1 is selected in the schematic – and cross-probe is selecting U3 in the PCB, select the '*Compiled Sheets'* tab of the schematic and you'll see it's U3 (device sheet says U1), as shown in Figure 164.

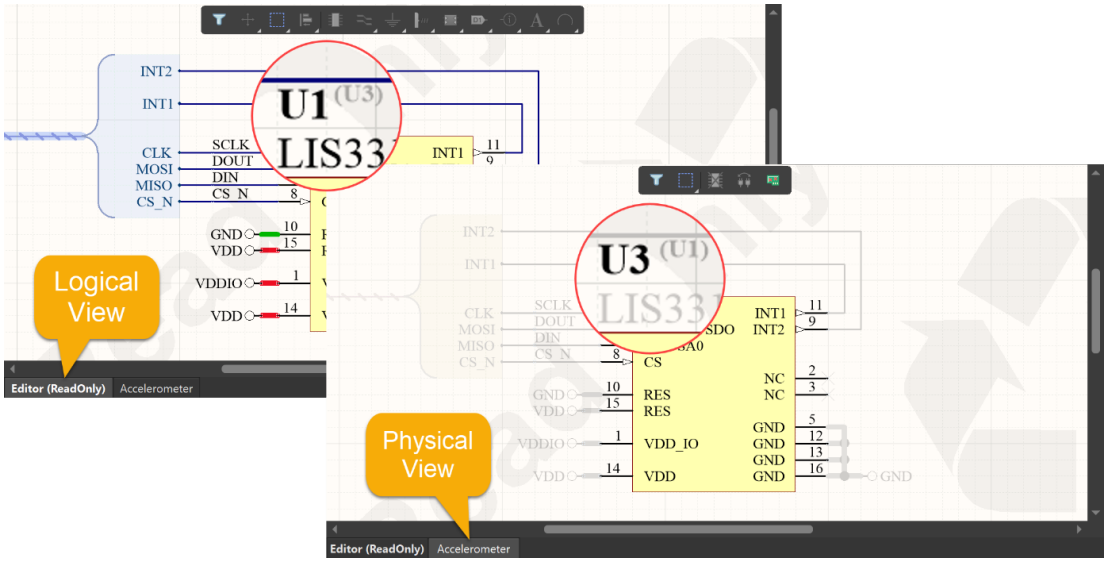


Figure 164. Logical and Physical View for a Schematic document

15. Press **Shift+C** to clear the mask.

For more information on **Device Sheets** and how they work, please refer to our online documentation. This exercise is only used to demonstration navigation techniques in the PCB.

14.5 Cross Select Mode

Tools » Cross Select Mode allows a PCB component to be selected directly from its schematic symbol counterpart. Used in conjunction with the PCB's **Tools » Component Placement » Reposition Selected Components** command, it enables a very powerful and effective way to go about component placement.

16. Verify that cross-select mode is enabled within the preferences.
    a) Open the Preferences, in the upper right corner of Altium Designer, and head to the *System branch,* and the *Navigation page*.
    b) Verify the **Cross Selection** checkbox is enabled. The *Cross Selection Mode* section also controls the selection scope.
    c) Click **OK** to close the preferences dialog.
17. To toggle the Cross Selection Mode from the menu use **Tools » Cross Select Mode**. If the icon has a blue outline around it, then Cross Select Mode in enabled as shown in Figure 165 below.

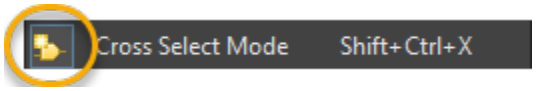


Figure 165. Cross Select mode enabled

18. Navigate to the *Projects* panel and double-click on `Microcontroller_STM32F101.SchDoc` to open the schematic document.
19. Select component `U2A`
20. Switch back to the `Bluetooth_Sentinel.PcbDoc` PCB document to see the `VFQFPN36_N` footprint selected.

14.6 3D Layout Mode

21. While still being in the active PCB document, select **View » 3D Layout Mode** or press the '**3**' key to enter 3D mode.

Numbers on the numerical keypad won't work,
use the ones above the qwerty keys.

a) Press the '**8'** key to change the 3D view to "*Orthogonal Rotation*" view.
b) Press the '**9'** key or '**0'** key for "*90 Degree Rotation*" or "*Zero Rotation*".
c) Press and hold the **Shift** key until you see an orb as shown in Figure 166 below.

Figure 166. Orb to navigate in the 3D environment

d) While still **holding Shift, press and hold** the **right mouse button** on one of the navigation objects in the orb shown in Figure 167 below (arrow, circle, dot) and now move the mouse to change the 3D view.

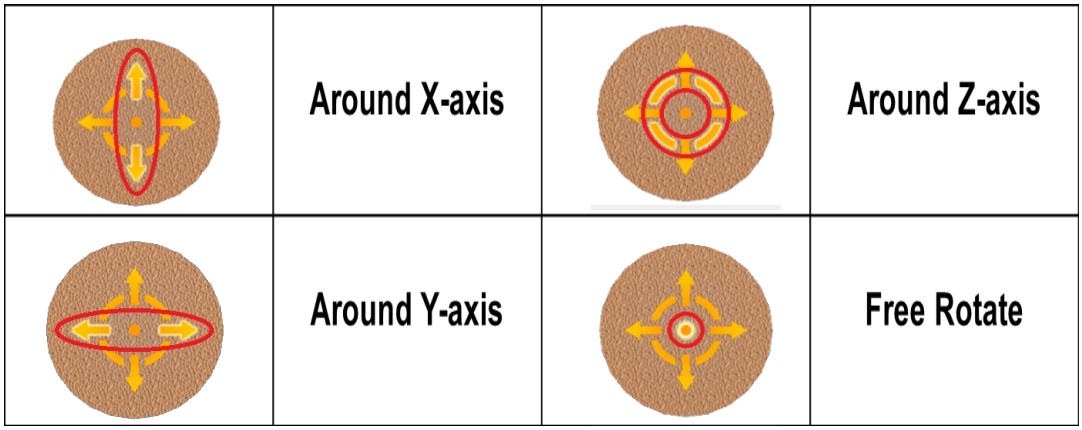


Figure 167. Orb controls for navigating in the 3D environment

22. Select **View » 2D Layout Mode** or press the '**2**' key to go back to the 2D view mode.

14.7 PCB Layers

The visible layer tabs will be displayed at the bottom of the workspace. These are the layers that are a part of your PCB design.

23. We can toggle through the different layers in multiple ways.
    a) Click the two small arrows beside the layer tabs to scroll through the tabs similar to what is shown in Figure 168.
    b) Click directly on different layer tabs change the current layer. When you're on a certain layer, this is what is called the Active layer.
    c) Press the "+", "-", or "*" shortcut key on the numerical keypad to change the active layer.
    d) Notice the Heads Up Display (HUD) in the top left corner will display the current layer name.

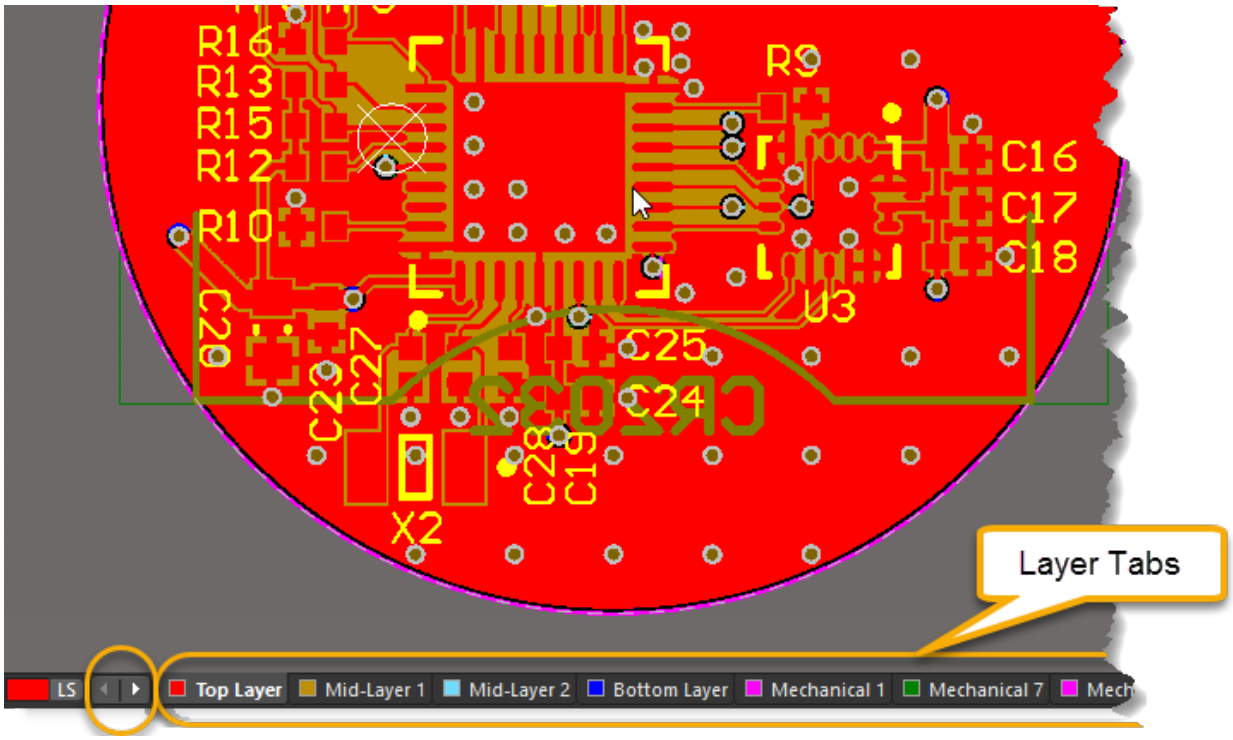


Figure 168. Layers tab showing all available layers

14.8 View Configurations

24. Open the *View Configuration* panel by clicking on the **Panels** button at the lower right side and selecting **View Configuration,** or the **L** shortcut key.
    a) The *View Configuration* panel with the tab *Layer & Colors* will open as shown in Figure 169 below.

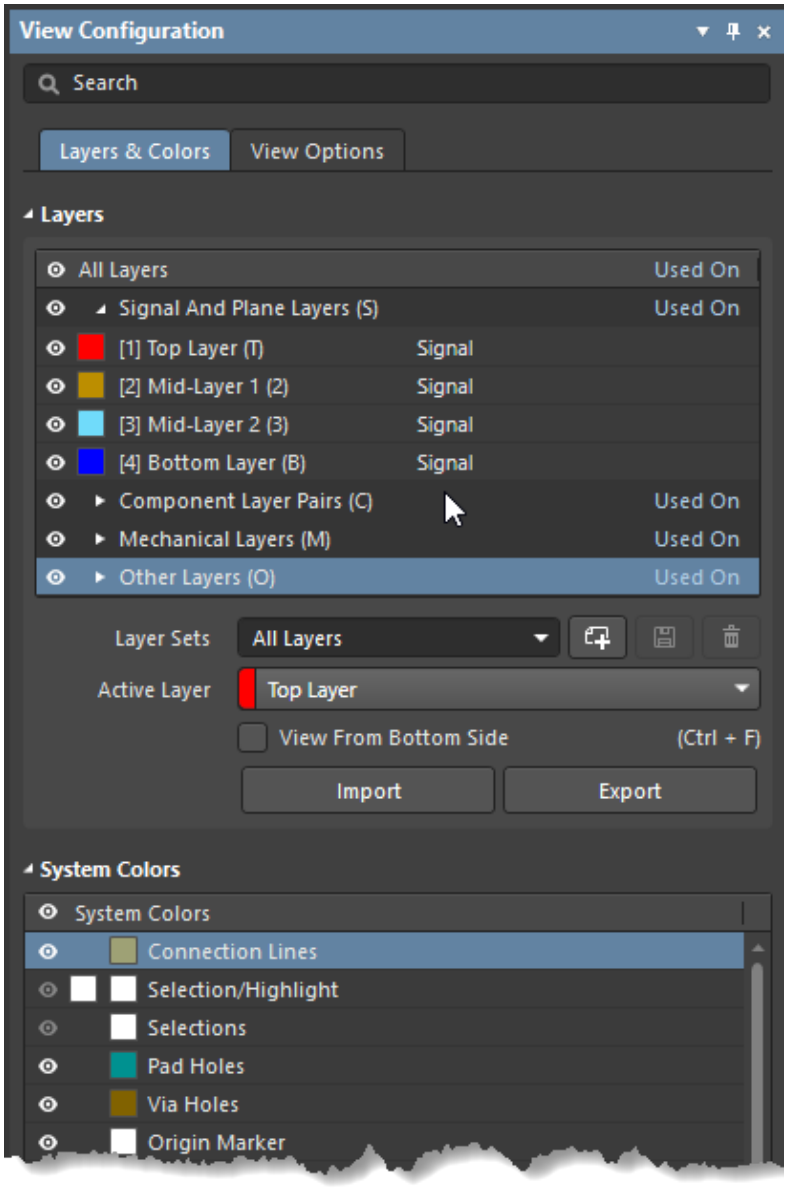


Figure 169. View Configuration panel

    b) Activate or deactivate the visibility for different layers with a click on the icon next to the layer color.
    c) Click on a color box to change the color of a particular layer.
    d) Select the *View Options* tab as shown in Figure 170 below.
    e) Under *Objects Visibility* click on the icon for polygons to hide the polygons.

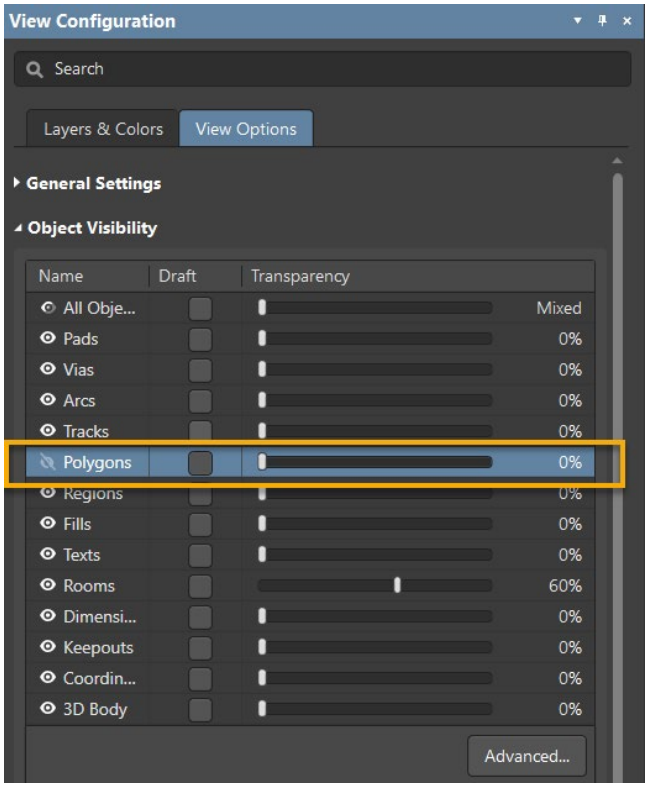


Figure 170. Object Visibility tab in the View Configuration panel

Although the polygons are not visible while hidden, they will still affect your design. They are still considered objects/obstacles for other features such as Interactive Routing. If you prefer to temporarily disregard them from your design, you can go to **Tools » Polygons Pours » Shelve Polygons.**

f) Under *Objects Visibility* click on the eye icon for Polygons to show the polygons again.

g) Move the slider on the right side to change the transparency for the polygon. You will also notice that you can manual change the % of transparency, beside the slider.

h) View the results in the PCB.

i) If you changed the color of a layer in Step *24.c)* above, change it back to its original color.

25. Experiment with other options and observe changes in the PCB view.

26. You can revert to the “Altium Standard 2D” configuration by selecting **Altium Standard 2D** from the *Configuration* drop-down list as shown in Figure 171 below.

Please note that changing the configuration back to Altium Standard 2D does not revert any changes that you’ve manually made to any layer colors.

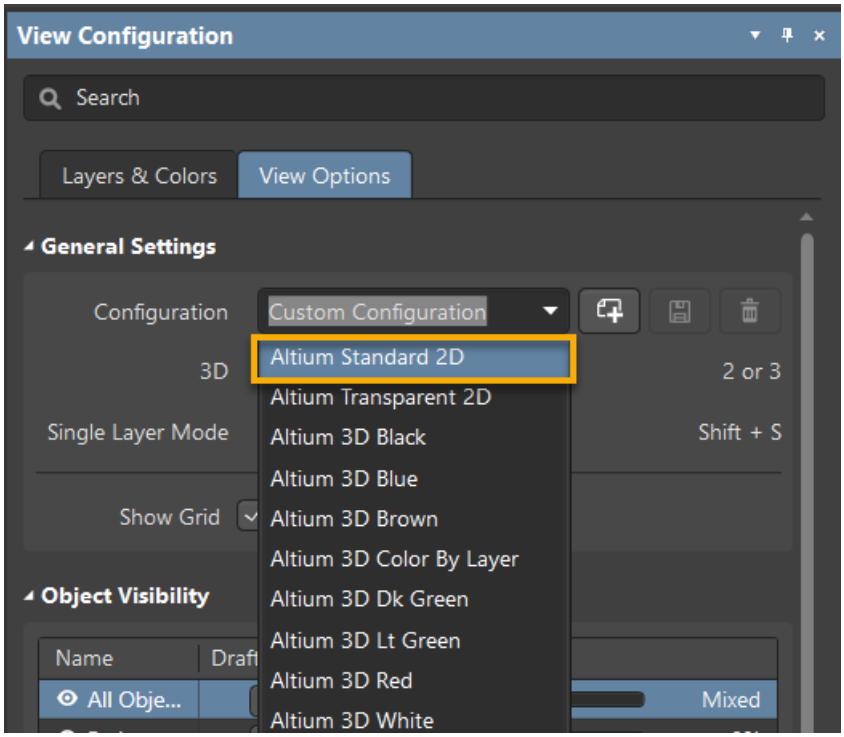


Figure 171. Different predefined Views

14.9 Board Insight Display

27. Open the *Preferences*, and head to the section *PCB Editor*:
    a) Select the *Board Insight Display* page.
    b) Enable all options under *Available Single Layer Modes* if not enabled already and press **OK**.
28. In the PCB document, toggle through the single layer modes by pressing **Shift + S**. Notice the differences between the modes as indicated in Figure 172, Figure 173,Figure 174 and Figure 175 on the following pages.

Notice in the HUD. It will show (Single) when you are in Single-Layer Mode. Single-Layer mode is a very useful viewing mode when you simply want to work on a specific layer and would like to hide the other layers that you're currently not working on.

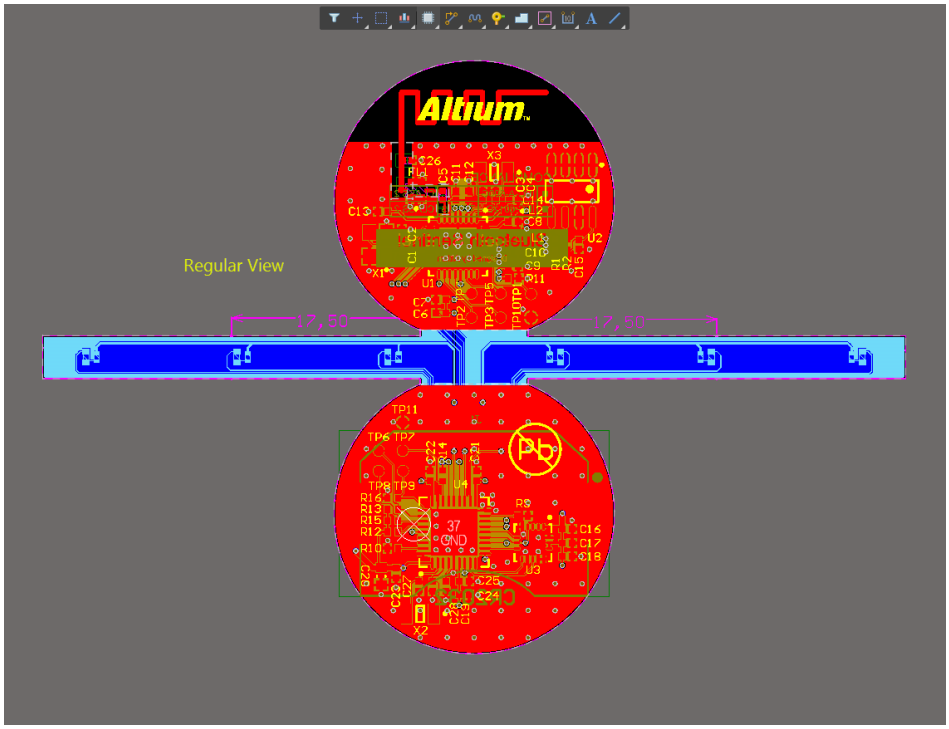


Figure 172. Regular 2D view

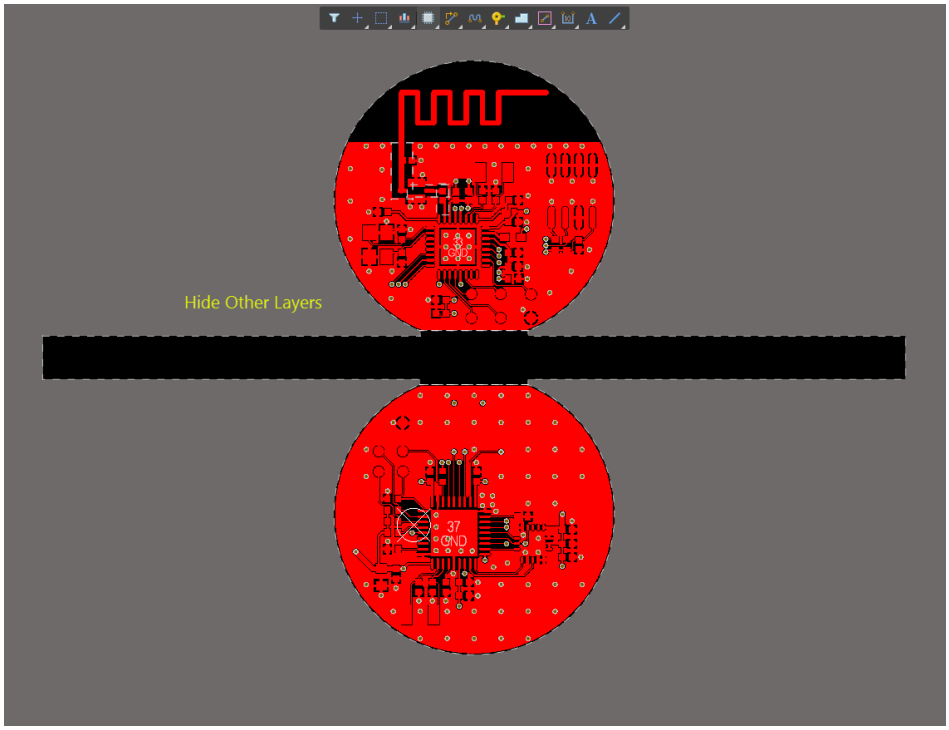


Figure 173. Single Layer Mode - Hide Other Layers

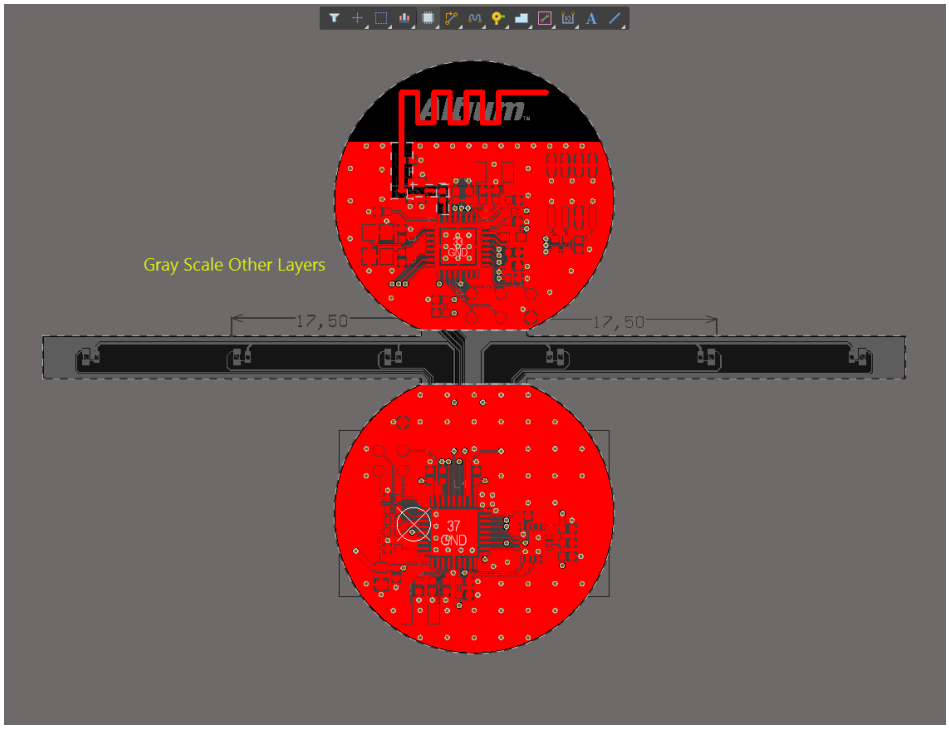


Figure 174. Single Layer Mode - Grayscale other layers

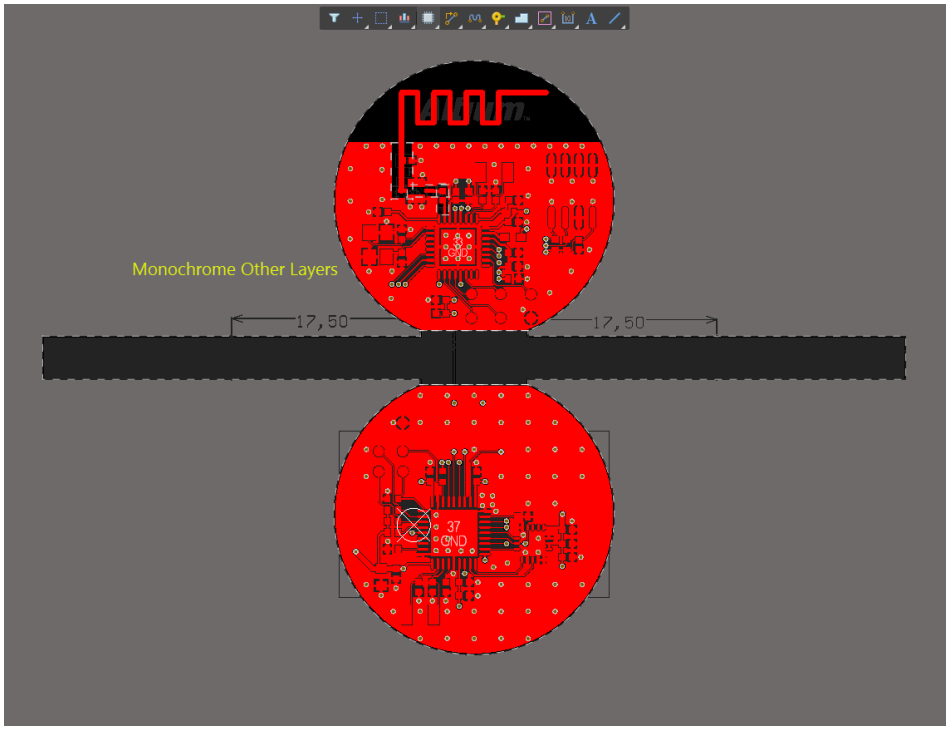


Figure 175. Single Layer Mode - Monochrome other layers

29. Return to the non-single layer mode using **Shift+S** until you see all the layers again.
30. Press **Shift+H** to toggle the HUD (Heads Up Display). We can also change the visibility in **Preferences » PCB Editor » Board Insight Modes » Display Heads Up Information**. Make sure the HUD is enabled when finished.

14.10 Display

31. Open the *Preferences* dialog and select the *Display* page from the *PCB editor* branch.
32. Under the *Highlighting Options* section:
    a) Enable all of the options.
    b) Press **OK** to exit the *Preferences* dialog.
33. Jump to component `C21` using the shortcut **J»C** and entering `C21` into Component Designator dialog.
34. If not already the active layer, change to the Top Layer as the active layer by selecting it from the Layer Tabs at the bottom of the Workspace. You may have to click the left or right arrows near the layer tabs to make it visible and select the Top Layer.
35. Enter Single Layer Mode by pressing **Shift+S.** All layers are now gray except the active layer (Top Layer).
36. Mask net `VBAT` by selecting Pad 2 from `C21`, or the via next to the Pad, using the command **Ctrl + left-click**. Notice that tracks connected to that Pad are now masked as well as shown in Figure 176 below. **Do Not** Clear the highlight.

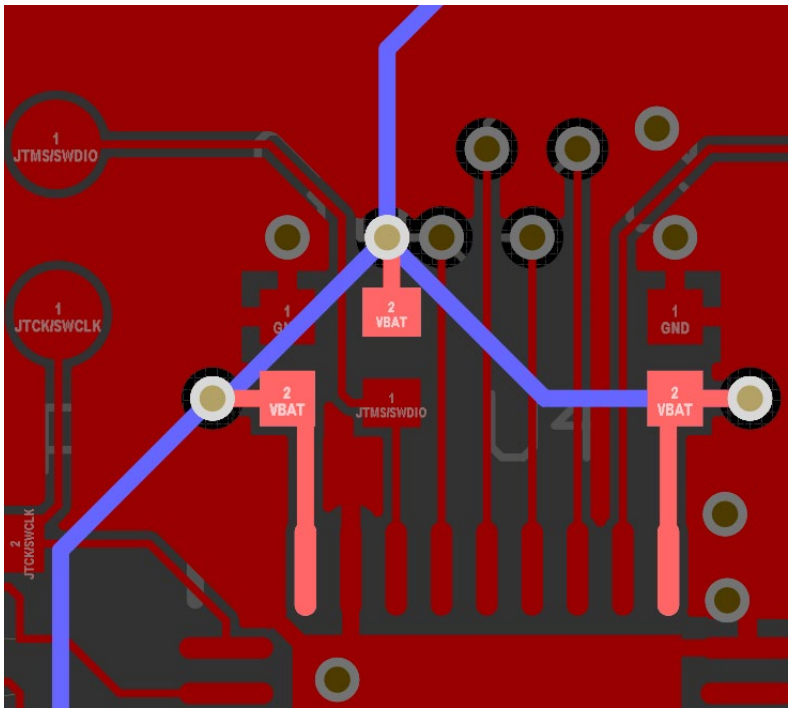


Figure 176. Highlight all primitives

37. Press **Shift+S** again**,** and now all other layers are hidden, instead of grey.
38. Open *Preferences* again, and head to *the Display* page under *PCB Editor*.
    a) Disable **Show All Primitives in Highlighted Nets**.
    b) Click **OK** to confirm. Notice that tracks on the Bottom Layer are no longer masked.
39. Clear selection by pressing **Shift + C**.
40. Exit single layer by executing **Shift + S** until all layers are showing.
41. Jump to component `U4` using the shortcut **J » C.**
42. Place a Route from `Pad U4-28` (Located in the upper left corner of the component) by selecting **Route » Interactive Routing**.
    a) Place the crosshair on `Pad U4-28` and select it. Notice that the pad and existing routing is now highlighted and the rest of the board is masked.
    b) Right-click twice to exit the routing mode.
43. Open the *Preferences* dialog- page *Display.*

a) Disable **Apply Mask During Interactive Editing** and disable **Apply Highlight During Interactive Editing.**

b) Click **OK** to return to the PCB.

44. In the PCB, the pad should no longer be highlighted.
45. Press **Shift+C** to clear the board.

Feel free to change the Preferences for Interactive Routing as you wish. These settings are based on personal preferences, and they will help you in the later modules of this course.

14.11 Board Insight Mode

46. Open the *Preferences,* section *PCB Editor*:
    a) Select the *Board Insight Modes* page.
    b) Ensure the *Display Heads Up Information* is active.
    c) Press **OK**.
47. Hover over component `U4` to see the Heads Up Display as shown in the top left corner as shown in Figure 177.

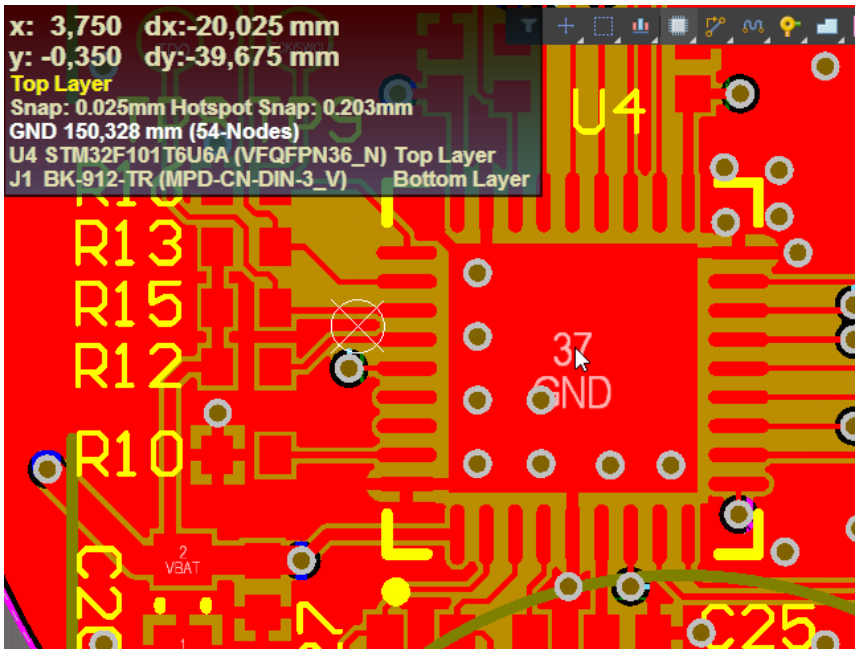


Figure 177. Heads Up Display

The *Heads-Up Display* can be disabled by pressing **Shift + H**

48. Close the project without saving.

Module 15: PCB Panels and Global Editing

15.1 Purpose

In this exercise, we'll explore the common panels used in the PCB environment such as the *Properties Panel, Filter* panel, *Find Similar Objects* and *List* panel *and PCB* panel.

Global Editing allows us to make a change to multiple objects at the same time. Global Editing tools are available in the schematic, schematic symbol editor, PCB editor, and footprint editors.

15.2 Preparation

1. Close any open PCB projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `AD20 Essentials Exercises\Module 15 - PCB Global Editing` folder.

15.3 Properties Panel

The *Properties* panel is used to modify properties for multiple objects easily. In this section, we will use the mouse and **Edit » Select** commands to select items then use the *Properties* panel to make changes in both the Schematic and PCB editors.

15.3.1 Modify PCB Traces

2. Open the `WCToppingGlobalEditing.PcbDoc` document and zoom in on `U1` as shown in Figure 178 below.

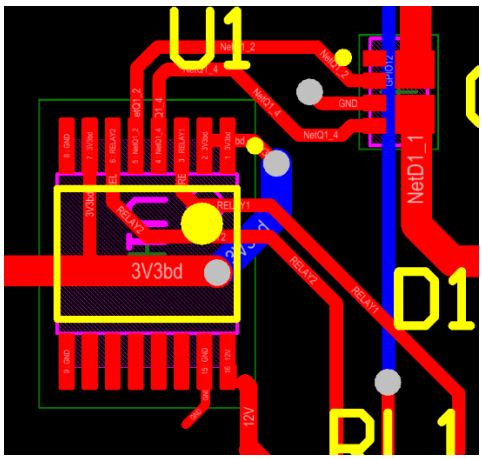


Figure 178. WCToppingGlobalEditing.PcbDoc near component U1

3. Ensure nothing is selected on the board. Then, open the *Properties* panel from the **Panel** button located in the bottom right corner of the editor. Ensure that the filter is set so that all the objects are highlighted in blue. Otherwise, click on **All-On**. The same pre-selection filter can also be accessed from the *Active Bar* as shown in Figure 179 below.

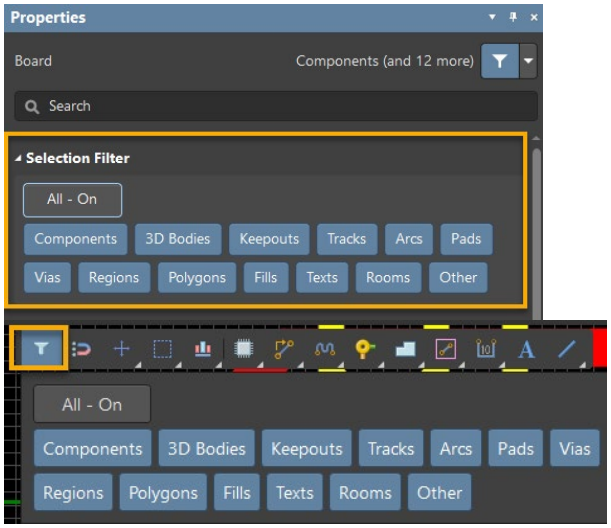


Figure 179. Selection filter with the scope set to "All-On"

4. Select multiple irregular traces quickly by using the **Edit » Select » Connected Copper** command, or the hotkeys **Ctrl+H** then hold **Shift** while clicking on the four traces, or by pressing the **Tab** button with a single trace selected to toggle the selection options, as shown in Figure 180 below. Right-click or hit **Esc** to terminate the command.

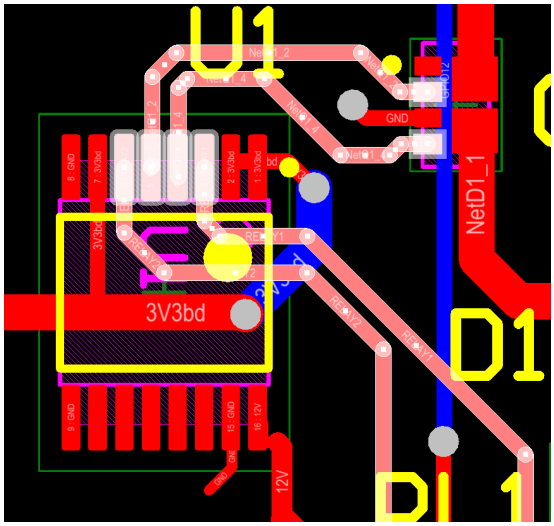


Figure 180. Selecting multiple traces with the Shift key and "Connected Copper" command

5. We now would like to change the widths of the selected tracks to 10mil. The *Properties* panel contains parameters and values of the selected objects. However, the visible values are represented by '*' since there are different values for the selected objects.
   a) As of right now, both tracks and pads are currently selected. Let's change the post-selection filter to only show properties of tracks by clicking on the down arrow button at the top-right of the panel (refer to Figure 181).
   b) Click on **All-Objects**, then ensure to only click **Track** from the post-selection filter (refer to Figure 181).

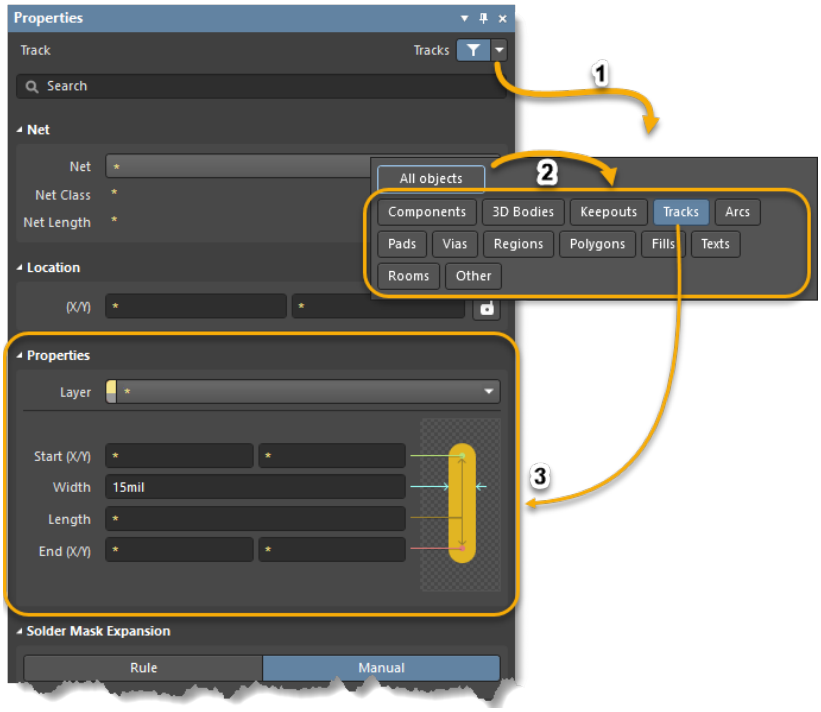


Figure 181. Properties panel with the post-selection filter set to Tracks only

6. In the *Properties* panel, locate the Width field and change the value from `15mil` to `10mil`. Hit **Enter**. The modified layout is then shown in Figure 182 below.

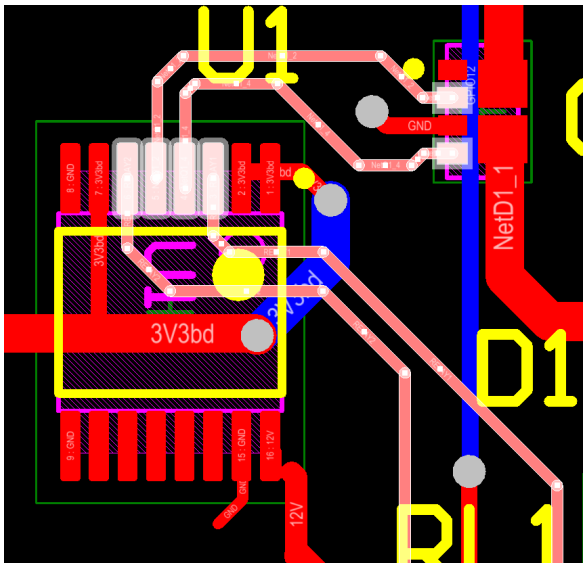


Figure 182. Board Layout with modified traces

In this exercise, we use the Global Editing workflow to modify the trace width. An excellent alternative is to re-apply the preferred width and clearance rules of an existing route using the command **Route » Retrace Selected.**

15.4 Filter Panel

The *PCB Filter* panel is another method to select items based on a query language. In the examples in this section, we will once again use the *Properties* panel to make changes after the appropriate elements have been selected by the *PCB Filter* panel.

15.4.1 Object Checkboxes

7. We will first familiarize ourselves with the checkbox options within the *PCB Filter* panel. Open the *PCB Filter* panel accessible via the **Panels** button in the bottom right corner of the editor.
8. Enable the **Track** checkboxes associated with **Net** and **Component** objects as well as the **Via** checkbox for **Net** and **Component** object as shown in Figure 183. Notice that the *Filter* field is populated with a query based on the combination of checkboxes you have selected.

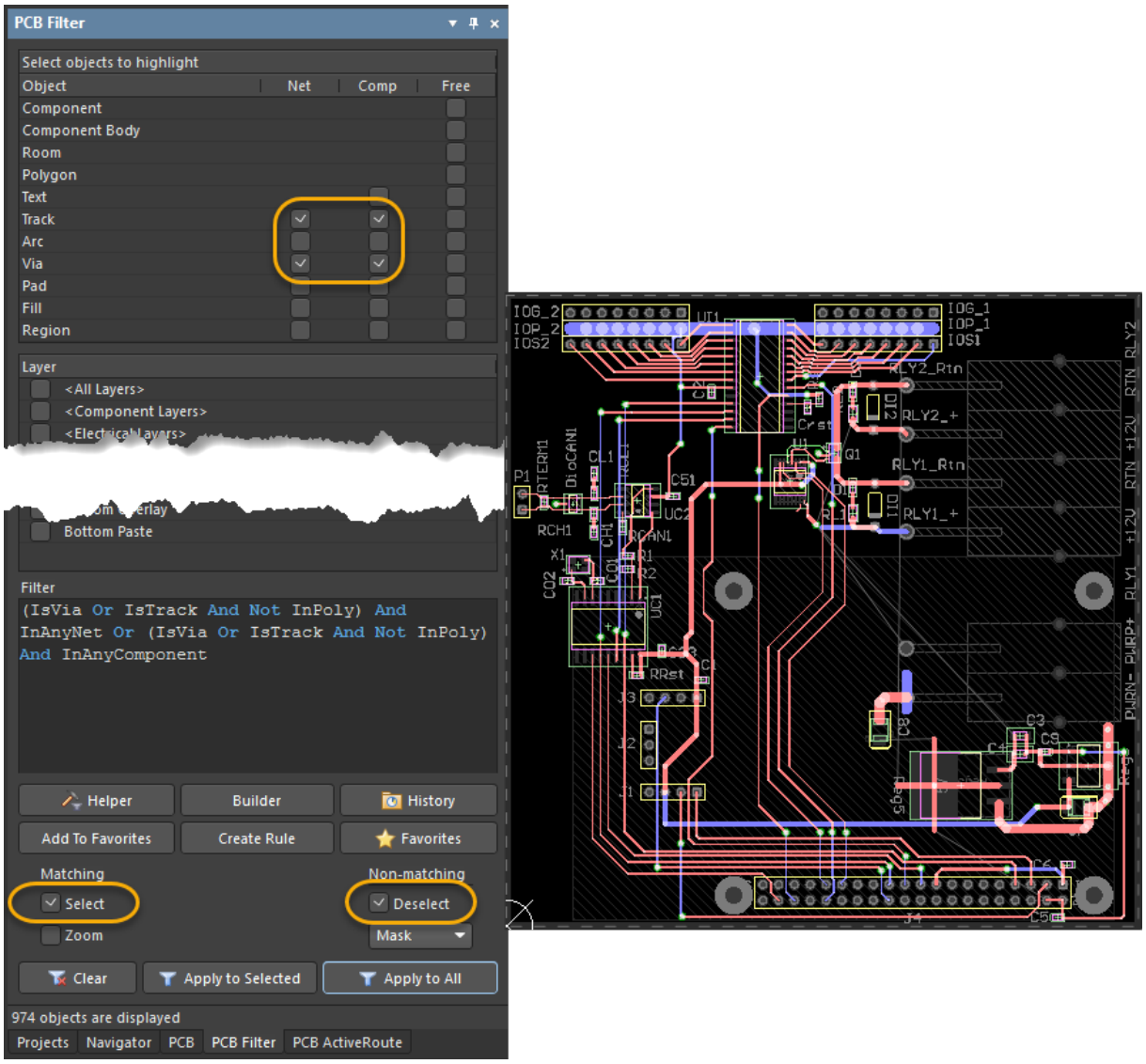


Figure 183. PCB Filter panel selections reflected in the PCB

9. Make sure that at the bottom of the *PCB Filter* panel you have *Matching* set to **Select**, *Non-matching* with **Deselect** enabled and with the mode set to **Mask**. Ensure that the filter in the *Properties* panel is set to **All objects.**
10. Click the **Apply to All** button in the bottom right corner of the panel to see the effect of the query to the design. You should have tracks highlighted as shown in Figure 183.
11. Disable the **Track on Comp** checkbox hit **Apply to All** again. Notice that the component outlines have now been excluded from the selection on the board.
12. Enable the **Top Layer** checkbox in the *Layer* pane then click **Apply to All**. Only tracks on the Top Layer will be highlighted.
13. Click the **Clear** button to reset the selection on the board.

14. Try enabling different combinations of the checkboxes then click **Apply to All** to see which objects are selected.
15. Right-click anywhere in the top pane of the *PCB Filter* panel and choose the **Uncheck All** checkbox to reset the query back to `Not All`. Uncheck any selections in the Layer pane as well.

15.4.2 Query Selection

16. In this section, we'll build a query using the *PCB Filter* panel to select components having designators starting with character `C`, and move them from the top to bottom layer.
17. To simplify the view, we will temporarily hide the tracks and vias. First, we will open the *View Configuration* panel by going to **View » Panels » View Configuration**. Select the *View Options* tab and set the *Tracks* and *Vias* to hidden as shown in Figure 184 below.

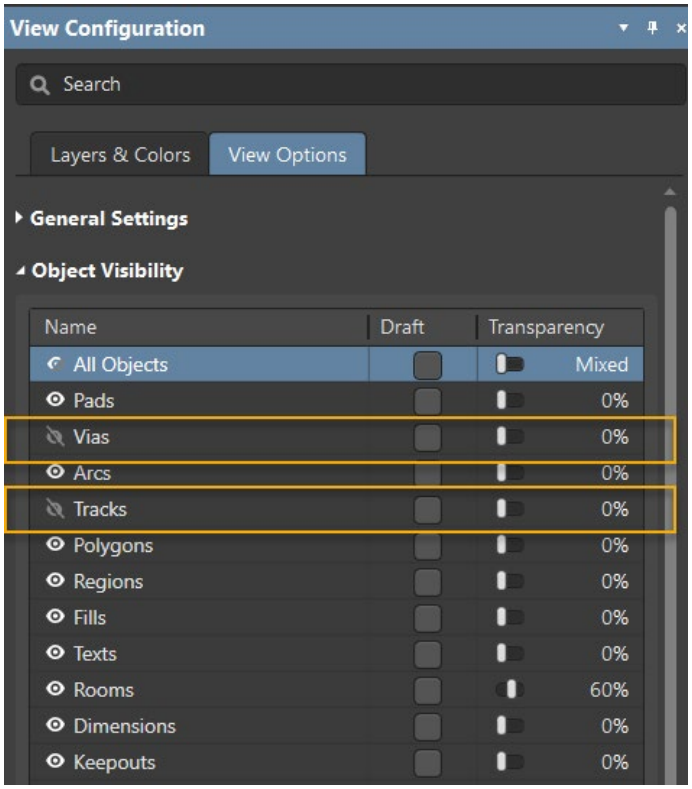


Figure 184. Hiding Tracks and Vias from the Properties panel

18. Open the *PCB Filter* panel and ensure that the "*Filter*" pane is set to (`Not All`.)
19. Enable the **Component "Free"** checkbox. `IsComponent` should appear in the filter pane. Click **Apply to All** to select all the components in the PCB.
20. Add the following text to the *Filter* pane: `and (Name like 'C*')` as shown in Figure 185 below then apply the changes. This naming syntax is important.
    The asterisk is a wildcard identifier. With this filter, any component that has a name beginning with the letter `'C'` will be selected. The selected components should now only be components with designators starting with letter "`C`" as shown in Figure 185 below.

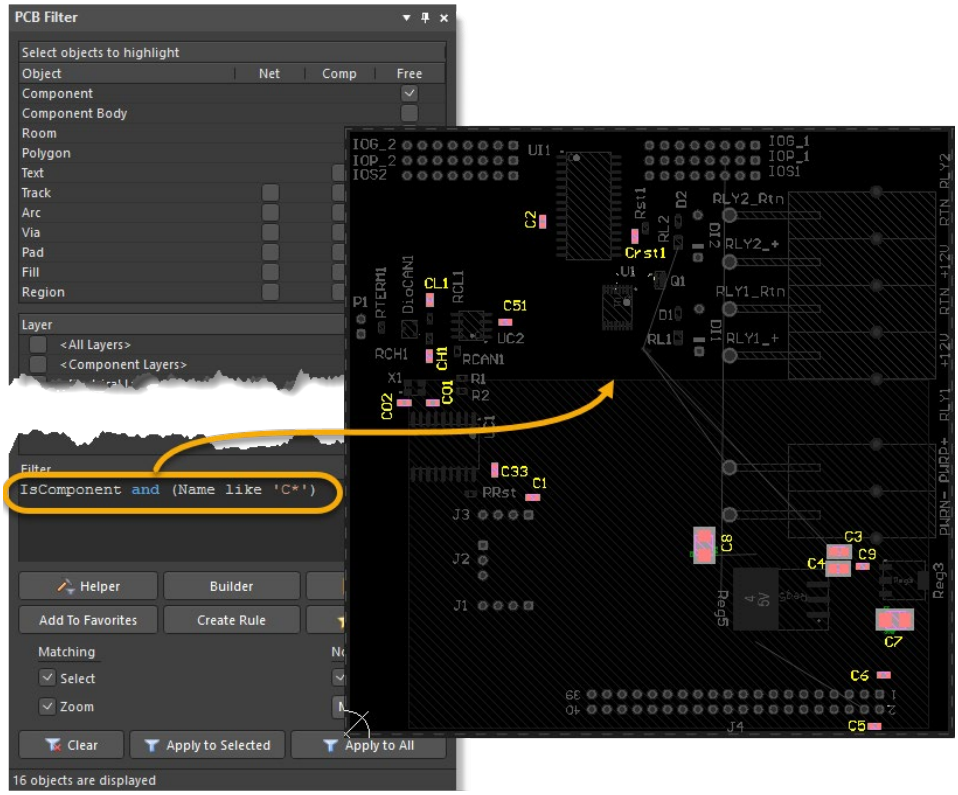


Figure 185. Components with designators starting with the letter "C" selected via the PCB Filter panel

21. Now modify the query from `'C*'` to `'C?'`.
22. After you apply the change, only components with a single character following the letter `'C'` will be selected.
23. Now that the nine capacitors are selected, open the *Properties* panel via the **Panels** button in the bottom right corner of the editor. Feel free to close the *PCB Filter* panel if desired.

24. Ensure that the filter option at the top is configured to show **Components**, as shown in Figure 186 below or else you will not see any information about the Components in the *Properties* panel.

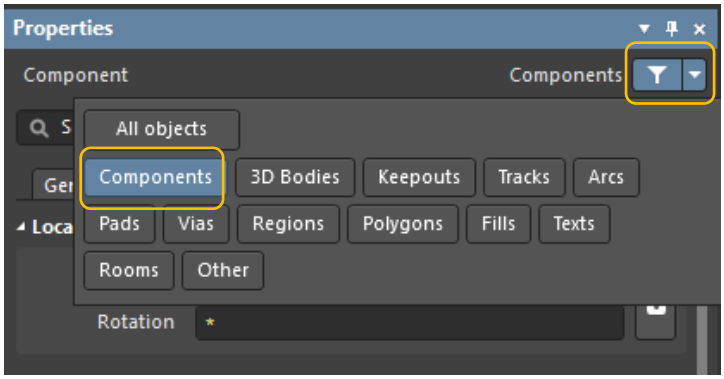


Figure 186. Filter to only Components

25. Then, locate the *Layer* attribute and change the value to **Bottom Layer**. The selected capacitors should now be blue indicating they have been flipped to the back side of the board as shown in Figure 187 below.

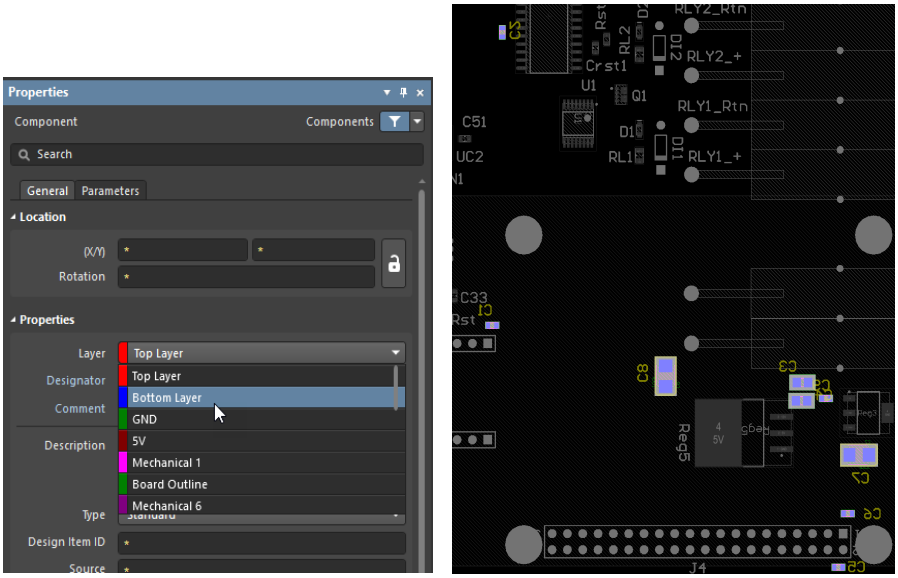


Figure 187. Nine capacitors flipped to the Bottom Layer

26. While the nine capacitors remain selected, change their layer back to **Top Layer** in the *Properties* panel.

15.5 Find Similar Objects

The *Find Similar Object (FSO)* dialog is another powerful way to select multiple items in either the schematic or PCB environments. It allows you to choose a particular element in the design and then identify one or more common parameters that will be used as the basis for global selection.

15.5.1 PCB Find Similar Objects Dialog

15.5.1.1 Selecting Vias

27. Let's first unhide the tracks and vias by going to the *View Configuration* panel (**View » Panels » View Configuration).** Open the *View Options* tab and activate the **Tracks** and **Vias**. We will now select vias of a specific size on this board and change their sizes.
28. Use **Shift+C** to verify that there are no objects currently selected or masked.
29. Select the via located near `UI1-14` and right-click to *Find Similar Objects* as shown in Figure 188 below.

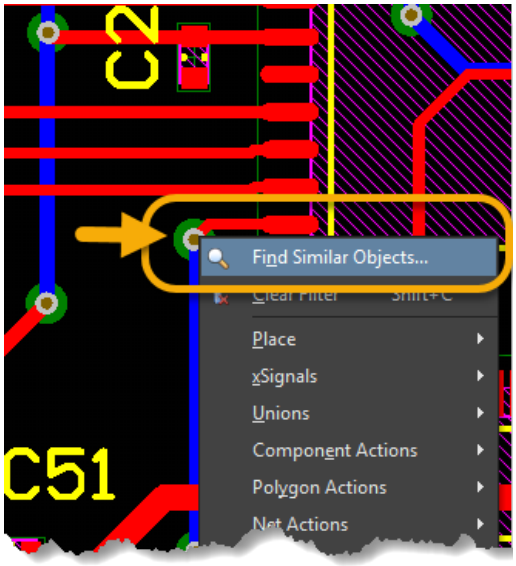


Figure 188. Right-click and select Find Similar Objects

30. The *FSO* dialog will appear, displaying via parameters as shown in Figure 189 below. Note that by default the *Object Kind* is set to a scope of **Same**. Enable **Mask** in the scope settings to mask the rest of the board.

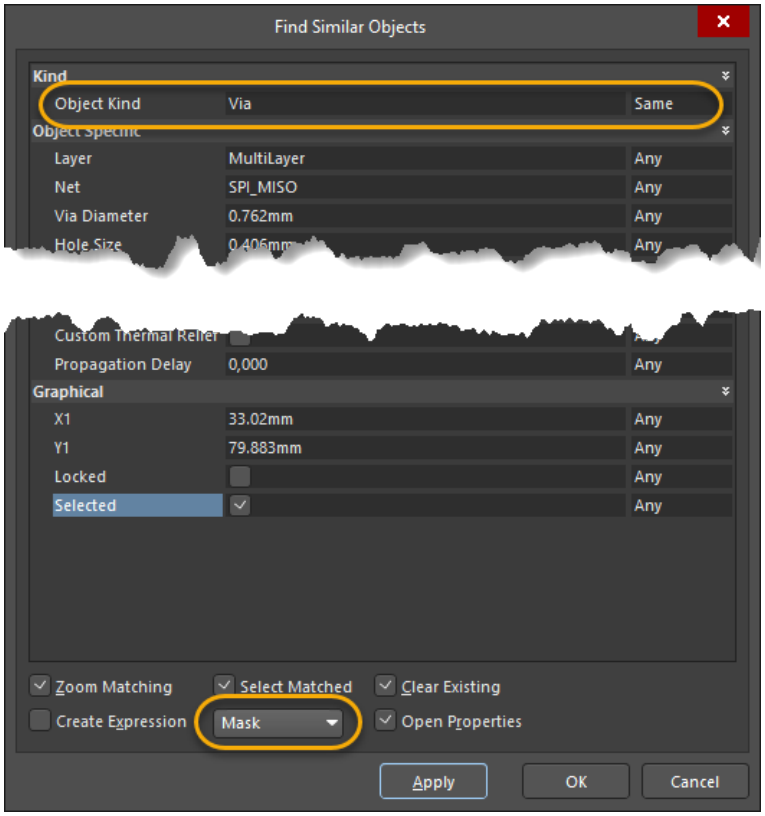


Figure 189. FSO for the selected Via

31. Click the **Apply** button to select all vias in the design as shown in Figure 190 below.

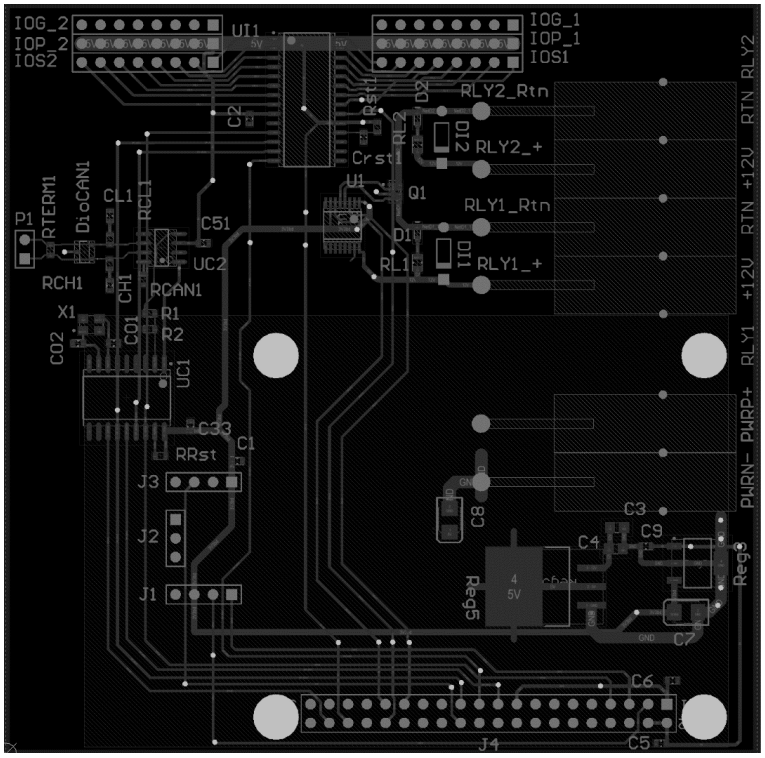


Figure 190. All Vias selected using the Find Similar Objects dialog

15.5.1.2 Changing Via Size

32. Instead of all vias, we only would like to change the via diameter of the 30mil vias. In the *Find Similar Objects* dialog, set the column of the *Via Diameter* parameter as shown in Figure 191 below.

33. Click the **Apply** button. Now only the vias of 30mil diameter are selected.

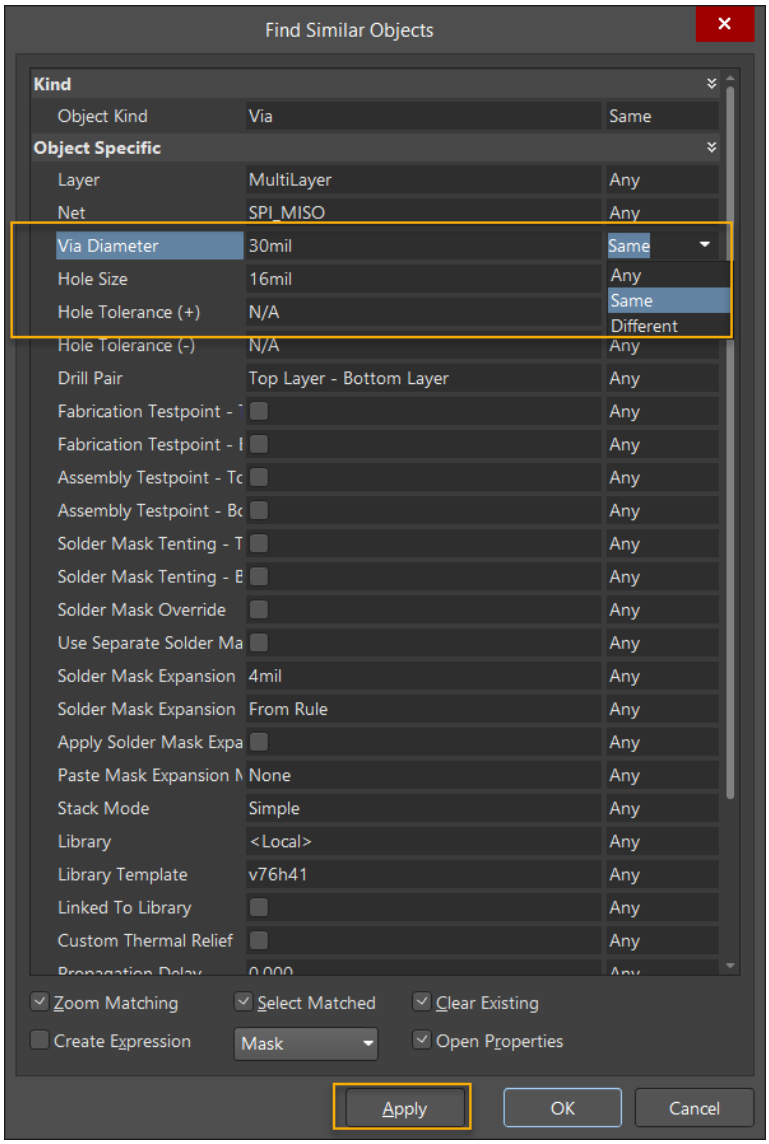


Figure 191. Find Similar Objects dialog to target specific Vias

34. In the *FSO* dialog click the **OK** button to close the dialog.
35. Change to the *Properties* panel.

36. Change the selection filter for the *Properties* panel to **Vias** as shown in Figure 192 below.
    a) Locate the *Via Diameter* field and change the value from `30mil` to `20mil`.
    b) Change *Hole Size* from `16mil` to `10mil`.
    c) Hit **Enter** and see the changes made on the selected vias.

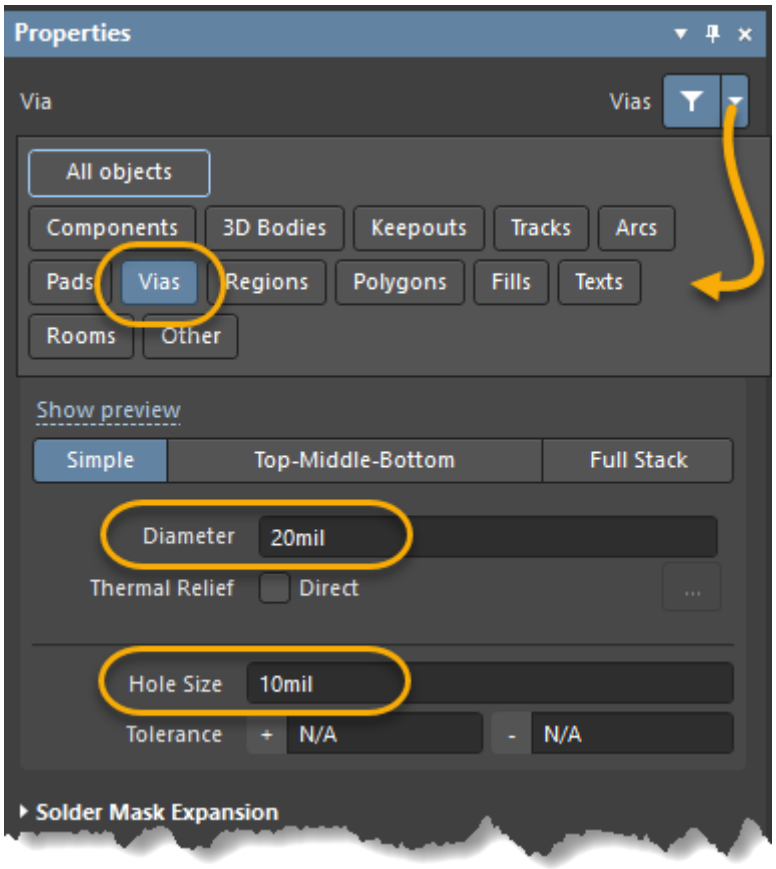


Figure 192. Properties panel with the new values for Via Diameter and Hole Size

37. Hit **Shift-C** to return to standard board view.

15.6 List Panel

The *PCB List* panel displays items from your design in a tabular format and is used for both selecting and editing.

15.6.1 Selecting Components

38. We will now view attributes of selected components in a tabular form and lock their positions, using the *PCB List* panel. **Open** the *PCB List* panel through the **Panels** access button in the bottom right corner of the editor.
39. Change the list settings at the top of the panel to match those shown in Figure 193. Note that all objects are included in the *PCB List* panel.

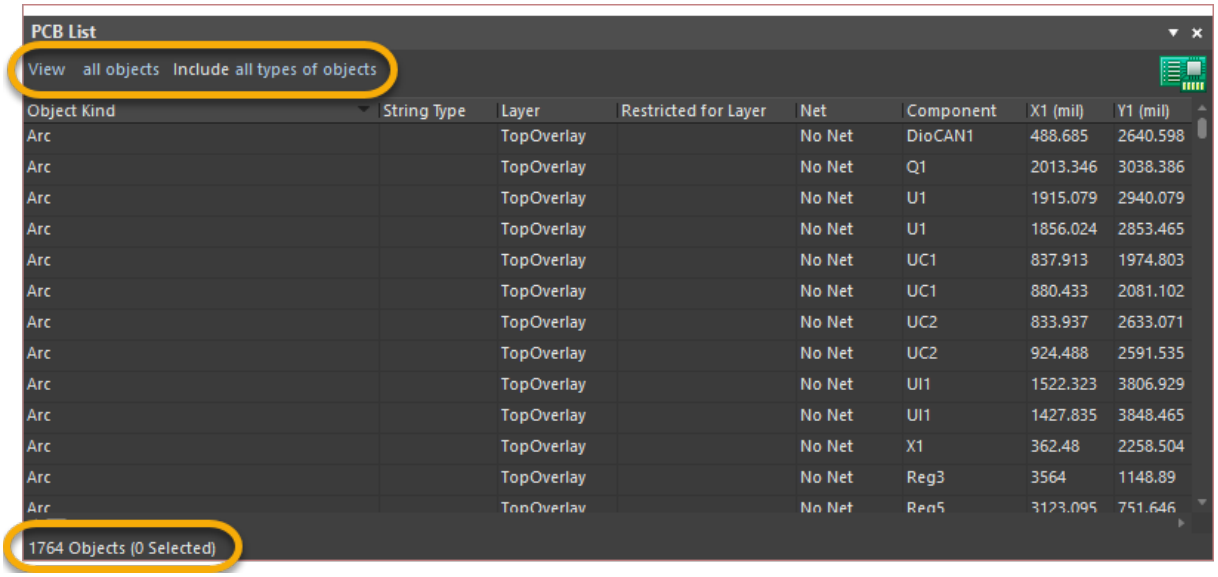


Figure 193. PCB List panel showing All Objects in the design

40. Open `Power Supply.SchDoc`.
41. Ensure **Cross Select Mode** is turned on, indicated by its blue highlight in **Tools » Cross Select Mode** (as shown in Figure 194 below)**.** If it's not enabled, click the icon to enable it.

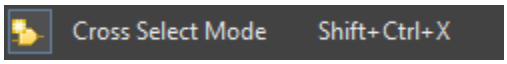


Figure 194. Cross Select mode enabled

42. Hold the **Shift** key to select all of the capacitors on `Power Supply.SchDoc`. Return to the PCB document and the five components should now be selected in the bottom right corner of the PCB.
43. In the *PCB List* panel, click on the "**View**" name hyperlink**,** and change it to **Edit** and change the "**all objects"** hyperlink to **selected objects.** The *PCB List* panel will now display only the selected components as shown in Figure 195 below.

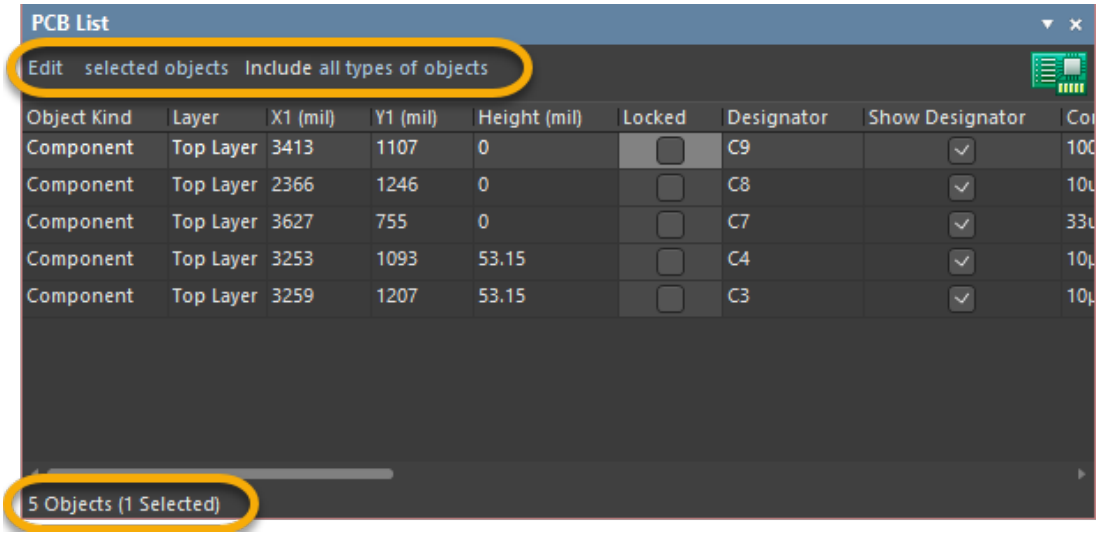


Figure 195. PCB List panel showing 5 components selected in the PCB.

15.6.2 Modifying Properties

Our objective is to lock all of these selected components on the board. We will do this by changing the *Locked* column in the *PCB List* panel.

44. Enable the **Locked** checkbox for the first component in the *PCB List* panel.
45. Click on the cell that you just locked and copy it with the **Ctrl+C** keys.

46. Select the locked cells for all the remaining four components by clicking on the first unchecked one, hold the **Shift** key then click on the last cell as shown in Figure 196 below.

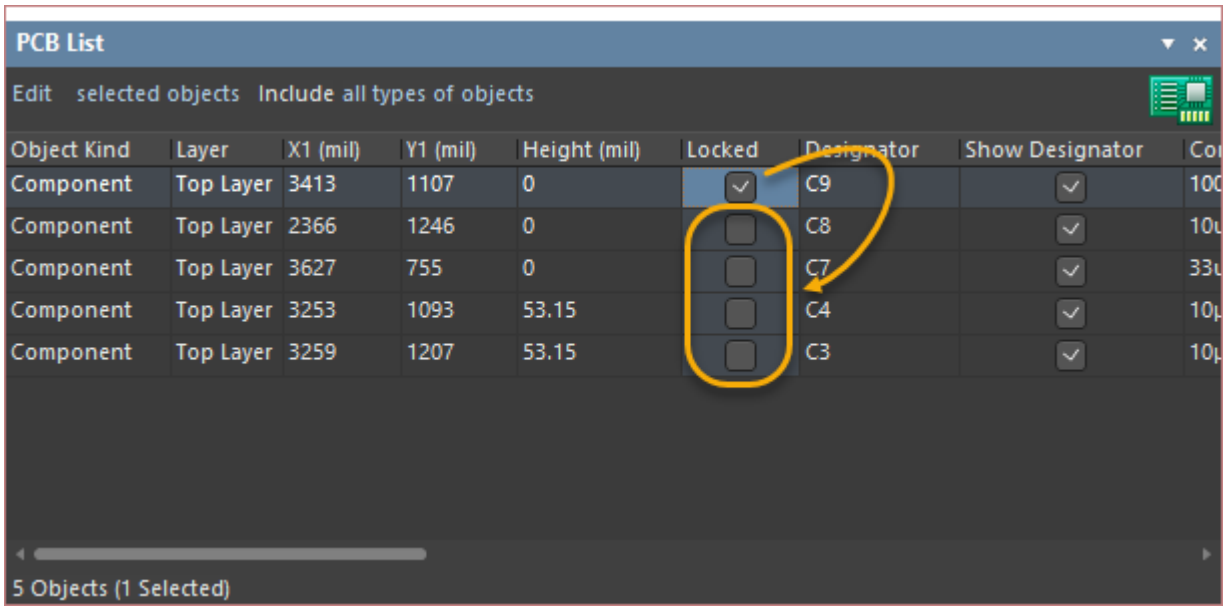


Figure 196. Copy & Paste to quickly change multiple rows in the PCB List panel at once

47. Use the **Ctrl+V** keys to paste the locked property quickly as shown in Figure 196 above.

The same method can be used to copy and paste multiple parameters in the *PCB List* panel at the same time.

48. Try to move any of the five components you just locked to confirm they will not move. Save and close all files.

Consider the circumstances where these batch replacement methods would help you with your designs.

Module 16: PCB Grids

16.1 Purpose

In this exercise, the configuration and the functionality for the Cartesian and Polar Grid will be reviewed. Both grids can be used as a local grid within the PCB editor for a specific document.

Designers need to precisely place and align components during PCB design and routing which is extremely difficult without the help of grids systems. Altium Designer supports multiple user-defined grids in both Cartesian and Polar forms, making component placement predictable and easy.

16.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `AD20 Essentials Exercises\Module 16 - PCB Grids` folder.
2. Open `WCTopping.PrjPcb` project file from the `Module 16 - PCB Grids` folder.
3. Open `WCTopping_Grid.PcbDoc`.

16.3 Predefined Default Grids

4. Press key **G** to open a list of predefined grid values.
5. Select one of the predefined grid values. The visible grid will change in the PCB, and the selected value will show in the Status Bar and the Heads-Up Information as shown in Figure 197 below.

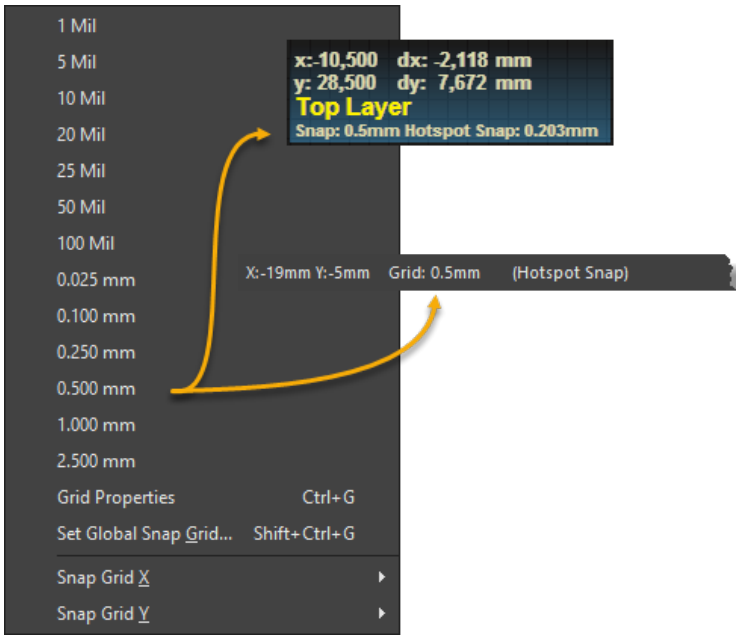


Figure 197. Grid selection dialog

16.4 Grid Manager

You can make your grid choice and tailor the settings to suit both your personal placement preferences and the immediate design requirements.

16.4.1 Global Board Snap Grid

6. Go to **Design » Import Changes from WCTopping.PrjPcb** and **Execute Changes.**

If some of the footprints fail to transfer to the PCB, refer to Module 11:Transfer to PCB to help you resolve those issues. If the footprints transferred successfully, you can continue with the next step.

7. Open the *Properties* panel by clicking **View » Panels » Properties.** The grid configuration can be seen inside the *Properties* panel, as shown in Figure 198 below. Close the other panes from the *Properties* panel or scroll down to the grid pane if needed to see the relevant information.

You can change your units from *mm* to *mil* using the **Q** hotkey, or **CTRL+Q** if you're in an action or panel.

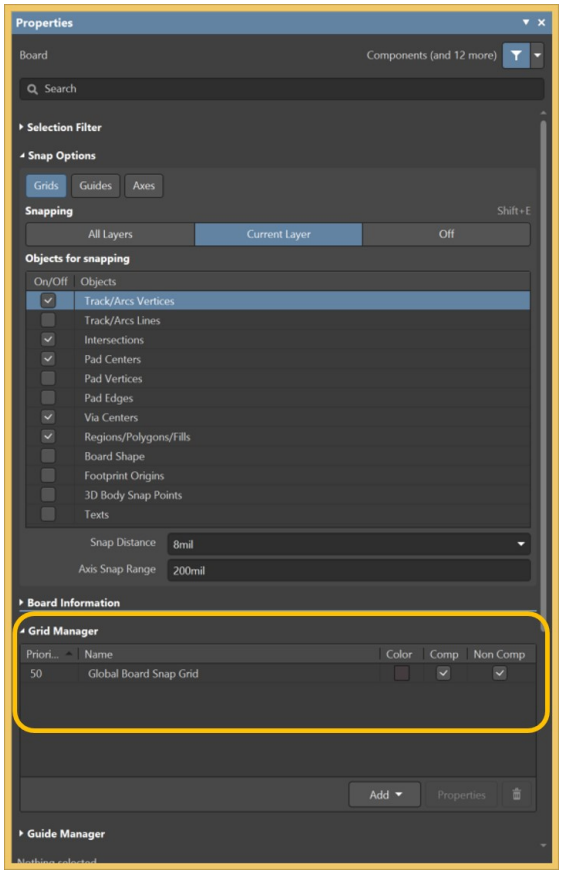


Figure 198. Properties panel with the Grid Manager

8. Double-click on the *Global Board Snap Grid* line in the *Grid Manager* pane to access parameters for the default grid as shown in Figure 199 below.

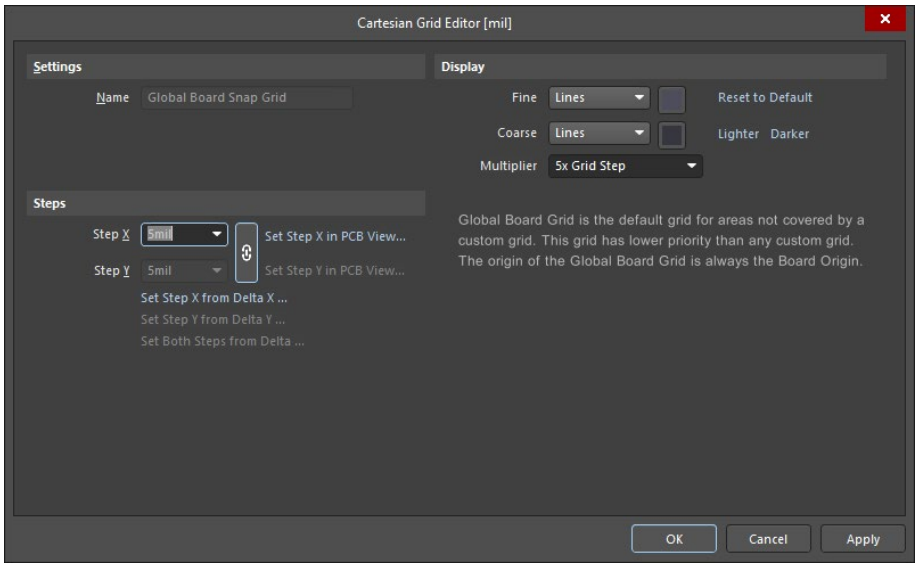


Figure 199. Default Grid Settings

You can access the Cartesian Grid Editor using the **Ctrl+G** hotkeys.

9. Change the *Step X* size to `10mil`, the *Multiplier* to `10x Grid Step` and the drawing configuration from `Dots` to `Lines`.

The *Multiplier* determines how many *Fine* units will be contained in a *Coarse* grid unit. One *Fine* unit is the value of the *Step* entered in the *Cartesian Grid Editor* dialog.

10. Click **OK** to accept changes.
11. Examine the new grid to see the changes as shown in Figure 200 below. Notice that there are 10 fine grids to each coarse grid and the smallest division is 10mil.

Figure 200. Modified default grid setting

16.4.2 Custom Cartesian Grid

12. Open the *Properties* panel.
13. Click the **Add** button in the *Grid Manager* dialog and select **Add Cartesian Grid**.
14. A grid named `New Cartesian Grid` will be added to the list in the *Grid Manager*, as shown in Figure 201 below.

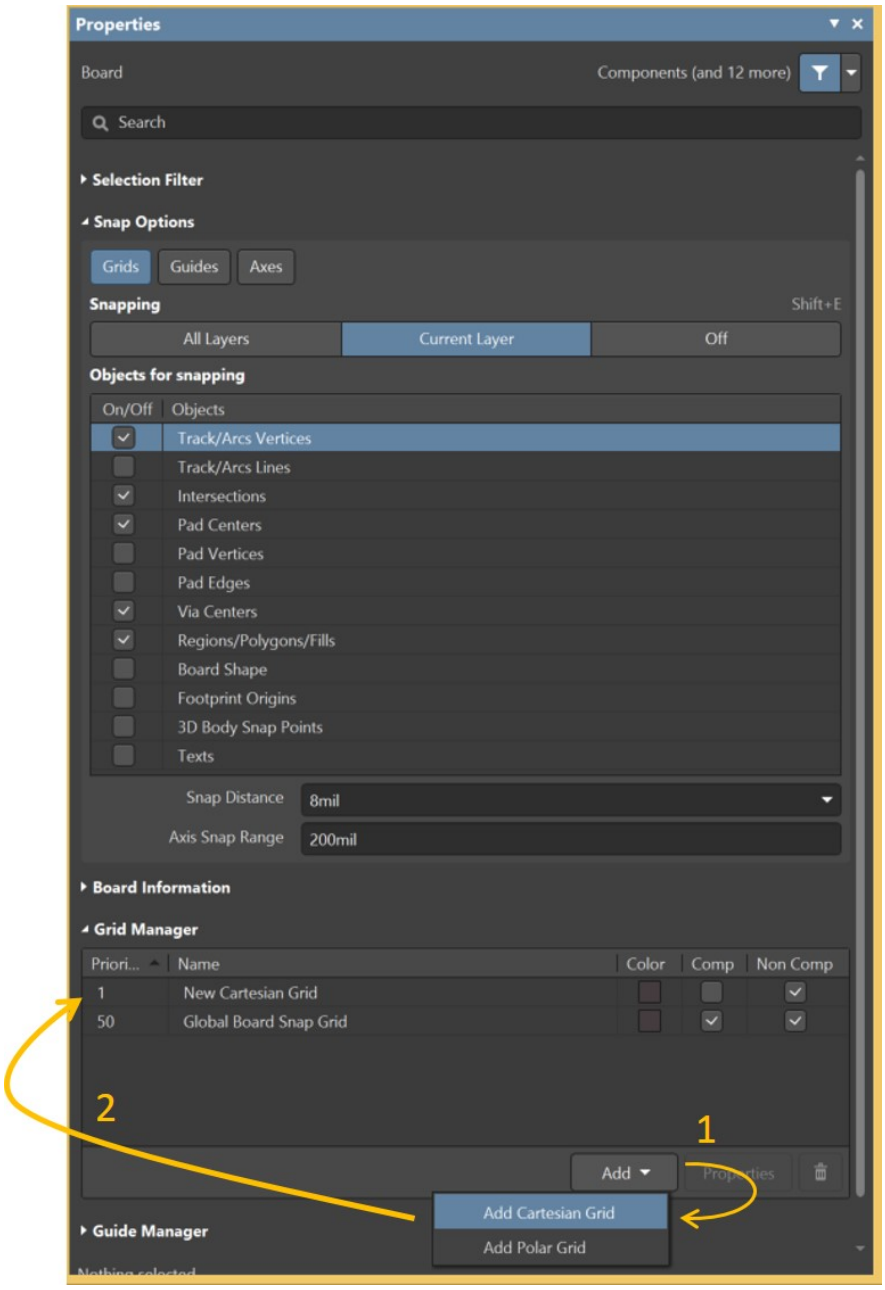


Figure 201. New grid added in the Grid Manager

15. Double click any field in the *New Cartesian Grid* entry to open the *Grid Editor* dialog as shown in Figure 202 below:

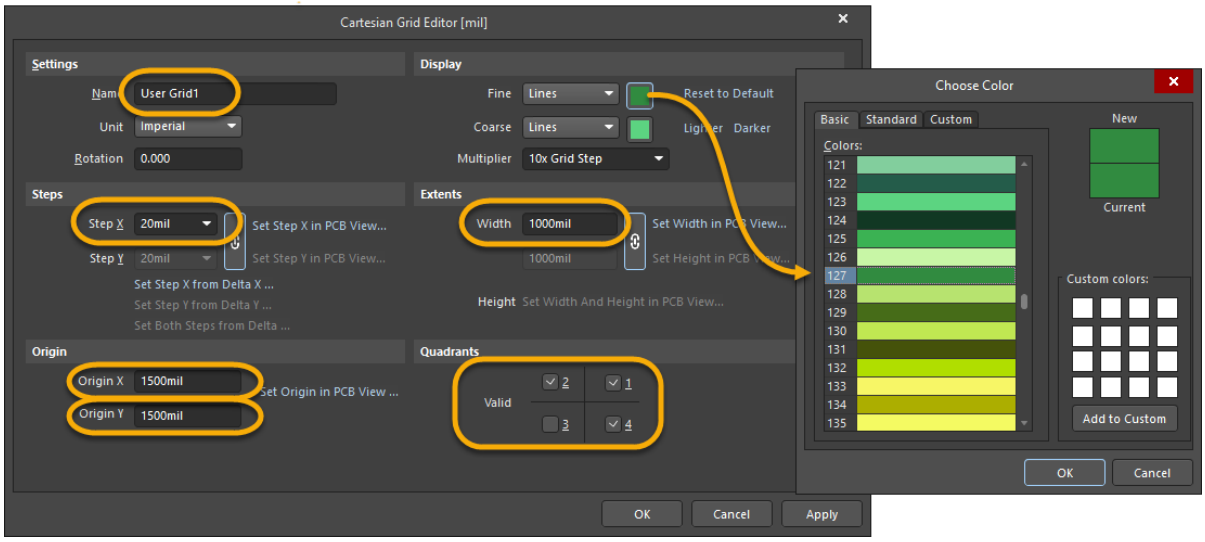


Figure 202. Cartesian Grid Editor

16. Edit the values for *Step*, *Extents*, *Quadrants* and *Display* to match those shown in Figure 202 above. The *Fine* color is set to `127` while *Coarse* color is set to `123`.
    a) Name: `User Grid1`
    b) Rotation: `0`
    c) Steps:`20mil`
    d) Width: `1000mil.`
    e) Origin X: `1500mil.`
    f) Origin Y: `1500mil.`
    g) Quadrants:1,2 and 4.
17. Click **OK** to return to the PCB editor and note the presence of the new grid. Zoom accordingly to see the fine grid detail as shown in Figure 203 below.

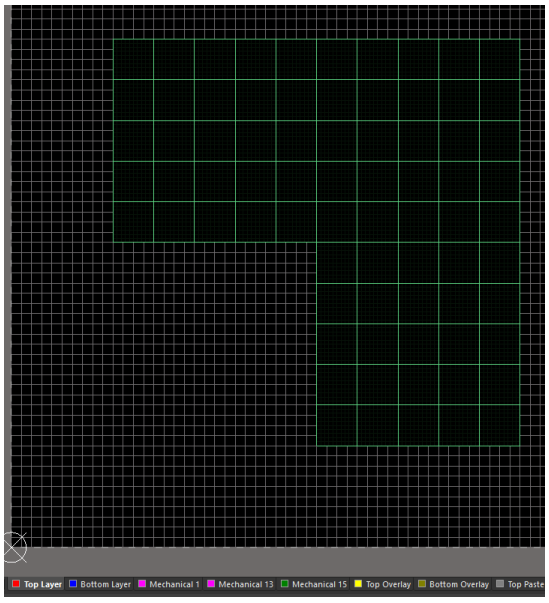


Figure 203. New Cartesian grid show in Quadrants 1,2 and 4 in the PCB

18. Find component `DI1` and drag it into the PCB space, to the right side of the new Cartesian grid. You will notice that the grid is no longer visible. It will reappear when the component is not selected.

19. Open the *Properties* panel and enable the **Comp** checkbox for the `UserGrid1` Cartesian grid as highlighted in Figure 204. This will allow us to place components on that grid.

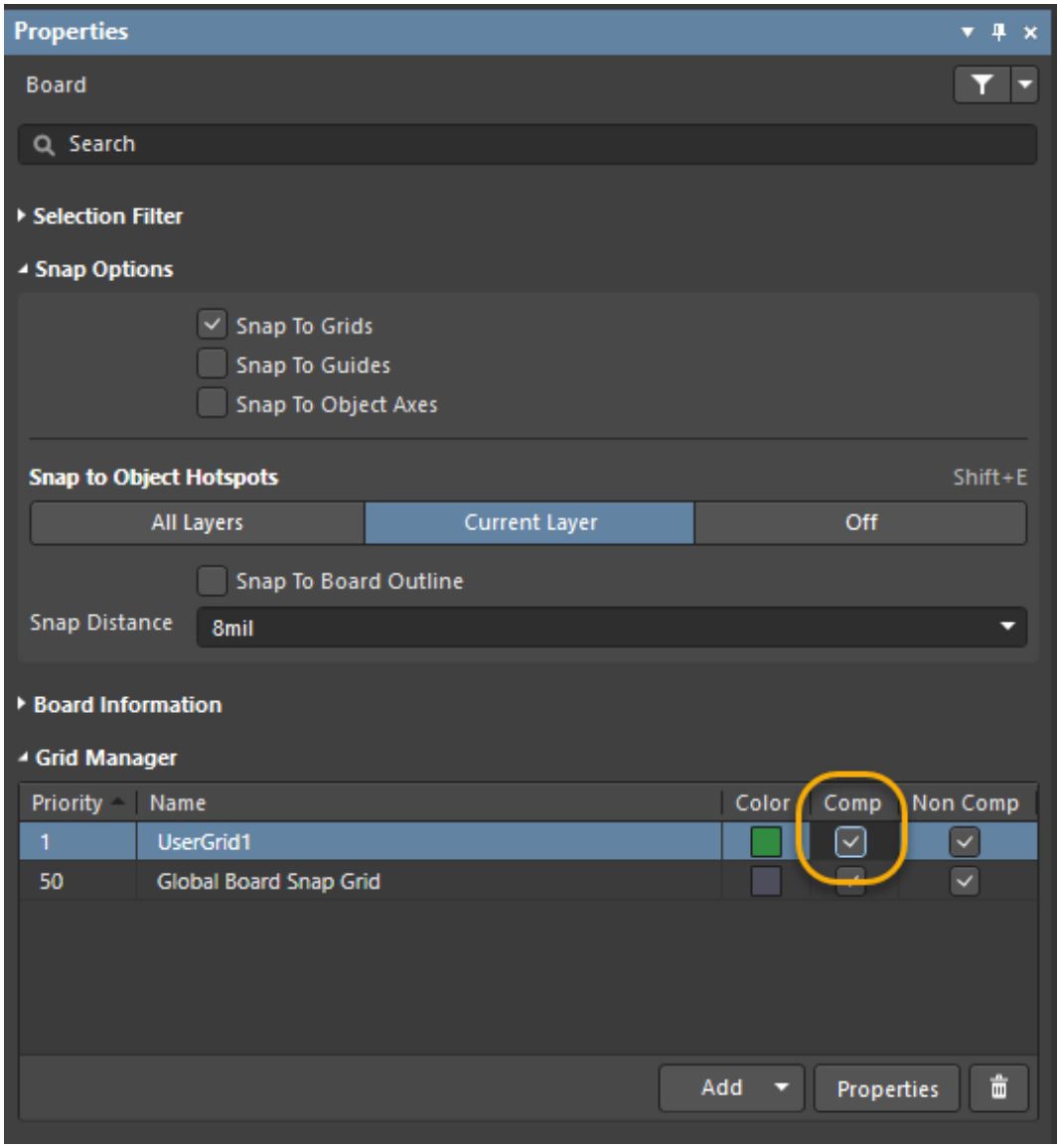


Figure 204. Enabling the Comp checkbox to be able to place components on the new grid

20. Move component `DI1` to see that the grid now remains visible.

16.4.3 Custom Polar Grid

21. Open the *Properties* panel again.
22. Click **Add** in the *Grid Manager* pane and select **Add Polar Grid**.
23. Enable the **Comp** checkbox to make the polar grid active for component operations.
24. Double-click the *New Polar Grid* row to open the *Polar Grid Editor*. Set the values as shown in Figure 205 below. The *Fine* and *Coarse* line colors are assigned to values `106` and `232` respectively.
    a) Name: `User Grid2`
    b) Angular Step: `5`
    c) Radial Step: `50mil`
    d) Start Angle: `0°`
    e) End Angle: `180°`
    f) Origin X: `1500mil.`
    g) Origin Y: `1500mil.`
    h) Radial Range min `0mil.`
    i) Radial Range Max: `1200mil`
25. Click **OK**.

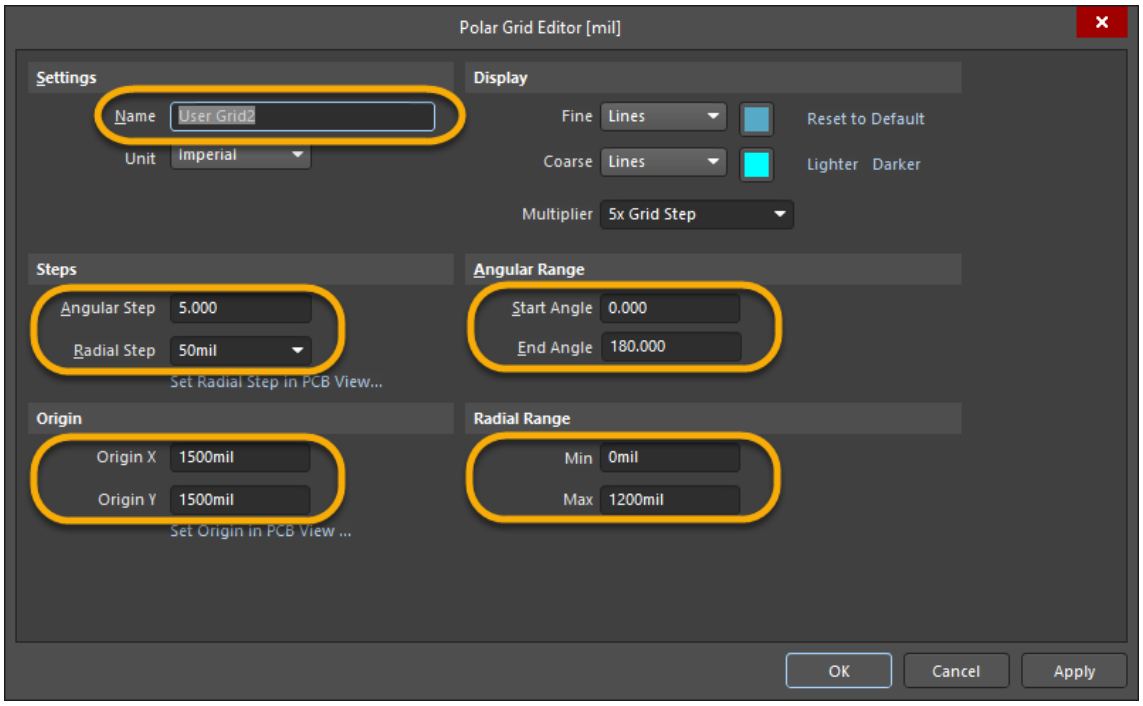


Figure 205. Polar Grid editor

26. Note the new Polar Grid in the PCB editor was generated over the Cartesian Grid we just generated.

27. The Grid Manager allows changing the priority for the overlapping grids if needed as shown in Figure 206 below.

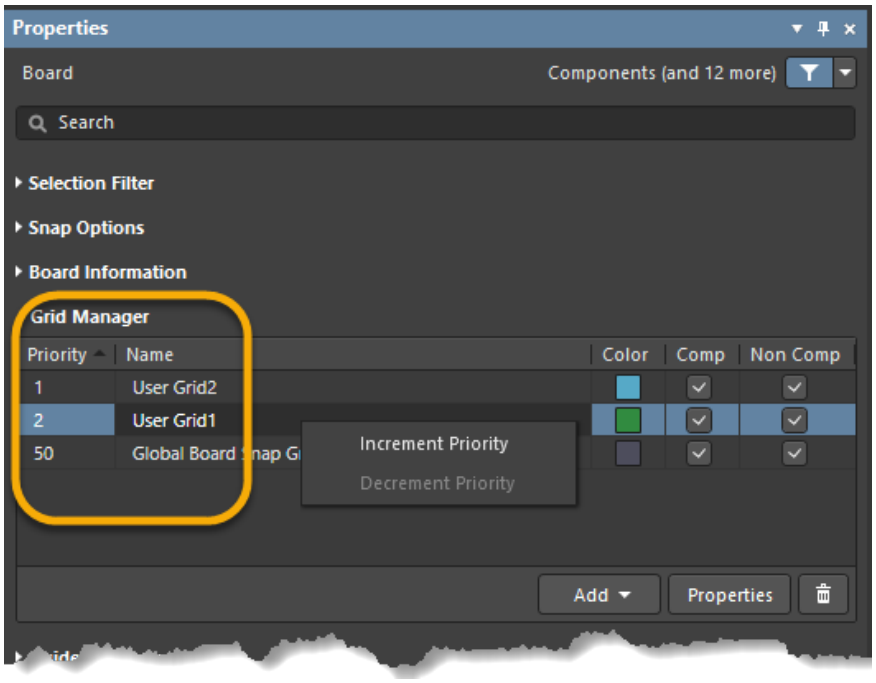


Figure 206. Priority for the Grids

28. Drag component `DI1` and `DI2` to the newly added polar grid. Test the rotations of the components by pressing **Spacebar** and **Shift + Spacebar** while moving the component to notice its alignment to the grid as shown in Figure 207 below.

For better visual experience, hide the rat's nests by going to **View » Connections » Hide All**.

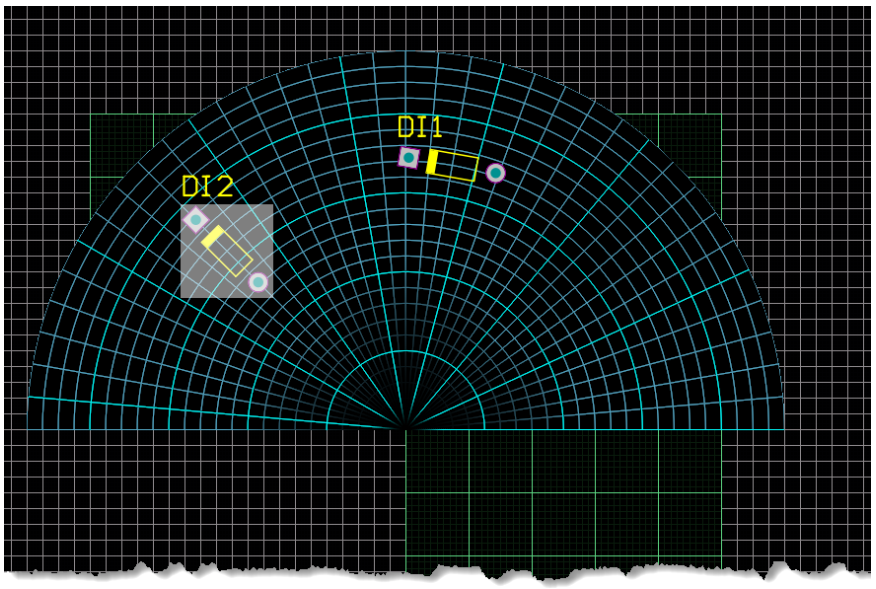


Figure 207. Aligning components to a Polar Grid

29. Save and close the project documents.

Module 17: PCB Layers

17.1 Purpose

In this exercise, we will examine the layers associated with your design and learn how to add plane layers to simplify power and ground connections used later.

17.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `AD20 Essentials Exercises\Module 17 - PCB Layers` folder.
2. Open the `WCTopping.PrjPcb` project file from the `Module 17 - PCB Layers` folder.

17.3 Adding a new PCB

3. Add a new PCB Document to the project by opening the *Projects* panel and right-clicking on the project name. From the right-click menu, select **Add New to Project » PCB** as seen in Figure 208 below.

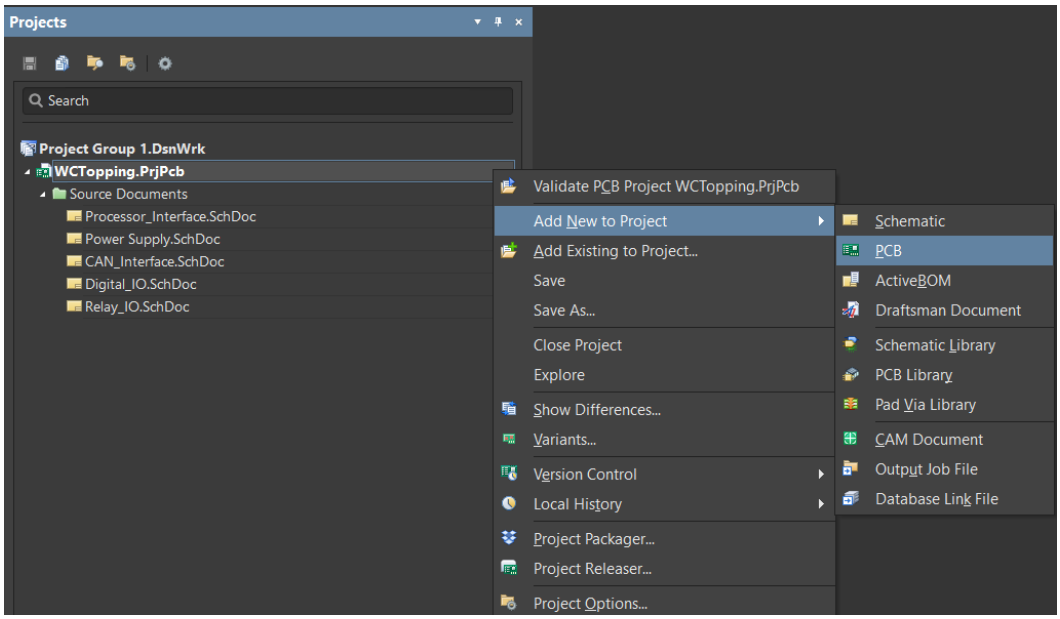


Figure 208. Add new PCB to the Project

4. Go to **File » Save All** and save the PCB document as `WCTopping.PcbDoc`.

17.4 Layer Stack Manager

17.4.1 Modify Default Layer Stack

5. With the `WCTopping.PcbDoc` as the active document, from the menu select **Design » Layer Stack Manager** or using the shortcut keys **D » K**. A new document will appear as shown in Figure 209 below. The *Properties* panel will also appear. Dock it along the left or right side of the workspace.

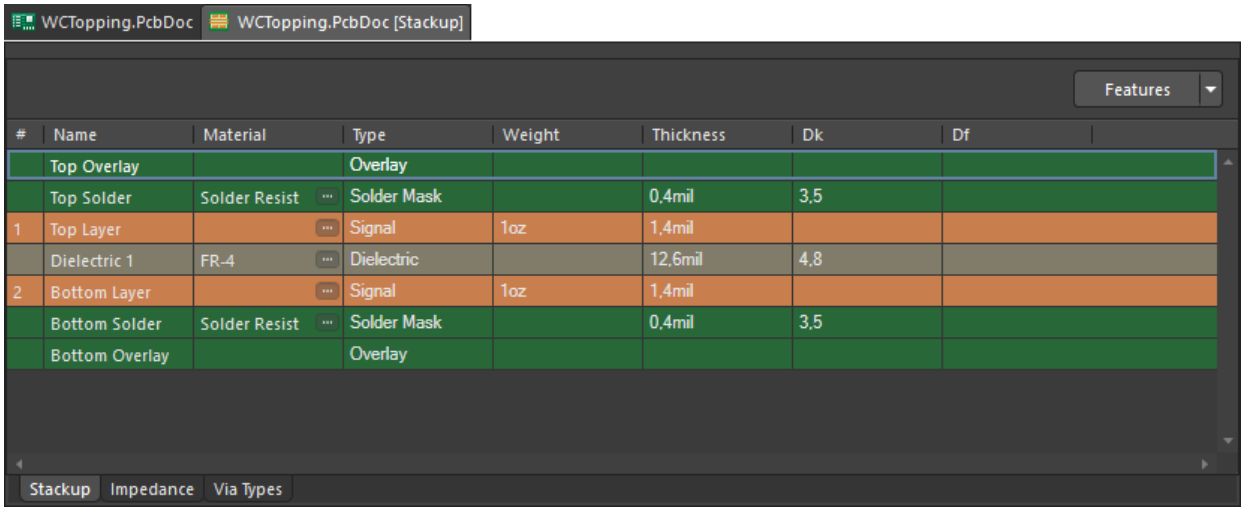


Figure 209. Layer Stack Manager

6. Select the layer *Top Layer*.
7. Right Click on *Top Layer*, select **Insert layer below » Plane** as shown in Figure 210 below.

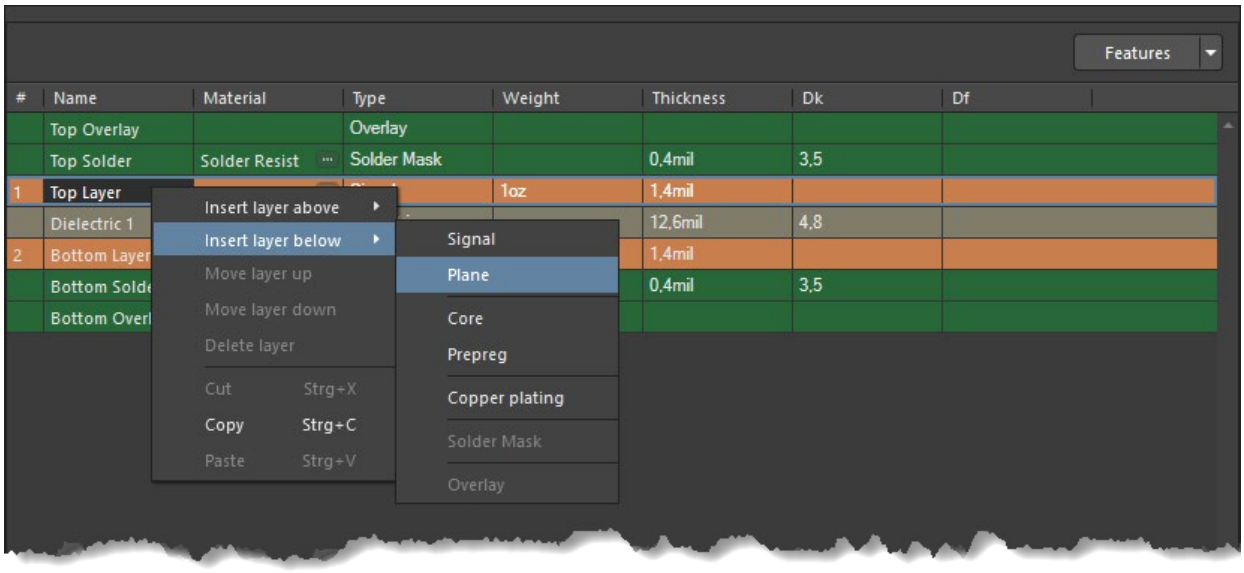


Figure 210. Adding Layers

8. As a default configuration, two Plane layers are added to create a symmetric stack as shown in Figure 211 below. Stack Symmetry can be enabled or disabled from the *Properties* panel.

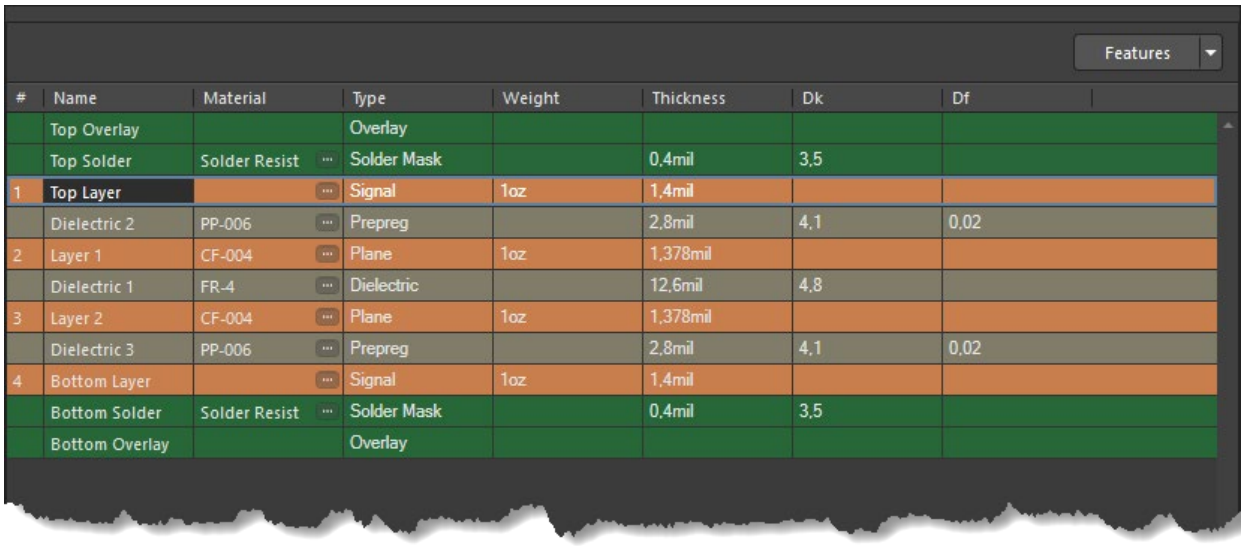


Figure 211. New Plane layers added

9. Click on the new *Layer 1* label created in the layer stack and rename it `GND`.

Edit text by selecting the cell then click the cell again with Left Mouse Button, or just press F2

10. Repeat the previous step to name the second Plane layer as `3V3`. The layer stack should look like Figure 212 below.

Figure 212. Layer Stack with renamed Plane Layers

11. Select one of the Plane layers that was just created, right-click and select **Delete Layer** to remove the Plane Layer. Repeat the step to remove the Dielectric 2/3.
12. Right click on *Top Layer*, select **Insert layer below » Signal**. As the default configuration, two signal layers are added, to create a symmetric stack.
13. Open the *Properties* panel. The *Properties* panel shows the layer information, e.g. the option for a symmetric stack.
14. Select the *Layer 1* and then *Dielectric 1*, observe the change in the *Properties* panel

15. Select the Material cell from *Dielectric 1* FR-4 ... . Click on ... to open the *Select Material* dialog. Select a new core material, e.g. Core-29 as shown in Figure 213 below.
16. Click **OK**.

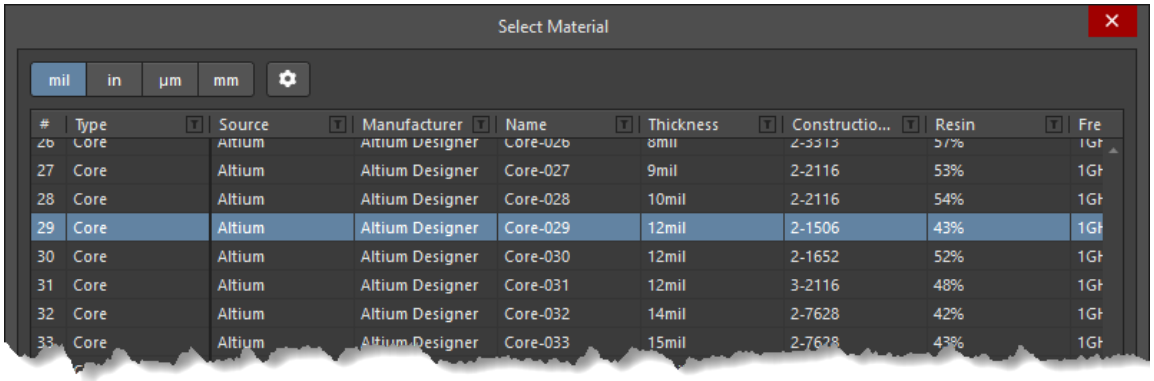


Figure 213. PCB Material database

17. Select Signal *Layer 2* and move the layer by right clicking the layer and selecting the **Move Layer Up** and **Move Layer Down** to observe the changes.
18. At the bottom of the *Layer Stack Manager* select the tab *Via Types*.
    By default, a through-hole Via from the Top Layer to Bottom Layer is available, as shown in Figure 214 below. Additional Vias like Blind or Buried vias, or uVias for HDI Designs can be added as well.

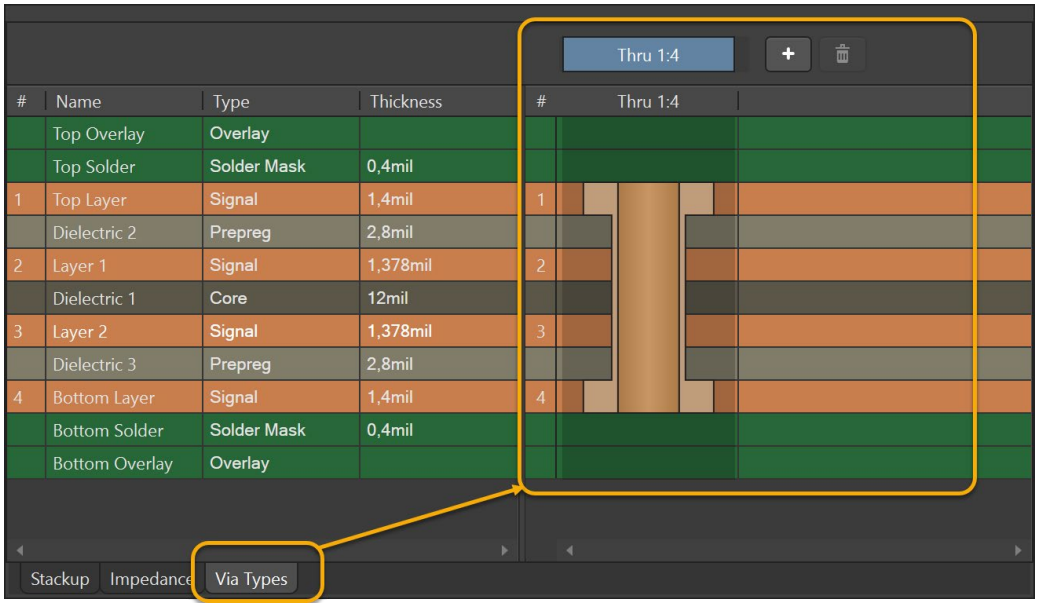


Figure 214. Layer Stack Manager with default via pairs

19. Close the *Layer Stack Manager* and save the changes.

17.5 Layer Stack Table

20. With the PCB Workspace in focus, press the **L** hotkey or select **View » Panels » View Configuration** to see all the layers for our design.
    a) Locate the `Mechanical 1` layer name in the list. If needed expand the section of the Mechanical Layers (**M**).
    b) Right-click and select **Edit Layer** and rename the layer to `Layer Stack`. You may also assign a *Layer Type* as shown in Figure 215 below. Click **OK** to close the dialog.

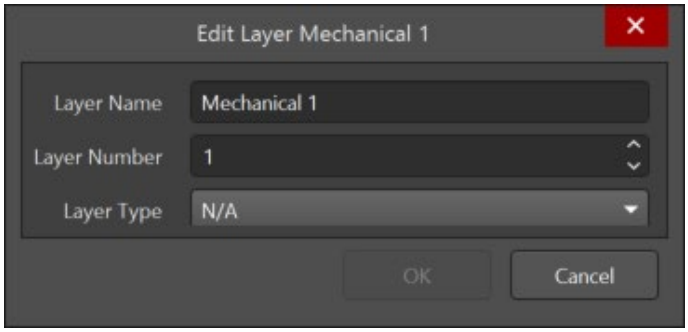


Figure 215. Renaming mechanical layers

21. With the PCB in focus, select from the menu **Place » Layer Stack Table** to add a table of the layer stack in the workspace.
22. Press **Tab** before placing the Layer Stack table.
23. Modify the following parameters as shown in Figure 216.
    a) Change **Layer** to `Layer Stack`
    b) Uncheck the **Show Board Map**
    c) Set **Line Width** to `8mil`
    d) Set **Text Height** to `30mil`
    e) Set **Stroke Width** to `5mil`

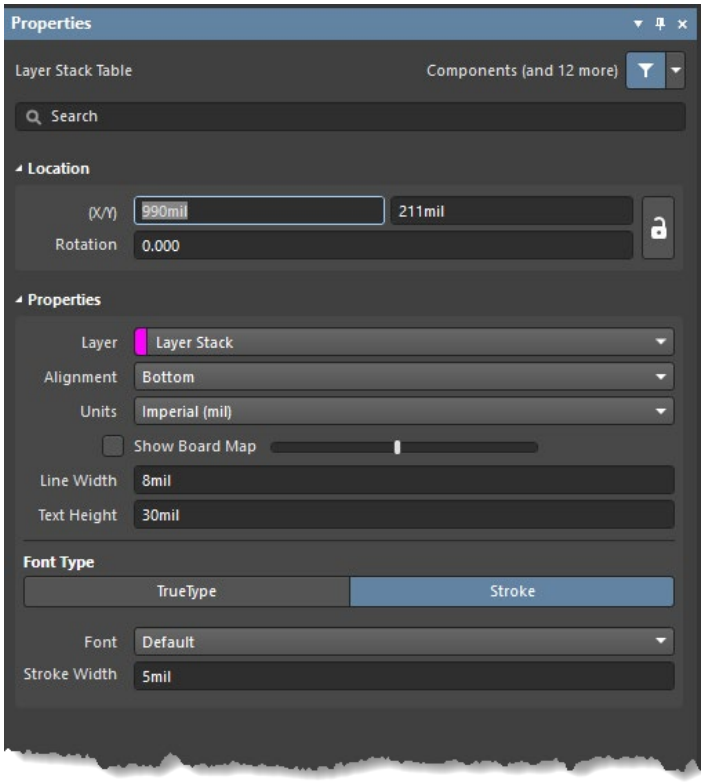


Figure 216. Layer Stack Table

24. Press the **Pause** button in the middle of the workspace and position the compact table below the board outline as shown in Figure 217 below. **Right Click** to exit Layer Stack Table placement.

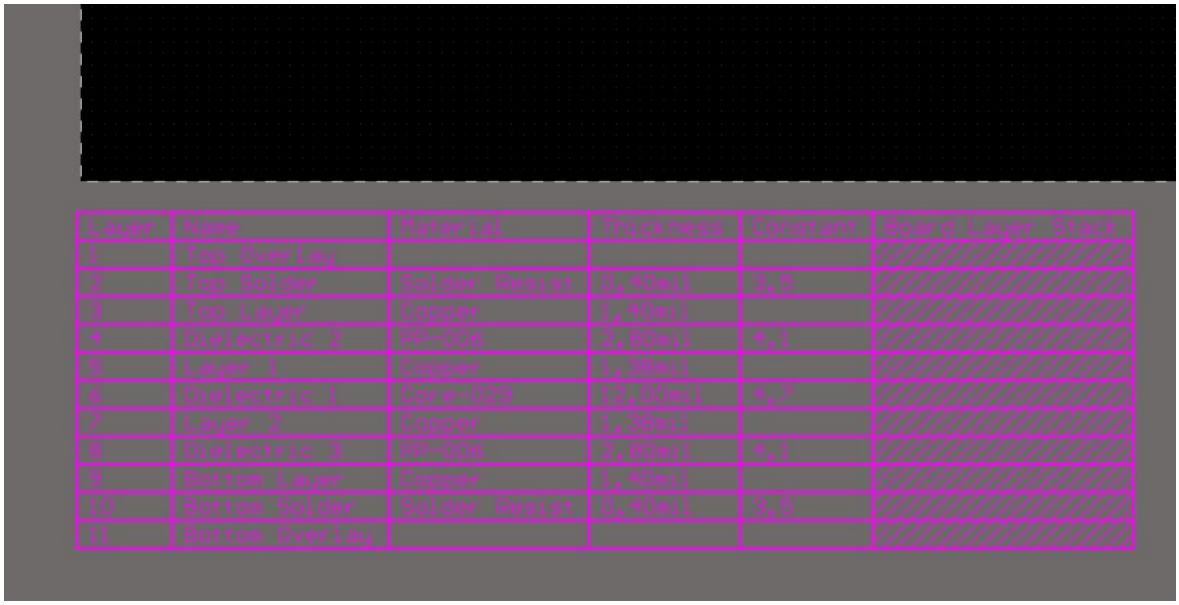


Figure 217. Place Layer Stack Table

25. Save and Close the Project and its documents.

Module 18: Shaping the PCB

18.1 Purpose

In this exercise, you will learn how to define a board shape using several methods.

The board shape or board outline is usually closed a polygon used to define the PCB boundaries. The board shape does not exist on a particular layer but rather defines the space where components and traces can be placed. In Altium Designer, the black area is the board shape and can be created and modified in several ways. This exercise will explore three of these methods.

18.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `Module 18 - Shaping the PCB` folder.
2. Open the `WCTopping.PrjPcb` project file located in the `Module 18 - Shaping the PCB` folder.

18.3 Board Planning Mode

3. Navigate to the *Projects* panel and double-click the icon for the PCB file `WCTopping.PcbDoc` with the left mouse so that the PCB document is open.
4. Set the origin to `2050mil` x `2150mil` by selecting the menu **Edit » Origin » Set**. Rather than clicking the location we want, press the **J » L** keys to open the *Jump Location* command.
5. Enter in the X and Y coordinates as shown in Figure 218 below.

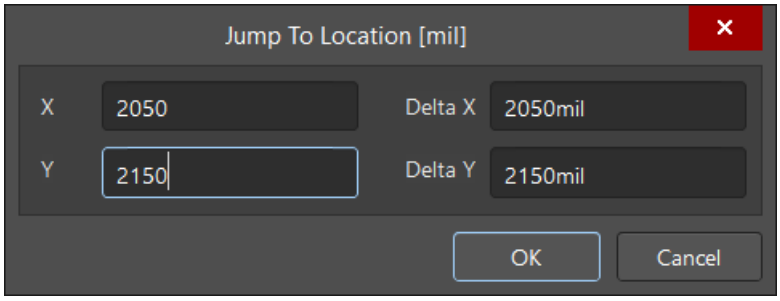


Figure 218. Jump to Location

6. Press **Enter <u>twice</u>** to set the origin at that location. A circle and X will appear at the new origin location as shown in Figure 219.
7. Change the grid setting to `500mils` by pressing **Ctrl + G** and entering `500mil` in the *Step X* field, and change the *Display* setting for *Fine* to *Lines* instead of *Dots*. Click **OK**.

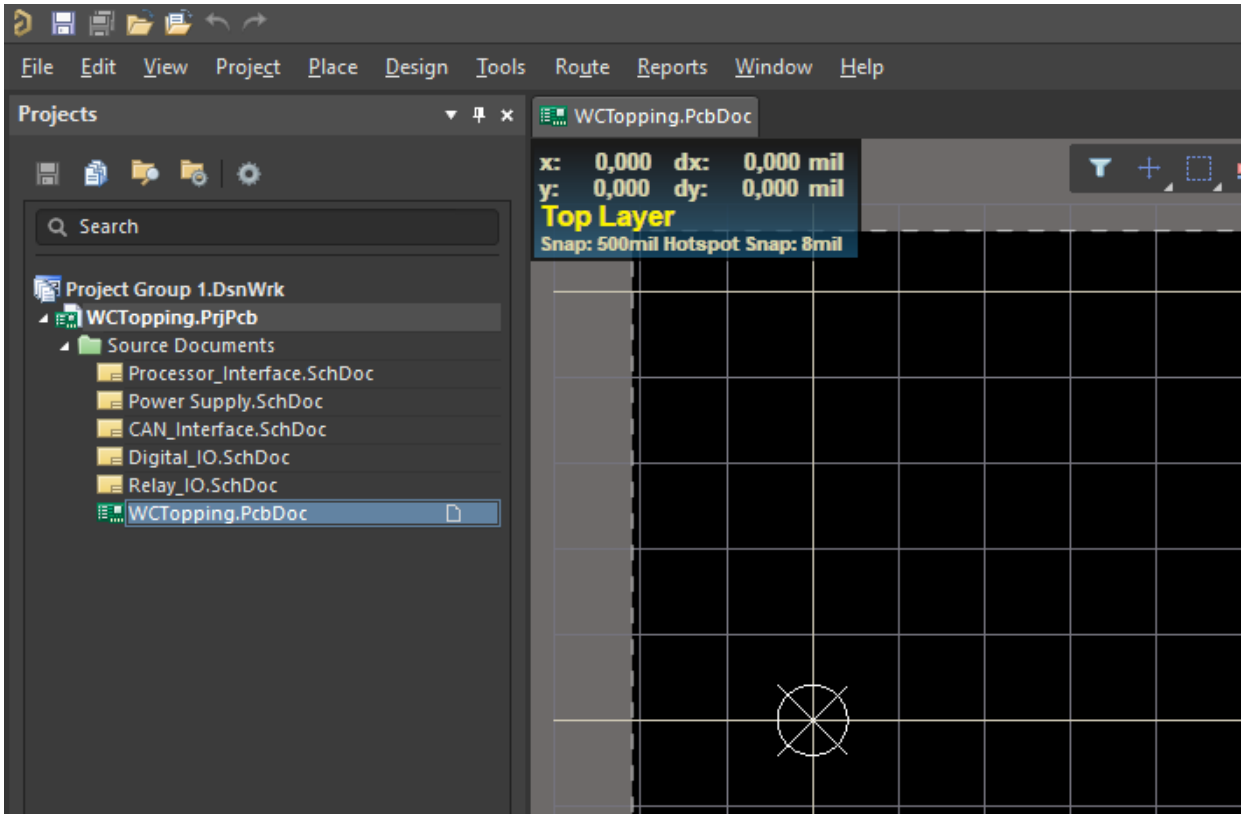


Figure 219. Defined origin on the grid

8. Change to the board planning mode with the command: **View » Board Planning Mode** (the shortcut to get to this mode is the **1** key).
9. Click on the menu **Design » Redefine Board Shape**, and then draw a closed rectangle starting at the origin (0,0). Use the X, Y position to guide you to draw a rectangle `3500mils` wide by `2000mils` tall. The *Heads-Up Display* or *Status Bar* will tell you the coordinates of where you're about to place the vertex. Be sure to left-click on every rectangle vertex and ending at the origin followed by a right click to end the command. The board should look like Figure 220.

If the corner mode is not 90° rectangle change it to 90° rectangle. You can change the corner mode by pressing **Shift +Spacebar**.
Examine the Status Bar in the bottom center of the window to see the current mode as you toggle through the five corner style choices.

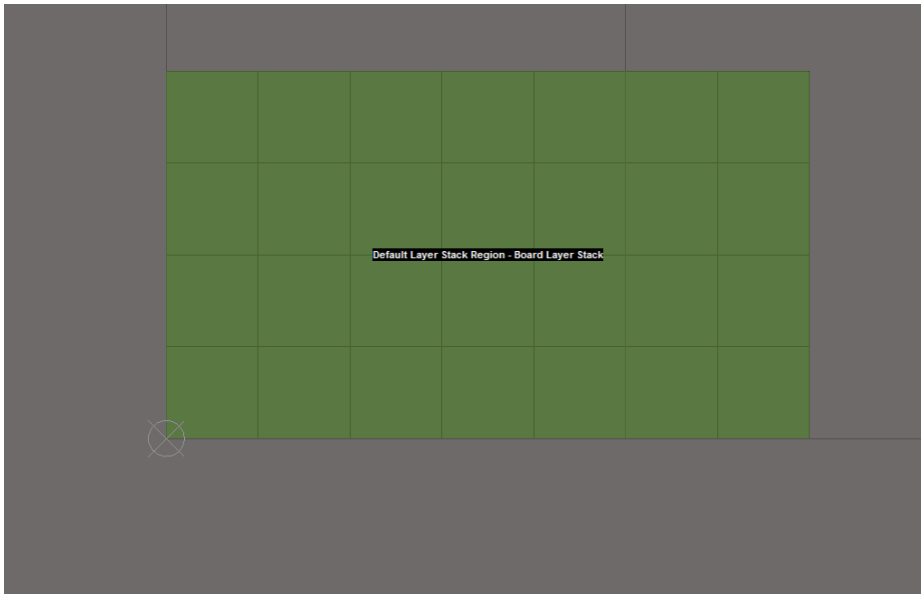


Figure 220. Board Planning Mode

Use the **Jump » Location** command to go to the X, Y locations quickly.

10. Select from the menu **Design » Edit Board Shape** then select and hold the right vertical sliding edge.
11. Drag the vertex to expand the rectangle to be `5000mils` wide.
12. Repeat with the top horizontal line to expand the rectangle to be `2500mils` tall.
13. Left click anywhere in an open space to exit the command, and your board should look like Figure 221.

Figure 221. Board Outline defined in Board Planning mode

14. From the menu, select **Design » Modify Board Shape** to get a crosshair.
15. Left-click at the coordinates below to modify the board shape. Use the *Heads-Up Display* or *Status Bar* to see the coordinates, or use the **J** » **L** shortcut keys.
    a) (5000, 1500)
    b) (2500, 1500)
    c) (2500, 1000)
    d) (5000, 1000)

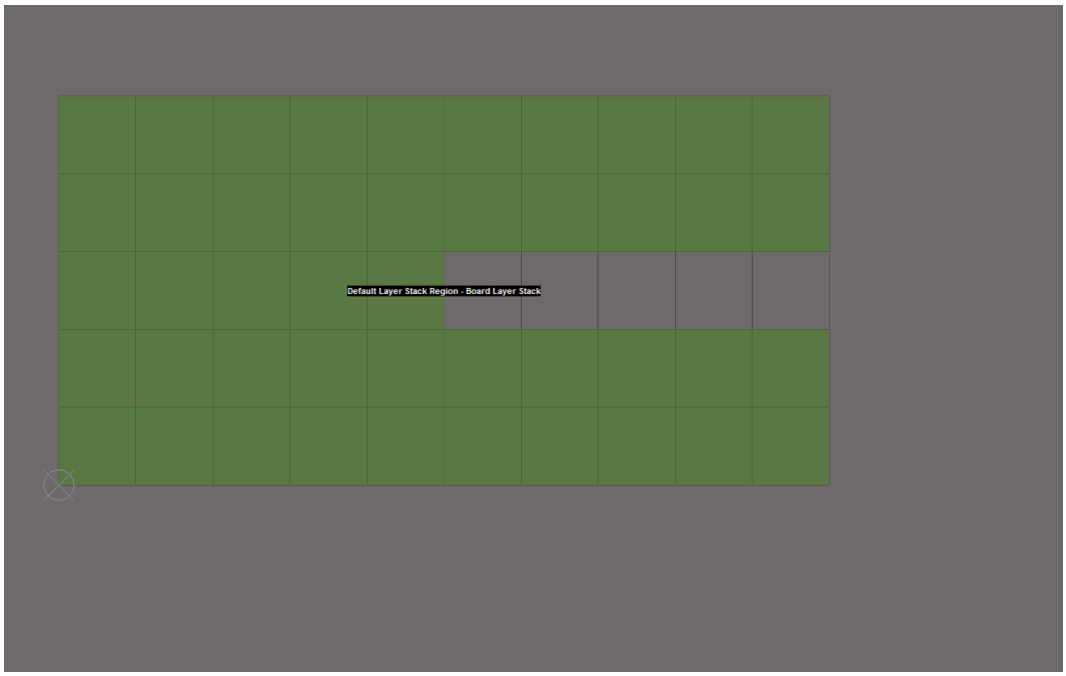


Figure 222. Board Outline modified in Board Planning Mode

16. Your board should now look like Figure 222.
17. Change back to **View » 2D Layout Mode** (or press the **2** key).
18. Change the *Step X* grid back to `5mils` in the *Grid Editor* using the keys **Ctrl + G**.

18.4 Define Board Shape from Selected Objects

18.4.1 Circle

19. Open the *View Configurations* panel by selecting the **View » Panels » View Configuration** or press the **L** hotkey.
20. Right-click layer `Mechanical 1` and **Edit Layer** as shown in Figure 223. Change the name to `Board Outline` and hit **Enter** to close the dialog.

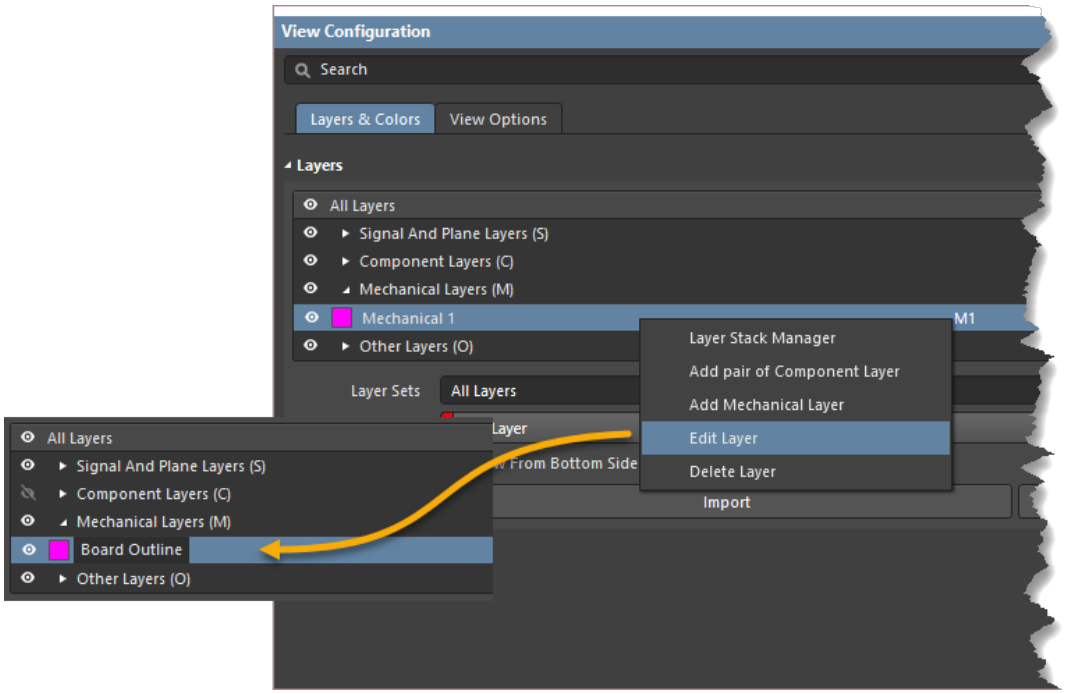


Figure 223. Renaming Mechanical Layer

21. Select *Board Outline* as the active layer in the Layer Bar at the bottom of the PCB environment.
22. Select **Place » Arc » Full Circle**
    a) Left-click anywhere near the middle of the existing PCB shape.
    b) Move the mouse either towards you or away from you to create enlarge the circle.
    c) Left-clicking again will set the size of the circle. To increase or decrease the size of the circle, select it and modify Radius in the *Properties* panel, if needed. A Radius of 1300mil should be fine.
    d) Press the right-mouse button to terminate the command. Your PCB should now be similar to Figure 224.

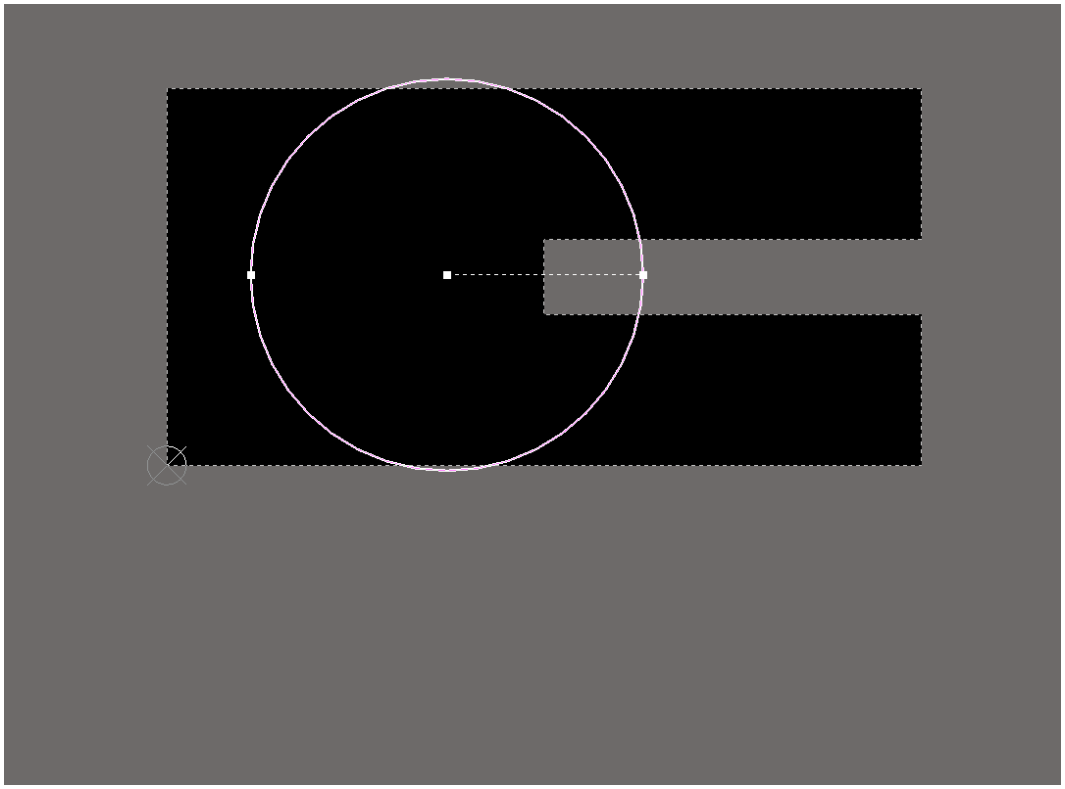


Figure 224. Placing a circle in the PCB workspace

23. The circle should be highlighted in white indicating it is selected (if it is not, press **Ctrl + A** to select it).
24. To create a circular board shape, go to the menu **Design » Board Shape » Define from Selected Objects**. The black area, which is the board outline, should now be limited to the circle as shown in Figure 225.

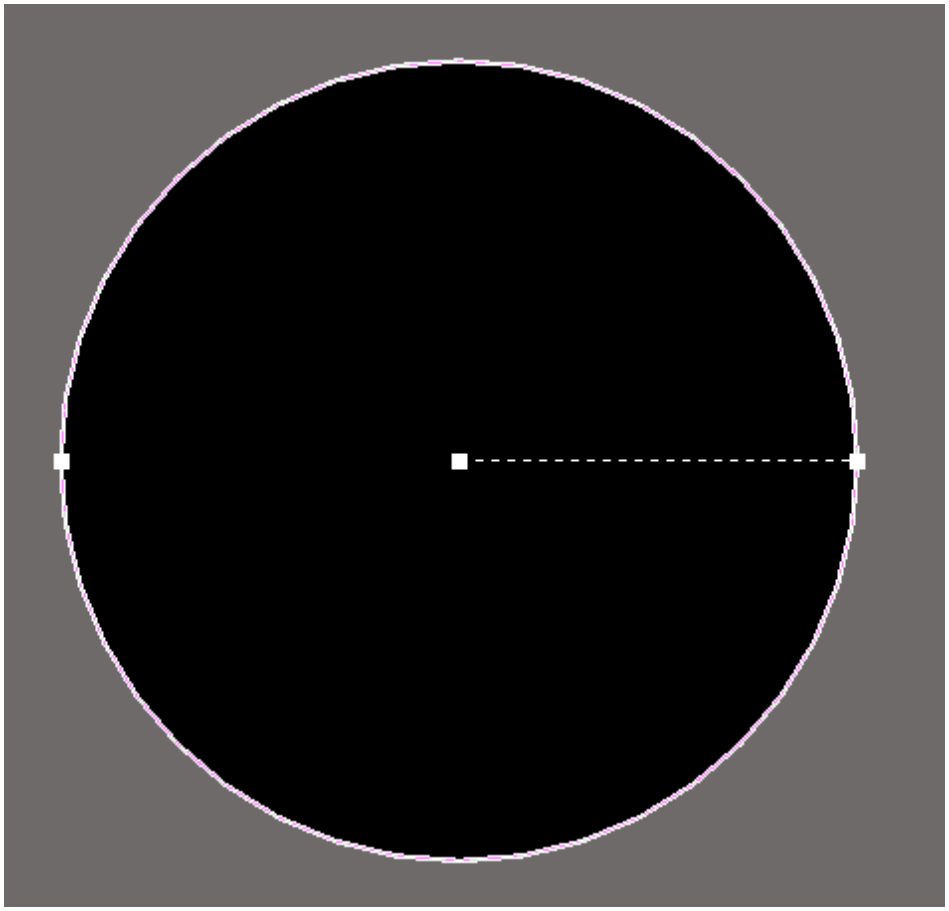


Figure 225. Defining the Board Shape from Selected Primitives

The display may be masked as a result of this action. Click the **Clear** button or in the bottom right corner of the editor or press **[Shift] + C** to return to normal view.

18.4.2 Rectangle

25. To simplify the drawing process, we will set a coarse grid. Press **Ctrl + G** to open the *Grid Editor* and change the Step value to `500mil`. Click **OK** to close the *Grid Editor*.
26. Click menu choices **Place » Line** and draw a closed rectangle of any shape inside of the existing circular PCB. Ensure that the rectangle is closed by looking for a white circle at the cursor when you are completing the polygon as shown in Figure 226.

If you're having issues placing the lines for the rectangle, this may be because you're in **Look Ahead Mode**. Hit the **1** key to disable this mode, which may help with line placement.

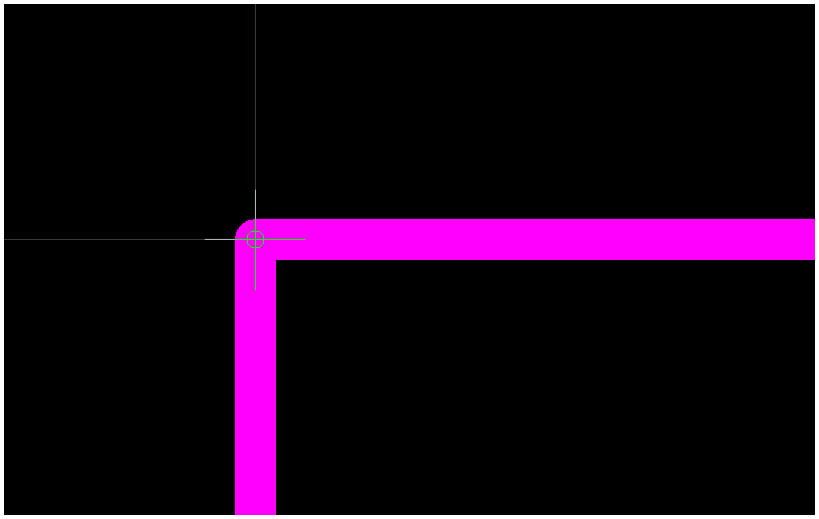


Figure 226. Drawing a rectangle with lines to define our Board Shape

26. After you complete the rectangle, right-click to terminate the command.

Try changing the corner style by pressing **[Shift]+[Spacebar]** while drawing the rectangle. Examine the *Status Bar* in the bottom center of the window to see the current mode as you toggle through the five corner style choices.

27. Right-click a second time to terminate the place line command.
28. Select the lines of the newly defined rectangle by **Shift** selecting all 4 lines, or using a selection rectangle.
29. Now we'll create a board shape to match your rectangle by selecting the menu **Design » Board Shape » Define from selected objects**. Your PCB should now be rectangular and similar to Figure 227.

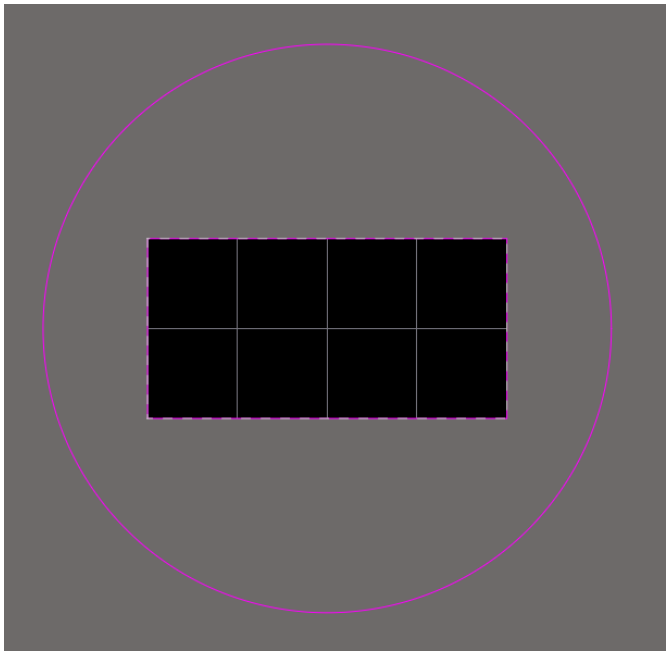


Figure 227. Example Board Shape defined from the rectangle

30. Since we changed our board shape, we can delete the outer circle by selecting it, and hit **Delete**. Feel free to view your PCB in 3D, by hitting the **3** key as shown in Figure 228.

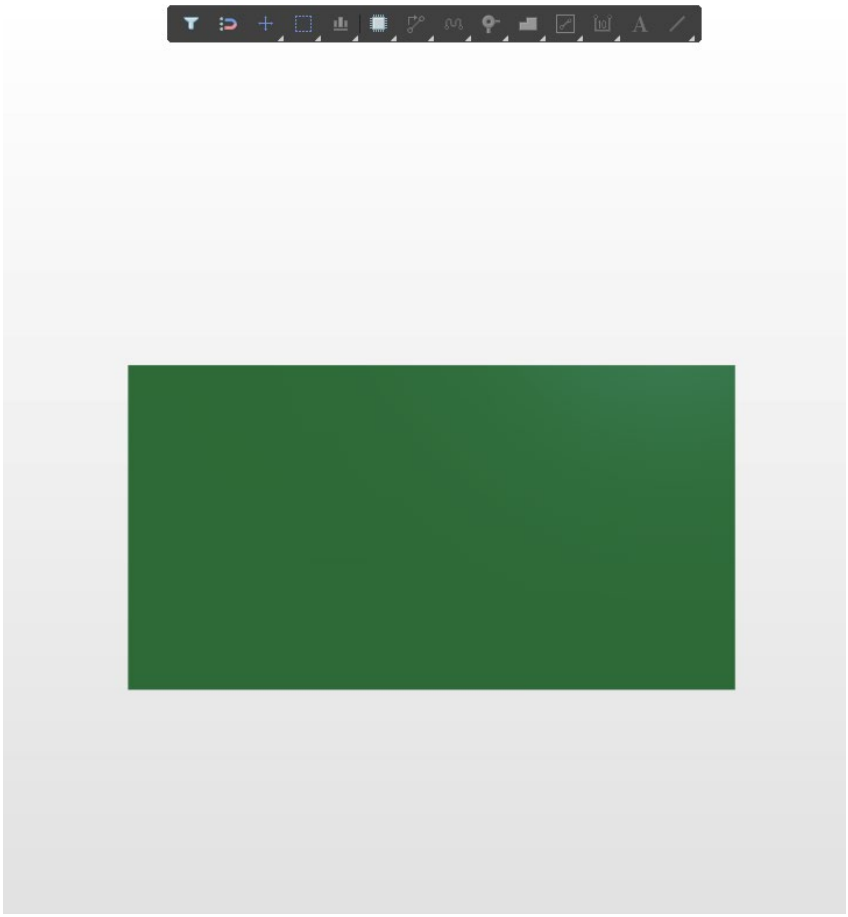


Figure 228. Example of a rectangular board shape in 3D

31. Return to 2D using the **2** key.
32. Remove the outline for the board by pressing **Ctrl + A** to select the outline, then press the **Delete** key to remove the outline.

18.5 Defining Board Shape from DXF/DWG

33. Select **File » Import » DXF/DWG** and browse to the project folder `Essentials Exercises\Shaping the PCB` for `outline.dxf` located in the exercise folder.
34. The *Import from AutoCAD* dialog box will open.
    Change the dialog box settings as shown in Figure 229.
    a) Scale: `mm`
    b) Layer mapping 0: `Not Imported`.
    c) Layer mapping 1: board_shape - `Board Outline` layer.
    d) Layer Mapping 2: Title Block - `Not Imported`.

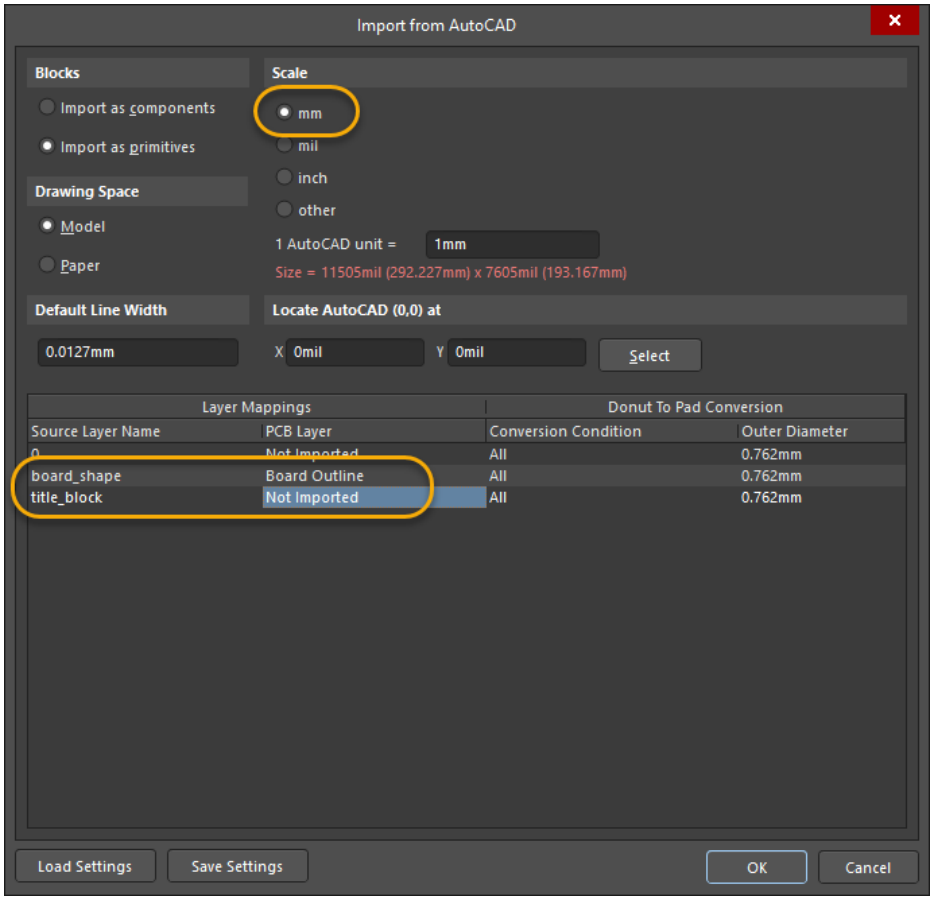


Figure 229. Import DXF dialog

To ensure the proper scale selection, check the Size = which will update based on the selected scale units and should match the dimensions you are expecting.

35. Click **OK.**

36. After clicking on **OK**, the board shape will be imported similar to Figure 230.

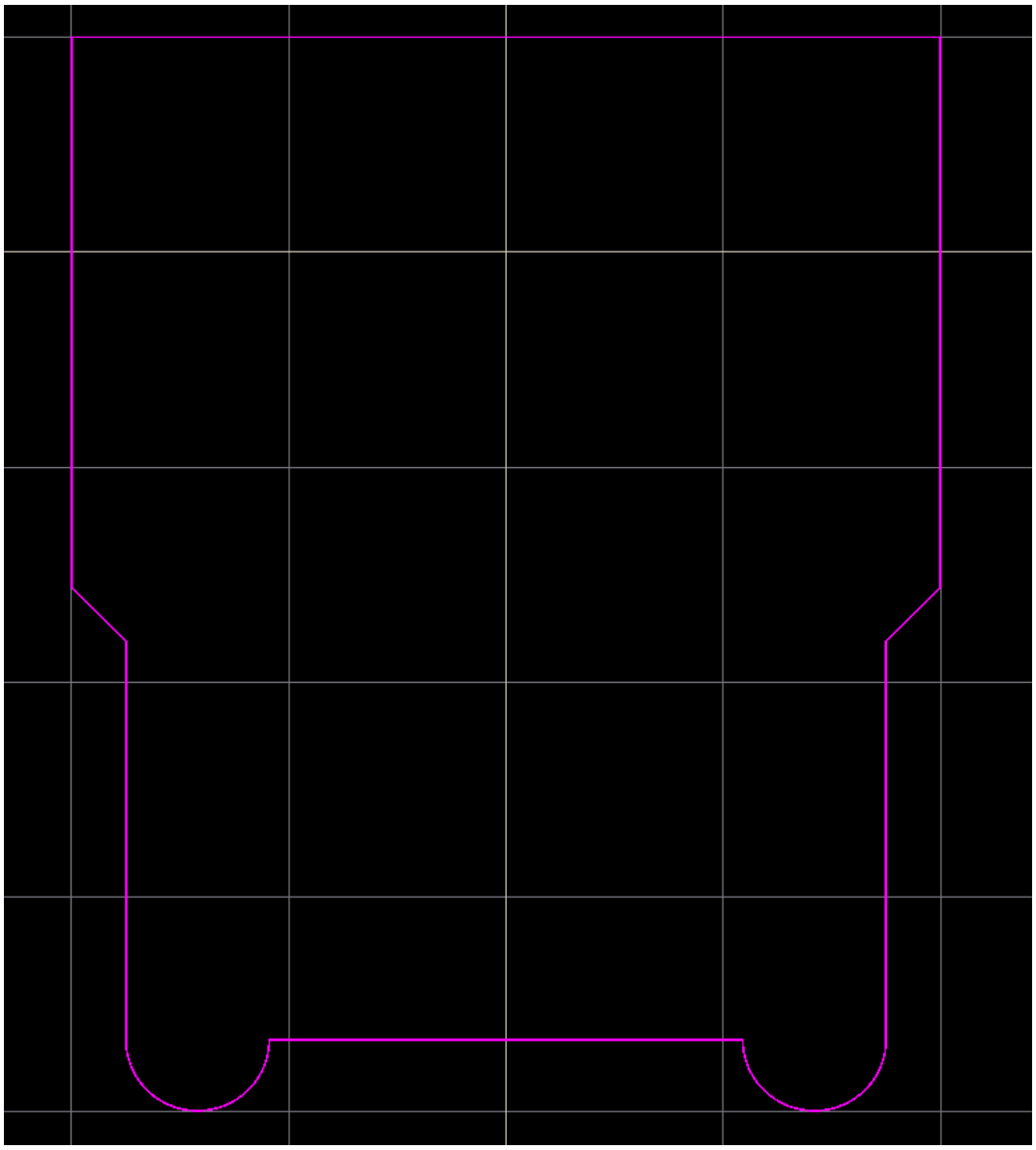


Figure 230. Imported DXF outline

37. Select all the line segments in the imported outline by pressing **Ctrl + A**.
38. Select **Design » Board Shape » Define from Selected Objects** to create the new board shape as shown in Figure 231.

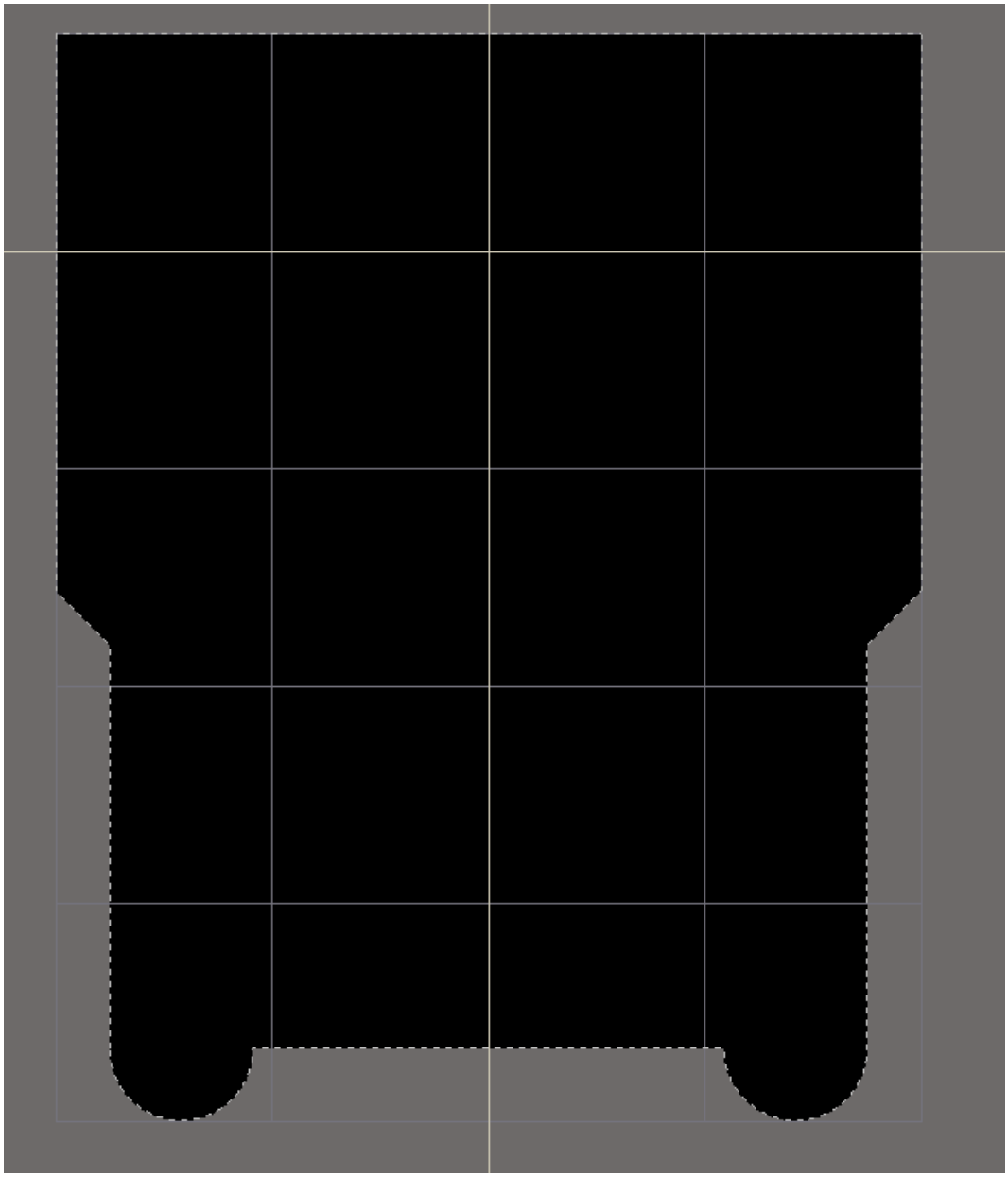


Figure 231. Final Board Shape created from a DXF outline

39. Save your changes and close the document.

Module 19: PCB Design Rules

19.1 Purpose

In this exercise, students will learn various methods to create PCB design rules. Rule priorities will also be defined, to ensure proper rule application on objects targeted by multiple rules.

Design rules are used to ensure the PCB layout meets the specified design requirements and highlights violations when items fall outside of their defined constraints. These rules cover various aspects of the design and collectively form an “instruction set” for the PCB editor to follow. Many of them can also be monitored in real-time by the online Design Rule Checker (DRC).

19.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `Module 19 - PCB Rules and Panels` folder.
2. Open `WCTopping.PrjPcb` from the `Module 19 - PCB Rules and Panels` folder.
3. Open the `Rules.PcbDoc` file from the *Projects* panel.

19.3 Basic Design Rule

19.3.1 PCB Rules and Constraints Editor

4. Go to **Design » Import Changes From WCTopping.PrjPcb** and verify that there are no differences reported. Close the *Comparator Results* dialog.
5. Go to **Design » Rules** (**D » R**) and the *PCB Rules and Constraints Editor* dialog will open.
6. The tree on the left lists the rule categories. Expand the *Electrical* rule category. Notice that there are 6 rule types within this category. This can be seen in Figure 232.
7. Expand the *Clearance* rule type to display all clearance rules currently defined. Notice that there is only a single, default rule called *Clearance***.** Select it to view its rule definition on the right of the window as shown in Figure 232.

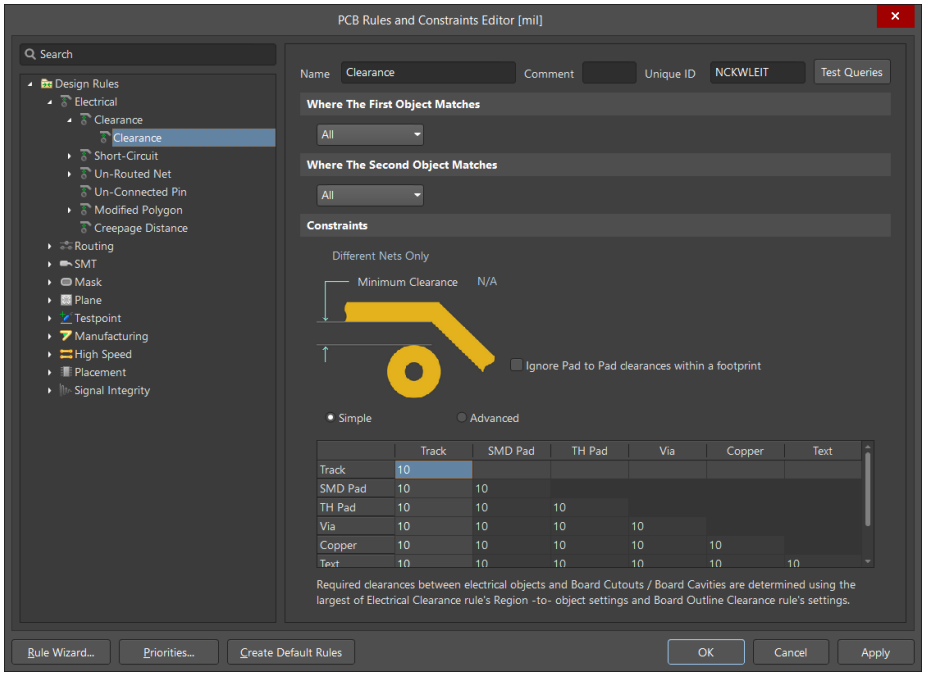


Figure 232. PCB Rules and Constraints Editor

8. We will now create a new rule which will target the 12V power net. Right-click on the *Clearance* rule type, just underneath *Electrical* and click **New Rule**. You should notice a new rule added with a default name given as `Clearance_1`.
9. Select the new rule `Clearance_1` on the left pane of the window to open the rule and assign each field according to Figure 233 below.
   a) Set the *Name* field to `PowerNets`.
   b) Set the first query (*Where the First Objects Matches*) to `Net`, and then `12V`.
   c) Set the *Minimum Clearance* constraint to `15mil`.

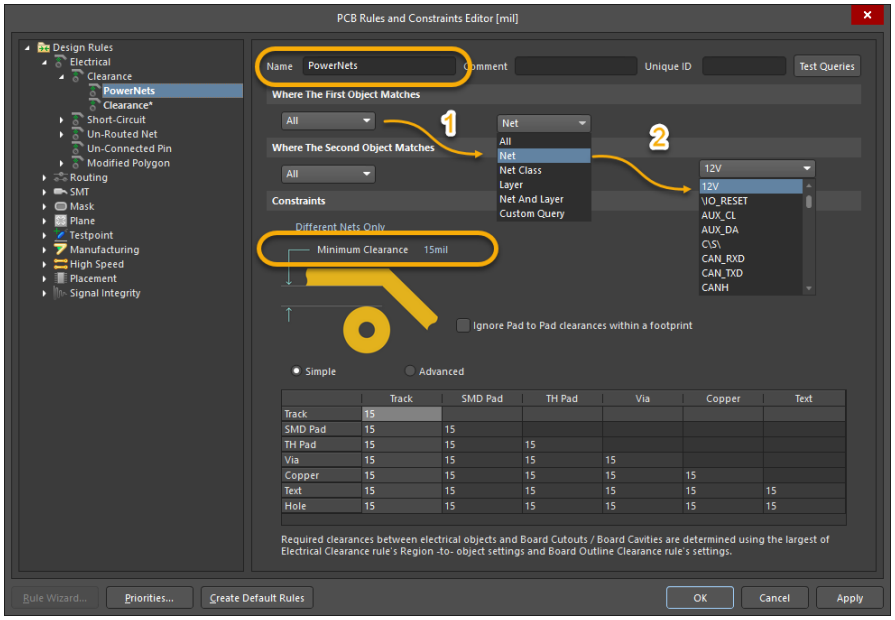


Figure 233. PowerNets clearance rule definition

19.3.2 Rule Priorities

10. Now that we have multiple clearance rules, let's ensure that their priorities are set correctly. Click on the **Priorities…** button on the bottom left of the *PCB Rules and Constraints Editor* window to open the *Edit Rule Priorities* window.

11. Ensure that the newly added rule's priority is number **1** as shown in Figure 234 below. If it's not, select that rule and click **Increase Priority**.
12. Click **Close** and then **OK** to accept changes and close all dialogs.

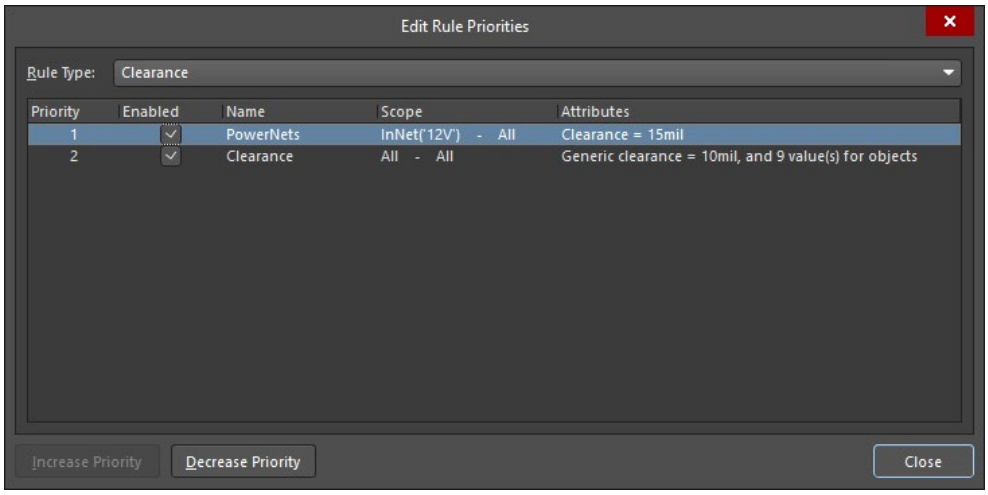


Figure 234. Rule Priorities window

To simplify the process of defining and managing rules, the idea is to first define the general rules that cover broad requirements of the PCB. Then, override these rules with specific rules in specific situations. Thus, it is a general practice to make the generic or "All-All" rule the least prioritized so that objects are applied with their specific rules.

19.4 Design Rule Using Net Classes

13. Instead of scoping objects of a single net in Design Rules, we can target objects from multiple nets by grouping them in Net Classes. We will now add a Net Class to the 12V net, and assign a Width rule to this Net Class. **Open** `Power Supply.SchDoc`.
14. Go to **Place » Directives » Parameter Set**. Your mouse cursor will contain the `Parameter Set` attached to it.
15. Before placing it, hit the **Tab** key to open the *Properties* panel.
16. Assign the fields according to the following steps, while also referring to the images in Figure 235, Figure 236 and Figure 237.
    a) Change the Label value to `PowerNets`

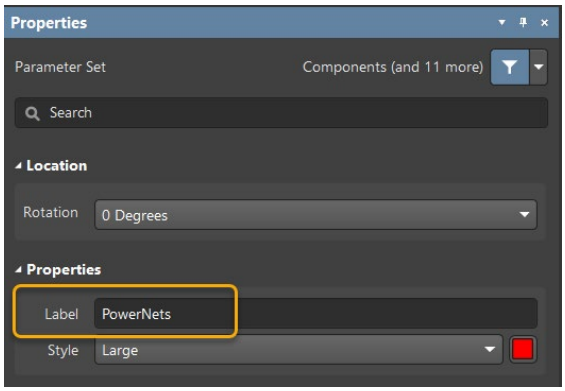


Figure 235. Change the Label value to "PowerNets"

    b) Under the *Parameters* pane, click **Add…** and add a **Net Class.**

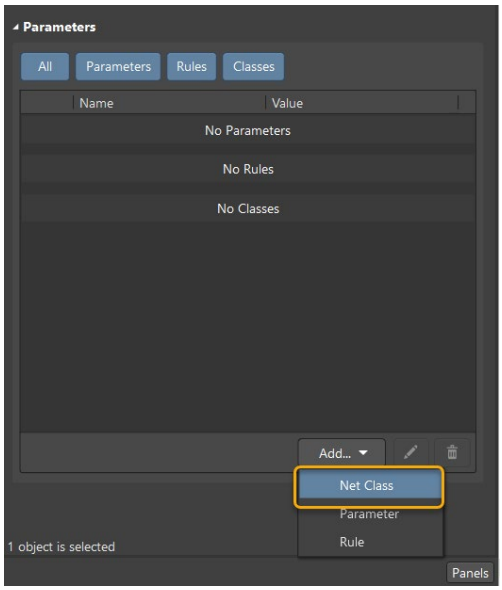


Figure 236. Adding a NetClass through the Parameters pane in the Properties panel

c) Name the new NetClass as `PowerNets`.

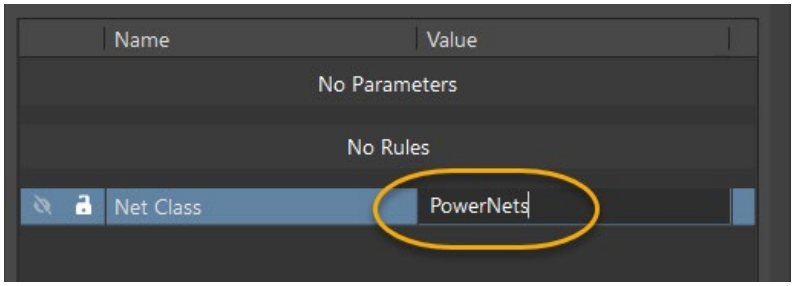


Figure 237. Changing the NetClass value to "PowerNets"

17. Click on the pause symbol to continue and place it on the *12V* power line as shown in Figure 238. Once placed, right-click to end the placement command.

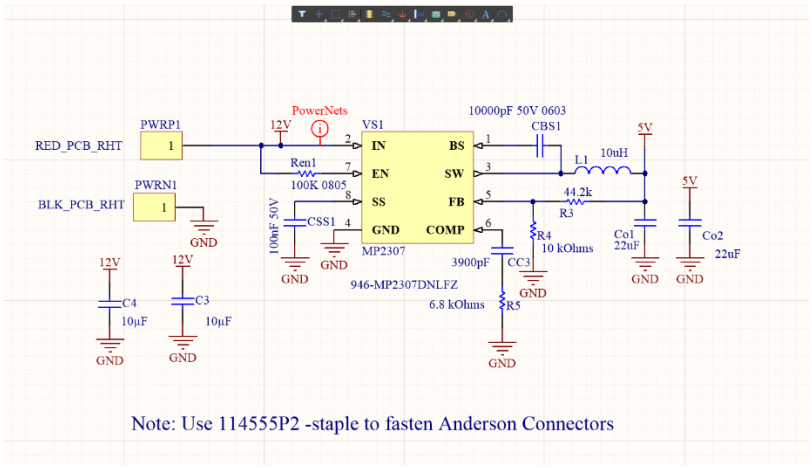


Figure 238. Placing the PowerNets directive

18. Perform a **Design » Update PCB Document Rules.PcbDoc**.
19. Click **Execute Changes** and **Close** as shown in Figure 239. The focus will automatically change to the PCB file.

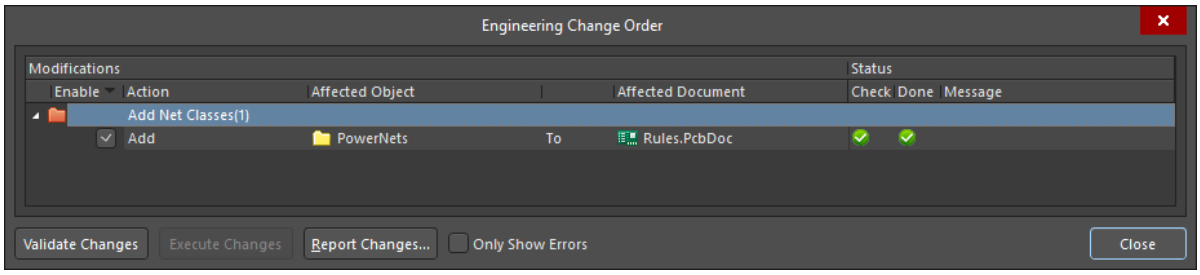


Figure 239. Engineering Change Order (ECO) showing the addition of the PowerNets NetClass

If you add a new net class directly from the PCB, they will not be present on the schematic. When running the Engineering Change Order, you may be prompted of a possible removal of the net class. Simply untick the box to prevent the removal within, or edit the **Project » Project Options » ECO** generation tab to prevent any removal of net classes.

20. We will now add a Width rule for the *PowerNets* Net Class that we just added. Go to **Design » Rules**, then navigate to the **Routing » Width** design rule.
21. Add a new *Width* rule and assign the name and constraints according to Figure 240.

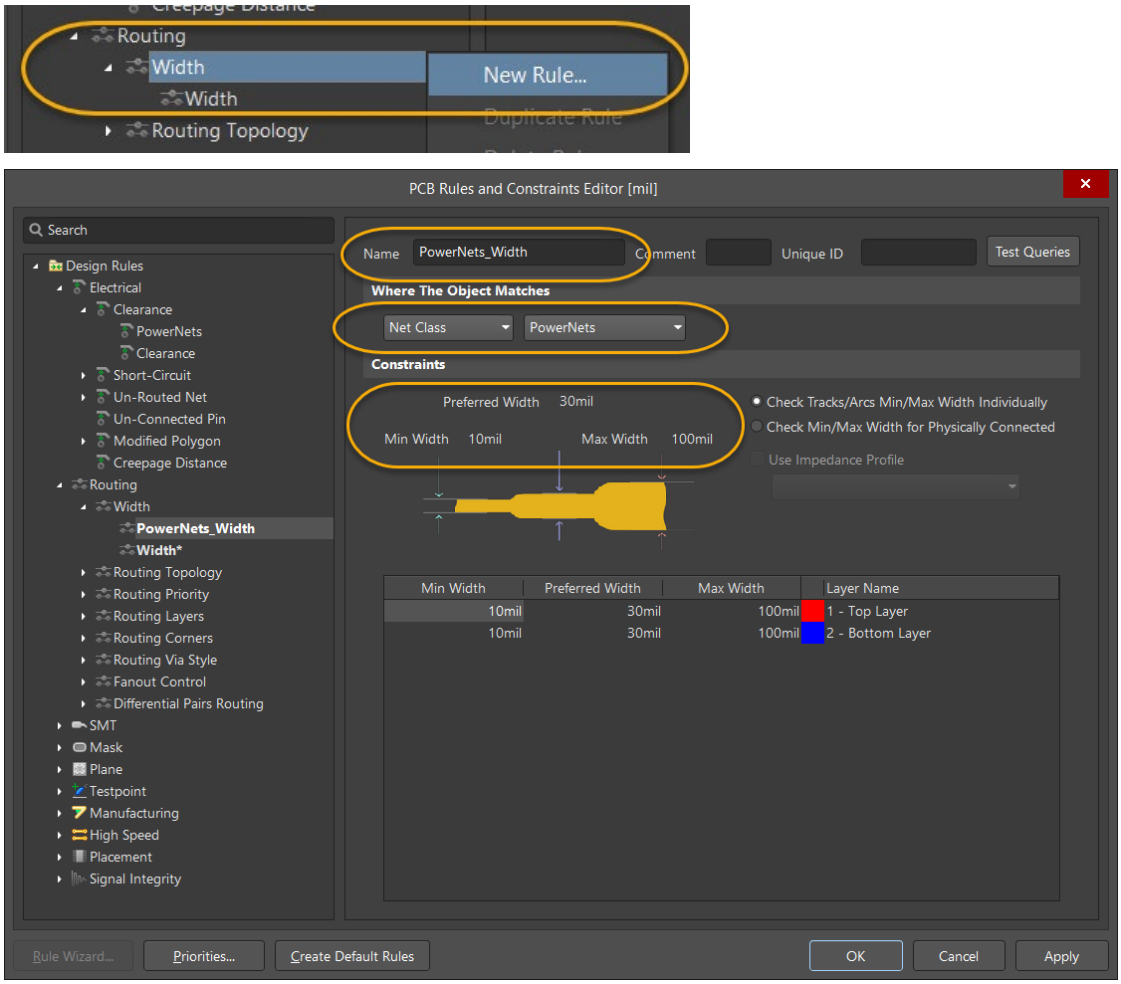


Figure 240. PowerNets Width Rule definition

22. Once again, ensure that the newly created *PowerNets_Width* rule is of the highest priority as described in Section 19.3.2 Rule Priorities.
23. Click **OK** to save and close *PCB Rules and Constraints Editor*.

19.5 Design Rule Using Custom Queries

19.5.1 Create Queries Using Find Similar Objects

We will now create a design rule which targets components of a specific footprint. We can build a custom query for this purpose, using **Find Similar Objects**.

24. While in the `Rules.PcbDoc`, right-click on component `RLY4` and click on **Find Similar Objects**.
25. Ensure that *Object Kind* and *Footprint* are set to **Same**, and the **Create Expression** checkbox is ticked as shown in Figure 241. Then, click **OK**.

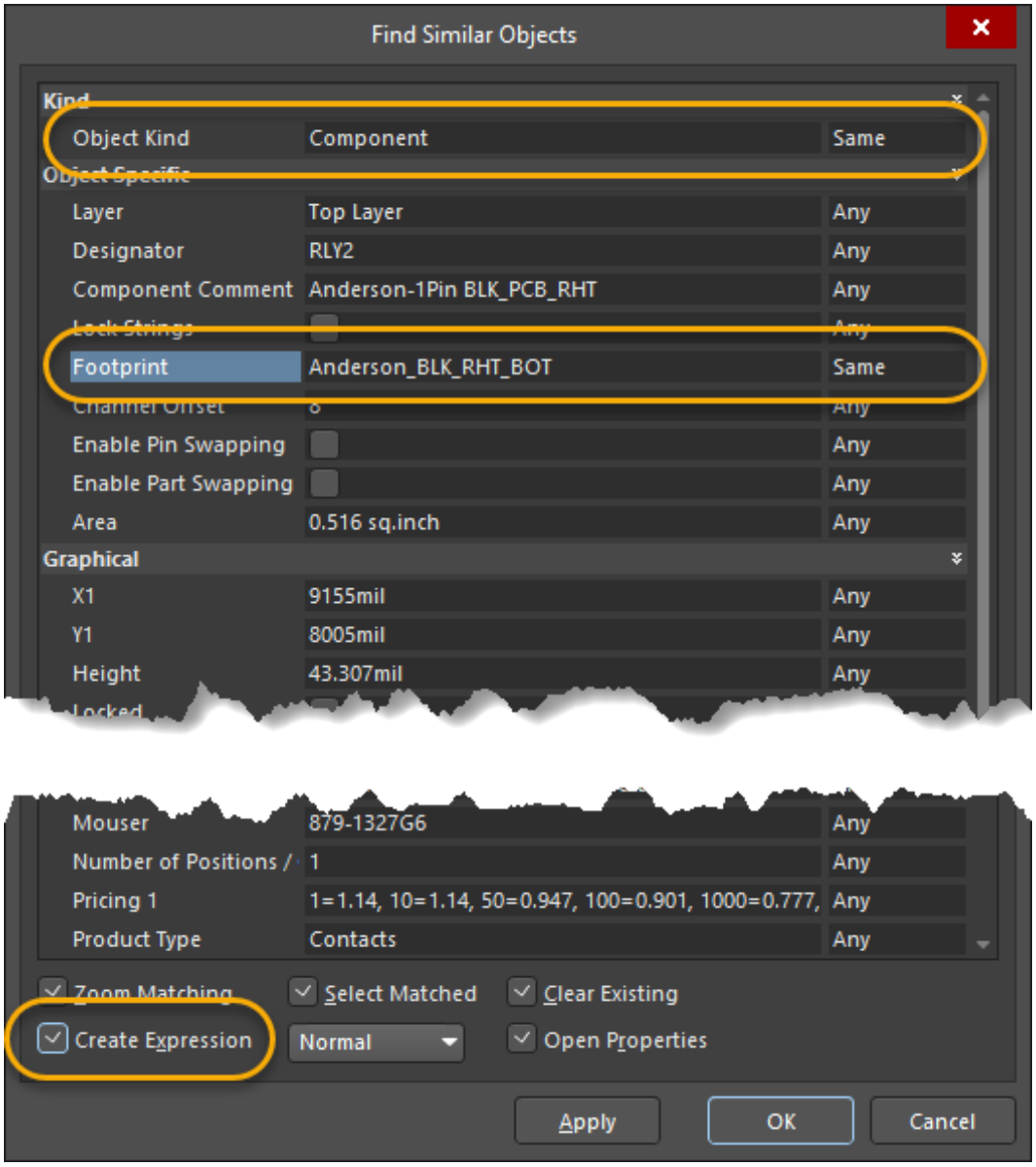


Figure 241. Find Similar Objects dialog

26. Open the *PCB Filter* panel by going to **Panels** button in the lower right corner and select **PCB Filter** or **View » Panels » PCB Filter**. Feel free to dock this panel wherever you wish.
27. Within this panel, the query created by running *Find Similar Objects* will be shown in the *Filter* pane. We'll now create a *Placement* rule that will apply to the selected components.
28. Within the *PCB Filter* panel, click on **Create Rule** near the bottom of the panel as shown in Figure 242.

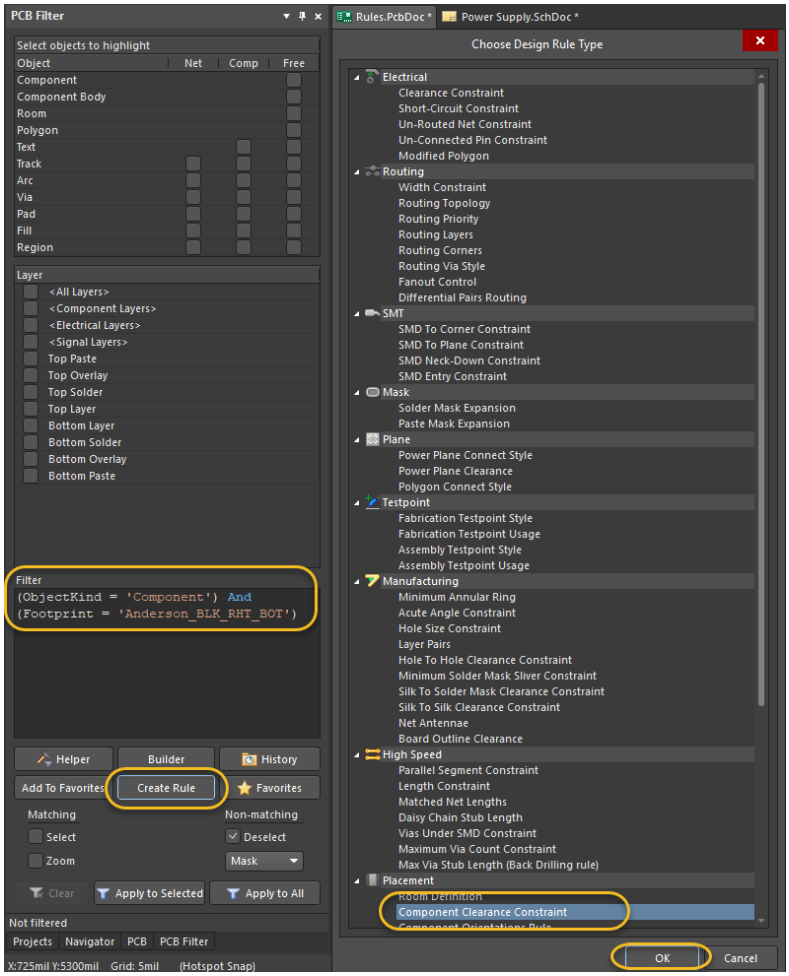


Figure 242. Creating a Clearance Rule from the PCB Filter panel

29. In the *Choose Design Rule Type* window, under **Placement**, click on **Component Clearance Constraint**. Click **OK**.
30. We will define a component clearance constraint between components of the `Anderson_BLK_RHT_BOT` footprint and components of the `Anderson_Red_RHT_BOT` footprint.
31. In the PCB Rules and Constraints Editor, define the rules through the following procedure. Refer to Figure 243 for the following steps:
    a) Copy the query from *Where The First Object Matches* field with **Ctrl+C**.
    b) Change the *Where The Second Object* Matches from **All** to **Custom query** and paste the query into the field using **Ctrl+V**.
    c) Then, change "`Anderson_BLK_RHT_BOT`" to "`Anderson_Red_RHT_BOT`". **Please keep in mind that the query is case sensitive**.
    d) Rename the rule to "`Anderson`"
    e) Set the minimum horizontal clearance to `100mil`.

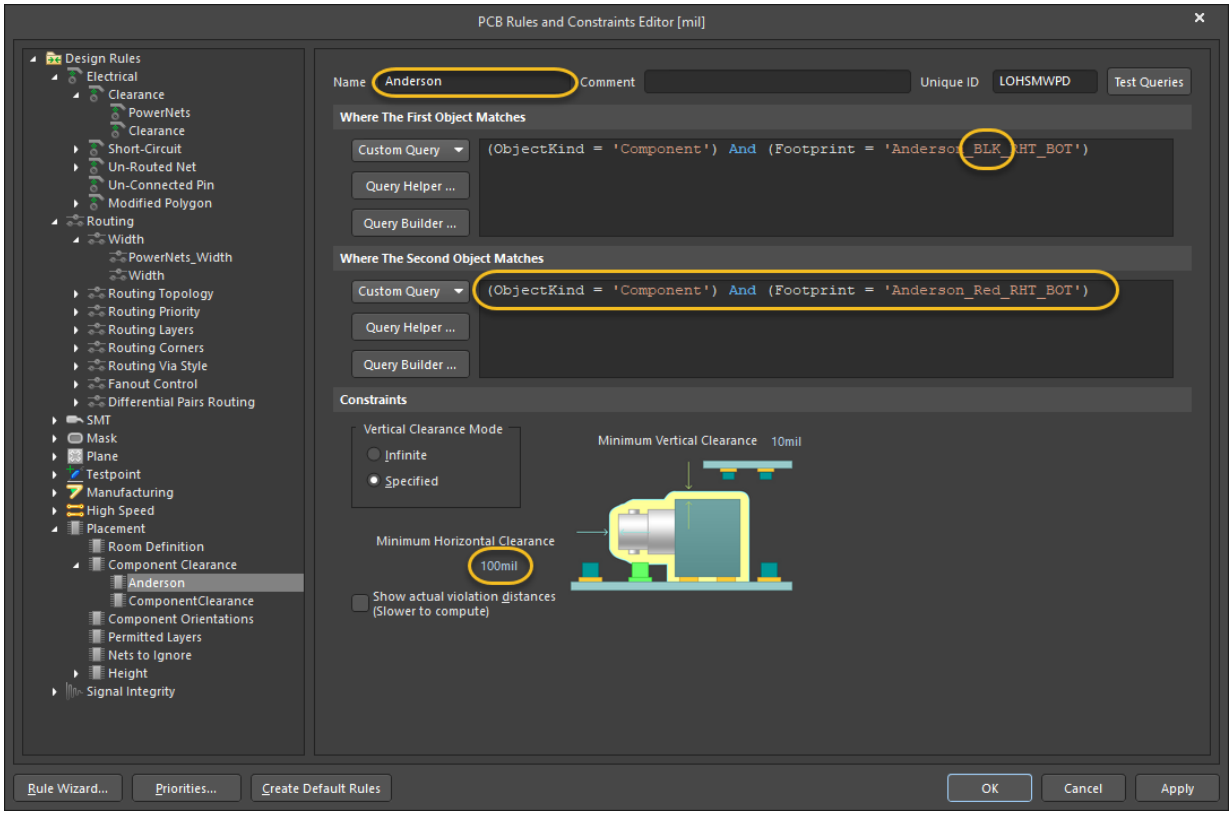


Figure 243. Anderson Component Clearance rule definition

32. Ensure that the newly created `Anderson` Component Clearance rule is of the highest priority as described in Section 19.3.2 Rule Priorities.
33. Click **Close** and then click **OK**.

19.5.2 Verifying Applied Design Rules in the PCB

34. Go to **Preferences**. Under the *PCB Editor* branch, open the *General* page. Ensure that **Online DRC** is enabled. Click **OK**. Hit **Shift+C** to clear any masks or selection.
35. Notice that 5 components (`RLY4`, `RLY3`, `PWRP1`, `PWRN1`, `RLY1`) have green highlights as they are violating the new *Component Clearance* constraint. Moving the components will remove or add the green highlight depending on the distance between each other.
36. Drag `RLY4` away from the other components to remove the violations on `RLY4` and `RLY3`.
37. Drag `PWRN1` towards `RLY4`; while dragging, hit **R** to change the component pushing mode to *Push:Avoid*, as indicated in the Heads-up Display. Drop it to the right of `RLY4`.
38. Lastly, drag `RLY1` towards `RLY4` and drop it to the left of `RLY4`.
    Notice the clearance allowed between `RLY1` and `RLY4` is larger than between `RLY4` and `PWRN1`, because of the *Anderson Component Clearance* rule defined earlier. This can be seen in Figure 244 below.

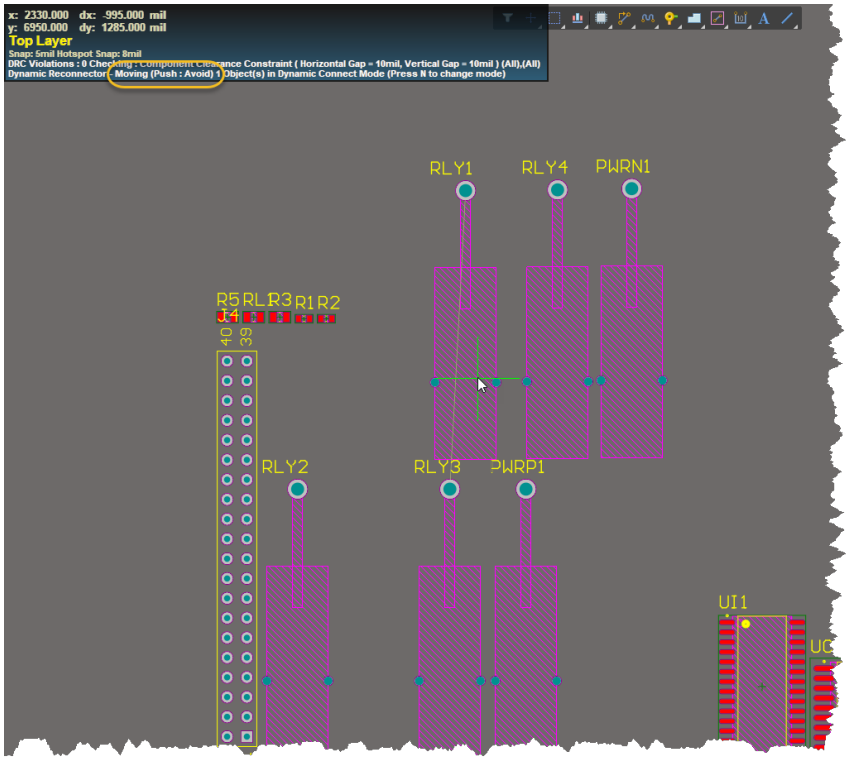


Figure 244. Applied Anderson Component Clearance constraint

19.6 Design Rule Wizard

41. We will now see how we can create design rules using the Design Rule Wizard.
42. Go to **Design » Rule Wizard…** and click **Next**.
43. Set:
    a) Name to `PwrWidth`
    b) Comment to `Need wider traces for power`
44. Select *Width Constraint* as shown in Figure 245 and click **Next**.

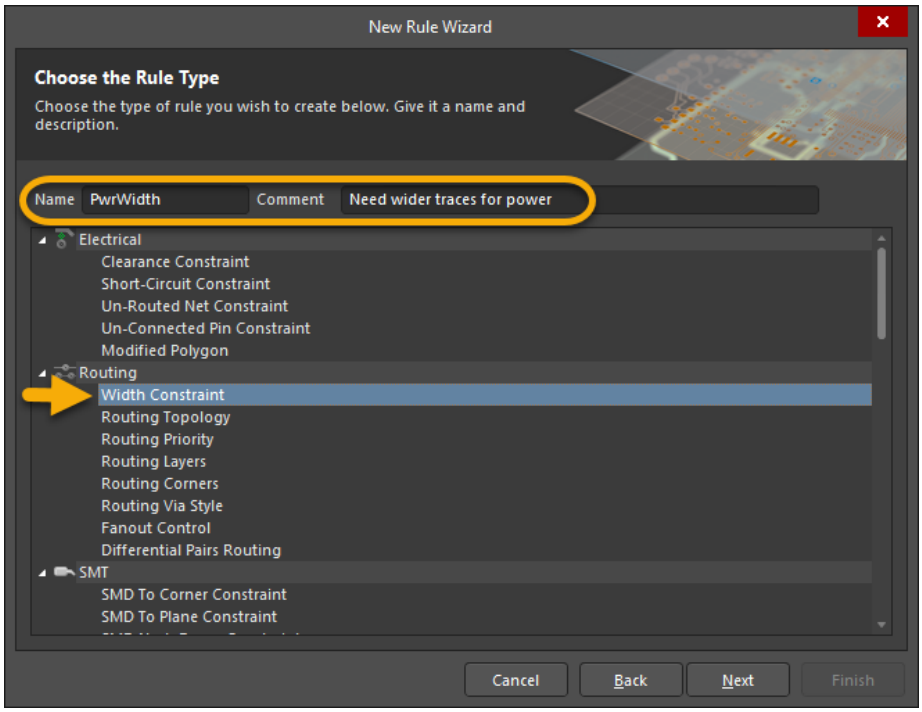


Figure 245. Rule Wizard rule type setup

45. Select *A Few Nets* for the *Rule Scope* and click **Next**.
46. Within the *Advanced Rule Scope*:
    a) Change the first *Belongs to Nets* condition value field to *5V*.
    b) Change the second *Belongs to Nets* condition value field to *12V*.
    c) Right click on the third *Belongs to Nets* condition and click on **Delete**. Repeat the step for the fourth condition as well.
    d) Ensure that the operator is set to **OR**. See Figure 246 for the final configuration of the rule.

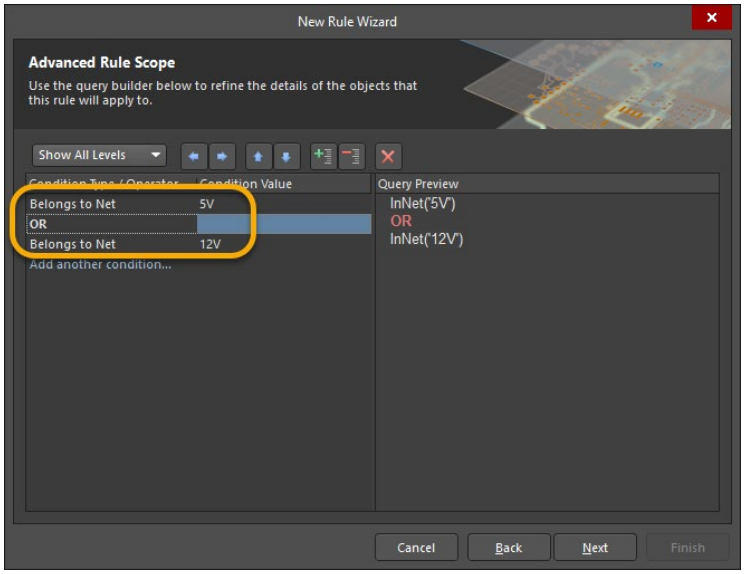


Figure 246. Rule Wizard configuration

47. Click **Next**. Verify that its priority is the highest on the list (#1).
48. Click **Next**. Check the **Launch main design rules dialog** checkbox, then click **Finish**.
49. The *PCB Rules and Constraints Editor* will appear with the new rule. We can then modify the minimum, preferred and maximum width constraints for this rule as shown in Figure 247.

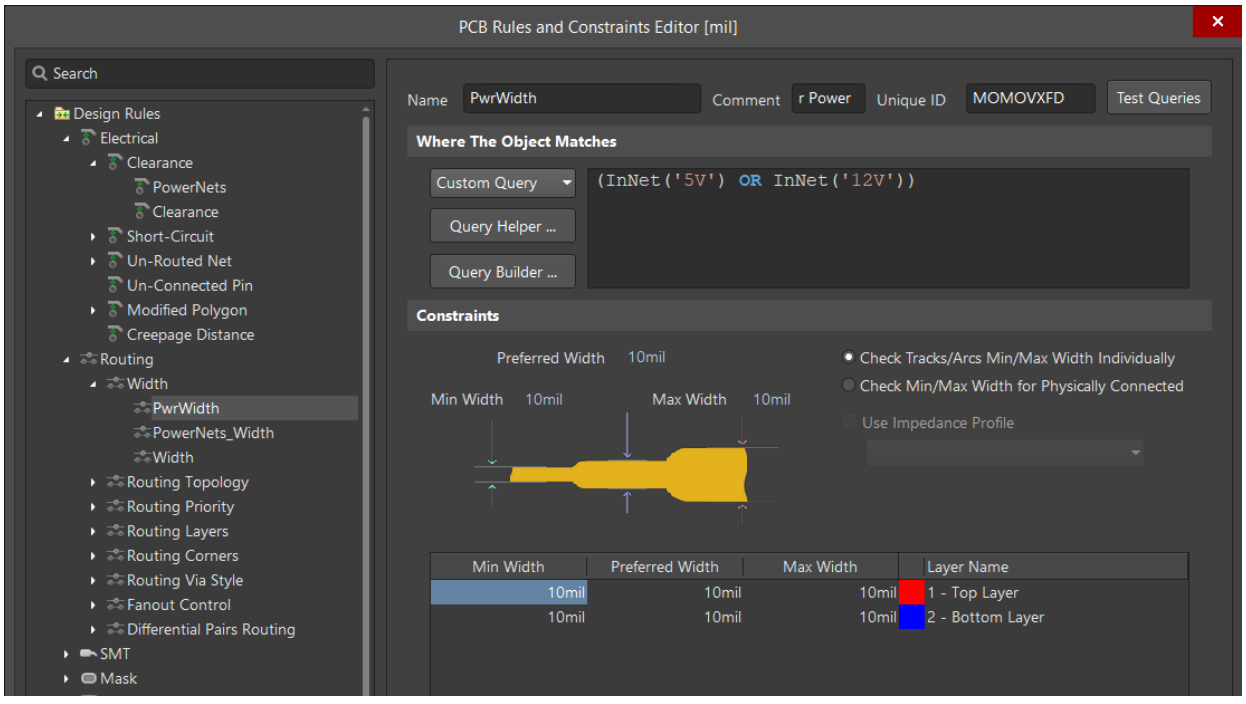


Figure 247. New PwrWidth rule created

The *PowerNets_Width* rule which we created in Section 19.4 Design Rule Using Net Classes is now invalid and superseded by the *PwrWidth* rule. Objects within the 12V nets will now only be applied by the *PwrWidth* rule.

50. Click **OK** to close the Design Rules dialog. Save and Close all documents.

Module 20: PCB Placement

20.1 Purpose

This exercise will allow you to practice different techniques for component placement.

Component placement directly determines the degree of difficulty for routing and the performance of the final product.

20.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `Module 20 - PCB Placement` folder.
2. Open `WCTopping.PrjPcb` from the `Module 20 - PCB Placement` folder.

20.3 Shortcuts

Rotate Counter Clockwise (shortcut **SPACE**)
Flip Layer (shortcut **L**)
Change Pushing Mode (shortcut **R**)
Net Line Connect Mode (shortcut **N**)
Reposition selected Components (shortcut **T » O » C**)
Arrange within rectangle (shortcut **T » O » I**)
Edit » Move » Move (shortcut **M » M)**
Edit » Move » Component (shortcut **M » C)**

20.4 Placement of Components Within Rooms

3. Open the PCB file `WCTopping.PcbDoc`.
   Notice all components are within the rooms placed outside of the board towards the right.
4. Test out the component behavior when a Room is moved using Figure 248 below as a reference.
   a) Drag one of the Rooms, such as `Relay_IO`. Notice the behavior of all components being moved simultaneously when the room is moved.
   b) Repeat for other rooms that have not yet been moved and notice the behavior.

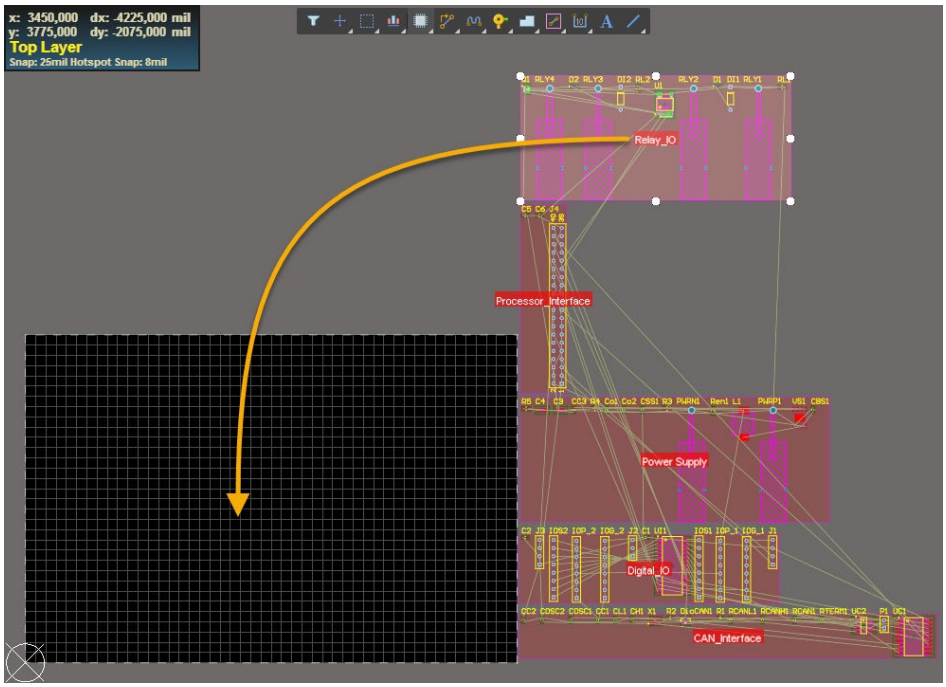


Figure 248. Moving Rooms and Footprints

5. Set the Pre-selection filter from the *Active Bar* to only activate *Rooms* as shown in Figure 249.
6. Then, draw a selection rectangle around all the rooms. Hit the **Delete** key to delete all of them, leaving the components remaining on the PCB document.

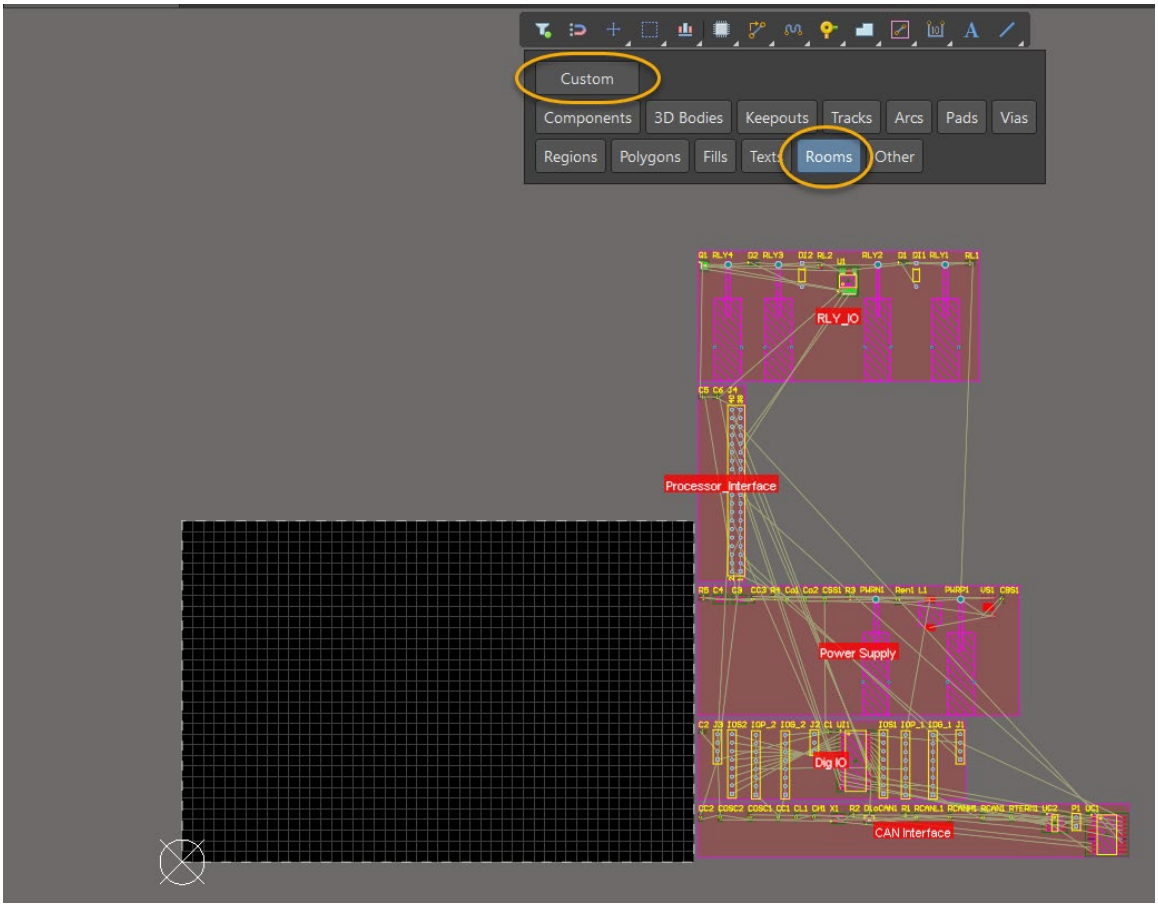


Figure 249. Set Pre-Selection filter to Rooms only

20.4.1 Placing within an Area

7. Enable **All-On** in the Pre-selection filter on the *Active Bar* shown in Figure 250.

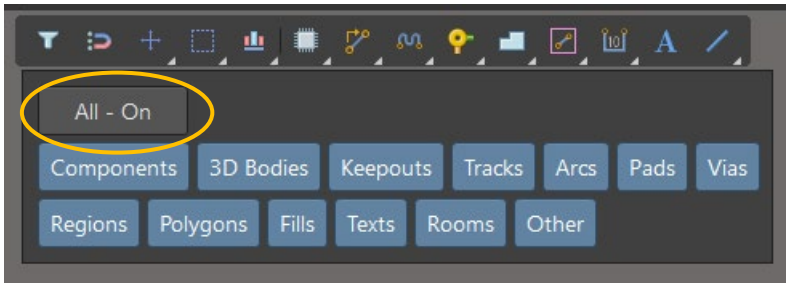


Figure 250. Enable All Objects in the Selection Filter

8. Select all the components in the design by drawing a rectangle around them.
9. Use the command **Tools » Component Placement » Arrange within Rectangle**.
   a) Define a rectangle of the entire board outline by:
      i) First click in the upper left corner of the board.
      ii) Next, click towards the bottom right corner.
      iii) Right Click to exit command.
   b) Compare your result with Figure 251 below. This illustrates how you can get any selected group of components within a specific area of your board to begin placement. It is not necessary for the drawn box to be large enough for components, a small box would do.

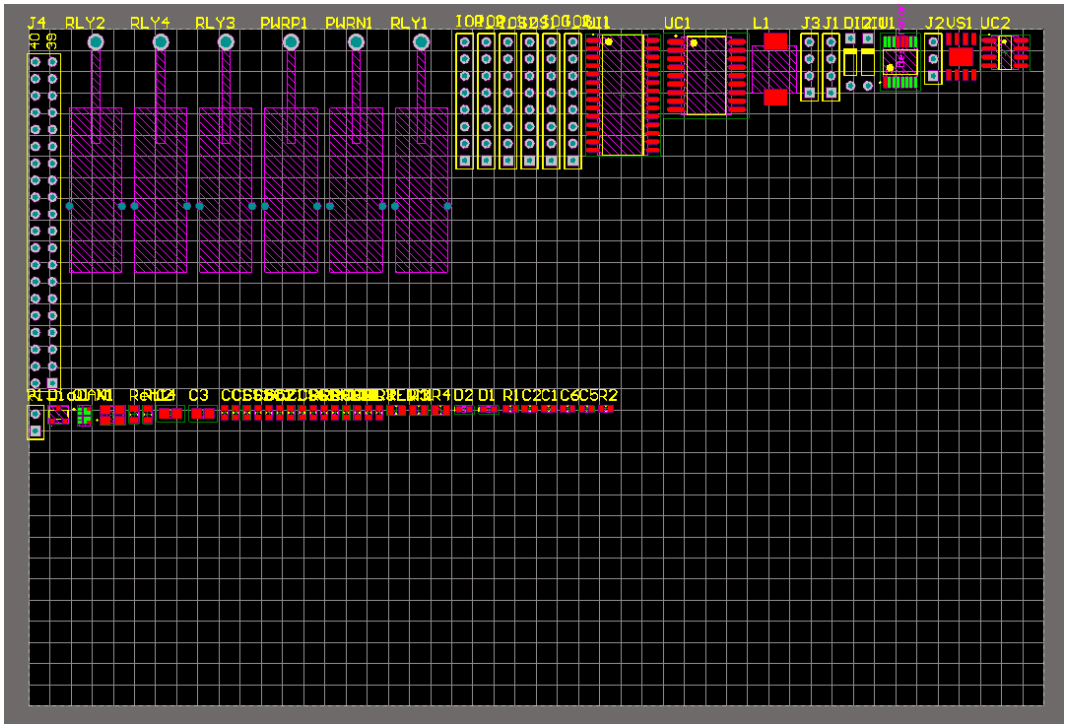


Figure 251. Component Placement within the board area

10. We'll now position all the components outside of the board.
    a) With all the components still selected, click **Tools » Component Placement » Arrange Outside Board.**
    b) **Shift+C** to clear component selection. See Figure 252 to see the end result.

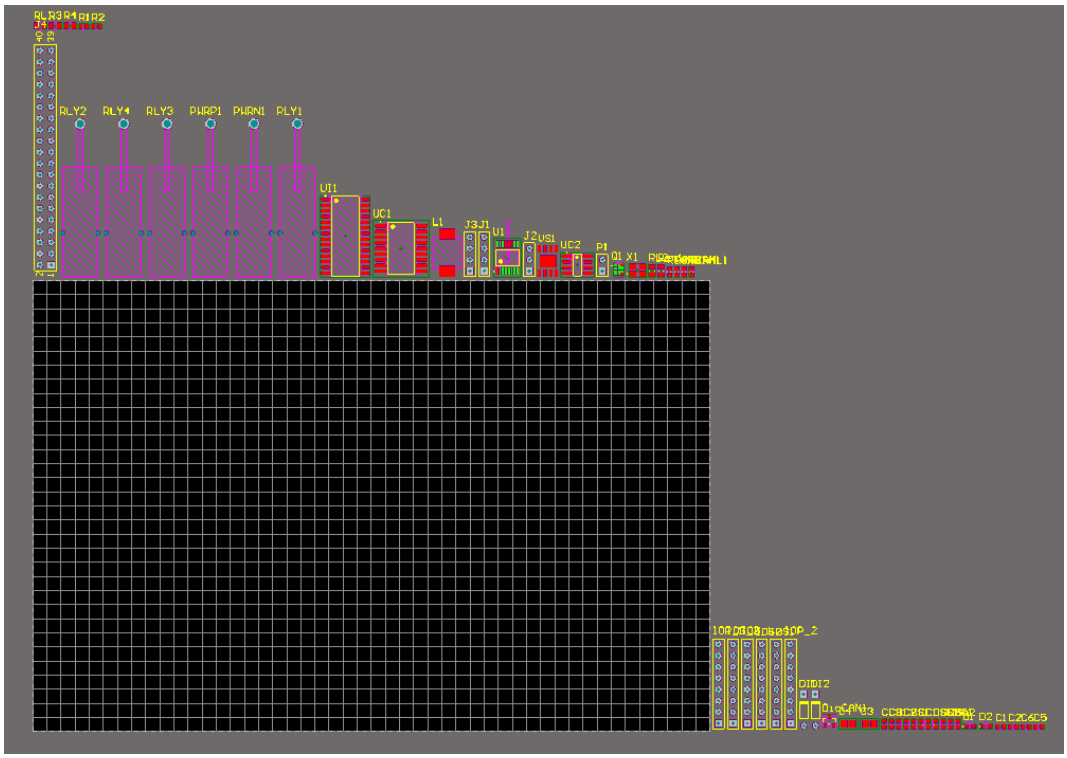


Figure 252. Components Arranged Outside of the board

20.4.2 Placing Individual Components

11. Click on the 40-pin header, component `J4`, and drag it to the position shown in Figure 253.

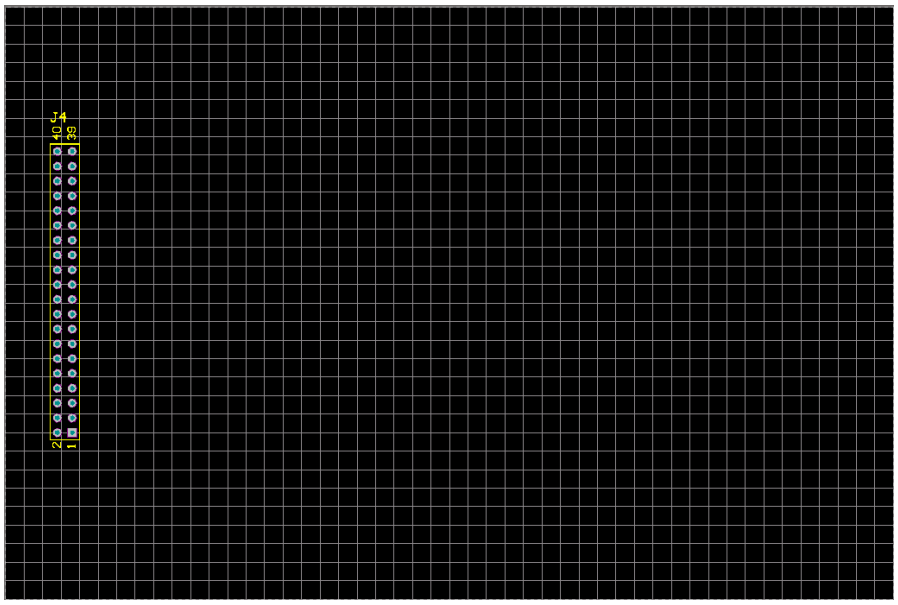


Figure 253. Manually place J4 within the board area

12. Continue placing components using the drag method.
    a) Use the Move command using the **Edit » Move » Move** (**M**, **M** hotkeys) and left-click on component `UI1` and move to its new position as shown in Figure 254.
    b) During component movement, release your left mouse. The component remains on the cursor.
    c) **Left-click** to position the component – no hold and drag required.

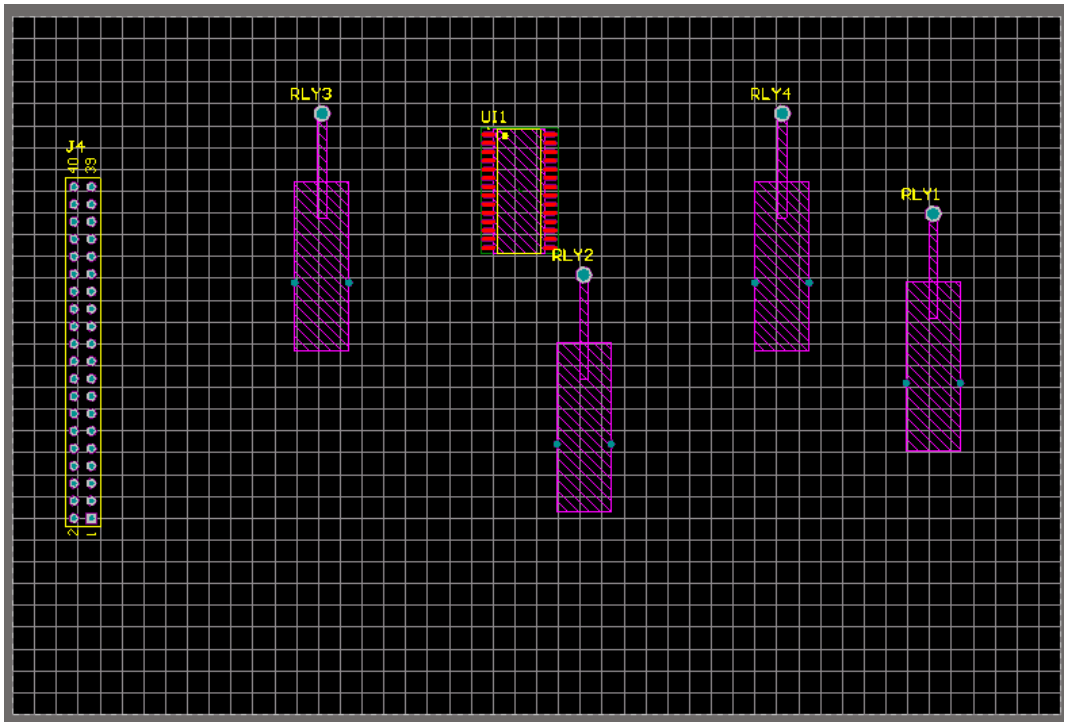


Figure 254. Placing other components with Edit » Move » Move

While moving a component, press **Spacebar** to rotate the component counter-clockwise and press **Shift + Spacebar** to rotate the component clockwise.

13. Position components `RLY1`, `RLY2`, `RLY3` and `RLY4` so that the result looks similar to Figure 254.

Connection lines will appear when dragging a component to show the same net connections between the footprints. While dragging a component, press **N** to toggle the visible component net connections ON/OFF. To hide all connection lines, use **View » Connections » Hide All.**

20.4.3 Cross Select Mode

Tools » Cross Select Mode used in conjunction with the PCB Editor's **Tools » Component Placement » Reposition Selected Components** command is a very effective way to place components by selecting them from the schematic.

14. Open *Preferences*, in the upper right corner of Altium Designer.
15. Navigate to the section *System - Navigation* and verify that the **Cross Selection** checkbox is enabled under *Cross Select Mode*.
16. Enable Cross Select Mode from **Tools » Cross Select Mode**. If the icon of this menu item already has a blue outline around it, then Cross Select Mode is already enabled as shown in Figure 255.

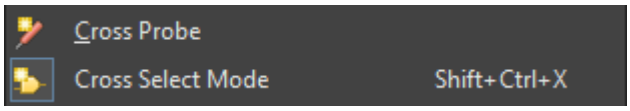


Figure 255. Blue highlighted icon indicating Cross Select mode is enabled

17. In `Processor_Interface.SchDoc`, select capacitors `C5` and `C6` sequentially.
    a) Hold down **Shift** while left-clicking on the component bodies as shown in Figure 256 below.
    b) The order you select them is important as this will determine the placement order in the PCB.

Be sure to click on the body of the capacitor rather than just the designator or comment. The component will not be selected in the PCB if the body of the symbol is not selected.
Change the Grid (**G key**) if it is difficult to select the body from the capacitor.

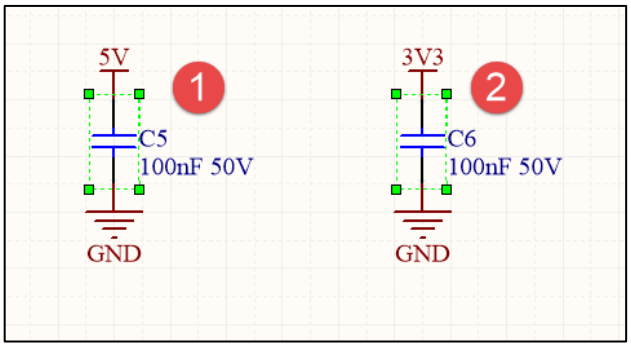


Figure 256. Selecting Capacitor symbol bodies

18. Bring into focus the `WCTopping.PcbDoc`. In the PCB Editor, make sure the corresponding footprints are currently selected. Access the command **Tools » Component Placement » Reposition Selected Components**. Component `C5` should automatically jump to your cursor for placement.
19. Click to place `C5` in the bottom left corner as shown in Figure 257, using the **Spacebar** to rotate.
20. Component `C6` should now automatically jump to your cursor. Place it in accordance with Figure 257.

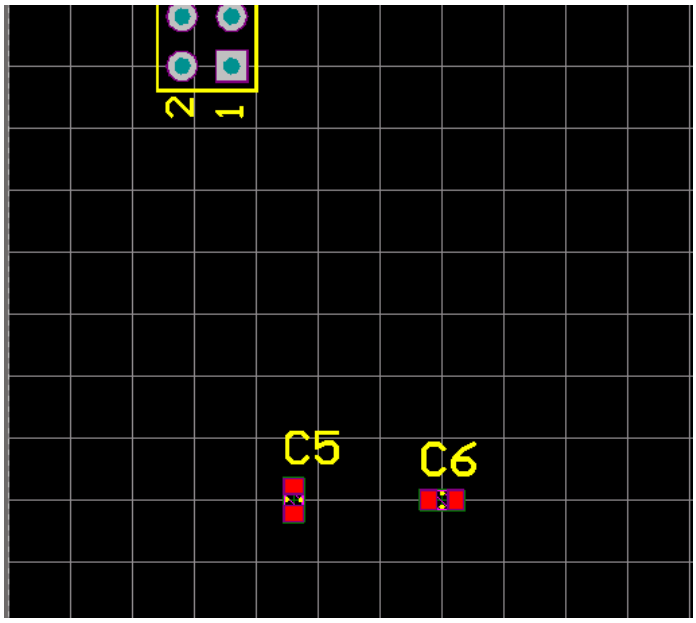


Figure 257. Positioning components C5 and C6 loosely

21. Now we will position the leftmost capacitor exactly in the position we want.
    a) Double click the capacitor, C5, to open its *Properties* panel dialog.
    b) Change the (x, y) position to (200mil, 100mil) as shown in Figure 258.
    c) Press **Enter** to accept the change.

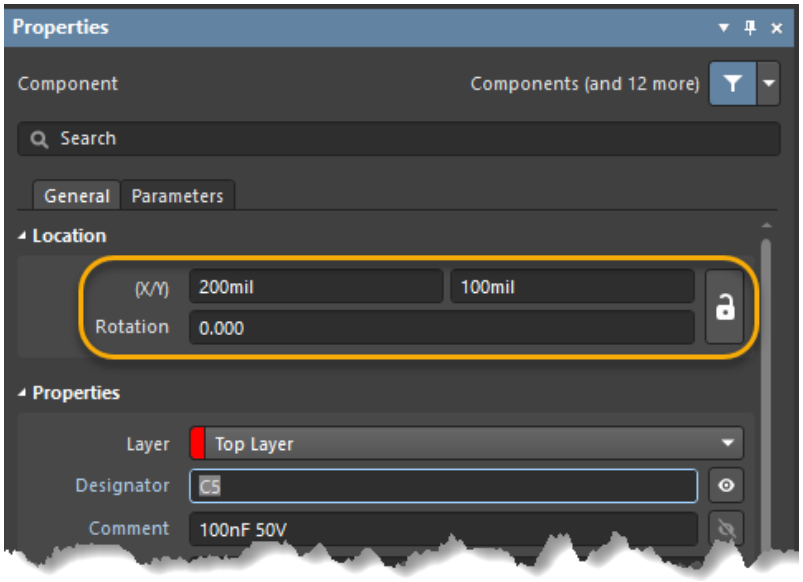


Figure 258. Placing C5 at a specific location on the board

22. We will now position the component C6 using the **Move** command.
    a) Invoke the Object Move command by pressing **M » M** hotkeys.
    b) Click on C6
    c) With the component now attached to the cursor, use the **Jump » New Location** command.
        i) Press the **J » L** hotkeys and enter (X, Y) values of (500mil, 300mil) as shown in Figure 259 below.
        ii) Select the **OK** button with the **TAB key,** press **Enter** to accept the jump values and press **Enter** again WITHOUT MOVING THE MOUSE to fix the position. **Right click** to exit the Move command

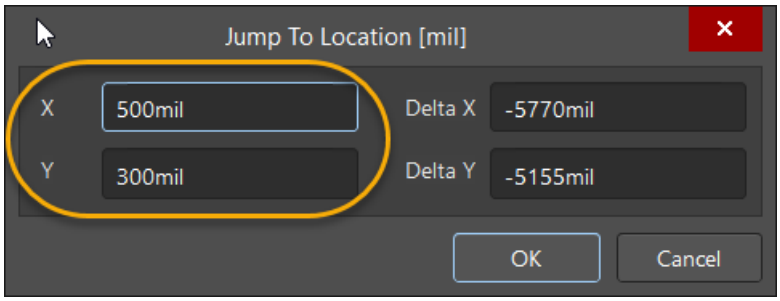


Figure 259. Jump to Location dialog

23. Place UI1 outside of the board.
24. Align the Relay Components:
    a) Select components RLY1, RLY2, RLY3, and RLY4.
    b) Use the command **Edit » Align » Distribute Horizontally**. The components should be placed approximately as shown in Figure 260.

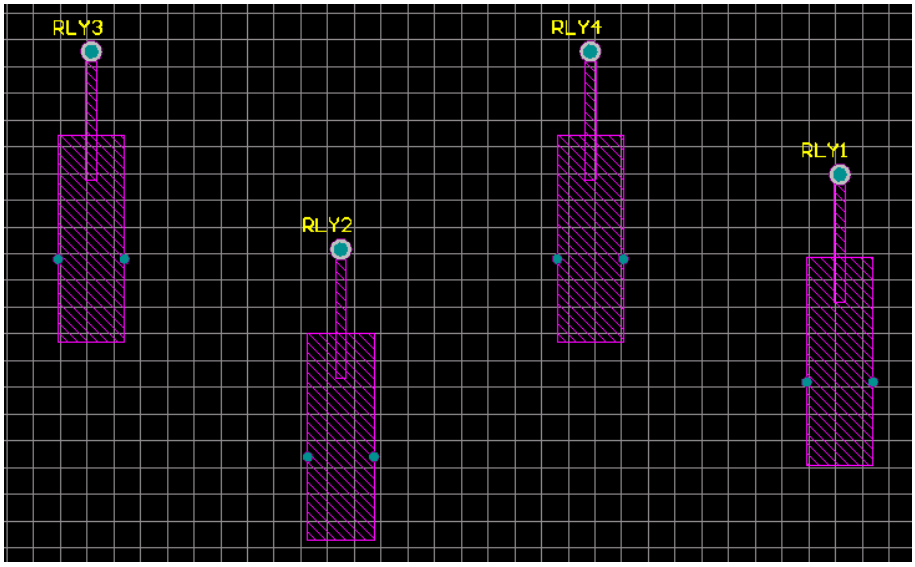


Figure 260. Components distributed evenly horizontally

25. With the components still selected, use **Edit » Align » Align Vertical Centers** then choose the component at either end as the primitive reference. Be sure to click between the pads. See Figure 261. Right-click to exit the 'Align' command.

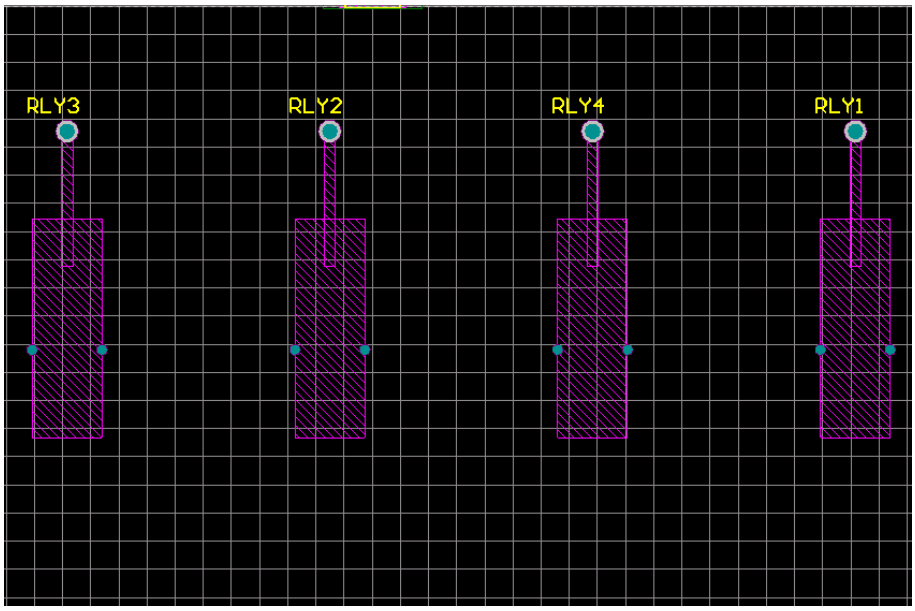


Figure 261. Components aligned with vertical centers

Alignment and move commands can be performed on component properties, such as Designators. Hold **Ctrl** while selecting to select.

20.4.4 Position Remaining Components

26. To place the remaining unplaced components (Please bear in mind that the PCB does not have to be completely placed, due to time restrictions):
    a) Use click and move method:
        i) Click in the center of a component and hold down the left mouse button.
        ii) The component will now move with your cursor.
        iii) Release the mouse button at the desired location.
    b) Placement from the *PCB* panel*:*
        i) Open the *PCB* panel.

ii) Set the Drop-down at the top to **Components**.

iii) For drag and drop, in the select boxes at the top of the panel, uncheck **Select** and check **Clear Existing**. Feel free to Uncheck Zoom if this makes your flow easier.

iv) Click on the *Component Class* `<Outside Board Components>`.

v) All outside board components will be listed in the *Components* pane.

vi) In the *Components section*:
Click a component and drag it directly to the right and the component will pop onto your cursor for placement.

vii) Release the mouse at the desired location.

c) Multiple Placement from the *PCB* panel*:*

i) Try selecting multiple components in the pane using **Ctrl + Click**.

ii) Drag one of the selected components to the right will begin a reposition selected components for the components that had been selected.

iii) Place the components in the PCB as they pop onto your cursor.

27. For the remainder of the components, in the *PCB* panel, enable the **Select** checkbox at the top. The *<Outside Board Components>* class should still be selected, so just click the **Apply** button. This will select the remaining components outside the board area.
28. Use **Tools » Component Placement » Reposition Selected Components** to sequentially load up the cursor with the unplaced parts. Place 5 or 6 footprints for practice. Right click to escape the *Move* command.
29. Refer to Figure 262 as a sample of a finished PCB component layout for the training.
30. Once complete, save and close all documents.

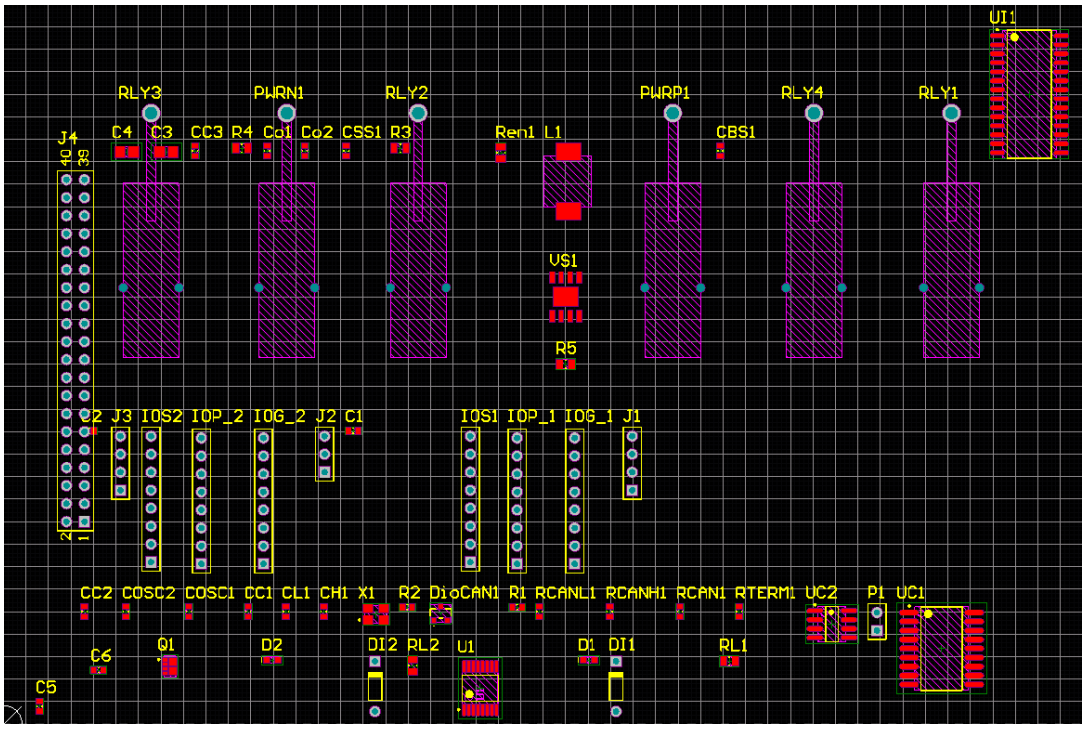


Figure 262. End result of the PCB layout

Module 21: PCB Routing

21.1 Purpose

In this exercise, you will practice routing using various PCB routing modes. Within each mode, routing options and tips will also be discovered.

Routing is essentially the process of making electrical connections through tracks and vias. Altium Designer provides a series of powerful interactive routing tools. The main aspects of Interactive Routing (Clearance, Width, Allowed Layers and Via Style) are driven and defined by your Design Rules.

21.2 Preparation

1. Go to **File » Open**, and head to the `Module 21 - PCB Routing` folder of the `AD 20 Essential Exercises` folder, and open `WCToppingRouting.PcbDoc`

21.3 Shortcuts

Routing Modes:
Interactive Routing (shortcut **U»T** or **CTRL+W**)
Interactive Differential Pair Routing (shortcut **U»I**)
Active Route (shortcut **SHIFT+A**)

During Routing:
Help (shortcut **F1** (Online Help)
Shortcuts List ~ (US) or **Shift+F1** (other regions)
Elbow Side (shortcut **Space**)
Corner Style (shortcut **Shift+Space**)
Routing mode (shortcut **Shift+R**)
Track Width (shortcut **Shift+W** or **3**)
Via Size (shortcut **Shift+V** or **4**)
Layer Change (shortcut **CTRL+L** or regular layer change method)
Remove Last Segment (shortcut **BACKSPACE**)

Unroute:
Unroute Board (shortcut **U»U»A**)
Unroute Net (shortcut **U»U»N**)

21.4 PCB Routing

21.4.1 Setting Preferences

2. Before routing, it's good to be aware of some of the preferences which can affect its behavior.
   a) Press **O - T** to open the *Preferences* dialog to the *PCB Editor* branch on the *Interactive Routing* page.
   b) Set the following options as follows:
      - *Automatically Terminate Routing*: **On**
      - *Automatically Remove Loops*: **On**
      - *Track Width Mode*: `Rule Preferred`
      - *Via Size Mode*: `Rule Preferred`
   c) Click **OK** to apply and close the *Preferences* dialog. In the lower right corner, go to **Panels » View Configuration**. In the *Layers & Colors* tab, hide the Mechanical Layers by clicking on the "eye" icon . You can close this panel.

21.4.2 Quick Routing for Simple Nets

In this section, we'll hold the **Ctrl** key and left-click to quickly complete simple, local routes.

3. Jump to component `UI1` by hitting **J » C** and type `UI1`. Without moving your cursor's location, zoom into `UI1`.
4. To start the Interactive Router, press the **U - T** shortcut keys, select the **Route » Interactive Routing** menu command or click the Interactive Routing button on the *Active Bar* .
5. Hold the **Ctrl** key, then left-click on one of the pads, e.g. `Pad 28 of UI1`. If the route is possible on the selected layer, it will be instantly routed.
6. Continue to hold the **Ctrl** key and try another net, e.g., `UI-27`, `UI1-26`, `UI-25`. Right-click or hit the **Escape** key to exit the interactive routing command.

21.4.3 Interactive Routing Nets

7. Zoom into the area of the board toward the upper left portion of the PCB shown in Figure 263 below. We will now route the connections between components `P1, RTERM1, DioCAN1, CH1, CL1, and UC2` through the following procedure:

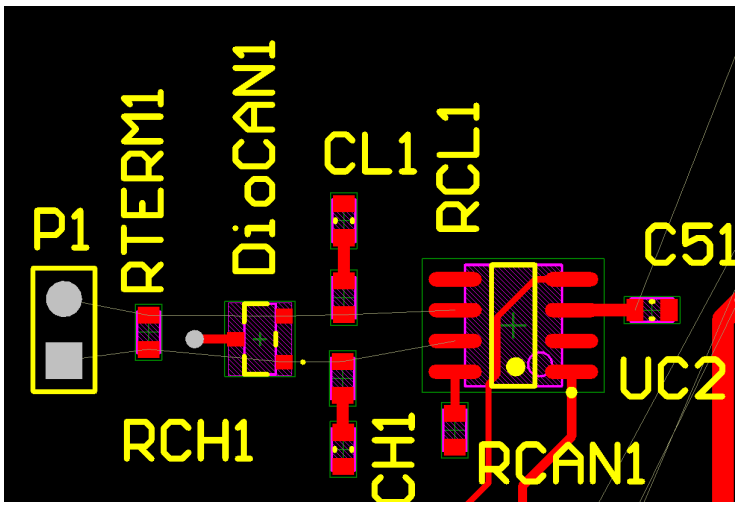


Figure 263. Connection lines (known as a Ratsnest) between un-routed components

   a) Start the route command with **Route » Interactive Routing** (or **U - T** hotkeys)
   b) The cursor will change to a command crosshair, indicating that you are in an interactive command.

c) Left-click on `Pad 2` of `P1` to begin the route. Move the mouse away from the pad to see the tracks the Interactive Router is planning to place.

d) Move your mouse to `Pad 1` of `RTERM1` and left-click to terminate the route. Upon terminating the route, we will still remain in the interactive routing mode and the editor is ready for the next net to route, as indicated by the active command crosshair.

8. While still in active interactive routing mode, left-click on `Pad 1` of `P1`.

a) Hit the **~ tilde** (US) or **Shift+F1** for other regions key to access all the shortcuts and commands while in this mode. Notice that 3 is the hotkey to cycle track-width source of this route as shown in Figure 264.

b) Press **Esc** to escape the shortcuts menu.

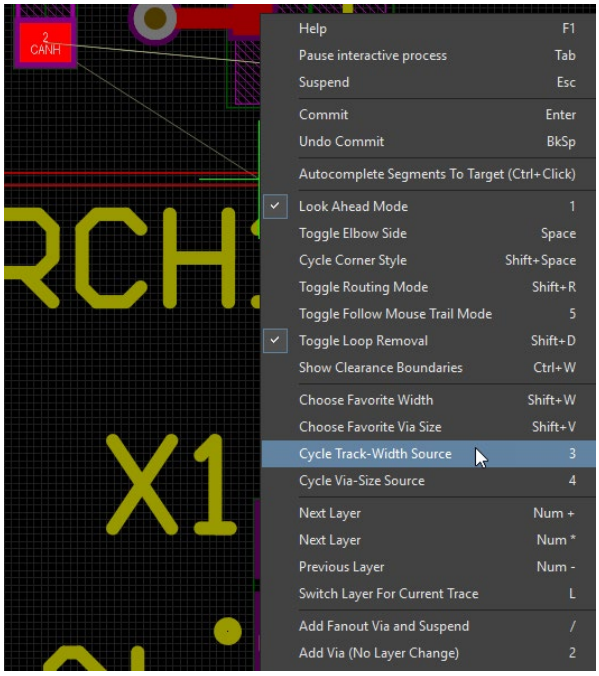


Figure 264. Next available option or shortcut from the Tilde menu

c) Hit the **3** key to cycle through the different trace width values set in the *Trace Width* design rule.

d) You can hit the **Spacebar** key to toggle the "elbow side" angle (Start/End toggle).

e) Hit **Shift + R** to cycle through the available Conflict Resolution modes.

f) Experiment with a few of the different routing modes, *Walkaround, HugNPush, Push, Ignore Obstacles* and *Stop at First Obstacle*. The current routing resolution mode is displayed in the *Status Bar* and the *Heads-Up Display* as shown in Figure 265.

Figure 265. Conflict Resolution modes displayed in the HUD

You can control which Conflict Resolution Modes are available by changing the preferences in the *PCB Editor* branch under the *Interactive Routing Page* and altering the preferences in the *Routing Conflict Resolution* section or by pressing **Tab** during a route.

9. Hit the **Backspace** key to remove the last placed vertex while still in the routing command.
10. During a route:
    a) Right-click once (or press **Esc**) to exit current route placement.
    b) Right-click again while no routing is active (or press **Esc**) to exit the Interactive Routing command completely.

21.4.4 Vias

21.4.4.1 Setting Via Sizes

To have multiple via sizes available during interactive routing, we'll modify the *Routing Via Style* Design Rule.

11. Open the *RoutingVias* design rule in **Design » Rules,** section *Routing - Routing Via Style*.
    a) Modify the *Via Diameter* and *Via Hole Size* values as shown in Figure 266. Note that the Maximum values are in the middle.
    b) Click **OK** to accept changes and close the dialog.

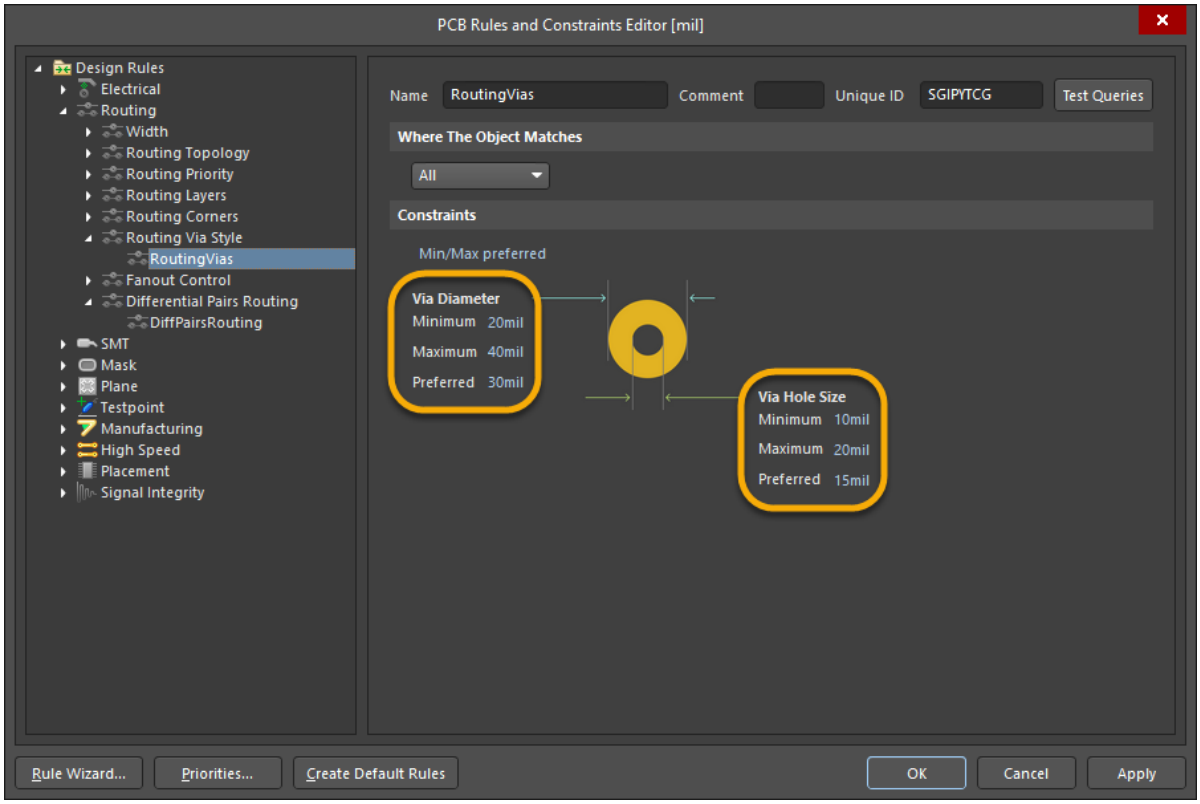


Figure 266. Changing the Routing Via Style design rule

21.4.4.2 Adding Vias

12. Start the interactive routing command by pressing **U - T**.
    a) Move the cursor and click on `Pad 13` of component `UC1`.
    b) Start routing and hit the **+ key** on the NumPad to drop a via and go to the next signal layer as shown in Figure 267. If you do not have a NumPad, hit the **~ tilde** key and click **next layer**.

c) Hit the **4** key to toggle through the three via sizes that you added in the design rule earlier.

d) After pressing the **L** key, a via should be added as shown in Figure 267. Click the left mouse button to place the via.

e) Continue finishing the route on the bottom layer. Once completed, hit **ESC** or right-click to end the command

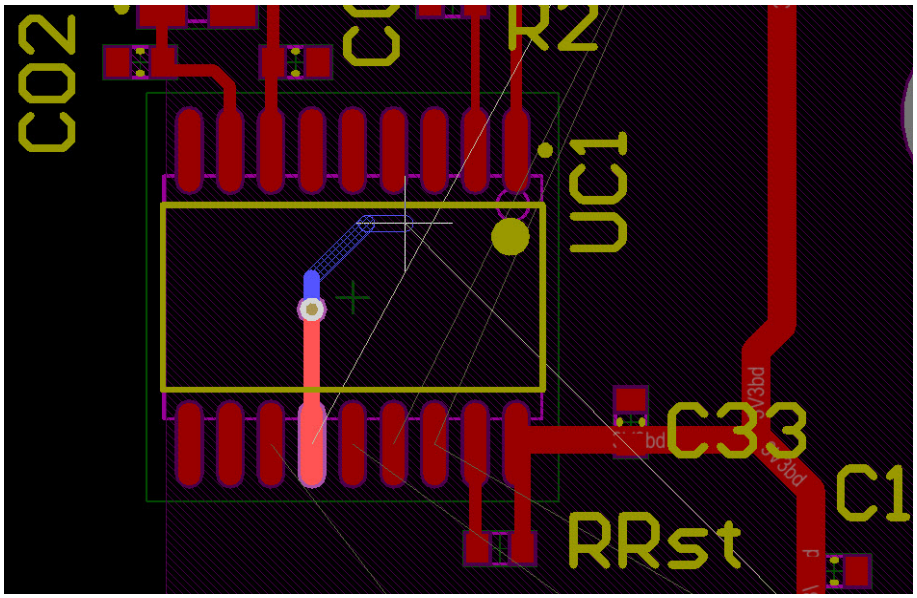


Figure 267. Changing Via Size while routing

The active layer will now be switched to the Bottom Layer, as shown on the current layer tab below. New routes will start on this active layer, unless if starting the route from a top-layer surface mount pad.

It is not necessary to memorize all the interactive routing options. Instead, hit the tilde (**~**) key located below the **Esc** button or the **F1** key to display all available options. This key works to display all possible command options within the PCB and Schematic Editors.

21.4.5 Multi-Trace Routing

To improve the routing efficiency for grouped signals, especially buses, you can use the Interactive Multi-Routing command.

13. Find `UC1` in the PCB document using the command **Jump » Component** (shortcut **J - C**).
14. Hold **Shift** and click on `Pads 12, 13, 14` and `16` to select multiple pads as shown in Figure 268.

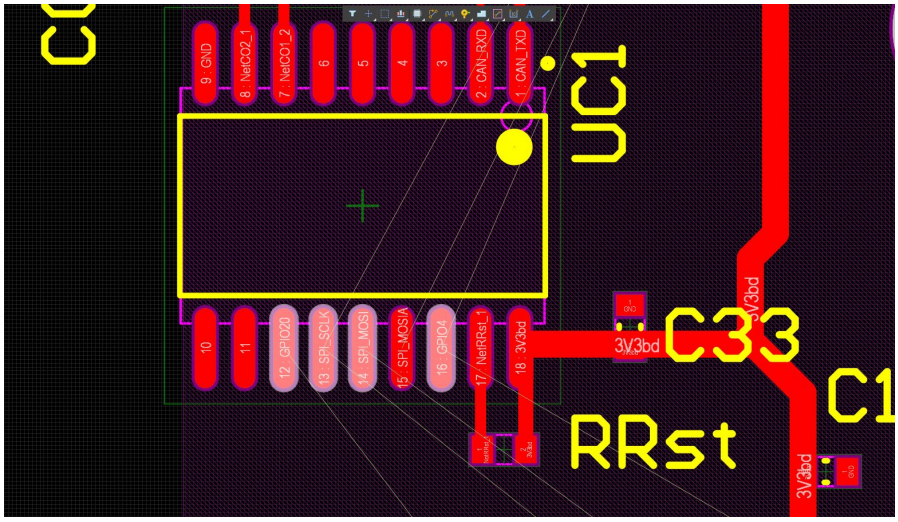


Figure 268. UC1 pads to be used for Multi-Trace routing

15. Start the multi-route from these pads using the command **Route » Interactive Multi-Routing** (**U » M**). The command crosshair will appear to indicate you are in an interactive command.
    a) Left-click either of the four pads to set the base track for the multi-route.
    b) Press **Tab** to open the *Properties* panel. Change the Bus Spacing to 10mil by pressing **From Rule (10mil)**. Also, while in the *Properties* panel, in the Interactive Routing Options, change the routing mode to "Walkaround Obstacles" mode. Hit the **Pause** button to resume routing.
    c) Move the cursor down towards `J4`, and you will see four tracks are routing simultaneously. If traces do not appear, press the **3** key to cycle the trace width until it changes to 10mil.
    d) Left-click to place vertical routes. Then, move the cursor and hover above the destination pad on the far right (`Pin 7` of `J4`) as shown in Figure 269. If you're satisfied with the routes, left-click to place them.

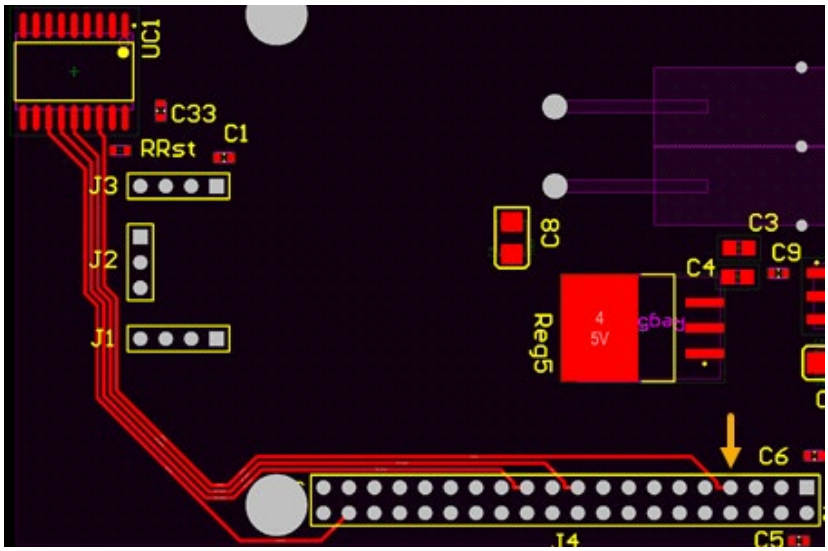


Figure 269. Multi-Trace Routing from UC1 to J4

During the Interactive Multi-Routing process, hit **Tab** and you can set the gap between the tracks. Or simply press shortcut **B** to reduce the gap, or **Shift + B** to increase the gap as you route. If there is not enough room, the multi-router will not complete the route.

21.4.6 Adjusting Tracks

16. If you need to adjust a route, rather than deleting it, we can route it again. *Automatic Loop Removal* will remove the old redundant tracks automatically. A good connection to try this on is the portion of tracks on the Top Layer between `J1` and `J4`.
    a) Start a route on the Top Layer from `J1 Pad 1` to `J4 Pad 5`.
    b) Re-route a more direct trace between the two pads.
    c) Once the route is finished, the Loop Removal removes the previously routed trace.
17. To drag a track to adjust it, exit the Interactive Router, and left-click the track to drag it.
18. If you do want to delete an entire track and start again, use the **Route » Un-route » Connection**, or just press **U - U - C** and select the track to remove. You can also left-click to select a segment of the trace and then press the **TAB** key to select all segments of that track on that layer

When moving or dragging tracks, we can change its behavior in the Preferences, under the *PCB Editor* branch and under the *Interactive Routing Page.*

21.4.7 Contour Follow Mode

This routing method allows you to quickly follow a shape already generated by the board edge, a cutout or even a through-hole pad or via.

19. In order to connect the 5V net from the J4 connector to the REG3 regulator, we are going to connect the net using C5.
    a) Locate Pad 2 on C5. Begin a route by using the shortcut **U »T**.
    b) Begin routing around the large via located nearest to C5, and as you approach the via, press **SHIFT + F** to enter Contour Follow Mode. This should cause the trace to follow the rounded edge of the via.
    c) Route around the edge of the via until you reach the top and click the Left Mouse Button to set an anchor point.
    d) Finish routing to Reg3 Pin 1 (5V) using all methods you have learned so far.

21.4.8 Track Glossing

Using some of the interactive and dynamic routing features may leave some undesired track segments or less effective route paths. We can use Track Glossing to clean up route paths after placement. Find a route on the PCB that you feel can use some cleanup.

20. To gloss a trace:
    a) Left-click to select a segment of the trace and then press the **TAB** key to select all segments of that track on that layer. (or **Edit » Select » Physical Connection** or **E » S » C**)
    b) Run Track Glossing by going to **Route » Gloss Selected**. See Figure 270 as an example.

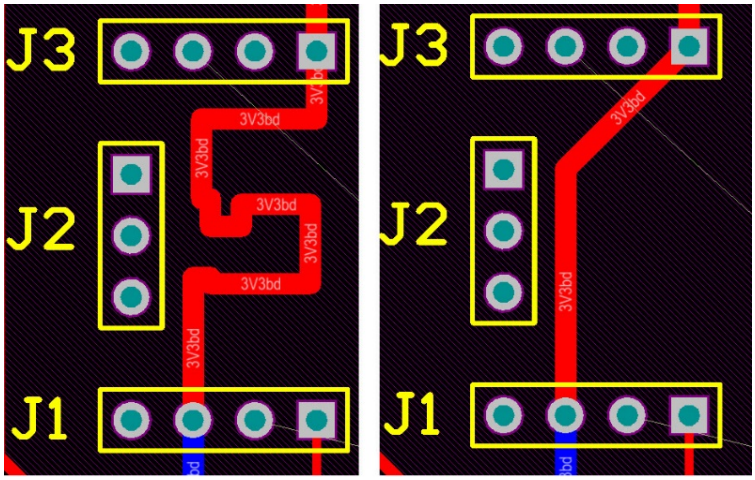


Figure 270. Gloss Tracks, Before (left) and After (right)

21.4.9 Active Route

21. Select the connection lines between `UI1(Pad 1 to 8)` and `IOS2(Pad 1 to 8)` by holding the **Alt** key followed by dragging your mouse from right to left around the connection lines to select them. This will create a selection rectangle and will highlight the connection lines as shown in Figure 271.

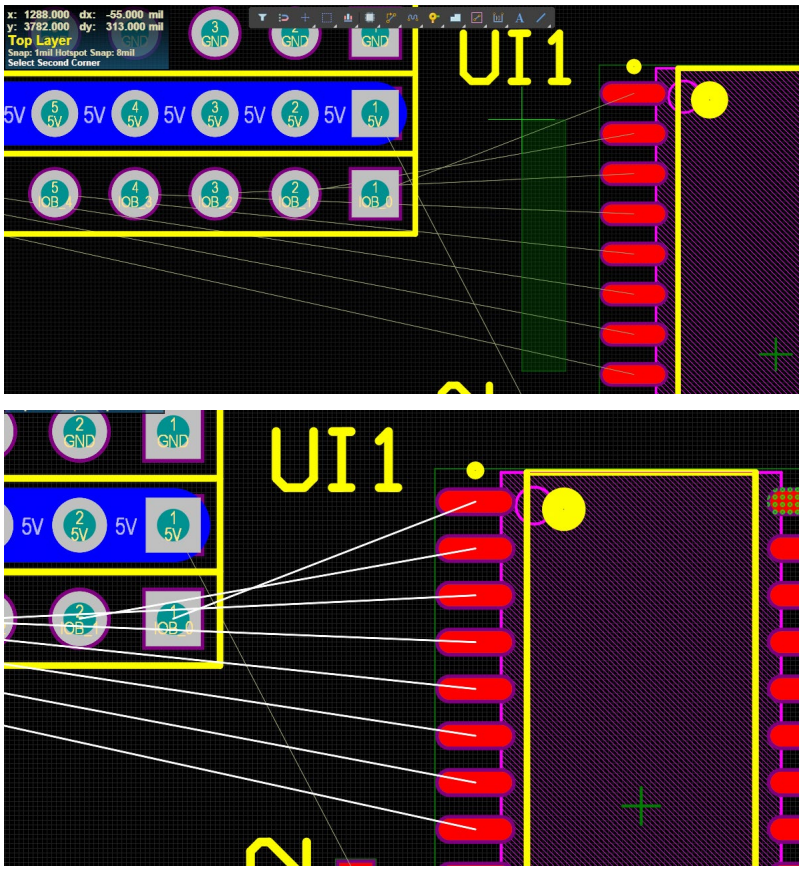


Figure 271. Select the Connection Lines to route the selected traces

22. Open the *PCB ActiveRoute* panel by going to **Panels** button in the lower right corner and select **PCB ActiveRoute** or **View » Panels » PCB ActiveRoute**.
23. Enable both checkboxes in the *PCB ActiveRoute* panel Layers section to allow ActiveRoute to utilize both the Top and the Bottom layer when the command is activated, as shown in Figure 272 below. However, in our case since we only have SMD Top-Layer pads on UI1 and ActiveRoute does not drop vias, routes will only be placed on the Top Layer.

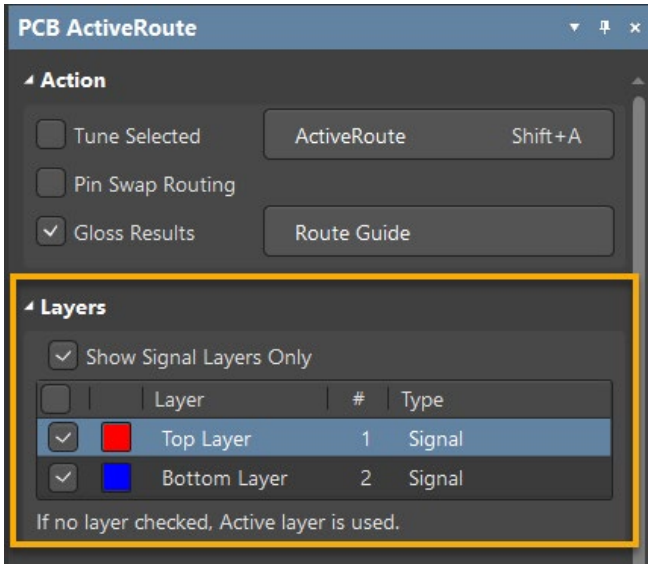


Figure 272. PCB ActiveRoute routing layer setup

24. Click the **ActiveRoute** button in at the top of the panel as shown in Figure 273, or select **Route » ActiveRoute**. The routes will be completed as shown in Figure 274.

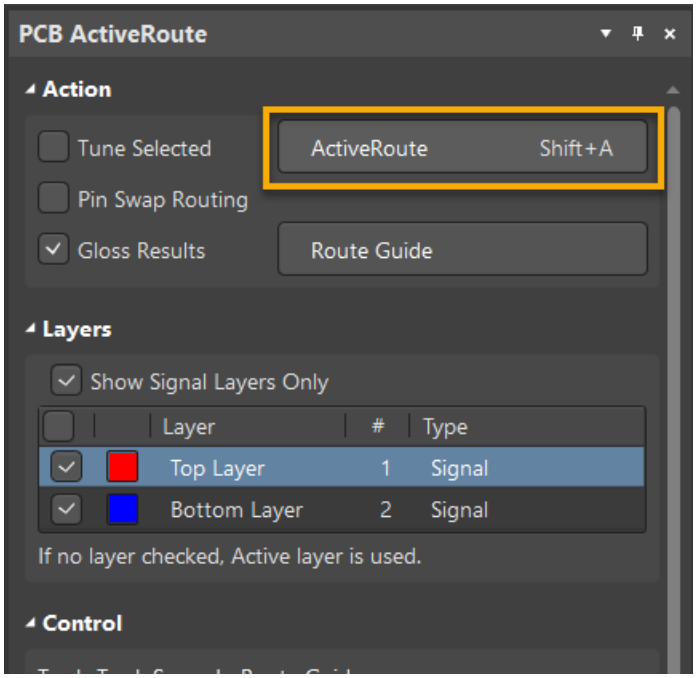


Figure 273. Click the ActiveRoute button to start the routing

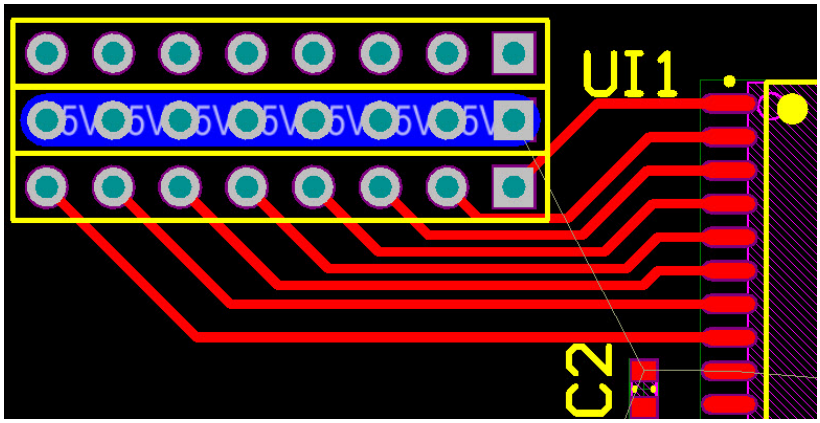


Figure 274. ActiveRoute completed

25. Please continue routing the PCB using the various routing techniques discussed in this exercise. Use Figure 275 below as a guide. Please bear in mind that the PCB does not have to be completely routed.

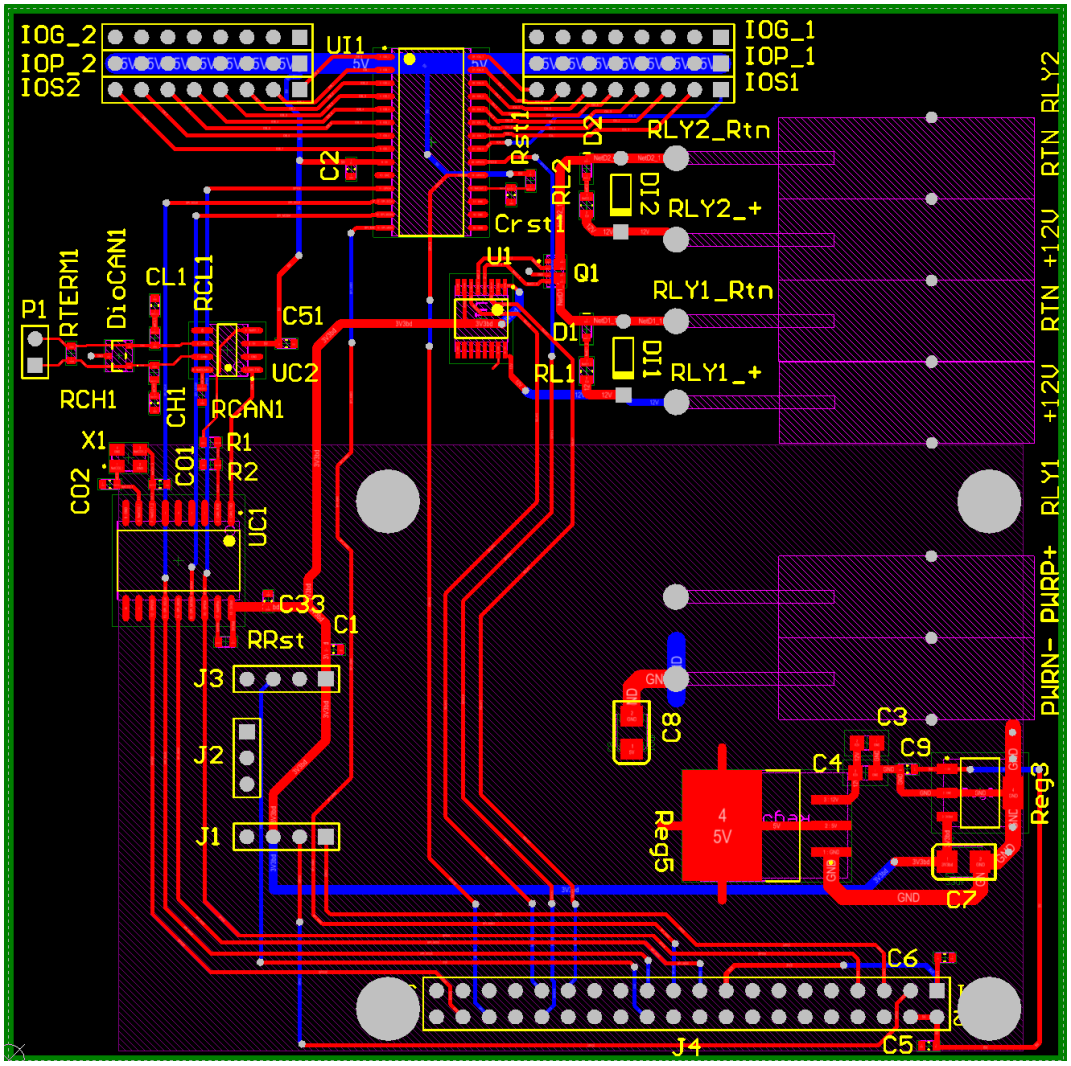


Figure 275. Routed PCB example

Routing Tips:
- Use **Ctrl + Click** on a net object to mask everything but the selected net for better visibility of target locations.
- Use HugNPush routing mode for routing close to the bus routes or other traces.
- Press **Shift + C** to clear applied mask after a route.
- Use the bracket keys to increase **[** and decrease **]** the applied mask and can be used during routing.
- Holding **Shift + Alt** while selecting connection lines will accumulate the selection.

Module 22: Polygon Pours

22.1 Purpose

In this exercise, students will learn to place and edit polygons pours and power planes.

22.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `Module 22 - Polygons and Power Planes` folder.
2. Open the `WCTopping.PrjPcb` project.
3. From the *Projects* panel double click on the file `WCTopping.PcbDoc` to open the PCB document.
4. Open *Preferences* and navigate to *PCB Editor - General*. Enable **Repour Polygons After Modification** under *Polygon Rebuild*. This removes the need to manually repour the polygons after every change. Click **OK** to close the *Preferences* dialog.

22.3 Placing a Polygon Pour

5. We will be placing a polygon on the bottom layer. Click on the **Bottom Layer** tab at the bottom of the PCB editor workspace to make it the active layer.
6. View the entire board area using the **V - D** hot keys.
7. Set the origin in the lower left corner of the PCB using the commands **Edit » Origin » Set**. Your cursor turns to a cross-hair. Zoom in to the lower left corner of the PCB and click to place the origin.
8. Set the grid spacing to `10mil` by pressing **G** and select `10mil`. Ensure to be on imperial units, as indicated on the status bar – else, hit the **Q** key to toggle units from metric.
9. If it is not visible, enable the *Heads Up Display (HUD)* by pressing **Shift + H** as shown in Figure 276 below. You can change HUD window opacity and options in *Preferences*, *PCB Editor* branch, *Board Insight Modes* page.

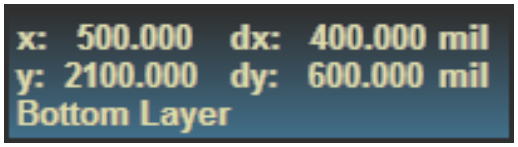


Figure 276. Heads Up Display toggled with the Shift+H keys

10. Access the command **Place » Polygon Pour** (hot keys **P - G**).
11. Press **Tab** to pause the command and open the *Properties* panel to access the *Polygon Pour* dialog as shown in Figure 277.
12. Be sure to assign it to the **5V** net and ensure the option is selected in the pull-down for **Pour Over All Same Net Objects**.

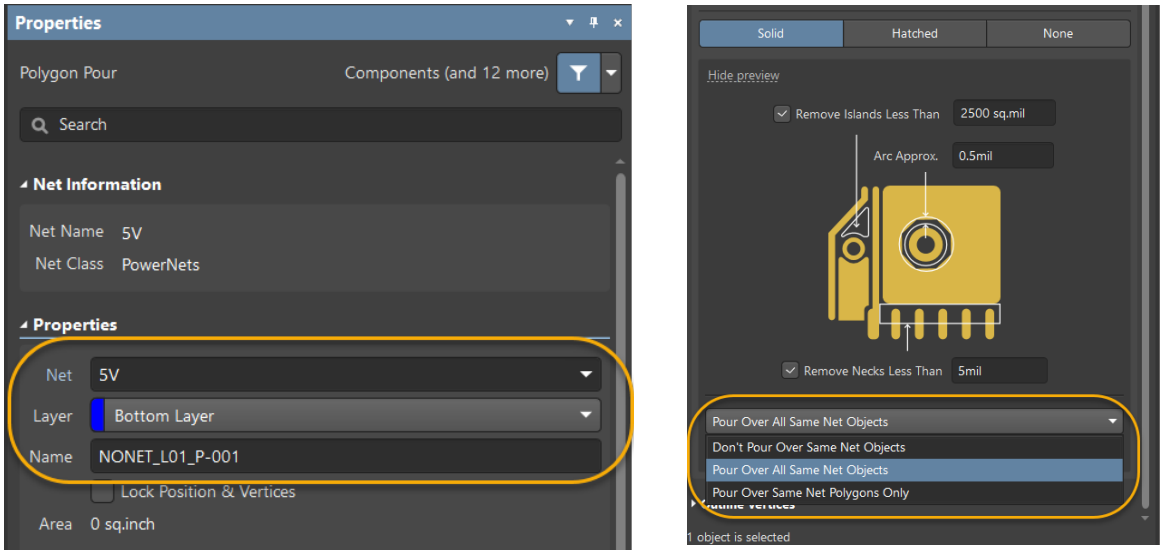


Figure 277. Polygon Pour dialog

13. Click on the pause symbol to continue placing the polygon. Your cursor will turn into a cross-hair indicating that you may begin drawing the polygon. Click on the board origin in the bottom left corner and the *HUD* should indicate co-ordinates (0,0) as shown in Figure 278.

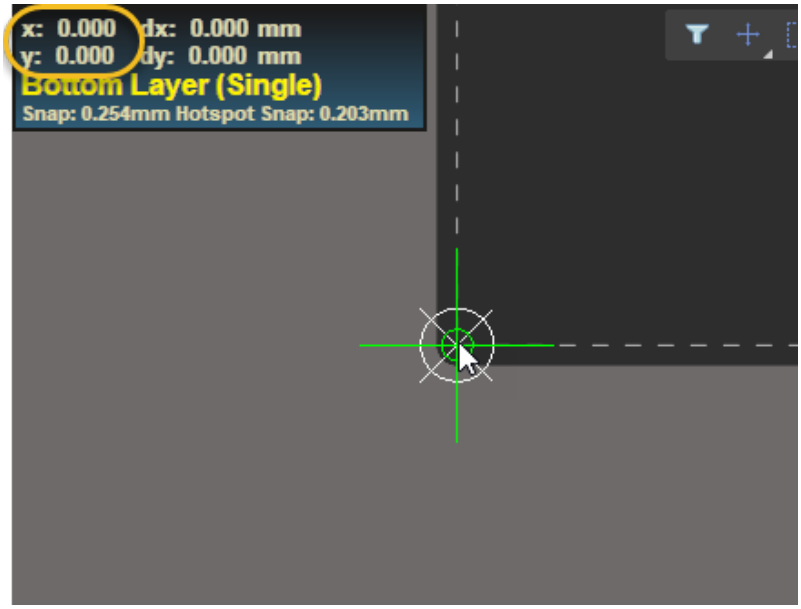


Figure 278. Heads-Up Display (HUD) showing (0,0) coordinates

14. Hit **Shift + Spacebar** until the *HUD* indicates *Line 90/90 Horizontal Start* to provide a right angle for the polygon. Explore changing the corner mode by pressing **Shift + Spacebar** until the HUD indicates *Line 90/90 Vertical Start with Arc (radius: 100mil).* While in this mode, use the **comma** '**,**' and **period** '**.**' keys to increase/decrease the radius value.

15. If necessary, hit **Spacebar** to change from vertical to horizontal if necessary.
16. Complete the rectangular polygon as shown in Figure 279.

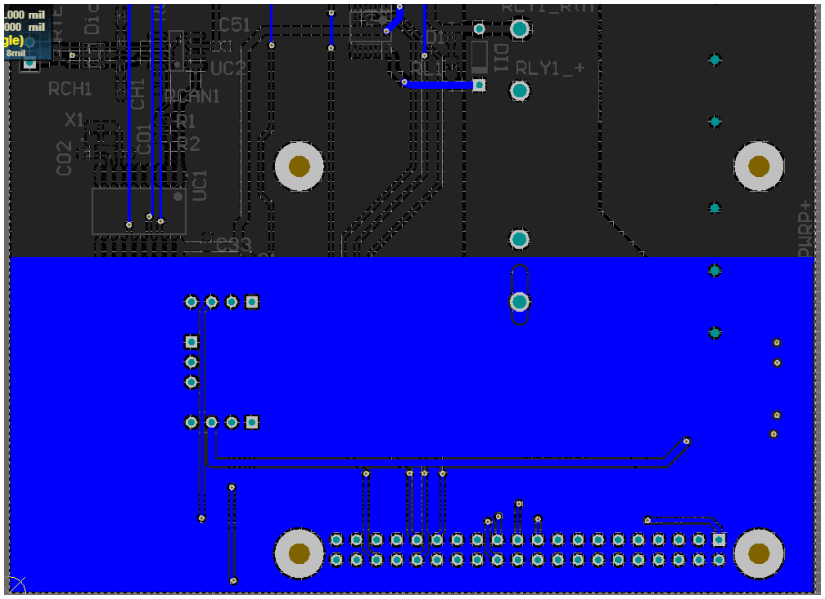


Figure 279. Polygon Pour added to the PCB

22.4 Polygon Properties

17. Double-click on the newly created polygon on the Bottom Layer to open its *Properties* panel.
18. Ensure that the post-selection filter in the *Properties* panel is reset to have all objects selected, as shown in Figure 280.

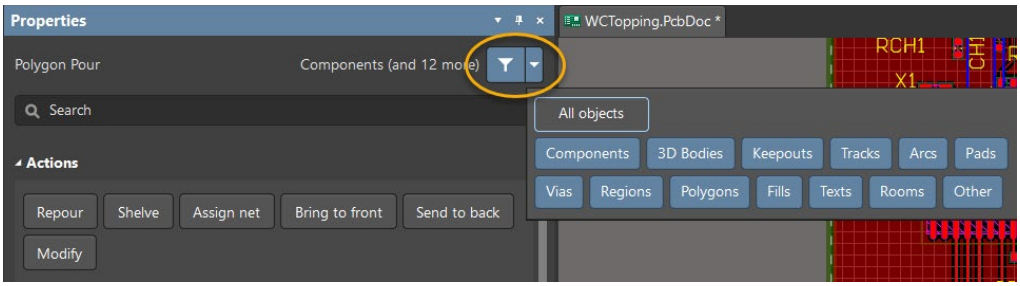


Figure 280. All Objects selected in the selection filter

19. Change the *Fill Mode* to **Hatched** as shown in Figure 281. After changing it to **Hatched**, the polygon should then repoured by clicking the "**Repour**" action at the top of the *Properties* panel.

If your Preferences are set to **Repour Polygons After Modification**, the Repour dialog shown below in the Properties panel will not be present, since the polygon will repour automatically.

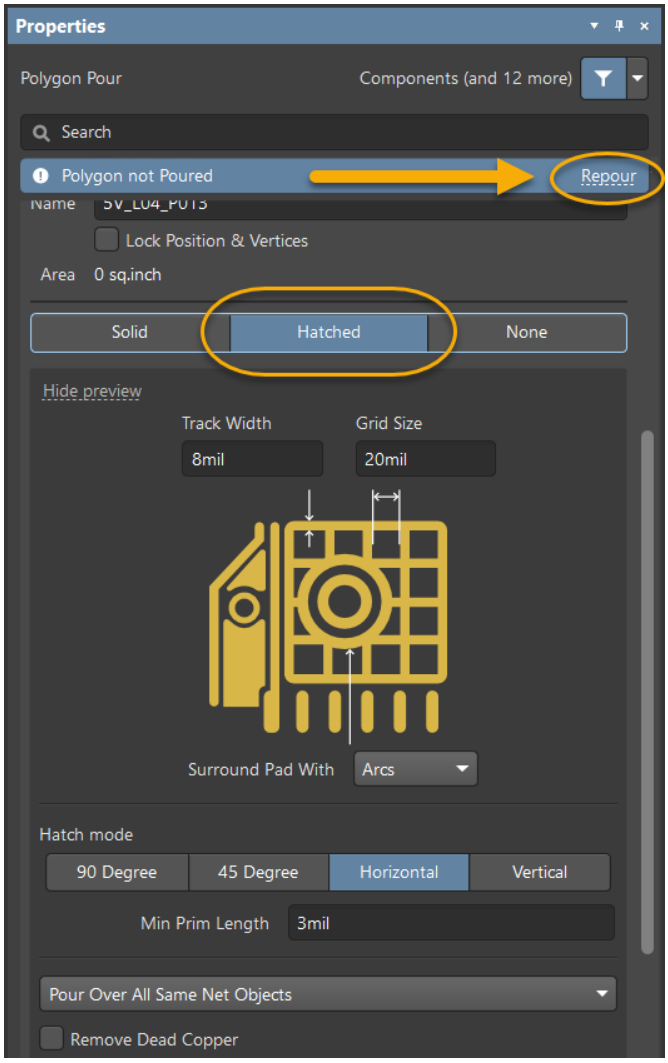


Figure 281. Changing the Polygon fill mode

20. From the **Place** menu, select **Polygon Pour Cutout**.
21. Using Figure 282 as a reference, draw a rectangular **Polygon Pour Cutout** on the existing Bottom Layer polygon we just created.
22. Once defined, right click to terminate the polygon outline and right-click once more to exit polygon pour cutout placement mode.
23. You should now see that once the polygon repours, there is a cutout and absence of copper within the cutout region similar to Figure 282. If your polygon does not repour automatically, make sure you go to **Tools » Polygon Pours » Repour All.**

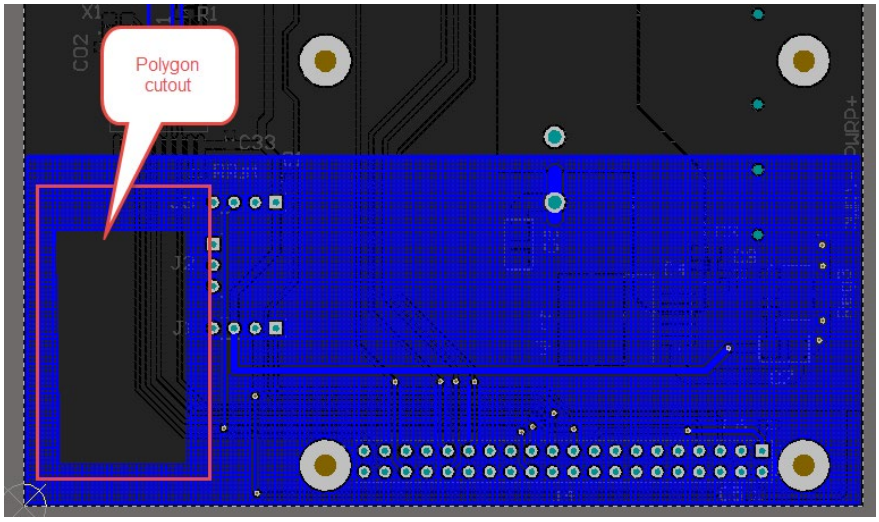


Figure 282. Polygon Pour Cutout

22.5 Polygon Pour Manager

24. Click on the Top Layer tab at the bottom of the PCB editor to set it as the active layer. Notice the multiple polygon pours on this layer and the specific clearances as shown in Figure 283. With multiple polygons overlapping, pour order or priority must be defined.

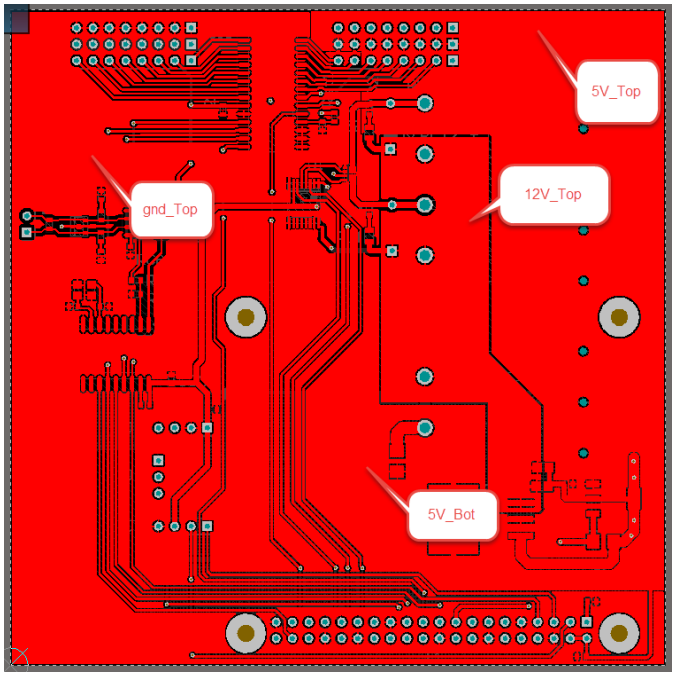


Figure 283. Polygon Pours on Top Layer

25. Open the polygon pour manager by going to **Tools » Polygon Pours » Polygon Manager**.
26. Within the Pour Order pane, we can change the pour order of the polygons. Select `12V_Top` polygon and click **Move Down** until it is moved all the way down to the bottom of the *Pour Order* list seen in Figure 284. Click **OK**.

The polygon that is higher in the pour order will pour first, and the polygons lower in the list will then pour around existing polygons with necessary clearance.

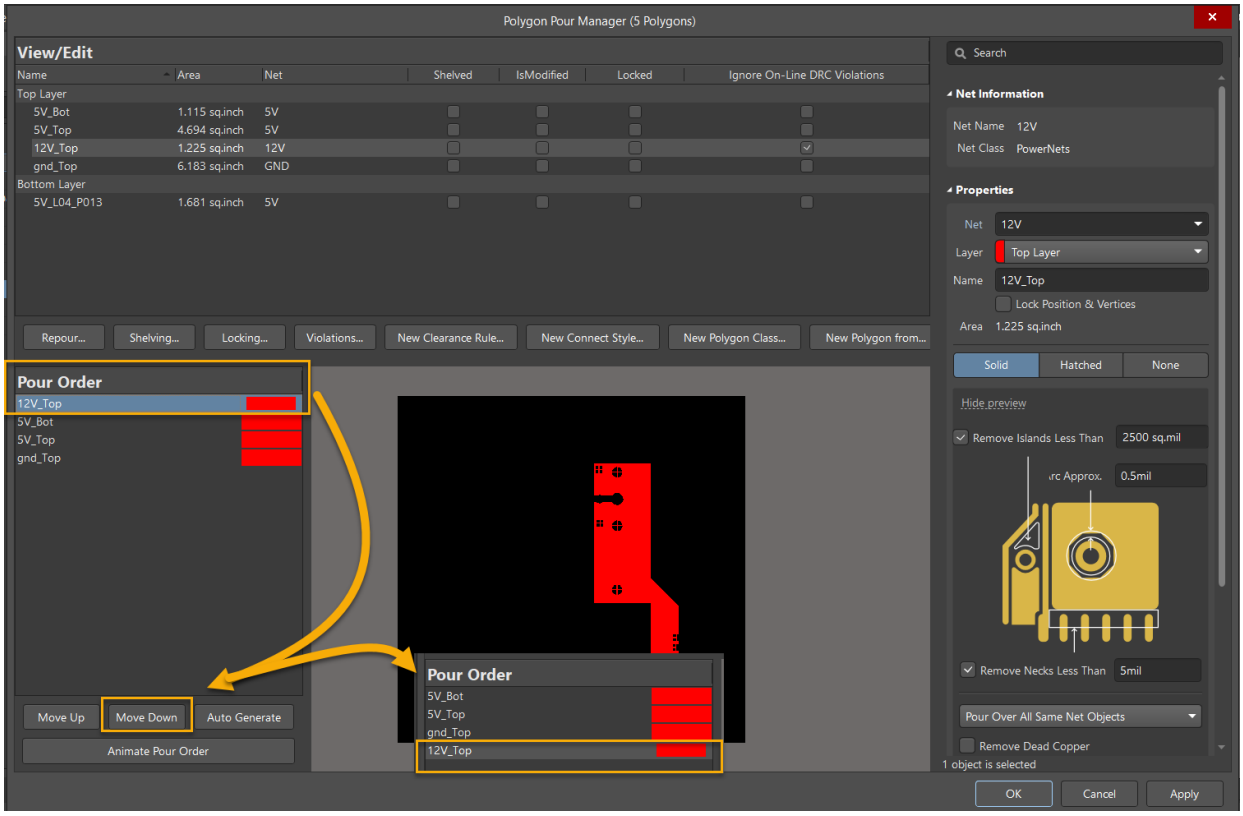


Figure 284. Polygon Pour Manager

27. In the PCB, notice that the `12V_Top` polygon is no longer visible when repoured since the *GND_Top* polygon which floods the entire board is higher in the pour order. This is a deliberate error and we should revert it back to the original order.
28. Open the **Polygon Pour Manager** again by going to **Tools » Polygon Pours » Polygon Manager** and set the `12V_Top` polygon back to the top of the pour order. Click **OK** to return to the PCB and ensure to see the `12V_Top` polygon poured on the board.

22.6 Flooding a Layer

29. We will now create a polygon which floods an entire layer of the board. Go back to **Tools » Polygon Pours » Polygon Manager**.
30. Click on **New Polygon from…** then **Board Outline** as seen in Figure 285. A new Polygon will be added in the Polygon Pour Manager.

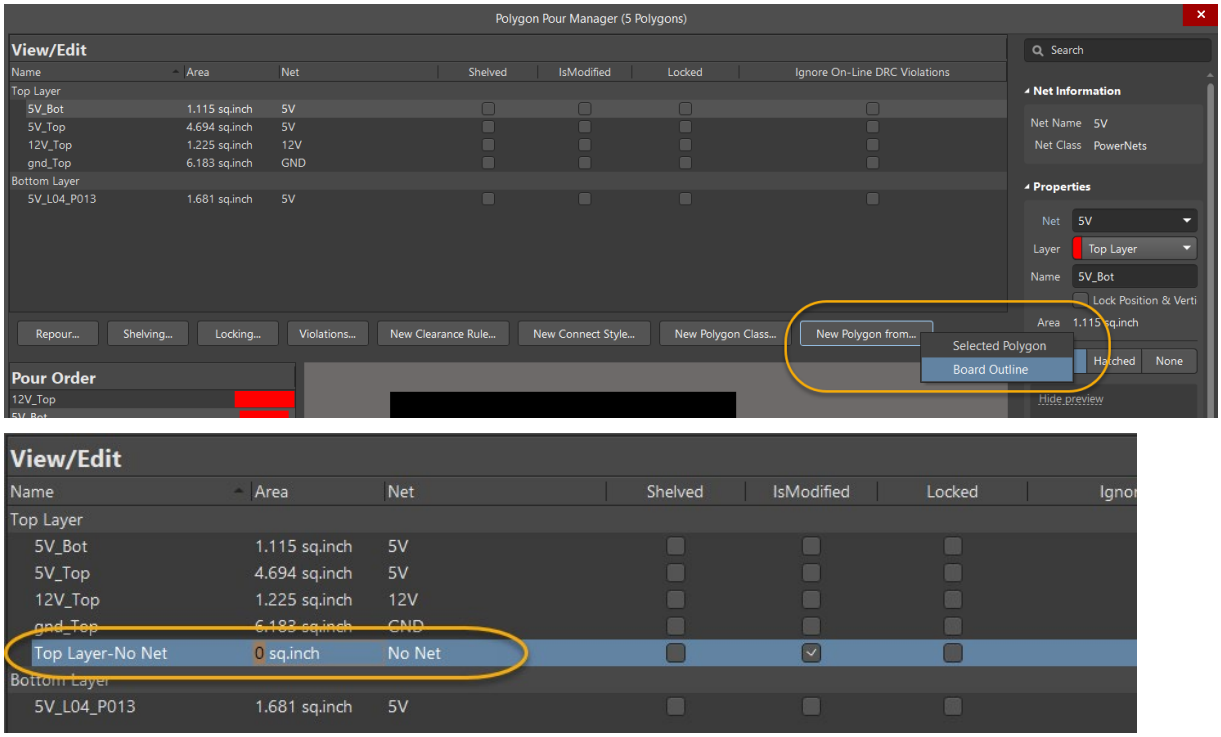


Figure 285. Creating a polygon from the board outline in the Polygon Manager

31. Match the settings of the newly created Polygon to what is shown in Figure 286. The shape of this polygon will be dependent on the first, Bottom Layer polygon that we created in a previous step. Your polygon may vary from what is shown below.

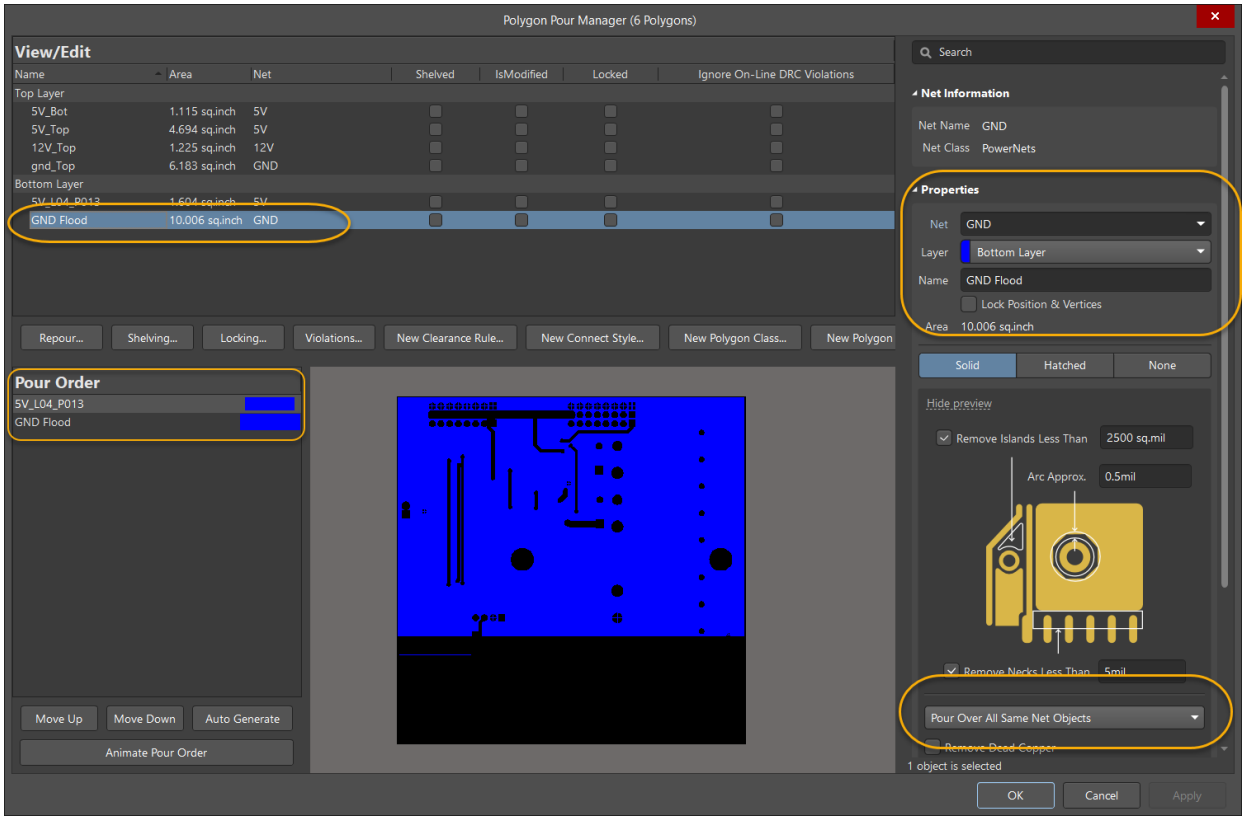


Figure 286. Polygon Pour dialog

32. Ensure that this newly created polygon is last in the pour order and click **OK** to close the *Polygon Manager*.
33. If you received green highlights showing a Violation, rebuild the polygon by going to **Tools » Polygon Pours » Repour All**

22.7 Editing Polygons

22.7.1 Editing an Edge

34. In the PCB editor, click on **Bottom Layer** from the layers tab to make it the active layer.
35. Navigate to the bottom of the board where you manually created the polygon earlier in this exercise.
36. Left-click on the polygon to select it. If the first click selects the Polygon for the entire board, click again.
37. Hover over the right vertical edge of the polygon (not at a vertex). The edge will be highlighted by a thick white line and the cursor will turn into a 2-way arrow as shown in Figure 287.

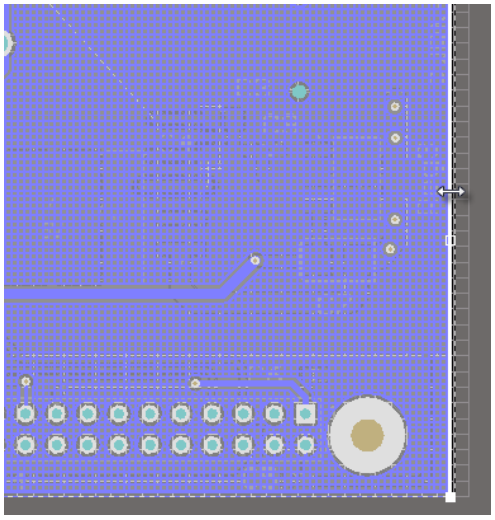


Figure 287. Hovering over the right edge of the polygon

38. Now click and hold the left mouse button; the cursor will become a crosshair and the *HUD* will show that the *Sliding Edge* mode is active as shown in Figure 288.

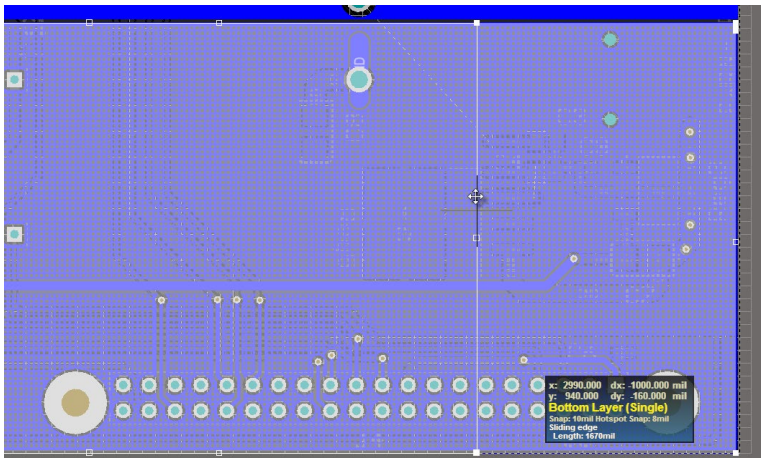


Figure 288. Moving the right edge of a polygon using the Sliding Edge mode

39. Slowly move the cursor to the left and the polygon edge slides to the left, becoming narrower. Release the mouse button. Notice that you are constrained to only the x-direction ensuring that the height of the polygon is unchanged.

22.7.2 Adding a Vertex

You can add a center vertex at any point on the edge of the polygon. Center vertices are also automatically added or removed as changes are made.

40. Select the same polygon as in Section 22.7.1 that we just modified. Hold **Ctrl**, position the cursor at the desired location for the center vertex.
41. Left-click and move the mouse button while holding **Ctrl**. The new vertex will appear in the desired location as a white square as shown in Figure 289.

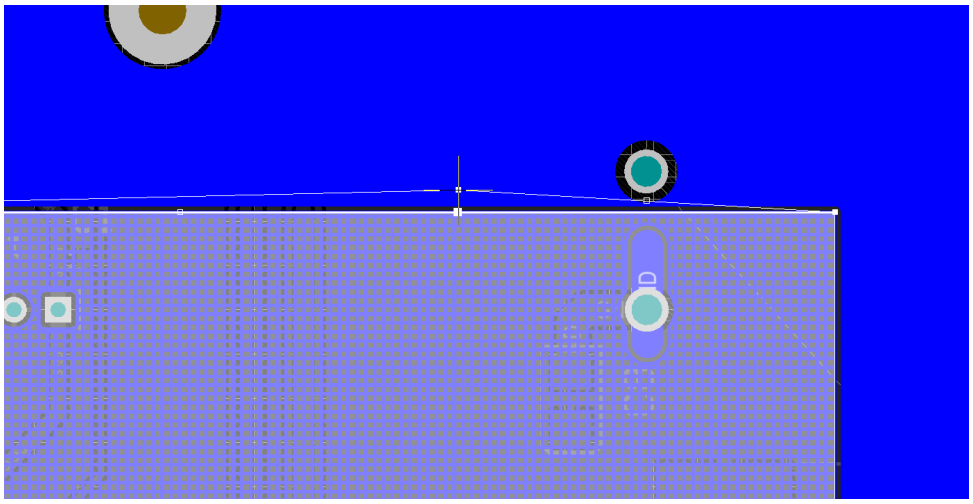


Figure 289. Adding a vertex to a polygon

22.7.3 Deleting a Vertex

42. Select the same polygon as in Section 22.7.1. Click and hold a corner vertex then press **Delete**. The cursor will turn into a crosshair, and the vertex will be removed reshaping the polygon as shown in Figure 290.

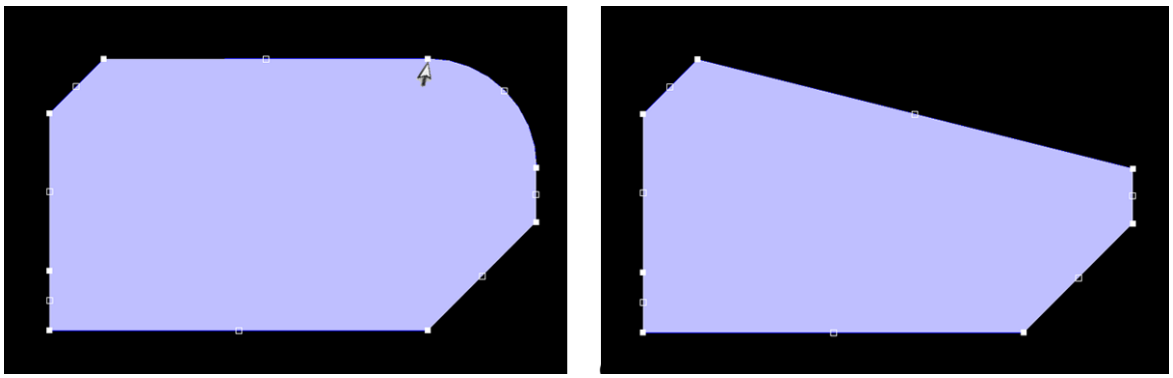


Figure 290. Deleting a vertex

22.8 Controlling the Connection Style - Polygon

43. As of right now, all polygons are connected with a relief connect style as shown between the pads of the headers on the top left of the board, and the GND polygon shown in Figure 291.

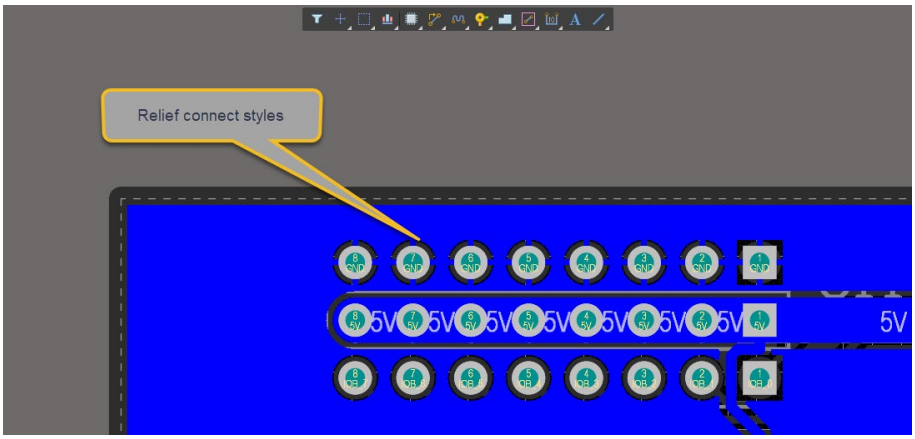


Figure 291. Relief connect style

44. We would like to change the connections from relief connect to direct connect. In the PCB, go to **Design » Rules**.
45. Click on *Polygon Connect Style* under *Plane* and select *Polygon Connect* to open the rule.
46. Set the *Connection Style* to **Direct Connect** as shown in Figure 292**.** Click **OK** to return to the board.

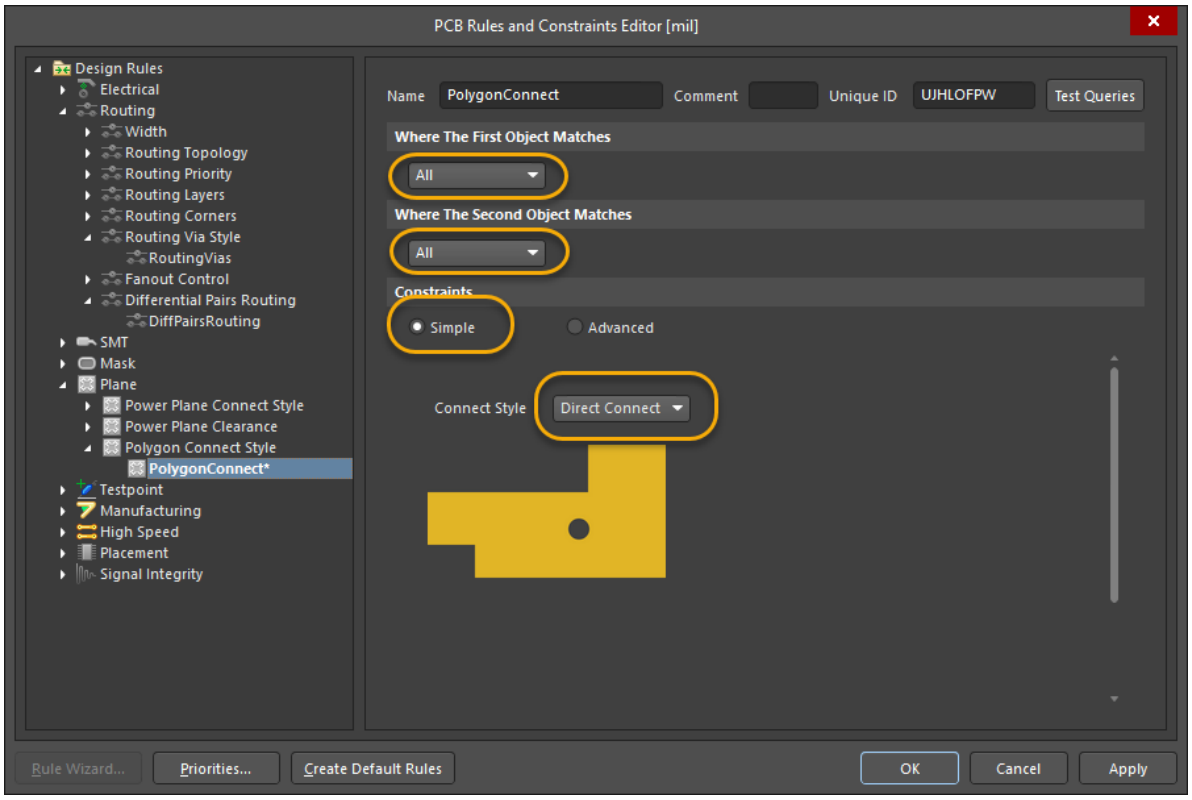


Figure 292. Polygon Connect Style rule

47. Go to **Tools » Polygon Pours » Repour All**. Note that the pads are now direct connected to the pour as shown in Figure 293.

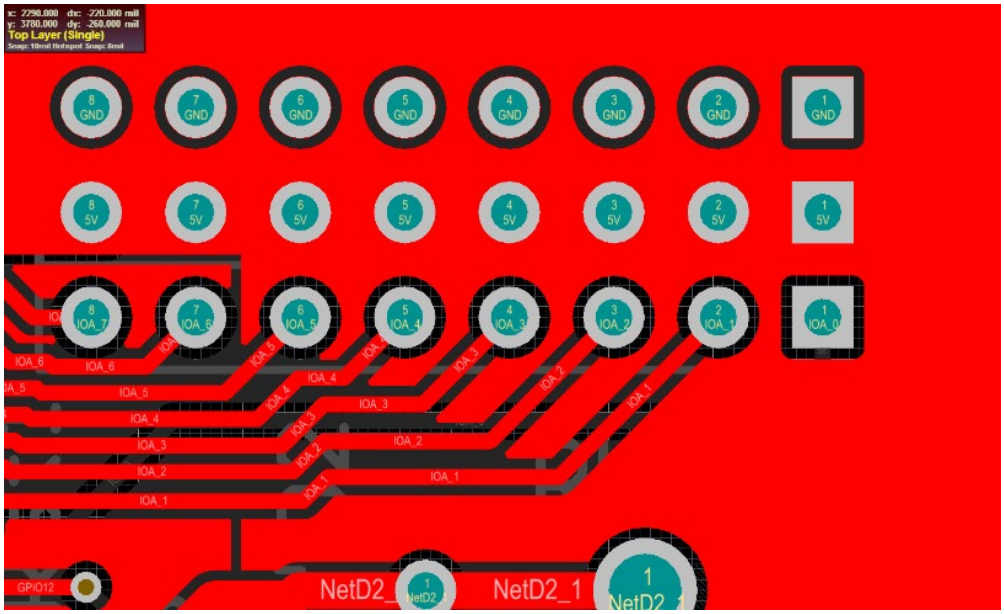


Figure 293. Repoured polygon

The design rule can be configured as both a simple and advanced mode. In Advanced mode, you can set the connection styles for through hole, surface mount and via connection.

48. To avoid design rule violations or automatic repour of polygons when routing on top of existing polygons, polygons should be shelved. Polygons can be shelved through a couple of different ways, e.g., by right-clicking on a polygon **» Polygon Actions » Shelve All**, or **Tools » Polygon Pours » Shelve Polygons** or in *Polygon Manager* window.
49. Go to **Tools » Polygon Pours » Shelve 6 Polygon(s).** Note the number will vary depending on the number of polygons on the board. Observe that all the polygons have disappeared.
50. Restore using **Tools » Polygon Pours » Restore 6 Shelved Polygon(s)**.

22.9 Power Planes

51. Select **GND** from the layers tab, which is an internal plane. Hit **Shift+S** to go into *Single Layer Mode.*
52. Double-click in the board area to open the *Split Plane* dialog appears as shown in Figure 294. Assign the net **GND** to the Power Plane.

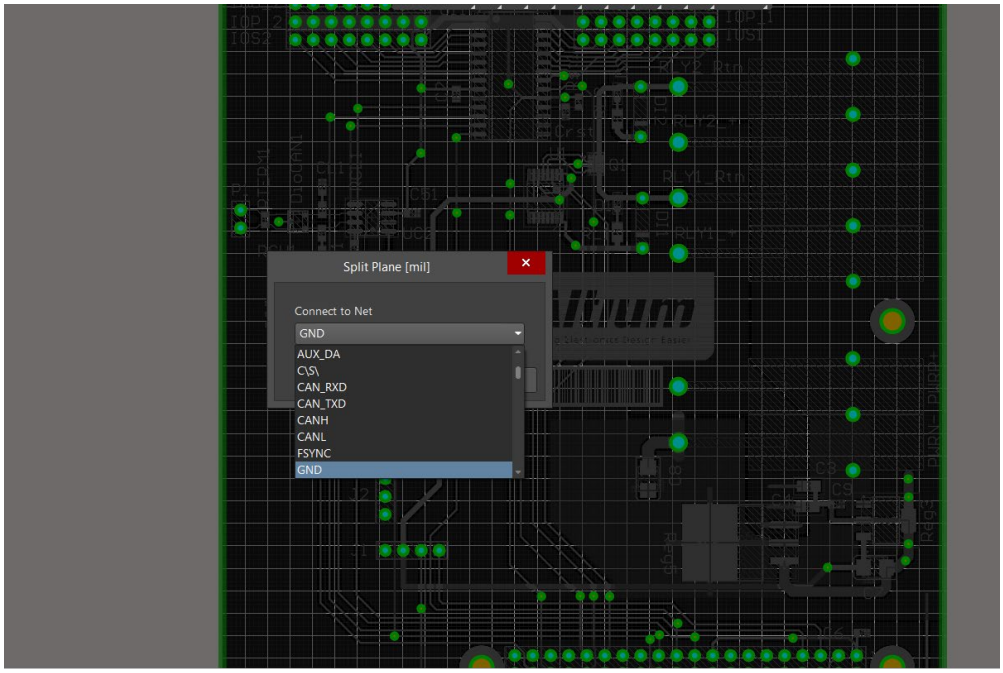


Figure 294. Changing the Power Plane net to GND

53. Now, click on **5V** from the layers tab and repeat the previous step to assign net 5V to the plane as shown in Figure 295.

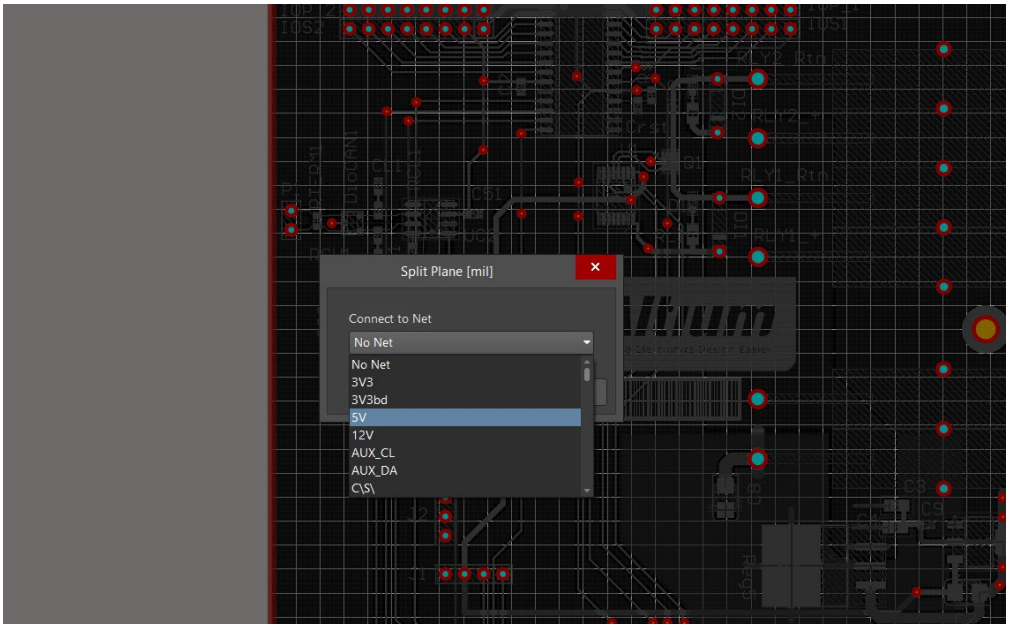


Figure 295. Changing the Power Plane net to 5V

22.9.1 Controlling the Connection Style – Power Plane

54. Like for the polygons, we can also control the connection style of pads/vias for a Plane. Go to **Design » Rules,** navigate to *Plane* and open the page *Power Plane Connect Style*. Open the default *PlaneConnect* rule. Ensure the fields in this dialog are set as the following:
    - Constraint: **Simple**
    - Connect Style: **Relief Connect**
    - Conductors: **4**
    - Conductor Width: `10 mil`
    - Air-Gap: `10 mil`
    - Expansion: `10 mil`
55. Click **OK** to close the window.
56. Note that the pads connected to **5V** on the top left of the board should be connected with a relief connect to the power plane as shown in Figure 296.

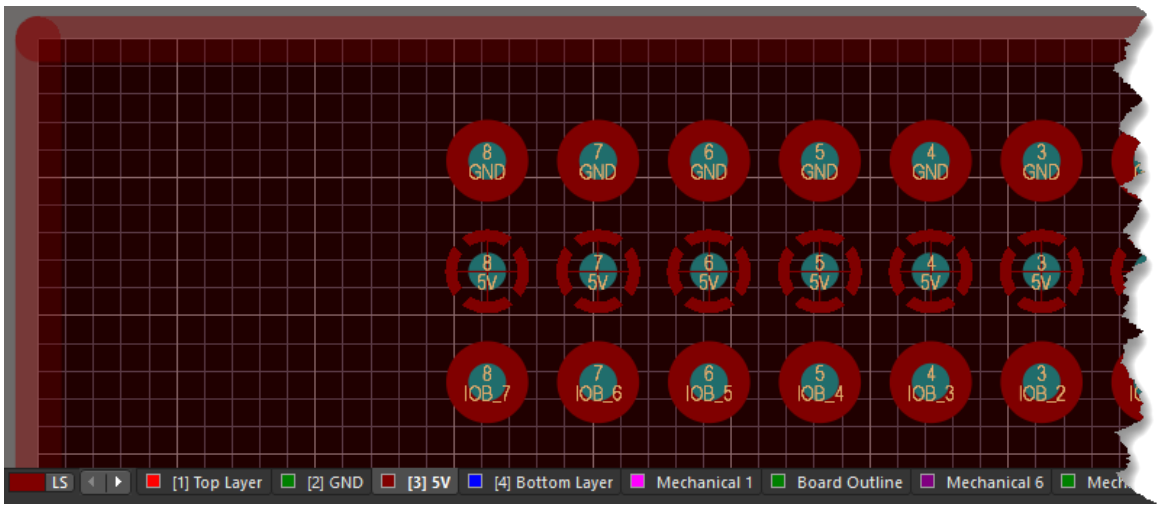


Figure 296. Relief Connect style for the 5V plane

If any additional rules are added for this category, ensure that the new rule has a higher priority than the ALL scoped rule. Otherwise, the new rule will be ignored. Once the **ALL** scoping query is encountered, it overrides lower priority rules for that category.

57. Save and close all files.

Module 23: DRCs on PCBs

23.1 Purpose

In this exercise, students will run a Design Rule Check. Using the PCB Rules and Violations panel, students will learn how to properly inspect and resolve violations.

23.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `AD20 Essentials Exercises\Module 23 - DRCs on PCBs` folder.
2. Open the `WCTopping.PrjPcb` project.
3. Navigate to the *Projects* panel and open `WCT.PcbDoc`.

23.3 Design Rule Check

4. Go to **Tools » Design Rule Check** to open the *Design Rule Checker*.
   a) Enable **Create Report File** and **Create Violations**.
   b) Click **Run Design Rule Check…** button on the bottom left as shown in Figure 297.

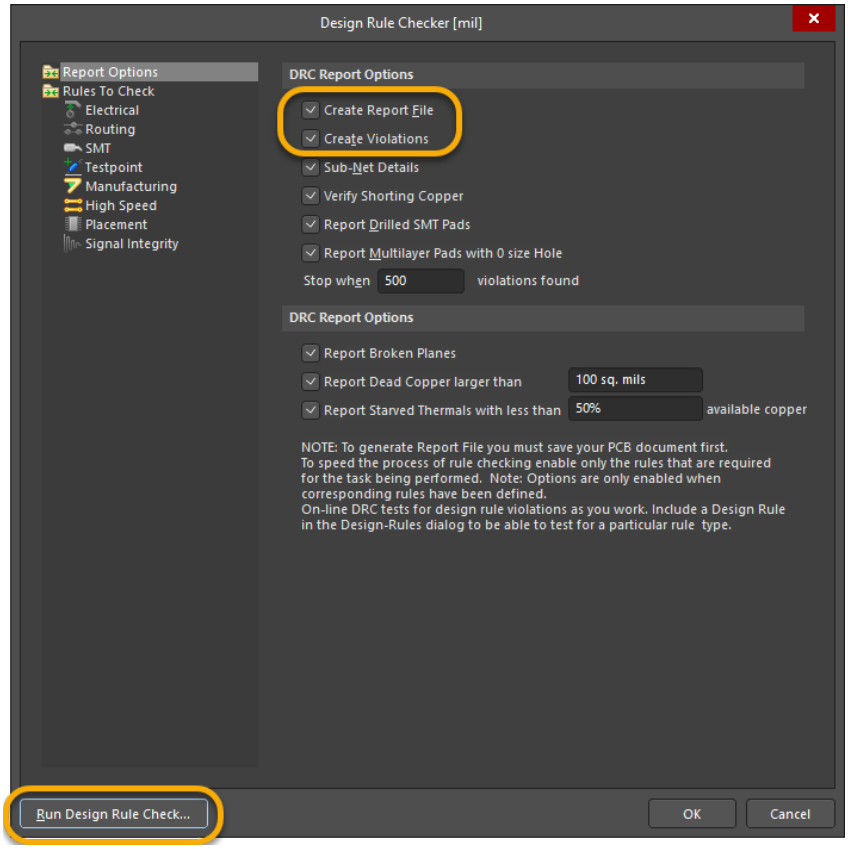


Figure 297. Design Rule Checker

5. The *Design Rule Verification Report* should generate, indicating 10 Rule Violations. Scroll down to the *Rule Violations* section to see the violated rules as shown in Figure 298.
   a) Un-Routed Net Constraint

b) Minimum Solder Mask Sliver
c) Silk to Solder Mask
d) Component Clearance Constraint

6. If the *Messages* panel popped-up, feel free to close it for now.

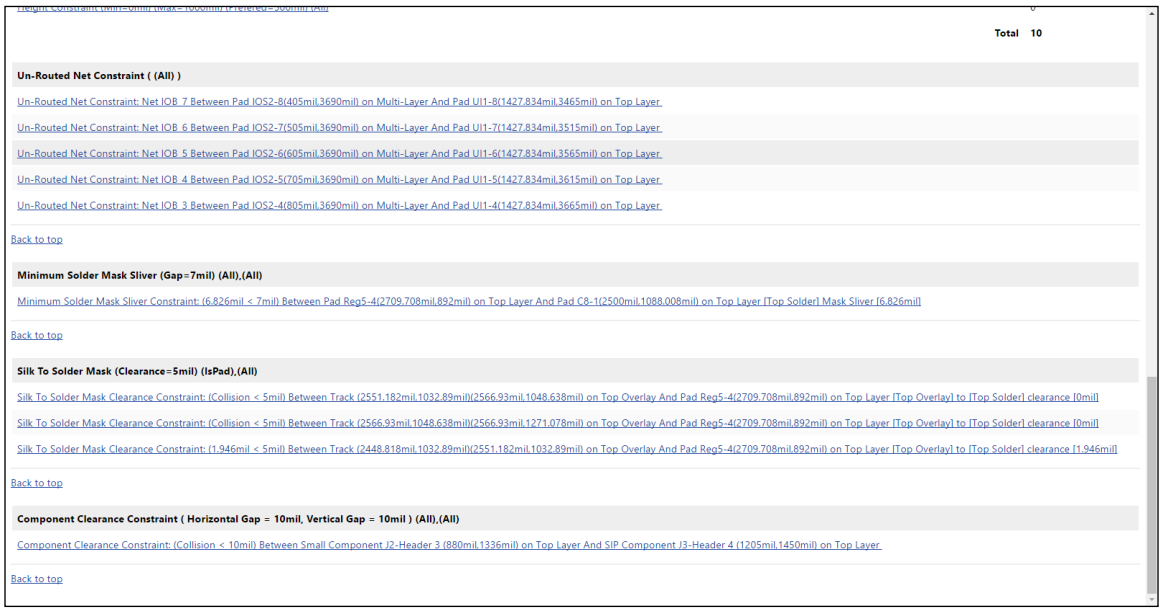


Figure 298. Rule violations in the report file

By default, the **Design Rule Verification Report** will only show a maximum of 500 violations. This can be changes in the *DRC Report Options* before running the report.

7. Click on the first violation under the *Un-Routed Net Constraint* list in the report. It should then open the PCB document and navigate to the violating area, which is near component `IOS2`.

23.4 PCB Rules and Violations Panel

8. Instead of using the report to navigate through the violations, we will use the *PCB Rules and Violations* panel while in the PCB. While in the `WCTopping.PcbDoc`, click the **Panels** button on the bottom right, and then select the *PCB Rules and Violations* panel from the menu as shown in Figure 299.

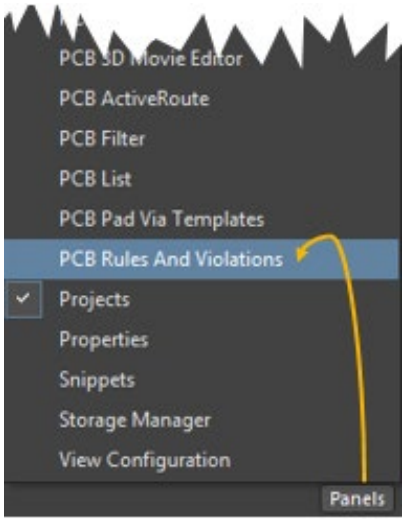


Figure 299. Opening the PCB Rules and Violations panel

9. This would open the *PCB Rules and Violations* panel. Dock this panel to the left of the workspace as shown in Figure 300.

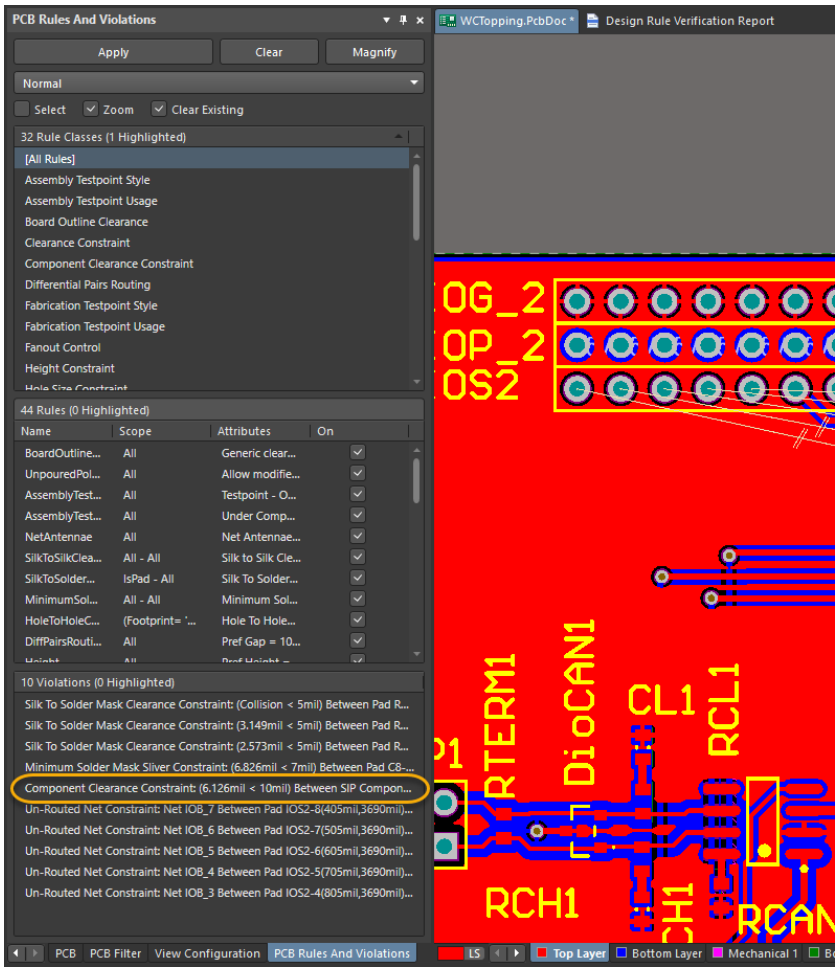


Figure 300. PCB Rules and Violations panel

10. Click on the **Component Clearance Constraint** violation, which is the first violation in the *Violations* list at the bottom of the *PCB Rules and Violations* panel. It should then navigate to the violating area. Instead of just zooming to the violating area, we would also like to perform a mask of the other objects, to improve clarity.
11. In the *PCB Rules and Violations* panel, change the highlight options from **Normal** to **Mask** as shown in Figure 301.

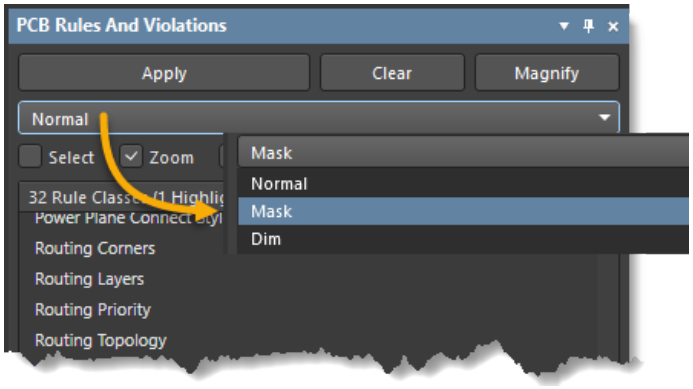


Figure 301. Mask control in the PCB Rules and Violations panel

12. Left-click on the **Component Clearance Constraint** violation again from *Violations* list. The workspace should zoom to the violating primitives and mask the rest of the board except for the primitives shown in Figure 302.

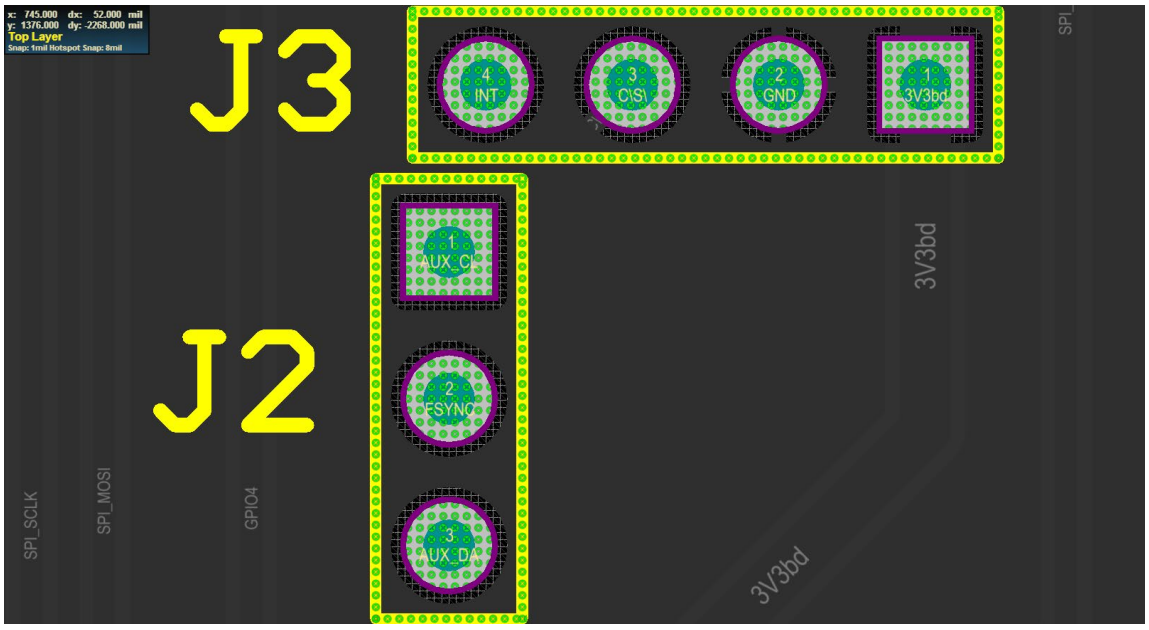


Figure 302. Violating components in the PCB

13. Enable the **Select** checkbox from the top of the *PCB Rules and Violations* panel and repeat the steps. Notice that the violating primitives will now be selected.
14. To clear all the masks and selection, click the **Clear** button from the *PCB Rules and Violations* panel or hit **Shift+C** on the keyboard.

23.5 Resolving Violations

Instead of running all the rules from the Design Rule Check window as described in the previous section, we can run only selected design rules from the *PCB Rules and Violations* panel.

15. Right-click on **Un-Routed Net Constraint** in the *Rule Classes* section of the *PCB Rules and Violations* as shown in Figure 303, and click **Run DRC Rule Class (Un-Routed Net Constraint)** option from the popup menu to run the Design Rule Check.
16. Five violations of this rule type should be highlighted. Find these listed at the bottom of the *PCB Rules and Violations* panel in the *Violations* section as shown in Figure 304.
17. In the *Violations* pane of the panel, click on the first violation in the list to jump to component `IOS2`. One of the pads from `IOS2` should be highlighted along with one pad of component `UI1`, and the rest of the board is masked similar to Figure 305.

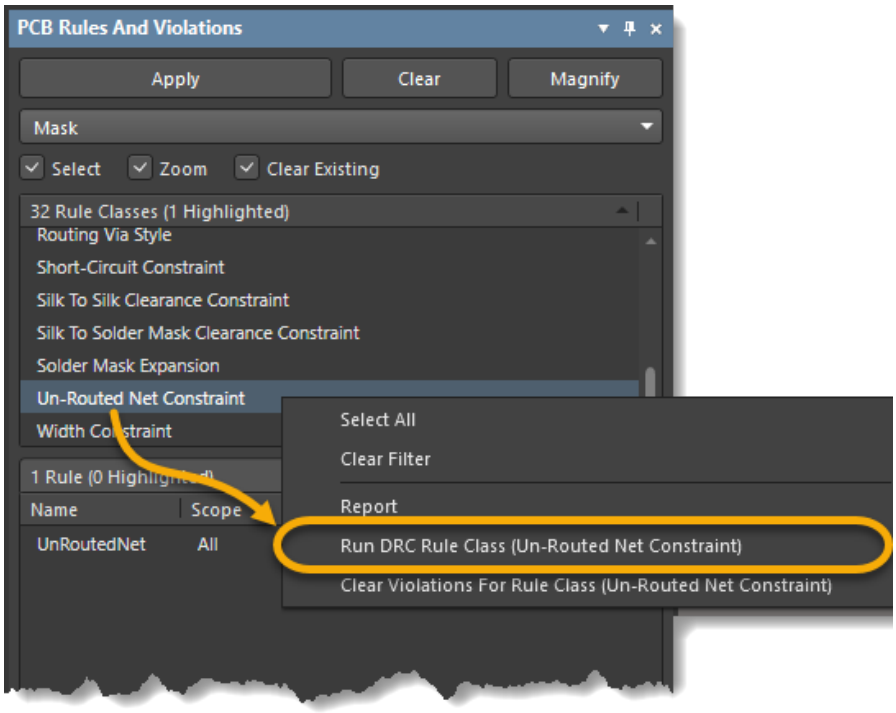


Figure 303. Rule Classes in the PCB Rules and Violations panel

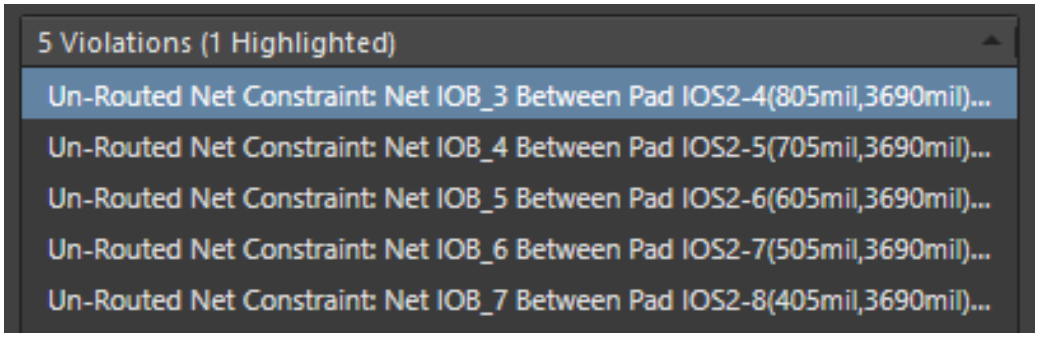


Figure 304. Un-Routed Net violations

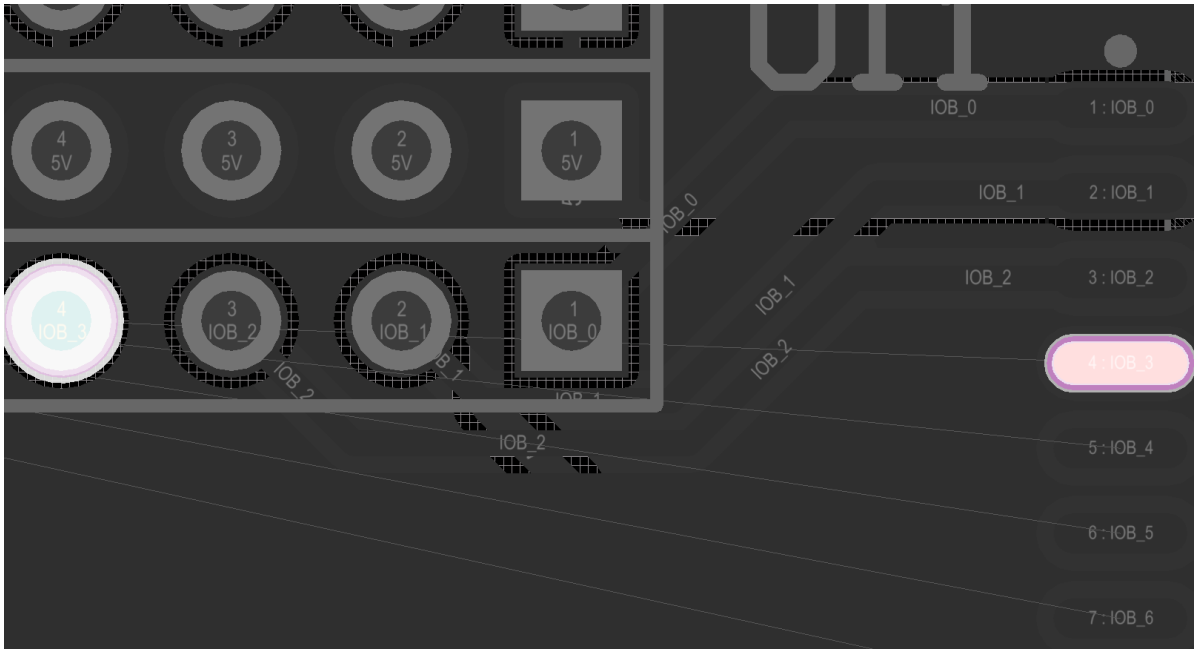


Figure 305. Highlighted violations

18. Since this is an **Un-Routed Net Constraint** violation, we will resolve it by routing a trace from the highlighted pad of `U1` to the highlighted pad of `IOS2` with **Route » Interactive Routing**.
19. Repeat the previous steps for the other 4 **Un-Routed Net Constraint** violations in the *PCB Rules and Violations* panel. Using **Shift+S** for Single Layer Mode may help you with routing.

Although we resolved the first Un-Route Net Constraint violation, it was not removed from the *Violations* panel. The Design Rule Check must be ran again for the Violations to update. You can run the DRC again by going to **Tools » Design Rule Check**, or run the DRC for that specific Rule from the *PCB Rules and Violations* panel.

20. Repour the polygon by going to **Tools » Polygon Pours » Repour All**. The polygon will then look similar to Figure 306 .

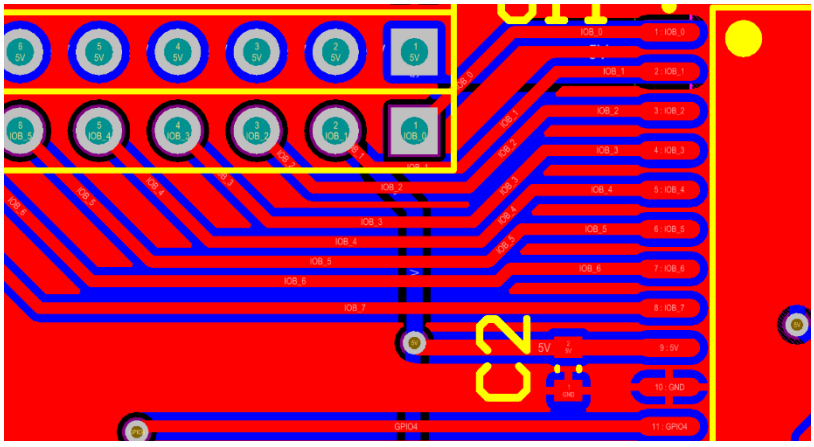


Figure 306. Routed nets with the repoured polygons

21. Go to **Tools » Design Rule Check** and click **Run Design Rule Check.**
22. The *Design Rule Verification Report* should now show five rule violations as shown in Figure 307.

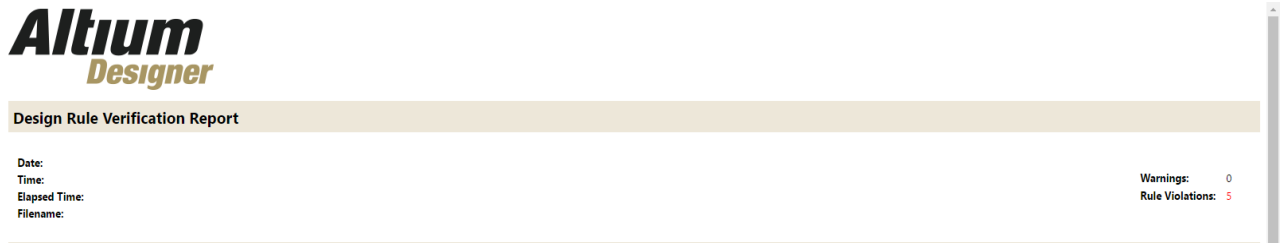


Figure 307. Design Rule report

23. Open the PCB document and select the **[All Rules]** in the *Rule Class* section from the top of the *PCB Rules and Violations* panel.
24. Select the **Component Clearance Constraint** violation from the *Violations* section at the bottom of the panel to zoom to the offending primitives, `J2` and `J3`.
25. Select the **Component Clearance Constraint** class from the *Rule Class* section to view its 3 rules as shown in Figure 308.

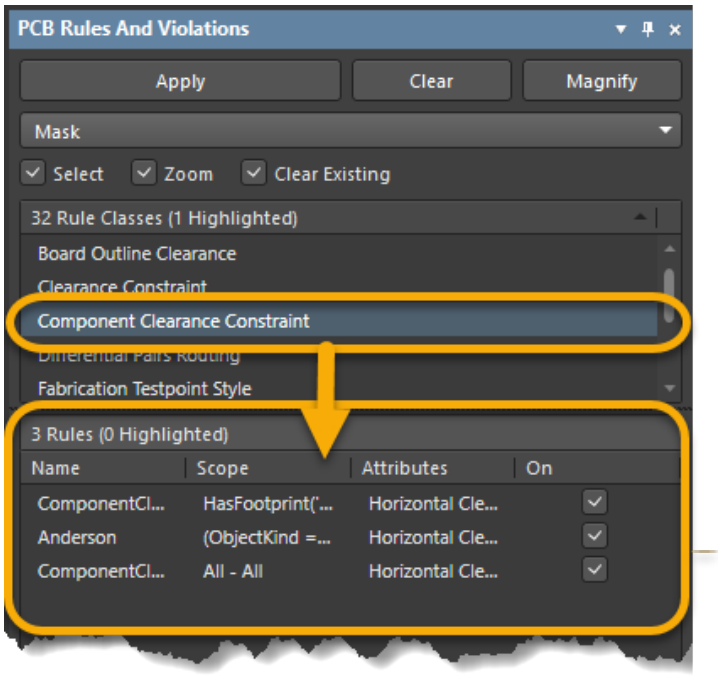


Figure 308. Component Clearance rule class

26. Double-click on the rule named *"Component Clearance"* with the Scope **All – All**.
27. The *Edit PCB Rule – Component Clearance* window should open.

28. Change the **Minimum Horizontal Clearance** from `10mil` to `5mil` as shown in Figure 309. Click **OK**.

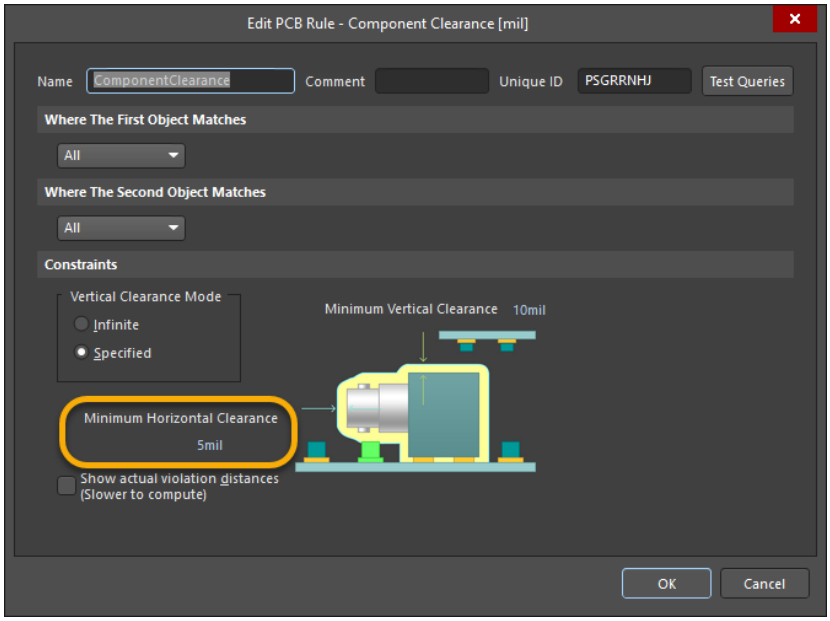


Figure 309. Editing the Component Clearance design rule

29. Run the Design Rule Check again from the **Tools** menu. The Design Rule Check should now indicate four violations. Return to the PCB document. Select **[All Rules]** from the *Rule Class* section to view the four violations.
30. Select each of the violation to notice the same pad highlighted. Select the Top Layer tab at the bottom of the workspace.
31. Click the **Clear** button from the *PCB Rules and Violations* panel.
32. Remove the **GND** tracks from component `C8`. Move component `C8` to the left so that it is no longer overlapping the violating pad of `Reg5`. Re-route `Pad 2` of component `C8` to the **GND** via that it was just connected to.
33. Repour all Polygons by going to **Tools » Polygon Pours » Repour All**.
34. Run the **Design Rule Check** and ensure there are 0 violations.
35. Save your changes and close the document.

Module 24: Draftsman

24.1 Purpose

In this exercise, students will learn to create a Draftsman document which can then be used for board production drawings for fabrication and assembly. Students will learn to place board views (assembly/fabrication/drill drawing), tables (drill table, layer stack legend, BOM) and other graphical notes or objects to improve clarity.

24.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `Module 24 - Draftsman` folder.
2. Open the `WCTopping.PrjPcb` project.
3. Open the `WCTopping.PcbDoc` and view it in 3D. In this exercise, we will learn to produce a board production drawing of this PCB using Draftsman document.

24.3 Creating a Draftsman Document

24.3.1 Creating Draftsman From a Template

4. Open a new Draftsman document with the command **File » New » Draftsman Document**.
5. Select the *Template* **Assy Drawing – Main views (rect. board 3x1)**. Ensure to point to the `WCTopping.PrjPcb` and `WCTopping.PcbDoc` as shown in Figure 310. Click **OK** to generate the Draftsman document based off the chosen Draftsman template.

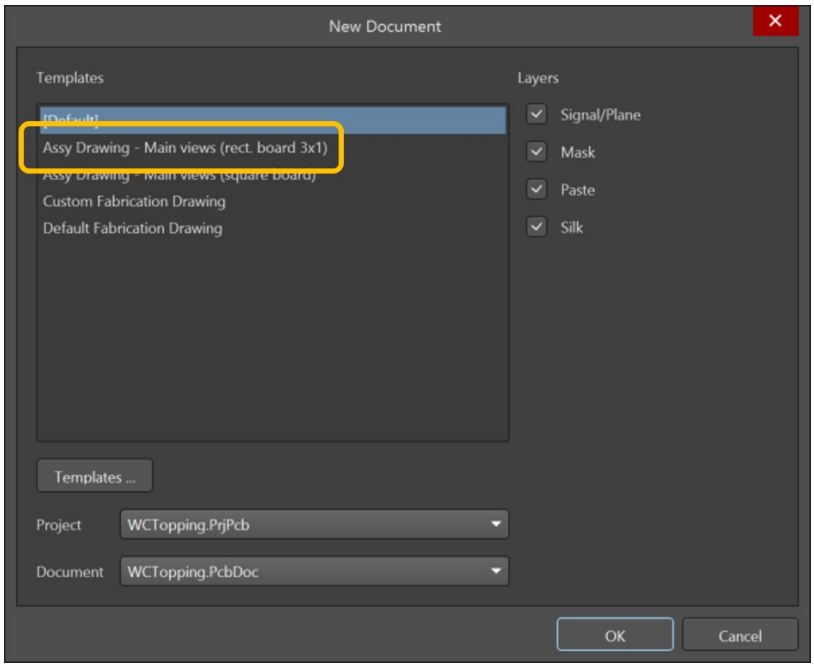


Figure 310. Create a new Draftsman document dialog

6. Dock the *Properties* panel if it appears in the middle of the screen.

7. The Draftsman document created will have multiple assembly views of the board, as pre-defined in the template. Go to **File » Save As** and save the Draftsman document as `WCTopping - Assembly View.PCBDwf` to the project folder.
8. Go to *Preferences*. Expand the *Draftsman* branch and open the *Templates* page. The path in this page currently points to the Altium Draftsman template folder by default. The path could be changed to a company-standard template folder if required.
9. Close the *Preferences* without making any changes.

24.3.2 Creating Draftsman Without a Template

10. Open another new Draftsman document with the command **File » New » Draftsman Document**. This time around, select **[Default]** as the template. Ensure to point to `WCTopping.PrjPcb` and `WCTopping.PcbDoc`.
11. Click **OK** to generate an empty Draftsman document.
12. Go to **File » Save As** and save the Draftsman document as `My Draftsman WCTopping.PCBDwf`.

24.4 Page Configuration

13. We will now define a sheet template for this Draftsman document. Open the *Properties* panel and open the *Page Options* tab. Under the *Formatting and Size* pane, select *Template* and from the Templates dropdown, select **ANSI B Landscape** as shown in Figure 311.

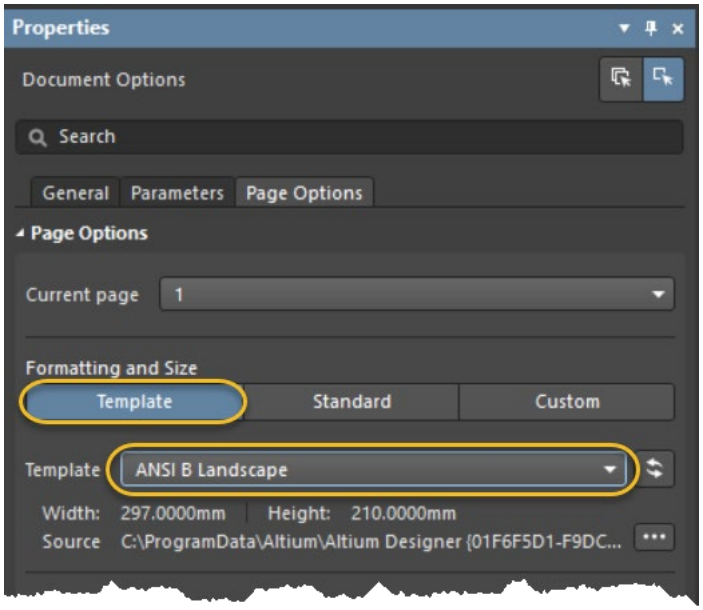


Figure 311. Changing the document template

14. Click on the *Parameters* tab. In this tab, any parameters associated with the PCB, System or Project will be listed. In the *Parameters* tab, locate the Parameter named *DocumentNumber*.
15. Change its value to *DOC1234* as shown in Figure 312.

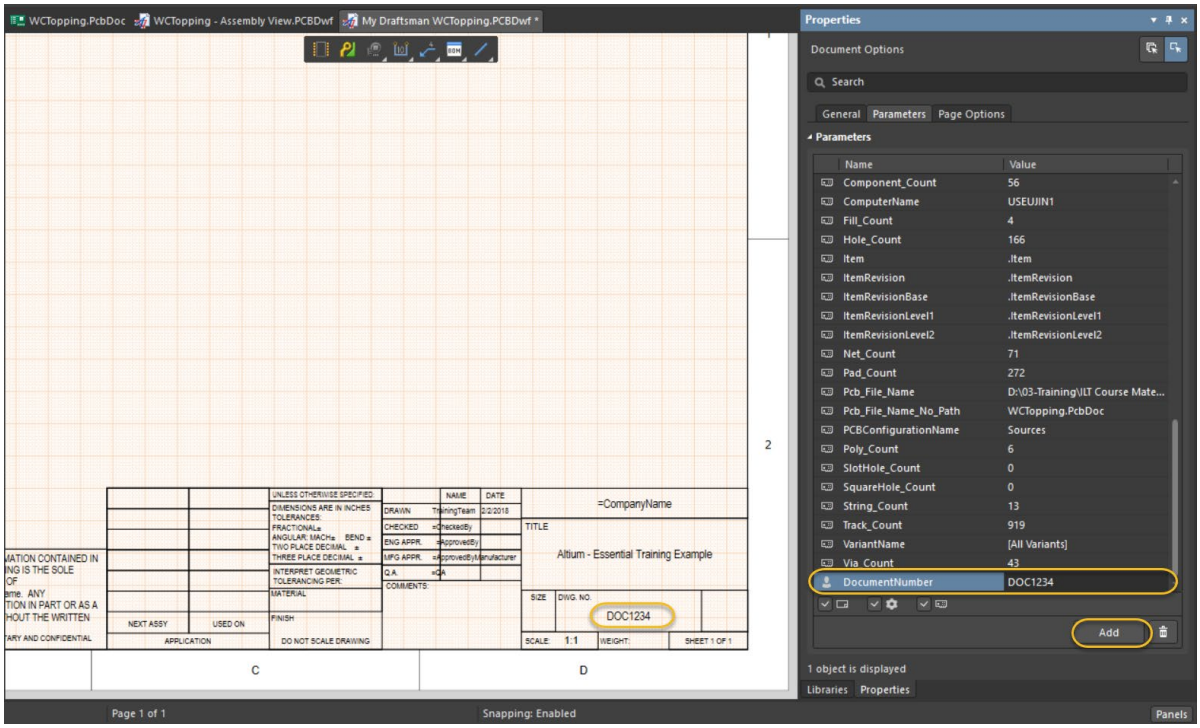


Figure 312. Adding a document parameter to the Draftsman sheet

24.5 Assembly View

16. We will first place assembly views or also known as mechanical views on our Draftsman document. Refer to Figure 313 for the following steps.
17. Go to **Place » Board Assembly View**. By default, the top-side view of the board will be provided. Place this view in the middle of the document.
18. **Right-click** on the placed assembly view and click **Copy**. A crosshair will be present, where you will need to select the view to **Copy** it Then, right-click on an empty space to the right of the Assembly View and click **Paste**.
19. Double-click on the pasted assembly view to open its *Properties* panel. In the *Properties* pane, change its *View side* to `Right` as shown in Figure 313.

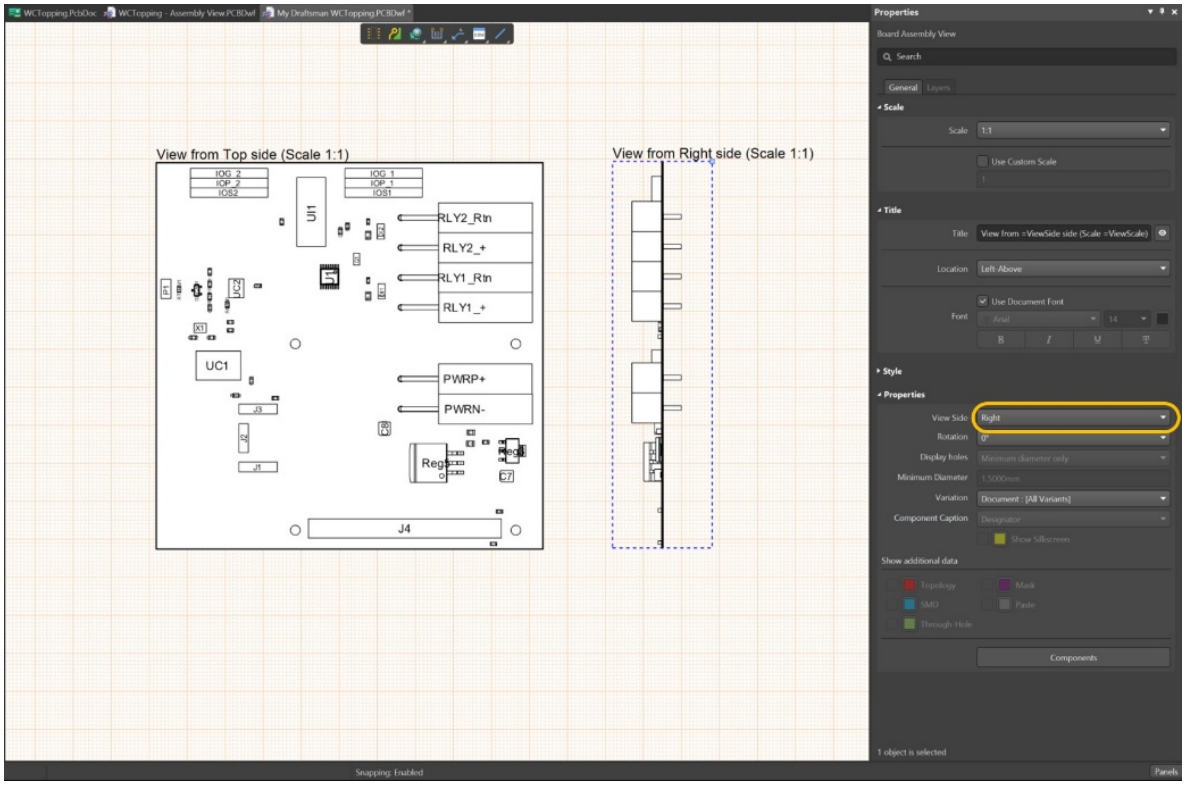


Figure 313. Assembly Views placed in the Draftsman document

20. Repeat Steps 18 and 19 above twice to place more Assembly Views.
21. Change their *View side* to `Left` and `Front`, respectively to place Left and Front assembly views onto the document. Move the views to the appropriate locations, as shown in Figure 314.

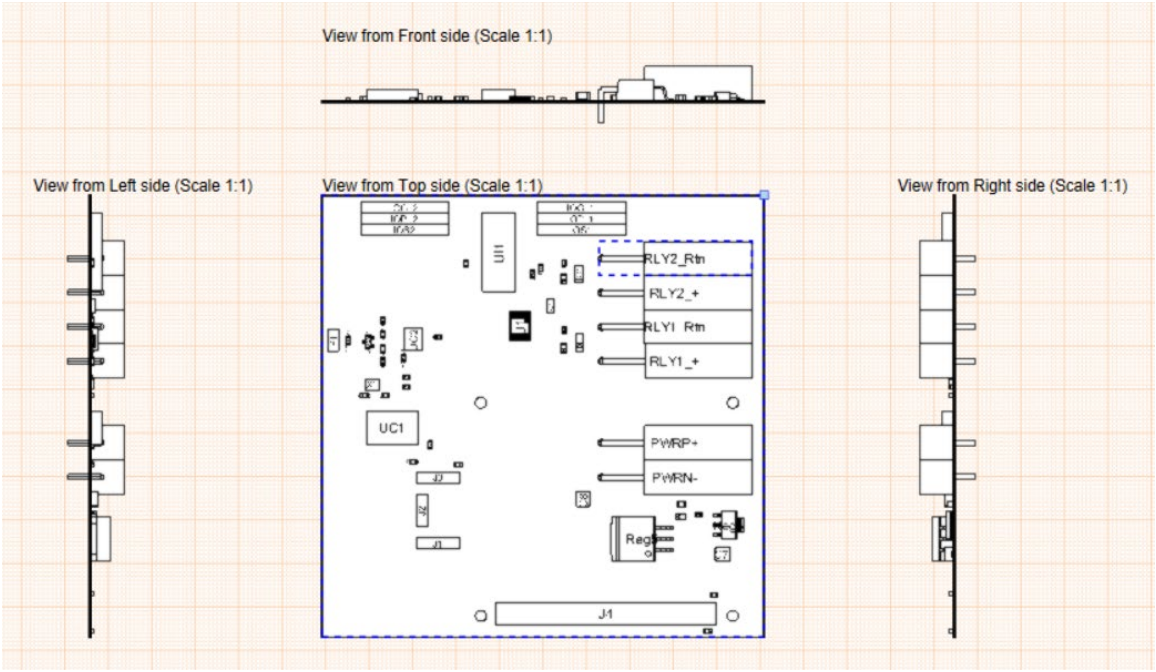


Figure 314. Multiple Assembly Views of the board

22. Double-click on the *Top Side* assembly view to open its *Properties* panel. Check on **Show Silkscreen** and **Show SMD** and **Through-Hole** pads as shown in Figure 315.

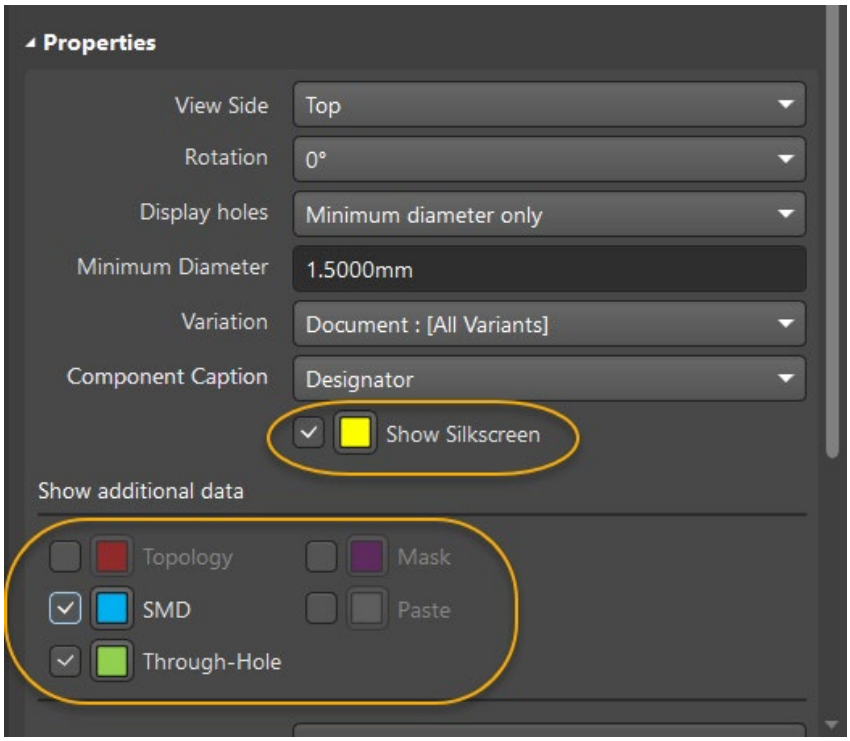


Figure 315. Show Silkscreen, SMD pads and Through-Hole Pads from the Properties panel

24.6 Fabrication View

23. We would now like to add fabrication views on a new sheet. To add a new sheet to this document, simply right-click anywhere on the document and click **Add Sheet**. Scroll down to see the new sheet that was added. A Draftsman document can contain multiple sheets.
24. To place fabrication views, go to **Place » Board Fabrication View**. By default, the top layer of the board will be provided. Place this view on the left half of the document. Use
25. Right-click on the placed fabrication view and click **Copy**. A crosshair will be present, where you will need to select the view to **Copy** it. Then, right-click on an empty space to the right of the fabrication view and click **Paste**.
26. Double-click on the pasted fabrication view to open its *Properties* panel. In the *Properties* pane, change its *Layer* from `Top Layer` to `Bottom Layer`.
27. Use Figure 316 as a reference to how your sheet should look.

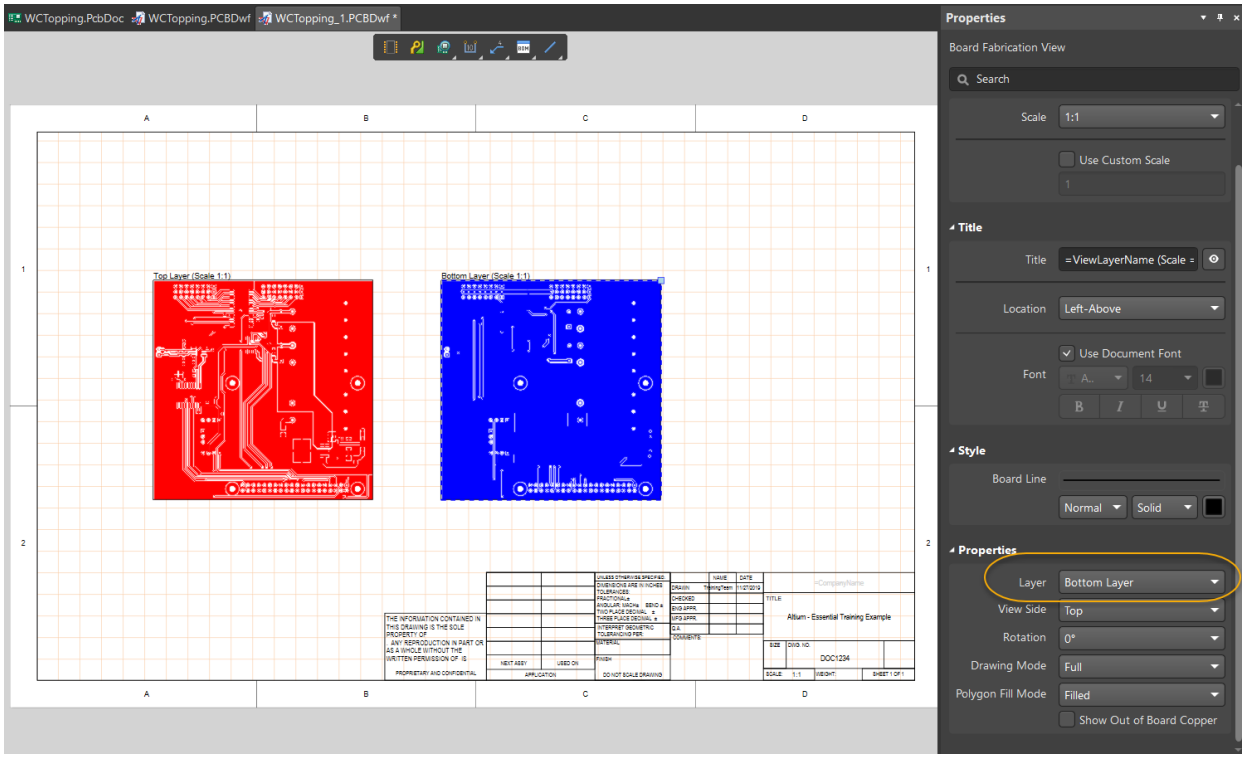


Figure 316. Fabrication View and Layer configuration in the Properties panel.

24.7 Additional Views and Dimensions

Detail, isometric and section views of the board can also be placed in the Draftsman document. Various dimensions can then be placed on any of these views.

28. Return to the first page of the Draftsman document containing the Assembly Views. Go to **Place » Additional Views » Board Isometric View** and place the view on the bottom left of the page. Use Figure 317 as a reference.
29. Currently, the components on the left of the board are too small and should be enlarged for easier visibility.
    a) We can place a detail view of this area by going to **Place » Additional Views » Board Detail View**.
    b) Left-click on any view to starting drawing a circle around the area which we wish to enlarge.
    c) Place this detail view in an empty area on the page. If required, it can also be scaled up or down in its *Properties* panel as shown in Figure 317.

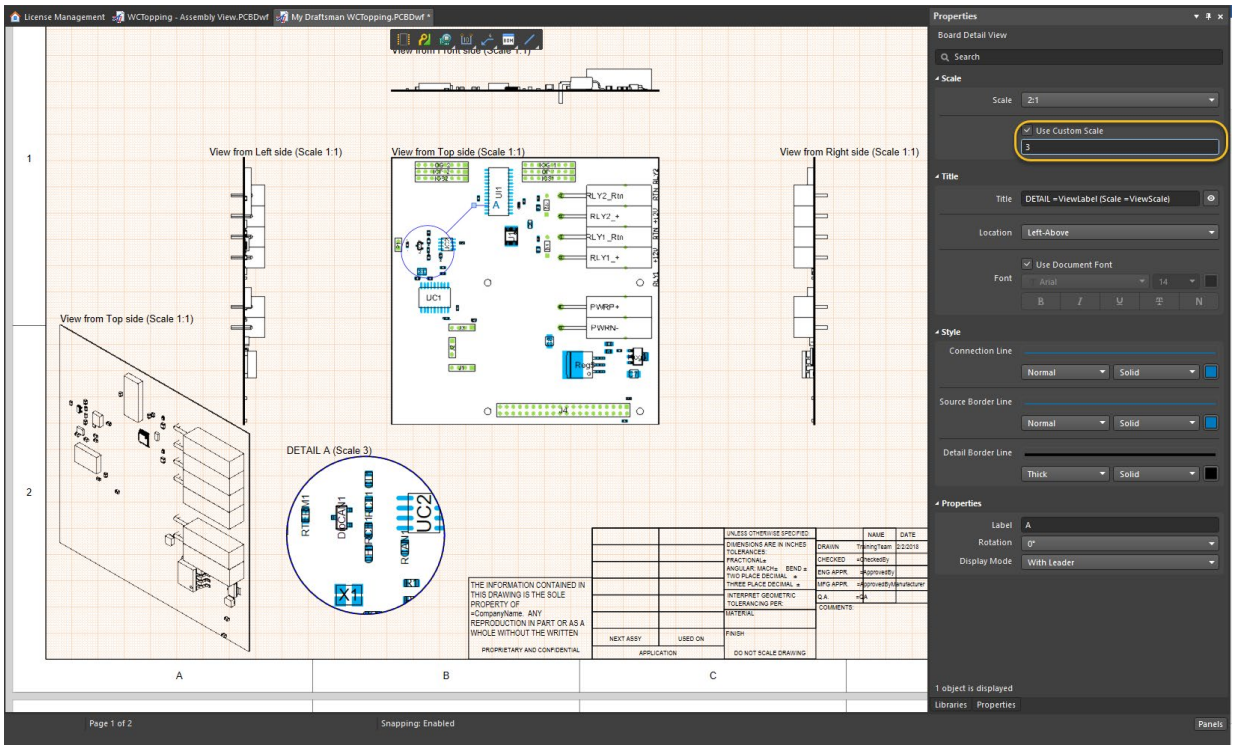


Figure 317. View placement and properties

30. Section views of the board can also be placed, for better board representation. Go to **Place » Additional views » Board Section View** and move the cursor towards the top side assembly view. Feel free to use Figure 318 as a placement guide.
    a) Hit **Spacebar** to toggle between horizontal and vertical cut
    b) Left-click to set the position of the cut line.
    c) Then, move the cursor on either side of the cut line to set the view direction. Left-click again to confirm the view direction.
    d) Left-click again to place the section view in the desired location on the top left of the page as shown in Figure 318.

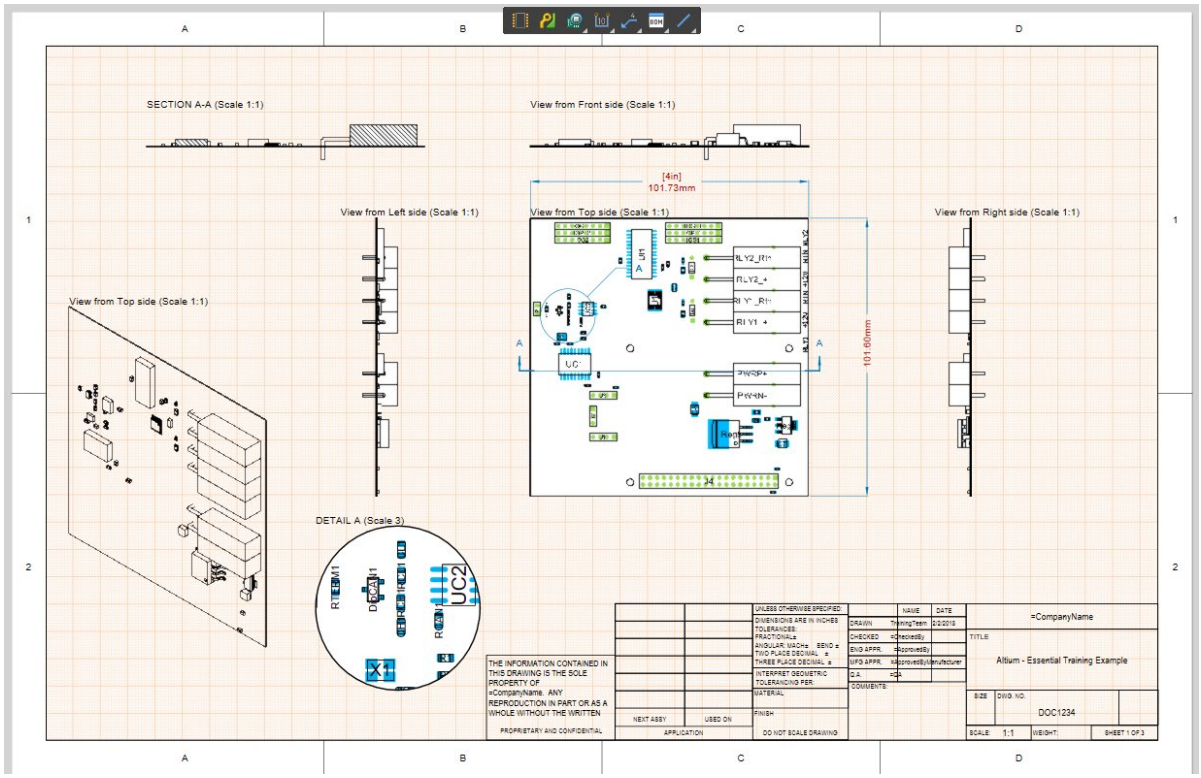


Figure 318. Assembly Views with additional detail, isometric and section views

31. We will now place linear dimensions on the top side assembly view by going to **Place » Linear Dimension**.
32. Then, left-click on the horizontal and vertical edges of the board shown in Figure 318.
33. To change its units, double-click on the dimension and configure it in the *Properties* panel.

24.8 Tables

34. We will add a third page to this document by right clicking anywhere on the document and click **Add Sheet**.
35. Scroll down all the way to the newly created page.
36. On this third page, place the following objects with the default configuration. Use Figure 319 as a reference.
    a) a Drill Drawing View **Place » Additional Views » Drill Drawing View**
    b) a Drill Table **Place » Drill Table**
    c) a Layer Stack Legend **Place » Layer Stack Legend**
    d) a BOM Table **Place » Bill of Materials**

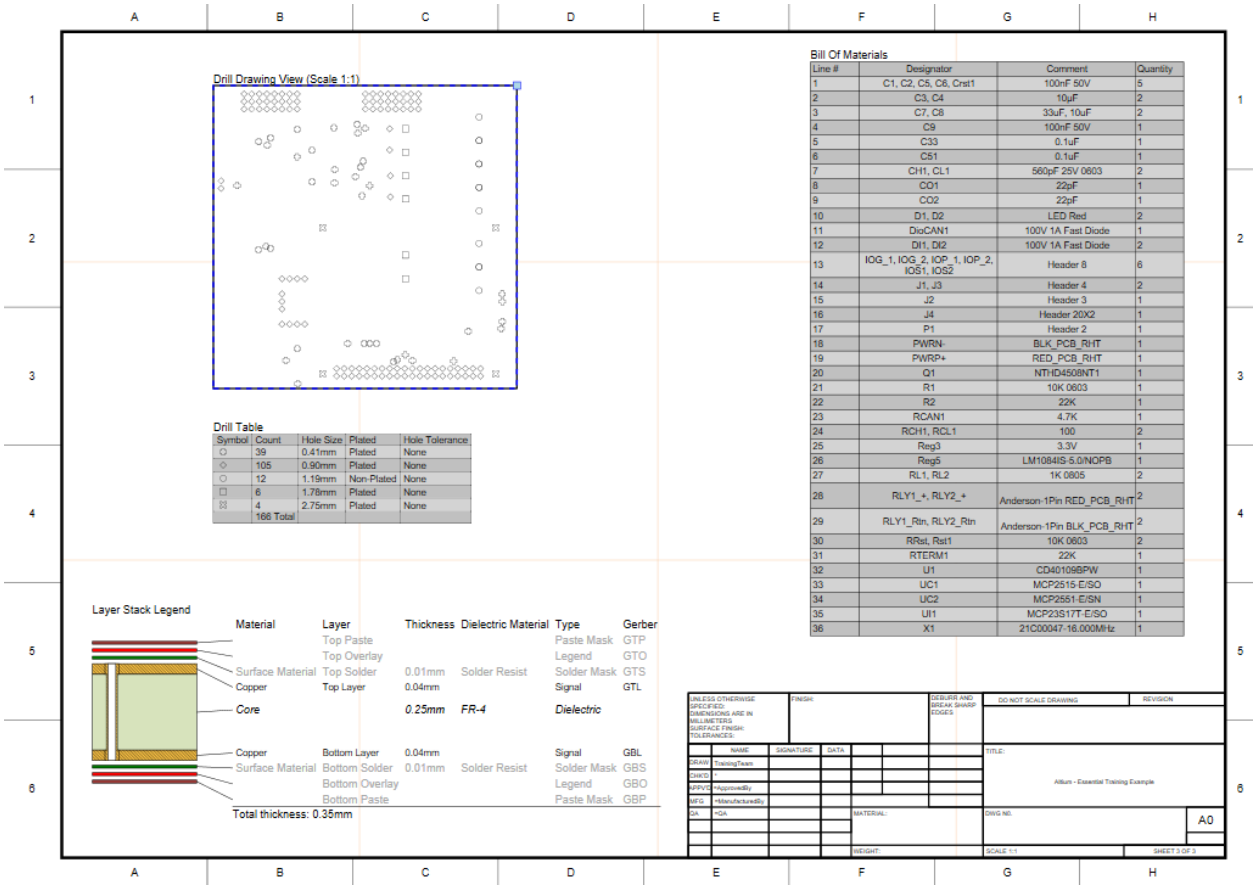


Figure 319. Drill Drawing, Drill Table, Layer Stack and BOM placed on the sheet

24.8.1 Optional

37. Feel free to add a fourth page to this document by right clicking anywhere on the document and click **Add Sheet**.
38. Scroll down all the way to the newly created page.
39. On this fourth page, you can place additional objects with the default configuration like:
    a) Board region
    b) Board Realistic
40. Close and save all projects and documents.

Module 25: PCB Fabrication Files

25.1 Purpose

In this exercise, students will learn to work with the Smart PDF as well as configure and export a Bill of Materials (BOM). Additionally, an Output Job is added to the project to manage outputs including documentation outputs, report outputs (BOM) and fabrication outputs like Gerber, NC-Drill.

25.2 Preparation

1. Close all open projects. For this exercise, we will be using the `WCTopping.PrjPcb` project located in the `Module 25 - PCB Fabrication Files` folder.
2. Open the `WCTopping.PrjPcb` project
3. Open any of the schematic document, such as `Digital_IO.SchDoc`.

25.3 Bill of Materials

4. To generate a Bill of Materials (BOM), we can go to **Reports » Bill of Materials** from the schematic environment. The BOM dialog will appear as shown in Figure 320.

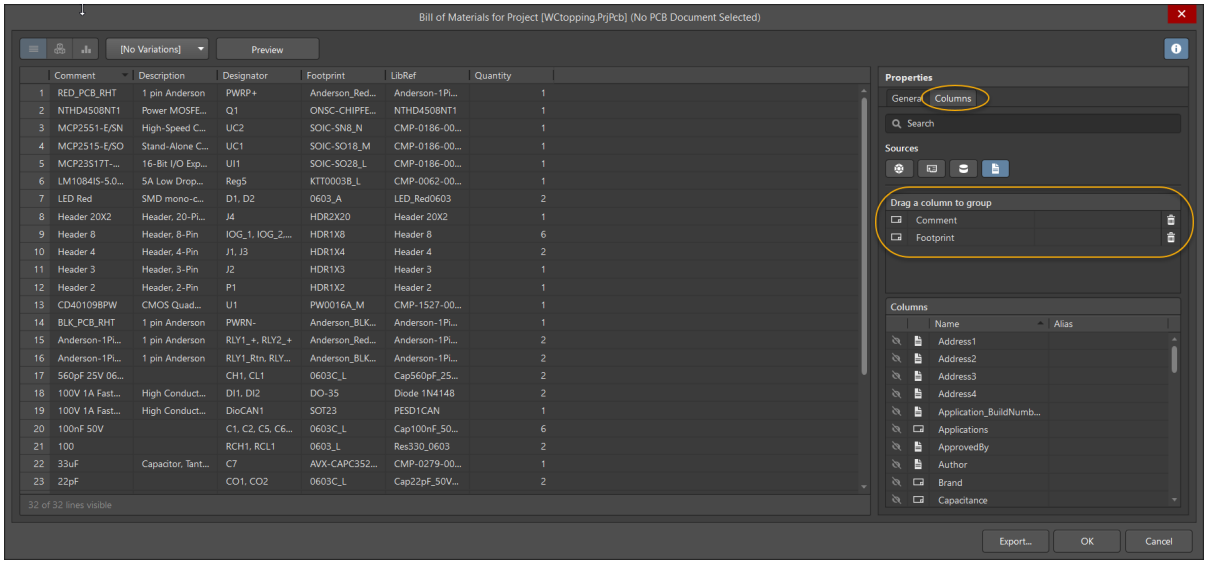


Figure 320. Bill of Materials dialog window

5. Currently, multiple components appear on a single row because they are grouped by the **Comment** and **Footprint** parameters as shown in the *Columns* tab of the *Properties* panel on the right. This is what we consider the *Grouped Columns* pane. Click the **Trash** icon next to the **Comment** and **Footprint** parameters and the BOM will have one component per row.
6. To control which parameters are available for your Bill of Materials, you can enable or disable the options under the *Sources* pane. If you hover over the icons under the *Sources* tab, it will show you where the parameters originate from.
7. Component parameters can be added or removed from your Bill of Materials, by clicking on the icon in the *Columns* pane. Columns can also be re-arranged by dragging their headers to the intended locations. This is all shown in Figure 321.

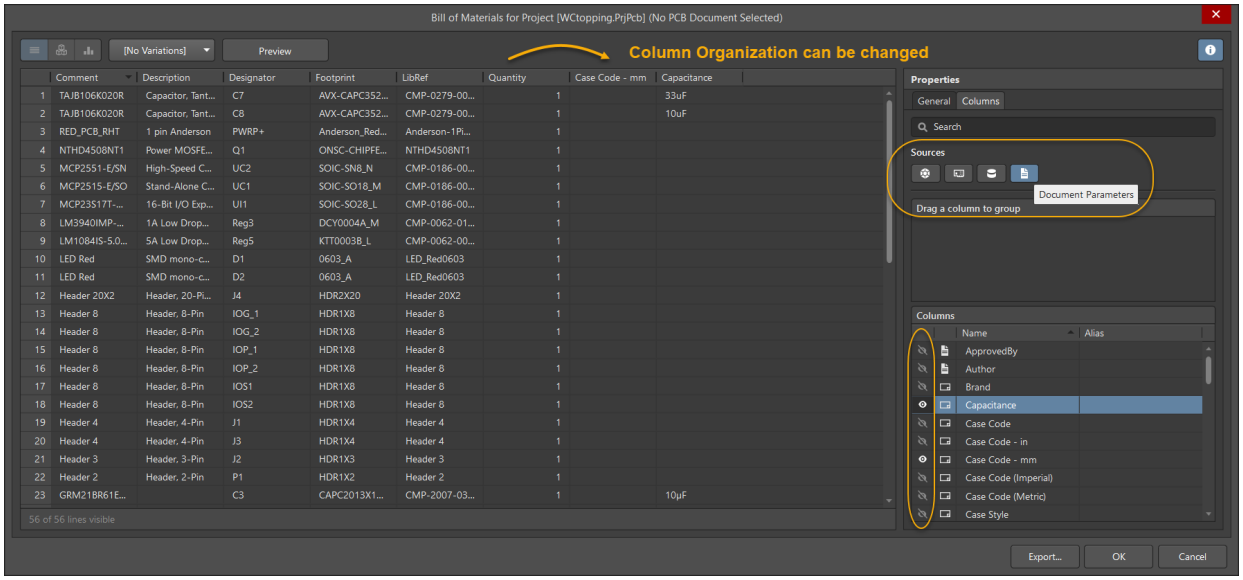


Figure 321. BOM with parameters from the PCB

8. Under *Export Options* from the *General* tab, select **Microsoft Excel Worksheet (*.xls)** as the file format as shown in Figure 322. Click **Export** to generate the report. Save the file to your `Desktop`. Close the generated Excel file upon opening.

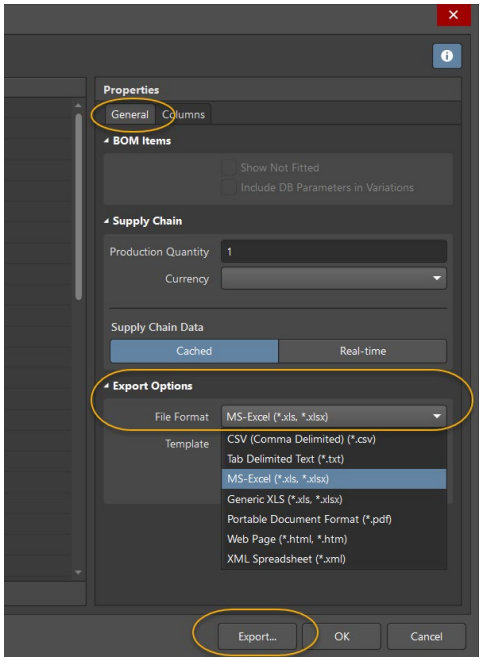


Figure 322. BOM export options

9. Click **OK** to save and close the BOM dialog.

25.4 Output Job File

An Output Job file allows users to add, configure, and reuse a variety of output configurations. In this exercise, you will define a single output job file containing all outputs we need for the working project. The outputs will include schematic and PCB printouts, validation reports, Gerbers, an NC Drill file report and a Pick & Place output.

25.4.1 Creating a New Output Job File

10. In the *Projects* panel, right-click on the project and go to **Add New to Project » Output Job File**. A new Output Job file will be added to the project as shown in Figure 323.

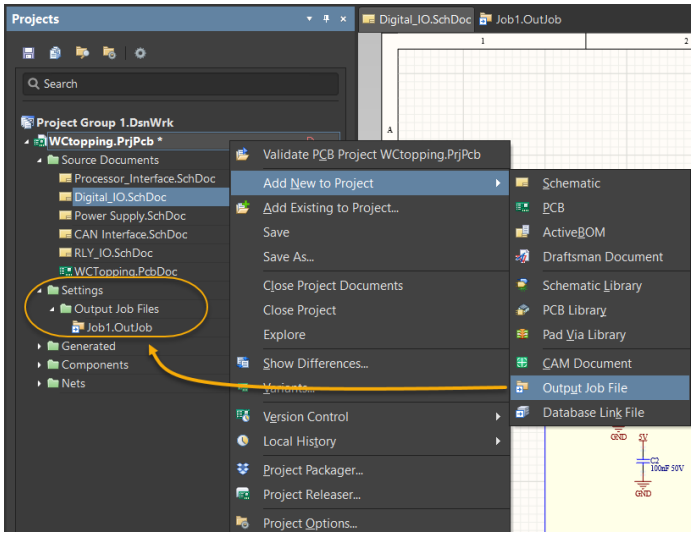


Figure 323. Output Job File added to the project

25.4.2 Defining Outputs

25.4.2.1 Documentation Outputs

11. In the *Outputs* pane, find the category for **Documentation Outputs**. Use Figure 324 as a guide for the following steps.
12. Click **[Add New Documentation Output] » PCB Prints » [PCB Document].**
13. Click **[Add New Documentation Output] » Schematic Prints » All SCH Documents.**

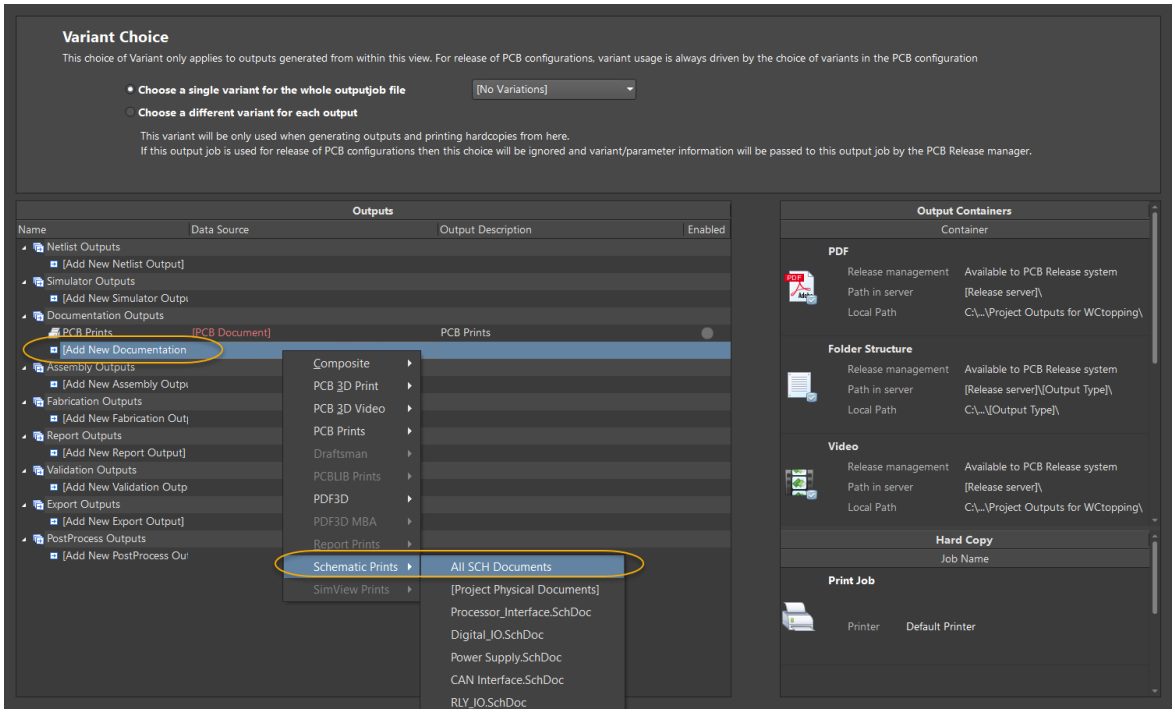


Figure 324. Output Job output configuration

14. Right-click on **Schematic Prints** output and click **Page Setup…**. Ensure that its *Color Set* option is set to `Color` as shown in Figure 325. Click **Close** to close the dialog.

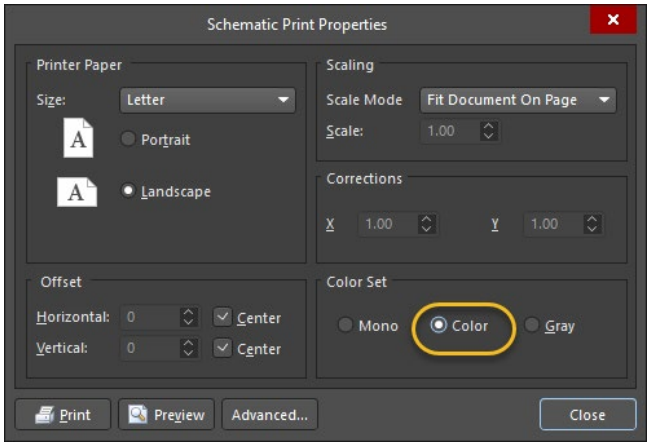


Figure 325. Schematic Print Page Setup

Defining the data source as the generic [SCH Document], [Project] or [PCB Document] instead of the specific file name would allow the Output Job file to be re-used in multiple projects as a Template.

25.4.2.2 Assembly Outputs

15. Find the category for **Assembly Outputs.** The following steps will be similar to what was done previously in Section 25.4.2.1 using Figure 324**.**
16. Click **[Add New Assembly Output] » Assembly Drawings » [PCB Document].**
17. Click **[Add New Assembly Output] » Generates Pick and Place Files » [PCB Document].**

25.4.2.3 Fabrication Outputs

18. Find the category for **Fabrication Outputs.** The following steps will be similar to what was done previously in Section 25.4.2.2 using Figure 324**.**
19. Click **[Add New Fabrication Output] » NC Drill Files » [PCB Document].**
20. Click **[Add New Fabrication Output] » Drill Drawings » [PCB Document].**
21. Click **[Add New Fabrication Output] » Gerber Files » [PCB Document].**
22. Double-click on the **Gerber Files** output to open its *Gerber Setup* dialog. Open the *Layers* tab. Click on **Plot Layers** and choose to plot **Used Layers** as shown in Figure 326. Click **OK**.

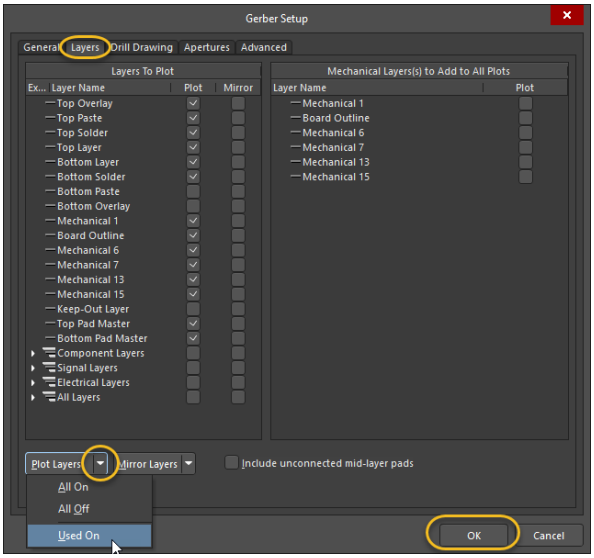


Figure 326. Gerber Setup

25.4.2.4 Report Outputs

23. Find the category for **Report Outputs.**
24. Click **[Add New Report Output] » Bill of Materials » [Project].**
25. Double click on the **Bill of Materials** output to open its BOM dialog. Here you can customize the grouping and desired columns as explored previously. Click **OK** to close the dialog.

25.4.2.5 Validation Outputs

26. Find the category for **Validation Outputs.**
27. Click **[Add New Validation Output] » Electrical Rules Check » [Project].**
28. Click **[Add New Validation Output] » Design Rules Check » [PCB Document].**

25.4.3 Defining Containers

25.4.3.1 PDF Container

29. In the *Output Containers* section on the right side of the Output Job File, click on **PDF** to select it as shown in Figure 327.
30. Then, click on the balloons in the *Enabled* column, corresponding to the outputs to map it to the PDF container. Ensure to click in the order shown in Figure 327.

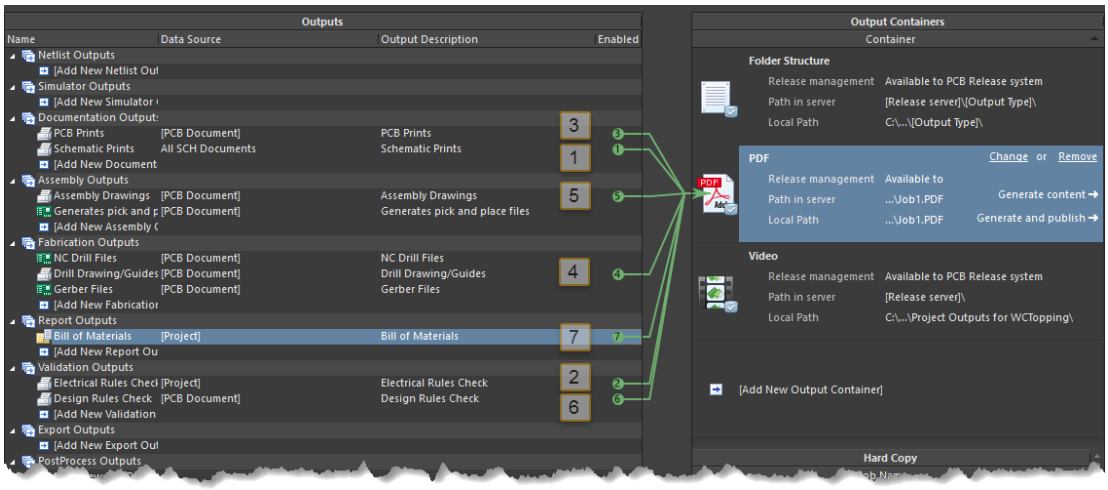


Figure 327. Output Job File with outputs linked to PDF container

25.4.3.2 Folder Container

31. In the *Output Containers* pane, select **Folder Structure**.
32. Then, click on the balloons in the *Enabled* column, corresponding to the outputs to map it to the Folder Structure container. Ensure to click in the order as shown in Figure 328.

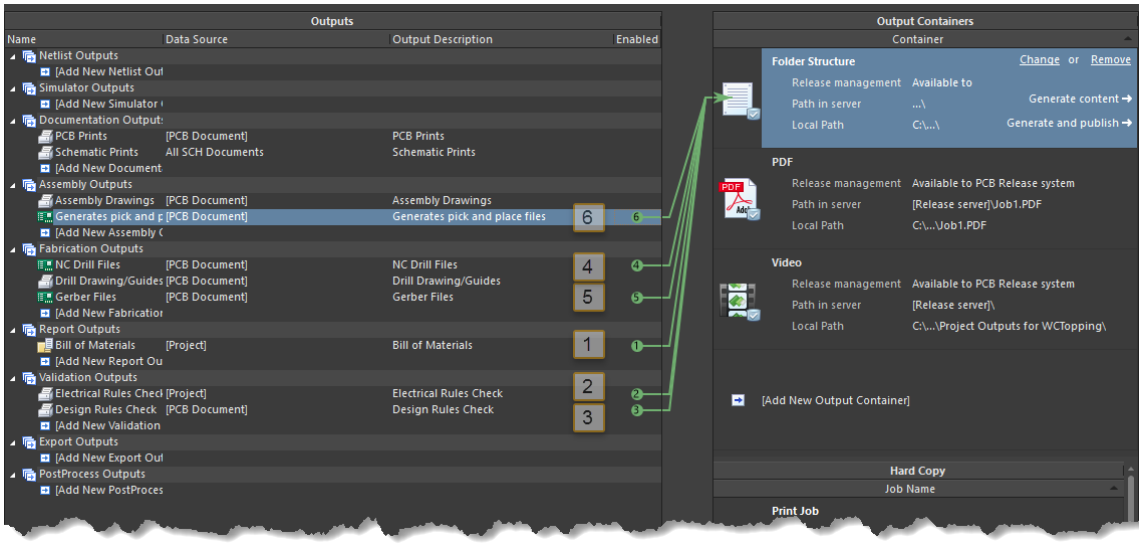


Figure 328. Output Job File with outputs linked to the Folder Structure Container

33. Save the output job file by going to **File » Save**. Save it with the default name, *Job1.Outjob.*
34. The outputs have now been created, configured and mapped to their corresponding containers. Next, we will generate these outputs.

25.4.4 Generating the Outputs

25.4.4.1 Generating the PDF

35. To create the PDF file of all seven outputs, highlight the PDF container and click on the **Generate Content** link as shown in Figure 329.
36. The PDF file will pop up after processing and will be saved to the `Project Outputs for WCTopping` project sub-folder.

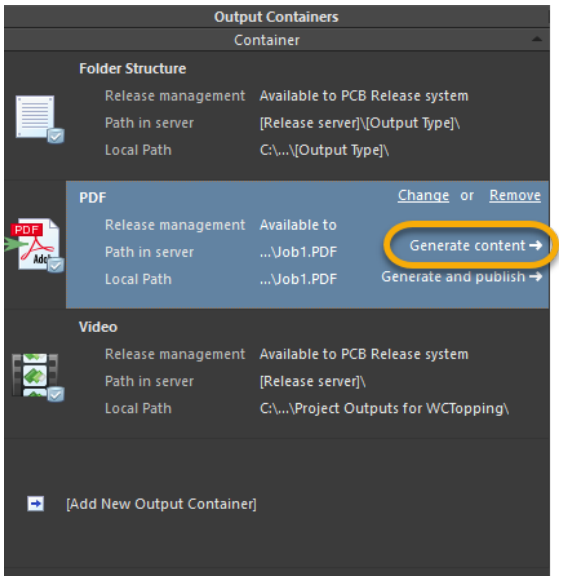


Figure 329. Generating the PDF outputs

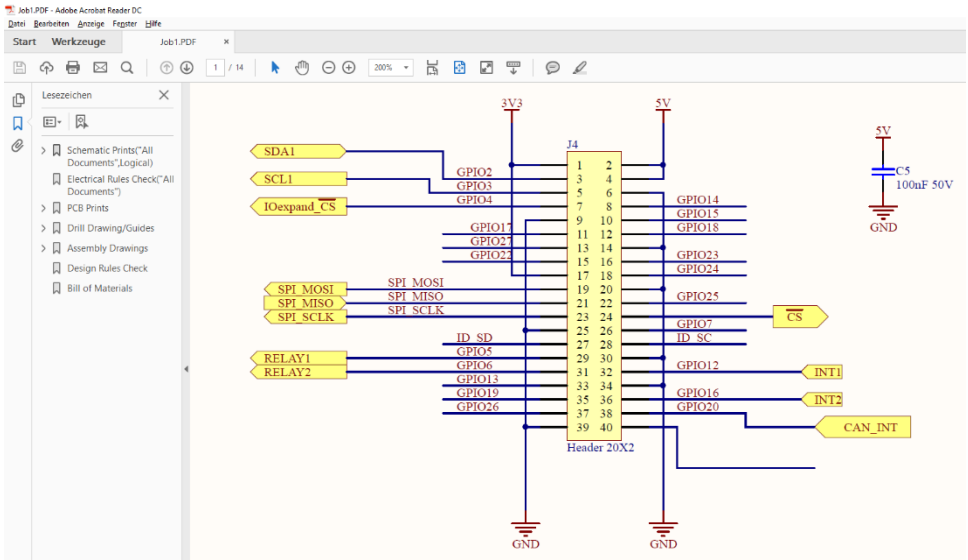


Figure 330. Generated PDF based on our outputs from the Output Job

25.4.4.2 Generating the Folder Structure

37. Click on the **Generate content** link in the *Folder Structure* section. Individual outputs will appear as separate tabs in the Altium Design workspace and as folders under the Generated section of the *Projects* panel as shown in Figure 331.
38. These files will also be saved to the `Project Outputs for WCTopping` project sub-folder.

If you wanted to save these Generated files in another folder, choosing the Output Container and clicking "**Change"** would allow you to specify the Output Folder.

39. Save and close all files.

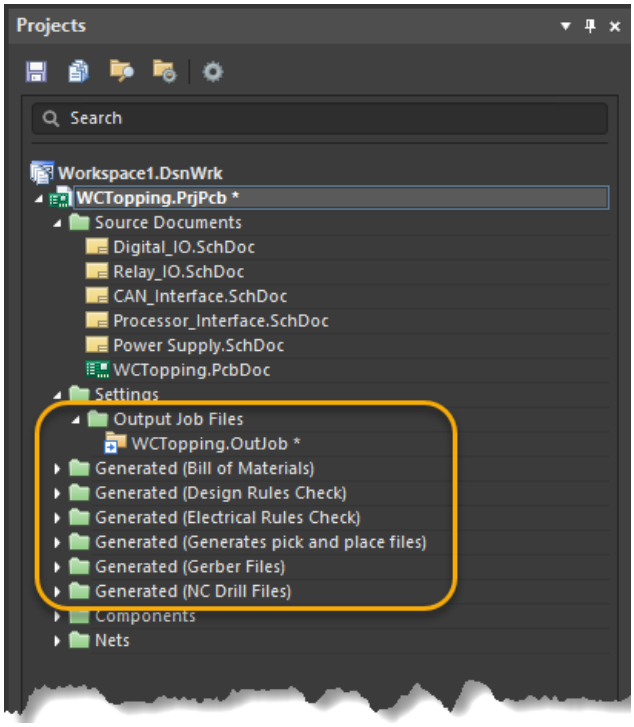


Figure 331. Folder Structure outputs generated from the Output Job

Module 26: Schematic Library

26.1 Purpose

In this exercise, students will learn the steps necessary to create a schematic symbol by creating a custom crystal symbol. Start this exercise will all files and projects closed.

26.2 Create a Crystal Symbol

26.2.1 Creating a Schematic Library

1. In your training folder, or wherever you have stored your training files, create a new folder called `My_Components`.
2. In Altium Designer, go to **File » New » Library » Schematic Library** to create a new schematic library, as shown in *Figure 332*.

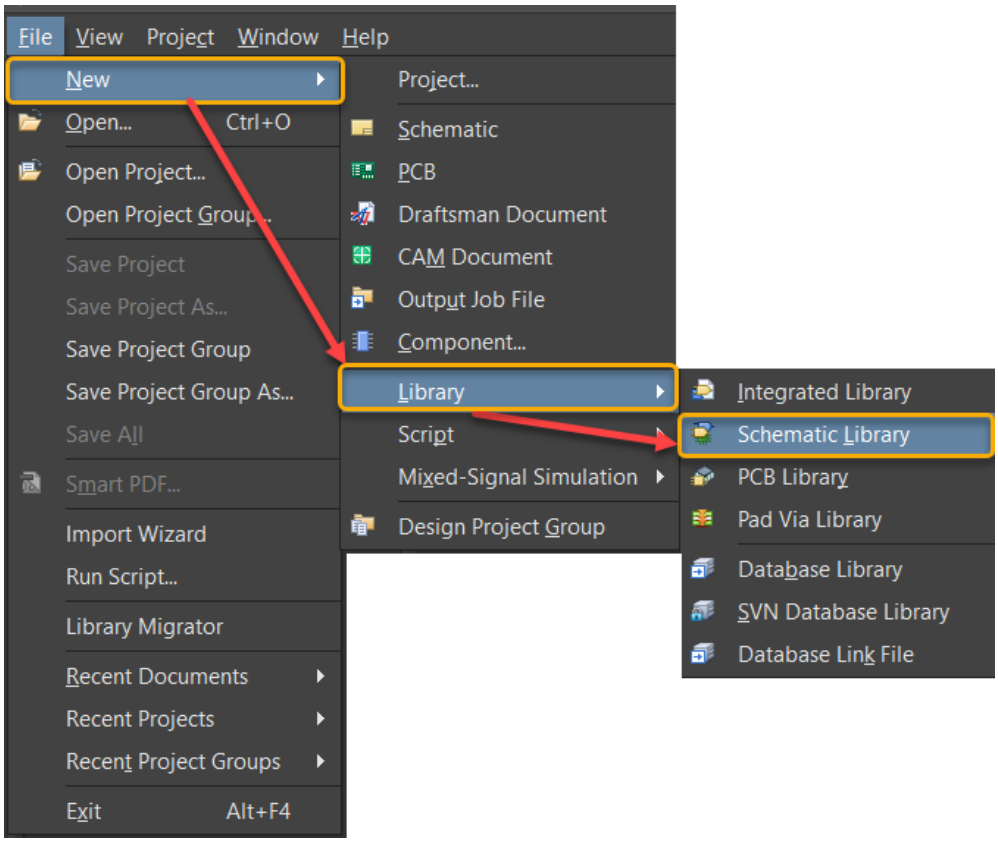


Figure 332. A new schematic library

3. Go to **File » Save As** and save the library as `My_Symbols.SchLib` in the newly created `My_Components` folder.

26.2.2 Symbol Properties

4. We are now ready to start creating the symbol. The completed crystal symbol that we will be building in this exercise will be similar to what is shown in Figure 333.

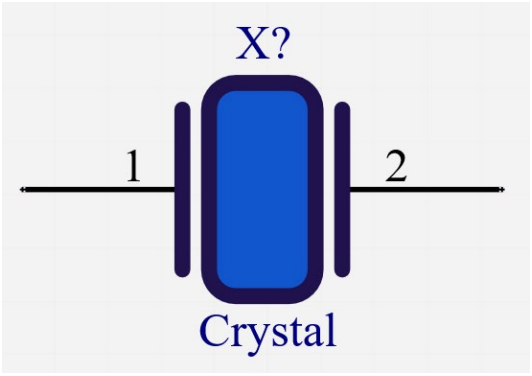


Figure 333. Completed crystal symbol for this exercise

5. By default, every new schematic library will contain a default symbol with the name `Component_1` as shown in Figure 334. If you do not see the *SCH Library* panel, go to **View » Panels » SCH Library**.

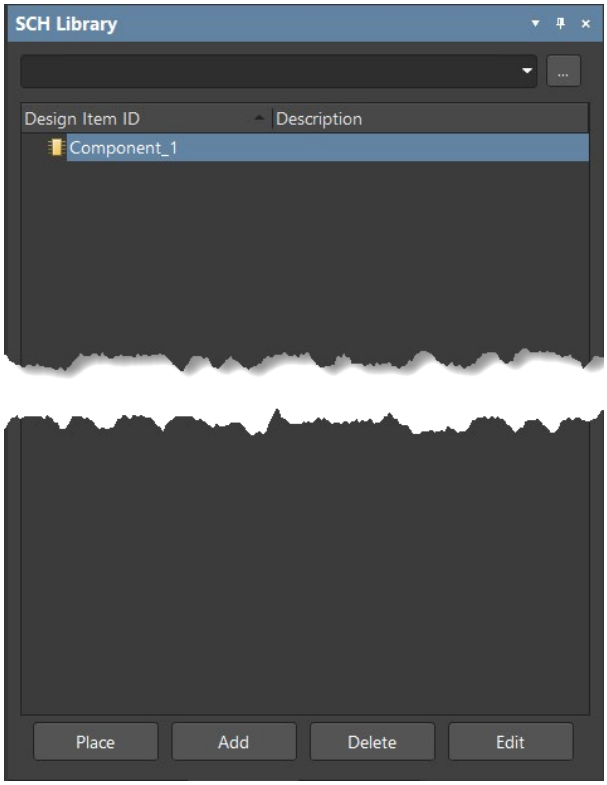


Figure 334. Default Component Name in SCH Library panel

6. Double-click on `Component_1` in the *SCH Library* panel to open its *Properties* panel as shown in Figure 335.
7. Locate the *Design Item ID* of the component and enter `Crystal` as the new name.
8. Enter `X?` in the *Designator* field. The `?` syntax will allow us to annotate the components in our design later on.
9. In the *Description* field, enter `50MHz 20pF 20ppm`.

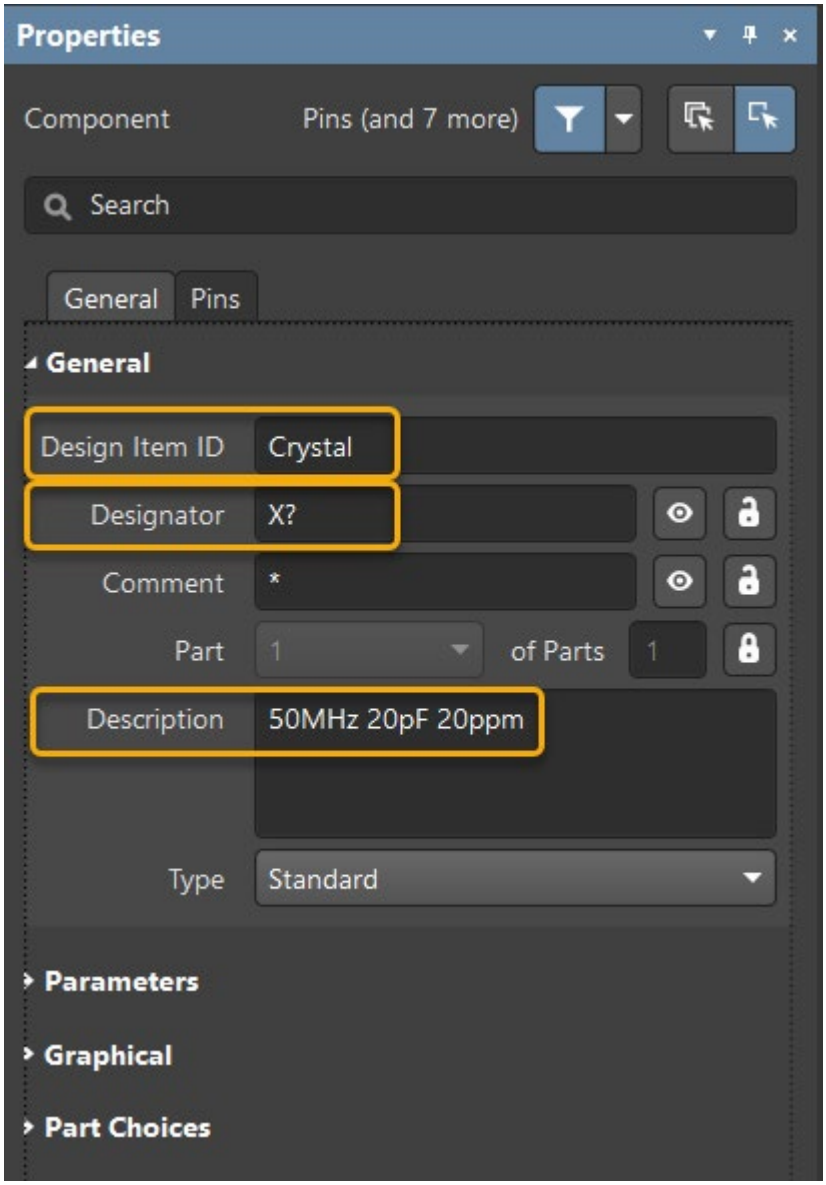


Figure 335. Changing the symbol Properties

10. Save the symbol library with the command **File » Save.** If you would like to view the Library Options, go to **Tools » Document Options** and view the options in the *Properties* panel.

26.2.3 Defining the Pins

A proper Snap Grid will simplify symbol creation. The ideal setting for this exercise is a grid of **50mil**.

11. Go to **Tools » Document Options....** to open the *Properties* panel. Change the Snap Grid to `50mil` as shown in Figure 336, if it is currently 10mil or 100mil. As an alternative, you can click anywhere in the schematic editor, and hit the **G** key to toggle between the preset grids.

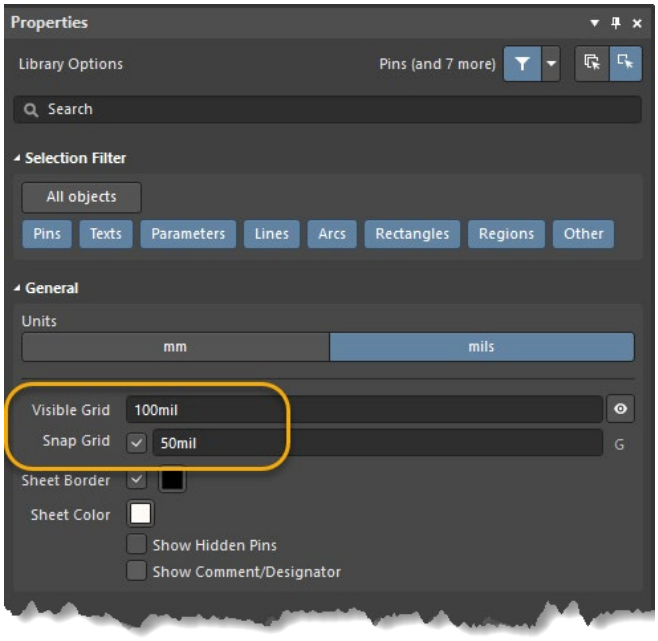


Figure 336. Snap Grid set to 50mil

12. Go to **Place » Pin** (or use the **P»P** shortcut keys) and hit **Tab** to open its *Properties* panel. Configure the settings to match those in the highlighted box of Figure 337. Hit the **Enter** key or the **Pause** icon after you've updated all fields to continue pin placement.

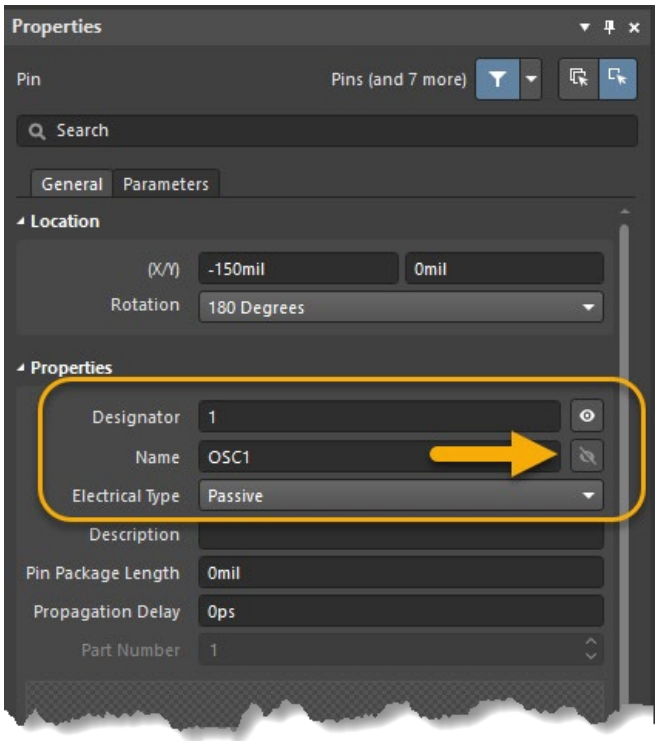


Figure 337. Properties of Pin 1 in the Properties panel

13. Notice that one end of the pin has a small “X” at one end as shown in Figure 338. This X cross-hair defines its electrical connection or hotspot. Use the **X** key to flip the pin horizontally while the pin is floating to flip its orientation. You can also use the **Spacebar** to help with the orientation if needed.

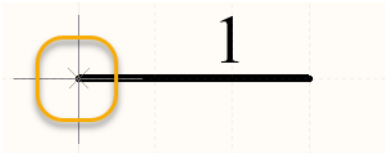


Figure 338. Pin with electrical hotspot on the left

14. Using Figure 339 as reference, place pin 1 to the left of the origin.
15. After placing Pin 1, another pin will appear on the cursor with the pin name and designator already incremented to 2. Ensure that the X cross-hair of this pin is on the right and the pin is positioned to the right of the origin as shown in Figure 339.
16. **Right-click** or hit **ESC** to exit pin placement mode.

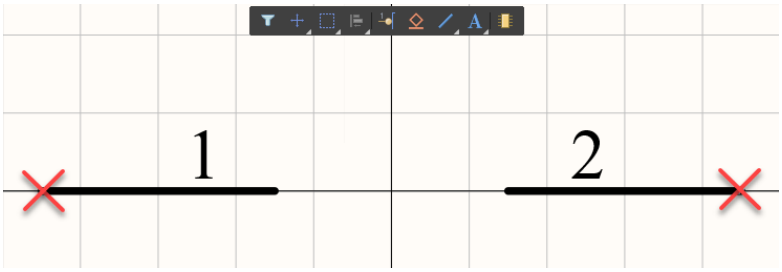


Figure 339. Position pin 2 to the right of pin 1 separated by a pin length (hotspot for image enlarged)

Each Pin has a single electrical hotspot on one end for a valid wire connection. When a placed pin is moved the cursor automatically jumps to the hotspot. Click and hold on each pin to confirm that the hotspot is positioned on the outer end of each pin, ready for wiring.

26.2.4 Adding Symbol Body

17. To place the two vertical lines on the end of each pin, use the command **Place » Line**.
18. Press the **Tab** key during placement to open the *Properties* panel and set the *Line Width* to **Medium** and choose a color of your choice. Press **Enter** or the **Pause** icon to continue the placement.
19. Draw the 2 lines similar to what is shown in Figure 340. Right-click to terminate each line segment. Right-click a second time when you are finished to terminate the command.

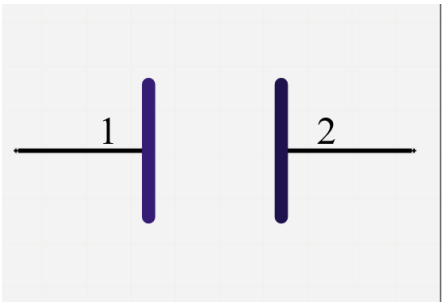


Figure 340. Adding lines to crystal symbol

20. Go to Preferences, *Schematic* branch, and *Defaults* page. Select the object *Round Rectangle* from the *Primitive* list and change both the *Corner X Radius* and *Corner Y Radius* to `50mil`. Click **OK** to apply this as default.

21. Go to **Place » Round Rectangle** command to add the central body for the crystal.
22. Hit the **Tab** key to open its *Properties* panel. Change the *Border* to **Medium** and select your choice of color as shown in Figure 341.

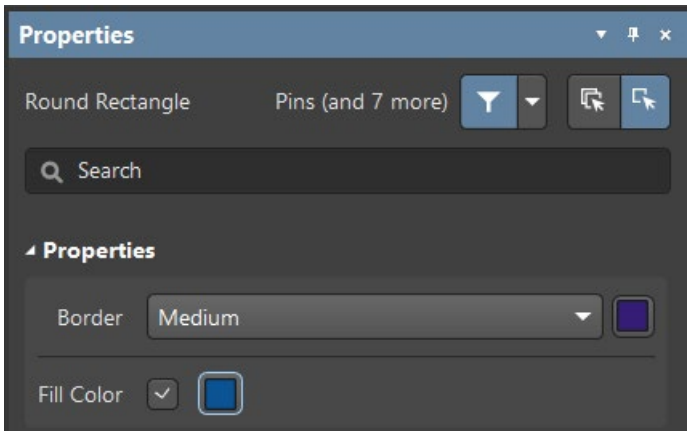


Figure 341. Round Rectangle dialog

23. Draw the rectangle in the center of the symbol as shown in Figure 342 by clicking in the top left corner, then the bottom right corner. Right-click to terminate the command.

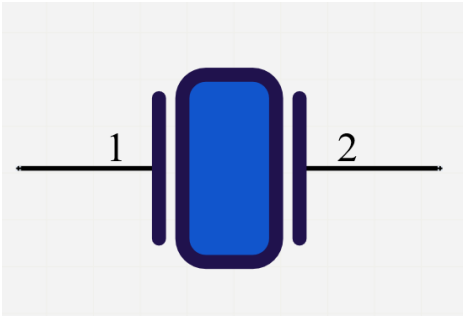


Figure 342. Adding the central body to the crystal

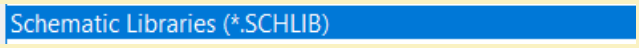
24. Save the library by going to **File » Save**.

26.3 Using the Symbol

26.3.1 Installing the Library

25. To install your new Schematic Library, open the *Components* panel by selecting **Components** from the **Panels** button in the bottom right corner of Altium Designer, or from **View » Panels » Components**.
26. Click the Hamburger Button and choose **"File-based Libraries Preferences…**".
27. The *Available File-based Libraries* dialog will open.
28. Click on the *Installed* tab to view currently installed system libraries.
29. Click the **Install…** button.
    a) Navigate to the `My_Components` folder that you created and select the `My_Symbols.SchLib` library.
    b) Click the **Open** button.

If your library file is not visible, be sure that the filter drop-down in the lower right corner of the file browser window is set to `Schematic Libraries(*.SCHLIB)`.



30. Your *Available file-based Libraries* panel should now be added to the list as shown in Figure 343.

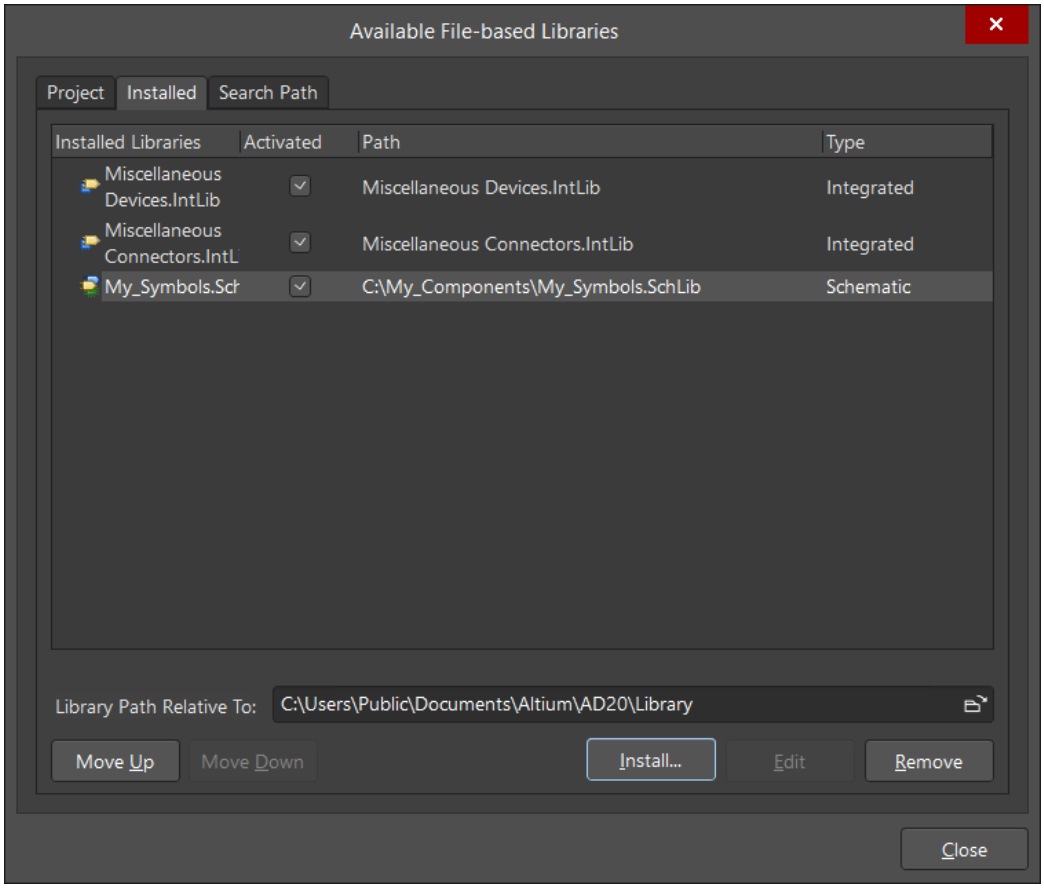


Figure 343. My_Symbols.SchLib added to the Installed Libraries list

31. Click the **Close** button in the *Available File-based Libraries* dialog.
32. Back in the *Components* panel, select `My_Symbols.SchLib` from the drop-down list. The crystal symbol should appear as shown in Figure 344.

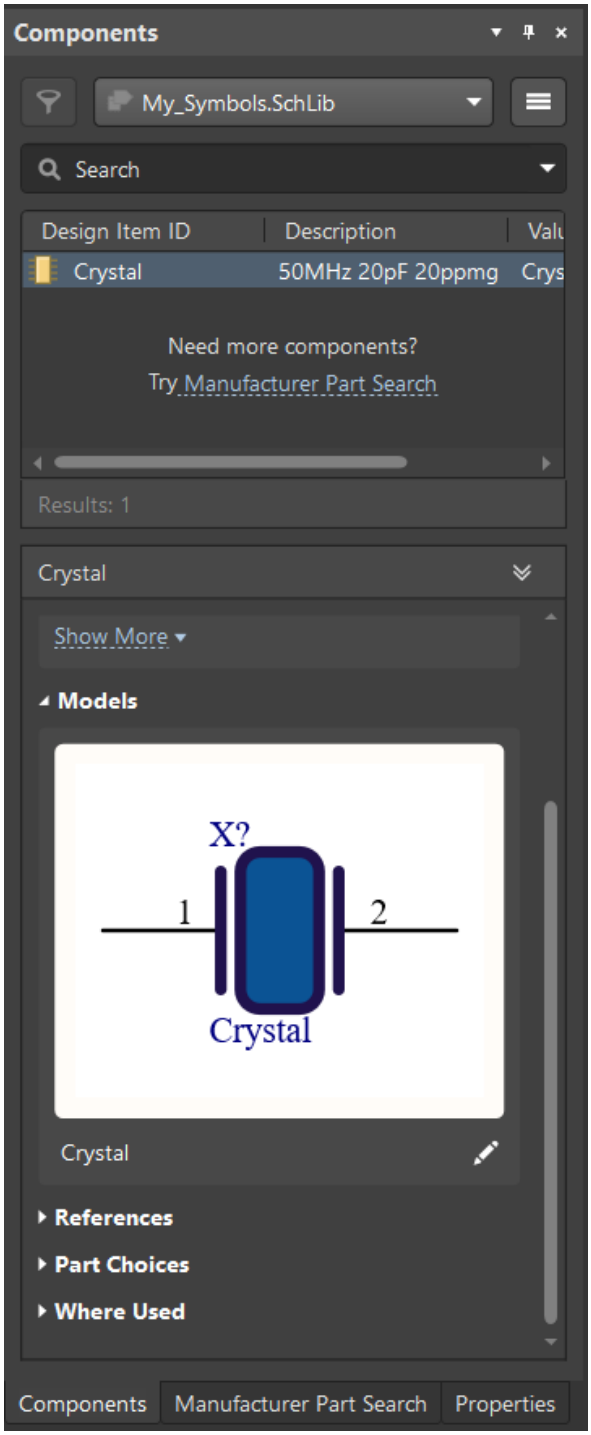


Figure 344. Crystal symbol appearing in Components panel

26.3.2 Adding Symbol to Schematic

33. Create a new schematic document by going to **File » New » Schematic.**
34. Place the crystal symbol in the schematic by right-clicking the component in the *Components* panel, and **Place Crystal**.
35. Left-click to place the component in the schematic. Then, right-click to end the placement command.

26.4 Revising the Symbol

26.4.1 Making a Change to the Symbol

36. Open the `My_Symbols.Schlib`. If you have closed the window, use the command **File » Open » Recent Documents » My_Symbols.SchLib**.
37. Double click the center body of the crystal and change the *Fill color* in the *Properties* panel for the Round Rectangle to **Yellow** as shown in Figure 345.
38. Save the library file.

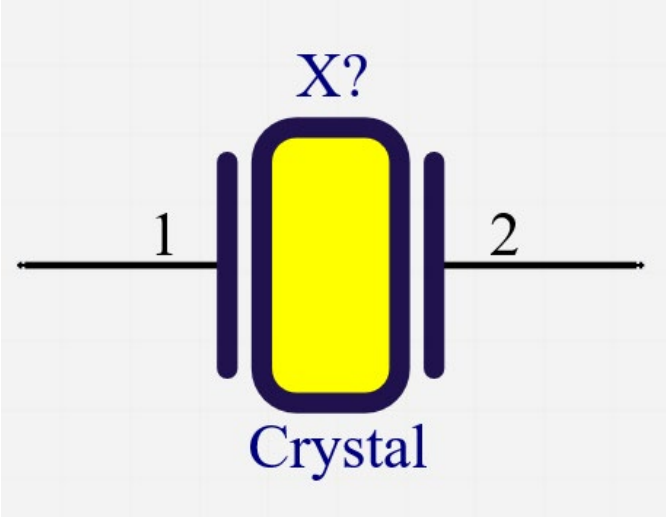


Figure 345. Updated crystal symbol

26.4.2 Updating the Schematic

39. Return to the schematic document created earlier. The symbol on the schematic is still the initial version before changing the fill color to yellow. To update this symbol:
    a) Go to **Tools » Update from Libraries.** Ensure to check on **Fully Replace symbols on sheet with those from library** and click the **Finish** button.
    b) Click **Execute Changes** in the *ECO* dialog, followed by **Close**. The symbol should then be updated with the newest version, with a yellow-filled rectangle.
40. Close the schematic document without saving.

Module 27: PCB Library

27.1 Purpose

In this exercise, students will learn the steps necessary to create a PCB footprint by creating a HC49SMD package for the crystal symbol created earlier.

27.2 Creating a PCB Library

1. Create a new PCB Library by going to **File » New » Library » PCB Library** as shown in Figure 346.

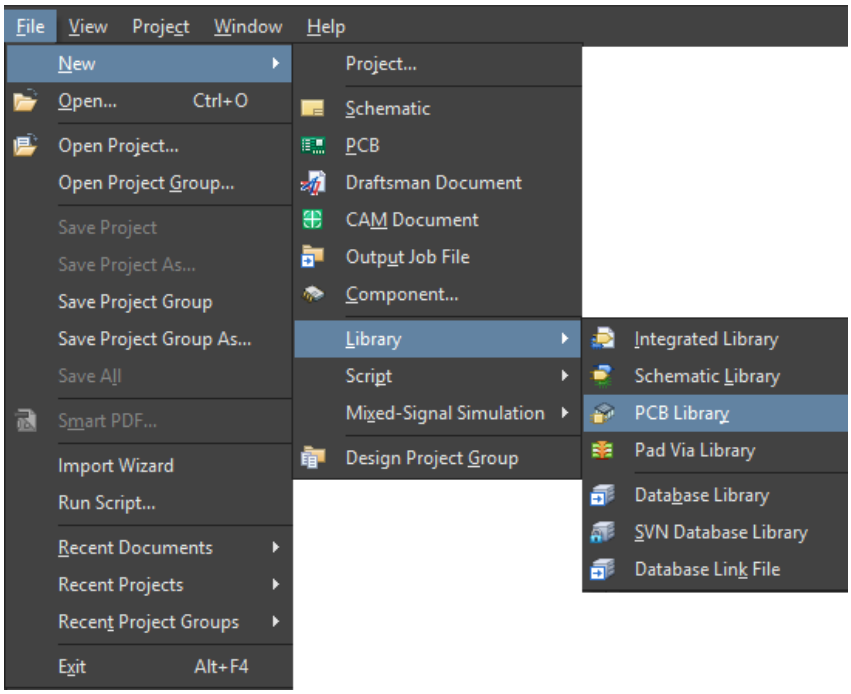


Figure 346. A new PCB Library

2. Go to **File » Save As** and save the library as `My_Footprints.PcbLib` in the `My_Components` folder created in the previous exercise.
3. By default, every new PCB footprint library will contain a default footprint with the name `PCBCOMPONENT_1` as shown in Figure 347. If you do not see the *PCB Library* panel, go to **View » Panels » PCB Library**.

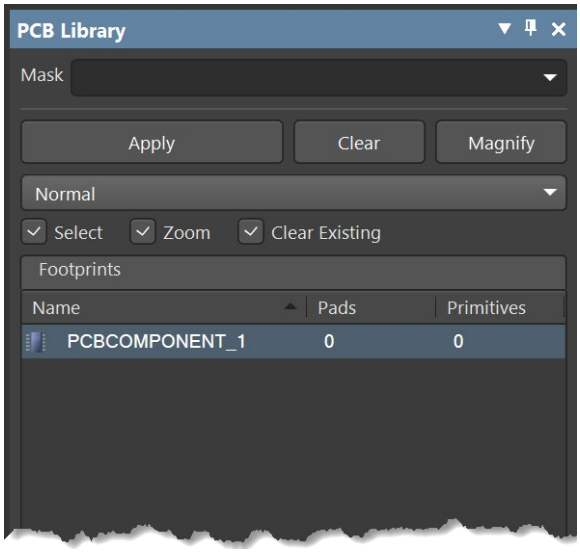


Figure 347. PCB Library panel with default component PCBCOMPONENT_1

4. Double-click on `PCBCOMPONENT_1` and in the pop-up window rename the component to `HC49SMD`, set the *Height* field to `4.5mm` and add the *Description* `HC49SMD-RoHS Compliant` as shown in Figure 348. Use the **CTRL+Q** shortcut keys to toggle between mil and mm units if needed. Click **OK** when finished.

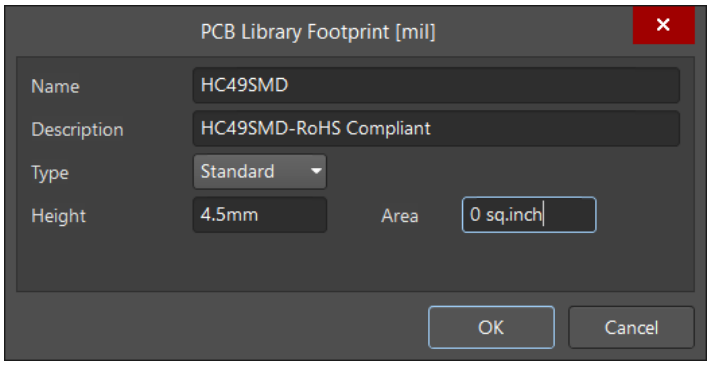


Figure 348. Configuration for the new Footprint

27.3 Footprint Creation

A preview of the footprint we will be creating in this exercise is shown below in Figure 349.

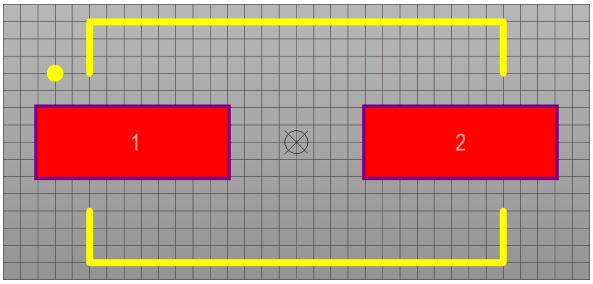


Figure 349. Crystal footprint that will be created in this exercise

27.3.1 Setting the Grid

5. The footprint uses metric dimensions so hit the **Q** key to toggle the workspace units to **mm** if needed. Hit the **G** key to display the list of preset grid settings and choose a value of **0.5mm** as shown in Figure 350.

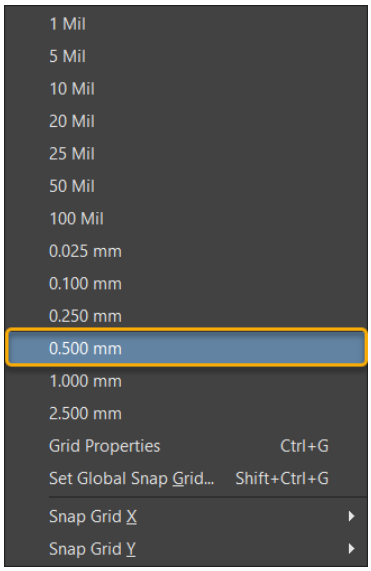


Figure 350. Preset grid

6. The grid value and the selected unit should now be visible in the lower left corner as shown in Figure 351.

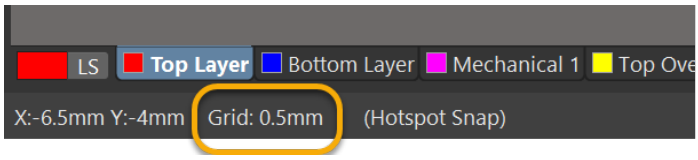


Figure 351. Grid setting in the bottom left corner of editor indicates 0.5mm

To change the current grid markers and color press **CTRL+G** to open the *Grid Editor*. Select the marker style **Lines, Dots**, or **Do Not Draw** and the color you prefer.

27.3.2 Adding Pads

7. To place a pad, go to **Place » Pad,** or use the **P » P** shortcut keys.
8. Hit the **Tab** key to open its *Properties* panel before placing it.
9. Change its *Designator* to `1` and ensure **Top Layer** is selected in the *Layer* option.
10. In the *Pad Stack* region, change *Shape* to **Rectangular** the pad dimensions to X*:* `5.5mm` Y: `2mm`. You may need to scroll down the *Properties* panel to enter the pad dimensions, as shown in Figure 352.

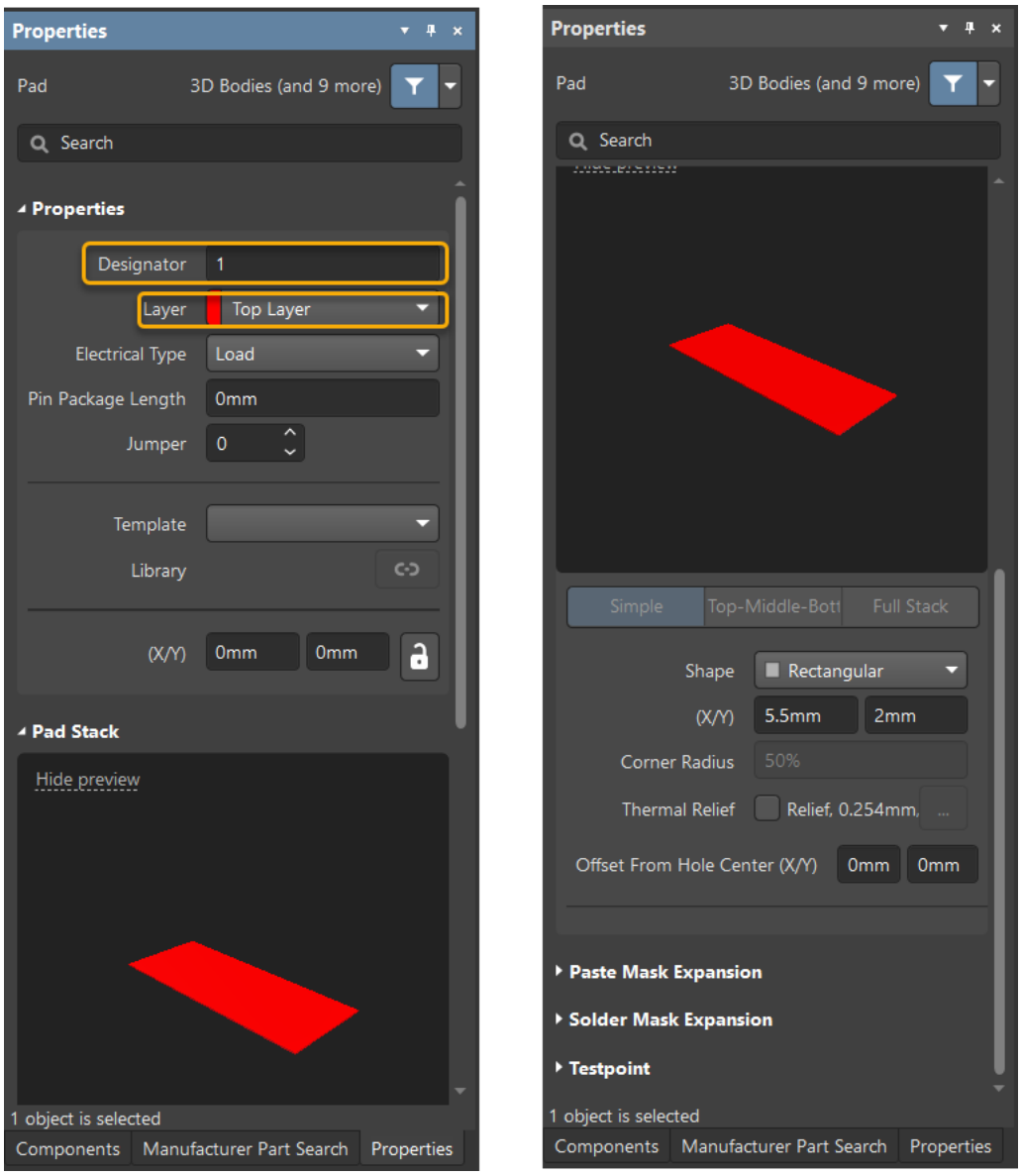


Figure 352. Pad Properties dialog with parameters for Pad 1

11. Hit **Enter** or click the pause symbol to continue placement.
12. Left-click to place Pad 1 on the origin.
    A second pad of the same shape and with Designator 2 is now attached to your cursor.
13. If it is not currently visible, turn on the *Heads Up Display (HUD)* by hitting **Shift + H.**
14. Slowly move the position of Pad 2 until the *HUD* displays an incremental *dx* value of **9.5mm** and a *dy* value of **0mm** as shown in Figure 353. Left-click to place Pad 2 at that position.

Figure 353. Heads Up Display showing incremental cursor position

15. Right-click or hit the **ESC** key to terminate the command. The placed pads and their properties should then be listed in the *PCB Library* panel, as shown in Figure 354.

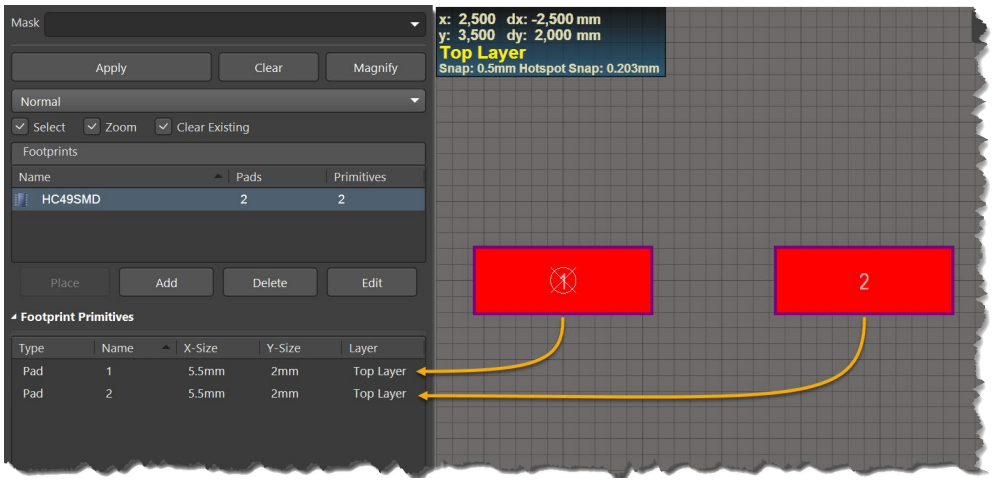


Figure 354. PCB Library panel showing two pads added for component HC49SMD

27.3.3 Setting Component Origin

There are three options to set an origin for a component: **Pin 1**, **Center** or **Location** as shown in Figure 355.

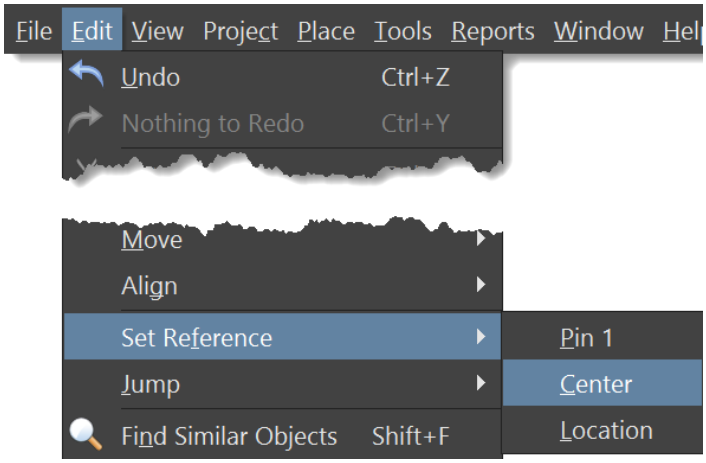


Figure 355. Setting Component Origin from the Edit menu

16. Assign the origin to the center of the existing pads by going to **Edit » Set Reference » Center**. The resulting circle and diagonal lines indicating the component origin is shown in Figure 356.

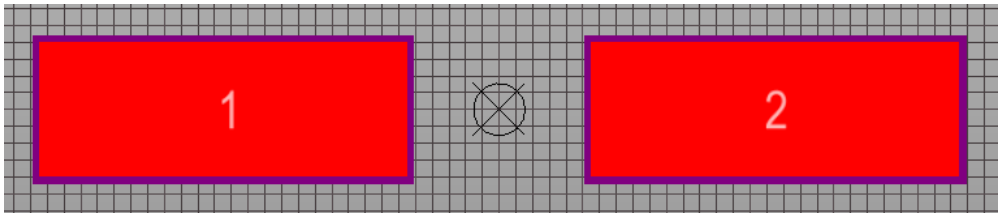


Figure 356. Component Origin set to Center

27.3.4 Adding Top Overlay Outline

17. Switch to the silkscreen layer by clicking on the **Top Overlay** tab at the bottom of the workspace.
18. Go to the **Place » Line** command and hit the **Tab** key to set the line width to `0.2mm` in the *Properties* panel. Press **Enter** or click the pause symbol to continue the line placement
19. Now that we have a component origin, we can use the *HUD* to position the component outline precisely. Move the cursor to the component origin at the center of the component and note that the *HUD* reports an *x* and *y* value of 0.

The command **Edit » Jump » New Location (J-L)** with x:`0` and y:`0` or **…Jump » Reference (Ctrl+END)** could help to find the correct position.

20. Using the *HUD* to guide you, move the cursor to location x: `-6.000` y: `2.000` and click to set the starting point for the line.
21. Move the cursor up and to the right to see the corner angle mode of the line. Press **Shift + Spacebar** to change the shape of the corner and **Spacebar** to change the orientation until the *HUD* displays *Line 90 StartPlace Line* as shown in Figure 357.

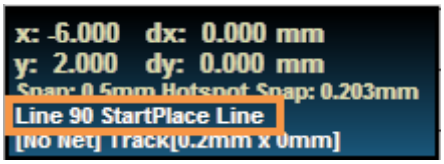


Figure 357. HUD displaying current line mode of 90 degrees

21. Move the cursor to the following positions and left click to set the vertices at each location:
    a) `(-6.0, 3.5)`
    b) `( 6.0, 3.5)`
    c) `( 6.0, 2.0)`
22. Right-click to terminate the line and right click again to terminate the command. The line should now appear as in Figure 358.

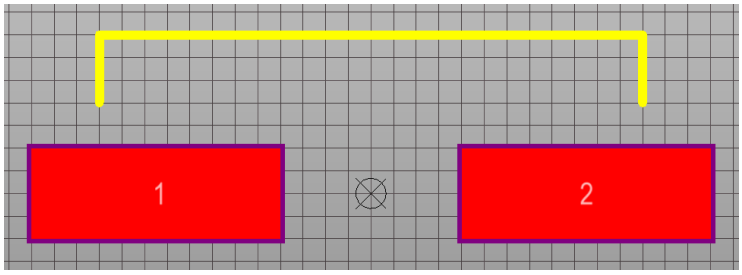


Figure 358. The top portion of the Overlay added to the component

23. We will Copy and Paste to add the lower silkscreen shape. In the PCB editor, Altium Designer requires a reference point for the copied object so that it can assign a local origin.
    a) Drag a selection rectangle around the yellow lines
    b) Press **Ctrl + C** to copy it to the clipboard.
    c) The cursor becomes a crosshair and a message is displayed at the bottom of the screen: *Select a reference point* as shown in Figure 359.
    d) Click on the Origin to set it as a reference point.

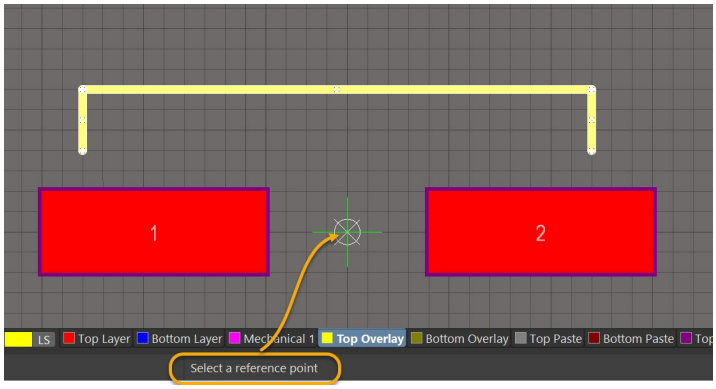


Figure 359. Copy the upper outline and set a reference point in preparation for pasting

e) Hit **Ctrl + V** to paste a copy of the outline. Notice that the cursor is attached outside of the horizontal line – the reference point you just selected.

f) Hit the **Spacebar** twice to rotate the objects, left-click to position the cursor on the origin so that the line is now symmetrically below the pads as shown in Figure 360. Left-click to place the lines.

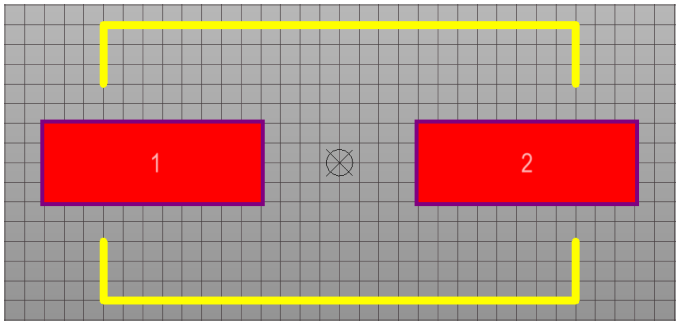


Figure 360. Copied outline

27.3.5 Add Pin 1 Marker

24. For soldering and debugging purposes, Pad 1 should also be visually marked. Use the command **Place » Full Circle** to add a circle near Pad 1. Click at **(-7.0, 2.0)** then move one grid unit to the right and click again to set the radius. Right-click to end the command once the circle is placed.
25. Double-click on the circle to open the *Properties* panel. Set the *Radius* to `0.12mm` and hit **Enter**. The component should now look like shown in Figure 361.

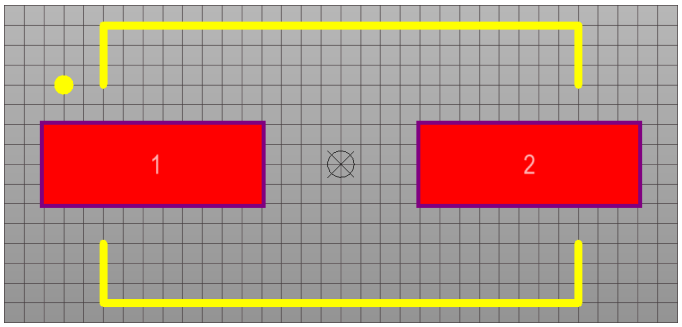


Figure 361. Footprint HC49SMD

26. Go to **File » Save** to save the library.

27.3.6 3D Crystal Body

29. Go to **Place » 3D Body** from the menu as shown in Figure 362.

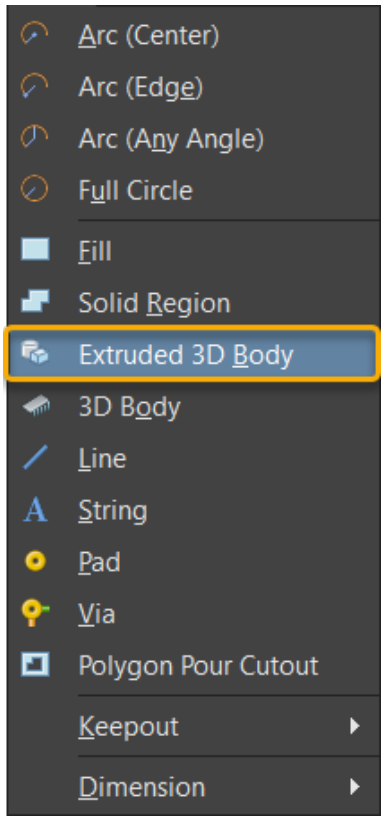


Figure 362. Extruded 3D Body

30. Hit the **Tab** key to open the *Properties* panel. Make the following changes as shown in Figure 363.
    a) Set *3D Model Type* to **Extruded.**
    b) In the *Identifier* field, enter: `HC49SMD`
    c) Choose a color for the extruded shape and change the opacity in the *Display* settings.
    d) Set *Overall Height* = `6mm`, *Standoff Height* = `1mm`.
    e) Press **Enter** or the Pause icon to start drawing the body.

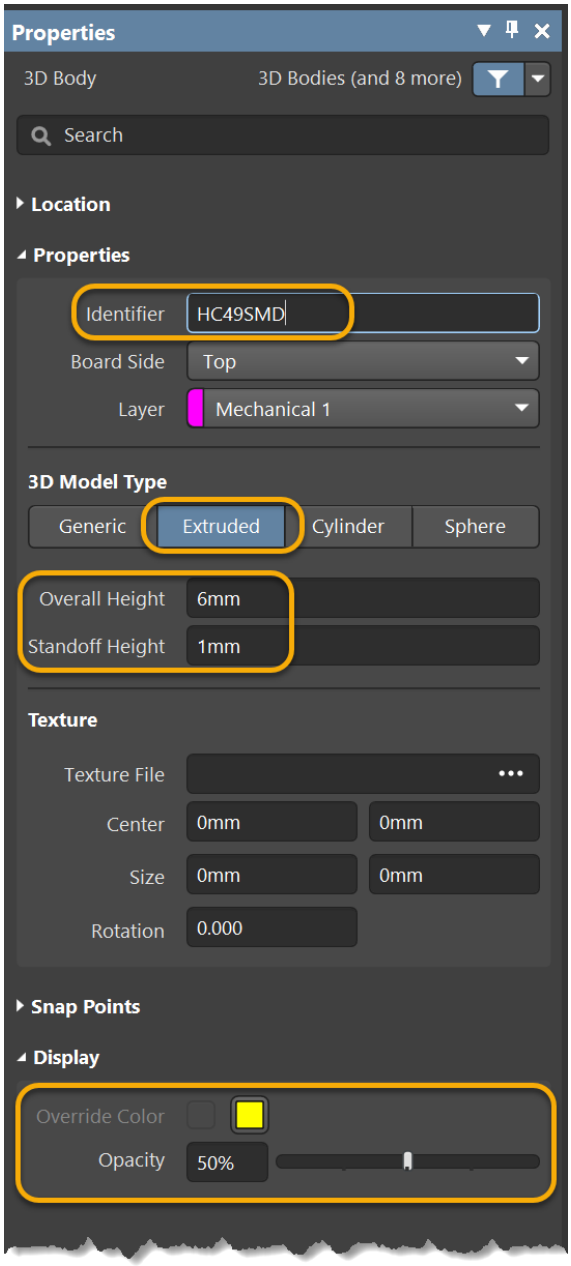


Figure 363. 3D Body Dialog for Crystal

31. Draw a rectangle to match the silkscreen outline in the editor.
    a) Similar to the placement of a polygon, click on each of the corner vertices.
    b) Right-click or hit the **ESC** key twice to finish the drawing of the body.
32. The 2D footprint should now look like Figure 364 in 2D, with the purple hatched lines indicating the 3D body.

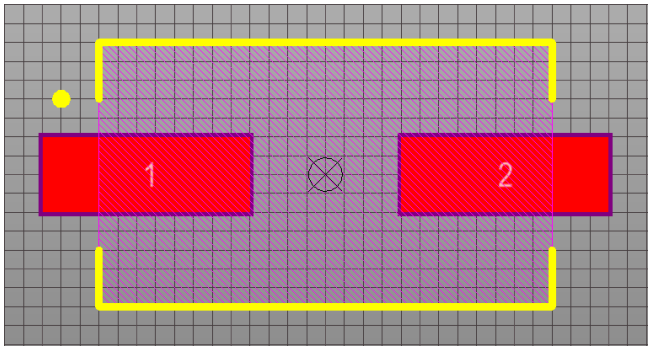


Figure 364. 2D Crystal Footprint

33. Hit the **3** key to view the body in 3D mode as shown in Figure 365. Hit the **2** key to go back into 2D mode.

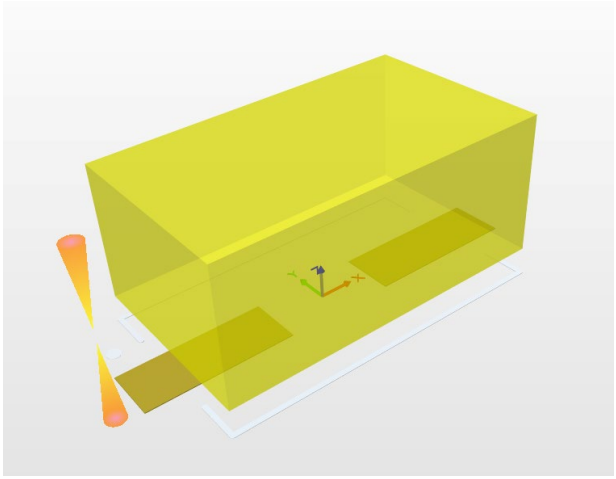


Figure 365. 3D Crystal Body and Footprint

34. Go to **File » Save** to save the library.

27.4 Use the Footprint

27.4.1 Installing the Footprint Library

35. To install your new PCB Library, open the *Components* panel by selecting **Components** from the **Panels** button in the bottom right corner of Altium Designer, or from **View » Panels » Components**.
36. Click the Hamburger button and choose **File-based Libraries Preferences…**.
37. Choose the *Installed* tab.
38. Click the **Install** button.
39. Navigate to the `My_Components` folder that you created and select the `My_Footprints.PcbLib` library.
40. Click the **Open** button. Your footprint library will now appear in the window as shown in Figure 366.

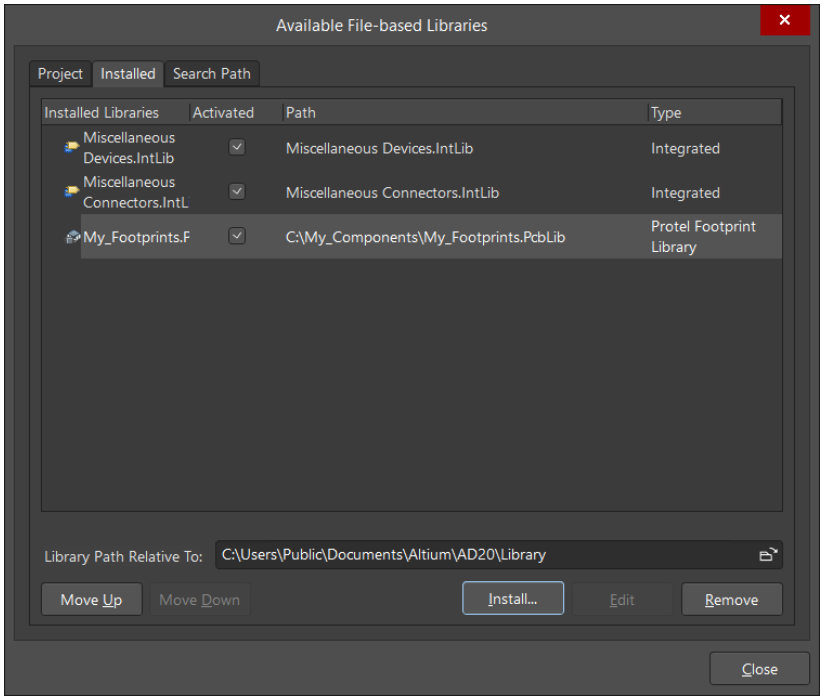


Figure 366. Available File-based Libraries dialog with My_Footprints.PcbLib added

41. Close the *Available File-based Libraries* dialog.

27.4.2 Adding Crystal Footprint to PCB Document

42. In the Components panel, click on the Hamburger icon and enable the **Footprints** checkbox as shown in Figure 367. left.

43. Select `My_Footprints.PcbLib` from the drop-down list as shown in as shown in Figure 367 right.

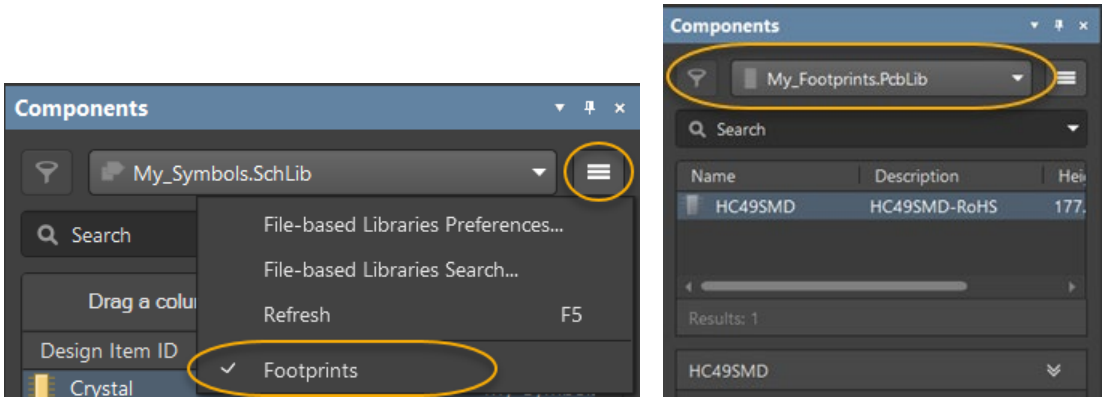


Figure 367. Enabling Footprints in the Components panel

44. Create a new PCB document by going to **File » New » PCB**.
45. From the *Components* panel, right-click on **HC49SMD** and place the footprint into the PCB.
46. Hit the **3** key to view the component in 3D and ensure that the 3D body is visible.
47. Close the PCB document without saving it.

Module 28: Component Creation

28.1 Purpose

In this exercise, students will learn to link the Crystal symbol with its footprint, both created in earlier exercises. A linked symbol and footprint is what we generally refer to as a unified component.

28.2 Linking Footprint to the Symbol

1. Open your `My_Symbols.SchLib` created earlier from the `My_Components` folder.
2. Open the Crystal symbol from the *SCH Library* panel. If you do not see the *SCH Library* panel, go to **View » Panels » SCH Library**.
3. In the *Model* pane at the bottom, click **Add Footprint** as shown in Figure 368.

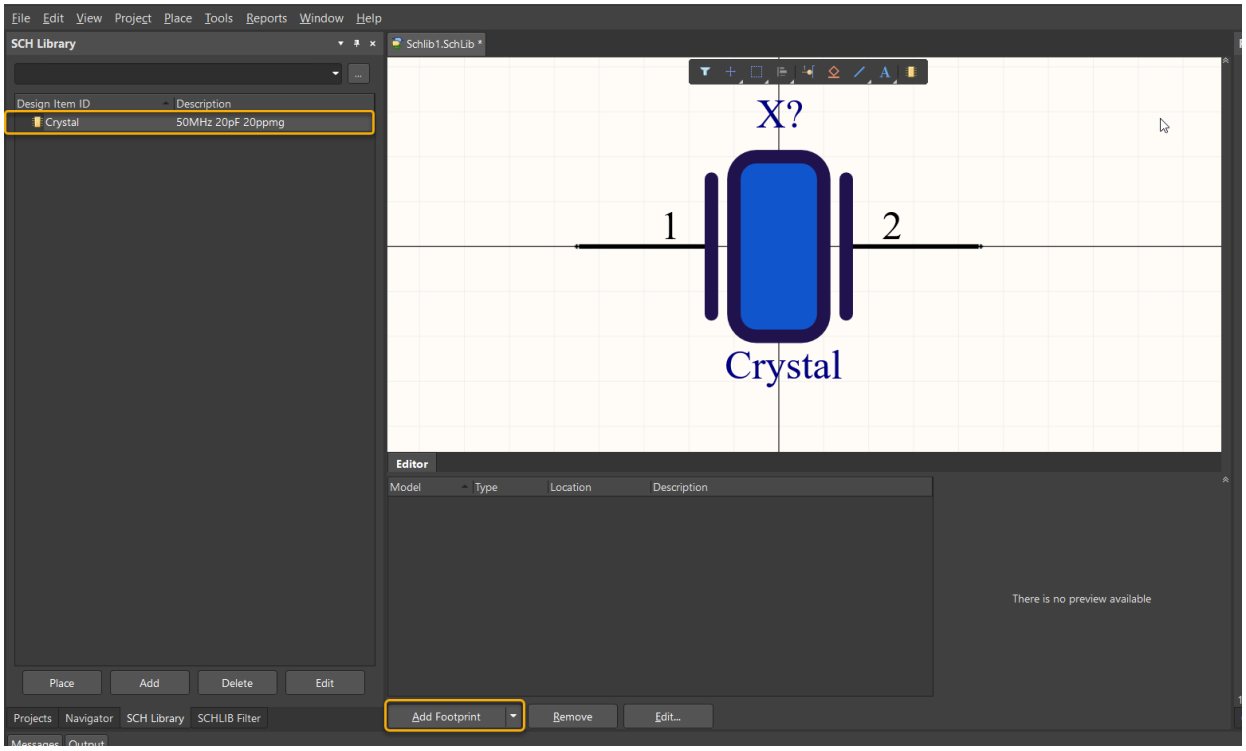


Figure 368. The Model pane in the Symbol Editor Workspace

4. The PCB Model dialog will open. Click **Browse…** and the *Browse Libraries* dialog will open as shown in Figure 369. Ensure to set `My_Footprints.PcbLib` as the active library. If not, click on the ••• button and install `My_Footprints.PcbLib`.
5. Select the footprint `HC49SMD`.

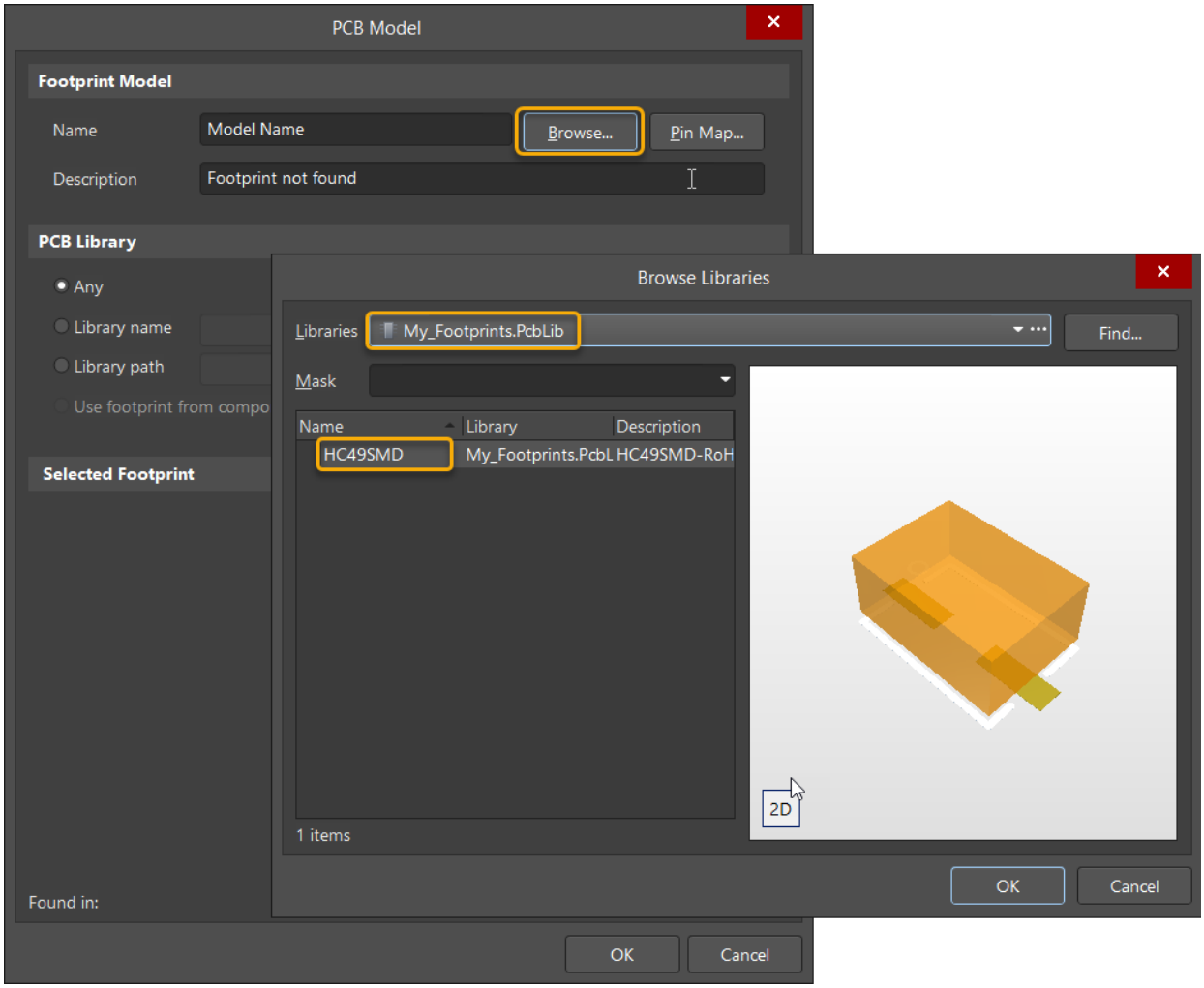


Figure 369. Link the HC49SMD footprint with the crystal symbol

In the *PCB Model* dialog, there are 3 PCB Library options (Any, Library name and Library path). If the Any option is chosen, it will search for the footprint that has matching name in all currently available libraries. Thus, it is recommended for either the library name or library path option to be used when working with passives or generic footprint names which may be duplicated in multiple installed PCB libraries.

6. Close the dialogs *Browse Libraries* and *PCB Model* with **OK**
7. Note the crystal footprint preview in the *Symbol Editor* and shown in Figure 370. The crystal footprint is now linked to the symbol.

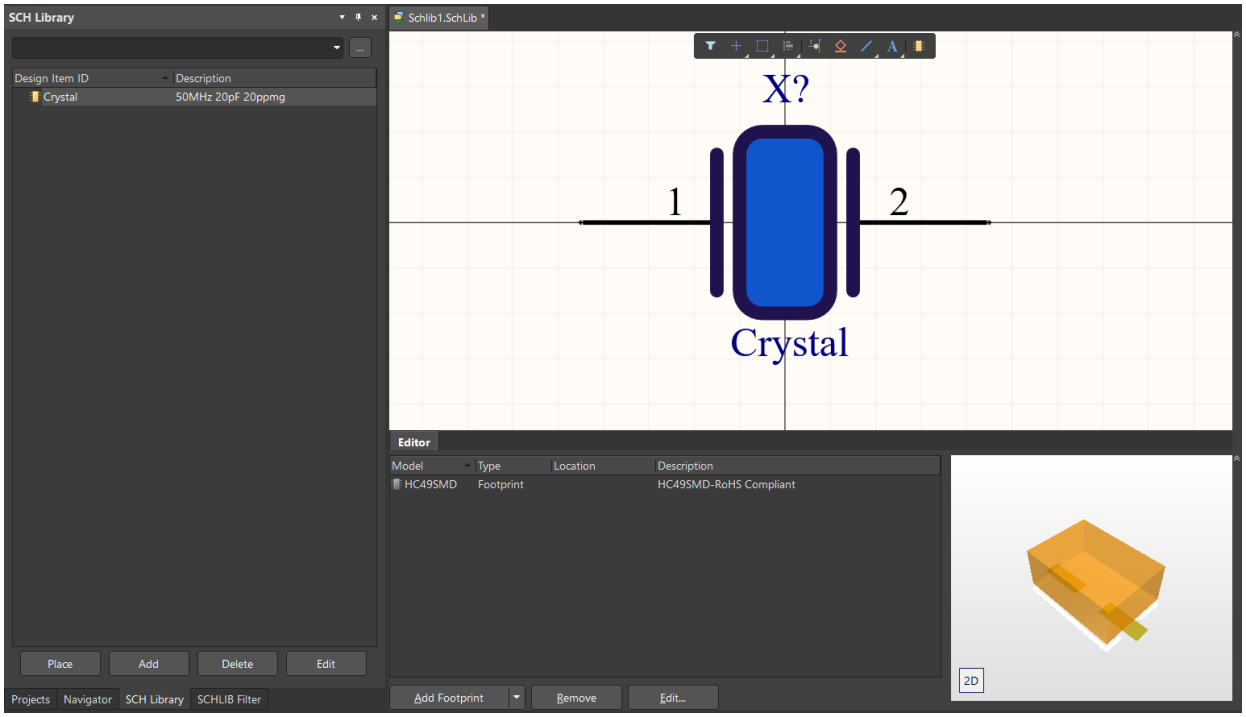


Figure 370. Symbol Editor with linked footprint

8. Save the schematic library by going to **File » Save**.
9. Close all documents.

28.3 Using Linked Component

Now we will test our linked component by adding it to a schematic, and then synchronize it with the PCB.

8. Create a new project by going to **File » New » Project**. Give it a name of your choice and click **Create.**
9. Right-click on newly created project from the *Projects* panel and add a new Schematic and a new PCB document.
10. Save the Project and its files by going to **File » Save All.**
11. Open the *Components* panel with the `My_Symbols.SchLib` active. You should see a preview for both the symbol and footprint of the crystal as shown in Figure 371

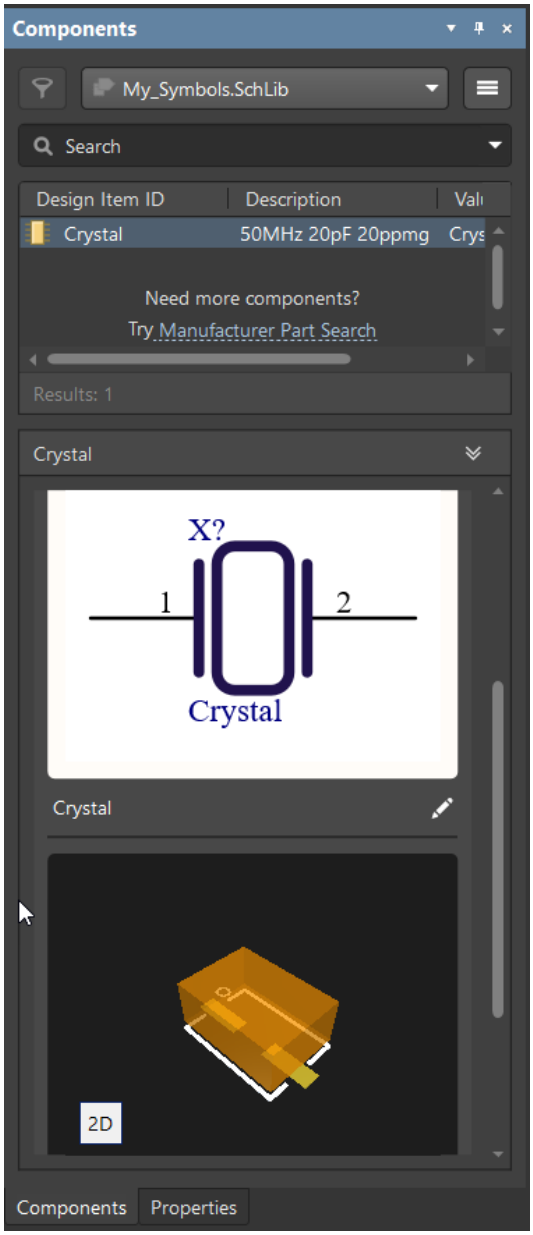


Figure 371. Components panel showing symbol and footprint for crystal component

12. With the schematic document active, right-click the Crystal from the *Components* panel and place it into the schematic. Right-click to end the component placement.
13. Annotate the component by going to **Tools » Annotation » Annotate Schematic Quietly.** Click **Yes** when prompted.
14. Update the PCB by going to **Design » Update PCB Document**. Disable the **Add Rooms** checkbox and click **Execute Changes** in the ECO. Click **Close** to close the *ECO* dialog.
15. Now the crystal footprint is visible in the PCB as shown in Figure 372.

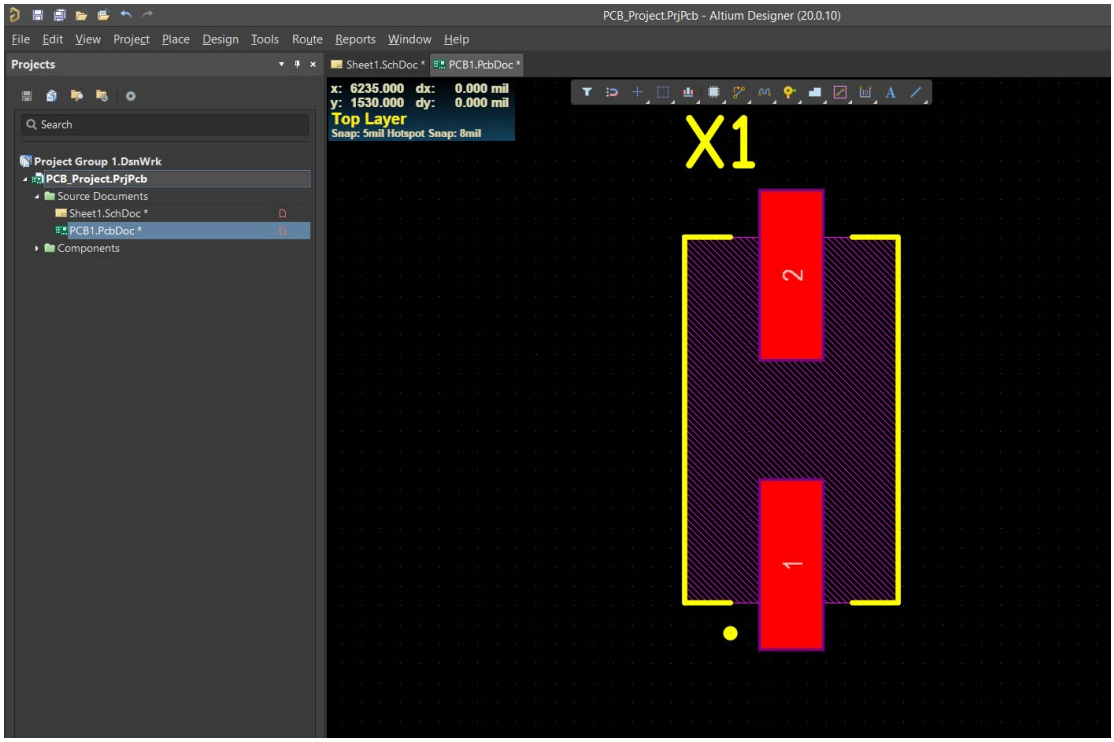


Figure 372. 2D Crystal in PCB Editor

15. Change to 3D view by hitting the **3** key and rotate the view by holding **Shift** and the **right mouse button** see Figure 373.

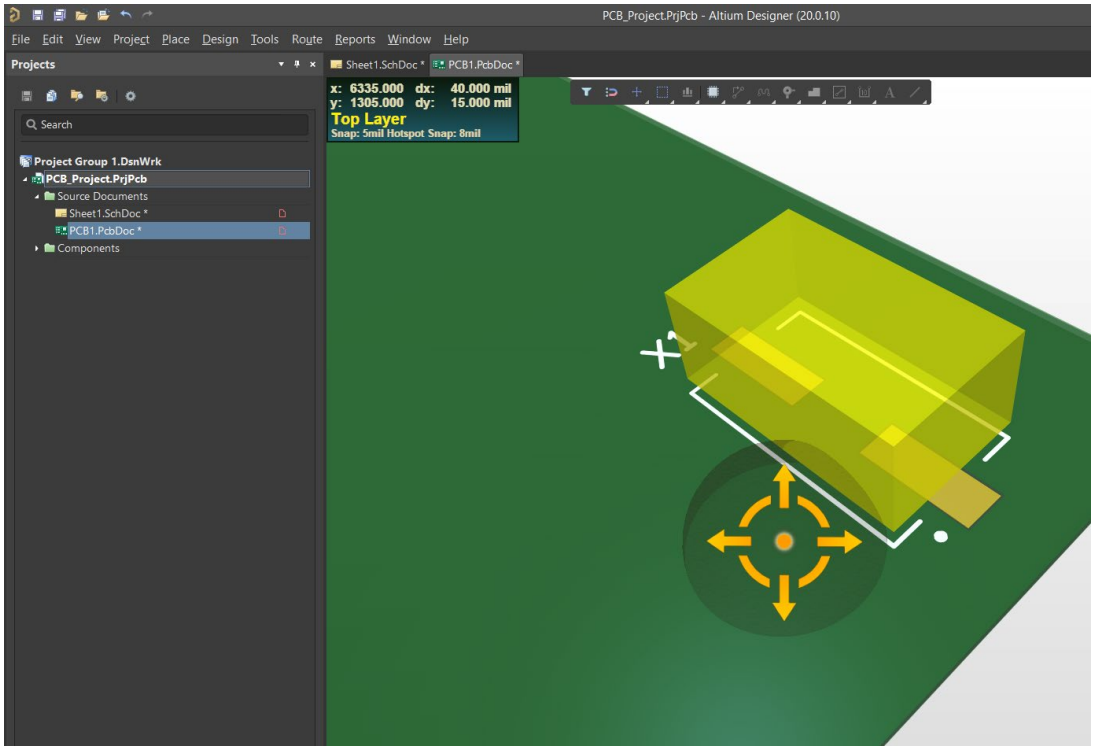


Figure 373. 3D Crystal in PCB Editor

16. Close all files without saving, including the project in the *Projects* panel.

Module 29: Integrated Library

29.1 Purpose

In this exercise, you will recreate the WCT Integrated Library by creating a library package project.

29.2 Preparation

1. Close all open projects and documents. For this exercise, we will be using files located in the `AD20 Essentials Exercises\Module 29 - Integrated Library Project` folder.

29.3 Create Integrated Library Project

2. Create an Integrated Library Package Project using the menus:
   File » New » Library » Integrated Library as in Figure 374. A new Integrated Library Project is created and added to the *Projects* panel.

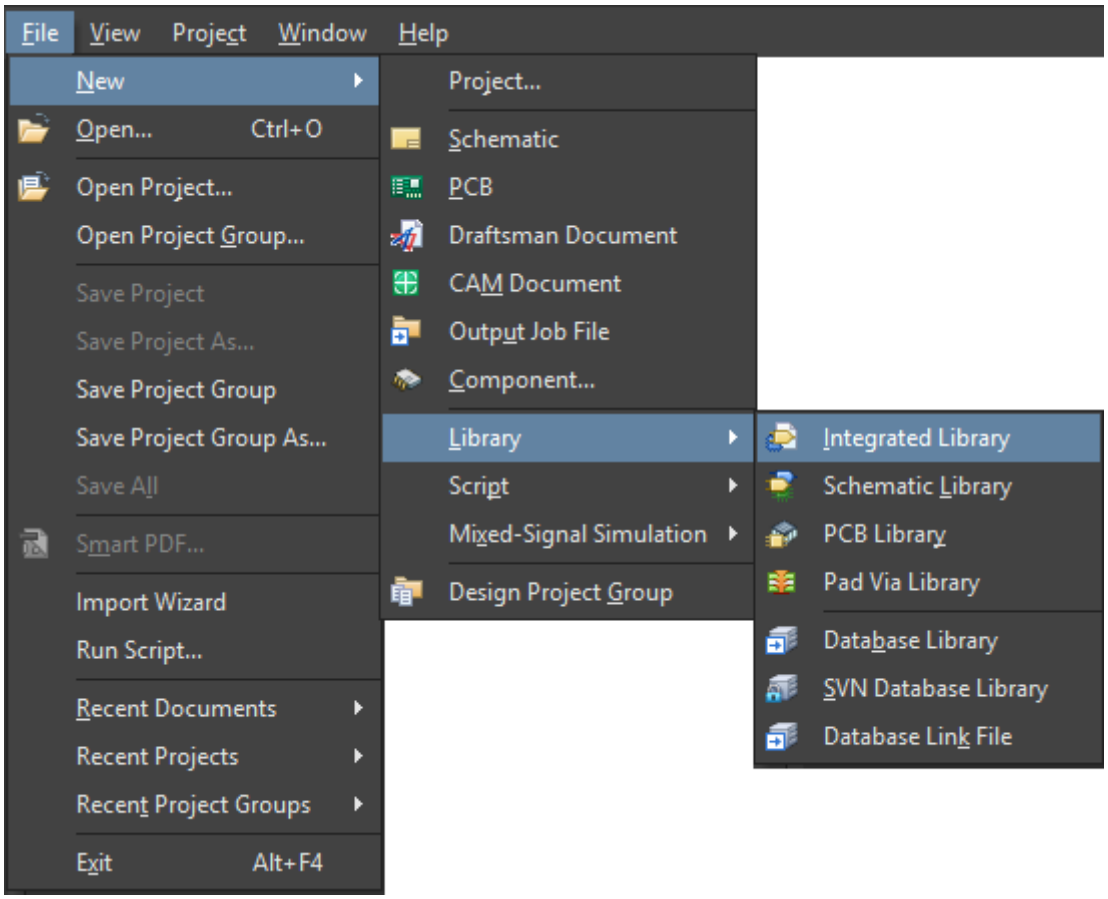


Figure 374. Create project dialog

3. Add the `WCT.SchLib` and `WCT.PcbLib` libraries
   a) Right-clicking on the `WCT.LibPkg` from the *Projects* panel, and **Add Existing to Project...** .
   b) Browse to the `Module 29 - Integrated Library Project` training exercise folder to find both the `WCT.SchLib` and `WCT.PcbLib` libraries.

Be sure that the drop-down filter in the bottom right is set to `All files.`
It is possible to select both using **Ctrl + Left-click**.

4. Save the project and give it a name of `WCT.` It should look like Figure 375.

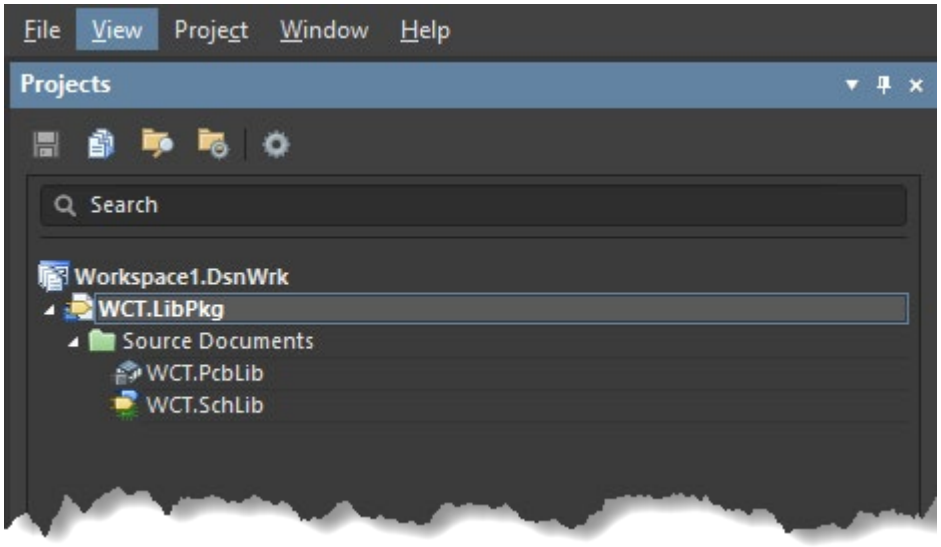


Figure 375. WCT.LibPkg Project

5. Compile the project by right-clicking the `WCT.LibPkg` from the *Projects* panel and **Compile Integrated Library WCT.LibPkg**.
6. The *Messages* panel will automatically appear if there are any errors. In this case, there is an intentional error named:
   Cap10uf25V_0603: Could not find port '3' on model '0603C_L' for pin '3'. (Figure 376).

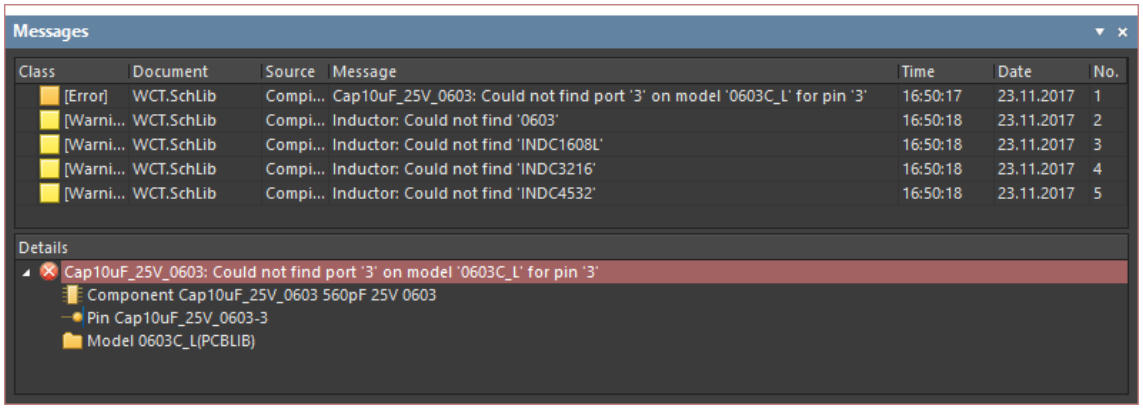


Figure 376. Error highlighted in Messages panel

7. Double-click the error in the *Messages* panel to jump to the violating symbol. From the *Properties* panel, notice that the *Pins* tab shows that the symbol has `Pin 1` and `Pin 3`, as shown in Figure 377.

If you have closed the *Messages* panel, it can be reopened from **View » Panels » Messages**. To open the *Properties* panel, use the menus **View » Panels » Properties** and select that symbol.

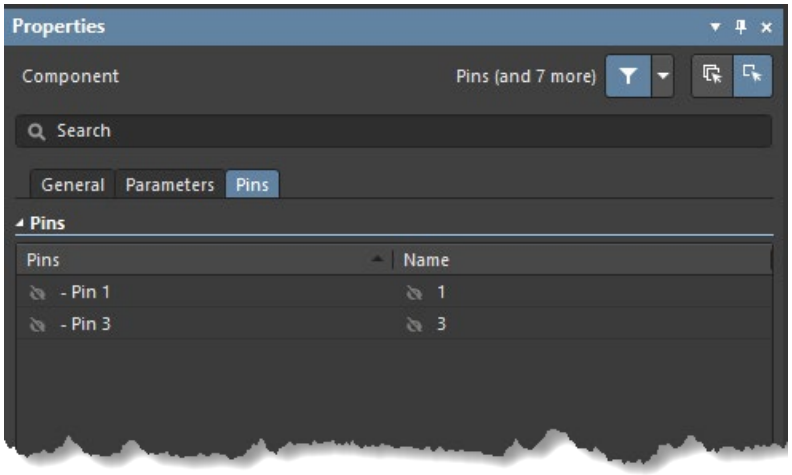


Figure 377. Pin information of the schematic symbol in the SCH library

8. From the *Projects* panel, double-click to open the `WCT.PcbLib`.
9. From the *PCB Library* panel, select the `0603C_L` footprint.
10. Notice that the footprint has `Pad 1` and `Pad 2` which do not match the schematic `Pin 1` and `Pin 3` of `Cap10uF_25V_0603` from the schematic library.
11. Set the `WCT.SchLib` as the active document and `Cap10uF_25V_0603` as the active symbol.
12. Select `Pin 3` in the workspace (pin on the right-side) and change the value for the Designator and Name to `2` as shown in Figure 378, so that it matches the pads of the `0603C_L` footprint.

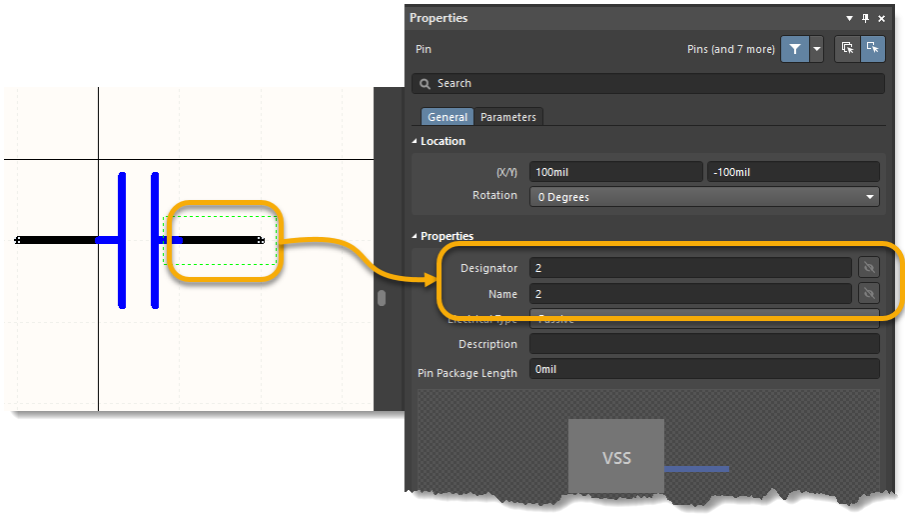


Figure 378. Pin properties

13. Save the `WCT.SchLib`.
14. Compile the `WCT.LibPkg` again by right-clicking on the project name in the *Projects* panel and select **Compile Integrated Library WCT.LibPkg**.

If there are no errors in the library, the Messages panel will not automatically appear. The messages panel can be opened and reviewed from **View » Panels » Messages**.

15. Notice that the error `Could Not Find Port` error has been resolved in the *Messages* panel but, there are still some warnings for `Could Not Find Footprint`, as shown in Figure 379.

16. Extra footprints have been linked to the symbols shown in these warnings, but the footprints were not included in the PCB library.

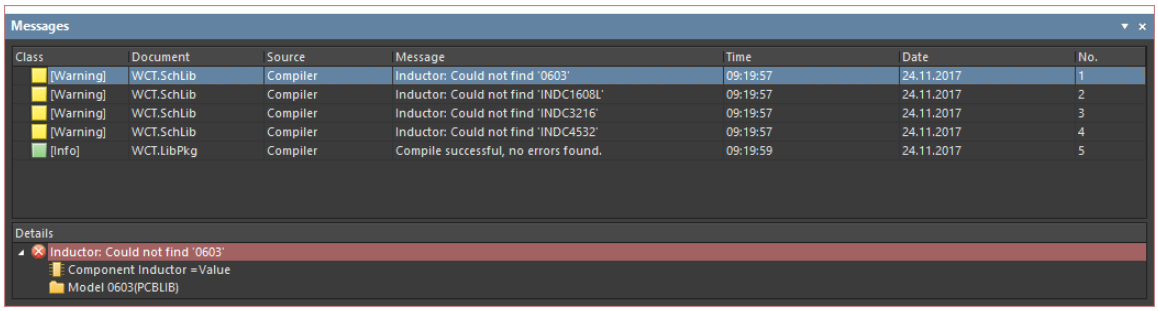


Figure 379. Removing extra models from the Models area of the Schematic Library Editor

17. Double-click each `Inductor: Could not find <Name>` warning in the *Messages* panel to jump to the symbol with the invalid footprint link as shown in Figure 380.
18. Remove the footprint model that is causing the Warnings, by pressing the **Remove** button. In this case, it's the `INDCxxxx` and `0603` footprint models.

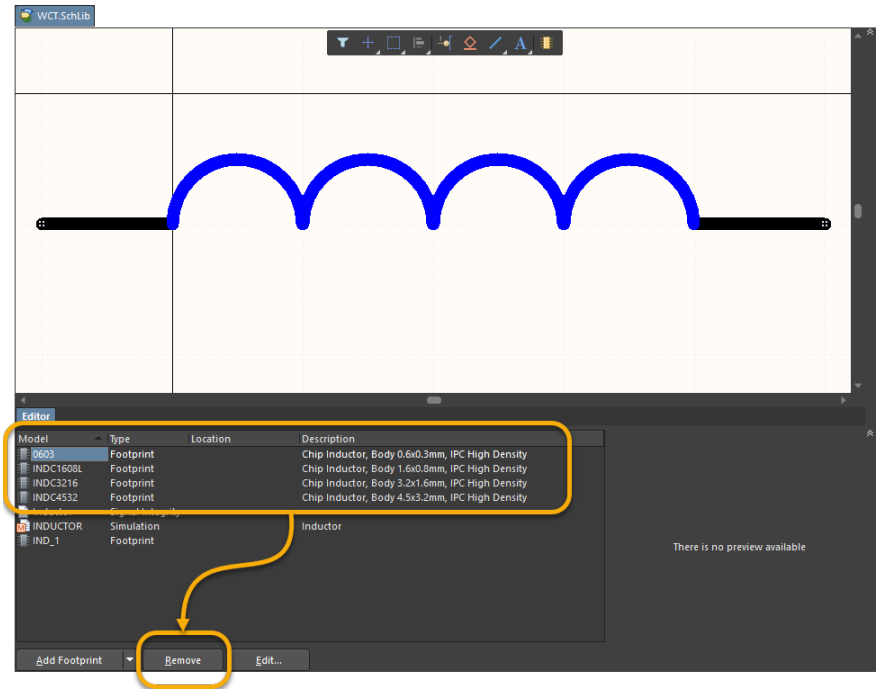


Figure 380. Remove links for footprints from SCH Library

19. Save the schematic library.
20. Compile the project again by selecting **Project » Compile Integrated Library WCT.LibPkg**.
21. Open the *Messages* panel, and you'll notice the errors & warnings associated with that component are no longer present.
22. Verify the Integrated Library has been generated in the `Integrated Libraries\Project Outputs for WCT` folder, or wherever the `WCT.LibPkg` was saved.
    a) Right-click on the project name in the *Project* panel and select **Explore** to browse the folder.
    b) Navigate to the Project Outputs subfolder to verify that it contains the `WCT.IntLib` file.

29.4 Using the New Library

When opening an Integrated Library, a prompt to **Extract Sources** or **Install Library** will open.

23. Close all projects and files in the *Projects* panel.
    a) Select **File » Open**.
    b) Navigate to the `Project Outputs for WCT` folder.
    c) Select the `WCT.IntLib` file and click **Open**. The popup window shown in Figure 381 will appear.
    d) Select **Install Library**. Note: The library will already have been installed as it gets added during the compile process.

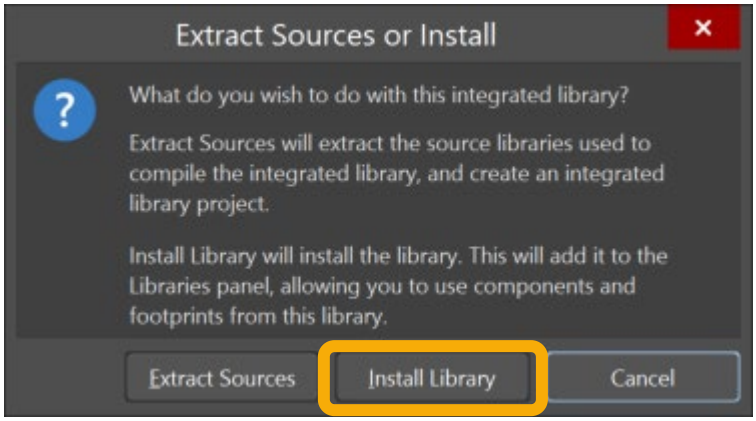


Figure 381. Opening the WCTopping.IntLib

If you select the **Extract Sources** button when you open an integrated library, Altium will create a new library project and decompile the file into the SchLib and PcbLib files, allowing you to edit the libraries.
You can then recompile these edited files to update the IntLib.

24. Open the *Components* panel and select the new library from the drop-down list. You will see the symbol and footprint previews and these components would be ready to use.

Congratulations on completing the Altium Designer Essentials Course

Thank you for choosing Altium Designer